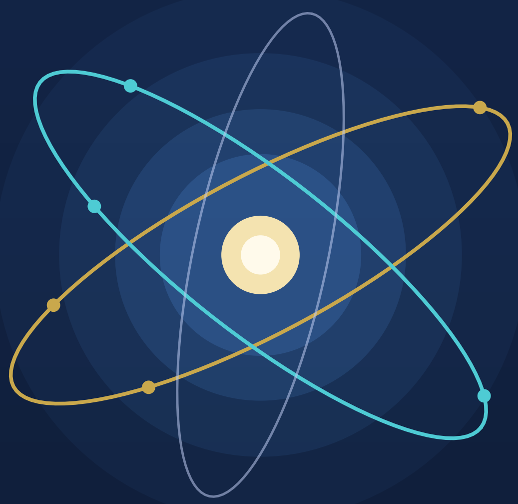

Nuclear and Particle Physics

Bharat Kumar

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Preface

This book grew out of the nuclear and particle physics course taught by the author. It expands each lecture into a self-contained chapter that can be read as a continuous narrative or used chapter by chapter alongside a formal course. The aim is straightforward: to help students build a clear physical picture of the nucleus and of the processes that govern it, without losing the mathematical tools they need for examinations, research projects, and further study.

The notes are written for a broad student audience. Advanced undergraduates who are meeting nuclear structure, radioactivity, and reactions for the first time should find a guided path from simple questions (how large is a nucleus? why is the binding energy per nucleon nearly constant?) to the standard results of the field. Postgraduate students and early researchers will find enough depth in the derivations, worked estimates, and thematic arcs (semi-empirical mass formula, shell model, α, β, and γ decay, fission and fusion) to use the book as a companion when revising fundamentals or preparing for more specialized reading. Wherever possible the presentation is intuition first and formalism second: the physical question is posed before the algebra is developed in detail.

The material follows a natural progression. Early chapters establish nuclear size, density, and binding. The semi-empirical mass formula and mass parabolas come next, followed by the deuteron and the nucleon–nucleon force, scattering, and meson exchange. The nuclear shell model, radioactivity, and nuclear reactions occupy the middle of the book; fission and fusion close the arc. Figures from the lecture slides are retained where they help, and historical notes with portraits appear when the origin of an idea is part of understanding it.

About the author. Bharat Kumar teaches nuclear and particle physics and works on the theory of dense nuclear matter and neutron stars. These notes reflect that classroom experience: they try to keep the student in view at every step, whether the reader is meeting the subject for the first time or returning to it with a research problem in hand.

Contents

Chapter 1

Nuclear Size and Charge

"Like the radius of an atom, the radius of a nucleus is not a precisely defined quantity."
K. S. Krane

1.1 Posing the problem: how big is a nucleus?

Nuclei are not hard spheres with sharp edges. The density of nucleons and the nuclear potential are relatively constant over short distances and then drop rapidly toward zero. It is therefore natural to characterise the nuclear shape with two parameters: a **mean radius**, where the density is half its central value, and a **skin thickness**, over which the density falls from near its maximum to near zero. A third parameter is needed later for nuclei that are not spherical [6].

A further subtlety is that the radius one measures depends on the experiment. Electromagnetic probes (electron scattering, muonic X rays, optical and X-ray isotope shifts) determine the distribution of *nuclear charge*, mainly the protons. Hadronic probes (α scattering, pionic atoms) determine the distribution of *nuclear matter*, all nucleons. The two distributions are similar but not identical, especially in heavy nuclei with neutron skins [6, 1].

The nuclear size scale is the femtometre (fermi),

$$1\,\mathrm{fm} = 10^{-15}\,\mathrm{m} = 10^{-13}\,\mathrm{cm}. \tag{1.1}$$

Empirically, charge radii satisfy

$$R \simeq r_0\, A^{1/3}, \qquad r_0 \approx 1.2\text{–}1.25\,\mathrm{fm}, \tag{1.2}$$

from electron scattering. That $A^{1/3}$ law is the geometric face of nearly constant central density: if $A/(\frac{4}{3}\pi R^3)$ is roughly constant, then $R \propto A^{1/3}$.

Definition

The constant r_0 depends on the surface definition. Common conventions give

$$r_0 \approx \begin{cases} 1.2 - -1.25\,\mathrm{fm} & \text{(charge / equivalent hard sphere)}, \\ 1.3 - -1.4\,\mathrm{fm} & \text{(strong interaction / } \alpha \text{ scattering)}. \end{cases}$$

Throughout this chapter, $r_0 \approx 1.2\,\text{fm}$ is used for charge radii unless stated otherwise.

Key idea

Wavelength criterion. Two related conventions must not be mixed. The de Broglie wavelength is $\lambda_{\text{dB}} = h/p = 2\pi\hbar/p$, whereas the reduced wavelength $\bar{\lambda} = \hbar/p$ sets the phase-resolution scale. The invariant criterion is $q\ell/\hbar \gtrsim 1$, where q is the momentum transferred to the nucleus. Requiring the stricter optical-style $\lambda_{\text{dB}} \lesssim 10\,\text{fm}$ gives $p \gtrsim 124\,\text{MeV/c}$. High-energy electron beams (100 MeV–1 GeV) provide exactly that [6].

Before writing a single scattering amplitude, it helps to recall why ordinary light cannot answer the size question.

1.1.1 A classroom analogy: chalk and wavelengths

Imagine a stick of chalk, a few centimetres long (Figure 1.1). With the naked eye (that is, with visible light), one can estimate its diameter without effort. Sound waves of ordinary frequencies cannot do the same job. Both light and sound are waves; the difference is their wavelength relative to the object being examined.

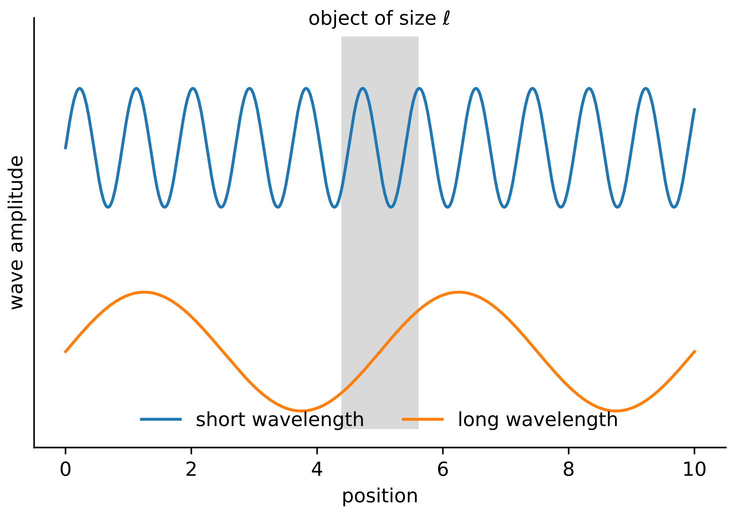


Figure 1.1: Resolution schematic. A short wavelength records structure on the scale ℓ; a much longer wavelength averages over it. Visible light (400–700 nm) has a wavelength far smaller than the diameter of the chalk, so the eye can resolve its size. Audible sound does not.

Visible light spans roughly 400–700 nm, four to seven orders of magnitude smaller than a centimetre-scale object. Audible sound, by contrast, lives in a very different regime. The wavelength of a wave of speed v and frequency ν is

$$\lambda = \frac{v}{\nu}. \tag{1.3}$$

Taking $v \approx 340\,\text{m/s}$ for air and $\nu = 1\,\text{kHz}$ as a representative audible tone, one finds $\lambda \approx 0.34\,\text{m}$—about thirty times larger than the chalk. A wave that long cannot "see" structure on the scale of a centimetre; it simply diffracts around the obstacle as if the obstacle were a point.

Key idea

To resolve a feature of size ℓ, a probe needs wavelength $\lambda \lesssim \ell$. Visible light works for chalk because $\lambda_{\text{light}} \ll \ell_{\text{chalk}}$. Audible sound fails because $\lambda_{\text{sound}} \gg \ell_{\text{chalk}}$. The same criterion decides whether electrons can resolve a nucleus.

The same logic, applied to the nucleus, is unforgiving. Early estimates from α-particle scattering already indicated nuclear radii of order a few fermis (10^{-15}–10^{-14} m). Visible light ($\lambda \sim 10^{-7}$ m) is hopelessly coarse. Even hard γ-rays fall short of the needed resolution. One needs a probe whose de Broglie wavelength is itself of nuclear size (a few fermis or less). High-energy electrons provide precisely that.

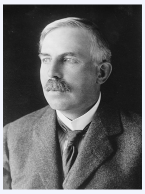
Remark

Electrons are preferred over protons or α particles for precision *charge*-density work because the electron–nucleus interaction is electromagnetic and calculable to high accuracy. Strong-interaction probes, while useful for hadronic form factors and neutron distributions, introduce model dependence that electron scattering largely avoids [2, 7]. Mass and charge radii need not agree at the percent level: Coulomb repulsion tends to push protons slightly farther out than neutrons in heavy nuclei.

1.2 Historical path: from Rutherford to Hofstadter

1.2.1 Rutherford scattering and a first radius

Historical note: Ernest Rutherford (1871–1937)



Ernest Rutherford was born in Brightwater, New Zealand, and trained at Canterbury College before moving to the Cavendish Laboratory under J. J. Thomson. After work on radioactivity in Montreal (where he classified α, β, and γ radiation and, with Soddy, formulated radioactive transformation), he took the chair at Manchester in 1907. There, with **Hans Geiger** and **Ernest Marsden**, he directed the α-particle scattering experiments that overthrew the Thomson "plum pudding" atom. In 1911 he proposed that essentially all the atomic mass and positive charge sit in a minute central *nucleus*, with electrons in the surrounding volume. He received the 1908 Nobel Prize in Chemistry for investigations into the disintegration of the elements and the chemistry of radioactive substances, ironically awarded just as his nuclear atom was taking shape. Later, at the Cavendish, he directed the laboratory in which Chadwick discovered the neutron (1932). Rutherford remains the founding figure of nuclear physics: every scattering formula in this chapter descends, conceptually, from his 1911 insight.

Rutherford's analysis of α-particle scattering established that the positive charge of the atom is concentrated in a tiny central nucleus. As long as the projectile remains outside the range of the nuclear force and samples only the Coulomb field of a point charge Ze, the classical differential cross section is

$$\left(\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}\right)_{\mathrm{R}} = \left(\frac{Z_1 Z_2 e^2}{8\pi\varepsilon_0\, m v_0^2}\right)^2 \frac{1}{\sin^4(\theta/2)} = \left(\frac{Z_1 Z_2 e^2}{16\pi\varepsilon_0\, E}\right)^2 \frac{1}{\sin^4(\theta/2)}. \tag{1.4}$$

The distance of closest approach for a head-on collision is

$$d_0 = \frac{Z_1 Z_2 e^2}{4\pi\varepsilon_0\, E}, \tag{1.5}$$

and for a general impact parameter the periapsis d and scattering angle are related by

$$d = \frac{d_0}{2}\left(1 + \frac{1}{\sin(\theta/2)}\right). \tag{1.6}$$

When d becomes smaller than a critical value d_N, the measured angular distribution begins to deviate from Equation (1.4). That deviation has two physical origins: (i) the finite spatial extent of the nuclear charge (only charge inside radius d contributes fully to the deflection), and (ii) the onset of the short-range strong interaction.

Historical note: James Chadwick (1891–1974)

Sir James Chadwick studied under Rutherford at Manchester and later worked with Hans Geiger in Berlin before internment in Germany during World War I. Returning to Cambridge, he became Rutherford's close collaborator at the Cavendish. In 1920 he systematically analysed α-particle scattering from a range of targets and helped establish that nuclear radii scale roughly as $A^{1/3}$, the first quantitative size law for nuclei. His lasting fame, however, comes from 1932: interpreting the penetrating radiation that Bothe, Becker, Joliot and Curie had attributed to hard γ-rays, Chadwick showed that it consisted of neutral particles of roughly proton mass: the *neutron*. That discovery completed the proton–neutron picture of the nucleus and earned him the 1935 Nobel Prize in Physics. Without the neutron, the charge densities we extract from electron scattering could not be converted into nucleon densities via A/Z.

Chadwick and later workers analysed such deviations for a series of targets and established the empirical law

$$R_N \simeq r_0\, A^{1/3}, \tag{1.7}$$

with $r_0 \sim 1.3\,\text{fm}$ from α scattering, slightly larger than the electromagnetic value, as expected when the strong force sets the scale [2].

Caution

As θ increases on $(0, \pi)$, $\sin(\theta/2)$ *increases*, so $\csc(\theta/2)$ *decreases*. The Rutherford cross section therefore falls with angle because of $1/\sin^4(\theta/2)$, not because $\sin(\theta/2)$ shrinks. A common slip in lecture notes reverses this statement.

1.2.2 Why electrons?

α particles and protons feel the poorly known nuclear force as soon as they penetrate the surface. Electrons do not: they couple only through the electromagnetic interaction, which is known to exquisite accuracy. High-energy electron scattering, pioneered by Robert Hofstadter and collaborators at Stanford in the 1950s, therefore became the definitive probe of nuclear charge distributions [7, 8, 2].

Historical note: Robert Hofstadter (1915–1990)

Robert Hofstadter was born in New York City and earned his Ph.D. at Princeton (1938) under Louis Ridenour and others, working on photoconductivity and infrared detectors. After wartime radar research and a period at Princeton, he joined Stanford University in 1950 as Associate Professor. There he seized the opportunity offered by the newly available high-energy electron beams from the Stanford linear accelerator (Mark III and related machines). From about 1953 onward, elastic and inelastic electron scattering became his principal research programme.

Directing a high-energy electron beam onto thin targets and measuring the angular distribution of scattered electrons, Hofstadter's group mapped the *charge form factors* of nuclei from light systems (carbon, oxygen) to heavy ones (gold, bismuth), and then of the proton and neutron themselves. The data revealed that nuclei have a roughly constant central density and a diffuse surface (the empirical basis of the Woods–Saxon profile), and that the proton is not a point particle but an extended object of rms radius $\sim 0.8\,\mathrm{fm}$. The term *fermi* (fm) for 10^{-15} m entered common use in this community; Hofstadter himself used it in his 1961 Nobel lecture, "The electron-scattering method and its application to the structure of nuclei and nucleons" (11 December 1961).

He shared the 1961 Nobel Prize in Physics with Rudolf Mössbauer: Hofstadter "for his pioneering studies of electron scattering in atomic nuclei and for his consequent discoveries concerning the structure of nucleons." He remained at Stanford for the rest of his career, was elected to the U.S. National Academy of Sciences (1958), and later received the National Medal of Science (1986). The oxygen data we discuss below (Phys. Rev. **113**, 666, 1959, with Ehrenberg, Meyer-Berkhout, Ravenhall and Sobottka) are a canonical product of that Stanford programme [7, 8].

1.3 The de Broglie condition and the energies required

Historical note: Louis de Broglie (1892–1987)

Prince Louis-Victor de Broglie, of the French noble house of Broglie, was trained first in history before turning to theoretical physics under his brother Maurice and later Paul Langevin. In his 1924 doctoral thesis at the Sorbonne, he proposed that every material particle of momentum p is associated with a wave of wavelength $\lambda = h/p$, extending Einstein's wave–particle duality of light to matter. The idea seemed speculative until Davisson and Germer (1927), and independently G. P. Thomson, observed electron diffraction by crystals, confirming the de Broglie relation experimentally.

De Broglie received the 1929 Nobel Prize in Physics for the discovery of the wave nature of electrons. His relation is the conceptual starting point of this chapter: only when λ_{dB} is of nuclear size (a few fm) can an electron beam resolve the charge distribution inside a nucleus. High-energy electron accelerators are, in that sense, machines built to make de Broglie's wavelength short enough.

Quantum mechanics equips every particle of momentum p with a wavelength

$$\lambda = \frac{h}{p} = \frac{2\pi\hbar}{p} \equiv \lambda_{\mathrm{dB}}. \tag{1.8}$$

Large momentum means short wavelength. For electrons accelerated to kinetic energies well above their rest energy, the kinematics become ultra-relativistic, and the connection between energy and wavelength is especially simple.

1.3.1 Relativistic kinematics

The total energy E of an electron of rest mass m_e and momentum p is

$$E = \sqrt{p^2c^2 + m_e^2c^4}. \tag{1.9}$$

The kinetic energy controlled by the accelerator is the excess over rest energy:

$$K = E - m_ec^2 = \sqrt{p^2c^2 + m_e^2c^4} - m_ec^2. \tag{1.10}$$

Caution

The rest energy is $m_ec^2 \approx 511\,\text{keV}$, *not* $m_e^2c^2$. The latter does not even have the dimension of energy. Always write $K = E - m_ec^2$.

For electrons, once $K \gg m_ec^2$—true already at a few tens of MeV—the rest mass may be neglected and

$$E \approx K \approx pc. \tag{1.11}$$

The de Broglie wavelength then collapses to

$$\lambda_{\text{dB}} \approx \frac{hc}{E} = \frac{2\pi\,\hbar c}{E}. \tag{1.12}$$

In nuclear units the conversion constants

$$hc \approx 1240\,\text{MeV}\cdot\text{fm}, \qquad \hbar c \approx 197.3\,\text{MeV}\cdot\text{fm} \tag{1.13}$$

are indispensable. Thus, to resolve structure on the scale of 10 fm,

$$E \approx \frac{1240\,\text{MeV}\cdot\text{fm}}{10\,\text{fm}} = 124\,\text{MeV}. \tag{1.14}$$

Example: electrons at 500 MeV

For $E = 500\,\text{MeV}$ one has $\lambda_{\text{dB}} \approx 2.48\,\text{fm}$, already small compared with the diameter of a heavy nucleus [2]. Using Equation (1.12),

$$\lambda_{\text{dB}} \approx \frac{1240\,\text{MeV}\cdot\text{fm}}{500\,\text{MeV}} = 2.48\,\text{fm}, \qquad \bar{\lambda} = \frac{\hbar c}{E} \approx 0.395\,\text{fm}.$$

The two numbers describe different, explicitly named quantities; they must not both be called λ.

Key idea

For phase resolution, choose a momentum transfer $q \gtrsim \hbar/\ell$; for the stricter full-wavelength criterion use $p \gtrsim h/\ell$. Thus resolving $\ell \sim 1\,\text{fm}$ requires momentum transfers of a few hundred MeV/c, while $\lambda_{\text{dB}} \lesssim 1\,\text{fm}$ requires $E \sim 1.2\,\text{GeV}$. Modern facilities (Jefferson Lab, MAMI, formerly SLAC) operate comfortably in this regime.

1.4 Elastic electron scattering: the oxygen example

A classic data set is the elastic scattering of 420 MeV electrons from oxygen, shown in Figure 1.2 [8, 7].

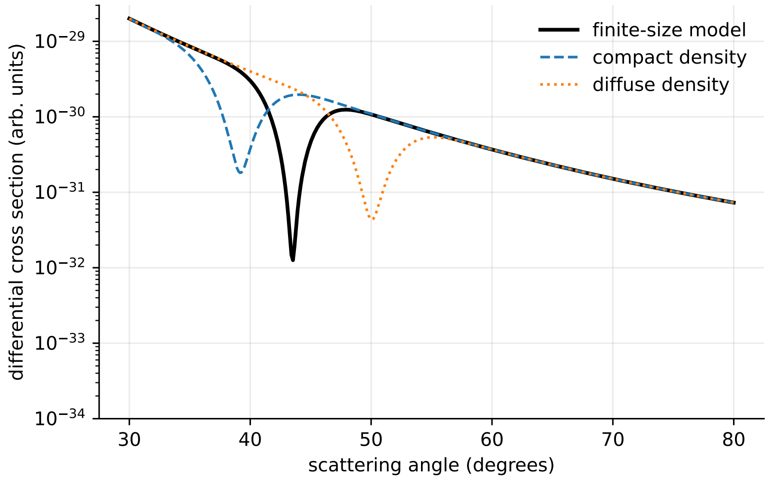


Figure 1.2: Differential cross section for elastic electron scattering from oxygen at 420 MeV, compared with harmonic-well charge distributions (a)–(c). The data display a clear diffraction minimum near 43°–44°. Original redraw of the trend in Ehrenberg et al., Phys. Rev. **113**, 666 (1959); the model curves are schematic rather than digitised data.

What does one actually measure? A beam of monoenergetic electrons strikes a thin target. Detectors placed at laboratory angle θ count the scattered electrons. The observable is the differential cross section $d\sigma/d\Omega$, proportional to the number of electrons scattered into a solid angle $d\Omega$ about θ.

Two features of Figure 1.2 are immediate.

1. **Overall fall-off.** The cross section drops by several orders of magnitude as θ increases from $\sim 30°$ to $\sim 80°$.

2. **Diffraction structure.** Superimposed on the fall-off is a deep minimum near 43°–44°, followed by a secondary maximum and a further decline.

The first feature is the relativistic generalisation of Rutherford scattering (Mott scattering for Dirac electrons). The second is the fingerprint of a finite nuclear size: diffraction of the electron wave by an extended charge distribution. Before developing that analogy, two experimental points deserve emphasis.

1.4.1 Selecting nuclear, not electronic, scattering

An oxygen target contains both a nucleus and eight atomic electrons. An incident electron can scatter from either. Scattering from an atomic electron transfers a large fraction of the beam energy (comparable masses in a nearly elastic collision), so the scattered electron emerges with substantially reduced energy. Scattering from the much heavier nucleus is essentially elastic: the recoiling nucleus takes little kinetic energy, and the scattered electron retains nearly its full beam energy.

Remark

In the laboratory one therefore applies an energy cut, retaining only events with $E' \approx E_{\text{beam}}$. That cut isolates elastic *nuclear* scattering and rejects the electron–electron background. The data of Figure 1.2 are of this elastic character.

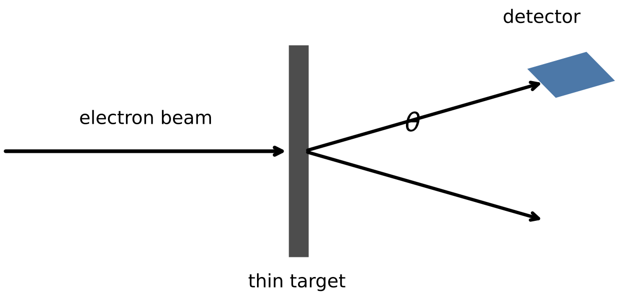


Figure 1.3: Left: schematic of an electron beam scattering from a thin sample into a detector at angle θ. Right: qualitative angular dependence of the detector reading, with a diffraction minimum at θ_{min}.

Even the Rutherford formula is not quantitatively applicable at 420 MeV. Electrons are ultra-relativistic ($E/m_e c^2 \sim 800$), so the proper point-charge baseline is the *Mott* cross section, the Dirac generalisation of Rutherford scattering. Finite nuclear size then multiplies the Mott result by the square of a form factor (Section 1.6).

Historical note: Nevill F. Mott (1905–1996)

Sir Nevill Francis Mott was one of the leading theoretical physicists of the twentieth century. Educated at Clifton College and St John's College, Cambridge, he worked in Copenhagen and Göttingen in the early years of quantum mechanics before returning to the United Kingdom. In 1929–1930, applying Dirac's relativistic theory of the electron to Coulomb scattering, he derived the cross section for a point charge that now bears his name: the *Mott formula*. It reduces to Rutherford scattering in the non-relativistic limit, but at high energy it includes spin and relativistic kinematics and becomes the natural baseline for electron–nucleus experiments.

Mott's career ranged far beyond nuclear scattering: he made foundational contributions to solid-state physics (metal–insulator transitions, the Mott insulator), shared the 1977 Nobel Prize in Physics with P. W. Anderson and J. H. Van Vleck for electronic structure of magnetic and disordered systems, and served as Cavendish Professor at Cambridge. In the present chapter, whenever we write $(\mathrm{d}\sigma/\mathrm{d}\Omega)_{\text{Mott}}$, we are using the point-charge Dirac result he obtained almost a century ago, the yardstick against which finite-size form factors are measured.

1.5 Diffraction analogy and a first estimate of the nuclear size

The diffraction minimum in Figure 1.2 is the nuclear counterpart of a familiar optical effect. Consider monochromatic, coherent light of wavelength λ incident on a single slit of width b (Figure 1.4). On a distant screen the intensity is

$$I(\theta) = I_0 \left(\frac{\sin\beta}{\beta}\right)^2, \qquad \beta = \frac{\pi b \sin\theta}{\lambda}. \tag{1.15}$$

The intensity vanishes whenever $\beta = m\pi$ for nonzero integer m, i.e. whenever

$$b \sin\theta = m\lambda, \qquad m = \pm 1, \pm 2, \ldots \tag{1.16}$$

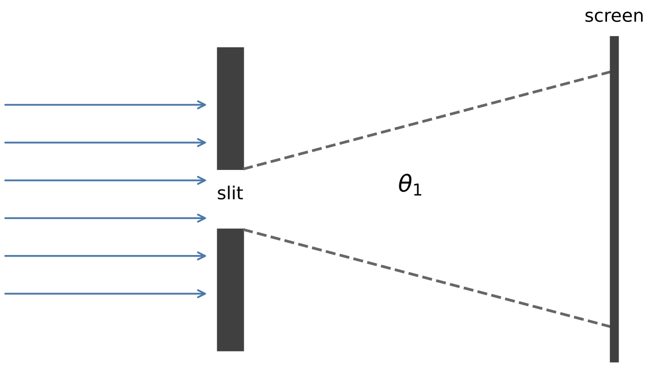


Figure 1.4: Geometry of single-slit diffraction. Plane waves of wavelength λ pass through a slit of width b and form an interference pattern on a distant screen ($L \gg b$). Intensity minima occur near $y_m = m\,\lambda L/b$ for integer $m \neq 0$.

For a *circular* aperture of diameter D, the first dark ring of the Airy pattern sits at

$$\sin\theta = \frac{1.22\,\lambda}{D}. \tag{1.17}$$

Treating the nucleus, very roughly, as a circular obstacle of diameter D, one may invert Equation (1.17) to estimate D from the observed diffraction minimum. Equivalently, the first minimum is often written condition as $\theta_{\min} \approx \lambda_{\mathrm{dB}}/D$ (in radians, for small angles), corresponding to a path difference of half a wavelength between opposite edges of the charge distribution [2].

Example: oxygen at 420 MeV

At $E = 420\,\mathrm{MeV}$,

$$\lambda \approx \frac{hc}{E} \approx \frac{1240\,\mathrm{MeV}\cdot\mathrm{fm}}{420\,\mathrm{MeV}} \approx 3.0\,\mathrm{fm}.$$

The first minimum in the oxygen data lies near $\theta \approx 45^\circ$. Equation (1.17) then yields

$$D = \frac{1.22\,\lambda}{\sin\theta} \approx \frac{1.22 \times 3.0\,\mathrm{fm}}{\sin 45^\circ} = \frac{1.22 \times 3.0\,\mathrm{fm}}{1/\sqrt{2}} \approx 5.2\,\mathrm{fm}.$$

The corresponding radius, $R = D/2 \approx 2.6\,\mathrm{fm}$, is a respectable first estimate for ^{16}O. The empirical rule $R \approx 1.2\,\mathrm{fm}\,A^{1/3}$ with $A = 16$ gives $R \approx 3.0\,\mathrm{fm}$—the same order of magnitude.

Remark

This optical analogy is intentionally crude. The nucleus is not a hard disk with a sharp edge; the electron is a charged Dirac particle, not a scalar optical wave; and the Coulomb field extends beyond the nuclear surface. A proper treatment replaces the slit formula by a Fourier transform of the charge density (the form factor). The estimate nonetheless shows that the angular location of the first diffraction minimum is a direct ruler for the nuclear size. As the beam energy rises, λ falls and the first minimum moves to smaller angles [2], exactly as seen in multi-energy data for gold and other targets.

Why is only one minimum visible in the oxygen data? For a second minimum one would need $b \sin\theta = 2\lambda$, or $\sin\theta = 2\lambda/D$. With $D \approx 5.2\,\mathrm{fm}$ and $\lambda \approx 3\,\mathrm{fm}$ that ratio exceeds unity: no real angle exists. Heavier nuclei (larger D) or higher beam energies (smaller λ) push λ/D down and bring additional diffraction minima into the physical domain. Lead or gold at hundreds of MeV to GeV energies display a rich oscillatory pattern; the number of observed minima tracks the zeros of the form factor of a uniform sphere [3].

1.6 From diffraction to charge density: the form factor

Electron scattering can do more than estimate a single radius. Because the interaction is electromagnetic and weak compared with the nuclear binding scale (at moderate momentum transfers), the Born approximation is an excellent starting point for light nuclei. The scattering amplitude then becomes proportional to the Fourier transform of the nuclear charge density: the *charge form factor*.

1.6.1 Plane-wave Born picture

Write the initial and final electron waves as plane waves of physical momenta $\boldsymbol{p}$ and $\boldsymbol{p}'$ (Coulomb distortion is discussed below):

$$\psi_i(\boldsymbol{r}) = e^{i\boldsymbol{p}\cdot\boldsymbol{r}/\hbar}, \qquad \psi_f(\boldsymbol{r}) = e^{i\boldsymbol{p}'\cdot\boldsymbol{r}/\hbar}. \tag{1.18}$$

For elastic scattering $|\boldsymbol{p}| = |\boldsymbol{p}'| \equiv p$; only the direction changes. Define the physical three-momentum transfer $\boldsymbol{q} = \boldsymbol{p} - \boldsymbol{p}'$. Squaring,

$$\begin{aligned} q^2 &= |\boldsymbol{p} - \boldsymbol{p}'|^2 = p^2 + p^2 - 2\boldsymbol{p}\cdot\boldsymbol{p}' = 2p^2(1-\cos\theta) \\ &= 4p^2 \sin^2(\theta/2), \end{aligned} \tag{1.19}$$

where the last step uses the half-angle identity $1 - \cos\theta = 2\sin^2(\theta/2)$. Therefore

$$\boldsymbol{q} = \boldsymbol{p} - \boldsymbol{p}', \qquad q = |\boldsymbol{q}| = 2p\sin(\theta/2). \tag{1.20}$$

The corresponding wave-vector transfer is $\Delta k = \boldsymbol{q}/\hbar$. Throughout this chapter q has momentum units; every Fourier phase therefore contains $qr/\hbar$.

The electron couples to the nuclear charge density $\rho(\boldsymbol{r}')$ through the Coulomb interaction. The perturbation may be written

$$H' = \int \rho(\boldsymbol{r}')\,\frac{-e}{4\pi\varepsilon_0\,|\boldsymbol{r} - \boldsymbol{r}'|}\,\mathrm{d}\tau'. \tag{1.21}$$

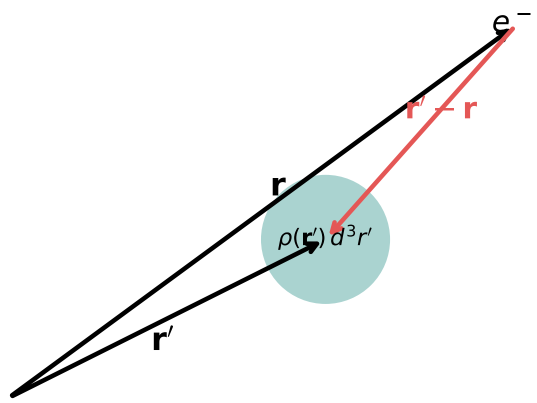


Figure 1.5: Geometry of the Coulomb interaction. An electron at $\boldsymbol{r}$ interacts with a nuclear charge element $\rho(\boldsymbol{r}')\,\mathrm{d}\tau'$ at $\boldsymbol{r}'$. The relative vector is $\boldsymbol{r}' - \boldsymbol{r}$; $\mathrm{d}\tau'$ is a nuclear volume element.

In first-order perturbation theory the transition amplitude is

$$\mathcal{M}_{fi} = \int \psi_f^*(\boldsymbol{r})\, H'\, \psi_i(\boldsymbol{r})\, \mathrm{d}\tau. \tag{1.22}$$

Substitute the plane waves and (1.21):

$$\begin{aligned}\mathcal{M}_{fi} &= \int \mathrm{d}\tau\, e^{-i\boldsymbol{k}'\cdot\boldsymbol{r}} \left[\int \mathrm{d}\tau'\, \rho(\boldsymbol{r}')\, \frac{-e}{4\pi\varepsilon_0\, |\boldsymbol{r}-\boldsymbol{r}'|}\right] e^{i\boldsymbol{k}\cdot\boldsymbol{r}} \\ &= \int \mathrm{d}\tau'\, \rho(\boldsymbol{r}') \left[\int \mathrm{d}\tau\, \frac{-e}{4\pi\varepsilon_0\, |\boldsymbol{r}-\boldsymbol{r}'|}\, e^{i\boldsymbol{q}\cdot\boldsymbol{r}/\hbar}\right],\end{aligned} \tag{1.23}$$

with $\boldsymbol{q} = \boldsymbol{p} - \boldsymbol{p}'$. The inner integral is the Fourier transform of the Coulomb kernel. Shifting $\boldsymbol{s} = \boldsymbol{r} - \boldsymbol{r}'$ shows that it equals

$$e^{i\boldsymbol{q}\cdot\boldsymbol{r}'/\hbar} \int \frac{-e}{4\pi\varepsilon_0\, s}\, e^{i\boldsymbol{q}\cdot\boldsymbol{s}/\hbar}\, \mathrm{d}^3 s = e^{i\boldsymbol{q}\cdot\boldsymbol{r}'/\hbar} \left(\frac{-e\hbar^2}{\varepsilon_0\, q^2}\right)$$

(in the convention where $\int e^{i\boldsymbol{q}\cdot\boldsymbol{s}/\hbar}/s\, \mathrm{d}^3 s = 4\pi\hbar^2/q^2$). Thus

$$\mathcal{M}_{fi} = \underbrace{\left(-\frac{e\hbar^2}{\varepsilon_0\, q^2}\right)}_{\text{point Coulomb / Mott kinematics}} \int \rho(\boldsymbol{r}')\, e^{i\boldsymbol{q}\cdot\boldsymbol{r}'/\hbar}\, \mathrm{d}\tau'. \tag{1.24}$$

The amplitude therefore factorises into a known kinematic piece times the Fourier transform of the charge density.

Definition: charge form factor

$$F(\boldsymbol{q}) = \frac{1}{Ze}\int \rho(\boldsymbol{r}')\, e^{i\boldsymbol{q}\cdot\boldsymbol{r}'/\hbar}\, \mathrm{d}\tau', \qquad F(0) = 1. \tag{1.25}$$

For a spherical ground state $\rho = \rho(r)$ only, choose the polar axis along $\boldsymbol{q}$ so that $\boldsymbol{q}\cdot\boldsymbol{r}' = qr\cos\theta'$. Then

$$\begin{aligned} F(q) &= \frac{1}{Ze}\int_0^\infty \rho(r)\, r^2\, \mathrm{d}r \int_0^{2\pi} \mathrm{d}\phi' \int_{-1}^{1} \mathrm{d}(\cos\theta')\, e^{iqr\cos\theta'/\hbar} \\ &= \frac{4\pi}{Ze}\int_0^\infty \rho(r)\, \frac{\sin(qr/\hbar)}{qr/\hbar}\, r^2\, \mathrm{d}r, \end{aligned} \tag{1.26}$$

because $\int_{-1}^{1} e^{ix\cos\theta}\, \mathrm{d}(\cos\theta) = 2\sin x/x$. Equivalently, writing the Fourier transform of a normalised charge density [2],

$$F(\Delta k) = \int \rho_e(\boldsymbol{r})\, e^{i\Delta k\cdot \boldsymbol{r}}\, \mathrm{d}^3 r, \qquad \Delta k = \boldsymbol{q}/\hbar,$$

(up to normalisation by the total charge Ze).

The measured cross section factors as

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} = \left(\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}\right)_{\text{point}} |F(q)|^2, \tag{1.27}$$

where the point-charge baseline is the Rutherford (non-relativistic) or Mott (relativistic Dirac) formula. All information about the spatial distribution of charge resides in $F(q)$. The momentum transfer that enters the Fourier phase has magnitude

$$q = |\boldsymbol{q}| = 2p\sin(\theta/2) \tag{1.28}$$

for elastic scattering at beam momentum p [1].

Caution

The form factor is defined as the ratio of the full scattering amplitude to the point-charge (Rutherford/Mott) amplitude, so that the Coulomb $1/q^2$ factors cancel by construction [6, 1]. Do not confuse $F(q)$ with the full Born amplitude, which still contains the photon propagator. Also keep the matrix-element convention consistent: $\mathcal{M}_{fi} = \langle f|H'|i\rangle$ with the final wave conjugated, and a fixed sign for $\boldsymbol{q} = \boldsymbol{k} - \boldsymbol{k}'$.

Key idea

Measuring $\mathrm{d}\sigma/\mathrm{d}\Omega$ as a function of angle (hence of q) determines $|F(q)|$. One may then either (i) extract $\rho(r)$ by a model-independent numerical inversion, or (ii) assume a parametric Woods–Saxon shape and fit its parameters to the data [1]. Both routes are used in practice; the parametric form is convenient for nuclear-model calculations.

Example: uniform sphere

For a uniform charge distribution of radius R_A,

$$\rho(r) = \begin{cases} \dfrac{Ze}{\frac{4}{3}\pi R_A^3}, & r \le R_A, \\ 0, & r > R_A. \end{cases}$$

Equation (1.26) becomes

$$F(q) = \frac{4\pi}{Ze}\frac{Ze}{\frac{4}{3}\pi R_A^3}\int_0^{R_A}\frac{\sin(qr/\hbar)}{qr/\hbar}\,r^2\,\mathrm{d}r = \frac{3}{R_A^3}\int_0^{R_A} r^2\,\frac{\sin(qr)}{qr}\,\mathrm{d}r. \tag{1.29}$$

Integrate by parts (or use $j_1(x) = \sin x/x^2 - \cos x/x$) with $x = qR_A/\hbar$:

$$F(q) = \frac{3}{(qR_A/\hbar)^3}\left[\sin(qR_A/\hbar) - (qR_A/\hbar)\cos(qR_A/\hbar)\right] = \frac{3j_1(qR_A/\hbar)}{qR_A/\hbar}. \tag{1.30}$$

The zeros of j_1 fix the diffraction minima. The first zero of $j_1(x) = 0$ solves $\tan x = x$ and lies at $x \approx 4.493$, so $q_1 R_A/\hbar \approx 4.49$; counting zeros at fixed energy is a quick estimate of R_A [3].

For any spherical density the small-q expansion follows from $\sin(qr/\hbar)/(qr/\hbar) = 1 - q^2r^2/(6\hbar^2) + \cdots$:

$$\begin{aligned} F(q) &= \frac{4\pi}{Ze}\int_0^\infty \rho(r)\left(1 - \frac{q^2r^2}{6\hbar^2} + \cdots\right)r^2\,\mathrm{d}r \\ &= 1 - \frac{q^2}{6\hbar^2}\langle r^2\rangle + \cdots, \end{aligned} \tag{1.31}$$

where $\langle r^2\rangle = (1/Ze)\int \rho(r)\,r^2\,\mathrm{d}^3r$. Thus the slope of $F(q)$ at the origin determines the rms charge radius without assuming a full density model. For the uniform sphere, $\langle r^2\rangle = \frac{3}{5}R_A^2$.

Coulomb distortion and heavy nuclei

At low energy, or for heavy nuclei ($Z\alpha$ not small), plane waves are a poor description of the electron in the nuclear Coulomb field. One then solves the Dirac equation in the electrostatic potential of a trial $\rho(r)$ (phase-shift / distorted-wave analysis) and adjusts ρ until the calculated cross section matches experiment. The qualitative content of Equation (1.27) survives; quantitative extractions for ^{208}Pb or ^{197}Au require this treatment [2].

1.7 Empirical charge densities: the Woods–Saxon profile

Once form-factor data are inverted, a remarkably simple picture emerges. Figure 1.6 shows proton (charge) densities for three representative nuclei: ^{16}O, ^{118}Sn, and ^{197}Au. Hofstadter's classic compilation for a broader range of nuclei (H through Bi) shows the same pattern [2]: a flat interior and a diffuse surface.

Several features are universal enough to deserve names.

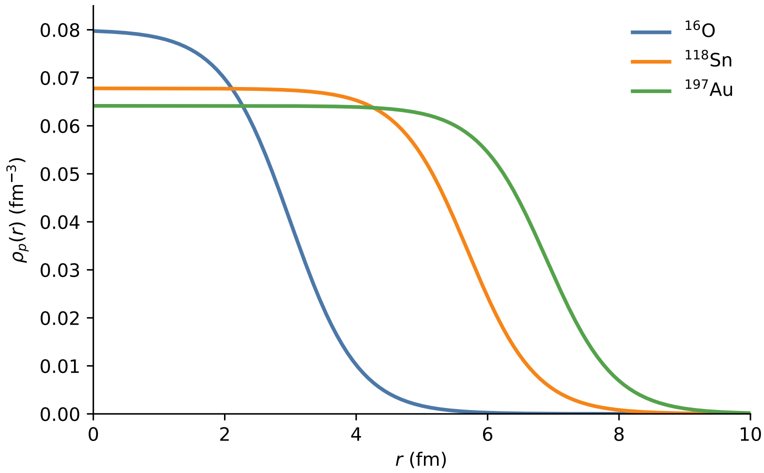


Figure 1.6: Woods–Saxon charge distributions for ^{16}O, ^{118}Sn, and ^{197}Au. The proton density $\rho_p(r)$ is nearly flat in the nuclear interior and falls smoothly through the surface. Heavier systems have larger half-density radii and slightly lower central charge densities.

1. **A flat interior.** From the centre out to a few fermis, $\rho_p(r)$ is roughly constant. Nuclei are not hollow; charge fills the volume.
2. **A diffuse surface.** The density does not terminate abruptly. It falls from $\sim 90\%$ to $\sim 10\%$ of its central value over a surface thickness of order 2.5 fm.
3. **Larger systems, larger radii.** The half-density radius grows with mass number, consistent with $R \propto A^{1/3}$.
4. **Mild A-dependence of the central charge density.** Lighter nuclei show a somewhat higher central ρ_p; heavier nuclei are slightly lower, consistent with Coulomb repulsion pushing protons outward. When one forms the *nucleon* density by scaling $\rho_A(r) \approx (A/Z)\,\rho_p(r)$, the central values collapse to a nearly universal $\rho_0 \approx 0.16 - -0.17\,1/\mathrm{fm}^3$.

Historical note: R. D. Woods and D. S. Saxon

The functional form that now bears their names was introduced in 1954 by **Roger D. Woods** and **David S. Saxon** at the University of California, Los Angeles, in a Physical Review paper on the elastic scattering of 20 MeV protons (Phys. Rev. **95**, 577, 1954). They needed a smooth, finite-range mean-field potential that was roughly constant in the nuclear interior and fell to zero over a surface of finite thickness: neither a pure square well nor an infinite oscillator. The form

$$V(r) = -\frac{V_0}{1 + \exp\big[(r-R)/a\big]}$$

did the job, and it quickly became the standard phenomenological optical and shell-model potential.
The same shape describes the empirical *charge density* extracted from electron scattering: a flat bulk and a diffuse surface. In the literature one meets both "Woods–Saxon" and "Saxon–Woods"; both refer to this two-parameter Fermi profile. Saxon later contributed broadly to nuclear reaction theory and computing in physics; the potential remains one of the most-cited phenomenological tools in nuclear structure.

Definition: Woods–Saxon (two-parameter Fermi) density

$$\rho(r) = \frac{\rho_0}{1 + \exp\big[(r - R)/a\big]}. \tag{1.32}$$

Here ρ_0 is the central density, R is the half-density radius ($\rho(R) = \rho_0/2$), and a is the diffuseness. The surface thickness t (distance from 90% to 10% of ρ_0) is related to a by

$$t = 4a \ln 3 \approx 4.40\, a. \tag{1.33}$$

A global analysis of scattering data finds that a is essentially independent of mass number,

$$a = 0.52\,\text{fm} \pm 0.01\,\text{fm}, \tag{1.34}$$

so the surface thickness is $t = 4a \ln 3 \approx 2.3\,\text{fm}$ [1]. The same functional form appears as the Woods–Saxon (or Saxon–Woods) mean field in the nuclear shell model [3, 4, 1].

What the charge densities show [1]

Comparing Woods–Saxon charge profiles for light and heavy nuclei one finds three regularities:

1. Larger nuclei have a larger half-density radius.
2. The surface width a (hence t) is nearly the same for all A.
3. The *central charge* density is higher in light nuclei than in heavy ones (Coulomb repulsion pushes protons outward as Z grows).

After rescaling to nucleon density by A/Z (Lecture 2), the central *nucleon* density becomes nearly universal.

Caution

Electrons couple to electric charge. What Figure 1.6 shows is therefore the *proton* (charge) density $\rho_p(r)$, not the full nucleon density. Neutrons are electromagnetically nearly invisible to ordinary (parity-conserving) electron scattering. The rescaling $\rho_A \approx (A/Z)\,\rho_p$ assumes similar proton and neutron profiles (reasonable for $N \approx Z$, increasingly questionable for heavy, neutron-rich systems).

1.7.1 Several definitions of "the" nuclear radius

Because the surface is diffuse, there is no unique nuclear radius. Three common definitions, following [2], are:

- R_m: radius at which the density has fallen to half its central value (the Woods–Saxon R).
- R_N: radius at which the density has essentially vanished (a practical outer edge).
- $R_{1/2}$: a half-density radius defined on a particular model of the edge.

They differ by $\sim 10--20\,\%$ and all scale roughly as $A^{1/3}$. The volume per nucleon implied by these radii lies in the range $\sim 4--9\,\mathrm{fm}^3$, corresponding to the saturation density above. When comparing data sets, one must therefore state which radius is quoted: an rms charge radius from electron scattering is not numerically identical to a reaction strong-absorption radius, even for the same nucleus [6, 1].

1.7.2 Charge radius and the $A^{1/3}$ law

Electron scattering determines not only the shape of $\rho(r)$ but also moments. The most frequently quoted is the root-mean-square (rms) charge radius,

$$\langle r^2 \rangle_{\rm ch}^{1/2} = \left(\frac{1}{Ze} \int r^2\, \rho(r)\, \mathrm{d}\tau \right)^{1/2}. \tag{1.35}$$

For a uniform sphere of radius R one has $\langle r^2 \rangle^{1/2} = \sqrt{3/5}\, R$; the Woods–Saxon surface corrects this at the few-percent level. From the low-q expansion Equation (1.31),

$$\langle r^2 \rangle = -6\, \hbar^2 \frac{\mathrm{d}F}{\mathrm{d}(q^2)}\bigg|_{q=0}, \tag{1.36}$$

so a careful measurement of $F(q)$ near the origin yields the rms radius model-independently. Empirically, the half-density (or equivalent hard-sphere) radius follows [1]

$$R_0 = 1.2\,\mathrm{fm} \times A^{1/3}. \tag{1.37}$$

This is exactly what one expects for systems of different mass but the *same* bulk density: volume $\propto A$ implies radius $\propto A^{1/3}$. Equation (1.37) is a central input to the liquid-drop model developed in later chapters.

Example: ^{12}C radius from form-factor data [3]

Expanding $F(q)$ at low q and fitting measured cross sections for ^{12}C yields $\langle r^2 \rangle^{1/2} \approx 2.44\,\mathrm{fm}$ for a uniform-sphere equivalent radius, consistent with $r_0 A^{1/3}$ at $A = 12$.

1.7.3 Which feature of $F(q)$ constrains which property?

The form-factor programme answers three nested questions about $\rho_{\rm ch}(r)$:

Observable in $F(q)$	Density property	Typical scale
Low-q slope of F	$\langle r^2 \rangle_{\rm ch}$	$r_{\rm rms} \sim 2$–$6\,\mathrm{fm}$
Location of diffraction minima	Overall size R	$R \approx 1.2\, A^{1/3}\,\mathrm{fm}$
Depth / washing of minima	Surface diffuseness a (skin t)	$a \approx 0.52\,\mathrm{fm}$, $t \approx 2.3\,\mathrm{fm}$

A sharp sphere produces deep zeros of $j_1(qR)$; a diffuse Woods–Saxon surface softens those minima but does not erase the $A^{1/3}$ growth of R [6, 1, 7]. Lecture 2 will convert the same profiles into a saturated *matter* density.

Takeaway

The Woods–Saxon profile encodes two great empirical facts of nuclear bulk structure: **saturation** (a constant central density of order $0.16/\text{fm}^3$) and a **surface of nearly universal thickness** ($t \approx 2.3\,\text{fm}$). Both reappear in the liquid-drop model, the nuclear equation of state, and the neutron skins of heavy nuclei.

1.8 What electrons do not see: neutrons and skins

Electrons couple to electric charge. In a nucleus, charge resides on protons (up to small meson-exchange and spin-orbit contributions). The neutron distribution $\rho_n(r)$ is invisible to ordinary electron scattering.

In light, $N = Z$ nuclei one may reasonably assume $\rho_n \approx \rho_p$. In heavy nuclei, where $N > Z$, a neutron skin develops: the neutron distribution extends farther than the proton distribution. The skin thickness

$$\Delta r_{np} = \langle r^2 \rangle_n^{1/2} - \langle r^2 \rangle_p^{1/2} \tag{1.38}$$

is a laboratory observable with a direct link to the density dependence of the nuclear symmetry energy, and, through that link, to the radii of neutron stars. Measuring Δr_{np} requires either hadronic probes (with their attendant model dependence) or parity-violating electron scattering, in which the Z^0 boson couples preferentially to neutrons. That programme (PREX, CREX, and their successors) lies beyond this chapter, but the conceptual foundation has been laid: once the proton distribution is known, the neutron distribution becomes the next frontier.

Remark

Complementary probes of the charge distribution include muonic atoms: because $m_\mu \approx 207\,m_e$, the muonic Bohr radius is ~ 200 times smaller than the electronic one, so the muon wave function overlaps the nucleus strongly and level shifts are sensitive to $\langle r^2 \rangle_{\text{ch}}$ [2]. Lecture 3 develops that finite-size spectroscopy in detail.

1.9 Summary and outlook

- Nuclear sizes are a few fermis. Probing them requires electron energies of hundreds of MeV so that $\lambda \lesssim R$.
- Rutherford/Chadwick α scattering first established $R \propto A^{1/3}$; Hofstadter electron scattering mapped the full charge density.
- Elastic electron scattering measures a diffraction pattern whose first minimum already yields a crude nuclear diameter, and whose full angular distribution determines the charge form factor $F(q)$.
- In the Born approximation, $\text{d}\sigma/\text{d}\Omega = (\text{d}\sigma/\text{d}\Omega)_{\text{Mott}}\,|F(q)|^2$, with $F(q)$ the Fourier transform of $\rho(r)$.
- Empirical densities are well described by a Woods–Saxon profile: flat interior, diffuse surface of nearly universal thickness, and radius scaling as $A^{1/3}$.

- Electrons map protons. Neutrons (and with them the symmetry energy and the neutron-star connection) require additional probes.

Next. Lecture 2 converts charge density to mass density and explains saturation ($R \propto A^{1/3}$, short-range force). Lecture 3 then remeasures the same radii through atomic and muonic finite-size shifts.

1.9.1 Diffraction, form factors, and central density

Elastic electron scattering at Stanford and elsewhere produces angular distributions reminiscent of optical diffraction; the charge form factor $F(q)$ is the Fourier transform of $\rho_{\text{ch}}(r)$, normalised so $F(0) = 1$. Inversion of precise $F(q)$ data shows that the *central nucleon density* is nearly universal, while light nuclei have more peaked central *charge* densities and heavy nuclei develop flatter interiors with a Coulomb-swollen proton profile [7, 8, 6, 1, 14].

1.10 Deeper physics: Mott baseline and reading $F(q)$

1.10.1 Rutherford / Mott baseline and finite-size reduction

For a point Coulomb centre the orbit itself supplies the cross section. Let

$$a = \frac{Zze^2}{4\pi\varepsilon_0}, \qquad E = \frac{1}{2}mv_0^2.$$

Conservation of energy and angular momentum for a repulsive hyperbolic orbit gives the impact-parameter relation

$$b = \frac{a}{2E}\cot\frac{\theta}{2}. \tag{1.39}$$

An azimuthally symmetric beam maps the incident annulus $2\pi b\,\mathrm{d}b$ into the solid angle $\mathrm{d}\Omega = 2\pi\sin\theta\,|\mathrm{d}\theta|$. Hence

$$\begin{aligned}\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} &= \frac{b}{\sin\theta}\left|\frac{\mathrm{d}b}{\mathrm{d}\theta}\right| = \left(\frac{a}{4E}\right)^2\csc^4\frac{\theta}{2} \\ &= \left(\frac{Zze^2}{16\pi\varepsilon_0 E}\right)^2\csc^4\frac{\theta}{2}.\end{aligned} \tag{1.40}$$

Using $mv_0^2 = 2E$, this is equivalently

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} = \left(\frac{Zze^2}{8\pi\varepsilon_0 mv_0^2}\right)^2\csc^4\frac{\theta}{2} \tag{1.41}$$

for non-relativistic projectiles of charge ze on a nucleus of charge Ze. A full quantum treatment of a pure $1/r$ potential reproduces the same result.

For a relativistic spin-$\frac{1}{2}$ electron, evaluating the Dirac current in the first Born approximation multiplies the spinless point-charge result by $1 - \beta^2\sin^2(\theta/2)$. Neglecting nuclear recoil,

$$\left(\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}\right)_{\text{Mott}} = \left(\frac{Z\alpha\hbar c}{2pv}\right)^2\frac{1-\beta^2\sin^2(\theta/2)}{\sin^4(\theta/2)}, \qquad \beta = \frac{v}{c}. \tag{1.42}$$

For $E \simeq pc$ and $\beta \simeq 1$, this becomes

$$\left(\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}\right)_{\mathrm{Mott}} \simeq \left(\frac{Z\alpha\hbar c}{2E}\right)^2 \frac{\cos^2(\theta/2)}{\sin^4(\theta/2)}.$$

The factor $\cos^2(\theta/2)$ is the helicity/spin correction; the remaining $\csc^4(\theta/2)$ is the Rutherford angular skeleton. Finite nuclear size then multiplies the appropriate point baseline by a form factor:

$$\left(\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}\right)_{\mathrm{exp}} = \left(\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}\right)_{\mathrm{point}} |F(\boldsymbol{q})|^2, \tag{1.43}$$

with $F(0) = 1$. At every nonzero angle, $|F|^2 < 1$: the nucleus is a softer scatterer than a point charge of the same Z [6, 1].

1.10.2 From $|F|^2$ back to $\rho_{\mathrm{ch}}(r)$

In the plane-wave Born approximation,

$$F(q) = \frac{1}{Ze} \int \rho_p(\boldsymbol{r})\, e^{i\boldsymbol{q}\cdot\boldsymbol{r}/\hbar}\, \mathrm{d}^3 r, \qquad q = 2p\sin(\theta/2). \tag{1.44}$$

A practical inversion pipeline is:

1. Divide the measured $\mathrm{d}\sigma/\mathrm{d}\Omega$ by the Mott (point) prediction to obtain $|F(q)|^2$.
2. Fit a parametric density (Woods–Saxon, or a model-independent Fourier–Bessel expansion) so that its Fourier transform matches $F(q)$ over the measured q range.
3. Quote moments: $\langle r^2 \rangle$ from the low-q curvature; R and a from the full pattern.

Diffraction minima at larger q constrain the surface diffuseness and possible shell oscillations in $\rho(r)$ [6, 1, 7].

Worked reading of a diffraction minimum

For a uniform sphere, the first zero of $j_1(qR)$ sits near $qR \approx 4.49$. With $q = 2p\sin(\theta/2)$ at fixed beam momentum, a measured first-minimum angle therefore returns R. Replacing the sharp edge by a Woods–Saxon skin of thickness $t \approx 2.3\,\mathrm{fm}$ fills in the zero partially but hardly moves its angular location: size and diffuseness play distinct roles in $F(q)$, as in Section 1.7.3.

Key idea

What Hofstadter measured. Not a single radius, but a function $\rho_{\mathrm{ch}}(r)$: central plateau, surface of thickness $\sim 2.3\,\mathrm{fm}$, and $R \propto A^{1/3}$. Form-factor zeros move to smaller angles as beam energy rises (de Broglie wavelength shrinks), exactly as in optical diffraction from a larger aperture relative to λ.

1.11 Physics depth: diffraction scale of $F(q)$

The Mott baseline of Lecture 1 tells you the cross section for a *point* charge. Every measured reduction $|F(q)|^2 < 1$ is therefore a statement about extended charge. The momentum transfer $q = (2E/c)\sin(\theta/2)$ (ultra-relativistic limit) is the ruler: when $qR \sim \pi$, the Fourier transform of a compact charge distribution develops its first diffraction minimum.

1.11.1 Worked estimate: first zero for a uniform sphere

For a sharp sphere of radius R,

$$F(q) = \frac{3j_1(qR)}{qR}, \qquad j_1(x) = \frac{\sin x - x\cos x}{x^2}. \tag{1.45}$$

The first nontrivial zero of j_1 sits near $qR \simeq 4.49$. Taking $R = 1.2\,A^{1/3}\,\mathrm{fm}$ for $^{12}\mathrm{C}$ gives $R \simeq 2.7\,\mathrm{fm}$, so

$$q_{\min} \simeq \frac{4.49}{R} \simeq 1.7\,\mathrm{fm}^{-1}.$$

At $E = 500\,\mathrm{MeV}$ one has $q \simeq (5.07\,\mathrm{fm}^{-1})\sin(\theta/2)$, hence $\theta_{\min} \sim 40°$. Raising E moves the same zero to smaller angle: the nucleus looks larger relative to the probe wavelength. A Woods–Saxon surface replaces the sharp zero by a shallow minimum and fills in $|F|^2$ between lobes, which is why real data never look like an ideal sphere.

Remark

Limits. The Born/PWBA reading $F(q) = \tilde{\rho}_{\mathrm{ch}}(q)/\tilde{\rho}_{\mathrm{ch}}(0)$ assumes a weak Coulomb distortion. For heavy targets and low E, distorted-wave analyses shift extracted radii by a few percent. Charge radii from electron scattering therefore quote the analysis framework, not only a single number R_{eq}.

Key idea

Physical meaning. Form-factor zeros encode a length. Skin thickness t damps high-q oscillations. Lecture 2's saturation density is the bulk plateau behind that surface; Lecture 3's isotope shifts track changes in $\langle r^2\rangle$ along a chain without remeasuring the whole $F(q)$.

1.11.2 Reading diffraction like optics

Optical Babinet intuition helps: a circular aperture of radius R has Airy zeros set by qR, while a soft edge (the nuclear surface) suppresses higher rings. Empirically, the equivalent uniform radius extracted from the first minimum and the rms radius from the small-q slope differ by ~ 0.2–$0.5\,\mathrm{fm}$ in heavy nuclei. Quoting only one without stating the definition confuses comparisons across experiments.

A second practical point is beam energy. At $E \sim 50\,\mathrm{MeV}$ the accessible q barely samples the surface; at several hundred mega-electronvolts one resolves interior oscillations of ρ_{ch}. Hofstadter's programme succeeded because both the Mott baseline and a wide q lever arm were available.

Students reconstructing $F(q)$ from a published ρ_{ch} should always overlay the Mott-reduced cross section, not only the form factor, so that normalisation systematics remain visible.

The Coulomb sum rule and longitudinal/transverse separations in quasi-elastic scattering extend the same electron probe beyond elastic form factors, but they lie outside this chapter's elastic focus. Keep the elastic lesson sharp: $F(q)$ is the Fourier transform of charge, zeros encode size, and surface thickness encodes diffuseness.

The multipole expansion of the charge density also clarifies magnetic versus charge scattering: elastic magnetic form factors (odd-mass targets) probe magnetisation densities and thereby connect to Schmidt moments in Lecture 14, while this chapter restricts attention to charge form factors of even-even systems whenever possible. Keeping charge and magnetisation channels separate prevents mis-reading a magnetic minimum as a charge radius. For classroom estimates, computing both the uniform-sphere zero and the Gaussian form factor $F = \exp(-q^2\langle r^2\rangle/6)$ at the same $\langle r^2\rangle$ shows how surface diffuseness alone moves strength from the diffraction minimum into the fills between lobes.

1.12 Further physics: light nuclei, rms radii, and halos

Electron scattering does not give a single number "the radius." Heavy nuclei have nearly flat central charge densities that fall over a skin of thickness $\sim 2.3\,\mathrm{fm}$. Light nuclei ($A \lesssim 20$) often have densities peaked near $r = 0$, so a hard-sphere radius is less useful than the rms radius

$$\langle r^2\rangle = \frac{\int r^2 \rho(r)\,\mathrm{d}^3r}{\int \rho(r)\,\mathrm{d}^3r}, \qquad r_{\mathrm{rms}} = \sqrt{\langle r^2\rangle}. \tag{1.46}$$

For a uniform sphere of radius R, $r_{\mathrm{rms}} = \sqrt{3/5}\,R$. Typical values from electron scattering are $r_{\mathrm{rms}}(^{12}\mathrm{C}) \approx 2.4\,\mathrm{fm}$, $r_{\mathrm{rms}}(^{16}\mathrm{O}) \approx 2.7\,\mathrm{fm}$, $r_{\mathrm{rms}}(^{208}\mathrm{Pb}) \approx 5.5\,\mathrm{fm}$, while the deuteron is anomalously large, $r_{\mathrm{rms}}(^{2}\mathrm{H}) \approx 2.1\,\mathrm{fm}$ for only two nucleons [14, 6].

Some loosely bound systems form **halo nuclei**: one or two valence nucleons orbit a compact core at large distance. An example is ^{11}Be (one neutron outside ^{10}Be), where the valence neutron rms radius can reach $\sim 6\,\mathrm{fm}$ while the core is only $\sim 2.5\,\mathrm{fm}$. Such extremes underline that $R = r_0 A^{1/3}$ is a bulk average, not a universal law for every light species [14].

1.12.1 Quantitative rms radii

Selected rms charge radii from electron scattering [14] (approximate values):

Nucleus	r_{rms} (fm)	equiv. $R = \sqrt{5/3}\,r_{\mathrm{rms}}$	$R/A^{1/3}$
^{1}H	0.84–0.88	~ 1.1	—
^{2}H	2.1	2.7	~ 2.2 (anomalous)
^{4}He	1.7	2.1	~ 1.3
^{12}C	2.5	3.2	~ 1.3
^{16}O	2.7	3.5	~ 1.4
^{40}Ca	3.5	4.5	~ 1.3
^{208}Pb	~ 5.5	~ 7.1	~ 1.2

Heavy nuclei approach $R \approx 1.2\, A^{1/3}$ fm. The deuteron is an outlier because its wave function extends far beyond the range of the NN force (Lecture 8).

1.12.2 Skin thickness and constant central density

Electron-scattering charge densities for medium and heavy nuclei share two features [6, 1]:

1. the central density $\rho(0)$ is nearly independent of A (saturation of density);
2. the surface is diffuse: the **skin thickness** t, defined as the distance over which ρ falls from 90% to 10% of its central value, is roughly

$$t \approx 2.3\,\text{fm},$$

almost independent of A.

A hard sphere would have $t = 0$. The finite t is the real-space counterpart of the gradual fall of the form factor $|F(q)|^2$ with momentum transfer. For a sharp sphere of radius R, the form factor involves spherical Bessel functions and has zeros at definite qR; a diffuse surface washes those minima out partially, but the diffraction pattern still fixes R [6, 7].

A fit of rms radii versus $A^{1/3}$ over the chart of the nuclides yields

$$R \approx R_0 A^{1/3}, \qquad R_0 \approx 1.2\text{–}1.25\,\text{fm} \tag{1.47}$$

($R_0 \approx 1.23\,\text{fm}$ is a common electron-scattering value) [6].

1.12.3 Borromean nuclei

A **Borromean** three-body system is bound, while every two-body subsystem is unbound. ^{6}He ($^4\text{He}+n+n$) is the classic nuclear example: the two valence neutrons bind only in the presence of the α core. Halo and Borromean systems show that weak binding and extended wave functions are generic near the drip lines [14].

1.13 Deeper reading: moments of ρ_{ch} from $F(q)$

Expanding $F(q)$ for small q connects diffraction language to rms radii used in atomic and muonic work (Lecture 3):

$$F(q) = 1 - \frac{q^2}{6}\langle r^2\rangle + \frac{q^4}{120}\langle r^4\rangle + \cdots . \tag{1.48}$$

A precision low-q measurement therefore determines $\langle r^2\rangle^{1/2}$ even when the first diffraction minimum is inaccessible. Conversely, mapping several minima constrains higher moments and the surface diffuseness.

1.13.1 Order-of-magnitude check

For lead, $\langle r^2\rangle^{1/2} \simeq 5.5\,\text{fm}$. At $q = 0.5\,\text{fm}^{-1}$ the quadratic term is $q^2\langle r^2\rangle/6 \simeq 0.25$, so $F \simeq 0.75$: already a large finite-size correction, long before the first zero. That is why even modest-q elastic data are nuclear-structure measurements, not mere Rutherford checks.

The same Fourier logic fails for neutrons: the photon couples to charge. Parity-violating scattering (previewed in the research frontier below) isolates the weak charge and thereby the neutron skin, linking Lecture 1's laboratory form factors to the symmetry energy that Lectures 2 and 6 carry into dense matter. When comparing two analyses, ask whether they report charge radius, diffraction radius, or equivalent uniform radius: those three numbers differ once the surface is nonzero.

1.13.2 From elastic form factors to skins

Because neutrons carry almost no charge, elastic $F(q)$ alone cannot fix R_n. Model-dependent hadronic probes and parity-violating asymmetries close that gap. A student exercise that combines a tabulated charge radius with an assumed skin $\Delta R = R_n - R_p$ and then recomputes the weak form factor at the PREX q illustrates how Lecture 1 data enter symmetry-energy fits used in Lectures 2 and 6.

Finally, always propagate the experimental q-calibration and radiative corrections when comparing two $F(q)$ data sets: apparent radius shifts of 0.05 fm are sometimes analysis artefacts rather than new structure.

The multipole expansion of the charge density also clarifies magnetic versus charge scattering: elastic magnetic form factors (odd-mass targets) probe magnetisation densities and thereby connect to Schmidt moments in Lecture 14, while this chapter restricts attention to charge form factors of even-even systems whenever possible. Keeping charge and magnetisation channels separate prevents mis-reading a magnetic minimum as a charge radius. For classroom estimates, computing both the uniform-sphere zero and the Gaussian form factor $F = \exp(-q^2\langle r^2\rangle/6)$ at the same $\langle r^2\rangle$ shows how surface diffuseness alone moves strength from the diffraction minimum into the fills between lobes.

Classroom reconstruction checklist

(1) Digitise or tabulate a published $\rho_{\mathrm{ch}}(r)$. (2) Compute $F(q)$ by numerical Fourier–Bessel transform. (3) Overlay Mott-reduced data if available. (4) Extract $\langle r^2\rangle$ from the small-q slope and compare with the first-minimum radius. (5) Vary the surface diffuseness at fixed rms radius and watch the higher diffraction lobes move. That five-step loop turns Lecture 1 from a plot into a reproducible analysis habit and prepares the isotope-shift discussion of Lecture 3, where only the lowest moments survive.

Error budgets in elastic analyses

Radiative corrections, beam-energy calibration, and target-thickness unfolding each shift apparent diffraction minima. A conscientious comparison of two $F(q)$ data sets therefore begins with the analysis notes, not only with overlaid points. When a modern phase-shift or optical-model Coulomb correction is applied, quote the framework beside any extracted R_{eq}.

Connection to weak probes

The weak charge density weights neutrons more heavily than protons. A parity-violating asymmetry at fixed q is therefore sensitive to R_n once the charge density is known from Lecture 1. Students can treat the charge form factor as an input and vary a two-parameter neutron density until the weak form factor matches a published asymmetry: that toy exercise shows why skins constrain the symmetry energy without yet solving the full many-body problem of Lectures 2 and 6.

Research frontier: neutron skins and future student projects

The elastic electron-scattering form factors of Lecture 1 remain the laboratory standard for nuclear charge densities: Fourier transforms of $\rho_{\text{ch}}(r)$ map diffraction minima to a length scale set by R_{eq}. That programme does not, however, determine the neutron distribution with comparable model independence, because the photon couples to charge. Modern parity-violating electron scattering isolates the weak charge, which is carried mainly by neutrons. The PREX-II measurement on ^{208}Pb reported a neutron skin $R_n - R_p = 0.283 \pm 0.071\,\text{fm}$ and thereby constrained the density dependence of the nuclear symmetry energy near saturation [40, 41]. Ab initio calculations that start from chiral nuclear forces typically predict a thinner skin and remain in tension with that extraction, so the laboratory-theory comparison is an active research question rather than a closed case [42]. What remains uncertain is how much of the discrepancy is experimental systematics, how much is missing many-body correlations, and how much is the mapping from a skin to the slope parameter L. Community roadmaps emphasise completing related experiments (CREX on ^{48}Ca, and future heavy-ion and electron facilities) because the same L that controls the Pb skin also enters crust thicknesses and tidal deformabilities of neutron stars [54].

Beyond the headline skin thickness, a student-ready research question is how to propagate the PREX covariance into a posterior on L without assuming a single Skyrme or relativistic-mean-field family. Comparing several EOS parametrisations side by side, with the same laboratory likelihood, is a concrete way to see model dependence rather than quoting one preferred central value.

Student directions. (1) Analytic estimate: recompute the plane-wave Born form-factor zero for a uniform sphere and compare with Woods–Saxon numerical form factors for $A = 12, 16, 208$. (2) Data project: using tabulated charge radii and the PREX skin, estimate R_n and compare with neutron-star crust-thickness scalings from Lecture 6. (3) Literature note: contrast PREX with CREX and state which observable each experiment constrains, citing the journal results rather than secondary blogs.

1.14 Problems with solutions

Problem 1.1 — Nuclear density. Estimate the nuclear mass density in $\text{g}\,\text{cm}^{-3}$, taking $R = r_0 A^{1/3}$ with $r_0 = 1.2\,\text{fm}$ and neglecting binding energy ($m_p \approx m_n \approx 1.67 \times 10^{-24}\,\text{g}$).

Solution 1.1. Volume $V = \frac{4}{3}\pi R^3$ with $R = r_0 A^{1/3}$ implies

$$\rho = \frac{A\, m_p}{\frac{4}{3}\pi (r_0 A^{1/3})^3} = \frac{3m_p}{4\pi r_0^3}.$$

Insert $m_p = 1.67 \times 10^{-24}\,\mathrm{g}$ and $r_0 = 1.2 \times 10^{-13}\,\mathrm{cm}$:

$$\begin{aligned} r_0^3 &= (1.2)^3 \times 10^{-39} = 1.728 \times 10^{-39}\,\mathrm{cm}^3, \\ 4\pi r_0^3 &\approx 2.17 \times 10^{-38}\,\mathrm{cm}^3, \\ \rho &= \frac{3 \times 1.67 \times 10^{-24}}{2.17 \times 10^{-38}} \approx 2.3 \times 10^{14}\,\mathrm{g\,cm^{-3}}. \end{aligned}$$

Nuclear matter is about 10^{14} times denser than water. The result is independent of A once saturation and $R \propto A^{1/3}$ are assumed.

Problem 1.2 — Radius from the form factor. At low momentum transfer the charge form factor admits

$$F(q^2) = 1 - \frac{q^2 \langle r^2 \rangle}{6\hbar^2} + \cdots .$$

Elastic electron scattering at $5°$ with $p = 720\,\mathrm{MeV}/c$ on $^{12}\mathrm{C}$ gives $\mathrm{d}\sigma/\mathrm{d}\Omega = 80\,\mathrm{mb/sr}$, while the Mott (point) cross section is about $99\,\mathrm{mb/sr}$. Estimate the rms charge radius of carbon.

Solution 1.2. The form factor is the ratio of measured to Mott cross sections:

$$|F|^2 = \frac{80}{99} \approx 0.808 \quad \Rightarrow \quad F \approx \sqrt{0.808} \approx 0.899$$

(positive at small q). Elastic kinematics give

$$q = 2p\sin(\theta/2) = 2 \times 720 \sin(2.5°) = 1440 \times 0.0436 \approx 62.8\,\mathrm{MeV}/c.$$

From $F = 1 - q^2\langle r^2 \rangle / [6(\hbar c)^2] + \cdots$,

$$\begin{aligned} \langle r^2 \rangle &= \frac{6(\hbar c)^2}{(qc)^2}(1 - F) = \frac{6 \times (197\,\mathrm{MeV\,fm})^2}{(62.8\,\mathrm{MeV})^2}(0.101) \\ &\approx \frac{6 \times 38809}{3944} \times 0.101 \approx 59.0 \times 0.101 \approx 5.95\,\mathrm{fm}^2, \end{aligned}$$

so

$$\sqrt{\langle r^2 \rangle} \approx 2.44\,\mathrm{fm}.$$

For a uniform sphere, $R = \sqrt{5/3}\,\sqrt{\langle r^2 \rangle} \approx 3.15\,\mathrm{fm}$, consistent with $r_0 A^{1/3}$ at $A = 12$.

Problem 1.3 — Diffraction estimate of radius. Electrons of energy $E = 180\,\mathrm{MeV}$ scatter elastically from $^{197}\mathrm{Au}$. Treating the nucleus as a hard sphere of radius $R = (1.18A^{1/3} - 0.48)\,\mathrm{fm}$, how many diffraction minima are kinematically accessible?

Solution 1.3. Uniform-sphere form-factor zeros solve $\tan x = x$ with $x = qR/\hbar$, near $x \approx 3\pi/2,\, 5\pi/2,\, 7\pi/2, \ldots$. Maximum $q \approx 2E/c$, so

$$x_{\max} = \frac{2E}{\hbar c} R \approx \frac{2 \times 180}{197} \times 6.4 \approx 11.7.$$

There are three zeros below $x_{\max}$ (up to $\sim 7\pi/2$). The angular distribution should show three clear minima.

Problem 1.4 — Uniform-sphere rms radius. For a uniformly charged sphere of radius R, show that $\sqrt{\langle r^2 \rangle} = \sqrt{3/5}\, R$. Evaluate R and the rms radius for ^{208}Pb with $R = 1.2\, A^{1/3}$ fm.

Solution 1.4. With $\rho = 3Ze/(4\pi R^3)$,

$$\langle r^2 \rangle = \frac{4\pi}{Ze} \int_0^R r^4 \rho \, \mathrm{d}r = \frac{3}{R^3} \int_0^R r^4 \, \mathrm{d}r = \frac{3}{5} R^2,$$

so $\sqrt{\langle r^2 \rangle} = \sqrt{3/5}\, R$. For $A = 208$, $R = 1.2 \times 208^{1/3} \approx 7.1\,\mathrm{fm}$ and $\sqrt{\langle r^2 \rangle} \approx 5.5\,\mathrm{fm}$ [6, 1].

Problem 1.5 — de Broglie wavelength for diffraction. Estimate the electron energy needed for a de Broglie wavelength $\lambda \sim 2\,\mathrm{fm}$ (use $p = \hbar/\lambda$ and $E \approx pc$ for ultrarelativistic electrons). Why is this scale natural for nuclear form-factor measurements?

Solution 1.5.

$$pc = \frac{\hbar c}{\lambda} \approx \frac{197\,\mathrm{MeV\,fm}}{2\,\mathrm{fm}} \approx 100\,\mathrm{MeV}.$$

Nuclear radii are a few fm, so diffraction structure in $F(q)$ appears when $q \sim \hbar/R$, i.e. hundred-MeV electrons [7, 6].

Chapter 2

Mass Density and Saturation

"The central nuclear charge density is nearly the same for all nuclei. Nucleons do not congregate near the center."

K. S. Krane

2.1 From charge density to mass density

Electron scattering maps the proton (charge) density $\rho_p(r)$. A central empirical fact is that this density is nearly the same in the interior of all nuclei: nucleons do not pile up at the centre. Empirically one fits

$$\rho_p(r) = \frac{\rho_p(0)}{1 + \exp\big[(r - R)/a\big]}, \tag{2.1}$$

with a nearly universal surface diffuseness $a = 0.52\,\mathrm{fm} \pm 0.01\,\mathrm{fm}$ and a half-density radius $R_0 = 1.2\,\mathrm{fm} \times A^{1/3}$ [1, 6].

Electrons do not couple to neutrons. The full nucleon density, often called the mass or matter density, must be reconstructed from the charge density together with a model of how neutrons are distributed, or measured with hadronic probes. That matter density, and the near constancy of its central value, is the subject of this chapter. It is also the microscopic reason the strong force must be short ranged: if every nucleon attracted every other nucleon, neither the density nor the binding energy would scale as they do [6].

Here $\rho_p(r)$ denotes the *proton number density* (nucleons per fm^3), not the electromagnetic charge density $e\,\rho_p$. Electron scattering determines ρ_p because the charge density is $e\,\rho_p$ for point-like protons (up to small meson-exchange corrections). It is equally important to know the distribution of *all* nucleons (neutrons and protons), because nuclear binding, the equation of state, and the liquid-drop model all refer to the total nucleon number density $\rho(r)$. (The true mass density is $m_N\rho$; nuclear physics usually quotes number densities.)

Since neutrons are uncharged, ordinary Coulomb scattering does not observe them. Under a standard working assumption, however, one can still infer $\rho(r)$ from $\rho_p(r)$.

2.1.1 The constant N/Z assumption

A nucleus with N neutrons and Z protons has overall ratio N/Z and mass number $A = N + Z$. It is usually assumed that the local neutron-to- proton density ratio is the same everywhere inside the nucleus:

$$\frac{\rho_n(r)}{\rho_p(r)} = \frac{N}{Z}. \tag{2.2}$$

The total nucleon density is the sum

$$\rho(r) = \rho_n(r) + \rho_p(r). \tag{2.3}$$

Eliminating ρ_n between these two equations gives immediately

$$\rho(r) = \rho_p(r)\left(1 + \frac{N}{Z}\right) = \rho_p(r)\,\frac{A}{Z}. \tag{2.4}$$

Definition: nucleon (mass) density from charge density

Under the assumption $\rho_n/\rho_p = N/Z$ everywhere,

$$\rho(r) = \frac{A}{Z}\,\rho_p(r).$$

Rescaling the measured proton density by A/Z converts a charge distribution into an estimate of the total nucleon density.

Caution

Equation (2.2) is an approximation, not a theorem. In heavy nuclei ($N > Z$) a *neutron skin* develops: $\rho_n(r)$ extends farther than $\rho_p(r)$. The constant-ratio assumption then fails in the surface. For the bulk and for light or near-symmetric nuclei it remains a useful first estimate; for precision neutron-skin physics one needs parity-violating electron scattering or hadronic probes (cf. Lecture 1).

2.1.2 What the rescaled densities look like

Figure 2.1 recalls the proton densities of ^{16}O, ^{118}Sn, and ^{197}Au. Applying Equation (2.4) produces the nucleon densities of Figure 2.2. Two features stand out at once.

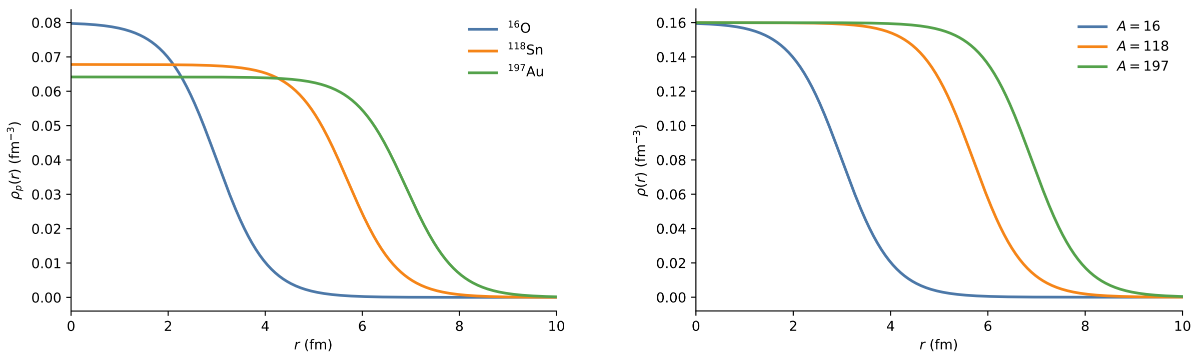


Figure 2.1: Proton densities $\rho_p(r)$ for ^{16}O, ^{118}Sn, and ^{197}Au.

Figure 2.2: Nucleon densities $\rho = (A/Z)\rho_p$ with common central ρ_0.

1. The half-density radii and surface widths are essentially the same as in the charge densities:

rescaling multiplies the vertical axis, not the radial shape.

2. All three nuclei now share virtually the *same central nucleon density*, $\rho_0 \approx 0.16--0.17\,1/\mathrm{fm}^3$.

Key idea

Saturation of nuclear matter: the central nucleon density is nearly independent of mass number [1]. Nuclei are not denser in the middle when they grow larger; they grow by adding a larger volume at fixed bulk density. Together with $B/A \approx$ constant, this strongly supports a saturating, short-range effective interaction. Density systematics alone do not prove a particular microscopic force: Pauli pressure, short-range repulsion, tensor correlations, and many-body forces all participate in setting the saturation point.

2.2 Woods–Saxon parameters and the nuclear surface

Electron scattering and mean-field phenomenology both use a diffuse edge: the nucleon density does not drop abruptly at R. The Woods–Saxon (two-parameter Fermi) form encodes a nearly flat interior and a surface of thickness set by the diffuseness a. Empirically $a \sim 0.5\,\mathrm{fm}$ across much of the chart, so the surface thickness $t \approx 4.4\,a \sim 2.3\,\mathrm{fm}$ is nearly universal while R grows as $A^{1/3}$ [1, 6].

The nucleon density is again described by a Woods–Saxon (two-parameter Fermi) profile,

$$\rho(r) = \frac{\rho_0}{1 + \exp\left[(r - R)/a\right]}, \tag{2.5}$$

where now ρ_0 is the central *nucleon* density, R is the half-density radius, and a is the diffuseness.

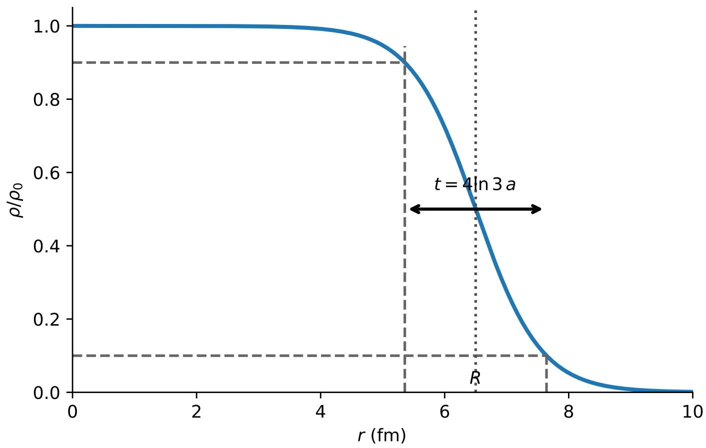


Figure 2.3: Geometry of the Woods–Saxon surface. The half-density radius R is the point where $\rho = \rho_0/2$. The surface thickness t, from $0.9\rho_0$ to $0.1\rho_0$, equals $4a\ln 3 \approx 4.4\,a$.

2.2.1 Surface thickness from first principles

To relate a to an observable surface thickness, evaluate the radii at which the density has fallen to 90% and 10% of ρ_0.

Setting $\rho = 0.9\rho_0$,

$$0.9\rho_0 = \frac{\rho_0}{1 + e^{(r-R)/a}} \implies e^{(r-R)/a} = \frac{1}{9} \implies \frac{r-R}{a} = -\ln 9 = -2\ln 3. \tag{2.6}$$

Hence

$$r_{90} = R - a\ln 9 = R - 2a\ln 3 \approx R - 2.20\,a. \tag{2.7}$$

Setting $\rho = 0.1\rho_0$,

$$0.1\rho_0 = \frac{\rho_0}{1 + e^{(r-R)/a}} \implies e^{(r-R)/a} = 9 \implies r_{10} = R + 2a\ln 3 \approx R + 2.20\,a. \tag{2.8}$$

The surface thickness is therefore

$$t \equiv r_{10} - r_{90} = 4a\ln 3 \approx 4.39\,a. \tag{2.9}$$

Remark

Lecture notes sometimes quote $t \approx 4.6a$ by approximating $\ln 9 \approx 2.3$. The exact factor is $4\ln 3 \approx 4.394$. A standard global fit gives $a = 0.52\,\mathrm{fm} \pm 0.01\,\mathrm{fm}$ [1], so

$$t = 4a\ln 3 \approx 4.39 \times 0.52\,\mathrm{fm} \approx 2.28\,\mathrm{fm}.$$

The nuclear "skin" is only a couple of fermis thick and is nearly independent of A. The redrawn Figure 2.3 uses the exact label $4\ln 3\,a$, so figure and derivation now follow the same convention.

Example: surface of a heavy nucleus

With $a = 0.52\,\mathrm{fm}$,

$$t \approx 2.3\,\mathrm{fm}.$$

For ^{197}Au, $R_0 = 1.2\,\mathrm{fm} \times 197^{1/3} \approx 7.0\,\mathrm{fm}$ [1]. The density is essentially flat for $r \lesssim 5\,\mathrm{fm}$ and has fallen to a few percent of ρ_0 by $r \approx 9--10\,\mathrm{fm}$, matching Figure 2.2.

The central core is uniformly populated with nucleons up to $r \sim R$; beyond that the density decreases smoothly. The nucleus does not have a sharp boundary but a diffuse quantum surface.

2.2.2 The $A^{1/3}$ law for the half-density radius

The mean nuclear radius R_0 increases with mass number. Saturation already implies the functional form. If the central density is A-independent, the volume that holds A nucleons must grow as

$$A = \tfrac{4}{3}\pi R^3\,\rho_0 \quad \Rightarrow \quad R = \left(\frac{3}{4\pi\rho_0}\right)^{1/3} A^{1/3} \equiv r_0\,A^{1/3}. \tag{2.10}$$

With $\rho_0 \approx 0.16\,\mathrm{fm}^{-3}$,

$$r_0 = \left(\frac{3}{4\pi \times 0.16}\right)^{1/3} \approx 1.14\,\mathrm{fm}, \tag{2.11}$$

close to the conventional fit value. This 1.14 fm is the radius of a sharp uniform sphere whose *mean* density is $0.16\,\mathrm{fm}^{-3}$; it is not numerically identical to a Woods–Saxon half-density radius, an rms radius, or a charge radius. Surface diffuseness and the chosen radius definition account for the familiar $r_0 \simeq 1.2\,\mathrm{fm}$ convention. Analysis of electron-scattering data yields the empirical expression [1]

$$R_0 = 1.2\,\mathrm{fm} \times A^{1/3}. \tag{2.12}$$

Combined with a constant central density, this is precisely the geometry of an incompressible liquid drop: volume $\propto A$, so radius $\propto A^{1/3}$. Equation (2.12) is a standard input to the liquid-drop model [1].

The mean inter-nucleon spacing at saturation follows from the same density:

$$d \sim \rho_0^{-1/3} \approx (0.16)^{-1/3}\,\mathrm{fm} \approx 1.8\,\mathrm{fm}, \tag{2.13}$$

comparable to the pion Compton wavelength that will appear in Lecture 11.

Takeaway

Three numbers characterise bulk nuclear matter as seen by electrons (rescaled to nucleons) [1]:

$$\rho_0 \approx 0.16/\mathrm{fm}^3, \quad R_0 = 1.2\,\mathrm{fm}\,A^{1/3}, \quad a = 0.52\,\mathrm{fm} \pm 0.01\,\mathrm{fm} \ \ (t \approx 2.3\,\mathrm{fm}).$$

They will reappear in the liquid-drop model, the semi-empirical mass formula, and the nuclear equation of state.

2.3 What uniform density teaches about the strong force

The nucleus exists because a strong attractive force counters the Coulomb repulsion among protons. Without that force the nucleus would explode. Nuclear attraction is far stronger than Coulomb repulsion at fermi distances, yet the density profile is *flat*, not centrally peaked.

One might have expected the opposite. In a self-gravitating body such as the Earth, density is highest at the core and decreases toward the surface: every mass element attracts every other, so the equilibrium is centrally condensed. Why is the nucleus different?

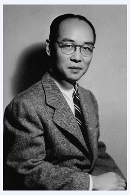
Key idea

Nuclear forces are *short-ranged.* A given nucleon interacts significantly only with its nearest neighbours, not with all other nucleons in the system. There is therefore no long-range "pull toward the centre," and the equilibrium density is set locally by the balance of short-range attraction and short-range repulsion (the hard core / Pauli pressure). The result is a saturated, roughly constant bulk density.

Remark

The range of the strong force is of order the pion Compton wavelength, $\hbar/(m_\pi c) \approx 1.4\,\mathrm{fm}$—comparable to the mean inter-nucleon spacing at saturation. Beyond two or three fermis the force has essentially died away. That is why adding nucleons increases the *volume* at fixed density rather than packing the centre more tightly, and why the surface thickness is a nearly universal few fermis set by the force range itself.

Historical note: Hideki Yukawa (1907–1981)



Hideki Yukawa, working in Osaka in 1934–1935, proposed that the short-range nuclear force is mediated by the exchange of a massive boson. The resulting potential,

$$V(r) \propto \frac{e^{-r/\lambda}}{r}, \qquad \lambda = \frac{\hbar}{mc},$$

falls exponentially with a range fixed by the mediator mass m. Yukawa estimated $m \sim 100\,\mathrm{MeV}/c^2$; the pion, discovered in 1947, has $m_\pi \approx 140\,\mathrm{MeV}/c^2$ and $\lambda \approx 1.4\,\mathrm{fm}$. He received the 1949 Nobel Prize in Physics, the first Nobel awarded to a Japanese scientist. Yukawa's insight is the microscopic reason the nuclear density saturates: the force simply does not reach across the whole nucleus.

2.4 Preview: finite-size spectroscopy

Electron scattering is not the only radius probe. For an atomic s state, the leading correction caused by replacing a point nucleus with a spherical charge distribution is

$$\Delta E_{ns} = \frac{2\pi}{3}\frac{Ze^2}{4\pi\varepsilon_0}\,|\psi_{ns}(0)|^2\langle r^2\rangle_{\mathrm{ch}} + \cdots . \tag{2.14}$$

The positive sign means that a finite nucleus softens the Coulomb singularity and raises the bound-state energy. This compact theorem, rather than a second uniform-sphere derivation, is all that is needed here. Lecture 3 derives it, reduces it to the uniform-sphere result $\langle r^2\rangle = 3R^2/5$, and explains how field shifts are separated from recoil (mass) shifts in real isotope spectroscopy.

2.5 Summary and outlook

- Electron scattering measures $\rho_p(r)$. Under $\rho_n/\rho_p = N/Z$, the nucleon density is $\rho = (A/Z)\,\rho_p$.
- Rescaled densities for different nuclei share a common central value $\rho_0 \approx 0.16/\mathrm{fm}^3$: nuclear matter saturates.
- The Woods–Saxon surface thickness is $t = 4a\ln 3$ with the usual $a = 0.52\,\mathrm{fm} \pm 0.01\,\mathrm{fm}$ ($t \approx 2.3\,\mathrm{fm}$); the half-density radius obeys $R_0 = 1.2\,\mathrm{fm}\,A^{1/3}$ [1].
- A flat bulk density is the geometric signature of a short-range strong force: nucleons feel only their neighbours.
- Atomic electrons (and especially muons) provide a second radius measurement via the finite-size shift $\Delta E \propto Z\,|\psi(0)|^2\,\langle r^2\rangle_{\mathrm{ch}}$.

Next. Lecture 3 specialises the finite-size shift to isotope differences and muonic atoms, recovering the same $R_0 \approx 1.2\,\mathrm{fm}$ from spectroscopy. Lectures 4–5 then leave size for binding energy and the SEMF that encodes saturation as a volume term.

2.5.1 Why $B \propto A$: saturation of the force

If every nucleon attracted every other nucleon equally, B would grow roughly as A^2. The empirical linearity $B \propto A$ therefore implies a short-range force that saturates: each nucleon binds mainly to nearest neighbours. The same physics keeps the central density from growing as A increases; high-energy NN phase shifts further show a short-range repulsion that prevents collapse [6, 14, 1].

2.6 Deeper physics: saturation density and short-range force

2.6.1 Numerical saturation density

If the mass density is roughly uniform inside $R = r_0 A^{1/3}$ with $r_0 \approx 1.2\,\mathrm{fm}$, the mean nucleon density is

$$\rho_0 = \frac{A}{\frac{4}{3}\pi R^3} = \frac{3}{4\pi r_0^3} \approx 0.14\text{–}0.17\,\mathrm{fm}^{-3}, \tag{2.15}$$

depending on the precise radius definition (half-density vs equivalent uniform sphere vs rms). A representative value is $\rho_0 \approx 0.16\,\mathrm{fm}^{-3}$ [1, 2, 4]. In mass units this is

$$\bar{\rho} \sim 2.3 \times 10^{17}\,\mathrm{kg\,m^{-3}} \sim 2 \times 10^{14}\,\mathrm{g\,cm^{-3}}, \tag{2.16}$$

about 10^{14} times liquid water: the densest stable bulk matter on Earth is ordinary nuclear matter.

2.6.2 Why density saturation implies a short-range force

If every nucleon interacted attractively with every other nucleon (infinite-range two-body force), the binding energy would grow as $\binom{A}{2} \propto A^2$ and the equilibrium density would not be A-independent. Observed $B \propto A$ (Lecture 4) and $\rho \approx$ const together require that each nucleon feels only a fixed number of neighbours: the force range is of order the internucleon spacing ~ 1–$2\,\mathrm{fm}$.

Light vs heavy surface fraction

With $t \approx 2.3\,\mathrm{fm}$ fixed and $R = 1.2\,A^{1/3}\,\mathrm{fm}$, a light nucleus ($R \sim 3\,\mathrm{fm}$) is mostly surface, while lead ($R \sim 7\,\mathrm{fm}$) is mostly bulk. That geometric fact foreshadows the SEMF: surface energy dominates the rise of B/A at low A; volume energy dominates the mid-mass plateau [1].

Remark

Mass vs charge distributions. Matter densities inferred from hadronic probes are similar in shape to charge densities, but not identical: neutron skins in heavy nuclei make the matter radius slightly larger than the charge radius. Lecture 1's electron probe measures charge; Lecture 2's saturation argument concerns the *matter* density that sets the SEMF volume term.

2.7 Physics depth: saturation density and force range

Lecture 2 established $\rho_0 \sim 0.16\,\mathrm{fm}^{-3}$ and $R \propto A^{1/3}$. Those two facts fix both a length and an energy scale for every later liquid-drop argument.

2.7.1 Worked estimate: Fermi momentum and kinetic scale

For a symmetric Fermi gas at density $n_0 = 0.16\,\mathrm{fm}^{-3}$,

$$k_F = (3\pi^2 n_0/2)^{1/3} \simeq 1.33\,\mathrm{fm}^{-1}, \qquad T_F = \frac{\hbar^2 k_F^2}{2m_N} \simeq 37\,\mathrm{MeV}. \tag{2.17}$$

The observed volume energy $\sim 16\,\mathrm{MeV}$ per nucleon then implies a potential contribution of order $-50\,\mathrm{MeV}$ per nucleon once kinetic energy is subtracted: attraction and Fermi motion nearly cancel. That cancellation is delicate; three-nucleon forces and the density dependence of the symmetry energy move the equilibrium point by only a few percent in n_0 but can change the pressure above saturation by much more (Lecture 6).

Internucleon spacing

The mean spacing $d \sim n_0^{-1/3} \simeq 1.8\,\mathrm{fm}$ is comparable to the pion Compton wavelength $\hbar/(m_\pi c) \simeq 1.4\,\mathrm{fm}$ (Lecture 11). Saturation therefore lives at the border between a long-range one-pion tail and a short-range repulsive core: neither a pointlike contact force nor an infinite-range $1/r$ force can produce an A-independent bulk density.

Remark

Assumptions. Uniform-density estimates ignore surface, Coulomb, and shell oscillations. They are excellent for order-of-magnitude pedagogy and poor as precision EOS inputs. Always state whether n_0 is a charge or matter density when comparing electron scattering with heavy-ion or astrophysical constraints.

2.7.2 Compressibility and the breathing mode

The incompressibility $K = 9n_0^2(\partial^2(E/A)/\partial n^2)_{n_0}$ sets the curvature of the energy per nucleon at saturation. Giant monopole resonance energies constrain $K \sim 220$–$260\,\mathrm{MeV}$ in symmetric matter. That number is the laboratory cousin of the speed of sound just above n_0. Soft EOS families with small K yield more compact stars at fixed mass; stiff families push radii outward. Quoting n_0 without K is like quoting a spring's equilibrium length without its spring constant.

Surface symmetry energy and the neutron-skin thickness further split isovector from isoscalar physics. A nucleus with a thick neutron skin looks like a laboratory droplet with an enhanced local asymmetry at the surface, connecting Lecture 1's radii to Lecture 2's saturation argument without yet invoking stars.

A Fermi-gas pressure $P = (2/5)nT_F$ at saturation is tens of mega-electronvolts per cubic femtometre, enormous on laboratory scales yet the baseline that gravity must overcome in Lecture 6. Adding interactions changes both the energy and the pressure; the slope of E/A versus n away from n_0 is the EOS. Students should plot a schematic E/A with a minimum at

n_0 and mark K as curvature: that single sketch organises saturation, giant resonances, and neutron-star radii.

Pressure sketch

From a quadratic expansion of E/A about n_0, derive $P \approx (K/9)(n - n_0)^2/n_0$ near saturation (schematic) and discuss why $P(n_0) = 0$. This short derivation links Lecture 2 to the EOS language of Lecture 6 without solving TOV equations.

2.8 Further physics: saturation and the uncertainty principle

Constant central density and $B/A \sim$ const are two faces of the same fact: each nucleon interacts mainly with nearest neighbours (**saturation** of the nuclear force). A quantum link makes that plausible. If A nucleons share a volume $\propto R^3$ with $R = r_0 A^{1/3}$, each occupies a linear size $\Delta x \sim r_0$. The uncertainty principle then gives a mean kinetic energy per nucleon of order

$$\langle T \rangle \sim \frac{\hbar^2}{2m_N r_0^2} \sim \text{tens of MeV}, \tag{2.18}$$

nearly independent of A. Because the kinetic energy per particle does not grow with system size, neither does the net binding per nucleon once nearest-neighbour attraction is fixed [14].

That argument fails for atoms: the long-range Coulomb field of the nucleus pulls electrons tighter as Z grows, so atomic size does not scale as $Z^{1/3}$ and the electron density rises with Z. Nuclear forces do not act that way: density saturates.

2.8.1 Saturation requires more than the Pauli principle

Nearest-neighbour counting explains why density and B/A do not grow with A, but a complete picture of saturation needs three ingredients [14]:

1. short-range attraction ($r \sim 1\,\text{fm}$);
2. a repulsive “hard core” at $r \lesssim 0.5\,\text{fm}$ (partly Pauli/quark structure);
3. the Pauli principle for nucleons as fermions.

Any pure power-law attraction would not saturate. The empirical Woods–Saxon density (flat interior, thin surface) is the geometric fingerprint of that three-part force.

2.8.2 Atoms vs nuclei

Atomic size does *not* grow as $Z^{1/3}$: the long-range Coulomb field of the nucleus increases electron density with Z. Nuclear density is approximately constant because the force is short-ranged and saturates. That contrast is the essential reason liquid-drop thinking works for nuclei and not for atoms [14, 5].

2.8.3 Fermi momentum and density at saturation

Treat protons and neutrons as independent Fermi gases in a volume $V = (4\pi/3)R^3$ with bulk density

$$n_0 \approx 0.15\text{–}0.17\,\mathrm{fm}^{-3} \tag{2.19}$$

[14, 1]. For $N \approx Z \approx A/2$ the Fermi energy of each species is

$$\varepsilon_F = \frac{\hbar^2}{2m}(3\pi^2 n)^{2/3} \approx 35\,\mathrm{MeV}, \tag{2.20}$$

corresponding to

$$p_F \approx 265\,\mathrm{MeV}/c, \qquad k_F = p_F/\hbar \approx 1.33\,\mathrm{fm}^{-1}. \tag{2.21}$$

The mean kinetic energy per nucleon in a free Fermi gas is $\frac{3}{5}\varepsilon_F \approx 21\,\mathrm{MeV}$. Neutron-scattering and optical-model analyses place the depth of the mean field near $U \sim -40\,\mathrm{MeV}$; the net volume binding is then of order $a_V \sim |U| - \frac{3}{5}\varepsilon_F \sim 15\text{–}20\,\mathrm{MeV}$, consistent with the SEMF volume coefficient [14].

These numbers fix the density and energy scales that Lectures 4–6 carry into the liquid-drop formula and into compact-star estimates.

2.9 Deeper reading: from n_0 to SEMF coefficients

Once n_0 and $r_0 = (3/4\pi n_0)^{1/3}$ are fixed, the SEMF volume term a_V is essentially the binding per nucleon in infinite symmetric matter. The surface term a_S measures the energy cost of the diffuse skin of thickness $t \sim 2.3\,\mathrm{fm}$ mapped in Lecture 1. A crude geometric estimate of the surface-to-volume ratio is $3t/R$ for light systems: that is why B/A rises steeply below $A \sim 50$ and flattens thereafter (Lecture 4).

2.9.1 Link to asymmetry and neutron stars

Replacing $N = Z$ by a neutron excess costs the symmetry energy $a_a(N-Z)^2/A$. The density slope of that symmetry energy near n_0 controls both laboratory neutron skins and the pressure of neutron-star matter just above saturation. Thus the pedagogical chain is tight: Lecture 1 measures charge radii, Lecture 2 fixes bulk density, Lecture 4 organises binding, and Lecture 6 asks what happens when gravity compresses the same saturated fluid far beyond n_0.

Keep the bookkeeping honest: quoting n_0 without an uncertainty on the compressibility K or the symmetry slope L understates how many EOS families remain allowed by the same laboratory saturation point.

2.9.2 Saturation as a many-body problem

In modern language, two-nucleon potentials fitted only to scattering underbind or overbind finite nuclei and misplace n_0 unless three-nucleon forces are included at the appropriate chiral order. Students need not master the EFT power counting here, but they should remember that the Fermi-gas estimate of Lecture 2 is a scale argument, not a derivation of the force. Lectures 7–11 rebuild the force from the deuteron upward; Lectures 4–6 spend the saturation scales those forces must reproduce.

A Fermi-gas pressure $P = (2/5)nT_F$ at saturation is tens of mega-electronvolts per cubic femtometre, enormous on laboratory scales yet the baseline that gravity must overcome in Lecture 6. Adding interactions changes both the energy and the pressure; the slope of E/A versus n away from n_0 is the EOS. Students should plot a schematic E/A with a minimum at n_0 and mark K as curvature: that single sketch organises saturation, giant resonances, and neutron-star radii.

Saturation checklist

Record n_0, $B/A|_{\text{vol}}$, K, and a schematic $S(n)$ slope L whenever you read an EOS paper. Recompute r_0 from your chosen n_0 and verify that $R = r_0 A^{1/3}$ matches Lecture 1 radii at the 5% level for medium-mass nuclei. Discrepancies larger than that usually mean you are mixing charge and matter radii or using a saturated density inconsistent with the SEMF fit of Lecture 4.

From density to pressure

Differentiate a schematic energy per nucleon $E/A = \frac{3}{5}T_F(n) + a(n) + b(n)^2$ to obtain pressure $P = n^2\partial(E/A)/\partial n$. At saturation $P = 0$ by definition; above saturation the same derivative becomes the EOS. Plotting $P(n)$ for two choices of K shows immediately why stellar radii move when incompressibility changes. Keep isoscalar (K) and isovector (L) derivatives distinct in that sketch.

Finite nuclei versus infinite matter

Surface and Coulomb energies vanish in infinite matter but dominate light nuclei. Extracting n_0 from finite nuclei therefore requires a model that removes those corrections. That is why modern determinations combine heavy-nucleus radii, giant resonances, and ab initio infinite-matter calculations rather than reading n_0 from a single A.

Saturation numbers to keep

Memorise four numbers with units: $n_0 \simeq 0.16\,\text{fm}^{-3}$, $B/A \simeq 16\,\text{MeV}$, $K \simeq 240\,\text{MeV}$, and $L \sim 40$–$80\,\text{MeV}$ (broad present range). Every EOS talk rewrites those symbols; students who know the numbers can detect inconsistent plots immediately. Recompute r_0 from n_0 whenever a paper uses a different saturation point, and adjust Lecture 1 radius comparisons accordingly.

Research frontier: saturation, symmetry energy, and student projects

Lecture 2 fixed the saturation density $n_0 \sim 0.16\,\text{fm}^{-3}$ and the volume energy scale from elementary Fermi-gas estimates of kinetic and potential contributions. Those order-of-magnitude arguments still teach why nuclei exist as saturated droplets, but they do not organise the hierarchy of two- and three-nucleon forces. Chiral effective field theory (EFT) now expands residual nuclear interactions systematically in momenta over a breakdown scale, so that saturation and the symmetry energy emerge from low-energy QCD symmetries rather than from a purely phenomenological well [43, 44]. The slope $L = 3n_0(dS/dn)_{n_0}$ of the symmetry energy near n_0 remains one of the most debated nuclear-matter parameters: laboratory skins, heavy-ion collective flow, and neutron-star radii pull in partly different directions, and modern reviews emphasise propagating those uncertainties into astrophysical equations of state [47, 40]. Three-nucleon forces shift the saturation point relative to pure two-nucleon Hamiltonians; omitting them typically misplaces both the binding energy per nucleon and the equilibrium density. Open questions include the size of higher-order chiral truncation errors at twice saturation density and the consistency between terrestrial constraints and multimessenger astrophysics.

A useful way to keep the uncertainty honest is to treat $S(n)$ not as a single curve but as a family of curves consistent with skins, dipole polarizability, and astrophysical radii. Student projects that recompute L from two published parametrisations and then ask which dense-matter observables move the most are closer to current practice than memorising one preferred number.

Student directions. (1) Analytic estimate: convert n_0 between fm^{-3} and $\mathrm{kg\,m}^{-3}$ and rederive $r_0 = (3/4\pi n_0)^{1/3}$. (2) Data project: from a published $S(n)$ parametrisation, extract L and compare with the PREX-implied range. (3) Literature survey: summarise how three-nucleon forces shift the saturation point relative to pure two-nucleon Hamiltonians [43].

2.10 Problems with solutions

Problem 2.1 — Saturation density. Using $R = 1.2\,A^{1/3}\,\mathrm{fm}$, compute the mean nucleon number density $\rho_0 = A/(\frac{4}{3}\pi R^3)$ in fm^{-3} and in $\mathrm{g\,cm}^{-3}$ (take $m_N = 1.67 \times 10^{-24}\,\mathrm{g}$).

Solution 2.1.

$$\rho_0 = \frac{3}{4\pi r_0^3} = \frac{3}{4\pi(1.2)^3} \approx 0.138\,\mathrm{fm}^{-3} \sim 0.14\text{–}0.17\,\mathrm{fm}^{-3}$$

depending on the precise radius definition. In mass units,

$$\rho_0 m_N \approx 2.3 \times 10^{14}\,\mathrm{g\,cm}^{-3},$$

as in Problem 1.1. This is the saturation density underlying the SEMF volume term.

Problem 2.2 — Neutron-star radius from density. A neutron star has mass $M \sim M_\odot = 2 \times 10^{30}\,\mathrm{kg}$ and density comparable to nuclear density $\rho \sim 2.3 \times 10^{14}\,\mathrm{g\,cm}^{-3}$. Estimate its radius.

Solution 2.2.

$$\frac{4}{3}\pi R^3 \rho = M \quad \Rightarrow \quad R = \left(\frac{3M}{4\pi\rho}\right)^{1/3} \approx \left(\frac{3 \times 2 \times 10^{33}\,\mathrm{g}}{4\pi \times 2.3 \times 10^{14}\,\mathrm{g\,cm}^{-3}}\right)^{1/3} \approx 13\,\mathrm{km}.$$

Nuclear saturation density plus solar mass already yields the observed $R \sim 10\,\mathrm{km}$ scale (Lecture 7).

Problem 2.3 — Surface thickness. For a Woods–Saxon profile with $a = 0.52\,\text{fm}$, compute the skin thickness t over which the density falls from 90% to 10% of its central value.

Solution 2.3.

$$t = 4a \ln 3 \approx 4 \times 0.52 \times 1.099 \approx 2.3\,\text{fm}.$$

The surface thickness is nearly independent of A, while $R \propto A^{1/3}$ grows: light nuclei are almost all surface; heavy nuclei have a thin skin on a large bulk.

Problem 2.4 — Mean free path vs radius. If the nucleon mean free path inside nuclear matter is $\lambda \sim 2\,\text{fm}$ and $R \sim 1.2\,A^{1/3}\,\text{fm}$, for what A is λ smaller than R? Interpret for the optical model.

Solution 2.4.

$$1.2A^{1/3} > 2 \quad \Rightarrow \quad A^{1/3} > \tfrac{5}{3} \quad \Rightarrow \quad A \gtrsim 5.$$

Already for light-to-medium nuclei nucleons typically undergo more than one collision while crossing the nucleus, motivating an optical (mean-field plus absorption) description of scattering [6, 1].

Problem 2.5 — Charge vs nucleon density. Electron scattering determines the charge density $\rho_{\text{ch}}(r)$. How does one estimate the nucleon density for an $N \approx Z$ nucleus, and why does the estimate degrade for heavy neutron-rich systems?

Solution 2.5. For $N \approx Z$ one rescales $\rho_A \approx (A/Z)\rho_{\text{ch}}$ if protons and neutrons share a similar radial profile. In heavy nuclei Coulomb pushes protons outward and excess neutrons form a skin, so proton and neutron densities differ and the simple A/Z rescaling is only approximate [1, 6].

Chapter 3

Finite Size and Isotope Shifts

"If we could only find a supply of atoms with 'point' nuclei we could measure ΔE and deduce R!"

K. S. Krane

3.1 Physical setup: what problem are we solving?

In solving the hydrogen atom one always assumes a point nucleus, $V = -Ze^2/(4\pi\varepsilon_0 r)$. Real nuclei are not points. Lectures 1–2 fixed the charge size from electron scattering: $R_0 \approx 1.2\,\mathrm{fm}\,A^{1/3}$ and a nearly universal skin $t \approx 2.3\,\mathrm{fm}$. The electron wave function penetrates to $r < R$, and inside the charge distribution the potential does *not* diverge as $1/r$. As a first approximation the nucleus may be taken as a uniformly charged sphere of radius R. Relative to a point nucleus, s-state energies then shift by an amount ΔE proportional to Z^4R^2 (derived below), which is precisely the finite-size correction introduced in Lecture 2.

No sample of point nuclei exists, so absolute finite-size shifts cannot be read off by simple subtraction. What can be measured cleanly is the difference between isotopes: same Z, different mass number and slightly different R. The isotope shift of K_α X-rays or optical lines then tests nuclear size, and in particular the $R \propto A^{1/3}$ law already suggested by electron scattering [6, 1]. Replacing the electron by a muon amplifies the same effect, because the muon Bohr orbit is about 207 times smaller. Optical isotope shifts further separate a mass (reduced-mass) contribution from a field (finite-size) contribution; heavy elements are field-dominated, which is why they diagnose nuclear charge radii most cleanly [6].

3.2 Building the perturbation H_1 carefully

3.2.1 Point nucleus

For a hydrogen-like ion the electron (charge $-e$) in the field of a *point* nucleus (charge Ze) has potential energy

$$U_{\text{point}}(r) = (-e)\cdot\left(\frac{1}{4\pi\varepsilon_0}\frac{Ze}{r}\right) = -\frac{Ze^2}{4\pi\varepsilon_0\, r}. \tag{3.1}$$

The minus sign means attraction: the energy is lowered when the electron approaches the nucleus. As $r \to 0$, $U_{\text{point}} \to -\infty$. That infinite well is unphysical for a real nucleus of finite size.

3.2.2 Finite uniform sphere: field and potential

Assume a sphere of radius R with total charge Ze spread uniformly. Gauss's law gives the electric field $E(r)$ (radial, outward for $Z > 0$):

Outside ($r > R$). All charge is enclosed, so

$$E(r) = \frac{1}{4\pi\varepsilon_0}\frac{Ze}{r^2}, \qquad V(r) = \frac{1}{4\pi\varepsilon_0}\frac{Ze}{r} \quad (r > R), \tag{3.2}$$

exactly as for a point charge (with $V(\infty) = 0$).

Inside ($r \le R$). Only the charge fraction r^3/R^3 is enclosed:

$$E(r) = \frac{1}{4\pi\varepsilon_0}\frac{Ze\,r}{R^3}. \tag{3.3}$$

The electrostatic potential is continuous at $r = R$. Integrating inward from the surface,

$$\begin{aligned}
V(r) &= V(R) + \int_r^R E(r')\,\mathrm{d}r' \\
&= \frac{1}{4\pi\varepsilon_0}\frac{Ze}{R} + \frac{1}{4\pi\varepsilon_0}\frac{Ze}{R^3}\int_r^R r'\,\mathrm{d}r' \\
&= \frac{1}{4\pi\varepsilon_0}\frac{Ze}{R} + \frac{1}{4\pi\varepsilon_0}\frac{Ze}{R^3}\cdot\frac{1}{2}(R^2 - r^2) \\
&= \frac{1}{4\pi\varepsilon_0}\frac{Ze}{2R}\left(3 - \frac{r^2}{R^2}\right).
\end{aligned} \tag{3.4}$$

Check continuity at $r = R$

At $r = R$,

$$V(R) = \frac{1}{4\pi\varepsilon_0}\frac{Ze}{2R}(3 - 1) = \frac{1}{4\pi\varepsilon_0}\frac{Ze}{R},$$

which matches the exterior formula. At $r = 0$,

$$V(0) = \frac{1}{4\pi\varepsilon_0}\frac{3Ze}{2R},$$

finite (no singularity).

The electron potential energy is $U = -eV$:

$$U_{\text{finite}}(r) = \begin{cases} -\dfrac{Ze^2}{8\pi\varepsilon_0 R}\left(3 - \dfrac{r^2}{R^2}\right), & r \le R, \\[2ex] -\dfrac{Ze^2}{4\pi\varepsilon_0 r}, & r > R. \end{cases} \tag{3.5}$$

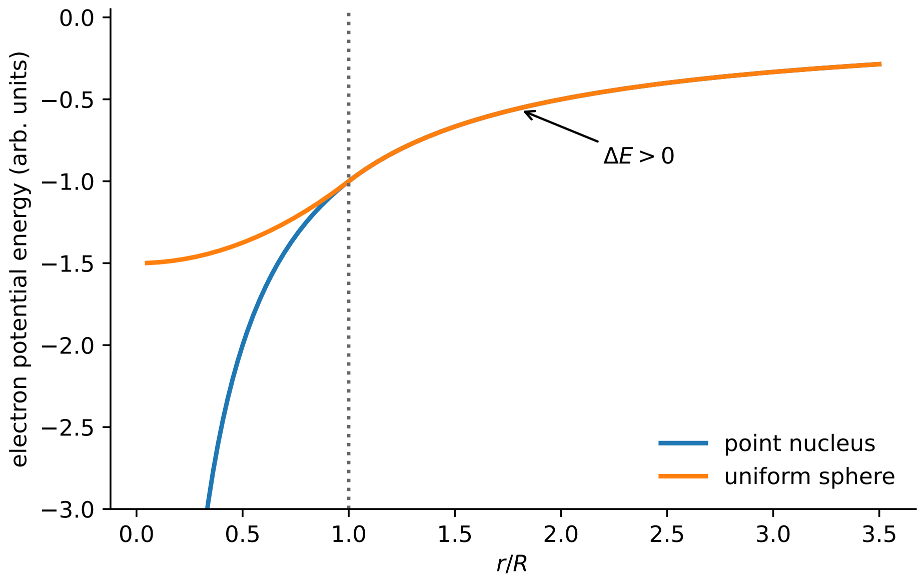


Figure 3.1: Electron potential energy for a point nucleus ($U \propto -1/r$, blue) and for a uniformly charged sphere of radius R (red). Outside the nucleus the two curves coincide. Inside, the finite-size well is shallower (less negative) because part of the nuclear charge lies outside the electron's position and attracts less strongly. The dashed lines are schematic $1s$ energies: raising the well raises the bound level from E_{point} to E_{finite} by $\Delta E > 0$ (weaker binding).

Remark

Compare U_{finite} and U_{point} at fixed $r < R$ in Figure 3.1. The finite-size well is *shallower* (less negative) because the electron, when inside the nucleus, does not feel the full Coulomb attraction of all the charge as if it were concentrated at a point. Raising the bottom of the well raises the **bound-state energy** of the electron, here the $1s$ energy E_{1s}. In the figure the dashed levels show this shift explicitly:

$$E_{\text{finite}} > E_{\text{point}} \qquad (\text{both negative; } |E_{\text{finite}}| < |E_{\text{point}}|),$$

so the finite-size correction is $\Delta E = E_{\text{finite}} - E_{\text{point}} > 0$.

Key idea

Reading Figure 3.1.

- At $r = 0$ the point potential dives to $-\infty$; the finite potential stays finite, $U(0) = -3Ze^2/(8\pi\varepsilon_0 R)$.
- For $0 < r < R$, red lies above blue: less attraction.
- For $r > R$, red = blue (all nuclear charge looks point-like).
- Bound-state energy E_{1s} (dashed) moves up with the well: weaker binding, $\Delta E > 0$.

3.2.3 Definition of the perturbation

Write the full Hamiltonian as $H = H_0 + H_1$, where H_0 is the usual point-nucleus hydrogen Hamiltonian and

$$H_1(r) \equiv U_{\text{finite}}(r) - U_{\text{point}}(r). \tag{3.6}$$

Outside the nucleus the two potentials agree, so $H_1 = 0$ for $r > R$. Inside,

$$\begin{aligned} H_1(r) &= -\frac{Ze^2}{8\pi\varepsilon_0 R}\left(3 - \frac{r^2}{R^2}\right) - \left(-\frac{Ze^2}{4\pi\varepsilon_0 r}\right) \\ &= -\frac{Ze^2}{8\pi\varepsilon_0 R}\left(3 - \frac{r^2}{R^2}\right) + \frac{Ze^2}{4\pi\varepsilon_0 r}. \end{aligned} \tag{3.7}$$

It is often convenient to factor $k \equiv Ze^2/(4\pi\varepsilon_0)$:

$$H_1(r) = k\left[\frac{1}{r} - \frac{3}{2R} + \frac{r^2}{2R^3}\right], \qquad r \le R. \tag{3.8}$$

Check $H_1(R) = 0$

At $r = R$,

$$\frac{1}{R} - \frac{3}{2R} + \frac{1}{2R} = \frac{2 - 3 + 1}{2R} = 0.$$

The perturbation vanishes smoothly at the nuclear surface. If a factor of two is dropped in the Coulomb term, this cancellation fails; that is a common algebraic trap.

Key idea

Only the interior region $r < R$ contributes to the energy shift. The entire calculation reduces to an integral of $H_1(r)$ weighted by the electron probability density inside the nucleus.

3.3 The $1s$ wave function, step by step

3.3.1 Normalised ground state

The hydrogen-like $1s$ wave function is

$$\psi_{1s}(r) = \frac{1}{\sqrt{\pi}}\left(\frac{Z}{a_0}\right)^{3/2} e^{-Zr/a_0}, \tag{3.9}$$

with

$$a_0 = \frac{4\pi\varepsilon_0 \hbar^2}{m_e e^2} \approx 0.529\,\text{Å} = 5.29 \times 10^{-11}\,\text{m}. \tag{3.10}$$

Why these factors?

- The exponential e^{-Zr/a_0} is the solution of the radial Schrödinger equation for $n = 1$, $\ell = 0$. Larger Z pulls the cloud inward (effective Bohr radius a_0/Z).

- The prefactor is fixed by normalisation $\int |\psi|^2\,\mathrm{d}^3r = 1$. For a purely radial function, $\int_0^\infty 4\pi r^2 |\psi|^2\,\mathrm{d}r = 1$.

Density at the origin. Set $r = 0$:

$$\psi_{1s}(0) = \frac{1}{\sqrt{\pi}}\left(\frac{Z}{a_0}\right)^{3/2}, \qquad |\psi_{1s}(0)|^2 = \frac{Z^3}{\pi a_0^3}. \tag{3.11}$$

The Z^3 dependence is crucial: heavier nuclei (larger Z) pack far more electron density near $r = 0$, so finite-size shifts grow rapidly with Z.

Caution

ψ_{1s} itself is proportional to $Z^{3/2}$, not Z^3. The quantity $Z^3/(\pi a_0^3)$ is $|\psi(0)|^2$, the probability density. Mixing the two is a frequent source of wrong powers of Z in ΔE.

3.3.2 First-order perturbation theory (what the formula means)

If the unperturbed eigenproblem is $H_0\psi_n = E_n^{(0)}\psi_n$ and we add a small H_1, the first-order energy correction is

$$\Delta E_n = \langle \psi_n | H_1 | \psi_n \rangle = \int \psi_n^*\, H_1\, \psi_n\, \mathrm{d}^3r. \tag{3.12}$$

For the real $1s$ wave function this is simply the expectation value of H_1 in the unperturbed ground state:

$$\Delta E = \int |\psi_{1s}(r)|^2 H_1(r)\,\mathrm{d}^3r. \tag{3.13}$$

Physical reading: average the extra potential energy H_1 with the probability density of finding the electron at each point. Since $H_1 = 0$ outside the nucleus, only the nuclear ball contributes.

3.4 Evaluating ΔE: full algebraic development

3.4.1 Write the integral in spherical coordinates

Because $|\psi_{1s}|^2$ and H_1 depend only on r,

$$\mathrm{d}^3r = 4\pi r^2\,\mathrm{d}r, \tag{3.14}$$

and

$$\Delta E = \int_0^R \underbrace{\frac{Z^3}{\pi a_0^3} e^{-2Zr/a_0}}_{|\psi_{1s}(r)|^2} H_1(r)\, 4\pi r^2\,\mathrm{d}r. \tag{3.15}$$

(The factor e^{-2Zr/a_0} comes from $|e^{-Zr/a_0}|^2 = e^{-2Zr/a_0}$.)

3.4.2 Approximation: the exponential is almost 1

We must integrate only up to $r = R$. Compare length scales:

Quantity	Typical size	Comment
Nuclear radius R	$\sim 1--7\,\mathrm{fm}$	few fm
Bohr radius a_0	$5.29\times10^{-11}\,\mathrm{m}$	atomic scale
Ratio R/a_0	$\sim 2\times10^{-5}$	extremely small

For any $r \le R$,

$$\frac{2Zr}{a_0} \le \frac{2ZR}{a_0} \sim 2Z\times10^{-5}. \tag{3.16}$$

Even for uranium ($Z = 92$),

$$\frac{2ZR}{a_0} \approx \frac{2(92)(7.4\,\mathrm{fm})}{5.29\times10^4\,\mathrm{fm}} \approx 2.6\times10^{-2} \ll 1. \tag{3.17}$$

Taylor expand:

$$e^{-2Zr/a_0} = 1 - \frac{2Zr}{a_0} + \cdots \approx 1, \tag{3.18}$$

with a relative error below a few percent even for the heaviest electronic ions, and much smaller for moderate Z. For high Z, relativistic Dirac wave functions are nevertheless required for precision work. Therefore, to an excellent approximation,

$$\Delta E \approx |\psi_{1s}(0)|^2 \int_0^R H_1(r)\,4\pi r^2\,\mathrm{d}r = |\psi_{1s}(0)|^2 \int_{r<R} H_1\,\mathrm{d}^3r. \tag{3.19}$$

Key idea

Because the nucleus is a speck on the atomic scale, the electron density hardly varies across it. Pull $|\psi(0)|^2$ out of the integral; only the volume integral of H_1 remains.

3.4.3 Volume integral of H_1 (line by line)

Start from Equation (3.8):

$$H_1(r) = k\left(\frac{1}{r} - \frac{3}{2R} + \frac{r^2}{2R^3}\right), \qquad k = \frac{Ze^2}{4\pi\varepsilon_0}. \tag{3.20}$$

Then

$$\begin{aligned} I \equiv \int_{r<R} H_1\,\mathrm{d}^3r &= \int_0^R H_1(r)\,4\pi r^2\,\mathrm{d}r \\ &= 4\pi k \int_0^R \left(r - \frac{3r^2}{2R} + \frac{r^4}{2R^3}\right)\mathrm{d}r. \end{aligned} \tag{3.21}$$

Integrate term by term:

$$\int_0^R r\,\mathrm{d}r = \frac{R^2}{2}, \tag{3.22}$$

$$\int_0^R \frac{3r^2}{2R}\,\mathrm{d}r = \frac{3}{2R}\cdot\frac{R^3}{3} = \frac{R^2}{2}, \tag{3.23}$$

$$\int_0^R \frac{r^4}{2R^3}\,\mathrm{d}r = \frac{1}{2R^3}\cdot\frac{R^5}{5} = \frac{R^2}{10}. \tag{3.24}$$

The first two contributions cancel:

$$\frac{R^2}{2} - \frac{R^2}{2} + \frac{R^2}{10} = \frac{R^2}{10}. \tag{3.25}$$

Therefore

$$I = 4\pi k\cdot\frac{R^2}{10} = \frac{2\pi}{5}\,k\,R^2 = \frac{2\pi}{5}\,\frac{Ze^2}{4\pi\varepsilon_0}R^2 = \frac{Ze^2R^2}{10\,\varepsilon_0}. \tag{3.26}$$

Why the cancellation?

The $1/r$ and constant pieces of H_1 rearrange the Coulomb singularity into a finite well. Their weighted integrals over the ball cancel, leaving only the r^2 piece of the interior potential, which produces the factor $R^2/10$. That is why $\Delta E \propto R^2$, not R^3: after cancellation one power of R is lost relative to a naive volume estimate.

3.4.4 Assemble ΔE

$$\begin{aligned}\Delta E = |\psi_{1s}(0)|^2\,I &= \frac{Z^3}{\pi a_0^3}\cdot\frac{2\pi}{5}\cdot\frac{Ze^2}{4\pi\varepsilon_0}R^2\\ &= \frac{Z^3}{\pi a_0^3}\cdot\frac{Ze^2R^2}{10\,\varepsilon_0}\cdot\frac{2\pi}{2\pi}\\ &= \frac{Z^4e^2R^2}{10\pi\,\varepsilon_0\,a_0^3}.\end{aligned} \tag{3.27}$$

Final formula: $1s$ finite-size shift (uniform sphere)

$$\boxed{\Delta E_{1s} = \frac{Z^4e^2R^2}{10\pi\,\varepsilon_0\,a_0^3} > 0}$$

How to read the powers.

- $Z^4 = Z\cdot Z^3$: one Z from the Coulomb strength Ze^2, and Z^3 from $|\psi(0)|^2$.
- R^2: residual after the integral cancellation discussed above.
- $1/a_0^3$: denser electron cloud for smaller Bohr radius.
- Positive sign: the well is shallower, so the level moves *up* (less bound).

Example: scaling with Z and R

Compare hydrogen ($Z = 1$, $R \sim 1\,\text{fm}$) with mercury ($Z = 80$, $R \sim 7\,\text{fm}$):

$$\frac{\Delta E(\text{Hg})}{\Delta E(\text{H})} \sim 80^4 \times 7^2 \approx 4.1 \times 10^7 \times 49 \approx 2 \times 10^9.$$

Hydrogen's shift is $\sim 10^{-10}$ of $|E_{1s}|$; mercury's absolute shift is still only $\sim 0.1\,\text{eV}$, but that is measurable against a K_α energy of $\sim 100\,\text{keV}$.

3.4.5 Link to the mean-square radius

For a general spherical charge density (not necessarily uniform), the leading result follows most cleanly in momentum space. Normalise its form factor by

$$F(\boldsymbol{q}) = \frac{1}{Ze} \int \rho_{\text{ch}}(\boldsymbol{r})\, e^{i\boldsymbol{q}\cdot\boldsymbol{r}/\hbar}\, \mathrm{d}^3 r = 1 - \frac{q^2 \langle r^2 \rangle_{\text{ch}}}{6\hbar^2} + O(q^4).$$

The Coulomb kernel is proportional to $\hbar^2/q^2$. Subtracting the point source cancels the pole:

$$\begin{aligned} \delta U(\boldsymbol{q}) &= -\frac{Ze^2\hbar^2}{\varepsilon_0 q^2}\left[F(\boldsymbol{q}) - 1\right] \\ &= \frac{Ze^2}{6\varepsilon_0}\langle r^2 \rangle_{\text{ch}} + O(q^2). \end{aligned} \tag{3.28}$$

A constant in momentum space is a contact interaction in position space,

$$\delta U(\boldsymbol{r}) = \frac{2\pi}{3}\,\frac{Ze^2}{4\pi\varepsilon_0}\,\langle r^2 \rangle_{\text{ch}}\,\delta^{(3)}(\boldsymbol{r}) + \text{higher moments}. \tag{3.29}$$

Taking its expectation value gives

$$\begin{aligned} \Delta E_{1s} &= |\psi_{1s}(0)|^2 \int \left[U_{\text{finite}}(\boldsymbol{r}) - U_{\text{point}}(\boldsymbol{r})\right] \mathrm{d}^3 r \\ &= \frac{2\pi}{3}\,\frac{Ze^2}{4\pi\varepsilon_0}\langle r^2 \rangle_{\text{ch}} |\psi_{1s}(0)|^2, \end{aligned} \tag{3.30}$$

where omitted terms contain higher nuclear moments and derivatives of the electronic density. For a uniform sphere,

$$\langle r^2 \rangle = \frac{\int_0^R r^2 \cdot 4\pi r^2\,\mathrm{d}r}{\frac{4}{3}\pi R^3} = \frac{3}{5}R^2. \tag{3.31}$$

Inserting this into Equation (3.30) recovers Equation (3.27):

$$\frac{2\pi}{3}\,k \cdot \frac{3}{5}R^2 \cdot \frac{Z^3}{\pi a_0^3} = \frac{2\pi}{5}\,kR^2 \cdot \frac{Z^3}{\pi a_0^3} = \frac{Z^4 e^2 R^2}{10\pi\varepsilon_0 a_0^3}. \tag{3.32}$$

So the uniform-sphere calculation is a special case of a more general theorem: the leading shift measures $\langle r^2 \rangle_{\text{ch}}$.

Writing E_{1s}^{point} for the unperturbed energy,

$$E_{1s} = E_{1s}^{\text{point}} + \Delta E_{1s}. \tag{3.33}$$

3.5 K_α X-rays: from levels to a measurable line

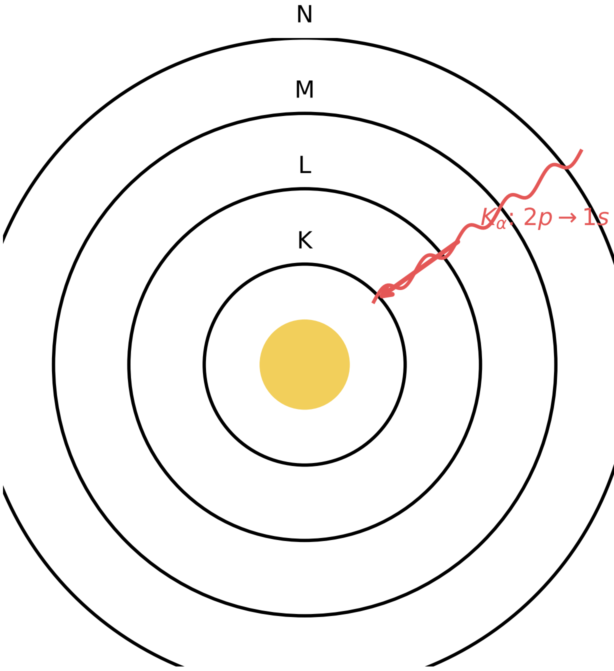


Figure 3.2: Atomic shells. A vacancy in the K shell ($1s$) is filled from a higher shell, emitting an X-ray. K_α corresponds mainly to $2p \to 1s$; K_β involves higher initial levels.

3.5.1 What is measured?

If a $1s$ electron is removed, an electron from a higher shell can drop into the vacancy. The energy difference is emitted as a characteristic X-ray (Figure 3.2). The K_α complex is dominated by $2p \to 1s$ transitions.

Historical note: Henry Moseley (1887–1915)

Henry Gwyn Jeffreys Moseley showed (1913–1914) that characteristic X-ray frequencies scale as $(Z-1)^2$, establishing Z as the true ordering parameter of the elements. That is why K_α energies are such a clean probe of nuclear charge and, through isotope shifts, of nuclear size.

3.5.2 Point-nucleus level energies

For a hydrogen-like ion,

$$E_n^{\text{point}} = -\frac{m_e Z^2 e^4}{8\varepsilon_0^2 h^2\, n^2} = -\frac{Z^2}{n^2}\,\text{Ry}, \qquad \text{Ry} \approx 13.6\,\text{eV}. \tag{3.34}$$

Thus

$$E_{1s}^{\text{point}} = -Z^2\,\text{Ry}, \qquad E_2^{\text{point}} = -\frac{Z^2}{4}\,\text{Ry} \quad \text{(includes } 2p\text{)}. \tag{3.35}$$

Photon energy for $2p \to 1s$. The electron falls from a less-bound level to a more-bound level. The photon energy is

$$\begin{aligned} E_{K_\alpha}^{\text{point}} &= E_2^{\text{point}} - E_{1s}^{\text{point}} \\ &= \left(-\frac{Z^2}{4} + Z^2\right)\text{Ry} = \frac{3}{4}\,Z^2\,\text{Ry}. \end{aligned} \tag{3.36}$$

(For real atoms, screening replaces Z by $\approx Z-1$, as in Moseley's law; the structure of the finite-size argument is unchanged.)

3.5.3 How ΔE_{1s} changes K_α

In non-relativistic theory, p-state wave functions vanish at $r=0$, so their finite-size shifts are much smaller than that of $1s$. Keeping only ΔE_{1s},

$$\begin{aligned} E_{K_\alpha} &= E_2^{\text{point}} - (E_{1s}^{\text{point}} + \Delta E_{1s}) \\ &= (E_2^{\text{point}} - E_{1s}^{\text{point}}) - \Delta E_{1s} \\ &= E_{K_\alpha}^{\text{point}} - \Delta E_{1s}. \end{aligned} \tag{3.37}$$

Sign: why K_α softens

- Finite size raises E_{1s} (makes it less negative).
- E_2 is almost unchanged.
- The gap $E_2 - E_{1s}$ therefore shrinks.
- The emitted X-ray has slightly lower energy.

Key idea

$$E_{K_\alpha} = E_{K_\alpha}^{\text{point}} - \frac{Z^4 e^2 R^2}{10\pi\,\varepsilon_0\,a_0^3}.$$

Larger $R \Rightarrow$ smaller E_{K_α}.

3.6 Isotope shifts: cancelling everything except R

3.6.1 Why isotopes?

Two isotopes of the same element have:

- the *same* Z (same Coulomb strength, same electronic structure to an excellent approximation);
- *different* A, hence different nuclear radii $R(A)$.

The dominant point-nucleus pieces cancel in a difference, but a measured isotope shift is not a pure radius observable without atomic-structure corrections.

Field shift versus mass shift

For a transition i, the leading isotope shift is

$$\delta\nu_i^{A,A'} = K_{i,\mathrm{MS}}\left(\frac{1}{A} - \frac{1}{A'}\right) + F_i\,\delta\langle r^2\rangle^{A,A'} + \cdots.$$

The first term is the normal plus specific *mass shift* (nuclear recoil and correlated electron motion); the second is the *field shift* caused by the changed charge distribution. In heavy atoms the field shift often dominates, but extracting a radius still requires calculated $K_{i,\mathrm{MS}}$ and F_i. Electron screening, relativistic wave functions, nuclear deformation, and higher moments also matter. In Dirac theory even $p_{1/2}$ states have nonzero nuclear overlap, so the claim that every p shift vanishes is only a non-relativistic leading-order statement [6, 2].

3.6.2 Algebra for two isotopes

Let the reference isotope have mass number A_0 and radius R_0, and another isotope have A and R. Then

$$E_{K_\alpha}(A_0) = E_{K_\alpha}^{\text{point}} - CR_0^2, \tag{3.38}$$
$$E_{K_\alpha}(A) = E_{K_\alpha}^{\text{point}} - CR^2, \tag{3.39}$$

where we abbreviated the positive constant

$$C \equiv \frac{Z^4 e^2}{10\pi\,\varepsilon_0\,a_0^3}. \tag{3.40}$$

Subtract:

$$\begin{aligned}\Delta E_{K_\alpha} &\equiv E_{K_\alpha}(A) - E_{K_\alpha}(A_0)\\ &= (E_{K_\alpha}^{\text{point}} - CR^2) - (E_{K_\alpha}^{\text{point}} - CR_0^2)\\ &= C(R_0^2 - R^2).\end{aligned} \tag{3.41}$$

Interpretation. If the second isotope is heavier, typically $R > R_0$, so $R_0^2 - R^2 < 0$ and $\Delta E_{K_\alpha} < 0$. The heavier isotope's K_α line lies at slightly lower energy. Experimental plots often display $-\Delta E_{K_\alpha}$ (a positive quantity that grows with A).

3.6.3 Insert $R_0 = 1.2\,\mathrm{fm}\,A^{1/3}$

Electron-scattering systematics give the half-density radius [1]

$$R = R_0 = 1.2\,\mathrm{fm} \times A^{1/3} \quad\Rightarrow\quad R^2 = (1.2\,\mathrm{fm})^2\,A^{2/3}. \tag{3.42}$$

Then

$$\Delta E_{K_\alpha} = C\,(1.2\,\mathrm{fm})^2(A_0^{2/3} - A^{2/3}) = \frac{Z^4 e^2}{10\pi\,\varepsilon_0\,a_0^3}(1.2\,\mathrm{fm})^2(A_0^{2/3} - A^{2/3}). \tag{3.43}$$

Equivalently,

$$-\Delta E_{K_\alpha} = C\,(1.2\,\mathrm{fm})^2(A^{2/3} - A_0^{2/3}). \tag{3.44}$$

What to plot

Fix a reference isotope A_0. For each other isotope A, compute $-\Delta E_{K_\alpha}$ and plot it against $A^{2/3}$.

- **If the points fall on a straight line**, the data support $R \propto A^{1/3}$.
- **The slope** equals $C \times (1.2\,\mathrm{fm})^2$ if the R_0 law holds; a measured slope is a consistency check on that same coefficient.

Takeaway

Experimental checklist:

1. Measure E_{K_α} for several isotopes of one element.
2. Form shifts relative to a reference isotope.
3. Plot $-\Delta E$ versus $A^{2/3}$.
4. Linearity $\Rightarrow R \propto A^{1/3}$; slope checks $R_0 = 1.2\,\mathrm{fm}\,A^{1/3}$ from form-factor data [1].

3.6.4 Mercury data

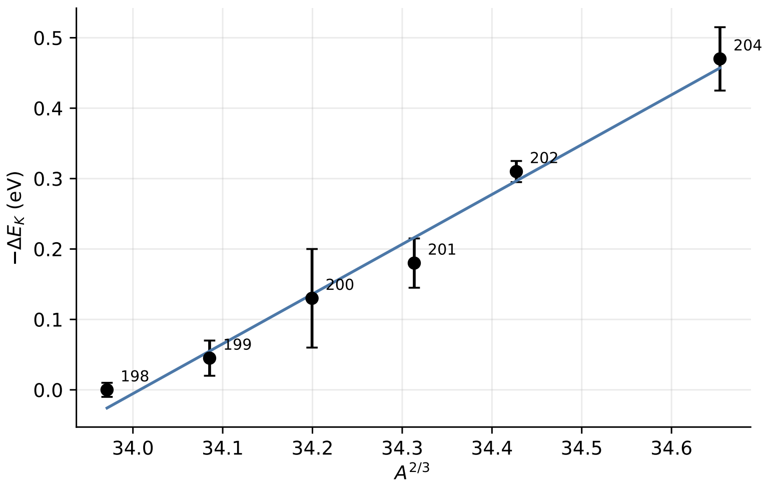


Figure 3.3: K_α isotope shifts in mercury ($Z = 80$), plotted as $-\Delta E$ versus $A^{2/3}$ relative to ^{198}Hg. Even and odd isotopes lie on nearly straight lines, as expected if $R \propto A^{1/3}$. Data redrawn after [6] (Lee et al., Phys. Rev. C **17**, 1859 (1978)).

Mercury ($Z = 80$) has several stable isotopes near $A \sim 200$. Figure 3.3 plots $-\Delta E$ against $A^{2/3}$ with ^{198}Hg as reference. The near-linearity is the experimental signature of $R \propto A^{1/3}$. Small odd–even staggering reflects pairing and slight deformation beyond the ideal spherical model.

Example: how small is the shift?

For mercury,

$$E_{K_\alpha} \sim 100\,\text{keV} = 10^5\,\text{eV}, \qquad |\Delta E| \sim 0.1\,\text{eV between nearby isotopes.}$$

The fractional change is

$$\frac{|\Delta E|}{E_{K_\alpha}} \sim 10^{-6}.$$

That is one part per million: tiny, but accessible to modern precision X-ray spectroscopy. The slope of the line, together with many other experiments, gives the standard $r_0 \approx 1.2\,\text{fm}$.

3.7 Muonic atoms: the same maths with a heavy lepton

Electronic shifts are parts per million. Nature provides a louder version of the same calculation: replace the electron by a muon.

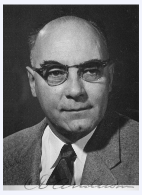
3.7.1 Muon basics

The muon μ^- has charge $-e$ and mass

$$m_\mu \approx 207\, m_e. \tag{3.45}$$

Lifetime $\tau_\mu \approx 2.2\,\mu\text{s}$ (rest frame), long enough to be captured into atomic orbits and emit X-rays before decaying ($\mu^- \to e^- \bar{\nu}_e \nu_\mu$).

Historical note: Carl D. Anderson (1905–1991)



Carl David Anderson discovered the positron (1932) and the muon in cosmic rays (1936, with Neddermeyer). The muon was first mistaken for Yukawa's meson; it is in fact a heavy lepton. Anderson's particle is the probe behind modern muonic-atom determinations of nuclear charge radii.

3.7.2 How the Bohr radius scales

The Bohr radius comes from balancing Coulomb attraction against kinetic energy. With reduced mass μ of the orbiting lepton and the nucleus,

$$a_\mu = \frac{4\pi\varepsilon_0\,\hbar^2}{\mu e^2} = a_\text{B}\,\frac{m_e}{\mu}. \tag{3.46}$$

For an electron on a heavy nucleus, $\mu \approx m_e$ and $a \approx a_\text{B}$. For a muon, $\mu \approx m_\mu$ (still $\ll m_\text{nucleus}$), so

$$a_\mu = a_0\,\frac{m_e}{m_\mu} \approx \frac{a_0}{207}. \tag{3.47}$$

Caution

The mass ratio is ≈ 207, not 27. Hence $a_\mu = a_0/207$. Using 27 underestimates how tightly the muon is bound by almost an order of magnitude in linear size.

3.7.3 How ΔE scales for a muon

Repeat Equation (3.27) with $a_0 \to a_\mu$ and $m_e \to m_\mu$ in the wave function. The structure is unchanged:

$$\Delta E^{(\mu)} = \frac{Z^4 e^2 R^2}{10\pi\,\varepsilon_0\,a_\mu^3}. \tag{3.48}$$

Compared with the electronic shift,

$$\frac{\Delta E^{(\mu)}}{\Delta E^{(e)}} = \left(\frac{a_0}{a_\mu}\right)^3 = \left(\frac{m_\mu}{m_e}\right)^3 \approx 207^3 \approx 8.9 \times 10^6. \tag{3.49}$$

So muonic finite-size shifts are roughly *ten million times* larger in the naive scaling (before accounting for the fact that for heavy nuclei the muon orbit is so small that the constant-ψ approximation and non-relativistic theory both need upgrading).

Example: muon K-shell size in lead

Electron K radius $\sim a_0/Z$. Muon K radius $\sim a_\mu/Z = a_0/(207\,Z)$. For lead ($Z = 82$),

$$\frac{a_\mu}{Z} \approx \frac{0.529\,\text{Å}}{207 \times 82} \approx 3\,\text{fm},$$

comparable to the nuclear radius. The muon wave function overlaps the nuclear charge distribution strongly; precision $\langle r^2 \rangle$ follows from measured muonic X-ray energies.

Key idea

Same formulae, different mass: muonic atoms turn a parts-per-million electronic effect into a large, accurately measurable shift, at the price of a more careful (often relativistic, numerical) treatment for heavy Z.

Remark

For heavy nuclei one no longer pulls $|\psi(0)|^2$ out as a constant: the muon $1s$ state varies over nuclear dimensions. One then solves the Dirac equation in the electrostatic potential of a realistic $\rho(r)$ and adjusts $\langle r^2 \rangle$ (or a full density model) to match the observed transitions. The conceptual origin of the shift is still the same H_1 we built in Section 3.2.

3.8 Worked summary of the mathematics

One-page derivation map

1. $U_{\text{point}} = -\dfrac{Ze^2}{4\pi\varepsilon_0 r}$.
2. Uniform sphere $\Rightarrow U_{\text{finite}} = -\dfrac{Ze^2}{8\pi\varepsilon_0 R}(3 - r^2/R^2)$ for $r \le R$.
3. $H_1 = U_{\text{finite}} - U_{\text{point}}$ (zero outside).
4. $\Delta E = \int |\psi_{1s}|^2 H_1 \, \mathrm{d}^3 r$.
5. $e^{-2Zr/a_0} \approx 1$ on $r \le R$ because $R \ll a_0$.
6. $\Delta E \approx |\psi(0)|^2 \int H_1 \, \mathrm{d}^3 r$.
7. $\int H_1 \, \mathrm{d}^3 r = \dfrac{2\pi}{5} k R^2$ after cancellation of two terms in the radial integral.
8. $|\psi(0)|^2 = Z^3/(\pi a_0^3) \Rightarrow \Delta E = \dfrac{Z^4 e^2 R^2}{10\pi\varepsilon_0 a_0^3}$.
9. $E_{K_\alpha} = E_{K_\alpha}^{\text{point}} - \Delta E$.
10. Isotope difference $\propto (R_0^2 - R^2)$; with $R = 1.2\,\text{fm}\,A^{1/3}$, plot vs $A^{2/3}$.
11. Muon: replace $a_0 \to a_0/207$; shift grows as 207^3.

3.9 Three experimental routes to nuclear size

1. **Electron scattering** (Lectures 1–2). Maps $F(q)$ and $\rho_p(r)$; yields $R_0 = 1.2\,\text{fm}\,A^{1/3}$ and $a = 0.52\,\text{fm} \pm 0.01\,\text{fm}$ [1].
2. **Electronic K_α isotope shifts** (this lecture). $\Delta E_{1s} = Z^4 e^2 R^2/(10\pi\varepsilon_0 a_0^3)$ softens K_α; differences test $A^{1/3}$ via linearity in $A^{2/3}$.
3. **Muonic atoms**. Same maths with $a_\mu = a_0/207$; much larger shifts and precise $\langle r^2 \rangle_{\text{ch}}$.

Takeaway

The algebra is lengthy, but the physics is simple: a finite nucleus softens the Coulomb singularity, raises s-state energies by an amount $\propto Z^4 R^2$, and that shift is visible in X-rays once you compare isotopes or switch to muons. The same $R_0 = 1.2\,\text{fm}\,A^{1/3}$ that appears in electron scattering [1] reappears in the slope of the isotope-shift plot.

Next. Lecture 4 leaves size for *energy*: binding energy, the B/A curve, and the liquid-drop SEMF. Lecture 5 then turns the same mass formula into mass parabolas and β stability.

3.9.1 Electronic, optical, and muonic isotope shifts

Electronic K X-ray and optical isotope shifts in chains such as Hg are tiny (parts in 10^6–10^7) yet resolve the predicted $A^{2/3}$ trend in $\langle r^2 \rangle$. Odd–even staggering of radii requires plotting odd-A and even-A isotopes separately. Muonic K X-rays amplify the same finite-size physics by roughly $(m_\mu/m_e)^3$ and make isotope shifts of order 0.1% visually obvious on the same figure scale [6, 1, 14].

3.10 Deeper physics: comparing size probes

Lectures 1–3 measure nuclear size three ways. The results agree at the percent level only when each method quotes a carefully defined moment of ρ_{ch} [6, 1, 2].

3.10.1 Electron scattering vs atomic shifts

Probe	What is measured	Sensitivity
Elastic e^- scattering	$\lvert F(q)\rvert^2 \to \rho_{\text{ch}}(r)$	Full radial profile; model-dependent at large q
Electronic K_α / optical isotope shift	$\Delta\langle r^2\rangle$ between isotopes	Relative radii; absolute scale needs calibration
Muonic atoms	Large finite-size shift $\propto m_\mu^3$	Absolute $\langle r^2\rangle_{\text{ch}}$ for many Z

Muonic 1s levels are far more sensitive than electronic ones because the Bohr radius scales as $1/m$: $m_\mu/m_e \approx 207$ compresses the orbit by the same factor and boosts the overlap with the nucleus by $\sim 207^3$. That is why muonic X-rays became a precision tool for charge radii once muon beams were available [2, 6].

3.10.2 Mean-square radius as the universal intermediate

All three methods ultimately quote moments of ρ_{ch}. For a uniform sphere of radius R,

$$\langle r^2\rangle = \tfrac{3}{5}R^2, \tag{3.50}$$

so $R_{\text{eq}} = \sqrt{5/3}\,\langle r^2\rangle^{1/2}$. Isotope-shift formulae depend on $\delta\langle r^2\rangle$, not on a model-dependent surface thickness. Electron scattering at low q measures the same $\langle r^2\rangle$ from the form-factor curvature (Lecture 1). Agreement among methods is a major empirical success of nuclear charge-density physics.

Order of magnitude for the 1s shift

For $Z \sim 80$ and $R \sim 7\,\text{fm}$, the electronic finite-size correction to the 1s level is only of order eV–keV on a keV–MeV scale, yet modern X-ray and laser spectroscopy resolve isotope differences cleanly. The same algebra with $m \to m_\mu$ yields keV to MeV shifts: muonic spectroscopy is a nuclear, not atomic, measurement in disguise [6].

Key idea

Absolute shift vs isotope difference. The absolute finite-size correction $\Delta E(A, Z)$ is entangled with atomic-structure uncertainties. Differences $\Delta E(A, Z) - \Delta E(A', Z)$ cancel most of that background and isolate $\delta\langle r^2\rangle$. Electron scattering and muonic atoms calibrate the absolute scale that isotope shifts then track along a chain.

3.11 Physics depth: finite-size energy and isotope differences

A pointlike nucleus would produce hydrogenic energies fixed by Z and reduced mass alone. Finite size spreads the proton charge over a ball of radius $\sim R$, lowering the electrostatic potential felt by an s-electron (or muon) inside the nucleus and shifting level energies.

3.11.1 Worked estimate: $1s$ shift scaling

In first-order perturbation theory the leading finite-size correction scales as

$$\Delta E_{1s} \propto Z^4 \langle r^2\rangle |\psi(0)|^2, \tag{3.51}$$

with $|\psi(0)|^2$ enormous for muonic atoms because $m_\mu/m_e \simeq 207$ shrinks the Bohr radius. For ordinary electronic K_α shifts the effect is parts in 10^4 or smaller; for muonic X-rays it reaches mega-electronvolt scales in heavy elements. Isotope shifts cancel most atomic-structure background and leave $\delta\langle r^2\rangle$ as the nuclear observable.

King-plot linearity intuition

Two optical transitions with field-shift factors F_i and mass-shift coefficients K_i give a linear King plot if only the leading $\langle r^2\rangle$ and mass terms matter. Curvature signals either higher nuclear moments (deformation, polarisability) or an incomplete electronic factor. Before claiming new physics, vary F within published uncertainties and recompute the residual.

Remark

Limits. Optical isotope shifts are differential; electron scattering and muonic atoms set the absolute radius ladder. Mixing the two without a common radius convention (point vs rms vs diffraction) produces apparent discrepancies that are only definitional.

3.11.2 Muonic versus electronic leverage

The muonic Bohr radius scales as $1/(Zm_\mu)$. For lead, the muon $1s$ wave function has substantial overlap with the nuclear volume, so the finite-size shift is a leading, not perturbative, piece of the X-ray energy. Electronic isotope shifts reverse the hierarchy: the nuclear piece is a small correction on top of a huge atomic transition, which is why differential measurements and King plots are essential.

Deformation adds a quadrupole contribution to $\langle r^2\rangle$ and can produce odd–even staggering beyond simple pairing. When an isotopic chain crosses a shape transition (Lecture 15), the slope of $\langle r^2\rangle$ versus N changes abruptly. Charge radii are therefore collective as well as single-particle observables.

Practical analysis often quotes the Barrett radius or a Fourier-Bessel fit rather than a single rms number. When converting a published isotope shift into $\delta\langle r^2\rangle$, write the field-shift formula with units and the electronic factor F explicitly, then propagate δF. A 10% error in F is common and can dominate the nuclear uncertainty budget for light chains where the field shift is not overwhelming.

Worked optical versus muonic comparison

Take a published optical $\delta\langle r^2\rangle$ along even–even cadmium or tin and convert to a differential radius. Compare the absolute radius at the anchor isotope with a muonic or elastic-electron value. Discrepancies at the 0.01 fm level are common and usually reflect electronic-factor or model-radius definitions rather than a failure of finite-size theory. Document definitions beside every number.

Physics-depth close

Isotope shifts are differential nuclear radii dressed in atomic clothing. The nuclear content is $\delta\langle r^2\rangle$; the atomic content is F and K. Separating them cleanly is the craft this chapter adds to the absolute form-factor programme of Lecture 1.

3.12 Further physics: separation energies and the isotope-shift ladder

Atomic size enters spectroscopy through level shifts; nuclear size also leaves fingerprints in the energies needed to remove a nucleon. The neutron and proton separation energies

$$S_n = [M(A-1,Z)+m_n-M(A,Z)]c^2, \qquad S_p = [M(A-1,Z-1)+m(^1\mathrm{H})-M(A,Z)]c^2 \quad (3.52)$$

are the nuclear analogues of ionisation energies: they probe valence nucleons. Sharp jumps in S_n or S_p at closed shells are shell signatures; smooth trends with A and Z follow the SEMF surface, Coulomb, and pairing terms (Lectures 4–5) [13, 6, 1].

Finite-size shifts of s levels and separation-energy systematics are complementary: both are sensitive to how tightly the outer nucleons (or the electron cloud near $r = 0$) feel the nuclear interior.

3.12.1 Mirror nuclei and a Coulomb-radius estimate

Binding-energy differences of mirror pairs provide an independent size check. For example,

$$B(^3\mathrm{H}) - B(^3\mathrm{He}) \approx 0.76\,\mathrm{MeV}$$

is largely the Coulomb energy of the two protons in ^{3}He. Writing

$$\Delta B \approx \frac{e^2}{4\pi\varepsilon_0}\left\langle\frac{1}{r_{12}}\right\rangle \sim \frac{e^2}{4\pi\varepsilon_0 R}$$

yields $R \sim 2\,\mathrm{fm}$, consistent with other light-nucleus radii [14]. Finite-size isotope shifts (this lecture) and mirror Coulomb differences are two faces of the same electrostatic physics.

3.12.2 Isotope-shift ladder: optical, K_α, and muonic

Three experimental ladders measure the same change of nuclear mean-square radius between isotopes A and A' [6, 1]:

1. **Optical isotope shifts** (valence transitions). The electron density at the nucleus is small, so shifts are tiny (parts in 10^6 of optical frequencies) but laser spectroscopy resolves them for long chains of radioactive isotopes.
2. K_α **X-ray isotope shifts.** The $2p \to 1s$ transition energy difference between isotopes is essentially the difference of $1s$ finite-size shifts, because p wave functions vanish at $r = 0$. A plot of ΔE_K versus $A^{2/3}$ is approximately linear for a sequence of isotopes [6].
3. **Muonic X rays.** The muon Bohr radius is $\sim m_e/m_\mu$ times smaller than the electronic one, so the muon spends a much larger fraction of its time inside the nucleus. Isotope shifts are correspondingly huge and dominate over many atomic-structure uncertainties.

For a uniform sphere the first-order $1s$ finite-size shift scales as

$$\Delta E_{1s} \propto Z\,|\psi_{1s}(0)|^2\,\langle r^2\rangle_{\rm ch} \propto Z^4 R^2, \tag{3.53}$$

so $\Delta E(A) - \Delta E(A') \propto R^2(A) - R^2(A')$. All three methods, after careful atomic-structure corrections, return R_0 consistent with electron scattering ($\sim 1.2\,\mathrm{fm}$).

3.12.3 Forward look: magic numbers in separation energies

Just as atomic ionisation energies jump at noble-gas Z, nuclear separation energies jump at magic N or Z:

$$N, Z = 2, 8, 20, 28, 50, 82, 126. \tag{3.54}$$

Lead isotopes show a clear discontinuity in S_n at $N = 126$ (^{208}Pb). The SEMF alone is smooth; the excess binding of magic and doubly magic nuclei relative to the liquid-drop fit is a shell-model signature that returns when single-particle potentials are discussed [14, 6]. For now it is enough that size probes and separation energies both organise the chart of nuclides before binding-energy systematics take over in Lecture 4.

3.13 Deeper reading: from $\langle r^2\rangle$ to the nuclear chart

Along an isotopic chain, $\langle r^2\rangle$ generally grows with N, but shell closures and deformation onsets produce slope changes and odd–even staggering. Those features are the spectroscopic fingerprints that Lectures 12–15 will reinterpret with single-particle and collective language.

3.13.1 Order-of-magnitude bridge

A change $\delta\langle r^2\rangle = 0.1\,\mathrm{fm}^2$ is a sub-percent effect on $R_{\rm rms} \sim 5\,\mathrm{fm}$, yet it is routinely resolved by laser spectroscopy. That sensitivity is why radioactive-beam isotope-shift programmes map charge radii far from stability where electron-scattering form factors are inaccessible. Lecture 1 still supplies the absolute calibration; Lecture 3 supplies the differential microscope; Lecture 5's mass parabolas ask a parallel question for binding rather than size.

3.13.2 Field shift versus mass shift

The mass shift contains normal and specific contributions from electron correlations. In light nuclei the mass shift can dominate; in heavy nuclei the field shift wins. Separating them requires either a King plot with two transitions or ab initio electronic factors. Nuclear physicists consuming isotope-shift radii should always ask which electronic factors were used and whether a King-plot nonlinearity was tested. That discipline prevents over-interpreting a radius kink as a shell closure when it might be an electronic-structure artefact.

Practical analysis often quotes the Barrett radius or a Fourier-Bessel fit rather than a single rms number. When converting a published isotope shift into $\delta\langle r^2\rangle$, write the field-shift formula with units and the electronic factor F explicitly, then propagate δF. A 10% error in F is common and can dominate the nuclear uncertainty budget for light chains where the field shift is not overwhelming.

Isotope-shift checklist

Write $\delta\nu = F\,\delta\langle r^2\rangle + K\,\mu$ (schematic) with units, cite the electronic factor source, and test King-plot linearity for two transitions. If nonlinearity remains after varying F within uncertainties, only then consider higher nuclear moments or speculative new interactions. Tie absolute anchors to muonic or electron-scattering radii so that a beautiful differential chain does not float in absolute scale.

Odd-even staggering and deformation

Odd–even staggering in charge radii often tracks pairing and the blocking of deformation. A sudden onset of deformation produces a kink in $\langle r^2\rangle(N)$ larger than any smooth liquid-drop expectation. Comparing radius kinks with $E(2^+)$ systematics (Lecture 15) is the standard way to decide whether a kink is a shell closure or a shape change.

Muonic ladder

Muonic X-ray cascades provide absolute radii for stable and long-lived targets that optical methods then use as anchors. When an optical chain is attached to a muonic absolute radius at one isotope, residual electronic-factor errors still affect isotope differences, but the absolute float is reduced. Document the anchor isotope explicitly in any radius table you compile.

Definition hygiene

Make a three-column table for one nucleus: point radius, rms radius, and diffraction radius, with references. The spreads are educational. Isotope-shift papers must state which electronic factors convert frequencies into $\delta\langle r^2\rangle$; copy those citations into your notes whenever you adopt a radius value for structure comparisons in later chapters.

Finite-size physics is therefore a conversion problem: electronic or muonic probes supply energies, nuclear models supply moments of charge, and the shared language is $\langle r^2 \rangle$ and its isotopic differences. Mastering that conversion is the practical skill Lecture 3 adds to Lecture 1's form factors.

In practice, rework one numerical estimate from the lecture with a different parameter set and compare the shift to the quoted experimental uncertainty. If the shift exceeds the uncertainty, the estimate is teaching a scale, not delivering a fit. That distinction keeps theoretical cartoons honest when they sit beside evaluated data tables and modern many-body calculations. Along the course arc, write one sentence linking this chapter backward and one forward: the discipline of those two sentences is as important as any single formula. Examination problems rarely ask for a literature review, but they do ask for the scale argument that literature reviews presuppose.

Research frontier: isotope shifts, King plots, and student projects

Finite-size corrections and isotope-shift ideas of Lecture 3 are now precision tools that sit at the border of atomic, nuclear, and particle physics. Modern laser and ion-trap spectroscopy measures optical isotope shifts at the hundred-hertz level; King plots of two transitions test the linearity expected from a simple field-plus-mass decomposition. Nonlinearities have been reported in Yb^+ and can arise either from higher-order nuclear structure (including deformation and nuclear polarisability) or from a hypothetical light boson mediating a new force between electrons and neutrons, so nuclear and particle interpretations must be disentangled carefully [57]. Evaluated charge-radius compilations remain the nuclear input for such analyses [58]. Muonic-atom X-rays continue to provide absolute size anchors for stable and long-lived species, while radioactive-molecule spectroscopy and trapped radioactive ions extend electroweak and size sensitivity at rare-isotope facilities [54]. Uncertainties that still dominate many King-plot interpretations include electronic factors F, specific mass-shift calculations, and incomplete knowledge of nuclear deformation along isotopic chains.

For nuclear-physics students, the practical skill is to separate electronic structure uncertainties from nuclear size uncertainties before claiming new physics. A careful King-plot analysis therefore begins with evaluated radii and documented electronic factors, then tests whether residual nonlinearities survive when nuclear deformation and higher-order field shifts are varied within published ranges.

Student directions. (1) Data project: construct a King plot from two published isotope-shift tables and fit the linear slope. (2) Analytic estimate: estimate $\delta\langle r^2 \rangle$ from a measured field shift using a textbook electronic factor F. (3) Literature comparison: compare optical and muonic determinations of $\langle r^2 \rangle$ for one even-even chain and discuss residual model dependence.

3.14 Problems with solutions

Problem 3.1 — Finite-size shift for a uniform sphere. Show that for a hydrogen-like $1s$ electron and a uniform nuclear charge of radius $R \ll a_0/Z$, the first-order energy shift relative to a point nucleus is

$$\Delta E \approx \frac{2}{5}\frac{Z^4 e^2}{4\pi\varepsilon_0}\frac{R^2}{a_0^3}$$

(up to the standard hydrogenic normalisation constants; [6]). Estimate ΔE for $Z = 80$, $R = 7\,\mathrm{fm}$.

Solution 3.1. Inside a uniform sphere the potential energy of an electron is

$$U_{\text{finite}}(r) = -\frac{Ze^2}{8\pi\varepsilon_0 R}\left(3 - \frac{r^2}{R^2}\right), \qquad r \le R,$$

while $U_{\text{point}} = -Ze^2/(4\pi\varepsilon_0 r)$. Their difference H_1 vanishes for $r > R$. With $|\psi_{1s}(0)|^2 = Z^3/(\pi a_0^3)$ nearly constant over the nucleus,

$$\int_{r<R} H_1\,\mathrm{d}^3 r = \frac{2\pi}{5}\frac{Ze^2}{4\pi\varepsilon_0}R^2,$$
$$\Delta E = |\psi(0)|^2 \int H_1\,\mathrm{d}^3 r = \frac{2}{5}\frac{Z^4 e^2}{4\pi\varepsilon_0}\frac{R^2}{a_0^3}.$$

(The factor $2/5$ is the uniform-sphere special case of the $\langle r^2\rangle = \frac{3}{5}R^2$ theorem.) For $Z = 80$, $R = 7\,\mathrm{fm} = 7\times10^{-15}\,\mathrm{m}$, $a_0 = 5.29\times10^{-11}\,\mathrm{m}$,

$$\frac{R^2}{a_0^3} \approx 3.3\times10^{-22}\,\mathrm{m}^{-1}, \qquad \Delta E \approx 7.8\,\mathrm{eV}$$

using $e^2/(4\pi\varepsilon_0) = 1.440\,\mathrm{MeV\,fm}$. This is small on a K-shell scale of tens of keV, but isotope *differences* are measurable. The non-relativistic estimate is not a precision result at $Z = 80$.

Problem 3.2 — Isotope shift and $A^{1/3}$. Two isotopes of the same element have mass numbers A and A' and radii $R = r_0 A^{1/3}$, $R' = r_0 (A')^{1/3}$. Argue that the difference of finite-size $1s$ shifts scales as

$$\Delta E(A) - \Delta E(A') \propto Z^4\big(A^{2/3} - A'^{2/3}\big).$$

Why is a plot of K_α X-ray isotope shifts versus $A^{2/3}$ expected to be approximately linear?

Solution 3.2. From Problem 3.1, $\Delta E \propto Z^4 R^2$ and $R^2 \propto A^{2/3}$. Hence

$$\delta(\Delta E) \propto Z^4\big(A^{2/3} - A'^{2/3}\big).$$

Plotting measured isotope shifts against $A^{2/3}$ (relative to a reference isotope) should be nearly linear with slope fixed by r_0. Deviations encode shell effects and deformation [6].

Problem 3.3 — Muonic enhancement. By what factor does the finite-size shift grow if the electron is replaced by a muon ($m_\mu/m_e \approx 207$), all else equal?

Solution 3.3. The Bohr radius scales as $1/\mu$, and $|\psi(0)|^2 \propto 1/a^3 \propto \mu^3$. Replacing m_e by m_μ therefore multiplies ΔE by

$$\Delta E_\mu = \left(\frac{m_\mu}{m_e}\right)^3 \Delta E_e \approx 207^3 \Delta E_e = 8.86 \times 10^6\, \Delta E_e \sim 9 \times 10^6\, \Delta E_e.$$

Caveat: for heavy nuclei the muon $1s$ orbit has size comparable to R, so the constant-$\psi(0)$ approximation and non-relativistic theory both need upgrading; the factor $\sim 10^7$ is the scaling estimate, not a precision prediction. Muonic K_α X-rays are a nuclear-size measurement, not an atomic one.

Problem 3.4 — Z^4 scaling of finite-size shift. From $\Delta E \propto Z\,|\psi(0)|^2\,R^2$ and the non-relativistic hydrogenic result $|\psi(0)|^2 \propto (Z/a_0)^3$, show that $\Delta E \propto Z^4 R^2$ for s states. Why do heavy nuclei dominate isotope-shift and muonic size measurements?

Solution 3.4.

$$|\psi(0)|^2 \propto Z^3 \quad \Rightarrow \quad \Delta E \propto Z \cdot Z^3 R^2 = Z^4 R^2.$$

The Z^4 factor amplifies the effect rapidly with Z, so Pb-region X-ray and muonic transitions are far more size-sensitive than light atoms [6, 1].

Problem 3.5 — Isotope field shift. Two isotopes of the same Z have charge radii differing by $\delta\langle r^2\rangle$. In the leading s-state approximation, how does the atomic transition energy shift between them? Which contribution dominates field shifts in heavy elements: mass or finite size?

Solution 3.5. The field (finite-size) isotope shift is proportional to $Z\,|\psi(0)|^2\,\delta\langle r^2\rangle$. In heavy elements the field shift dominates the reduced-mass (mass) shift, which is why optical and X-ray isotope shifts diagnose nuclear charge radii [6, 1].

Chapter 4

Binding Energy and the SEMF

"Attempting to understand this curve of binding energy leads us to the semiempirical mass formula."

K. S. Krane

Lectures 1–3 fixed the nuclear *size*: charge densities, saturation, and finite-size shifts. This lecture turns to nuclear *energy*: how tightly nucleons stick together, and how that energy varies with A and Z. Binding energy is the mass defect of the nucleus written as an energy. The semi-empirical mass formula (SEMF) is a five-parameter model of $B(A, Z)$ that organises the chart of nuclides and the B/A curve [6, 1, 13, 14, 2].

Key idea

Binding energy is the mass defect of the nucleus written as an energy. The SEMF is a five-parameter model of $B(A, Z)$ that organises the chart of nuclides and the B/A curve.

4.1 Why not every (Z, N) combination is a nucleus?

Not every combination of protons and neutrons forms a stable (or even bound) nucleus. Ten free protons do not assemble into a long-lived $Z = 10$, $N = 0$ system; ten free neutrons do not form a stable droplet either. Here it is important to separate *existence* from *stability*. A nuclide may be bound against immediate nucleon emission yet radioactive against beta decay, alpha decay, or fission. The chart of nuclides is therefore a narrow ribbon in the (N, Z) plane: outside that ribbon, systems either beta decay toward lower-mass isobars or, beyond the drip lines, cannot bind the last nucleon at all [14, 6, 1].

Stability is a competition among three physical agents, each with a different range and different A, Z dependence:

- **short-range strong attraction** among nearest neighbours, which wants as many nucleons packed at saturation density as possible and scales roughly with the nuclear volume;
- **long-range Coulomb repulsion** among protons, which wants fewer protons at fixed A and grows roughly as Z^2/R;
- **quantum statistics and pairing**: the Pauli principle gives separate proton and neutron

Fermi seas and penalises a large imbalance $N - Z$, while an additional attractive pairing correlation favours even proton and neutron numbers.

No single force wins everywhere. Light nuclei hug $N = Z$ because asymmetry energy dominates Coulomb. Heavy nuclei drift to $N > Z$ because Coulomb grows faster with Z than the asymmetry cost of a moderate neutron excess. Far from the ribbon, the last neutron or proton is unbound: those loci are the neutron and proton drip lines predicted semi-quantitatively by the SEMF and mapped experimentally with radioactive beams [14, 1].

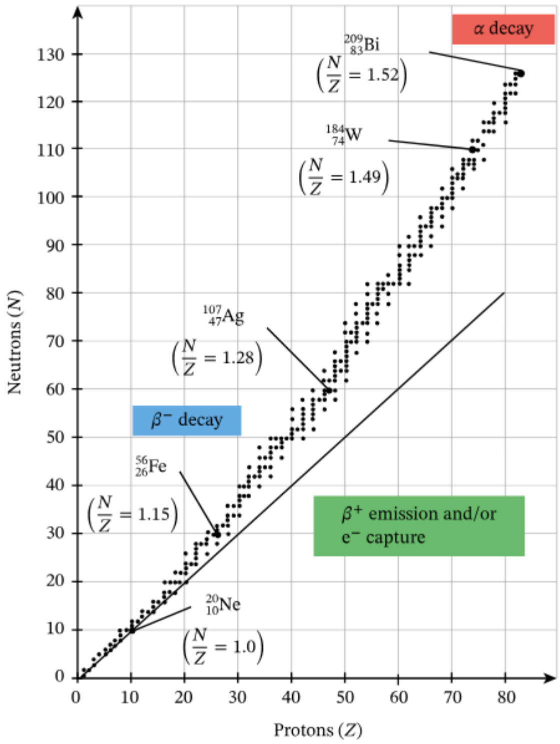


Figure 4.1: Chart of nuclides in the N–Z plane (valley of stability). Light nuclei prefer $N \approx Z$; heavy nuclei need $N > Z$. Nuclei off the valley return by $\beta^{\pm}$ (or electron capture) and, for heavy species, also by α emission; the drip lines mark where nucleon separation energies vanish.

Figure 4.1 is the experimental map we must eventually explain. Nuclei with a measured mass m are lighter than the free nucleons from which they are assembled. That mass defect is the **binding energy** B. A systematic study of $B(A, Z)$, and especially of the average B/A, leads to a few-parameter model of nuclear binding: the liquid-drop picture and the semi-empirical mass formula (SEMF) [6, 1, 2, 14]. The SEMF will not predict shell closures or spectroscopic details, but it organises why the valley exists, why $B/A \sim 8\,\text{MeV}$, and why fusion and fission both release energy.

4.2 Total energy of a nucleus: derivation from rest masses

The preceding survey explains why energy comparison is the natural next step: a proposed (Z, N) combination can persist only when no allowed route leads to a lower-energy set of products. Before constructing the SEMF, we must therefore distinguish three energies that are often compressed into a single formula:

1. the rest energy of Z free protons and N free neutrons;
2. the smaller rest energy of the bound nucleus;
3. the rest energy of the neutral atom whose mass is actually tabulated.

The derivation below keeps all three ledgers explicit.

4.2.1 Separated nucleons versus a bound nucleus

Imagine first that every constituent is at rest and separated by arbitrarily large distances, so that their mutual interactions vanish. Taking this separated configuration as the reference threshold gives

$$E_{\text{free}} = Zm_pc^2 + Nm_nc^2. \tag{4.1}$$

The rest masses in Equation (4.1) already include each nucleon's internal quark and gluon energy. What is absent is the energy associated with their relative motion and their mutual nuclear and Coulomb interactions. For a nonrelativistic nuclear Hamiltonian this bookkeeping may be displayed schematically as

$$H = Zm_pc^2 + Nm_nc^2 + T_{\text{rel}} + V_{\text{strong}} + V_{\text{C}} + \cdots . \tag{4.2}$$

Here $T_{\text{rel}} > 0$, whereas the expectation value of the attractive part of V_{strong} is negative. Coulomb repulsion raises the energy. A bound state exists only when all contributions after the rest masses combine to a negative eigenvalue.

When the free nucleons assemble into a nucleus, energy is carried away by photons, emitted particles, or recoil. If the final nucleus is left in its ground state, the maximum released energy is the binding energy $B(A, Z)$. Conversely, precisely this energy must be supplied to reach the separated-nucleon threshold. Energy conservation may therefore be written

$$E_{\text{nuc}} + B = E_{\text{free}}. \tag{4.3}$$

Since the nucleus at rest has $E_{\text{nuc}} = M_{\text{nuc}}(A, Z)c^2$, it follows step by step that

$$M_{\text{nuc}}(A, Z)c^2 = Zm_pc^2 + Nm_nc^2 - B(A, Z), \tag{4.4}$$

$$B(A, Z) = \Big[Zm_p + Nm_n - M_{\text{nuc}}(A, Z)\Big]c^2. \tag{4.5}$$

Thus the *total internal-energy shift* of a bound nucleus is negative when the zero is chosen at separated nucleons:

$$E_{\text{internal}} = E_{\text{nuc}} - E_{\text{free}} = \langle T_{\text{rel}} + V_{\text{strong}} + V_{\text{C}} + \cdots \rangle = -B. \tag{4.6}$$

The binding energy B is quoted as a positive number, while the energy shift of the bound state is $-B$. These two sign conventions describe the same physics. It would be incorrect to identify $-B$ with the potential energy alone: confined nucleons have positive kinetic energy, so the attractive potential must be more negative than $-B$.

Why the minimum of total energy determines stability

At fixed $A = N + Z$, beta processes convert a neutron into a proton or a proton into a neutron. They therefore move between neighbouring isobars. The comparison must include every particle in the initial and final states. Neutral atomic masses do this bookkeeping conveniently:

$$Q_{\beta^-} = [M_{\text{at}}(A, Z) - M_{\text{at}}(A, Z + 1)]c^2, \tag{4.7}$$

$$Q_{\beta^+} = [M_{\text{at}}(A, Z) - M_{\text{at}}(A, Z - 1) - 2m_e]c^2, \tag{4.8}$$

$$Q_{\text{EC}} \simeq [M_{\text{at}}(A, Z) - M_{\text{at}}(A, Z - 1)]c^2, \tag{4.9}$$

apart from small electron-binding and recoil corrections. A channel is energetically open only when its Q value is positive. Thus the stable member of an isobaric chain lies at an allowed minimum of the *atomic* mass, subject also to selection rules. Lecture 5 develops this idea into the mass parabola $M(A, Z)$.

4.2.2 Binding is necessary but not sufficient for stability

Equation (4.5) answers one specific question: is the nucleus below the threshold of Z free protons plus N free neutrons? If $B > 0$, complete breakup into free nucleons costs energy. That does *not* prove that the nuclide is stable. A different partition, such as an α particle plus a daughter nucleus, may have a still lower total energy; a neighbouring isobar may be accessible through beta decay; and a heavy nucleus may lower its energy by fission. For any proposed decay

$$P \longrightarrow 1 + 2 + \cdots,$$

the decisive quantity is

$$Q = \left(M_P - \sum_f M_f \right) c^2. \tag{4.10}$$

Positive Q means that energy conservation permits the decay, with Q becoming kinetic energy or radiation. Whether the decay is rapid then depends on the interaction involved, angular-momentum and parity selection rules, and possible barriers. Consequently:

- **bound** means below a specified breakup threshold;
- **energetically stable** means no allowed final state has lower total energy;
- **long-lived** may instead mean that a lower-energy channel exists but is strongly hindered, as in alpha decay or spontaneous fission.

4.2.3 Why experiments use atomic masses

Bare nuclear masses are less convenient experimentally than neutral atomic masses. Add the rest energy of Z free electrons to both sides of Equation (4.4):

$$M_{\text{nuc}}c^2 + Zm_ec^2 = Z(m_p + m_e)c^2 + Nm_nc^2 - B. \tag{4.11}$$

Neither side is yet an atomic mass, because binding the electrons also lowers the energy. Let $B_e(Z,A) > 0$ denote the total electronic binding energy of the neutral atom. Its mass satisfies

$$M_{\rm at}(A,Z)c^2 = M_{\rm nuc}(A,Z)c^2 + Zm_e c^2 - B_e(Z,A). \tag{4.12}$$

For a ground-state hydrogen atom,

$$m(^1\mathrm{H})c^2 = (m_p + m_e)c^2 - B_\mathrm{H}, \qquad B_\mathrm{H} = 13.6\,\mathrm{eV}, \tag{4.13}$$

and hence

$$(m_p + m_e)c^2 = m(^1\mathrm{H})c^2 + B_\mathrm{H}. \tag{4.14}$$

Substitute Equations (4.12) and (4.13) into Equation (4.11). The exact atomic-mass relation is

$$\begin{aligned} M_{\rm at}(A,Z)c^2 = {} & Zm(^1\mathrm{H})c^2 + Nm_n c^2 - B(A,Z) \\ & + ZB_\mathrm{H} - B_e(Z,A). \end{aligned} \tag{4.15}$$

Solving this equation for the nuclear binding energy gives

$$\begin{aligned} B(A,Z) = {} & \Big[Zm(^1\mathrm{H}) + Nm_n - M_{\rm at}(A,Z)\Big]c^2 \\ & + ZB_\mathrm{H} - B_e(Z,A). \end{aligned} \tag{4.16}$$

This is the complete derivation of the nuclear binding energy from tabulated atomic masses. Notice that electron rest masses have not merely been discarded: they cancel because each proton on the separated side is paired with one electron to form a hydrogen atom. Only the much smaller difference between electronic *binding* energies remains.

4.2.4 Controlled approximation used in nuclear physics

The electronic correction in the second line of Equation (4.16) is small on a nuclear scale. The hierarchy is

$$\begin{aligned} & B_\mathrm{H} = 13.6\,\mathrm{eV}, \qquad B_e(Z,A) \sim \mathrm{keV}\text{–}1\,\mathrm{MeV} \quad \text{(medium to very heavy atoms)}, \\ & B_{\rm nuc} \sim \mathrm{MeV}\text{–}\mathrm{GeV}. \end{aligned} \tag{4.17}$$

Even a binding energy of tens of MeV is millions of times larger than the hydrogen ionisation energy. In heavy nuclei the nuclear binding is typically hundreds of MeV to more than a GeV, whereas total electronic binding remains at most of order MeV. Therefore, unless high-precision mass differences are required, one neglects $ZB_\mathrm{H} - B_e$ and obtains

$$\boxed{B(A,Z) \simeq \Big[Zm(^1\mathrm{H}) + Nm_n - M_{\rm at}(A,Z)\Big]c^2.} \tag{4.18}$$

The approximation does not mean that electron binding is zero. It means that the difference of electronic binding terms is much smaller than the nuclear binding being extracted. Precision mass evaluations restore that correction when necessary.

4.3 Practical mass quantities

Section 4.2 has already established the physical meaning of B, the exact conversion between nuclear and atomic masses, and the approximation in Equation (4.18). We now use that result rather than derive it again. Throughout the remainder of this lecture, $M_{\mathrm{at}}(A, Z)$ denotes a neutral atomic mass, and $N = A - Z$.

4.3.1 Mass units and a numerical example

Atomic masses are normally quoted in unified atomic mass units. The mass–energy conversion is

$$1\,\mathrm{u}\,c^2 = 931.494\,\mathrm{MeV} \approx 931.5\,\mathrm{MeV}. \tag{4.19}$$

Thus a seemingly small mass defect of $10^{-2}\,\mathrm{u}$ corresponds to nearly 10 MeV, which is a natural nuclear-energy scale. When the SEMF is converted into an atomic-mass formula in this lecture, we use

$$M_{\mathrm{at}}(A, Z) \simeq Zm(^1\mathrm{H}) + Nm_n - \frac{B(A, Z)}{c^2}.$$

This is an atomic-mass convention for applying the SEMF, with the electronic correction derived in Equation (4.16) neglected; the binding-energy formula itself is a model of the nucleus and does not depend on that bookkeeping choice [4].

Mass defect and B

For the α particle, the relevant mass defect is approximately $\Delta m = 0.0304\,\mathrm{u}$. Therefore

$$B = \Delta m\, c^2 \approx (0.0304\,\mathrm{u})(931.494\,\mathrm{MeV/u}) \approx 28.3\,\mathrm{MeV}.$$

Dividing by four gives $B/A \approx 7.07\,\mathrm{MeV}$, exceptionally large for so light a system. This strong binding helps explain the prominence of alpha clustering and alpha-decay products (see Problem 4.1) [6, 1, 2].

4.3.2 Mass excess

Many tables quote the **mass excess** (also called mass defect in some older tables)

$$\Delta(A, Z) \equiv [M_{\mathrm{at}}(A, Z) - A\,\mathrm{u}]c^2 \tag{4.20}$$

in MeV rather than the atomic mass itself [1, 6]. Because $A \cdot \mathrm{u}$ is an integer number of mass units, differences of mass excesses are differences of mass energies. Binding energies and Q values can be written entirely in terms of Δ's. For example, once $\Delta(^AX)$, $\Delta(^1\mathrm{H})$, and $\Delta(n)$ are known, (4.18) becomes an arithmetic combination of tabulated MeV values. We keep the mass form (4.18) in the main text for transparency.

4.3.3 Separation energies

Just as atomic ionisation energies probe the outermost electron, nucleon separation energies probe the outermost nucleon. The neutron and proton separation energies are

$$S_n(A,Z) = B(A,Z) - B(A-1,Z) = \left[M_{\mathrm{at}}(A-1,Z) + m_n - M_{\mathrm{at}}(A,Z)\right]c^2, \tag{4.21}$$
$$S_p(A,Z) = B(A,Z) - B(A-1,Z-1) = \left[M_{\mathrm{at}}(A-1,Z-1) + m(^1\mathrm{H}) - M_{\mathrm{at}}(A,Z)\right]c^2. \tag{4.22}$$

The atomic-mass bookkeeping is deliberate: $m(^1\mathrm{H})$ appears in S_p rather than m_p, so that electron masses cancel exactly as they do in B [6]. Odd–even staggering in S_n is direct evidence for pairing (pairing section below). Shell closures appear as jumps in separation energies (shell model, later). In the one-nucleon criterion, the neutron and proton drip lines occur when S_n or S_p reaches zero: adding one more nucleon no longer increases B, so the system is unbound against that emission channel. Pairing can make two-nucleon separation energies relevant as well [14, 1].

4.4 The curve of B/A: what must be explained

Because B grows roughly linearly with A, it is standard to plot the average binding energy per nucleon,

$$\frac{B}{A} = (\text{total binding energy})/(\text{number of nucleons}), \tag{4.23}$$

against A. This curve is the central experimental object of nuclear structure at the bulk level: it compresses thousands of measured masses into one smooth trend plus small ripples. Attempting to understand it leads to the SEMF, in which a few general parameters characterise how B varies with A and Z [6]. If B were strictly proportional to A, the plot would be a horizontal line at about 8 MeV; every deviation from that line is a clue about the nuclear force, nuclear geometry, or quantum statistics [1, 13].

4.4.1 Features of the experimental curve

Several remarkable features stand out on the experimental B/A curve [6, 1, 14, 13, 2, 5, 4]:

1. **Nearly constant B/A.** Except for the very lightest nuclei,

   $$7.7\,\mathrm{MeV} \lesssim B/A \lesssim 8.8\,\mathrm{MeV}$$

   for stable species with $12 \lesssim A \lesssim 225$, or more loosely "between 7 and 9 MeV per nucleon." That is why one often says "about 8 MeV per nucleon." To within about 10%, most nuclei share this scale [6].

2. **Broad maximum in the iron–nickel region** ($A \sim 56$–62): the most tightly bound nuclei *per nucleon.* The smooth liquid-drop curve peaks near ^{56}Fe; precise atomic-mass data give the highest B/A at ^{62}Ni, with ^{58}Fe and ^{56}Fe only slightly lower [4, 14]. For energy release in stars and reactors, "iron peak" is the right mental picture even though the exact maximum nuclide is nickel-62.

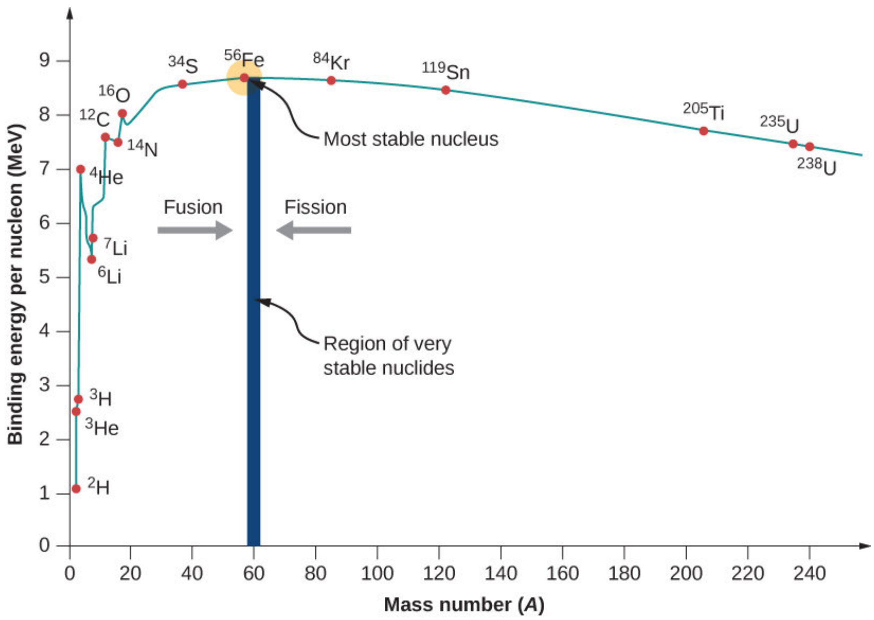


Figure 4.2: Binding energy per nucleon versus mass number. The curve is roughly flat near 8 MeV, peaks in the iron–nickel region, and falls slowly for heavy nuclei. Light $A = 4n$ species (^{4}He, ^{12}C, ^{16}O) sit above the smooth trend. Modern mass tables place the absolute maximum of B/A at ^{62}Ni, with ^{58}Fe and ^{56}Fe very close; the iron peak is the practical reference for fusion and stellar burning [6, 4].

3. **Rise at low A.** Light nuclei have a larger fraction of surface nucleons, so B/A climbs as A increases and the surface/volume ratio falls.

4. **Slow decline for heavy A.** Coulomb repulsion among many protons gradually wins; B/A falls toward uranium and beyond.

5. **Two ways to release energy.** Climbing the curve from below ($A \lesssim 56$) is fusion; climbing from above ($A \gtrsim 56$) is fission. Both increase B/A and therefore release energy. The light side of the curve is the domain of stellar nuclear burning; the heavy side is the domain of fission and α decay [14, 5, 13].

6. **Fine structure.** Odd–even staggering (pairing) and extra binding of light $A = 4n$ nuclei (α clustering: ^{4}He, ^{12}C, ^{16}O) sit on top of the smooth trend. Notably, ^{8}Be lies slightly *below* two α particles and therefore decays into them [2]. The SEMF captures the smooth part, not every shell or cluster peak.

4.4.2 Which SEMF term explains which feature?

The rest of this lecture builds the SEMF term by term. It helps to know in advance how each piece of the curve will be accounted for:

Feature on B/A	**SEMF term(s) responsible**
Nearly flat $\sim 8\,\mathrm{MeV}$ for $12 \lesssim A \lesssim 225$	Volume $+a_V A$: short-range saturation gives $B \propto A$ and $B/A \sim a_V$ before corrections.
Rise from very light A	Surface $-a_S A^{2/3}$: at small A the surface/volume ratio is large, so $\lvert B_S\rvert/A \propto A^{-1/3}$ is a big penalty that falls as A grows.
Broad maximum near iron–nickel	Competition of surface (favours larger A) and Coulomb (favours smaller Z, hence limits growth of A for a given valley N/Z).
Slow decline past the peak	Coulomb $-a_C Z(Z-1)/A^{1/3}$: $\lvert B_C\rvert/A$ grows for heavy systems with many protons.
Valley of stability ($N \approx Z$ at light A, $N > Z$ at heavy A)	Coulomb plus asymmetry $-a_\mathrm{a}(N-Z)^2/A$: Coulomb alone would push $Z \to 0$; asymmetry restores $N \sim Z$ when Coulomb is weak and permits controlled neutron excess when it is strong.
Odd–even ripples on B/A	Pairing δ: even–even nuclei gain a few hundred keV per nucleon; odd–odd lose a similar amount.
Spikes at ^{4}He, ^{12}C, ^{16}O	Beyond the SEMF: α clustering and shell structure (later chapters).

The liquid-drop model explains the *smooth* envelope. Shell and cluster effects are superimposed as corrections of order $1\,\mathrm{MeV}$ or less per nucleon on top of the $\sim 8\,\mathrm{MeV}$ bulk scale [6, 1].

Key idea

First clue about the force. If every nucleon attracted every other nucleon with a long-range force, we would expect $B \propto A(A-1) \sim A^2$ and $B/A \propto A$. Experiment gives $B \propto A$ and $B/A \sim$ const. Therefore each nucleon interacts only with its nearest neighbours. Combined with constant density (Lecture 2), that is the signature of a short-range force of range comparable to the internucleon spacing [6, 1].

4.4.3 Why $B \propto A$ is surprising (and important)

A long-range force among A particles produces $\binom{A}{2} \sim A^2/2$ pairs. Gravity and Coulomb are of that type (every mass attracts every mass; every charge feels every charge). The strong residual force is not: saturated nuclear matter has fixed density and fixed coordination number, so the total binding scales with the number of particles, not the number of pairs [6, 1, 13].

From electron scattering we already know that the nuclear density is roughly constant, so each nucleon has about the same number of neighbours and contributes roughly the same amount to the binding energy [6]. The same linear behaviour is consistent with the short-range conclusions drawn from nucleon-density systematics in the previous chapters [1].

That observation is the starting point of the liquid-drop model.

4.5 The liquid-drop idea

The liquid-drop model is a *semi-empirical* model of $B(A, Z)$ [1, 6, 12, 14, 2]:

- **semi:** the algebraic form of each term is motivated by physics (geometry, electrostatics, quantum statistics);
- **empirical:** the numerical coefficients are fixed by fitting measured masses.

The idea goes back to Bohr (1935): the short range of nuclear forces, together with additivity of volumes and of binding energies, invites a liquid-drop description [14]. The same empirical fact drives the analogy from another direction: the nearly homogeneous mass density inside the nucleus suggests comparing the nucleus with a liquid drop in which the main part of the interaction is only between nearest neighbours [2]. In practice the model describes how the nuclear binding energy depends on neutron and proton number, and it is an effective way to organise the relative stability of nuclides [1].

Why a "drop"? Nuclei and classical liquid drops share

1. many constituents;
2. nearly constant interior density (incompressibility at leading order; volume $\propto A$);
3. a well-defined surface with surface tension;
4. short-range cohesion among neighbours (like molecules in a water droplet, or van der Waals cohesion in a classical liquid).

One must *not* identify the nuclear force with van der Waals forces; only the geometry of a saturating short-range interaction is borrowed [1]. The analogy is useful for energetics, not for microscopic dynamics.

The first three SEMF terms (volume, surface, Coulomb) would appear in the energy of any charged liquid droplet. The last two (asymmetry, pairing) are quantum and have no pure classical analogue: they encode Pauli filling of two fermion species and the tendency of like nucleons to form $J = 0$ pairs [6, 14, 13]. In that sense the SEMF is a first attempt to apply nuclear models to systematic behaviour: liquid-drop physics for the gross collective features, and shell/quantum ingredients for the last two terms [6].

Throughout we assume a spherical nucleus to leading order (a good approximation for most ground states near stability) and a nuclear volume directly related to A [1, 4]. Deformations and shell corrections refine that picture later; they are not needed to understand the smooth B/A curve.

4.5.1 Historical development: Gamow, Weizsäcker, Bethe

The picture of the nucleus as a charged liquid drop was developed in the early 1930s by **George Gamow**. The quantitative mass formula was written down by **Carl Friedrich von Weizsäcker** in 1935 ("Zur Theorie der Kernmassen") as the **semi-empirical mass formula**, and refined the same year by **Hans Bethe** and collaborators. The formula is therefore often called the **Bethe–Weizsäcker** (or SEMF) formula [12, 6, 1, 14, 4]. The numerical coefficients are chosen to approximate observed ground-state binding energies; other than a_C (which is largely calculable from electrostatics), they are fixed empirically [14, 1].

Historical note: George Gamow (1904–1968)

George Gamow (born Georgiy Antonovich Gamov in Odessa) was a Soviet-American theoretical physicist and cosmologist. In the late 1920s he explained α decay as quantum tunnelling through the Coulomb barrier. In the early 1930s he developed the *liquid-drop* picture of the nucleus: nucleons packed at roughly constant density, held by a short-range attraction analogous to the cohesion of a liquid, with a surface and a Coulomb self-energy of the proton charge. That geometric intuition is the starting point of the SEMF volume, surface, and Coulomb terms. After leaving the Soviet Union he worked in the United States (George Washington University, later Colorado), co-authored the Alpher–Bethe–Gamow paper on Big Bang nucleosynthesis, and wrote celebrated popular books (*One, Two, Three... Infinity*, the Mr Tompkins series). The liquid-drop model of nuclear binding is one of his lasting contributions to nuclear physics.

Historical note: Carl Friedrich von Weizsäcker (1912–2007)

Carl Friedrich von Weizsäcker, German physicist and later philosopher, quantified Gamow's liquid-drop ideas in 1935 as the **semi-empirical mass formula**: binding energy written as a sum of volume, surface, Coulomb, and asymmetry contributions, with coefficients to be fixed by experiment [12]. His paper "Zur Theorie der Kernmassen" (Z. Phys. **96**, 431, 1935) established the algebraic structure still used today. He also worked on stellar energy generation (with Bethe) and, after the war, on the philosophy of science and the responsibility of scientists. In this course the SEMF is the Weizsäcker formula in modern notation, with standard numerical coefficients from global mass fits.

Historical note: Hans A. Bethe (1906–2005)

Hans Albrecht Bethe, German-American theoretical physicist, refined and popularised the mass formula in 1935 (hence *Bethe–Weizsäcker*). He received the 1967 Nobel Prize in Physics for the theory of energy production in stars (the CNO cycle and related work). Across a long career at Cornell he shaped nuclear and particle theory, nuclear astrophysics, and the theory of neutron stars. In the present lectures his name attaches both to the SEMF coefficients and to the stellar/compact-object applications of binding-energy systematics (Lectures 4–5).

4.6 SEMF terms derived in detail

The previous sections introduced the five SEMF terms and their historical origin. We now study each term in turn: packing for volume and surface, electrostatics for Coulomb, Fermi-gas (and shell) filling for asymmetry, and separation-energy systematics for pairing [13, 6, 1, 14, 2]. The A and Z dependence of each term reflects simple physical properties even though most numerical coefficients must be fitted [14].

4.7 Volume term: why $B_V = a_V A$

The volume term is the backbone of the B/A curve. If it were the only contribution, every nucleus would have $B/A = a_V \sim 15$–$16\,\text{MeV}$ independent of A and Z. All other SEMF terms are corrections that pull the actual curve away from that flat ceiling.

4.7.1 Short-range force and nearest-neighbour counting

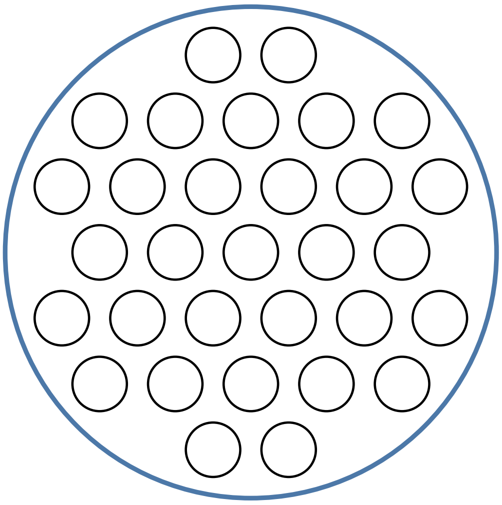


Figure 4.3: Volume (interior) nucleon. The open central circle is fully surrounded; black arrows show short-range attractive bonds to nearest neighbours (red).

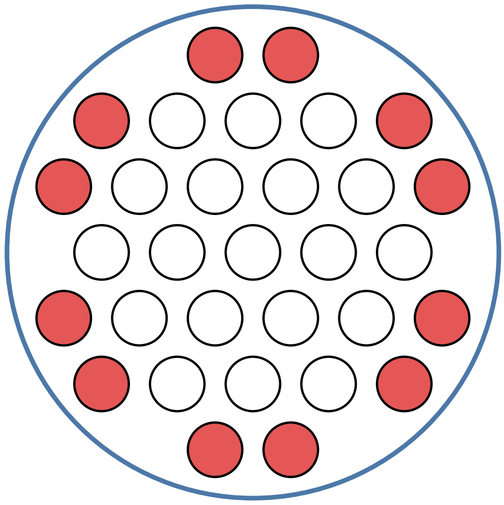


Figure 4.4: Surface nucleons (red). Some neighbour directions point outside the nucleus. Missing bonds reduce the binding energy.

Step 1 (range). The residual strong force has range $\sim$ 1–2 fm, comparable to the mean nucleon spacing but much smaller than the radius of a heavy nucleus. Therefore an interior nucleon interacts only with a fixed number of nearest neighbours, not with all other $A - 1$ nucleons [6, 1, 13, 2].

Step 2 (saturated density). Electron scattering shows that the central density is nearly independent of A. Each interior nucleon therefore has roughly the *same* number of neighbours, call it n_{nn} (order ten). Constant nucleon density likewise implies $V \propto A$ and a binding contribution proportional to A for medium-to-heavy nuclei [2].

Step 3 (bond sharing). Let $U > 0$ be the attractive energy associated with one nearest-neighbour bond. Each bond is shared by two nucleons, so the contribution of bonds to the binding energy of one interior nucleon is

$$\frac{B}{A}\Big|_{\mathrm{bulk}} \approx \frac{1}{2}\, n_{\mathrm{nn}}\, U. \tag{4.24}$$

For A nucleons,

$$B_V = a_V\, A, \qquad a_V \equiv \tfrac{1}{2}\, n_{\mathrm{nn}}\, U. \tag{4.25}$$

Equivalently, without bond counting: nucleons are closely packed and the force is short ranged, so each nucleon contributes in the same way independently of the total number; the force is therefore proportional to A, not to $A(A-1)/2 \sim A^2$ [13].

In three-dimensional close packing of equal spheres, $n_{\mathrm{nn}} = 12$, so $a_V = 6U$. Fitting nuclear masses gives

$$a_V \approx 15.5\text{–}15.8\,\mathrm{MeV} \tag{4.26}$$

[6, 1, 14, 13], hence a pedagogical bond scale $U \sim a_V/6 \sim 2.6\,\mathrm{MeV}$. That U is an effective

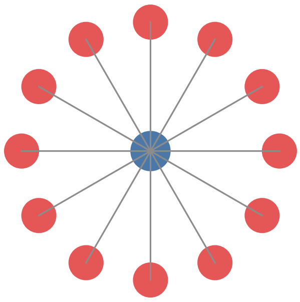


Figure 4.5: Idealised close packing: a central nucleon with twelve nearest neighbours (fcc/hcp coordination number 12).

number after kinetic energy is subtracted (see below), not a bare NN potential depth. By itself the volume term would give a flat $B/A \sim 16\,\text{MeV}$; all other SEMF terms reduce that ceiling toward the observed $\sim 8\,\text{MeV}$ [14].

4.7.2 Why not A^2? (long-range would fail)

If every nucleon attracted every other, the number of pairs would be

$$\binom{A}{2} = \frac{A(A-1)}{2} \propto A^2, \tag{4.27}$$

so $B \propto A^2$ and $B/A \propto A$. Experiment shows B/A is nearly flat ($\sim 8\,\text{MeV}$) for medium-heavy nuclei. That single fact forces nearest-neighbour (volume) scaling and rules out long-range counting for the strong force [6, 1].

Checklist for the volume term

1. Force range $\ll R \Rightarrow$ only nearest neighbours.
2. Constant density $\Rightarrow$ fixed coordination number.
3. $B_V = a_V A$ (linear in A, not A^2).
4. a_V is the largest SEMF coefficient: bulk attraction sets the energy scale.

4.7.3 Why $a_V \sim 16\,\text{MeV}$, not $\sim 50\,\text{MeV}$

Scattering and few-body work suggest that the potential energy of a nucleon in nuclear matter is of order $U_{\text{pot}} \sim -40$ to $-50\,\text{MeV}$. The SEMF volume coefficient is much smaller. The resolution is that the fitted $a_V \approx 15.5\,\text{MeV}$ is smaller than the binding of a nucleon to its neighbours inferred from the strength of the nuclear interaction ($\sim 50\,\text{MeV}$): the total binding is the difference between that attraction and the kinetic energy of the nucleon itself [13]. Nucleons are fermions and fill Fermi levels up to $\varepsilon_F \sim 35\,\text{MeV}$. The mean kinetic energy per nucleon is $\frac{3}{5}\varepsilon_F \sim 21\,\text{MeV}$. The *net* volume binding is

$$a_V \sim |U_{\text{pot}}| - \tfrac{3}{5}\varepsilon_F \sim 15\text{–}20\,\text{MeV}, \tag{4.28}$$

in agreement with the fitted a_V [13, 14]. Thus a_V already encodes the competition between short-range attraction and Fermi kinetic energy. A mean-potential estimate that yields $\sim 8\,\text{MeV}$ per nucleon after filling oscillator shells is the same physics in a shell-model language [14].

4.8 Surface term: why $-a_S A^{2/3}$

The surface term explains why B/A is *low* for the lightest nuclei and rises toward mid-mass (Section 4.4.2). Interior nucleons are fully coordinated; surface nucleons miss outward-pointing bonds. At small A almost every nucleon is a surface nucleon, so the fractional penalty $|B_S|/A \propto A^{-1/3}$ is large.

Step 1. Not every nucleon is an interior nucleon. Those on the surface (Figure 4.4) have fewer neighbours and are less tightly bound [1, 6, 13, 4]. In liquid-drop language: internal nucleons feel isotropic interactions, whereas nucleons near the surface feel forces coming only from the inside; the correction is therefore a surface-tension term [14]. Surface nucleons experience a smaller force, so the term is negative [4].

Step 2. Constant density $\Rightarrow R = r_0 A^{1/3}$. The surface area of a sphere is

$$S = 4\pi R^2 = 4\pi r_0^2\, A^{2/3} \propto A^{2/3}. \tag{4.29}$$

Equivalently, $S \propto V^{2/3}$ for a spherical nucleus [1]. The number of under-bound surface nucleons therefore scales as $A^{2/3}$.

Step 3. The volume term $a_V A$ *overcounts* the binding of surface nucleons (it pretends they have a full set of neighbours). One must subtract a correction proportional to the surface area:

$$B_S = -\,a_S\, A^{2/3}, \qquad a_S \approx 16.8\text{–}17.8\,\text{MeV}. \tag{4.30}$$

The coefficient a_S is of the same order as a_V, as expected if both come from the same short-range bonds; empirically $a_S \sim 13\text{–}18\,\text{MeV}$ [13, 14]. Together with Coulomb, the surface term limits the maximum A for which stable nuclei can exist: surface tension is literally what gives the model its name [5].

Effect on B/A.

$$\frac{B_S}{A} = -\frac{a_S}{A^{1/3}}. \tag{4.31}$$

At small A, almost all nucleons are on the surface, so B/A is strongly reduced. As A grows, surface/volume $\propto A^{-1/3}$ falls and B/A rises toward the volume ceiling. That is the main reason the experimental binding curve climbs through the light nuclei (Figure 4.10).

After volume and surface alone,

$$B \approx a_V A - a_S A^{2/3}. \tag{4.32}$$

Equation (4.32) depends only on A. It cannot explain Z dependence, the fall of B/A past iron, or the valley of stability. Three more terms are required.

Remark

A Fermi-gas treatment of kinetic energy in a finite box also produces a surface piece $\propto A^{2/3}$ (missing states near free surfaces). The empirical a_S combines that kinetic surface correction with the missing-bond (potential) surface tension [14].

4.9 Coulomb term: why $-a_C Z(Z-1)/A^{1/3}$

Unlike the nuclear force, Coulomb repulsion is long range: every proton feels every other proton. That is why the Coulomb piece scales as $Z(Z-1)$, not as Z. It is also the main reason B/A *falls* slowly past the iron peak (Section 4.4.2) and why heavy stable nuclei need a neutron excess to dilute the proton charge [1, 6, 13, 2]. Basic electrostatics gives a potential energy for a uniform spherical distribution of Z charges proportional to $Z(Z-1)/R$; with $R \propto A^{1/3}$ this becomes the familiar SEMF Coulomb term [1]. The analytic coefficient for a sharp sphere with $r_0 = 1.2\,\mathrm{fm}$ is $a_C = 0.72\,\mathrm{MeV}$; one often still leaves a_C slightly free because the real charge density is diffuse near the nuclear surface [1, 6].

4.9.1 Combinatorial count of proton pairs

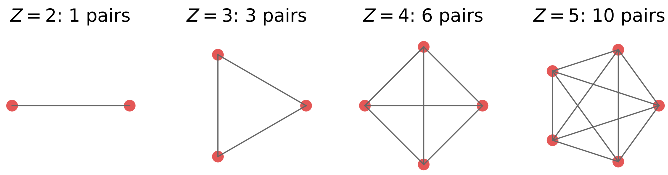


Figure 4.6: Pairwise interactions among Z protons: 2 protons $\to$ 1 link; 3 $\to$ 3; 4 $\to$ 6; 5 $\to$ 10. In general $\binom{Z}{2} = Z(Z-1)/2$. Because Coulomb is long range, every pair contributes.

Step 1 (long range). Number of distinct *pp* pairs:

$$\frac{Z(Z-1)}{2}. \tag{4.33}$$

Using $Z(Z-1)$ rather than Z^2 avoids counting a proton's self-repulsion [1, 6].

Step 2 (distance scale). Typical inter-proton distances scale with the nuclear radius $R \propto A^{1/3}$. Hence the Coulomb energy scales as

$$E_C \propto \frac{Z(Z-1)}{R} \propto \frac{Z(Z-1)}{A^{1/3}}. \tag{4.34}$$

4.9.2 Continuum formula: the factor 3/5

A controlled calculation treats the nucleus as a uniformly charged sphere of charge Ze and radius R. Assemble the sphere by bringing in thin charged shells from infinity.

At intermediate radius $r \le R$, the charge already present is

$$q(r) = Ze\,\frac{r^3}{R^3}, \tag{4.35}$$

and the potential at the surface of that sphere is $q(r)/(4\pi\varepsilon_0 r)$. The work to add a shell of charge $\mathrm{d}q = Ze \cdot 3r^2\,\mathrm{d}r/R^3$ is

$$\mathrm{d}W = \frac{q(r)}{4\pi\varepsilon_0 r}\,\mathrm{d}q = \frac{1}{4\pi\varepsilon_0}\frac{Ze\,r^2}{R^3} \cdot Ze \cdot \frac{3r^2}{R^3}\,\mathrm{d}r = \frac{(Ze)^2}{4\pi\varepsilon_0}\frac{3r^4}{R^6}\,\mathrm{d}r. \tag{4.36}$$

Integrating from $r = 0$ to $r = R$,

$$\begin{aligned} E_C &= \frac{(Ze)^2}{4\pi\varepsilon_0}\frac{3}{R^6}\int_0^R r^4\,\mathrm{d}r = \frac{(Ze)^2}{4\pi\varepsilon_0}\frac{3}{R^6}\cdot\frac{R^5}{5} \\ &= \frac{3}{5}\frac{1}{4\pi\varepsilon_0}\frac{(Ze)^2}{R}. \end{aligned} \tag{4.37}$$

(The factor 3/5 is pure geometry of a uniform sphere [13, 2, 1].)

With $R = r_0 A^{1/3}$ and $r_0 \approx 1.2\,\mathrm{fm}$,

$$\begin{aligned} a_C &= \frac{3}{5}\frac{e^2}{4\pi\varepsilon_0\,r_0} = \frac{3}{5}\frac{\hbar c\,\alpha}{r_0} \\ &\approx \frac{3}{5}\frac{197\,\mathrm{MeV\,fm} \times 1/137}{1.2\,\mathrm{fm}} \approx 0.72\,\mathrm{MeV}. \end{aligned} \tag{4.38}$$

In the SEMF one usually keeps the discrete-pair refinement

$$B_C = -\,a_C\,\frac{Z(Z-1)}{A^{1/3}}, \qquad a_C \approx 0.71\text{–}0.72\,\mathrm{MeV}. \tag{4.39}$$

(Replacing Z^2 by $Z(Z-1)$ subtracts the spurious self-energy of each proton.) Fits often leave a_C slightly free because the real charge density is not a sharp uniform sphere [1].

Physics consequences.

- $B_C < 0$: Coulomb always reduces binding relative to pure strong attraction.
- Along the valley, $Z \sim A/2$ to $A/2.5$, so $|B_C|/A$ grows roughly as $A^{2/3}$. That drives the slow *decline* of B/A for heavy nuclei.
- Coulomb alone favours $Z \to 0$ at fixed A. Nature does not: the asymmetry term restores $N \sim Z$.

Example: Coulomb cost for heavy nuclei

For $A \sim 238$, $Z \sim 92$,

$$\frac{B_C}{A} \sim -0.72 \cdot \frac{92 \cdot 91}{238^{4/3}} \approx -4.1\,\mathrm{MeV},$$

a multi-MeV reduction of B/A compared with a light nucleus. Volume and surface alone cannot produce that heavy-end decline.

After volume, surface and Coulomb,

$$B = a_V A - a_S A^{2/3} - a_C\frac{Z(Z-1)}{A^{1/3}}. \tag{4.40}$$

Maximising B at fixed A still pushes $Z \to 0$. The next term stops that pathology.

4.10 Valley of stability: what Coulomb + asymmetry must explain

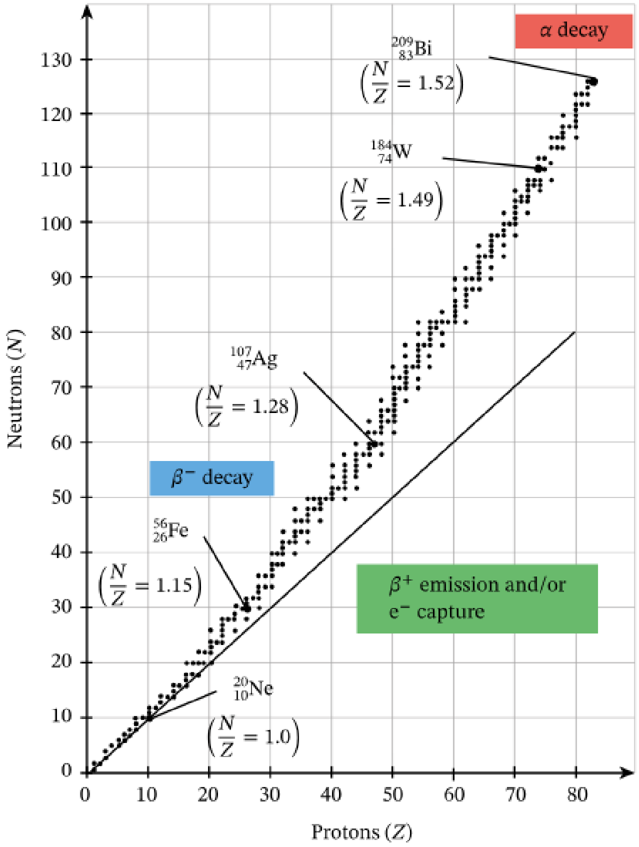


Figure 4.7: Chart of β-stable nuclides in the N–Z plane. Light systems hug $N = Z$. Heavier systems drift to $N > Z$ ($N/Z \approx 1.15$ for Fe, ≈ 1.28 for Ag, ≈ 1.49 for W, ≈ 1.52 for Bi). Neutron-rich: β^-. Proton-rich: β^+ / EC. Very heavy: α decay.

Figure 4.7 is the experimental map that Coulomb and asymmetry must jointly explain [6, 1, 13, 5, 14]:

1. **Light nuclei, $N \approx Z$.** Nuclear force is (approximately) charge independent. Pauli filling of two Fermi seas then prefers equal proton and neutron numbers when Coulomb is still weak. On the Segrè chart, stable nuclides hug $N \approx Z$ for $Z \lesssim 20$ [5].

2. **Heavy nuclei, $N > Z$.** Coulomb $\propto Z^2/A^{1/3}$ grows with Z. Extra neutrons dilute the proton density without adding Coulomb cost, so the valley bends above the $N = Z$ line ($N/Z \approx 1.15$ for Fe, ≈ 1.5 near Bi). Without an asymmetry term the SEMF would allow "stable isotopes of hydrogen with hundreds of neutrons" [6].

3. **Decay regions.** Neutron rich $\Rightarrow \beta^-$; proton rich $\Rightarrow \beta^+$ or EC. Very heavy systems also α decay when Coulomb has lowered B/A enough.

Without the asymmetry term, volume and surface alone place no restriction on N/Z, while Coulomb alone would drive every isobar toward $Z \to 0$. Nature does neither: the next term restores $N \sim Z$ at small A and permits only a controlled neutron excess at large A [1, 13].

Key idea

The valley is not a separate SEMF term. It is the observable balance of Coulomb (wants fewer protons) against asymmetry (wants $N = Z$).

4.11 Asymmetry (symmetry) term: why $-a_a(N - Z)^2/A$

Volume, surface, and Coulomb would allow any N/Z ratio at fixed A. Coulomb alone would drive $Z \to 0$. The asymmetry term restores $N \approx Z$ when Coulomb is weak and permits only a *controlled* neutron excess when Coulomb is strong. Together with Coulomb it shapes the valley of stability (Figure 4.7) and the mass parabolas of Lecture 5.

4.11.1 Physical picture

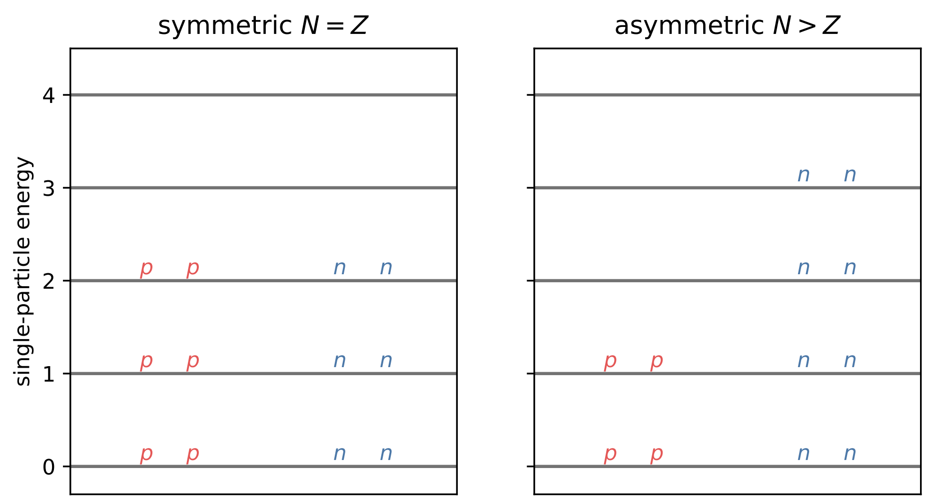


Figure 4.8: Left: symmetric filling $Z = N = A/2$. Right: asymmetric filling with extra neutrons. Excess neutrons must occupy higher single-particle levels, raising the total kinetic energy and lowering B.

Protons and neutrons are distinct fermions. Each species fills its own ladder of single-particle levels (Pauli principle) [6, 13, 14, 2, 4].

- If $N = Z$, both ladders are filled to the same height: minimal kinetic energy for given A.
- If $N > Z$, the neutron ladder is filled higher and the proton ladder lower. The extra neutrons sit in higher orbitals, so the total kinetic energy rises even though A is fixed.
- Higher energy of the system $\Rightarrow$ *smaller* binding energy B.

You cannot put excess neutrons into empty low proton levels: they are different particles. This correction (and pairing) go beyond a purely classical liquid drop: they require the quantum exclusion principle and, ultimately, the shell model [13]. A simple shell-level cartoon makes the scaling plausible: if adjacent shells are spaced by a constant δ, moving a proton up into a neutron orbital costs an energy that grows with the imbalance and yields the form $(N - Z)^2/A$ after averaging [4]. The same quadratic form follows from two Fermi gases (next subsection) [2].

4.11.2 Scaling: why $(N - Z)^2/A$

Define the neutron excess

$$\Delta \equiv N - Z = A - 2Z. \tag{4.41}$$

Two independent arguments give the same functional form.

(A) Toy ladder with spacing δ. Move $\Delta/2$ nucleons from the proton sea to the neutron sea. Each moved nucleon is raised by an energy of order $\delta \cdot (\Delta/2)$ relative to the Fermi surface, so the excess energy scales as

$$\Delta E \sim \delta \cdot \frac{\Delta^2}{4}. \tag{4.42}$$

In a large system the level density near the Fermi surface grows with A, so $\delta \sim 1/A$. Therefore

$$\Delta E \propto \frac{(N - Z)^2}{A}. \tag{4.43}$$

(B) Two Fermi gases (standard derivation). Let protons and neutrons be free Fermi gases in volume $V \propto A$, densities $\rho_p = Z/V$, $\rho_n = N/V$. The kinetic energy of one species is

$$E_{\text{kin},\tau} = \frac{3}{5} N_\tau \frac{\hbar^2}{2m} (3\pi^2 \rho_\tau)^{2/3} = \frac{3}{5} N_\tau \, \varepsilon_{F,\tau}. \tag{4.44}$$

Write $N = (A/2)(1 + x)$, $Z = (A/2)(1 - x)$ with $x = (N - Z)/A = \Delta/A$. Because $\varepsilon_{F,\tau} \propto \rho_\tau^{2/3} \propto N_\tau^{2/3}$,

$$E_{\text{kin}} = C' \left[\left(\frac{A}{2}\right)^{5/3} (1 + x)^{5/3} + \left(\frac{A}{2}\right)^{5/3} (1 - x)^{5/3} \right],$$

with C' fixed by the saturation Fermi energy. Expand $(1 \pm x)^{5/3} = 1 \pm \frac{5}{3}x + \frac{5}{9}x^2 + \cdots$: the odd powers cancel and

$$\begin{aligned} E_{\text{kin}} &= 2C' \left(\frac{A}{2}\right)^{5/3} \left(1 + \frac{5}{9}x^2 + \cdots\right) \\ &= CA + \frac{C}{3} \frac{(N - Z)^2}{A} + \cdots , \end{aligned} \tag{4.45}$$

with $C = \frac{3}{5}\varepsilon_F$ at $N = Z$ ($\varepsilon_F \approx 35\,\text{MeV}$ at saturation density). The linear (volume) piece is absorbed into a_V. The quadratic piece *raises* the energy of the nucleus and therefore *lowers* the binding:

$$B_{\text{a}} = -\, a_{\text{a}} \frac{(N - Z)^2}{A} = -\, a_{\text{a}} \frac{(A - 2Z)^2}{A}. \tag{4.46}$$

Empirical fits give

$$a_{\text{a}} \approx 23\text{–}24\,\text{MeV} \tag{4.47}$$

[6, 1, 14]. The pure kinetic estimate $C/3 \sim \varepsilon_F/3 \sim 12\,\text{MeV}$ is only about half of the empirical value: the rest comes from isovector potential energy (the force also prefers $N = Z$). The *form* $(N - Z)^2/A$ is already fixed by the Fermi gas.

Caution

Some modern fits write $-a'(Z - A/2)^2/A$ with $a' \approx 94\,\text{MeV}$. Because $(Z - A/2)^2 = (N - Z)^2/4$, that is the same physics with a coefficient four times larger. Always check the algebraic form before copying a number [4].

Checklist for the asymmetry term

1. Vanishes when $N = Z$.
2. Grows as $(N - Z)^2$: quadratic cost of imbalance.
3. Falls as $1/A$: large systems tolerate imbalance more easily (so heavy nuclei can sit at $N > Z$).
4. Together with Coulomb, fixes N/Z along Figure 4.7.

Key idea

Asymmetry energy is mainly a kinetic (Pauli) penalty, reinforced by isovector mean fields. It is not a new fundamental force. Figure 4.8 is the picture to keep: extra neutrons climb the neutron ladder.

After Coulomb and asymmetry,

$$B = a_V A - a_S A^{2/3} - a_C \frac{Z(Z-1)}{A^{1/3}} - a_\mathrm{a} \frac{(N-Z)^2}{A}. \tag{4.48}$$

4.11.3 Most stable Z at fixed A (preview; full derivation in Lecture 5)

Ignore pairing, treat A fixed, and minimise the atomic mass (or maximise B) with respect to Z. Balancing Coulomb against asymmetry yields the schematic result

$$Z^* \approx \frac{A/2}{1 + \kappa A^{2/3}}, \qquad \kappa \sim \frac{a_C}{4a_\mathrm{a}} \approx 0.0075, \tag{4.49}$$

so $Z^* \approx A/2$ for light nuclei and $Z^* < A/2$ ($N > Z$) for heavy ones: the bend of Figure 4.7 [14, 1, 6, 13]. The same stationarity condition is the Z_min formula that Lecture 5 will derive carefully, including the small $m_n - m(^1\mathrm{H})$ shift [6].

Remark

Lecture 5 starts from $M(A, Z)$, differentiates carefully with respect to Z (including $m_n - m(^1\mathrm{H})$), obtains the working formula $Z \approx A/[2(1 + 0.0078\,A^{2/3})]$, and connects that minimum to mass-parabola figures for odd and even A.

4.12 Pairing term: why δ depends on even/odd N, Z

Pairing is the smallest of the five SEMF terms in magnitude (typically $\sim 1\,\text{MeV}$ total, or $\lesssim 0.1\,\text{MeV}$ per nucleon), but it has outsized consequences for stability. It produces the odd–even staggering on B/A and, at fixed even A, splits the mass surface into two parabolas in Lecture 5. Empirically there are far more stable even–even nuclei than odd–odd ones; the pairing term is the liquid-drop encoding of that fact [6, 1].

4.12.1 Physical idea

Like nucleons (two protons or two neutrons) tend to form $J = 0$ pairs (opposite angular momenta). A paired configuration lies lower than an unpaired occupancy of the last orbital [6, 1, 13, 2]. Hence:

- even N (or Z): pairs complete $\Rightarrow$ extra binding;
- odd N (or Z): one unpaired nucleon $\Rightarrow$ less binding.

The odd–odd case is especially clear: when both Z and N are odd, one gains binding by converting one of the odd protons into a neutron (or vice versa) so that it can form a pair with its formerly odd partner [6]. With all other factors fixed, one must therefore *subtract* from B for odd–odd configurations and *add* for even–even ones [13]. Equivalently, two protons (or two neutrons) can occupy the same orbital with opposite spins, while an odd–odd nucleus always has an unpaired nucleon forced into the next higher level [2].

4.12.2 Experimental evidence: separation energies

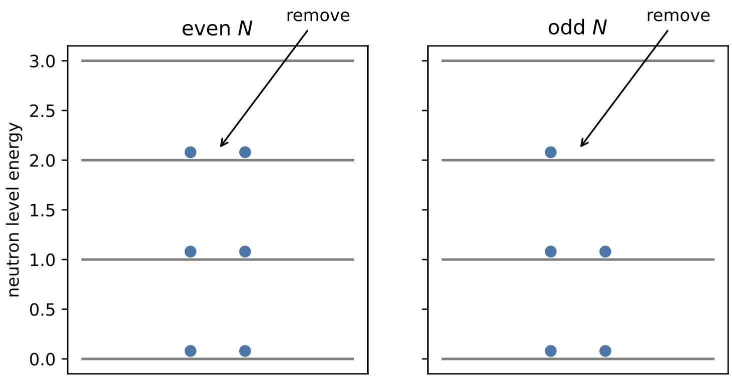


Figure 4.9: Schematic neutron separation: energy cost S_n. Even N: a pair must be broken. Odd N: the unpaired neutron is cheaper to remove.

$$S_n(A, Z) = [M(A-1, Z) + m_n - M(A, Z)]c^2, \tag{4.50}$$

$$S_p(A, Z) = [M(A-1, Z-1) + m_p - M(A, Z)]c^2. \tag{4.51}$$

Plotting S_n versus N shows a sawtooth: S_n is larger for even N (break a pair) than for odd N (remove the unpaired neutron). The same stagger appears in S_p versus Z [6, 1].

4.12.3 Census of stable nuclei

Configuration	Approx. number of β-stable nuclides
Even–even (Z even, N even)	$\sim$ 165–167
Odd A (exactly one of Z, N odd)	$\sim$ 100–110
Odd–odd (Z and N both odd)	only 4 (light)

The four stable odd–odd nuclei are ^{2}H, ^{6}Li, ^{10}B, and ^{14}N [6, 1]. Heavier odd–odd systems almost always β decay so that the unpaired proton and neutron become a like pair in an even–even daughter.

4.12.4 Pairing term in the SEMF

Taking odd-A nuclei as the baseline,

$$\delta = \begin{cases} +\,a_P\,A^{-3/4} & \text{even–even (more bound)}, \\ 0 & \text{odd } A, \\ -\,a_P\,A^{-3/4} & \text{odd–odd (less bound)}. \end{cases} \tag{4.52}$$

Throughout Lectures 4–5 we use the single convention $a_P = 34\,\mathrm{MeV}$ with $A^{-3/4}$. Other texts may instead use $\sim 12\,A^{-1/2}\,\mathrm{MeV}$; coefficients from those two conventions must never be interchanged [6, 1, 14, 2].

Caution

Sign conventions differ between sources (some write δ with the opposite overall sign in B vs M). Always verify that even–even nuclei gain binding and odd–odd nuclei lose it, and check whether a given formula fits B or M [1].

4.13 The full Weizsäcker / Bethe–Weizsäcker formula

Semi-empirical mass formula

$$B(A,Z) = a_V A - a_S A^{2/3} - a_C \frac{Z(Z-1)}{A^{1/3}} - a_{\mathrm{a}} \frac{(N-Z)^2}{A} + \delta, \tag{4.53}$$

with $N = A - Z$ and δ from Equation (4.52).

Equivalently, in atomic-mass form [4, 5, 6],

$$m(A,Z) = Z\,m(^1\mathrm{H}) + N\,m_n - B(A,Z)/c^2, \tag{4.54}$$

which connects the SEMF directly to tabulated masses. The Coulomb piece is sometimes written as $a_C Z^2/A^{1/3}$ (the large-Z approximation to $Z(Z-1)$); we keep $Z(Z-1)$ in (4.53) [14, 1, 6].

Coeff.	Typical value	Role	Derivation key
a_V	15.5–15.8 MeV	bulk nearest-neighbour net attraction	short range + saturation; net of Fermi KE
a_S	16.8–17.8 MeV	missing surface bonds	$S \propto R^2 \propto A^{2/3}$
a_C	≈ 0.72 MeV	proton self-energy	pairs $Z(Z-1)/2$; sphere 3/5
a_a	23–24 MeV	$N \neq Z$ kinetic (+ potential) penalty	two Fermi gases $\Rightarrow (N-Z)^2/A$
a_P	12–34 MeV	even–odd pairing	S_n stagger; even–even preference

A standard coefficient set that reproduces B/A well is [6, 1]

$$\begin{aligned} a_V &= 15.5\,\mathrm{MeV}, \quad a_S = 16.8\,\mathrm{MeV}, \qquad a_C = 0.72\,\mathrm{MeV}, \\ a_\mathrm{a} &= 23\,\mathrm{MeV}, \quad a_P = 34\,\mathrm{MeV} \qquad (\text{with } A^{-3/4}). \end{aligned} \tag{4.55}$$

Other widely used fits give $a_V = 15.75$, $a_S = 17.8$, $a_C = 0.71$, $a_\mathrm{a} = 23.7$ MeV [14], or nearly the same numbers (15.76, 17.81, 0.711, 23.7) [5], or $a_V \approx 15.8$, $a_S \approx 18.3$, $a_C \approx 0.71$ with pairing $\sim A^{-1/2}$ [4]. Differences of a few percent are fit choices (mass range, shell nuclei included or not), not new physics.

Remark

The SEMF is semi-empirical: the *forms* come from physics; the *numbers* are fixed by global fits to measured masses. It does not replace the shell model for magic numbers and spectroscopy, but it organises the bulk of the binding curve [6, 1]. Its importance is not that it predicts exotic new phenomena, but that it is a first successful application of nuclear models to a systematic nuclear property [6].

4.14 How the terms stack on B/A

Read Figure 4.10 from the top down at fixed A [2, 1, 13]:

How to read the B/A stack

1. **Volume (top).** Flat near $a_V \sim 15$–16 MeV. Independent of A if density and coordination number are fixed.
2. **Minus surface.** Large at small A; shrinks as $A^{-1/3}$. Explains the *rise* of net B/A from light nuclei toward iron.
3. **Minus Coulomb.** Grows with $Z^2/A^{4/3}$. Explains the *fall* of B/A past iron and the need for neutron excess.
4. **Minus asymmetry.** Extra cost when $|N-Z|$ is large. Moderate on the valley; large off it (drives β decay).
5. **Net B/A.** Peak near $A \sim 56$–62; fusion below the peak, fission above. Pairing is a small

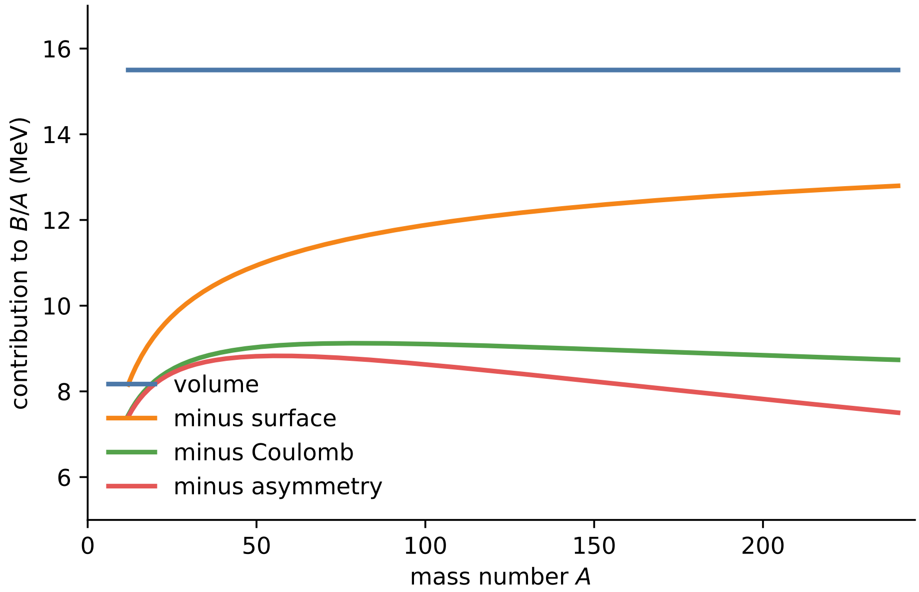


Figure 4.10: Decomposition of mean binding energy per nucleon. Top: volume ceiling. Subtract surface: rising light-A side. Subtract Coulomb: downward bend at large A. Asymmetry removes a further slice for neutron-rich heavy systems. Red: net B/A (pairing not drawn).

even–odd ripple on top.

Numerical sketch: ^{56}Fe

Use $a_V = 15.5$, $a_S = 16.8$, $a_C = 0.72$, $a_\mathrm{a} = 23$ (MeV) for ^{56}Fe ($A = 56$, $Z = 26$, $N = 30$):

$$\frac{B_V}{A} = 15.5,$$
$$\frac{B_S}{A} = -\frac{16.8}{56^{1/3}} \approx -4.4,$$
$$\frac{B_C}{A} = -0.72 \cdot \frac{26 \cdot 25}{56^{4/3}} \approx -2.18,$$
$$\frac{B_\mathrm{a}}{A} = -23 \cdot \frac{(4)^2}{56^2} \approx -0.12.$$

Sum ≈ 8.8 MeV before pairing: the right order of magnitude compared with the experimental peak near 8.8 MeV. Coefficients are global fits, not exact for every nuclide.

Worked stacking: ^{100}Mo **and** ^{208}Pb

Take the same coefficients and valley charges $Z \approx A/[2(1 + 0.0078A^{2/3})]$: for $A = 100$, $Z \approx 43$; for $A = 208$, $Z \approx 82$.

Medium mass (^{100}Mo, $Z = 42$ for the stable isobar):

$$\frac{B_V}{A} = 15.5, \quad \frac{B_S}{A} = -\frac{16.8}{100^{1/3}} \approx -3.6, \quad \frac{B_C}{A} \approx -0.72 \cdot \frac{42 \cdot 41}{100^{4/3}} \approx -2.67,$$

$$\frac{B_{\rm a}}{A} \approx -23 \cdot \frac{16^2}{100^2} \approx -0.59 \quad \Rightarrow \quad B/A \approx 8.62\,\text{MeV}.$$

The surface penalty is already moderate; Coulomb and asymmetry together remove $\sim$ 3.26 MeV/nucleon from the volume ceiling.

Heavy nucleus (^{208}Pb, $Z = 82$, $N = 126$):

$$\frac{B_S}{A} \approx -\frac{16.8}{208^{1/3}} \approx -2.8, \quad \frac{B_C}{A} \approx -0.72 \cdot \frac{82 \cdot 81}{208^{4/3}} \approx -3.88,$$

$$\frac{B_{\rm a}}{A} \approx -23 \cdot \frac{44^2}{208^2} \approx -1.03 \quad \Rightarrow \quad B/A \approx 7.75\,\text{MeV (before pairing)}.$$

Surface is a smaller fraction of A, but Coulomb and asymmetry now dominate: the net B/A has fallen $\sim$ 1.1 MeV below the iron peak even though the volume term is unchanged. That is the liquid-drop reason heavy nuclei are less tightly bound per nucleon and why fission can release energy by moving toward mid-mass daughters.

Vertical slices at small and large A

At $A \sim 20$: surface slice is thick; Coulomb slice is thin; net B/A still climbing.

At $A \sim 200$: surface slice is thinner; Coulomb slice is thick; asymmetry is visible; net B/A has fallen below the iron maximum.

That is the physics of stellar fusion (building toward iron) and of fission reactors (splitting uranium toward the peak).

4.15 Term-by-term derivation summary

Term	**Form**	**Why that form**
Volume	$+a_V A$	short range + fixed neighbours $\Rightarrow$ linear in A, not A^2; net of Fermi KE.
Surface	$-a_S A^{2/3}$	surface nucleons under-bound; $S \propto R^2 \propto A^{2/3}$.
Coulomb	$-a_C Z(Z-1)/A^{1/3}$	long-range pairs $Z(Z-1)/2$; $R \propto A^{1/3}$; sphere factor 3/5.
Asymmetry	$-a_{\rm a}(N-Z)^2/A$	two Fermi seas; expand $E_{\rm kin}$ for $N \neq Z$; $\delta \sim 1/A$.
Pairing	δ(ee/A/oo)	like-nucleon pairs; S_n stagger; almost no stable odd–odd.

4.15.1 How the five SEMF terms shape B/A

The five SEMF terms organise the B/A curve: volume sets the $\sim$ 8 MeV plateau, surface corrects light nuclei, Coulomb and asymmetry shape the heavy fall-off, and pairing splits even–even from odd–odd. Plotting each contribution versus A (as in standard textbooks) makes clear why iron-group nuclei maximise B/A and why heavy nuclei become Coulomb-unstable [6, 1, 14, 13].

4.16 Deeper physics: SEMF coefficients and the B/A landscape

4.16.1 Standard coefficient sets

Fitting the liquid-drop formula to measured masses yields coefficient sets that agree at the 10% level. One convenient set is [1]

Coefficient	Typical value (MeV)	Role
a_V	15.5	volume attraction
a_S	16.8	surface deficit
a_C	0.72	Coulomb (3/5 sphere)
a_a	23.2	asymmetry / symmetry
a_P	34	pairing ($\pm A^{-3/4}$ or $\pm A^{-1/2}$)

A modern alternative quotes $a_V \approx 15.8$, $a_S \approx 18.3$, $a_C \approx 0.71$, $a_a \approx 23.2$–94.8 depending on the exact $(N-Z)^2/A$ normalisation, and $a_P \approx 11$–$12\,\mathrm{MeV}$ for an $A^{-1/2}$ pairing form [4]. One derives $a_C = 3e^2/(5 \cdot 4\pi\varepsilon_0 r_0)$ from the uniform-sphere Coulomb integral and recovers $a_C \sim 0.7\,\mathrm{MeV}$ for $r_0 \sim 1.2\,\mathrm{fm}$.

Caution

Coefficients are *fit parameters* to terrestrial masses near the valley of stability. Using them far from stability (drip lines) or at extreme A (neutron stars) is an extrapolation: useful for orientation, not a substitute for a microscopic EOS.

4.16.2 Features of the B/A curve that the SEMF must reproduce

1. **Rise at low A:** surface term $\propto A^{-1/3}$ dominates; light nuclei are surface-rich.
2. **Broad maximum near $A \sim 56$–62** (iron–nickel): competition of surface (wants larger A) and Coulomb (wants smaller Z, hence limits growth).
3. **Slow decline for heavy A:** Coulomb $\propto Z^2/A^{1/3}$ grows; fission becomes favourable.
4. **Odd–even staggering:** pairing; even–even nuclei sit higher in B/A.
5. **Light $A = 4n$ peaks:** α-clustering (^{4}He, ^{12}C, ^{16}O) exceeds the smooth SEMF; shell and cluster physics beyond the liquid drop [2, 6].

4.16.3 Mass defect and Q values

Any reaction or decay Q value is a difference of binding energies (or masses). Fusion of light nuclei and fission of heavy nuclei both move the system toward the iron peak and release energy:

$$Q = [B(\text{products}) - B(\text{reactants})] > 0 \tag{4.56}$$

when the products lie higher on the B/A curve. The SEMF thus organises not only stability but also the energetics of reactors and stars [1, 4].

4.17 Physics depth: SEMF term scales and competition

The liquid-drop SEMF writes

$$B = a_V A - a_S A^{2/3} - a_C \frac{Z(Z-1)}{A^{1/3}} - a_a \frac{(A-2Z)^2}{A} + \delta, \tag{4.57}$$

with typical magnitudes $a_V \sim 16\,\mathrm{MeV}$, $a_S \sim 18\,\mathrm{MeV}$, $a_C \sim 0.7\,\mathrm{MeV}$, $a_a \sim 23\,\mathrm{MeV}$. Each term has a distinct A, Z scaling that students should feel numerically, not only symbolically.

4.17.1 Worked estimate: Coulomb vs surface for $A = 120$

Take $Z = 50$. Then

$$E_C \sim a_C Z^2/A^{1/3} \simeq 0.7 \times 2500/4.9 \simeq 360\,\mathrm{MeV},$$

while $E_S \sim a_S A^{2/3} \simeq 18 \times 24 \simeq 430\,\mathrm{MeV}$. Volume binding $a_V A \sim 1920\,\mathrm{MeV}$ still dominates, but Coulomb and surface are comparable: that near-cancellation foreshadows fission for heavier systems (Lecture 23), where increasing Z^2/A tips the balance.

Key idea

Physical meaning of pairing δ. Pairing is a few-mega-electronvolt correction, tiny against volume binding, yet it decides even–odd mass staggering and which isobar is β-stable (Lecture 5). Microscopic pairing gaps later reappear as seniority and BCS language in the shell model.

Remark

Assumptions. The SEMF is a smooth fit. Shell gaps of several MeV near magic numbers are residuals, not failures of the idea. Extrapolating SEMF coefficients to drip lines without microscopic corrections misestimates separation energies by mega-electronvolts.

4.17.2 Competition map across the chart

For light nuclei, surface wins over Coulomb and B/A rises with A. Near $A \sim 56$, volume binding is maximised relative to the penalties, which is why the iron region is the endpoint of exothermic fusion. Beyond, Coulomb grows faster than surface decreases (relative to volume), and B/A slowly declines: fission and α decay become exothermic for the heaviest species. The SEMF is therefore a single formula that organises stellar burning, fission reactors, and α chains.

Pairing $\delta \sim 12/\sqrt{A}\,\mathrm{MeV}$ (rough scale) is negligible in absolute binding yet decisive for odd–even effects. Shell corrections of several MeV near magic numbers are the main reason global SEMF fits leave mega-electronvolt residuals: they are features to be added, not noise to be ignored.

Evaluate each SEMF term for ^{16}O, ^{56}Fe, and ^{208}Pb in a short table: the pattern of which term dominates the correction to $a_V A$ is the whole pedagogical point of the liquid-drop expansion. Coulomb in lead is already several hundred mega-electronvolts; asymmetry is smaller but decides the neutron excess. Those two sentences are the bridge from binding systematics to fission and to the valley of β stability.

Worked term-by-term table

For ^{208}Pb, evaluate volume, surface, Coulomb, and asymmetry terms with standard coefficients and verify that the sum approaches the empirical binding energy within several hundred megaelectronvolts before shell corrections. Repeat for ^{16}O and notice the surface fraction's rise. That numerical habit is the surest way to remember why each SEMF piece exists.

Physics-depth close

Term-by-term SEMF arithmetic is the fastest way to anticipate whether a process is exothermic. If you can estimate Coulomb and asymmetry changes for a proposed reaction, you already understand most of the energy budget before opening a mass table.

Physics-depth close

Term-by-term SEMF arithmetic anticipates whether a process is exothermic. Estimating Coulomb and asymmetry changes is most of the energy budget before opening a mass table.

4.18 Further physics: scales, drip lines, and limits of the SEMF

The main text derived each SEMF term. The notes below fix the *scales* of the coefficients and turn B/A into fusion, fission, and drip-line physics [6, 1, 14, 13, 2, 4, 5].

4.18.1 Coefficient sets

Coefficient	Set A	Set B	Set C
a_V (MeV)	15.5	15.75–15.76	≈ 15.8
a_S (MeV)	16.8	17.8	≈ 18.3
a_C (MeV)	0.72	0.71	≈ 0.71
a_a (MeV)	23	23.7	≈ 23
pairing form	$\pm a_P A^{-3/4}$	$\pm 33.6\,A^{-3/4}$	$\pm a_P A^{-1/2}$

[6, 1, 14, 5, 4]. Differences of a few percent are fit choices (which mass table, whether magic nuclei are weighted), not new physics. Always match the algebraic form of the asymmetry and pairing terms before copying a number: some fits write $(Z - A/2)^2/A$ with a coefficient four times larger than $(N - Z)^2/A$ [4, 5].

4.18.2 Fermi-gas origin of volume, surface, and asymmetry

At saturation density $n_0 \approx 0.16\,\mathrm{fm}^{-3}$ with $N \approx Z \approx A/2$ [14],

$$\varepsilon_F \approx 35\,\mathrm{MeV}, \qquad \tfrac{3}{5}\varepsilon_F \approx 21\,\mathrm{MeV}, \qquad p_F \approx 265\,\mathrm{MeV}/c. \tag{4.58}$$

A mean potential depth $U \sim -40\,\mathrm{MeV}$ then yields a net volume coefficient

$$a_V \sim |U| - \tfrac{3}{5}\varepsilon_F \sim 19\,\mathrm{MeV}, \tag{4.59}$$

close to the empirical 15–16 MeV. Expanding the kinetic energy of two Fermi gases for small $\delta = (N-Z)/A$ gives a kinetic asymmetry coefficient $\sim \varepsilon_F/3 \sim 12$ MeV; the empirical $a_\mathrm{a} \approx 23$ MeV is roughly twice as large, so about half of asymmetry is potential (isovector) [14, 2].

Finite-size counting of states in a box also produces a kinetic surface term $\propto A^{2/3}$ of order 16 MeV, comparable to the empirical a_S [14]. The empirical surface coefficient therefore receives both this kinetic surface piece and the missing-bond (potential) surface tension of the liquid drop.

Thus the liquid-drop volume, surface, and asymmetry terms are not free inventions: they are the bulk, surface, and isovector pieces of a saturated Fermi liquid of nucleons.

4.18.3 Fermi-gas expansion for the asymmetry term

The kinetic energy of N_n neutrons and N_p protons in a common volume V is

$$E_K = \frac{3}{5}\,\varepsilon_F(N_n + N_p) \tag{4.60}$$

only when $N_n = N_p$. For $N_n \neq N_p$ one must use separate Fermi momenta. Setting $N_n = (A/2)(1+x)$ and $N_p = (A/2)(1-x)$ with $x = \Delta N/A$ and expanding $(1\pm x)^{5/3} = 1 \pm \frac{5}{3}x + \frac{5}{9}x^2 + \cdots$ yields [2]

$$E_K = CA + \frac{C}{3}\,\frac{(\Delta N)^2}{A} + \cdots, \tag{4.61}$$

with $C = \frac{3}{5}\varepsilon_F$ at symmetry. The first piece is absorbed into a_V; the second is precisely the asymmetry energy with a calculable kinetic coefficient. This is the derivation behind (4.46).

4.18.4 Fusion and fission Q values from B/A

Because B/A peaks near $A \sim 56$, energy is released when light nuclei fuse upward toward iron and when heavy nuclei fission downward toward iron [14, 5, 13, 6, 4]. For a reaction

$$X_1 + X_2 \to Y \qquad \text{or} \qquad X \to Y_1 + Y_2,$$

the Q value is the change in total binding energy (equivalently, the negative of the change in rest mass). The light side of the curve is the domain of stellar nuclear burning; the heavy side is the domain of fission and α decay (practically important mainly for $A \gtrsim 212$, because lifetimes are otherwise long) [14]. The SEMF, or measured mass excesses, turns the qualitative B/A argument into a number.

4.18.5 Drip lines from separation energies

Requiring the last added proton or neutron to be bound,

$$S_p > 0, \qquad S_n > 0,$$

defines the proton and neutron drip lines on the (N, Z) plane. Those lines are the boundary beyond which the last nucleon is unbound; most other combinations β or α decay toward long-lived species [14]. The SEMF predicts the drip lines semi-quantitatively; radioactive-beam experiments map the nucleus by nucleus [14, 1].

4.18.6 Pairing and mass parabolas (link to Lecture 5)

For even A, the pairing term splits the mass surface into two parabolas: even–even lower, odd–odd higher, separated by $\sim 2|\delta|$. That geometry is why some even–even nuclei can only decay by double β emission. For odd A, $\delta = 0$ and a single parabola remains [14, 6, 13, 5]. Lecture 5 makes those parabolas the main object of study.

4.18.7 What the SEMF cannot do

A few limitations should be kept in view:

- Magic numbers and shell gaps produce local spikes in B and in separation energies that the smooth SEMF misses [6, 14].
- Light $A = 4n$ cluster peaks (α, ^{12}C, ^{16}O) sit above the liquid-drop curve [2, 4].
- Coefficients fitted near the valley of stability are an extrapolation at the drip lines and a poor substitute for a microscopic EOS in neutron stars (Lecture 6) [4, 1].

The SEMF remains the right first model for the *smooth* landscape of nuclear binding.

4.19 Deeper reading: Q values from SEMF differences

Reaction and decay Q values are differences of binding energies. In the SEMF, many volume pieces cancel, leaving surface, Coulomb, and asymmetry changes as the leading drivers. That is why α decay favours heavy nuclei (large Coulomb gain for $Z \to Z - 2$) and why fusion of light nuclei releases energy while fusion of heavy nuclei does not (Lecture 25).

4.19.1 Link forward

Neutron-star crusts (Lecture 6) are assemblages of nuclei whose masses are still organised by the same SEMF bookkeeping, plus lattice and electron contributions. Rare-isotope mass measurements (Lecture 5) test how far the smooth drop formula can be trusted before shell evolution and continuum coupling dominate. When you quote a SEMF Q value, state whether pairing δ was included: for odd-A chains it can flip which neighbour is energetically favoured.

4.19.2 Fitting discipline

When students fit a_V, a_S, a_C, a_a to a mass subset, they should hold out a validation region (for example, all $A > 200$) and report the extrapolation error. Machine-learning residuals can reduce rms error on known nuclei while still failing at the drip line. The SEMF remains the right first model precisely because its failures are physically labelled: surface, Coulomb, asymmetry, pairing, and then shells.

Evaluate each SEMF term for ^{16}O, ^{56}Fe, and ^{208}Pb in a short table: the pattern of which term dominates the correction to $a_V A$ is the whole pedagogical point of the liquid-drop expansion.

Coulomb in lead is already several hundred mega-electronvolts; asymmetry is smaller but decides the neutron excess. Those two sentences are the bridge from binding systematics to fission and to the valley of β stability.

SEMF checklist

Build a short table of each SEMF term for three nuclei spanning light, iron-peak, and actinide regions. Identify which correction dominates the departure from $a_V A$. Then compute one α-decay Q and one fission-rough Q from SEMF differences to preview Lectures 16 and 23. Keep pairing visible in the table even when it is numerically small.

Residual patterns

After a global SEMF fit, plot residuals versus N and Z. Magic ridges appear as systematic underbinding or overbinding corridors. Those ridges are the empirical motivation for the shell model of Lectures 12–13. Pairing residuals alternate with odd/even A. A fit that absorbs shell structure into inflated a_S or a_a will look better globally and fail locally near closures: inspect residuals before trusting extrapolations.

Astrophysical Q values

Approximate neutron-capture Q values along a chain by SEMF differences of neighbouring isotopes. Near the drip line the answers become meaningless without continuum and shell corrections, which is precisely why Lecture 5 emphasises precision masses. Use SEMF Q values only as starting guesses for network sensitivity studies.

Drop-model forecasting limits

Use a SEMF fit to predict masses for $A = 120$ (interpolation) and for a neutron-rich $A = 120$ isotope far from stability (extrapolation). Compare with AME or a microscopic mass model. The interpolation error is small; the extrapolation error can be mega-electronvolts. That contrast is the practical justification for rare-isotope mass programmes emphasised in Lecture 5 and for not using SEMF alone in r-process networks.

The SEMF's lasting value is not millimetric mass accuracy; it is a labelled decomposition of binding into volume, surface, Coulomb, asymmetry, and pairing. Every later chapter that quotes a Q value is silently using that decomposition or a microscopic correction to it.

In practice, rework one numerical estimate from the lecture with a different parameter set and compare the shift to the quoted experimental uncertainty. If the shift exceeds the uncertainty, the estimate is teaching a scale, not delivering a fit. That distinction keeps theoretical cartoons honest when they sit beside evaluated data tables and modern many-body calculations. Along the course arc, write one sentence linking this chapter backward and one forward: the discipline of those two sentences is as important as any single formula. Examination problems rarely ask for a literature review, but they do ask for the scale argument that literature reviews presuppose.

Research frontier: mass models beyond the SEMF and student projects

The liquid-drop SEMF of Lecture 4 is still the pedagogical backbone of nuclear binding: volume, surface, Coulomb, asymmetry, and pairing terms explain the broad trends of B/A. Quantitative modern work, however, combines microscopic shell corrections, Bayesian coefficient fits, and machine-learning residuals trained against evaluated mass tables such as AME2020 [27]. Far from stability the smooth SEMF extrapolation fails: shell evolution, continuum coupling, and deformation require microscopic models constrained by rare-isotope data [53]. Community planning documents treat high-precision masses as enabling science for nucleosynthesis pathways and for neutrino-related nuclear matrix elements, because Q values enter both r-process waiting-point estimates and $0\nu\beta\beta$ phase space [54]. What remains uncertain is how far global mass models can be trusted beyond the last measured point, and how to quote honest uncertainties when several models disagree by mega-electronvolts near the drip lines. Machine-learning approaches can reduce rms residuals, but they must be audited for extrapolation artefacts.

In practice, a modern mass-model paper reports not only an rms residual on known nuclei but also a validation protocol for extrapolation. Students can reproduce a simple leave-one-out test on AME2020 subsets: hide a region of the chart, refit or retrain, and measure the prediction error. That exercise shows why drip-line forecasts need conservative uncertainty bars.

Student directions. (1) Computational fit: fit a_V, a_S, a_C, a_a to a subset of AME2020 even-even masses and quote rms residuals. (2) Data project: identify ten nuclei where $|M_{\text{exp}} - M_{\text{SEMF}}|$ exceeds 5 MeV and correlate them with magic numbers. (3) Critical essay: critique using SEMF coefficients near the drip line for r-process waiting-point estimates.

4.20 Problems with solutions

Problem 4.1 — Binding energy of the α particle. Atomic masses (u): $m(^1\text{H}) = 1.007825$, $m(n) = 1.008665$, $m(^4\text{He}) = 4.002603$. Compute B and B/A for the α particle. Use $c^2 = 931.5\,\text{MeV/u}$.

Solution 4.1.

$$\begin{aligned} B &= [2m(^1\text{H}) + 2m(n) - m(^4\text{He})]c^2 \\ &= (2 \times 1.007825 + 2 \times 1.008665 - 4.002603)\,\text{u}\,c^2 \\ &= 0.030377 \times 931.5\,\text{MeV} \approx 28.3\,\text{MeV}. \end{aligned}$$

$$B/A \approx 7.07\,\text{MeV}.$$

The α is exceptionally tightly bound for a light nucleus (cluster peak on the B/A curve) [6, 1].

Problem 4.2 — Coulomb coefficient a_C. Using classical electrostatics for a uniformly charged sphere of radius $R = r_0 A^{1/3}$ with $r_0 = 1.2\,\text{fm}$, estimate the SEMF Coulomb coefficient in the term $a_C Z^2/A^{1/3}$. Take $\hbar c \approx 197\,\text{MeV fm}$ and $\alpha \approx 1/137$.

Solution 4.2.

$$E_C = \frac{3}{5}\frac{(Ze)^2}{4\pi\varepsilon_0 R} = \frac{3}{5}\frac{\alpha\,\hbar c}{r_0}\frac{Z^2}{A^{1/3}}.$$

$$a_C = \frac{3}{5}\frac{\alpha\,\hbar c}{r_0} \approx \frac{3}{5}\frac{197/137}{1.2} \approx 0.72\,\text{MeV},$$

matching the usual fit [3, 6].

Problem 4.3 — B/A of ^{56}Fe from the SEMF. Use $a_V = 15.5$, $a_S = 16.8$, $a_C = 0.72$, $a_\text{a} = 23\,\text{MeV}$ (pairing neglected). Estimate B/A for ^{56}Fe ($Z = 26$).

Solution 4.3.

$$\frac{B}{A} = a_V - \frac{a_S}{A^{1/3}} - a_C\frac{Z(Z-1)}{A^{4/3}} - a_\text{a}\frac{(A-2Z)^2}{A^2}.$$

With $A = 56$, $Z = 26$, $A^{1/3} \approx 3.83$, $A - 2Z = 4$:

$$\begin{aligned} a_V &= 15.5, \\ a_S/A^{1/3} &\approx 4.39, \\ a_C Z(Z-1)/A^{4/3} &\approx 0.72 \times 650/(56 \times 3.83) \approx 2.18, \\ a_\text{a}(A-2Z)^2/A^2 &\approx 23 \times 16/3136 \approx 0.12. \end{aligned}$$

$$B/A \approx 15.5 - 4.39 - 2.18 - 0.12 \approx 8.8\,\text{MeV},$$

near the experimental iron-region maximum [6, 14].

Problem 4.4 — Volume vs surface for light and heavy nuclei. With $a_V = 15.5\,\text{MeV}$ and $a_S = 16.8\,\text{MeV}$, compare the volume and surface contributions to B/A for $A = 16$ and $A = 208$.

Solution 4.4. $B_V/A = a_V = 15.5\,\text{MeV}$ (independent of A); $|B_S|/A = a_S A^{-1/3}$.

A	$a_S A^{-1/3}$ (MeV)	surface/volume
16	$16.8/2.52 \approx 6.7$	$\sim 43\%$
208	$16.8/5.92 \approx 2.8$	$\sim 18\%$

Light nuclei are surface-dominated; heavy nuclei are bulk-dominated [13].

Problem 4.5 — Coulomb energy of ^{208}Pb. Estimate the Coulomb self-energy of ^{208}Pb ($Z = 82$, $R = 1.2\,A^{1/3}\,\text{fm}$) with the uniform-sphere formula. Give E_C in MeV and E_C/A.

Solution 4.5.

$$R \approx 1.2 \times 5.92 \approx 7.1\,\text{fm}, \quad E_C = a_C\frac{Z^2}{A^{1/3}} \approx 0.72\frac{82^2}{5.92} \approx 820\,\text{MeV}, \quad \frac{E_C}{A} \approx 3.9\,\text{MeV}.$$

Coulomb is the main reason B/A falls for heavy nuclei [1, 6].

Problem 4.6 — Asymmetry energy of ^{208}Pb. For ^{208}Pb ($N = 126$, $Z = 82$) with $a_\mathrm{a} = 23\,\mathrm{MeV}$, compute the asymmetry contribution to B and to B/A.

Solution 4.6.

$$E_\mathrm{a} = a_\mathrm{a}\frac{(N-Z)^2}{A} = 23\frac{44^2}{208} = 23 \times 9.31 \approx 214\,\mathrm{MeV}, \qquad \frac{E_\mathrm{a}}{A} \approx 1.0\,\mathrm{MeV}.$$

Neutron excess costs hundreds of MeV in total binding, but only about 1 MeV per nucleon at lead [14].

Problem 4.7 — Q value for $d + d \to t + p$. Binding energies: $B(^2\mathrm{H}) = 2.23\,\mathrm{MeV}$, $B(^3\mathrm{H}) = 8.48\,\mathrm{MeV}$. Compute the Q value of

$$^2\mathrm{H} + {}^2\mathrm{H} \to {}^3\mathrm{H} + p.$$

Also estimate the temperature needed for two deuterons to approach to 1.4 fm against Coulomb repulsion ($k_B = 8.6 \times 10^{-5}\,\mathrm{eV/K}$).

Solution 4.7. Energy release equals the increase in total binding:

$$Q = B_t - 2B_d = 8.48 - 2 \times 2.23 = 4.02\,\mathrm{MeV}.$$

Coulomb barrier at $R = 1.4\,\mathrm{fm}$:

$$E_\mathrm{cm} \simeq \frac{\alpha\,\hbar c}{R} \approx \frac{197/137}{1.4} \approx 1.03\,\mathrm{MeV}.$$

For two equal deuterons in the centre-of-mass frame, each carries half of this kinetic energy. A crude classical thermal threshold follows from $k_B T \sim E_\mathrm{cm}$:

$$T \sim \frac{1.03 \times 10^6\,\mathrm{eV}}{8.6 \times 10^{-5}\,\mathrm{eV/K}} \approx 1.2 \times 10^{10}\,\mathrm{K}.$$

This is a *classical* overestimate: quantum tunnelling through the Coulomb barrier permits fusion at much lower stellar temperatures [3].

Problem 4.8 — Both β modes of ^{64}Cu. Using the SEMF with $a_C = 0.697\,\mathrm{MeV}$, $a_\mathrm{a} = 23.3\,\mathrm{MeV}$, and $M_n - M_p - m_e = 0.782\,\mathrm{MeV}$, show that ^{64}Cu ($A = 64$, $Z = 29$) can β^- decay to ^{64}Zn and β^+ decay to ^{64}Ni. Estimate Q_{β^-} and Q_{β^+} (pairing: even–even final nuclei gain $\delta = \pm 34A^{-3/4}\,\mathrm{MeV}$ for even–even / odd–odd).

Solution 4.8.

$$Q_{\beta^-} = (M_n - M_p - m_e) + [B(64,30) - B(64,29)],$$

$$Q_{\beta^+} = (M_p - M_n - m_e) + [B(64,28) - B(64,29)].$$

$A^{1/3} = 4$ and $\delta_0 = 34/64^{3/4} = 1.503\,\text{MeV}$. Volume and surface cancel. For the $Z = 29 \to 30$ branch,

$$B(64,30) - B(64,29) = -\frac{0.697}{4}(60) - \frac{23.3}{64}(4^2 - 6^2) + 2\delta_0 \approx 0.180\,\text{MeV},$$

whereas for $Z = 29 \to 28$,

$$B(64,28) - B(64,29) = +\frac{0.697}{4}(56) - \frac{23.3}{64}(8^2 - 6^2) + 2\delta_0 \approx 2.569\,\text{MeV}.$$

Atomic-mass bookkeeping gives $(m_n - m(^1\text{H}))c^2 = +0.782\,\text{MeV}$ for β^-, while the free-mass term for β^+ is $(m(^1\text{H}) - m_n)c^2 - 2m_ec^2 = -1.804\,\text{MeV}$. Therefore

$$Q_{\beta^-} \approx 0.782 + 0.180 = 0.962\,\text{MeV} > 0, \quad Q_{\beta^+} \approx -1.804 + 2.569 = 0.765\,\text{MeV} > 0.$$

Both branches are open; maximum $e^\pm$ kinetic energies equal the corresponding Q values up to recoil. The calculation also shows why pairing cannot be omitted: it lowers both even–even daughters relative to the odd–odd parent [3].

Problem 4.9 — Dominant SEMF term for heavy nuclei. For ^{238}U, which SEMF term grows fastest with Z and drives fission instability? Estimate the Coulomb energy scale $a_C Z^2/A^{1/3}$ with $a_C \approx 0.7\,\text{MeV}$.

Solution 4.9. The Coulomb term $\propto Z^2/A^{1/3}$ grows fastest among the bulk terms for large Z. With $A = 238$, $Z = 92$,

$$\frac{a_C Z^2}{A^{1/3}} \approx 0.7 \times \frac{8464}{6.2} \approx 950\,\text{MeV}$$

total Coulomb energy (not per nucleon); per nucleon it is $\sim 4\,\text{MeV}$, comparable to surface and asymmetry and the driver of fissility [6, 1].

Chapter 5

Most Stable Z and Mass Parabolas

"For constant A, the mass formula represents a parabola of M vs Z."

K. S. Krane

5.1 Atomic mass from the SEMF

Lecture 4 constructed the binding energy $B(A, Z)$ and explained how its five terms shape the B/A curve. Stability against β decay is decided not by B alone but by the *atomic* mass $M(A, Z)$: at fixed mass number A, the isobar with the lowest mass cannot lower its energy by converting a neutron into a proton (or vice versa). Writing M as a function of Z at fixed A produces the mass parabolas of this lecture [6, 1, 14].

Key idea

From binding energy to atomic mass.

$$M(A, Z)\,c^2 = [Z\,m(^1\mathrm{H}) + (A - Z)\,m_n]c^2 - B(A, Z).$$

β decay changes Z by ± 1 at fixed A, so it moves along an isobaric slice of this surface. The SEMF turns that slice into a parabola in Z.

The SEMF binding energy is

$$B(A, Z) = a_V A - a_S A^{2/3} - a_C \frac{Z(Z-1)}{A^{1/3}} - a_\mathrm{a} \frac{(N-Z)^2}{A} + \delta(A, Z), \tag{5.1}$$

with $N = A - Z$ and

$$\delta = \begin{cases} +a_P A^{-3/4} & \text{even–even,} \\ 0 & \text{odd } A, \\ -a_P A^{-3/4} & \text{odd–odd.} \end{cases} \tag{5.2}$$

The atomic mass energy is

$$M(A, Z)\,c^2 = [Z\,m(^1\mathrm{H}) + (A - Z)\,m_n]c^2 - B(A, Z). \tag{5.3}$$

Using $m(^1\mathrm{H})$ rather than m_p folds the electron mass into the atomic ledger, which is what β

energetics require. Equivalently one may write the free-nucleon piece as $(A-Z)m_n+Z(m_p+m_e)$; the difference $m(^1\mathrm{H})-(m_p+m_e)$ is the electron binding in hydrogen and is negligible at the MeV scale used here [6, 1].

Key idea

Which SEMF terms matter at fixed A? When A is held constant, the volume term $a_V A$ and the surface term $a_S A^{2/3}$ are the same for every isobar and drop out of mass *differences*. What remains are:

- the free-nucleon mass piece, linear in Z through $Z\,m(^1\mathrm{H})+(A-Z)\,m_n$;
- Coulomb, quadratic in Z;
- asymmetry, quadratic in Z through $(A-2Z)^2$;
- pairing δ, constant on each even/odd Z ladder.

Lecture 4 fixed the bulk physics of B/A; Lecture 5 extracts the Z dependence that controls β stability.

Substitute (5.1) into (5.3). Because $M\,c^2$ contains $-B$, every binding term changes sign:

$$\begin{aligned} M(A,Z)\,c^2 =& [Z\,m(^1\mathrm{H})+(A-Z)\,m_n]c^2 \\ & -a_V A + a_S A^{2/3} + a_C\frac{Z(Z-1)}{A^{1/3}} + a_\mathrm{a}\frac{(A-2Z)^2}{A} - \delta. \end{aligned} \tag{5.4}$$

Here $N-Z=A-2Z$ has already been used in the asymmetry piece.

5.2 Why $M(A,Z)$ is a parabola in Z

Rewrite each Z-dependent contribution as a polynomial in Z. First the free-nucleon (atomic) piece:

$$[Z\,m(^1\mathrm{H})+(A-Z)\,m_n]c^2 = A\,m_n c^2 + [m(^1\mathrm{H})-m_n]c^2\,Z. \tag{5.5}$$

Next the asymmetry energy. Expand $(A-2Z)^2 = A^2-4AZ+4Z^2$:

$$\begin{aligned} a_\mathrm{a}\frac{(A-2Z)^2}{A} &= \frac{a_\mathrm{a}}{A}(A^2-4AZ+4Z^2) \\ &= a_\mathrm{a}A - 4a_\mathrm{a}Z + \frac{4a_\mathrm{a}}{A}\,Z^2. \end{aligned} \tag{5.6}$$

For Coulomb, distribute $Z(Z-1)=Z^2-Z$:

$$a_C\frac{Z(Z-1)}{A^{1/3}} = \frac{a_C}{A^{1/3}}\,Z^2 - \frac{a_C}{A^{1/3}}\,Z. \tag{5.7}$$

Volume $-a_V A$ and surface $+a_S A^{2/3}$ depend only on A. Pairing δ is constant on a given even/odd ladder of Z and will be kept separate from the smooth polynomial.

Insert (5.5)–(5.7) into (5.4):

$$
\begin{aligned}
M(A,Z)\,c^2 = &\Big(A\,m_n c^2 - a_V A + a_S A^{2/3} + a_\mathrm{a} A\Big) \\
&+ \Big([m(^1\mathrm{H}) - m_n]c^2 - 4a_\mathrm{a} - \frac{a_C}{A^{1/3}}\Big)Z \\
&+ \Big(\frac{a_C}{A^{1/3}} + \frac{4a_\mathrm{a}}{A}\Big)Z^2 - \delta.
\end{aligned}
\tag{5.8}
$$

This is the quadratic form

$$
M(A,Z)\,c^2 = \alpha(A) + \beta(A)\,Z + \gamma(A)\,Z^2 - \delta,
\tag{5.9}
$$

with coefficients (at fixed A)

$$
\alpha(A) = A\,m_n c^2 - a_V A + a_S A^{2/3} + a_\mathrm{a} A,
\tag{5.10}
$$

$$
\beta(A) = [m(^1\mathrm{H}) - m_n]c^2 - 4a_\mathrm{a} - \frac{a_C}{A^{1/3}},
\tag{5.11}
$$

$$
\gamma(A) = \frac{a_C}{A^{1/3}} + \frac{4a_\mathrm{a}}{A}.
\tag{5.12}
$$

Both contributions to γ are positive for physical coefficients, so $\gamma(A) > 0$: the parabola opens upward. The constant $\alpha(A)$ shifts the whole curve uniformly and does not affect which Z is most stable. Nuclei far from the vertex have higher mass and can β decay toward the minimum [6, 1].

Differentiating (5.9) (pairing held fixed) gives

$$
\frac{\partial}{\partial Z}\big[M(A,Z)\,c^2\big] = \beta(A) + 2\gamma(A)\,Z.
\tag{5.13}
$$

Setting the derivative to zero yields the continuous location of the vertex,

$$
Z_\mathrm{min} = -\frac{\beta(A)}{2\gamma(A)}.
\tag{5.14}
$$

Substitute (5.11)–(5.12):

$$
\begin{aligned}
Z_\mathrm{min} &= \frac{-[m(^1\mathrm{H}) - m_n]c^2 + 4a_\mathrm{a} + a_C A^{-1/3}}{2\big(a_C A^{-1/3} + 4a_\mathrm{a}/A\big)} \\
&= \frac{\big(m_n - m(^1\mathrm{H})\big)c^2 + a_C A^{-1/3} + 4a_\mathrm{a}}{2a_C A^{-1/3} + 8a_\mathrm{a}/A},
\end{aligned}
\tag{5.15}
$$

which is the same expression obtained by differentiating (5.4) term by term in the next section.

Geometrically, Coulomb raises the mass of proton-rich isobars; asymmetry raises the mass when $|N - Z|$ is large. The minimum is the compromise charge where those penalties balance, moderated by the small linear term from nucleon mass differences and the linear piece of the Coulomb expansion.

The most stable isobar is the integer Z nearest the minimum of that parabola.

Key idea

At fixed A, Coulomb repulsion and asymmetry energy together make the mass a quadratic in Z. Minimising mass maximises stability against β decay.

5.3 Most stable Z: differentiation of the mass formula

Ignore pairing for the moment (exact for odd A; for even A the pairing shifts two parallel parabolas but does not move their common $Z_{\min}$ at leading order). Differentiate (5.4) with respect to Z at fixed A.

The free-nucleon term contributes

$$\frac{\partial}{\partial Z}\big[Z\,m(^1\mathrm{H}) + (A-Z)\,m_n\big]c^2 = \big[m(^1\mathrm{H}) - m_n\big]c^2. \tag{5.16}$$

For Coulomb,

$$\frac{\partial}{\partial Z}\Big[a_C\frac{Z(Z-1)}{A^{1/3}}\Big] = \frac{a_C}{A^{1/3}}\,(2Z-1). \tag{5.17}$$

For asymmetry, with $u = A - 2Z$ so $\partial u/\partial Z = -2$,

$$\frac{\partial}{\partial Z}\Big[a_\mathrm{a}\frac{(A-2Z)^2}{A}\Big] = \frac{a_\mathrm{a}}{A}\cdot 2(A-2Z)\cdot(-2) = -\frac{4a_\mathrm{a}}{A}(A-2Z). \tag{5.18}$$

Volume, surface, and (for now) pairing do not contribute. Adding these derivatives,

$$\frac{\partial}{\partial Z}\big[M(A,Z)\,c^2\big] = \big[m(^1\mathrm{H}) - m_n\big]c^2 + a_C\frac{(2Z-1)}{A^{1/3}} - a_\mathrm{a}\frac{4(A-2Z)}{A}. \tag{5.19}$$

Set the derivative to zero and rearrange. Writing $[m(^1\mathrm{H}) - m_n]c^2 = -[m_n - m(^1\mathrm{H})]c^2$,

$$a_C\frac{(2Z-1)}{A^{1/3}} - \big[m_n - m(^1\mathrm{H})\big]c^2 = 4a_\mathrm{a}\frac{(A-2Z)}{A}. \tag{5.20}$$

Expand the left- and right-hand sides of (5.20):

$$a_C\frac{2Z}{A^{1/3}} - \frac{a_C}{A^{1/3}} - \big[m_n - m(^1\mathrm{H})\big]c^2 = 4a_\mathrm{a} - \frac{8a_\mathrm{a}}{A}\,Z, \tag{5.21}$$

where the asymmetry identity

$$\frac{4a_\mathrm{a}}{A}(A-2Z) = 4a_\mathrm{a} - \frac{8a_\mathrm{a}}{A}\,Z$$

has been used. Move every term that multiplies Z to the left and every constant to the right:

$$\frac{2a_C}{A^{1/3}}\,Z + \frac{8a_\mathrm{a}}{A}\,Z = \big[m_n - m(^1\mathrm{H})\big]c^2 + \frac{a_C}{A^{1/3}} + 4a_\mathrm{a}, \tag{5.22}$$

and therefore

$$Z_{\min} = \frac{(m_n - m(^1\mathrm{H}))c^2 + a_C A^{-1/3} + 4a_\mathrm{a}}{2a_C A^{-1/3} + 8a_\mathrm{a}/A}. \tag{5.23}$$

This matches (5.15) and is the exact stationary condition for the SEMF mass parabola [6, 1]. The integer Z closest to $Z_{\min}$ is the smooth-SEMF candidate; pairing, shell structure, and evaluated atomic masses decide the actual beta-stable member.

5.3.1 Numerical size of the free-mass term

From atomic mass units,

$$m_n - m(^1\mathrm{H}) = 1.008665 - 1.007825 = 0.000840\,\mathrm{u}, \tag{5.24}$$

so

$$\delta_{nH} \equiv \left[m_n - m(^1\mathrm{H})\right]c^2 \approx 0.000840 \times 931.5 \approx 0.782\,\mathrm{MeV}. \tag{5.25}$$

With the usual coefficients $a_C = 0.72\,\mathrm{MeV}$ and $a_\mathrm{a} = 23\,\mathrm{MeV}$, for $A \sim 100$ one has

$$\frac{a_C}{A^{1/3}} \approx 0.16\,\mathrm{MeV}, \qquad 4a_\mathrm{a} = 92\,\mathrm{MeV}, \qquad \frac{4a_\mathrm{a}}{A} \approx 0.92\,\mathrm{MeV}.$$

Thus δ_{nH} is not negligible next to $a_C/A^{1/3}$, but both are small compared with the asymmetry scale $4a_\mathrm{a}$ that dominates the numerator of (5.23) [6, 14].

5.3.2 Working formula used in the lecture

In the numerator of (5.23), drop δ_{nH} and $a_C A^{-1/3}$ relative to $4a_\mathrm{a}$. The relative error is of order $(0.782 + 0.16)/92 \sim 1\%$ at $A \sim 100$. Then

$$Z_\mathrm{min} \approx \frac{4a_\mathrm{a}}{2a_C A^{-1/3} + 8a_\mathrm{a}/A}. \tag{5.26}$$

Divide numerator and denominator by $8a_\mathrm{a}/A$. The pieces are

$$\frac{4a_\mathrm{a}}{8a_\mathrm{a}/A} = \frac{4a_\mathrm{a} \cdot A}{8a_\mathrm{a}} = \frac{A}{2}, \tag{5.27}$$

$$\frac{2a_C A^{-1/3}}{8a_\mathrm{a}/A} = \frac{2a_C A^{-1/3} \cdot A}{8a_\mathrm{a}} = \frac{a_C}{4a_\mathrm{a}}\,A^{2/3}, \tag{5.28}$$

$$\frac{8a_\mathrm{a}/A}{8a_\mathrm{a}/A} = 1, \tag{5.29}$$

so

$$Z_\mathrm{min} \approx \frac{A/2}{1 + \frac{a_C}{4a_\mathrm{a}}\,A^{2/3}} = \frac{A}{2\left(1 + \frac{a_C}{4a_\mathrm{a}}\,A^{2/3}\right)}. \tag{5.30}$$

With $a_C/a_\mathrm{a} = 0.72/23 \approx 0.0313$,

$$\frac{a_C}{4a_\mathrm{a}} = \frac{0.72}{4 \cdot 23} = \frac{0.72}{92} \approx 0.0078,$$

and therefore

$$Z_\mathrm{min} \approx \frac{A}{2\left(1 + 0.0078\,A^{2/3}\right)}. \tag{5.31}$$

The same approximated expression can be written without the factor of two in the denominator structure:

$$Z_\mathrm{min} \approx \frac{A}{2 + (a_C/(2a_\mathrm{a}))\,A^{2/3}}, \tag{5.32}$$

because

$$2\Big(1 + \tfrac{a_C}{4a_\mathrm{a}}A^{2/3}\Big) = 2 + \tfrac{a_C}{2a_\mathrm{a}}A^{2/3}, \tag{5.33}$$

and with the same coefficients $a_C/(2a_\mathrm{a}) = 0.72/46 \approx 0.0157$. A closely related empirical fit uses 0.0075 instead of 0.0078 in (5.31) [14].

Key idea

Light vs heavy nuclei. For small A, $0.0078\,A^{2/3} \ll 1$, so $Z_\mathrm{min} \approx A/2$ ($N \approx Z$): oxygen, carbon, calcium. For large A the Coulomb correction grows, $Z_\mathrm{min} < A/2$, and long-lived nuclei become neutron rich ($N/Z \sim 1.5$ for lead). Equation (5.31) gives $Z/A \approx 0.41$ for heavy species, matching observation [6, 1]. Without Coulomb, (5.23) reduces to $Z_\mathrm{min} = A/2$ up to the tiny δ_{nH} shift: every extra neutron in a heavy stable nucleus is a Coulomb effect competing with asymmetry.

A	Z_min (approx.)	Nearest stable Z	N/Z
16	7.6	8 (^{16}O)	1.0
40	18.3	18 (^{40}Ar); ^{40}Ca also stable	1.22 at $Z = 18$
56	25.1	26 (^{56}Fe)	1.2
125	52.3	52 (^{125}Te)	1.4
208	81.6	82 (^{208}Pb)	1.5

Values of Z_min come from (5.31). At $A = 40$ the smooth formula places the minimum near argon, not calcium. ^{40}Ca is nevertheless stable because its doubly magic shell closure supplies binding absent from the SEMF. The nearest integer to a smooth minimum is therefore a guide, not an evaluated-mass verdict.

5.4 Odd A: one parabola and β decay toward the minimum

For odd A, $\delta = 0$. There is a *single* mass parabola. Nuclei with Z far from Z_min have large mass differences to their neighbours and decay by $\beta^\pm$ or electron capture; nuclei near the minimum are stable or long-lived.

5.4.1 β energetics on the parabola

Along an odd-A isobar the available energy for a single β step is the atomic mass difference of neighbouring isobars. For β^- ($Z \to Z+1$),

$$Q_{\beta^-} = \big[M(A,Z) - M(A,Z+1)\big]c^2. \tag{5.34}$$

For β^+ ($Z \to Z-1$), the nuclear process creates a positron, so two electron masses must be accounted for when atomic masses are used:

$$Q_{\beta^+} = \big[M(A,Z) - M(A,Z-1)\big]c^2 - 2m_e c^2. \tag{5.35}$$

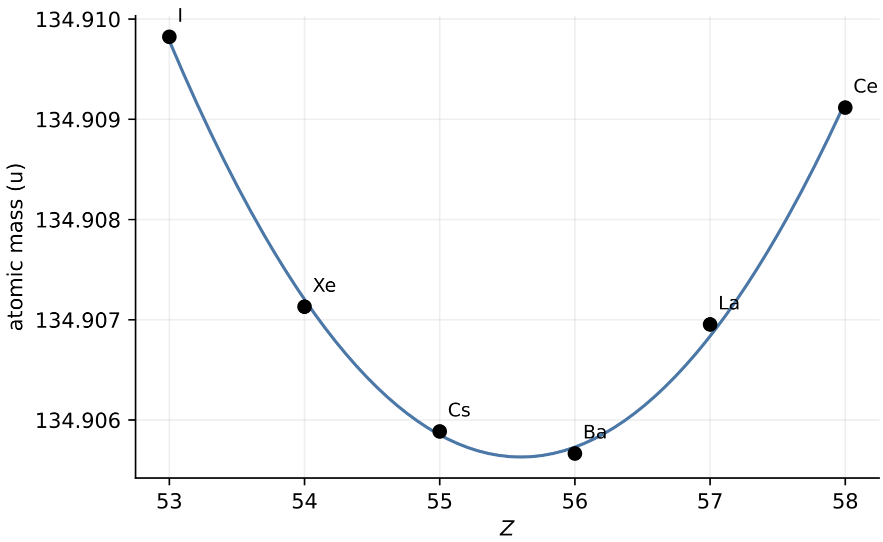


Figure 5.1: Mass parabola for odd $A = 135$ (data-generated from evaluated atomic masses; see tabulated values in [1]). One minimum near $Z = 56$; one (or few) β-stable isobar(s). Decay energy increases as one moves away from the minimum [1, 6].

Electron capture has a slightly lower threshold than β^+ (approximately (5.35) without the $2m_ec^2$ subtraction, apart from atomic-electron binding).

Apply the odd-A convention $Mc^2 = \alpha + \beta Z + \gamma Z^2$. First expand the neighbour:

$$\begin{aligned} M(A, Z+1)\,c^2 &= \alpha + \beta(Z+1) + \gamma(Z+1)^2 \\ &= \alpha + \beta Z + \beta + \gamma(Z^2 + 2Z + 1) \\ &= \alpha + \beta Z + \gamma Z^2 + \beta + 2\gamma Z + \gamma. \end{aligned} \tag{5.36}$$

Subtract from $M(A, Z)\,c^2$:

$$M(A, Z)\,c^2 - M(A, Z+1)\,c^2 = -\beta - 2\gamma Z - \gamma = -\beta - \gamma(2Z+1). \tag{5.37}$$

From (5.14), $Z_{\min} = -\beta/(2\gamma)$, so $\beta = -2\gamma Z_{\min}$ and $-\beta = 2\gamma Z_{\min}$. Therefore

$$Q_{\beta^-} = 2\gamma Z_{\min} - \gamma(2Z+1) = 2\gamma\big(Z_{\min} - Z - \tfrac{1}{2}\big). \tag{5.38}$$

Hence $Q_{\beta^-} > 0$ whenever $Z \le Z_{\min} - 1$ (integer steps below the vertex). Because $\gamma > 0$, the magnitude grows linearly as one climbs the sides of the parabola: β Q values increase away from $Z_{\min}$ [6, 1].

Key idea

Direction of β decay from geometry alone.

$Z < Z_{\min}$	neutron rich	β^- $(n \to p + e^- + \bar{\nu}_e)$,
$Z > Z_{\min}$	proton rich	β^+ or electron capture,
$Z \approx Z_{\min}$	stable / long-lived	no allowed single β.

The SEMF fixes the *direction* and approximate Q; Fermi theory fixes spectra and lifetimes.

5.5 Even A: pairing splits the mass surface into two parabolas

For even A, the pairing term shifts even–even nuclei *down* and odd–odd nuclei *up* by $|\delta| = a_P A^{-3/4}$ in the binding-energy convention of (5.1). Because $M\,c^2$ contains $-\delta$, the mass of an even–even isobar is lowered by $|\delta|$ and that of an odd–odd isobar is raised by $|\delta|$. The vertical separation between the two parabolas is therefore $2|\delta|$. The mass surface becomes two parallel parabolas sharing the same $Z_{\min}$ but offset in energy [6, 1, 14]:

$$M_{\rm ee}\,c^2 = \alpha + \beta Z + \gamma Z^2 - |\delta|, \tag{5.39}$$

$$M_{\rm oo}\,c^2 = \alpha + \beta Z + \gamma Z^2 + |\delta|. \tag{5.40}$$

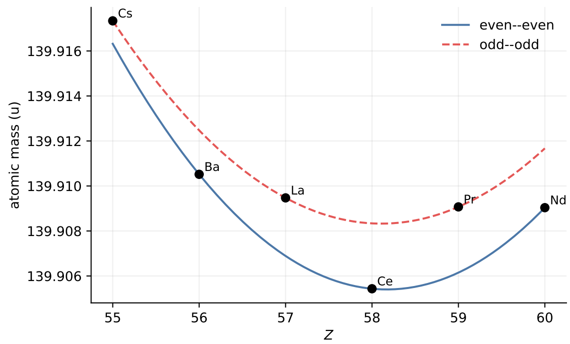


Figure 5.2: Even $A = 140$: pairing splits even–even (lower) and odd–odd (upper) chains. One β-stable even–even isobar near $Z = 58$.

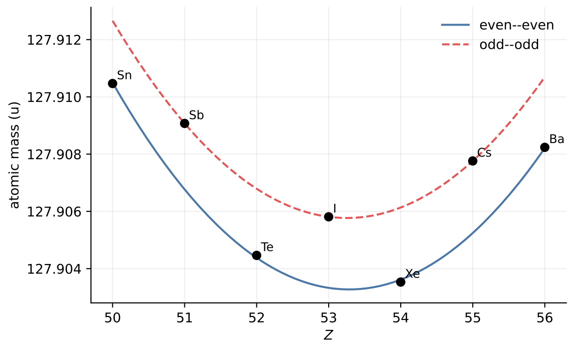


Figure 5.3: Even $A = 128$: two even–even isobars ($Z = 52, 54$) lie on the lower parabola; the odd–odd member can decay both ways. Some even–even chains are stable against *single* β but decay very slowly by double β (e.g. tellurium isotopes).

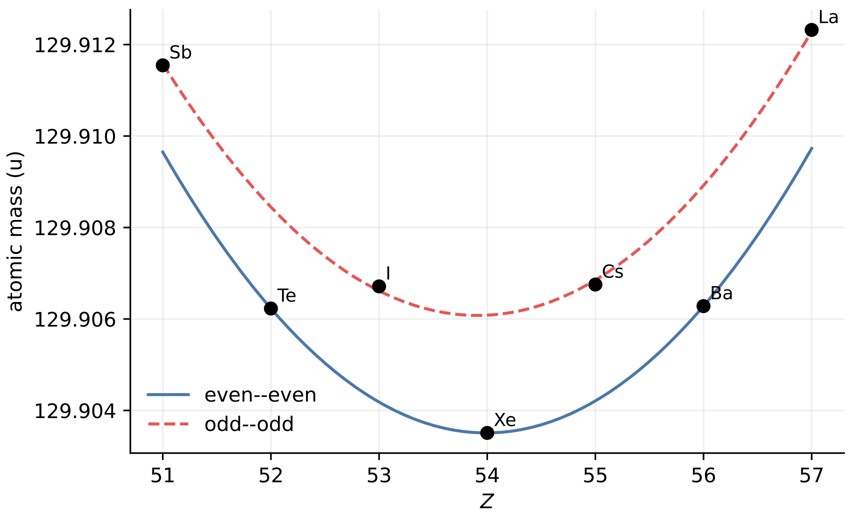


Figure 5.4: Even $A = 130$: three even–even nuclides ($Z = 52, 54, 56$) on the lower parabola can all be stable against single β decay relative to their odd–odd neighbours. Long-term evolution may still proceed by double β where $Q_{2\beta} > 0$.

5.5.1 Two effects absent from odd-A chains

1. **Bidirectional β decay of odd–odd nuclei.** An odd–odd isobar near the minimum of the *upper* parabola can have both neighbours on the lower parabola at lower mass. Then both β^- and β^+/EC are open. Classic examples: ^{128}I and ^{64}Cu [6, 1, 3].

2. **Double β decay of even–even nuclei.** If single β from an even–even nucleus would land on a higher odd–odd isobar, that single step is energetically forbidden, yet a transition to a lower even–even neighbour two units of Z away can still be allowed. With atomic masses,

$$Q_{2\beta^-} = [M(A, Z) - M(A, Z + 2)]c^2. \tag{5.41}$$

On the even–even parabola alone (pairing cancelled between parent and daughter), the same algebra that produced (5.38) with a unit step gives, for a double step,

$$Q_{2\beta^-} = -\beta - \gamma[(Z + 2)^2 - Z^2] = -\beta - \gamma(4Z + 4) = 2\gamma(Z_{\min} - Z - 1).$$

The nucleus may then decay only by **double beta decay** ($2\nu\beta\beta$, or the still-unseen $0\nu\beta\beta$). Candidates include ^{76}Ge, ^{128}Te, ^{130}Te, and ^{136}Xe, where single β is blocked by pairing but $Q_{2\beta} > 0$ [6, 1, 4]. Observationally, such nuclei can be extremely long lived: ^{128}Te has a $2\nu\beta\beta$ half-life of order 10^{24} yr.

Key idea

Pairing does not invent new forces; it rearranges which isobars are allowed to β decay. Odd A: one parabola, one valley. Even A: two parabolas, possible double-β and two-way decays.

5.5.2 Why light odd–odd nuclei can be stable

For small A the parabola is narrow ($\gamma(A)$ is larger). The pairing offset can then place an odd–odd nucleus low enough to be stable against single β. The four stable odd–odd species ^{2}H, ^{6}Li, ^{10}B, ^{14}N live in that regime; heavier odd–odd nuclei almost always find an open β path to an even–even daughter [1, 6]. Equivalently, at low A the Coulomb and asymmetry curvatures are large compared with $2|\delta|$, so the upper parabola can still dip below neighbouring even–even masses near $N = Z$. That geometric accident disappears once Coulomb stretches the valley and pairing no longer compensates.

5.6 β Q values from the SEMF (worked structure)

Start again from the atomic ledger (5.3). The β^- mass difference is

$$\begin{aligned}[M(A, Z) - M(A, Z + 1)]c^2 = [Z\, m(^1\mathrm{H}) + (A - Z)\, m_n]c^2 - [(Z + 1)\, m(^1\mathrm{H}) + (A - Z - 1)\, m_n]c^2 \\ - B(A, Z) + B(A, Z + 1).\end{aligned} \tag{5.42}$$

The free-nucleon difference collapses to $[m_n - m(^1\mathrm{H})]c^2 = \delta_{nH}$, so

$$Q_{\beta^-}(A, Z) = \delta_{nH} + B(A, Z + 1) - B(A, Z). \tag{5.43}$$

Insert the SEMF (5.1). Volume and surface cancel at fixed A. The remaining pieces are Coulomb, asymmetry, and pairing.

Write $\Delta X \equiv X(A,Z) - X(A,Z+1)$ for a term that enters M with a plus sign in (5.4). The Coulomb contribution is

$$\begin{aligned}\Delta_{\mathrm{C}} &= a_C A^{-1/3}\big[Z(Z-1) - (Z+1)Z\big] \\ &= a_C A^{-1/3}\big[Z^2 - Z - Z^2 - Z\big] = a_C A^{-1/3}(-2Z) = -\,2a_C A^{-1/3}\,Z. \end{aligned} \tag{5.44}$$

For asymmetry, set $u = A - 2Z$. Then the neighbour has $A - 2(Z+1) = u - 2$, and

$$\begin{aligned}(A-2Z)^2 - (A-2Z-2)^2 = u^2 - (u-2)^2 = u^2 - \big(u^2 - 4u + 4\big) = 4u - 4 \\ = 4(A-2Z) - 4 = 4(A-2Z-1). \end{aligned} \tag{5.45}$$

Therefore

$$\Delta_{\mathrm{a}} = \frac{a_{\mathrm{a}}}{A}\,4(A-2Z-1) = \frac{4a_{\mathrm{a}}}{A}(A-2Z-1). \tag{5.46}$$

For odd A (pairing zero),

$$Q_{\beta^-} = \delta_{nH} + \Delta_{\mathrm{C}} + \Delta_{\mathrm{a}}. \tag{5.47}$$

For even A, add the pairing difference $-\delta(A,Z) + \delta(A,Z+1)$. Moving toward Z_{min} lowers mass: Coulomb and asymmetry enter with opposite roles on the two sides of the valley.

Worked estimate: Q_{β^-} for $^{135}\mathrm{Cs}$

For odd $A = 135$, pairing vanishes and the SEMF gives a single parabola. With $a_C = 0.72$, $a_{\mathrm{a}} = 23$ (MeV), $Z_{\mathrm{min}} \approx 56$ so the valley sits near barium ($Z = 56$). Consider neutron-rich $^{135}\mathrm{Cs}$ ($Z = 55$):

$$Q_{\beta^-} = \big[M(135,55) - M(135,56)\big]c^2.$$

Volume and surface cancel at fixed A. From (5.44)–(5.46) with $Z = 55$,

$$A^{1/3} = 135^{1/3} \approx 5.13, \qquad \Delta_{\mathrm{C}} = -2\cdot 0.72\cdot\frac{55}{5.13} = -0.72\cdot\frac{110}{5.13} \approx -15.4\,\mathrm{MeV},$$

$$A - 2Z = 135 - 110 = 25, \qquad \Delta_{\mathrm{a}} = \frac{4\cdot 23}{135}(25-1) = 23\cdot\frac{96}{135} \approx 16.36\,\mathrm{MeV}.$$

Altogether

$$Q_{\beta^-} \approx 0.782 - 15.44 + 16.36 \approx 1.70\,\mathrm{MeV}.$$

The evaluated atomic-mass difference gives only $Q_{\beta^-} \approx 0.269\,\mathrm{MeV}$ [27]. This is not a contradiction: $^{135}\mathrm{Cs}$ lies very close to the parabola minimum, where a sub-MeV shell residual is comparable to the entire finite difference. The SEMF captures the direction and rough scale; evaluated masses, not the smooth formula, determine whether a near-valley decay is open and its precision Q value. The positive sign confirms β^- decay of neutron-rich $^{135}\mathrm{Cs}$ toward $^{135}\mathrm{Ba}$.

Use evaluated masses for decay decisions

The continuous SEMF vertex answers "where is the valley approximately?" It does not certify an integer nuclide as stable. For neighbouring isobars compute

$$Q_{\beta^-} = [M_{\rm at}(A, Z) - M_{\rm at}(A, Z + 1)]c^2$$

(and the corresponding β^+ or electron-capture expression) from an evaluated mass table such as AME2020. Pairing and shell corrections that are small in a total mass can dominate this near-cancellation [27].

Physics checklist for an isobar

1. Compute $Z_{\rm min}(A)$ from (5.31) (or (5.23)).
2. For odd A: one parabola; the integer Z nearest $Z_{\rm min}$ is stable; others β decay toward it with Q increasing up the sides via (5.38).
3. For even A: draw two parabolas split by $\sim 2|\delta|$; identify even–even minima and any odd–odd bidirectional cases; mark possible double-β pairs with (5.41).
4. Q_β follows from mass differences on that surface, with no weak matrix element required for the *sign* of the decay.

5.7 Connecting the valley of stability

Plotting the integer nearest to $Z_{\rm min}(A)$ for each A traces the valley of stability on the (N, Z) plane: $N \approx Z$ at low A, bending to $N > Z$ at high A. Nuclei below the valley (neutron excess) decay by β^-; nuclei above it decay by β^+ or electron capture. Lectures 4–5 therefore close the liquid-drop programme: binding energy and its Z landscape, including the β arrows that keep real nuclei near the valley. Lecture 6 follows the same $Z_{\rm min}(A)$ under electron degeneracy in neutron-star crusts; Lectures 18–19 turn the same Q values into spectral endpoints and ft values [6, 1, 13].

5.7.1 Even–even preference and double β candidates

Counts of β-stable nuclides favour even–even species strongly; odd–odd nuclei are almost never stable except for the four light cases already discussed. On even-A mass parabolas the pairing offset allows double-β candidates when a single β step climbs the upper parabola while a $\Delta Z = 2$ jump still lowers the mass [1, 6, 5, 13].

5.8 Problems with solutions

Problem 5.1 — Evaluate $Z_{\rm min}$. Using $Z_{\rm min} = A/[2(1 + 0.0078\, A^{2/3})]$, compute $Z_{\rm min}$ for $A = 100$ and $A = 200$. For each, give N/Z at that charge.

Solution 5.1. For $A = 100$, $A^{2/3} \approx 21.54$,

$$Z_{\min} = \frac{50}{1 + 0.0078 \times 21.54} = \frac{50}{1.168} \approx 42.8 \quad \Rightarrow \quad \frac{N}{Z} \approx \frac{57.2}{42.8} \approx 1.34.$$

For $A = 200$, $A^{2/3} \approx 34.2$,

$$Z_{\min} = \frac{100}{1 + 0.0078 \times 34.2} = \frac{100}{1.267} \approx 78.9 \quad \Rightarrow \quad \frac{N}{Z} \approx \frac{121}{79} \approx 1.53.$$

Heavy nuclei are neutron rich [6].

Problem 5.2 — Free-mass difference. Show that $(m_n - m(^1\mathrm{H}))c^2 \approx 0.782\,\mathrm{MeV}$ using $m_n = 1.008665\,\mathrm{u}$, $m(^1\mathrm{H}) = 1.007825\,\mathrm{u}$, and $c^2 = 931.5\,\mathrm{MeV/u}$. Where does this number enter $Z_{\min}$?

Solution 5.2.

$$(1.008665 - 1.007825) \times 931.5 = 0.000840 \times 931.5 \approx 0.782\,\mathrm{MeV}.$$

It appears in the numerator of the exact $Z_{\min}$ formula (5.23) as the free-mass drive toward protons; it is small compared with $4a_{\mathrm{a}} \approx 92\,\mathrm{MeV}$ [6, 14].

Problem 5.3 — Q_{β^-} sign on the parabola. At fixed odd A, $M(Z)$ is a parabola with minimum at $Z_{\min}$. Show that $Q_{\beta^-} = [M(Z) - M(Z+1)]c^2 > 0$ for all integers $Z < Z_{\min} - 1/2$, so neutron-rich isobars β^- decay toward the minimum.

Solution 5.3. Use the chapter convention $Mc^2 = \alpha + \beta Z + \gamma Z^2$ ($\delta = 0$) with $\gamma > 0$. Then

$$[M(Z) - M(Z+1)]c^2 = -\beta - \gamma(2Z + 1).$$

The minimum is at $Z_{\min} = -\beta/(2\gamma)$ (5.14), so $-\beta = 2\gamma Z_{\min}$ and

$$Q_{\beta^-} = 2\gamma(Z_{\min} - Z - \tfrac{1}{2}),$$

as in (5.38). For integer $Z \leq Z_{\min} - 1$ this is positive: $Q_{\beta^-} > 0$. Geometry alone forces β^- on the neutron-rich side [6, 1].

Problem 5.4 — Pairing split and double β. Even–even and odd–odd mass parabolas at fixed even A are split by $2|\delta|$ with $|\delta| = a_P A^{-3/4}$. For $A = 128$ and $a_P = 34\,\mathrm{MeV}$, estimate the vertical split $2|\delta|$. Explain in one sentence why an even–even nucleus can be stable against single β yet unstable against double β.

Solution 5.4.

$$A^{3/4} = 128^{0.75} \approx (2^7)^{0.75} = 2^{5.25} \approx 38, \quad |\delta| \approx 34/38 \approx 0.9\,\mathrm{MeV}, \quad 2|\delta| \approx 1.8\,\mathrm{MeV}.$$

Single β would land on the higher odd–odd parabola, so $Q_\beta < 0$ for that step, while a jump of $\Delta Z = 2$ to a lower even–even neighbour can still have $Q_{2\beta} > 0$ [6, 1].

Problem 5.5 — Both β modes. For odd–odd ^{64}Cu near mid-mass, explain qualitatively why both β^- (to ^{64}Zn) and β^+ (to ^{64}Ni) can be open, using the two-parabola picture.

Solution 5.5. ^{64}Cu sits on the upper (odd–odd) parabola. Both even–even neighbours ^{64}Ni and ^{64}Zn lie on the lower parabola and can be lower in mass than ^{64}Cu. Then both Q_{β^-} and Q_{β^+} are positive: bidirectional decay. Quantitative SEMF estimates give $Q \sim 1\,\mathrm{MeV}$ for both branches [3, 6].

Problem 5.6 — Odd–odd light stability. Name the four stable odd–odd nuclei and explain, using the mass-parabola picture, why they can be stable while heavy odd–odd nuclei generally are not.

Solution 5.6. ^{2}H, ^{6}Li, ^{10}B, ^{14}N. At small A the parabola is narrow (large γ) and pairing can leave an odd–odd nucleus below neighbouring single-β daughters; at large A Coulomb stretches the valley and an open β path almost always exists [6, 1].

Chapter 6

SEMF and Neutron Stars

"The same short-range force that saturates $B/A \sim 8\,\mathrm{MeV}$ cannot bind pure neutrons; only gravity at stellar mass can."

after Weizsäcker; Bethe (SEMF)

6.1 The puzzle: pure neutrons fail on Earth

The SEMF was built for ordinary nuclei near the valley of stability. If one asks what happens for pure neutron matter, $Z = 0$, the volume and asymmetry terms no longer cancel in a favourable way: the nuclear terms alone leave pure neutrons unbound. Laboratory nuclei never form a stable drop of neutrons.

Nature nevertheless produces compact stellar remnants that are mostly neutrons in the bulk, with masses of order a solar mass and radii of order ten kilometres. Gravity, negligible on the fermi scale of a single nucleus, becomes the leading long-range attraction at $A \sim 10^{56}$–10^{57}. This chapter extrapolates the liquid-drop idea to that extreme and recovers the solar-mass, ten-kilometre scale of neutron stars as an order-of-magnitude estimate [1, 4, 6].

6.1.1 Nuclear landscape and the neutron drip line

Lecture 5 fixed the most stable Z at fixed A. The chart of nuclides (Figure 6.1; cf. Lectures 4–5) also shows, for each Z, a maximum N that still binds. Beyond that **neutron drip line**,

$$S_n(A, Z) = [M(A-1, Z) + m_n - M(A, Z)]c^2 \leq 0, \tag{6.1}$$

and the last neutron is unbound (Lecture 4 separation-energy language). Even the dineutron is unbound; few-neutron systems are unstable. Protons, and the Coulomb–asymmetry balance of Lectures 4–5, are essential for ordinary nuclei.

From the SEMF alone, pure neutron matter ($Z = 0$, $N = A$) has

$$\frac{B}{A} \approx a_V - a_\mathrm{a} \approx 15.5 - 23 \approx -7.5\,\mathrm{MeV} < 0 \tag{6.2}$$

(neglecting surface for large A). Asymmetry wins: liquid-drop nuclear physics alone *forbids* a stable pure-neutron drop of any laboratory size.

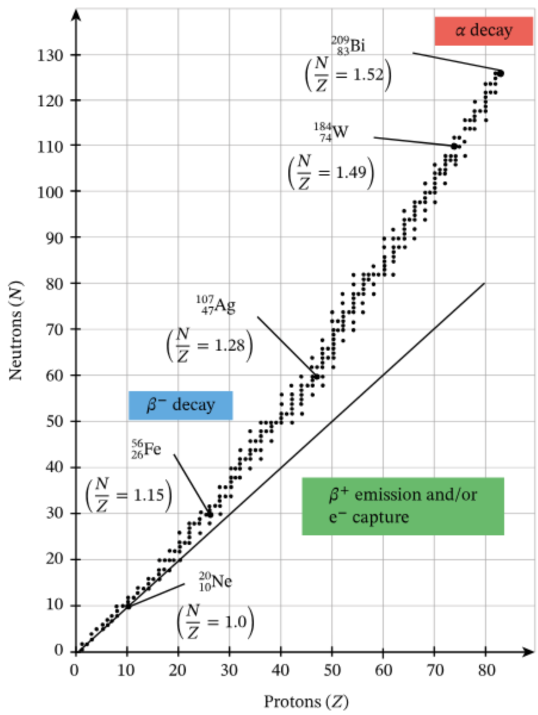


Figure 6.1: Nuclear landscape (valley of stability). For each Z only a limited band of N is bound. Beyond the neutron drip line, $S_n \leq 0$ and neutrons unbind. Laboratory nuclei never reach pure-neutron bulk matter.

6.1.2 Neutron stars exist anyway

Nature produces compact remnants that are mostly neutrons in the bulk (with a thin outer crust of nuclei and a small proton fraction in the core). Observed scales:

Quantity	Typical value
Mass	$\sim (1\text{–}2)\, M_\odot$
Radius	$\sim 10\text{–}15\,\text{km}$
Mean density	$\sim (1\text{–}3)\, \rho_0$ (nuclear saturation)
Nucleon number	$A \sim 10^{57}$ for $M \sim M_\odot$

That is $\sim 10^{55}$ times more nucleons than uranium, in a city-sized volume. The SEMF question: *how can such an object be bound when few neutrons are not?*

6.2 Birth of a neutron star

A compact sketch is enough for this nuclear course [4]:

1. Gas clouds collapse under gravity; fusion begins when the core is hot and dense (H $\to$ He in the Sun).
2. In stars several times heavier than the Sun, fusion proceeds through He, C, O, ... up to the iron-group peak of B/A (Lecture 4). Fusion past iron does not release net energy.

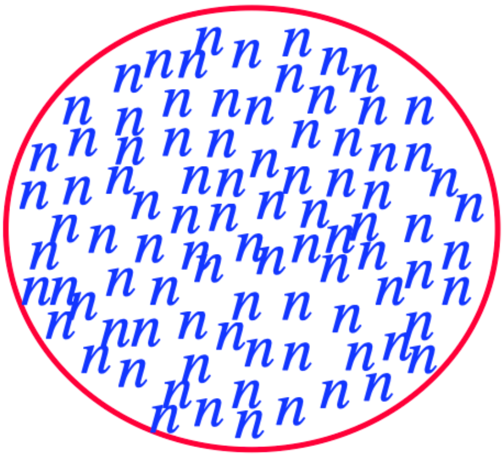


Figure 6.2: Schematic pure-neutron bulk (lecture cartoon). The star is not a giant laboratory nucleus: gravity and quantum degeneracy dominate the global energy and support balance.

3. When nuclear burning can no longer support the core, gravity wins: rapid collapse, bounce, and a core-collapse supernova that ejects the envelope.

4. In the residual core, electron Fermi energies become so high that inverse β decay (electron capture)

$$p + e^- \to n + \nu_e$$

converts most protons into neutrons. Neutrinos escape; the remnant cools into a neutron star.

Why inverse β wins at high density. In free space, $m_nc^2 - m_pc^2 \approx 1.3\,\text{MeV}$ and free neutrons β^- decay. Inside a dense Fermi sea of electrons, capturing an electron from a high Fermi level can lower the total energy of the system even though the free n is heavier. The chemical potentials rearrange until β equilibrium

$$\mu_n = \mu_p + \mu_e \tag{6.3}$$

is reached, typically with a proton fraction of only a few percent in the core: *mostly* neutrons, not purely $Z = 0$.

6.3 Scales: density, number of neutrons, and forces

6.3.1 Number of nucleons

One solar mass is $M_\odot \approx 2 \times 10^{30}\,\text{kg}$. With $m_n \approx 1.67 \times 10^{-27}\,\text{kg}$,

$$A_\odot \approx \frac{M_\odot}{m_n} \approx \frac{2 \times 10^{30}}{1.67 \times 10^{-27}} \approx 1.2 \times 10^{57}. \tag{6.4}$$

A $1.4\,M_\odot$ neutron star therefore contains of order 10^{57} nucleons.

6.3.2 Mean density and the nuclear connection

Nuclear saturation density from Lectures 1–2 is

$$n_0 \approx 0.16\,\mathrm{fm}^{-3} = 1.60 \times 10^{44}\,\mathrm{m}^{-3}. \tag{6.5}$$

This is a *number* density. Multiplication by the nucleon mass gives

$$\rho_0 = m_n n_0 \approx (1.6749 \times 10^{-27}\,\mathrm{kg})(1.60 \times 10^{44}\,\mathrm{m}^{-3}) \approx 2.68 \times 10^{17}\,\mathrm{kg\,m}^{-3}. \tag{6.6}$$

If a star of mass $M = 1.4\,M_\odot$ has radius $R = 12\,\mathrm{km}$,

$$\bar{\rho} = \frac{3M}{4\pi R^3} \approx 3.8 \times 10^{17}\,\mathrm{kg\,m}^{-3} \approx 1.4\,\rho_0. \tag{6.7}$$

Neutron stars are *nuclear-density objects* writ large: the same saturation that fixed $R = r_0 A^{1/3}$ for Pb reappears as a kilometre-scale radius for $A \sim 10^{57}$.

One uniform-density radius, written two ways

For a uniform sphere, $A = n_0(4\pi R^3/3)$, so

$$R = \left(\frac{3A}{4\pi n_0}\right)^{1/3} = r_0 A^{1/3}, \qquad r_0 \equiv \left(\frac{3}{4\pi n_0}\right)^{1/3} \approx 1.14\,\mathrm{fm}.$$

For $A = 10^{57}$, both forms give $R \approx 11.4\,\mathrm{km}$. Using the rounded nuclear-size coefficient $r_0 = 1.2\,\mathrm{fm}$ instead gives $12\,\mathrm{km}$. These are the same uniform-density estimate, not independent radius prescriptions.

6.3.3 Why gravity was dropped in Lectures 4–5

For two nucleons a fermi apart,

$$\frac{Gm_n^2}{r} \sim 10^{-36}\,\mathrm{MeV} \qquad (r \sim 1\,\mathrm{fm}),$$

negligible next to a_V, a_C, or a_a. For $A \sim 10^{57}$ nucleons in a sphere of radius $\sim 10\,\mathrm{km}$, every nucleon feels the gravitational field of the whole star. Total gravitational self-energy scales as $GM^2/R \sim A^{5/3}$ at fixed density and can outweigh a fixed MeV-per-nucleon nuclear deficit.

Key idea

Laboratory nuclei: drop G. Neutron-star scale: keep G. The SEMF structure is reused as energy bookkeeping; gravity is the new long-range term.

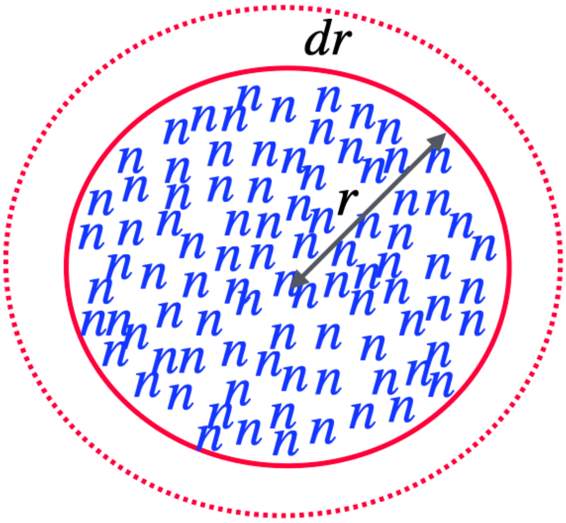


Figure 6.3: Assembly of a uniform sphere by thin shells. A shell of thickness dr at radius r feels the potential of the mass already inside r. Integrating gives $U_G = -3GM^2/(5R)$.

6.4 Gravitational self-energy of a uniform sphere

6.4.1 Derivation (same logic as Coulomb 3/5)

Uniform density $\rho = M/(\frac{4}{3}\pi R^3)$. Enclosed mass at radius r:

$$m(r) = M\,\frac{r^3}{R^3}. \tag{6.8}$$

Shell mass:

$$\mathrm{d}m = 4\pi r^2\,\mathrm{d}r\,\rho = \frac{3M}{R^3}\,r^2\,\mathrm{d}r. \tag{6.9}$$

Potential of the interior on the shell:

$$\phi(r) = -\frac{G\,m(r)}{r} = -\frac{GM\,r^2}{R^3}. \tag{6.10}$$

Work to bring $\mathrm{d}m$ from infinity:

$$\mathrm{d}U = \phi\,\mathrm{d}m = -\frac{3GM^2}{R^6}\,r^4\,\mathrm{d}r. \tag{6.11}$$

Integrate $0 \to R$:

$$U_G = -\frac{3GM^2}{5R}. \tag{6.12}$$

Compare with the Coulomb self-energy of a uniform charged sphere (Lecture 4):

$$E_\mathrm{C} = \frac{3}{5}\frac{k_e(Ze)^2}{R}, \qquad U_G = -\frac{3}{5}\frac{GM^2}{R}.$$

Same geometric factor 3/5; opposite sign because gravity is attractive.

Binding language. $U_G < 0$ lowers the total energy relative to dispersed matter at infinity. In nuclear $B > 0$ language (Lecture 4), gravity contributes $+|U_G| = 3GM^2/(5R)$ to the binding budget.

Physics checklist for Figure 6.3

1. Shell-by-shell assembly: identical strategy to electrostatics.
2. Scaling: $U_G \propto M^2/R \propto A^{5/3}$ at fixed density.
3. Real stars are stratified; the GM^2/R scaling remains the right order of magnitude.

6.5 A labelled toy model: SEMF plus gravity

6.5.1 Modified binding energy

This section is a **toy EOS/energy model**, not a microscopic equation of state for neutron-star matter. Start from the SEMF of Lectures 4–5 and add the magnitude of the Newtonian gravitational self-energy as an extra contribution to the *binding* energy (positive when the system is more tightly held together than free nucleons):

$$B = a_V A - a_S A^{2/3} - a_C \frac{Z(Z-1)}{A^{1/3}} - a_{\mathrm{a}} \frac{(N-Z)^2}{A} + \delta + \frac{3GM^2}{5R}. \tag{6.13}$$

In a total-energy ledger the gravitational self-energy is negative ($U_G = -3GM^2/5R$); writing $+|U_G|$ in B is the same physics packaged as binding. The pure-neutron SEMF exercise below uses this convention only as an order-of-magnitude cartoon.

6.5.2 Idealised pure-neutron limit

Take $Z = 0$, $N = A$, $M = m_n A$, and impose the same uniform density throughout:

$$R = r_0 A^{1/3}, \qquad r_0 = \left(\frac{3}{4\pi n_0}\right)^{1/3} \approx 1.14\,\mathrm{fm}.$$

- Coulomb vanishes;
- asymmetry: $-a_{\mathrm{a}}(N-Z)^2/A = -a_{\mathrm{a}}A$;
- surface $\propto A^{2/3}$ negligible vs volume for huge A (surface/volume $\to 0$; same reason bulk beats surface in macroscopic matter);
- pairing $\propto A^{-3/4}$ negligible at $A \sim 10^{56}$.

Then

$$\frac{3GM^2}{5R} = \frac{3Gm_n^2}{5r_0}\,A^{5/3}, \tag{6.14}$$

and

$$B \approx (a_V - a_{\mathrm{a}})\,A + \frac{3Gm_n^2}{5r_0}\,A^{5/3}. \tag{6.15}$$

Nuclear piece alone is negative.

$$a_V - a_{\mathrm{a}} \approx -7.5\,\mathrm{MeV}.$$

That is the SEMF version of "you cannot build a stable nucleus from neutrons alone."

Gravity can reverse the sign. The second term grows as $A^{5/3}$. For large enough A it overcomes the 7.5 MeV per-nucleon deficit.

Caution

What this calculation is and is not. It is a pedagogical SEMF extrapolation: an existence argument for a bound pure-neutron system at stellar A. It is *not* modern neutron-star structure. Real support against collapse is neutron **degeneracy pressure** (and nuclear repulsion at short distance) balanced by gravity in general-relativistic hydrostatic equilibrium (Tolman–Oppenheimer–Volkoff equation). Gravity is attractive; degeneracy provides the outward pressure. Here gravity enters only as extra binding in an energy ledger, good for order-of-magnitude A and M, not for precision $M(R)$ curves or the maximum mass.

6.6 Boundedness: critical A, mass, and numerical estimate

Require $B \geq 0$. Divide (6.15) by A:

$$(a_V - a_\mathrm{a}) + \frac{3Gm_n^2}{5r_0} A^{2/3} \geq 0. \tag{6.16}$$

With $a_V - a_\mathrm{a} \approx -7.5\,\mathrm{MeV}$,

$$A^{2/3} \geq \frac{7.5\,\mathrm{MeV}}{C}, \qquad C \equiv \frac{3Gm_n^2}{5r_0}. \tag{6.17}$$

Order of magnitude for C. In SI units,

$$\begin{aligned} G &= 6.67 \times 10^{-11}\,\mathrm{m^3\,kg^{-1}\,s^{-2}}, \\ m_n &= 1.67 \times 10^{-27}\,\mathrm{kg}, \\ r_0 &= 1.14\,\mathrm{fm} = 1.14 \times 10^{-15}\,\mathrm{m}. \end{aligned}$$

Then

$$\begin{aligned} \frac{Gm_n^2}{r_0} &= \frac{6.67 \times 10^{-11}\,(1.67 \times 10^{-27})^2}{1.14 \times 10^{-15}} \approx 1.63 \times 10^{-49}\,\mathrm{J} \\ &\approx 1.02 \times 10^{-36}\,\mathrm{MeV}, \end{aligned}$$

so

$$C = \frac{3}{5}\frac{Gm_n^2}{r_0} \sim 6 \times 10^{-37}\,\mathrm{MeV}.$$

Requiring $CA^{2/3} \gtrsim 7.5\,\mathrm{MeV}$ gives

$$A^{2/3} \gtrsim \frac{7.5}{6 \times 10^{-37}} \sim 1.25 \times 10^{37},$$

and raising both sides to the 3/2 power,

$$A \gtrsim \left(1.25 \times 10^{37}\right)^{3/2} = \left(1.25 \times 10^{37}\right)\sqrt{1.25 \times 10^{37}} \sim 10^{56}.$$

Thus

$$A \gtrsim 10^{56} \tag{6.18}$$

(lecture estimate; the prefactor depends on the exact gap $a_{\mathrm{a}} - a_V$ and on r_0).

Mass.

$$M \approx m_n A \sim 10^{56} \times 1.67 \times 10^{-27}\,\mathrm{kg} \sim 1.7 \times 10^{29}\,\mathrm{kg} \sim 0.1\,M_\odot$$

for $A \sim 10^{56}$. Using $A \sim 10^{57}$ (closer to observed $1\,M_\odot$) matches solar-mass neutron stars. The SEMF+gravity argument recovers the *solar-mass order of magnitude*: that is its success [4].

Why $A = 2$, 100, or even 10^{10} still fail

For any laboratory or even planetary A, the gravity term per nucleon is far below 1 eV. Then

$$\frac{B}{A} \approx a_V - a_{\mathrm{a}} < 0:$$

unbound. Only stellar A makes $CA^{2/3}$ competitive with 7.5 MeV.

Scale	A	Nuclear B/A	Gravity vs nuclear
Deuteron	2	$\sim +1.1\,\mathrm{MeV}$ (needs $p + n$)	irrelevant
Lead	~ 200	$\sim +8\,\mathrm{MeV}$	irrelevant
"Neutron drop" (lab)	any small	$\sim -7.5\,\mathrm{MeV}$ if $Z = 0$	still unbound
Neutron star	$\sim 10^{57}$	nuclear + $\lvert U_G \rvert$	gravity decisive

6.7 Physics beyond the SEMF cartoon

The liquid-drop-plus-gravity estimate answers only whether a compact neutron configuration can be bound relative to dispersed nucleons. It does not specify how pressure balances gravity, how the crust stratifies, or what appears above a few times nuclear saturation density. Those questions need the equation of state and general-relativistic hydrostatics (TOV). The remarks below keep the same Fermi-gas and asymmetry language as Lectures 4–5 while marking where the cartoon ends [6, 1, 4].

6.7.1 What actually supports a neutron star

In hydrostatic equilibrium, gravity pulls inward and a pressure gradient holds the star up. For white dwarfs the pressure is electron degeneracy; for neutron stars it is primarily **neutron degeneracy pressure** (Fermi gas of neutrons at nuclear density), stiffened at high density by repulsive nuclear forces. The SEMF calculation never writes that pressure: it only asks whether the total energy is lower than that of dispersed nucleons. Both viewpoints are useful; they answer different questions (global binding vs local force balance).

6.7.2 Internal structure (qualitative)

- **Atmosphere / outer crust:** ordinary (exotic) nuclei in a lattice with electrons; density rises from $\mathrm{g\,cm^{-3}}$ toward nuclear.
- **Inner crust:** neutron-rich nuclei coexist with a neutron fluid (past the drip density).
- **Outer core:** uniform *npe* matter (neutrons, protons, electrons) in β equilibrium; mostly neutrons.
- **Inner core:** uncertain (hyperons? deconfined quarks?): the regime where laboratory SEMF coefficients are least trustworthy.

6.7.3 Link back to asymmetry and the Fermi gas

The same Pauli filling that produced the asymmetry term $-a_\mathrm{a}(N-Z)^2/A$ (Lecture 4) is the microscopic origin of Fermi kinetic energy. In pure neutron matter the kinetic energy per nucleon is large; the SEMF packages that cost into a_a. Gravity does not cancel the Pauli principle: it adds a different energy scale that grows with system size.

Key idea

SEMF alone: pure neutrons unbound ($a_V < a_\mathrm{a}$). SEMF + gravity: a bound system appears only for $A \gtrsim 10^{56}$, with $M \sim M_\odot$ and $R \sim 10\,\mathrm{km}$ as order-of-magnitude scales. Few neutrons cannot form a star; a stellar number of neutrons can. Realistic stars are not pure neutron and are not held by this energy ledger: they are charge-neutral, β-equilibrated matter supported by an equation of state in hydrostatic (TOV) equilibrium.

6.8 Lessons and course connection

1. Asymmetry energy is real: unbound pure-neutron drops and the need for $N \sim Z$ in light nuclei share one SEMF term.
2. Coefficients a_V, a_a are terrestrial fits; using them at $A = 10^{57}$ is a bold but instructive extrapolation.
3. Gravity changes the ranking of terms: surface, Coulomb, and pairing drop out; volume, asymmetry, and GM^2/R remain.
4. Saturation density links fm and km: $R = r_0 A^{1/3}$ from electron scattering predicts neutron-star radii at the order-of-magnitude level.

Takeaway

Course arc (Lectures 1–6).

- **1–3:** Nuclear size and density (fm, saturation).
- **4–5:** Binding energy, SEMF, most stable Z, mass parabolas.
- **6:** Extreme SEMF: pure neutrons + gravity $\Rightarrow$ neutron-star mass/radius order of

magnitude, with pointers to β equilibrium and TOV structure.

Next (Lecture 7). Leave the macroscopic star and return to the microscopic force that underlies a_V: the two-nucleon problem and the deuteron, the only bound NN system.

6.8.1 From white dwarfs to neutron-star remnants

White dwarfs are supported by electron degeneracy; more massive remnants cross into regimes where electrons are captured and the core becomes a giant neutron-rich Fermi sea—a neutron star of solar mass and few-kilometre radius. Type-II supernova models connect that remnant to an intense burst of neutrinos (as in SN 1987A) and to r-process neutron captures on seed nuclei near iron [14, 5, 4].

6.9 Deeper physics: from SEMF cartoon to realistic neutron stars

The pure-neutron SEMF+gravity estimate recovers solar-mass and ten-kilometre scales, but it is not a structure calculation. Three further layers make the cartoon honest without leaving the spirit of this course.

6.9.1 Chemical equilibrium and the proton fraction

In a neutron-star core, weak interactions maintain

$$n \leftrightarrow p + e^- + \bar{\nu}_e \tag{6.19}$$

in equilibrium with charge neutrality $n_p = n_e$ (and muons once $\mu_e > m_\mu c^2$). Chemical equilibrium requires

$$\mu_n = \mu_p + \mu_e. \tag{6.20}$$

Near saturation one may parameterise the energy per nucleon of asymmetric matter as

$$\frac{E}{A} = \varepsilon_0(n_b) + S(n_b)\,(1 - 2x_p)^2 + \cdots, \tag{6.21}$$

with proton fraction $x_p = n_p/n_b$ and symmetry energy $S(\rho)$. Differentiating and imposing (6.20) with ultra-relativistic electrons ($\mu_e = \hbar c\,(3\pi^2 n_e)^{1/3}$) yields a few-percent x_p near n_0: matter is neutron rich but not pure neutron. The SEMF $Z = 0$ exercise is therefore a pedagogical limit, not a literal composition. The asymmetry energy still sets the energy cost of that neutron richness: the same a_a (and its density dependence) that shaped the valley of stability controls the pressure of neutron-star matter [4, 1].

6.9.2 Fermi gas: momentum, energy, and degeneracy pressure

In a volume V the number of neutron spin states with momentum less than p_F is

$$N = 2 \cdot \frac{V}{(2\pi\hbar)^3} \cdot \frac{4\pi}{3} p_F^3 = \frac{V p_F^3}{3\pi^2\hbar^3}. \tag{6.22}$$

Hence

$$p_F = \hbar(3\pi^2 n)^{1/3}, \qquad n = N/V. \tag{6.23}$$

At nuclear saturation density $n \approx n_0 \approx 0.16\,\mathrm{fm}^{-3}$,

$$p_F c \approx \hbar c\,(3\pi^2 n_0)^{1/3} \sim 300\,\mathrm{MeV},$$

so the neutrons are mildly relativistic. In the nonrelativistic limit the kinetic energy density and pressure of a Fermi gas are

$$\varepsilon_{\mathrm{kin}} = \frac{3}{5}\, n\, \frac{p_F^2}{2m_n}, \tag{6.24}$$

$$P = \frac{2}{3}\,\varepsilon_{\mathrm{kin}} = \frac{(3\pi^2)^{2/3}}{5} \frac{\hbar^2}{m_n}\, n^{5/3}. \tag{6.25}$$

The corresponding mean degeneracy energy per neutron is

$$\frac{E_{\mathrm{deg}}^{\mathrm{NR}}}{N} = \frac{3}{5}\frac{p_F^2}{2m_n}, \qquad E_{\mathrm{deg}}^{\mathrm{NR}} = \frac{3\alpha^2\hbar^2}{10m_n}\frac{N^{5/3}}{R^2}, \qquad \alpha \equiv \left(\frac{9\pi}{4}\right)^{1/3}, \tag{6.26}$$

where $n = 3N/(4\pi R^3)$. Thus the chain is explicit:

$$n \longrightarrow p_F = \hbar(3\pi^2 n)^{1/3} \longrightarrow E_{\mathrm{deg}}^{\mathrm{NR}} \longrightarrow P_{\mathrm{deg}}^{\mathrm{NR}} \propto n^{5/3}.$$

In the ultra-relativistic limit, the filled Fermi sphere instead gives

$$\frac{E_{\mathrm{deg}}^{\mathrm{UR}}}{N} = \frac{3}{4} p_F c, \qquad E_{\mathrm{deg}}^{\mathrm{UR}} = \frac{3\alpha\hbar c}{4}\frac{N^{4/3}}{R}, \qquad P_{\mathrm{deg}}^{\mathrm{UR}} = \frac{1}{3}\varepsilon_{\mathrm{kin}} \propto n^{4/3}. \tag{6.27}$$

These are ideal-gas **toy EOS limits**; interactions are essential near and above saturation density.

That kinetic pressure supports the star against gravity (analogous to electron degeneracy in white dwarfs, but with the neutron mass and nuclear densities). Nuclear potentials modify both energy and pressure: attractive at low density, repulsive at high density. The SEMF volume and asymmetry terms are bulk averages of that physics near n_0; they are not a substitute for $P(\varepsilon)$. Pure neutron matter is not self-bound at zero pressure in realistic calculations: stellar gravity and the EOS together make the compact object.

6.9.3 Energy minimisation and the equilibrium scale

For the same uniform sphere, combine Equation (6.26) with $U_G = -3Gm_n^2 N^2/(5R)$:

$$E_{\mathrm{toy}}^{\mathrm{NR}}(R) = \frac{3\alpha^2\hbar^2}{10m_n}\frac{N^{5/3}}{R^2} - \frac{3Gm_n^2}{5}\frac{N^2}{R}. \tag{6.28}$$

The stationary condition $\mathrm{d}E/\mathrm{d}R = 0$ gives

$$R_{\mathrm{eq}}^{\mathrm{NR}} = \frac{\alpha^2 \hbar^2}{G m_n^3} N^{-1/3}. \tag{6.29}$$

Equivalently, the virial relation is $2E_{\mathrm{deg}}^{\mathrm{NR}} + U_G = 0$. Pauli energy rising as R^{-2} therefore balances gravity rising as R^{-1}.

In the ultra-relativistic limit both terms scale as R^{-1}:

$$E_{\mathrm{toy}}^{\mathrm{UR}}(R) = \frac{1}{R}\left[\frac{3\alpha\hbar c}{4} N^{4/3} - \frac{3Gm_n^2}{5} N^2\right]. \tag{6.30}$$

There is no finite-radius minimum in this idealised Newtonian limit. The coefficient changes sign at

$$N_{\mathrm{crit}}^{2/3} = \frac{5\alpha\hbar c}{4Gm_n^2}, \tag{6.31}$$

which exposes a limiting mass scale. Nuclear interactions and general relativity determine the actual stellar sequence.

6.9.4 Hydrostatic equilibrium: Newtonian then TOV

Global binding does not replace local force balance. In Newtonian gravity a spherical shell of thickness $\mathrm{d}r$ is in equilibrium when the pressure gradient balances the gravitational force on the enclosed mass $m(r)$:

$$\frac{\mathrm{d}P}{\mathrm{d}r} = -\frac{Gm(r)\,\rho(r)}{r^2}, \qquad \frac{\mathrm{d}m}{\mathrm{d}r} = 4\pi r^2 \rho(r). \tag{6.32}$$

In general relativity the structure of a static spherical star obeys the Tolman–Oppenheimer–Volkoff (TOV) equations. With energy density $\varepsilon(r)$ (including rest mass) and pressure $P(r)$,

$$\frac{\mathrm{d}P}{\mathrm{d}r} = -\frac{G}{c^2}\frac{(\varepsilon + P)(m + 4\pi r^3 P/c^2)}{r^2(1 - 2Gm/(rc^2))}, \tag{6.33}$$

$$\frac{\mathrm{d}m}{\mathrm{d}r} = 4\pi r^2 \frac{\varepsilon}{c^2}. \tag{6.34}$$

Compared with (6.32), three relativistic corrections appear:

1. $\varepsilon + P$ replaces ρc^2 (pressure gravitates);
2. $m + 4\pi r^3 P/c^2$ replaces m (pressure contributes to the enclosed source);
3. the redshift factor $1 - 2Gm/(rc^2)$ in the denominator steepens the gradient near the Schwarzschild scale.

Boundary conditions: regular centre $m(0) = 0$, $P(0) = P_c$; the stellar surface is the radius R where $P(R) = 0$, and the gravitational mass is $M = m(R)$. The input is a cold equation of state $P(\varepsilon)$; the output is a mass–radius curve $M(R)$ and a maximum mass. Observed $M_{\max} \gtrsim 2\,M_\odot$ pulsars require a sufficiently stiff EOS at a few times ρ_0; GW170817 tidal deformability requires that the EOS not be too stiff near $2\rho_0$. The liquid-drop estimate $R \sim 10\,\mathrm{km}$, $M \sim M_\odot$ is the right order of magnitude; precision astrophysics needs the full EOS ladder [4].

The Newtonian equation is recovered when

$$P \ll \varepsilon, \qquad \varepsilon \simeq \rho c^2, \qquad \frac{2Gm}{rc^2} \ll 1, \qquad \frac{4\pi r^3 P}{mc^2} \ll 1.$$

Then Equation (6.33) reduces to $\mathrm{d}P/\mathrm{d}r = -Gm\rho/r^2$, and Equation (6.34) reduces to $\mathrm{d}m/\mathrm{d}r = 4\pi r^2 \rho$.

Key idea

Three layers of understanding.

1. SEMF + gravity: order-of-magnitude A, M, R (this lecture).
2. Fermi gas + β equilibrium: composition and degeneracy pressure.
3. Realistic EOS + TOV: precision $M(R)$, maximum mass, tides.

This course develops layers 1–2 with honest pointers to layer 3.

6.9.5 Link to laboratory symmetry energy

The slope of the symmetry energy $S(\rho)$ at saturation,

$$L = 3n_0 \frac{\mathrm{d}S}{\mathrm{d}n_b}\bigg|_{n_0}, \tag{6.35}$$

correlates with the neutron-skin thickness of ^{208}Pb and with neutron-star radii. Lecture 1's charge radii plus parity-violating electron scattering (PREX/CREX) therefore constrain the same physics that sets R_{NS} [1].

6.10 Physics depth: neutron-star mass and radius scales

Lecture 6 recovered the kilometre / solar-mass scale from Newtonian gravity plus nuclear saturation. The argument is robust enough to survive general relativity at the factor-of-two level and fragile enough that the dense-matter EOS still moves radii by kilometres.

6.10.1 Worked estimate: uniform-density star

For constant density ρ and mass $M = (4\pi/3)R^3\rho$, the Newtonian estimate $GM^2/R \sim (M/m_N)\,E_{\mathrm{nuc}}$ with $E_{\mathrm{nuc}} \sim 10\,\mathrm{MeV}$ yields $R \sim 10\,\mathrm{km}$ for $M \sim M_\odot$. Replacing the energy scale by the Fermi energy of a pure neutron gas at n_0 gives the same order. The lesson is dimensional: nuclear density times solar mass must produce kilometres.

$$R \sim \left(\frac{3M}{4\pi m_N n_0}\right)^{1/3} \sim 12\,\mathrm{km} \quad \text{for} \quad M = 1.4\,M_\odot. \tag{6.36}$$

Key idea

What the estimate cannot do. It cannot predict tidal deformability, maximum mass, or a possible phase transition in the core. Those require an EOS $P(\varepsilon)$ integrated in the TOV equations. Laboratory n_0, K, and L anchor only the low-density part of that EOS.

Remark

Assumptions. Newtonian gravity, uniform density, and a single nuclear energy scale. Real stars have crusts, density gradients, and possible exotic cores. Use the estimate to sanity-check published M–R curves, not to replace them.

6.10.2 Crust versus core scales

The outer crust is a lattice of nuclei immersed in electrons; densities climb from terrestrial solids to $\sim 10^{11}\,\mathrm{g\,cm^{-3}}$ before neutrons drip. The inner crust adds a neutron gas. Only above $\sim 0.5\,n_0$ does uniform matter take over. Radius measurements are sensitive to both the crust thickness (correlated with L) and the core stiffness. A change of 1 km in radius at fixed mass is a major nuclear-physics statement, not a small correction.

Maximum masses $\gtrsim 2\,M_\odot$ require a sufficiently stiff core EOS. Softening from hyperons or a phase transition must still allow those masses, which is a strong filter on exotic models. The Lecture 6 dimensional estimate explains why the numbers are kilometres and solar masses; it does not explain why some EOS families are already ruled out.

Convert nuclear saturation density into $2.8 \times 10^{14}\,\mathrm{g\,cm^{-3}}$ and compare with white-dwarf central densities: the gap of several orders of magnitude is why neutron stars require nuclear physics while white dwarfs are still atomic/ionic. Also estimate the Newtonian binding energy GM^2/R for $1.4\,M_\odot$, 12 km: it is a few times 10^{53} erg, comparable to a core-collapse neutrino burst energy scale, linking the structure lecture to multimessenger astrophysics.

Worked compactness

Compute $GM/(Rc^2)$ for $M = 1.4\,M_\odot$ and $R = 11, 12, 13\,\mathrm{km}$. The dimensionless compactness changes by $\sim 10\%$ across that radius range, enough to move tidal deformabilities substantially. This arithmetic shows why radius uncertainties of 1 km are scientifically decisive even though the Lecture 6 scale estimate only aimed at order-of-magnitude kilometres.

Physics-depth close

Dimensional analysis explains why neutron stars are kilometres in radius and solar masses in mass. Nuclear physics enters by fixing n_0 and the EOS stiffness that move those numbers by $\mathcal{O}(1)$, which is exactly the precision multimessenger astronomy now demands.

6.11 Further physics: nuclear density as the NS density scale

Electron scattering yields a bulk nucleon number density and corresponding mass density

$$n_0 \approx 0.16\,\mathrm{fm}^{-3}, \qquad \rho_0 = m_n n_0 \approx 2.68 \times 10^{14}\,\mathrm{g\,cm}^{-3} \tag{6.37}$$

[14, 1]. A self-gravitating ball of that density with mass $M \sim M_\odot$ has radius

$$R \sim \left(\frac{3M}{4\pi\rho_0}\right)^{1/3} \sim 10\text{–}15\,\mathrm{km}, \tag{6.38}$$

the same order obtained from $R = r_0 A^{1/3}$ with $A \sim 10^{57}$. Thus the laboratory saturation density is the bridge between fermi-scale nuclear physics and kilometre-scale compact stars. Realistic neutron-star models replace the pure-neutron SEMF cartoon by β-equilibrated matter and the TOV equation, but they still pass near ρ_0 in the outer core [14, 4].

6.11.1 Drip lines and pure neutrons

The neutron drip line is where $S_n = 0$. Pure neutron matter ($Z = 0$) lies far beyond that line for any laboratory A: the asymmetry energy costs $\sim a_\mathrm{a} \approx 23\,\mathrm{MeV}$ per nucleon relative to $N = Z$ matter, while volume attraction is only $\sim 16\,\mathrm{MeV}$ net. Gravity at $A \sim 10^{57}$ supplies an extra bulk binding $\propto A^{5/3}/R$ that laboratory nuclei never feel [14, 4].

Compact-star densities still pass near the laboratory saturation number density $n_0 \sim 0.16\,\mathrm{fm}^{-3}$ in the outer core; that is why electron-scattering radii and neutron-star radii are conceptually linked.

6.11.2 Gravitational scale and Schwarzschild comparison

For $M = M_\odot$ the Schwarzschild radius is

$$R_\mathrm{S} = \frac{2GM}{c^2} \approx 3\,\mathrm{km}. \tag{6.39}$$

Observed neutron-star radii ~ 10–$14\,\mathrm{km}$ are only a few times R_S: general relativity is essential for the structure equations, even though the pedagogical SEMF+gravity estimate of this lecture is Newtonian. White dwarfs, supported by electron degeneracy at much lower density, have $R \sim R_\oplus$ and weaker relativity; the larger critical mass scale for neutron stars reflects the larger Fermi energy of nucleons compared with electrons at white-dwarf densities [14].

6.11.3 Asymmetry energy and the crust

In the outer crust of a neutron star, nuclei coexist with a degenerate electron gas; as density rises, nuclei become more neutron rich until free neutrons drip out ($S_n \to 0$). That threshold is the same drip-line physics as in the laboratory SEMF, now under pressure and in β equilibrium. The bulk asymmetry coefficient a_a and the density dependence of the symmetry energy control the proton fraction in the core and the thickness of the crust: links between nuclear mass fits and compact-star observables [14, 4].

6.11.4 Fermi energy in neutron matter

At $n \sim n_0$, a pure neutron gas would have a Fermi energy of order tens of MeV (comparable to the laboratory ε_F scale of Lecture 2). β equilibrium $n \leftrightarrow p + e + \bar{\nu}_e$ then fixes a small proton fraction so that $\mu_n = \mu_p + \mu_e$. The SEMF asymmetry term is a schematic way of writing the cost of $N \neq Z$; modern equations of state refine that cost as a function of density, but the order of magnitude is set by the same nuclear saturation physics measured on Earth.

6.12 Deeper reading: from SEMF to stellar pressure

In the crust, nuclei from Lectures 4–5 coexist with a degenerate electron gas; the neutron drip line appears when S_n hits zero. In the outer core, uniform *npe* matter takes over, and the pressure is dominated by neutron degeneracy plus interactions. The symmetry energy slope L raises the pressure of neutron-rich matter just above n_0, correlating with both neutron skins (Lecture 1) and stellar radii.

6.12.1 Multimessenger checklist

When reading a mass–radius posterior, ask three nuclear questions: (1) Was n_0 fixed to laboratory saturation? (2) How was L constrained (skins, dipole polarisability, heavy-ion flow)? (3) Is the high-density extension nucleonic or does it allow a phase transition? Gravitational-wave tidal deformability and X-ray pulse-profile radii answer astrophysical questions; they do not erase the need for those nuclear anchors. Lectures 23–24 later remind us that the same nuclei can also fission when assembled in terrestrial or astrophysical extremes.

6.12.2 Laboratory anchors checklist

Useful laboratory anchors include: saturation density and binding from masses and radii; incompressibility from giant resonances; symmetry energy and its slope from skins, dipole polarisability, and heavy-ion flow; high-density constraints from hard photon or meson production (more model dependent). Astrophysical anchors include pulsar masses, NICER radii, and GW170817 tidal deformability. A responsible EOS paper states which anchors were imposed as priors and which were used only as posterior checks.

Convert nuclear saturation density into $2.8 \times 10^{14}\,\mathrm{g\,cm^{-3}}$ and compare with white-dwarf central densities: the gap of several orders of magnitude is why neutron stars require nuclear physics while white dwarfs are still atomic/ionic. Also estimate the Newtonian binding energy GM^2/R for $1.4\,M_\odot$, $12\,\mathrm{km}$: it is a few times $10^{53}\,\mathrm{erg}$, comparable to a core-collapse neutrino burst energy scale, linking the structure lecture to multimessenger astrophysics.

Neutron-star checklist

(1) Convert n_0 to $\mathrm{g\,cm^{-3}}$. (2) Estimate uniform-density R for $1.4\,M_\odot$. (3) List laboratory anchors (n_0, K, L) separately from astrophysical anchors (masses, radii, tides). (4) State whether a published EOS allows a phase transition. This checklist prevents treating a multimessenger posterior as if it had no nuclear priors.

Tidal deformability intuition

For a given mass, a larger radius usually implies a larger tidal deformability Λ, because the star is less compact. Soft EOS families that shrink radii also suppress Λ, which is why GW170817's tidal constraint reshaped the nuclear EOS landscape alongside radius measurements. The Lecture 6 dimensional estimate cannot predict Λ, but it explains why the relevant compactness $GM/(Rc^2)$ is $\mathcal{O}(0.15)$ rather than planetary.

Crust thickness and L

A larger symmetry-energy slope L typically thickens the crust and increases the radius at fixed mass for many EOS parametrisations. Neutron skins of ^{208}Pb correlate with that slope. Laboratory skins and astrophysical radii are therefore not independent anecdotes; they are linked through the same isovector physics introduced in Lectures 1–2.

Reading an M–R curve

On a published mass–radius plot, mark $1.4\,M_\odot$ and read off radius ranges for soft and stiff EOS bands. Convert the radius difference into a compactness difference. Then ask which nuclear priors (n_0, K, L) were imposed. If the paper does not say, the posterior is harder to interpret for nuclear physics purposes even if the astrophysics is carefully done.

Neutron stars convert nuclear saturation into astrophysical kilometres. Keeping laboratory anchors and multimessenger observables on separate mental lists is the clearest way to read contemporary EOS papers without confusion.

In practice, rework one numerical estimate from the lecture with a different parameter set and compare the shift to the quoted experimental uncertainty. If the shift exceeds the uncertainty, the estimate is teaching a scale, not delivering a fit. That distinction keeps theoretical cartoons honest when they sit beside evaluated data tables and modern many-body calculations. Along the course arc, write one sentence linking this chapter backward and one forward: the discipline of those two sentences is as important as any single formula. Examination problems rarely ask for a literature review, but they do ask for the scale argument that literature reviews presuppose.

Research frontier: multimessenger neutron-star EOS and student projects

Lecture 6 used Newtonian gravity and SEMF bookkeeping to recover the kilometre/solar-mass scale of neutron stars. That estimate remains an excellent order-of-magnitude check, but quantitative astrophysics now combines general relativistic structure equations with laboratory nuclear constraints. Since GW170817, tidal deformability from gravitational waves, NICER pulse-profile radii, and massive pulsars jointly constrain the dense-matter equation of state, while laboratory symmetry-energy and neutron-skin data supply the low-density anchor [49, 48, 47, 40]. Whether the core contains only nucleonic matter or hosts a phase transition (hyperons, quark matter, or condensates) remains unsettled; the Bayesian combination of laboratory and astrophysical likelihoods is a major research industry. Uncertainties that still dominate EOS posteriors include the high-density symmetry energy, the possible presence of first-order phase transitions, and systematic modelling of the neutron-star crust. The pedagogical lesson of the lecture survives: without nuclear saturation density and a stiffness measure near and above n_0, even the most precise multimessenger data cannot be interpreted.

A clean student project is to hold the low-density nuclear constraints fixed and vary only the high-density extension, then ask which multimessenger observable moves. That exercise makes the division of labour between laboratory nuclear physics and astrophysical inference explicit, and it shows why PREX, NICER, and GW170817 are complementary rather than redundant.

Student directions. (1) Analytic estimate: recompute the uniform-density radius $R = r_0 A^{1/3}$ for $A = 10^{57}$ and compare with a published $1.4\,M_\odot$ radius range. (2) Conceptual link: relate PREX's L constraint qualitatively to crust thickness and tidal deformability. (3) Term paper: compare GW170817 tidal limits with NICER radius posteriors, citing at least one review [48].

6.13 Problems with solutions

Problem 6.1 — Nucleon number of a solar-mass neutron star. Estimate A for a neutron star of mass $M = 1.4\,M_\odot$ ($M_\odot = 2 \times 10^{30}$ kg, $m_n = 1.67 \times 10^{-27}$ kg).

Solution 6.1.

$$A \approx \frac{M}{m_n} = \frac{1.4 \times 2 \times 10^{30}}{1.67 \times 10^{-27}} = \frac{2.8 \times 10^{30}}{1.67 \times 10^{-27}} \approx 1.7 \times 10^{57}.$$

Order of magnitude: $A \sim 10^{57}$.

Problem 6.2 — Radius from nuclear density. Convert $n_0 = 0.16\,\mathrm{fm}^{-3}$ to mass density using $m_n = 1.6749 \times 10^{-27}$ kg. Then use that density and $M = M_\odot$, recompute R and compare with $R = r_0 A^{1/3}$ for $A = M/m_n$, where $r_0 = (3/4\pi n_0)^{1/3}$.

Solution 6.2. The conversion is

$$\rho_0 = m_n n_0 = (1.6749 \times 10^{-27})(0.16 \times 10^{45}) = 2.68 \times 10^{17}\,\mathrm{kg\,m^{-3}} = 2.68 \times 10^{14}\,\mathrm{g\,cm^{-3}}.$$

From density:

$$R = \left(\frac{3M}{4\pi\rho}\right)^{1/3} = \left(\frac{3 \times 2 \times 10^{33}\,\mathrm{g}}{4\pi \times 2.68 \times 10^{14}\,\mathrm{g\,cm^{-3}}}\right)^{1/3} \approx 12.1\,\mathrm{km}.$$

(cf. Problem 2.2). Also

$$A = \frac{M_\odot}{m_n} \approx 1.19 \times 10^{57}, \qquad r_0 = \left(\frac{3}{4\pi n_0}\right)^{1/3} \approx 1.14\,\mathrm{fm}.$$

Therefore

$$R = 1.14\,\mathrm{fm} \times (1.19 \times 10^{57})^{1/3} = 12.1\,\mathrm{km}.$$

The two expressions agree identically because r_0 was derived from the same uniform density.

Problem 6.3 — SEMF pure neutrons unbound. With $a_V = 15.5\,\mathrm{MeV}$ and $a_\mathrm{a} = 23\,\mathrm{MeV}$, show that pure neutron matter ($Z = 0$) has negative binding per nucleon from nuclear terms alone, and estimate the critical A at which $CA^{2/3}$ with $C = 3Gm_n^2/(5r_0) \sim 6 \times 10^{-37}\,\mathrm{MeV}$ can compensate.

Solution 6.3. For $Z = 0$, $N = A$,

$$\frac{B}{A} \approx a_V - a_\mathrm{a} = 15.5 - 23 = -7.5\,\mathrm{MeV}$$

(unbound). Adding the gravity piece $+CA^{2/3}$ per nucleon and setting $CA^{2/3} = 7.5\,\mathrm{MeV}$,

$$A^{2/3} = \frac{7.5}{6 \times 10^{-37}} \sim 1.3 \times 10^{37}, \qquad A \sim (1.3 \times 10^{37})^{3/2} \sim 10^{56}.$$

This is the stellar scale of Lecture 6. (The precise C comes from $3GM^2/(5R)$ with $M = Am_n$, $R = r_0 A^{1/3}$.)

Problem 6.4 — Fermi momentum at saturation. Using $p_F = \hbar(3\pi^2 n)^{1/3}$ and $n = n_0 = 0.16\,\mathrm{fm}^{-3}$, compute $p_F c$ in MeV and comment on whether the neutrons are nonrelativistic.

Solution 6.4.

$$(3\pi^2 n)^{1/3} = \left(3\pi^2 \times 0.16\right)^{1/3} \approx (4.74)^{1/3} \approx 1.68\,\mathrm{fm}^{-1},$$
$$p_F c = \hbar c \cdot 1.68 \approx 197 \times 1.68 \approx 330\,\mathrm{MeV}.$$

Since $m_n c^2 \approx 940\,\mathrm{MeV}$, one has $p_F c/(m_n c^2) \sim 0.35$: mildly relativistic. The nonrelativistic pressure formula is a first estimate; a realistic EOS must include relativistic kinematics and nuclear interactions.

Problem 6.5 — Degeneracy–gravity balance. For N neutrons in a uniform sphere, use $p_F = \alpha\hbar N^{1/3}/R$, $\alpha = (9\pi/4)^{1/3}$. (a) Derive the nonrelativistic degeneracy energy and minimise $E = E_\mathrm{deg} - 3Gm_n^2 N^2/(5R)$. (b) Show why the ultra-relativistic limit produces a limiting mass rather than a finite equilibrium radius in this toy model.

Solution 6.5. The filled Fermi sphere has mean kinetic energy $3p_F^2/(10m_n)$, hence

$$E_\mathrm{deg}^\mathrm{NR} = \frac{3\alpha^2\hbar^2}{10m_n}\frac{N^{5/3}}{R^2}.$$

Setting $\mathrm{d}E/\mathrm{d}R = 0$ gives

$$R_\mathrm{eq} = \frac{\alpha^2\hbar^2}{Gm_n^3}N^{-1/3}, \qquad 2E_\mathrm{deg}^\mathrm{NR} + U_G = 0.$$

Ultra-relativistically,

$$E_{\mathrm{deg}}^{\mathrm{UR}} = \frac{3\alpha\hbar c}{4}\frac{N^{4/3}}{R},$$

so both kinetic and gravitational terms scale as R^{-1}. Their coefficients determine stability; equality gives $N_{\mathrm{crit}}^{2/3} = 5\alpha\hbar c/(4Gm_n^2)$. A realistic maximum mass requires interactions and TOV gravity.

Problem 6.6 — Why laboratory pure-neutron drops fail. Using the SEMF schematic $B/A \approx a_V - a_a$ for $Z = 0$, explain why pure neutron matter is unbound in the laboratory even though neutron stars exist.

Solution 6.6. With standard coefficients $a_V < a_a$, the per-nucleon nuclear binding for $Z = 0$ is negative: asymmetry wins over volume. Neutron stars are bound by gravity at $A \sim 10^{56}$–10^{57}, which contributes a huge negative gravitational energy per nucleon absent in laboratory droplets [6, 1, 4].

Chapter 7

The Deuteron and Nuclear Force

"For nuclear physicists, the deuteron should be what the hydrogen atom is for atomic physicists."

K. S. Krane

7.1 Why the deuteron is special

A deuteron is a bound proton–neutron system: the nucleus of deuterium (^{2}H). It is the simplest bound state of nucleons, and therefore an ideal laboratory for the nucleon–nucleon (NN) interaction. There is no bound *pp* or *nn* analogue: the Coulomb repulsion plus the slightly weaker *pp* strong force leave the dineutron and diproton unbound, so the deuteron also teaches the spin and isospin selectivity of the low-energy force (triplet *np* binds; singlet does not) [6, 1].

In atomic physics, the hydrogen spectrum taught the Coulomb force and quantum electrodynamics. In nuclear physics one would like a similar study for the deuteron: measure transitions between excited states and read off the force. **Unfortunately there are no bound excited states of the deuteron.** The system is so weakly bound that any excitation puts the proton and neutron into the continuum. All information must therefore come from the ground-state binding energy and size, from ground-state multipoles (spin, magnetic moment, quadrupole moment), and from scattering of free nucleons [6, 2].

Key idea

The deuteron is the hydrogen atom of nuclear physics, with one crucial difference: it has only a ground state. Every number measured is a property of that single state, so the force must be extracted carefully and honestly.

7.2 Binding energy: three independent measurements

The binding energy B_d is known to high precision. Three independent methods agree [6].

7.2.1 Definition

Working always with *atomic* masses (as in Lectures 4–5),

$$B_d = [m(^1\text{H}) + m(n) - m(^2\text{H})]c^2. \tag{7.1}$$

With $c^2 = 931.50\,\text{MeV/u}$, mass differences in atomic mass units convert directly to MeV.

7.2.2 Method 1: mass spectrometry

Mass doublets involving carbon and deuterium compounds fix $m(^2\text{H})$ to better than $10^{-8}\,\text{u}$. Combined with the hydrogen and neutron masses, one finds

$$B_d = 2.22463 \pm 0.00004\,\text{MeV}. \tag{7.2}$$

7.2.3 Method 2: radiative capture

When a free proton and neutron form a deuteron,

$$p + n \to {}^2\text{H} + \gamma, \tag{7.3}$$

the photon energy (minus a small recoil correction) equals B_d. The measured value is $2.224589 \pm 0.000002\,\text{MeV}$, in excellent agreement with the mass-spectroscopic result.

7.2.4 Method 3: photodisintegration

The reverse process,

$$\gamma + {}^2\text{H} \to p + n, \tag{7.4}$$

has a threshold equal to B_d (again corrected for recoil). Experiment gives $2.224 \pm 0.002\,\text{MeV}$.

How weak is "weakly bound"?

Typical mid-mass nuclei have $B/A \sim 8\,\text{MeV}$ (Lecture 4). The deuteron has $B/A \approx 1.11\,\text{MeV}$. It sits far down on the rising part of the B/A curve: almost unbound. That single fact will control everything we calculate below.

For course work we adopt the rounded value used in the original lecture,

$$B_d \approx 2.225\,\text{MeV}, \tag{7.5}$$

which is consistent with all three determinations at the level needed for the square-well analysis.

7.3 A simple model: the square well

7.3.1 Why a square well?

The true NN potential is complicated: a repulsive core at short distance, a strong attraction at $\sim 1\,\text{fm}$, a pion-exchange tail at larger r, and spin- and tensor-dependent pieces (Lectures 7–9).

To learn the *scale* of the force we represent the interaction by a three-dimensional square well in the relative coordinate $r = |\boldsymbol{r}_p - \boldsymbol{r}_n|$:

$$V(r) = \begin{cases} -V_0, & r < R, \\ 0, & r > R. \end{cases} \tag{7.6}$$

Here r is the proton–neutron separation and R is the *radial range*: the attraction acts for separations $r < R$. It is neither a nucleon radius nor the deuteron rms charge radius, and it is not a diameter. We adopt the explicit toy-model convention $R = 2.00\,\mathrm{fm}$; the measured B_d then fixes V_0.

The square well is pedagogically ideal because interior and exterior solutions are elementary, matching reduces to a cotangent condition, and the same R later reappears as the hard-sphere scale of low-energy scattering (Lecture 10). Modern potentials replace the flat bottom with a soft core and a Yukawa tail, but they must still reproduce the same B_d and scattering lengths [6, 1].

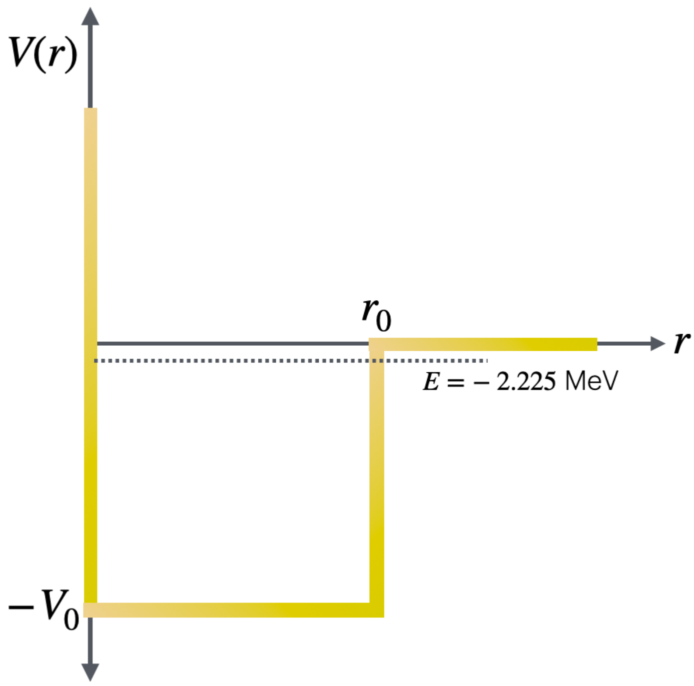


Figure 7.1: Square-well model of the NN potential (lecture figure). Inside $r < R$ the potential is $-V_0$; outside it vanishes. The deuteron energy $E = -B_d$ lies just below threshold.

7.3.2 Reduced mass and the radial equation

The two-body problem with equal masses $m_p \approx m_n \approx m_N$ reduces to a one-body problem with reduced mass

$$\mu = \frac{m_p m_n}{m_p + m_n} \approx \frac{m_N}{2}. \tag{7.7}$$

For a central potential we write the wave function as $\psi(\boldsymbol{r}) = Y_{\ell m}(\hat{r})\, u(r)/r$. The radial function $u(r)$ obeys

$$-\frac{\hbar^2}{2\mu}\frac{\mathrm{d}^2 u}{\mathrm{d}r^2} + \left[V(r) + \frac{\ell(\ell+1)\hbar^2}{2\mu r^2}\right] u = Eu. \tag{7.8}$$

Assumption (to be checked later): the ground state has $\ell = 0$, by analogy with the hydrogen $1s$ state. Then the centrifugal term vanishes and (7.8) looks exactly like a one-dimensional Schrödinger equation on the half-line $r > 0$.

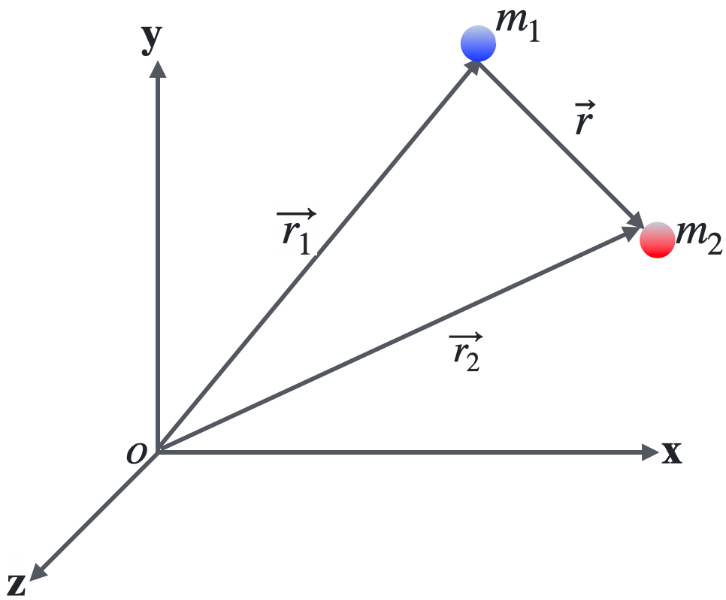


Figure 7.2: Relative coordinate of proton and neutron. The square well acts on this separation; the centre of mass moves freely.

7.3.3 Interior and exterior solutions

With $E = -B_d < 0$, define

$$k = \frac{1}{\hbar}\sqrt{2\mu(V_0 - B_d)} \qquad (r < R), \tag{7.9}$$

$$\gamma = \frac{1}{\hbar}\sqrt{2\mu B_d} \qquad (r > R). \tag{7.10}$$

Equivalently,

$$k = \frac{\sqrt{2\mu(V_0 + E)}}{\hbar}, \qquad E = -B_d.$$

For $\mu \approx m_N/2$, the useful numerical coefficient is $\hbar^2/(2\mu) \approx 41.5\,\mathrm{MeV\,fm^2}$.

Inside the well the general solution is

$$u(r) = A\sin(kr) + B\cos(kr). \tag{7.11}$$

Boundary condition at the origin: ψ must remain finite, so $u(0) = 0$. That forces $B = 0$:

$$u(r) = A\sin(kr), \qquad r < R. \tag{7.12}$$

Outside the well,

$$u(r) = Ce^{-\gamma r} + De^{+\gamma r}. \tag{7.13}$$

Boundary condition at infinity: u must not explode, so $D = 0$:

$$u(r) = Ce^{-\gamma r}, \qquad r > R. \tag{7.14}$$

7.3.4 Matching: the transcendental condition

The physical content of the bound-state problem is fixed by the finite range of the force and by quantum mechanics. Outside the well there is no attraction, so a bound system ($E = -B_d < 0$) can exist only because the wave function tunnels under the continuum; the exterior solution must decay exponentially. Inside the well the nucleons feel a constant attraction $-V_0$, so the radial function oscillates. Matching the two regions at $r = R$ quantises the allowed (V_0, R) pairs that reproduce the measured binding energy [6, 13].

Write

$$V(r) = \begin{cases} -V_0 & r < R, \\ 0 & r > R, \end{cases} \qquad E = -B_d < 0.$$

For $r < R$ the radial Schrödinger equation becomes

$$-\frac{\hbar^2}{2\mu}\frac{\mathrm{d}^2 u}{\mathrm{d}r^2} - V_0 u = Eu \quad \Rightarrow \quad \frac{\mathrm{d}^2 u}{\mathrm{d}r^2} = -k^2 u,$$

with wave number

$$k = \frac{\sqrt{2\mu(V_0 - B_d)}}{\hbar} = \frac{\sqrt{2\mu(V_0 + E)}}{\hbar}, \tag{7.15}$$

where the second form uses $E = -B_d$ so that $V_0 - B_d = V_0 + E$ (not $V_0 + |E|$ written as a separate definition: since $E = -B_d$, one has $V_0 + E = V_0 - B_d$). Both expressions are identical; they are not two different physical definitions. The general interior solution is $u_{\mathrm{I}}(r) = A\sin(kr) + B\cos(kr)$. Because $\psi = u/r$ must remain finite at the origin, $u(0) = 0$, which forces $B = 0$:

$$u_{\mathrm{I}}(r) = A\sin(kr). \tag{7.16}$$

Physically this is the statement that there is no source of probability current at $r = 0$; the relative wave function vanishes at vanishing separation in the s-wave radial problem.

Outside the well, $V = 0$ and $E = -B_d$, so

$$\frac{\mathrm{d}^2 u}{\mathrm{d}r^2} = \gamma^2 u, \qquad \gamma = \frac{\sqrt{2\mu B_d}}{\hbar}.$$

The exterior solution is $u_{\mathrm{II}}(r) = Ce^{-\gamma r} + De^{+\gamma r}$. The growing exponential is discarded so that the state is normalisable:

$$u_{\mathrm{II}}(r) = Ce^{-\gamma r}. \tag{7.17}$$

The decay length $1/\gamma$ is large when B_d is small: weak binding means the nucleons spend much of their time *outside* the range of the force, which is why the deuteron rms radius exceeds the naive well radius.

Continuity of ψ and of $\mathrm{d}\psi/\mathrm{d}r$ at the edge of a finite square well requires continuity of u and u' at $r = R$:

$$A\sin(kR) = Ce^{-\gamma R}, \tag{7.18}$$

$$Ak\cos(kR) = -C\gamma e^{-\gamma R}. \tag{7.19}$$

Dividing these two equations eliminates the unknown amplitudes A and C and yields the matching condition

$$k\cot(kR) = -\gamma. \tag{7.20}$$

Equation (7.20) is transcendental: once R and B_d are fixed, it determines the depth V_0 that can support the observed bound state. That is the quantitative link between the measured binding energy and the strength of the residual force.

7.3.5 Amplitude matching and normalisation

The first matching equation gives

$$C = A\sin(kR)e^{\gamma R}. \tag{7.21}$$

For the convention $\int_0^\infty |u(r)|^2\,\mathrm{d}r = 1$, the common interior amplitude is therefore fixed by writing $C = A\sin(kR)e^{\gamma R}$ and evaluating each piece:

$$\int_0^R \sin^2(kr)\,\mathrm{d}r = \int_0^R \frac{1-\cos(2kr)}{2}\,\mathrm{d}r = \frac{R}{2} - \frac{\sin(2kR)}{4k}, \tag{7.22}$$

$$\int_R^\infty e^{-2\gamma r}\,\mathrm{d}r = \frac{e^{-2\gamma R}}{2\gamma}, \tag{7.23}$$

so that

$$\begin{aligned} 1 &= A^2\int_0^R \sin^2(kr)\,\mathrm{d}r + C^2\int_R^\infty e^{-2\gamma r}\,\mathrm{d}r \\ &= A^2\left[\frac{R}{2} - \frac{\sin(2kR)}{4k}\right] + A^2\sin^2(kR)\,e^{2\gamma R}\cdot\frac{e^{-2\gamma R}}{2\gamma} \\ &= A^2\left[\frac{R}{2} - \frac{\sin(2kR)}{4k} + \frac{\sin^2(kR)}{2\gamma}\right]. \end{aligned} \tag{7.24}$$

Thus A is the inverse square root of the bracket and C follows from continuity. The probability beyond the force range is

$$P_> = \frac{A^2\sin^2(kR)}{2\gamma}. \tag{7.25}$$

As $B_d \to 0^+$, $\gamma \to 0^+$, the exterior term dominates the norm: the state becomes spatially large even though the force range stays fixed. At threshold $\gamma = 0$, matching requires $kR = \pi/2$ for the first state, or $V_0R^2 = \pi^2\hbar^2/(8\mu)$.

7.4 Numerical result for the adopted toy range

Electron scattering gives an rms charge radius of the deuteron of about 2.1 fm. That is *not* the force range: it is a charge-weighted rms distance from the centre of mass. The pn separation, matter radius, charge radius, and square-well range are distinct definitions. For the adopted teaching model $R = 2.00\,\text{fm}$, with $B_d = 2.2246\,\text{MeV}$ and $\hbar^2/(2\mu) \approx 41.5\,\text{MeV fm}^2$, one solves (7.20) as follows.

Define the dimensionless quantities

$$\xi \equiv kR, \qquad \eta \equiv \gamma R = R\sqrt{\frac{2\mu B_d}{\hbar^2}}. \tag{7.26}$$

For $R = 2.00\,\text{fm}$,

$$\eta = 2.00\sqrt{\frac{2.2246}{41.5}} \approx 2.00 \times 0.232 \approx 0.463.$$

The matching condition $k\cot(kR) = -\gamma$ becomes $\xi\cot\xi = -\eta$, or

$$\xi\cot\xi = -0.463. \tag{7.27}$$

The relevant root in $(\pi/2, \pi)$ is $\xi = 1.820$. Then

$$V_0 - B_d = \frac{\hbar^2 k^2}{2\mu} = \frac{\hbar^2}{2\mu}\frac{\xi^2}{R^2} = 41.5 \cdot \frac{(1.820)^2}{(2.00)^2} \approx 34.4\,\text{MeV}, \tag{7.28}$$

$$V_0 = 34.4 + 2.2246 \approx 36.6\,\text{MeV}. \tag{7.29}$$

$$V_0 \approx 36.6\,\text{MeV} \qquad (R = 2.00\,\text{fm}). \tag{7.30}$$

Equation (7.20) constrains a curve in the (R, V_0) plane, not a unique pair: one observable (B_d) cannot fix two potential parameters. Our quoted depth is conditional on the labelled $R = 2.00\,\text{fm}$ convention; a shorter range requires a deeper well. The residual NN attraction sampled by the deuteron is tens of MeV deep on a fermi scale, barely enough to bind [6, 13].

Key idea

Barely bound. The energy $E = -B_d$ lies just below the top of the well. If the force were slightly weaker, or the range slightly smaller, there would be no bound state at all. The wave function barely "turns over" inside the well so that it can match a decaying exponential outside [6].

Why this matters for the Sun and the early universe

Deuterium formation is the first step in the solar *pp* chain and in Big Bang nucleosynthesis. If no stable two-nucleon bound state existed, heavier nuclei would not form in the early universe and stellar hydrogen burning would not begin as it does. The weak binding of the deuteron is a threshold fact about the world we inhabit [6].

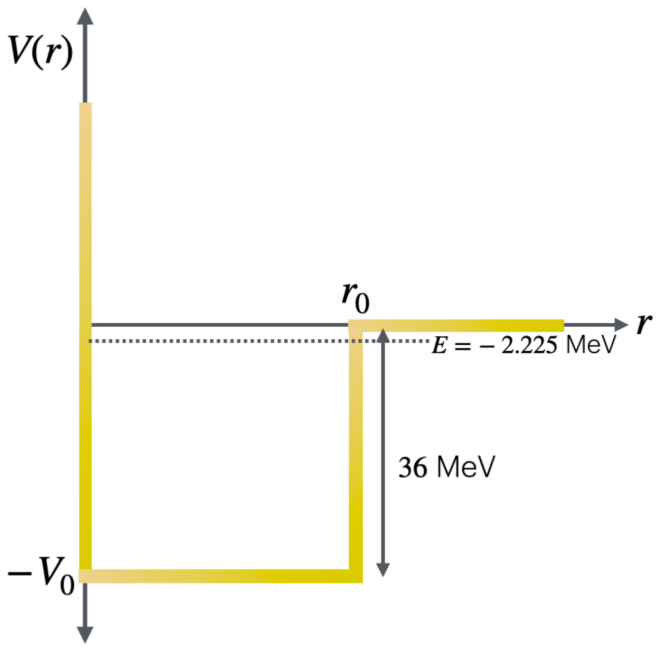


Figure 7.3: Lecture figure emphasising the depth V_0 needed to bind the deuteron. The bound level sits only $\sim 2.2\,\text{MeV}$ below the continuum, so most of the well depth is cancelled by kinetic energy of confinement.

Remark

Because the binding is weak, the probability of finding the nucleons *outside* the range R is large. The exponential tail extends far; low-energy NN scattering is correspondingly sensitive to the asymptotic normalisation. A large fraction of the time is spent outside the well [2].

7.5 Residual force: why nucleons attract at all

Before leaving the central model, place the force in a modern context. Nucleons are not elementary: $p = uud$, $n = udd$. The fundamental strong interaction (QCD) confines colour. Colour-neutral nucleons still interact at short distance through a residual force, much as neutral atoms form molecules through residual electromagnetic multipoles.

Yukawa's meson exchange (1935) explains the long-range tail: the force range is of order $\hbar/(m_\pi c) \sim 1.4\,\text{fm}$. The square well is a teaching caricature of that attraction plus short-range repulsion. Lectures 7–9 will show that even this picture is incomplete: the force depends on spin and contains a non-central (tensor) piece.

7.6 What we have learned, and what we have not

1. The deuteron binding energy is $B_d \approx 2.224\,\text{MeV}$, measured three independent ways.
2. There are no bound excited states: the system is barely bound.
3. The labelled teaching choice $R = 2.00\,\text{fm}$ requires depth $V_0 \approx 36.6\,\text{MeV}$; shorter range needs deeper V_0.

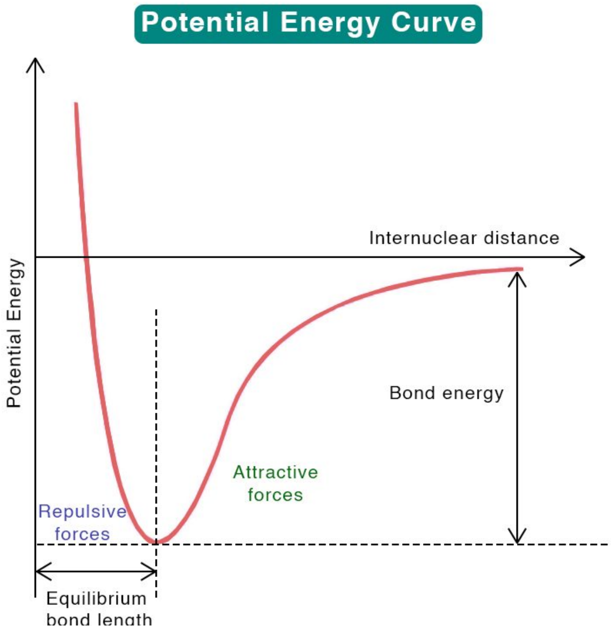


Figure 7.4: Molecular analogy: short-range repulsion, attractive well, finite range. Colour-neutral nucleons interact through residual colour forces on the fermi scale in a similar qualitative way.

4. That depth sets the scale of the NN attraction underlying the SEMF volume term a_V (Lecture 4).

5. We have *assumed* $\ell = 0$. Spin, magnetic moment, and quadrupole moment will test that assumption (Lectures 8–9).

Takeaway

In this lecture and the next.

- This lecture: binding energy and a central well $\Rightarrow$ force range and depth.
- Next lecture: I^π and μ_d $\Rightarrow$ spin dependence and a few-percent D-wave.
- Lecture after: $Q_d \neq 0 \Rightarrow$ tensor (non-central) force.

Only after those three steps is the qualitative NN force complete for PH5001.

7.6.1 Spin dependence: why only the triplet np binds

The deuteron has no particle-stable excited state: $B_d \simeq 2.224\,\text{MeV}$ is small compared with typical $B/A \sim 8\,\text{MeV}$. The absence of bound nn and pp systems, together with a bound triplet np, is the classic low-energy demonstration that the nuclear force is spin dependent (singlet unbound, triplet bound) rather than a mere Coulomb effect [6, 14, 1].

7.7 Deeper physics: weak binding and the asymptotic wave function

7.7.1 No bound excited states

Unlike the hydrogen atom, the deuteron has **no bound excited states**. Any excitation exceeds B_d and produces a free $p + n$ pair in the continuum. All structure information must therefore come from the ground-state multipoles (Lectures 8–9) and from scattering (phase shifts), not from a discrete spectrum [6, 2].

7.7.2 Wave function mostly outside the well

With the lecture convention $V_0 = 36.6\,\text{MeV}$ and $R = 2.00\,\text{fm}$, the matching condition $k\cot(kR) = -\gamma$ places the eigenenergy just below threshold. The exterior decay constant $\gamma = \sqrt{2\mu B_d}/\hbar$ is small ($\gamma^{-1} \approx 4.3\,\text{fm}$), so the exponential tail extends far: the two nucleons spend a large fraction of the time *outside* the range of the force. That is why the rms charge radius ($\sim 2.1\,\text{fm}$ from electron scattering) is not a direct measure of the force range, and why low-energy NN scattering is so sensitive to the asymptotic normalisation.

Range vs depth degeneracy

Fixing only B_d leaves a curve of allowed (R, V_0). Scattering lengths and effective ranges (Lecture 10) break that degeneracy by adding low-energy continuum information. A single bound-state energy never determines a unique potential.

Remark

Exterior dominance and capture. Because $P_> \sim 0.5$–0.7, radiative np capture at thermal energies and low-energy np scattering are controlled by the same long tail that sets r_{rms}. A deep interior well can be replaced by a different V_0 without changing low-energy observables if the asymptotic γ and normalisation are held fixed. That is why modern analyses quote scattering lengths and asymptotic normalisation constants alongside B_d [6, 14].

7.7.3 Square well as a teaching potential

Realistic potentials (Yukawa one-pion tail, hard or soft core at short distance, tensor and spin–orbit terms) are more complex. The square well is retained because:

- it is analytically solvable for $L = 0$;
- it returns $V_0 = 36.6\,\text{MeV}$ for the labelled $R = 2.00\,\text{fm}$ choice, the correct scale;
- it makes the near-threshold character of the bound state obvious.

Lectures 8–9 show why a pure central well is still incomplete: spin and quadrupole data demand a tensor component.

7.8 Square-well matching: from algebra to exterior probability

7.8.1 Interior and exterior solutions

The lecture derives the bound-state condition by matching $u(r) = r\psi(r)$ and its derivative at $r = R$. Inside the well ($E < 0$, $V = -V_0$) the solution is oscillatory,

$$u_{\text{in}}(r) = A\sin(kr), \qquad k = \frac{\sqrt{2\mu(V_0 - B_d)}}{\hbar}, \tag{7.31}$$

while outside it decays exponentially with

$$u_{\text{out}}(r) = Be^{-\gamma r}, \qquad \gamma = \frac{\sqrt{2\mu B_d}}{\hbar}. \tag{7.32}$$

Continuity of u and u' at $r = R$ eliminates A/B and yields $k\cot(kR) = -\gamma$. Physically, the interior wave must "turn over" just enough that the logarithmic derivative matches the decay slope imposed by B_d. Because $B_d \ll V_0$, one has $kR \approx \pi/2$ (just below the first quarter-wave threshold) and $\gamma R \ll 1$: the bound state sits in the corner of parameter space where the well is deep but the eigenenergy is shallow.

7.8.2 A worked estimate of the exterior fraction

Take $B_d = 2.22\,\text{MeV}$, $\mu = m_N/2$, and $\hbar c = 197\,\text{MeV fm}$. Then

$$\gamma^{-1} = \frac{\hbar}{\sqrt{2\mu B_d}} \approx \frac{197}{\sqrt{938 \times 2.22}} \approx 4.3\,\text{fm}, \tag{7.33}$$

comparable to the deuteron rms radius. With $R = 2.00\,\text{fm}$ and a typical normalisation, the probability outside the well is

$$P_> \approx \frac{\displaystyle\int_R^\infty u^2\,\mathrm{d}r}{\displaystyle\int_0^\infty u^2\,\mathrm{d}r} \sim 0.5\text{–}0.7, \tag{7.34}$$

depending on the precise V_0 fit. Most of the wave function therefore samples the force *beyond* its nominal range: low-energy NN physics is controlled by the asymptotic tail, not by the interior where V_0 is large. That is the same halo logic seen later for light nuclei (Lecture 1 further material), but here it appears in the simplest two-body bound state.

Threshold arithmetic

The quantum condition for one s-wave bound state, $V_0R^2 > \pi^2\hbar^2/(8\mu) \approx 109\,\text{MeV fm}^2$, separates bound from unbound channels. The triplet deuteron clears it ($\sim 140\,\text{MeV fm}^2$); the singlet ${}^1S_0$ channel falls just short ($\sim 93\,\text{MeV fm}^2$). Spin dependence is therefore not a matter of sign reversal but of depth crossing a quantitative threshold [6, 14].

Remark

Limits of the square well. The model ignores the tensor force (Lecture 9), spin–orbit terms, and the long-range one-pion tail (Lecture 11). It also hides the fact that realistic potentials have a repulsive core at $r \lesssim 0.5\,\mathrm{fm}$. Its value is pedagogical transparency: one sees directly how weak binding inflates r_{rms} and why scattering data (Lecture 10) are mandatory to fix the potential beyond a single binding energy.

7.8.3 Reading the matching condition graphically

Define $f(kR) = kR\cot(kR) + \gamma R$. A bound state exists when $f(kR) = 0$. As $B_d \to 0$, $\gamma \to 0$ and the solution approaches $kR = \pi/2$ from below. Increasing B_d increases γR and pulls kR slightly below $\pi/2$, deepening the interior oscillation. Plotting $V_0(R)$ for fixed B_d produces a family of acceptable wells: shallow and wide, or deep and narrow. Only additional data (phase shifts in Lecture 10, tensor observables in Lectures 8–9) select a member of that family. The square well therefore teaches *qualitative* weak-binding physics and *quantitative* threshold arithmetic, but not a unique microscopic potential [2, 6].

Key idea

Weak-binding checklist (Lecture 7).

- Matching: $k\cot(kR) = -\gamma$ with $\gamma = \sqrt{2\mu B_d}/\hbar$.
- Exterior: $P_> \sim 0.5$–0.7, $r_{\mathrm{rms}} > R$.
- Threshold: $V_0R^2 \gtrsim 109\,\mathrm{MeV\,fm}^2$ for one s-wave bound state.
- Beyond B_d: need μ_d, Q_d, and a_t (Lectures 8–10).

7.9 Further physics: only one bound NN state

There is only one bound $A = 2$ nucleus, the deuteron, and it has no bound excited states. Its binding energy is only $B_d \approx 2.225\,\mathrm{MeV}$, so any excitation puts the system into the $p + n$ continuum [14, 6, 13]. That single fact forces the entire multipole programme of Lectures 7–9: structure information must come from ground-state moments and from scattering, not from a discrete spectrum.

Modern NN potentials also include three-nucleon forces, which have no classical two-body analogue and are hard to fix from scattering alone [13, 4]. For PH5001 the two-body square well and the tensor force extracted from the deuteron remain the essential lessons.

At the QCD level, the residual nucleon–nucleon force is the long-distance face of a theory in which the strong coupling *grows* toward large distance (confinement) and becomes weak at short distance (asymptotic freedom) [15, 4]. Yukawa's pion sets the range ~ 1–$2\,\mathrm{fm}$ of the residual force used in this chapter.

7.9.1 Bound-state condition for a square well

For an s-wave square well of depth V_0 and range R,

$$V(r) = \begin{cases} -V_0 & r < R, \\ 0 & r > R, \end{cases} \tag{7.35}$$

the radial function $u(r) = r\psi(r)$ oscillates inside the well and decays as $e^{-\kappa r}$ outside, with

$$k = \frac{\sqrt{2\mu(V_0 + E)}}{\hbar}, \qquad \kappa = \frac{\sqrt{-2\mu E}}{\hbar}, \qquad E = -B_d < 0. \tag{7.36}$$

Matching u and u' at $r = R$ yields

$$k\cot(kR) = -\kappa. \tag{7.37}$$

The first bound state appears when the exterior decay constant vanishes, $\kappa \to 0^+$, so $\cot(kR) \to 0^-$ and

$$kR = \frac{\pi}{2} \qquad \text{(threshold)}. \tag{7.38}$$

At $E = 0$ one has $k = \sqrt{2\mu V_0}/\hbar$, hence at least one bound state requires

$$V_0 R^2 > \frac{\pi^2\hbar^2}{8\mu} \approx 109\,\mathrm{MeV\,fm}^2 \tag{7.39}$$

($\mu \approx m_N/2$). Fitting the deuteron gives approximately

$$V_0 R^2(s{=}1) \approx 140\,\mathrm{MeV\,fm}^2, \tag{7.40}$$

while low-energy np scattering in the singlet channel yields

$$V_0 R^2(s{=}0) \sim 90\text{–}95\,\mathrm{MeV\,fm}^2 \tag{7.41}$$

just *below* threshold. Typical phenomenological values are $V_0 \approx 47\,\mathrm{MeV}$, $R \approx 1.7\,\mathrm{fm}$ (triplet) and a shallower, wider singlet well [14, 6].

Because B_d is small, the exterior decay length $1/\kappa$ is long. Normalising the asymptotic wave $u(r) \sim Ae^{-\kappa r}$ for $r > R$, the probability outside the well is

$$P_> = \frac{\int_R^\infty A^2 e^{-2\kappa r}\,\mathrm{d}r}{\int_0^\infty u^2\,\mathrm{d}r} \sim \frac{A^2 e^{-2\kappa R}/(2\kappa)}{\text{total norm}}, \tag{7.42}$$

which is of order 50%–70% for the deuteron: most of the probability density sits *outside* the range of the force, and the rms radius exceeds R.

7.9.2 No *nn* or *pp* bound states and charge independence

Charge independence of the strong force, together with the Pauli principle, explains why only pn binds: the $s = 1$, $\ell = 0$ state is allowed for pn but forbidden for identical nn or pp; the $s = 0$ channel is unbound for all three pairs [14]. Mirror binding differences such as ^{3}H vs ^{3}He are then mostly Coulomb.

The naive reading "attractive in $s = 1$, repulsive in $s = 0$" is wrong: both channels are attractive; only the triplet is deep enough to bind [14, 6].

7.9.3 Yukawa potential and pion range

A more realistic long-range form is

$$V(r) = g\,\frac{\hbar c}{r}\,e^{-r/r_0}, \qquad r_0 = \frac{\hbar}{m_\pi c} \approx 1.4\,\text{fm}. \tag{7.43}$$

Yukawa predicted the pion mass from the force range [9, 14, 5]. In Born approximation the Fourier transform of the Yukawa potential is the propagator

$$M(q^2) \propto \frac{1}{q^2 + m_\pi^2 c^2}, \tag{7.44}$$

which reduces to a contact interaction when $m_\pi \to \infty$ (zero range). Short-range repulsion ("hard core") and multi-meson exchanges complete modern NN potentials [14, 15].

The same Yukawa logic, applied to the weak interaction with a heavy mediator, explains why the weak force looks feeble at low energy even though its underlying coupling is not tiny: $G_F \propto g^2/m_W^2$ [5]. For the strong residual force the light pion makes the range nuclear, not subnuclear.

7.9.4 Photodisintegration and the binding energy

The deuteron binding energy is measured to high precision from the threshold for

$$\gamma + {}^2\text{H} \to p + n, \tag{7.45}$$

and from the reverse radiative capture. The photon energy must supply at least $B_d \approx 2.224\,\text{MeV}$ (plus a small centre-of-mass correction). That B_d is only $\sim 0.1\%$ of the rest energy of the deuteron: the system is barely bound, which is why the wave function leaks so far outside the well [6, 14].

7.9.5 Why scattering is needed

With only one bound state, the deuteron samples the NN force mainly in the ${}^3S_1$–${}^3D_1$ channel at a fixed average separation. Nucleon–nucleon scattering (especially on hydrogen targets, to avoid multiple scattering inside a heavy nucleus) maps other partial waves, energies, and spin orientations. Phase shifts then constrain the full potential used in few-body and many-body calculations [6, 4].

7.9.6 Triplet vs singlet wells in numbers

Phenomenological square-well fits that also use low-energy scattering give distinct wells in the two spin channels [14]:

Channel	V_0 (MeV)	R (fm)	V_0R^2 (MeV fm^2)
Triplet $s = 1$ (bound)	46.7	1.73	≈ 140
Singlet $s = 0$ (unbound)	12.5	2.79	≈ 93.5
Threshold for one bound state			109

Both channels are attractive; only the triplet clears the quantum threshold $V_0R^2 > 109\,\text{MeV fm}^2$. That is the quantitative content of "spin-dependent nuclear force" at the square-well level.

7.9.7 Normalisation and the asymptotic constant

The exterior wave $u(r) \sim Ae^{-\gamma r}$ defines the **asymptotic normalisation constant** A, which sets the absolute scale of the np capture cross section and of the deuteron photo-disintegration amplitude. Low-energy NN scattering fixes the *ratio* of phase shifts (Lecture 10) but not A without a normalisation convention. In practice A is tied to the deuteron rms radius and to $\eta_D = a_D/a_S$, the D/S ratio at large r (Lecture 9). The square well makes the point transparent: a shallow bound state forces a long tail ($\gamma^{-1} \gg R$), so observables sensitive to the wave function at $r \sim 5$–$10\,\text{fm}$ (the electromagnetic radius, capture at thermal energies) depend on the *exterior* shape even though the binding energy is fixed by the interior matching.

Rms radius from weak binding

A crude estimate treats the deuteron as a sphere of radius $r_{\text{rms}} \approx 2.1\,\text{fm}$. Compare with the force range $R \sim 2\,\text{fm}$: the rms radius exceeds the well radius because $P_> \sim 0.5$–0.7. For a deeply bound state ($B \sim 30\,\text{MeV}$), $\gamma^{-1} \lesssim 1\,\text{fm}$ and r_{rms} would track R. The deuteron is the opposite limit, foreshadowing halo nuclei in Lecture 1 further material.

7.9.8 Bridge to Lectures 8–10

Lecture 7 fixes the *existence* and *weakness* of the NN bound state. Lecture 8 asks what magnetic moment that state carries; Lecture 9 asks whether the charge distribution is spherical. Neither question is answered by B_d alone. Lecture 10 adds continuum phase shifts in the singlet and triplet channels, breaking the (R, V_0) degeneracy of the square well. Together, bound and scattering data constrain the same underlying potential, with the deuteron sampling only the triplet, near-threshold corner of the full parameter space [6, 14].

7.9.9 Levinson threshold and the singlet channel

Levinson's theorem links the number of bound states to $\delta_0(0)$. The triplet channel has one bound state (deuteron), so $\delta_0^{(t)}(0) = +\pi/2$ and $a_t > 0$. The singlet channel has none, so $\delta_0^{(s)}(0) = 0$ and $a_s < 0$ with $|a_s| \gg R$: the virtual $^1S_0$ state sits just above threshold (Lecture 10). The square-well threshold $V_0R^2 \gtrsim 109\,\text{MeV fm}^2$ quantifies how close the singlet well falls short ($\sim 93\,\text{MeV fm}^2$). Spin dependence is therefore a quantitative depth difference, not a sign flip of the force.

7.9.10 Capture cross section and the long tail

Radiative np capture at thermal energies proceeds through an s-wave np collision and emission of a real photon. The cross section scales with the square of the asymptotic wave function at the origin and with $1/v$ (standard $1/v$ law for slow neutrons). Because the deuteron wave function extends to $\sim 5\,\mathrm{fm}$, the capture rate is enhanced relative to a compact bound state of the same B_d. This is another manifestation of exterior dominance: weak binding inflates the tail, which controls both scattering lengths (Lecture 10) and radiative processes [6, 14].

7.10 Deeper reading: matching, asymptotic normalisation, and halos

The square-well deuteron of Lecture 7 is solved by matching logarithmic derivatives at the well radius R. Writing $\gamma = \sqrt{2\mu B_d}/\hbar$ for the exterior decay constant, the asymptotic wave function $u(r) \propto e^{-\gamma r}$ dominates the normalisation when B_d is small. The exterior probability

$$P_> = \int_R^\infty |u|^2\,\mathrm{d}r \Big/ \int_0^\infty |u|^2\,\mathrm{d}r \tag{7.46}$$

approaches unity as $B_d \to 0$: that is the halo limit. For the physical deuteron, $B_d = 2.224\,\mathrm{MeV}$ and $P_>$ is already large, which is why low-energy np scattering and the deuteron radius are tightly linked through the scattering length (Lecture 10).

A one-parameter well that fits only B_d cannot predict both the radius and the D-wave quadrupole moment (Lectures 8–9). Effective field theory reorganises that underdetermination into a power counting with truncation errors. The pedagogical moral remains: weak binding amplifies long-distance physics and suppresses sensitivity to short-distance details, until observables that probe the tensor force or short-range currents are considered.

Connect forward: the same exterior dominance appears in halo nuclei near drip lines (Lecture 13) and in the need for continuum coupling in reactions (Lecture 21). Connect back: saturation in Lecture 2 requires attractions strong enough to bind bulk matter, yet the deuteron remains only weakly bound because of the spin–isospin structure of the force.

Effective range and asymptotic normalisation

The asymptotic normalisation coefficient (ANC) of the deuteron wave function controls peripheral transfer reactions that attach a proton or neutron to a target (Lecture 21). Because the exterior wave is fixed by B_d and the scattering length, ANCs are more stable than short-range spectroscopic factors when the reaction is peripheral. Students should connect the Lecture 7 matching condition to that exterior normalisation before treating transfer as a pure structure measurement.

Tensor force preview

Even before Lecture 9, note that a central square well cannot generate a quadrupole moment. Any observable sensitive to tensor structure (Q_d, some polarisation observables) requires physics beyond the toy well. Keep a list of which deuteron observables are long-distance dominated (B_d, radius, scattering length) and which need interior tensor physics (μ_d details, Q_d, photodisintegration at higher energy).

Three-body force teaser

The triton and ^{3}He binding energies cannot be reproduced from two-body forces alone fitted to NN scattering. That fact, matured in Lecture 11, already warns that the deuteron's success as a two-body system does not license a pure two-body description of bulk matter (Lecture 2).

Numerical matching exercise

Choose $R = 2.0\,\text{fm}$ and solve for the well depth V_0 that reproduces $B_d = 2.224\,\text{MeV}$ with reduced mass $\mu = m_N/2$. Then repeat for $R = 1.5$ and $R = 2.5\,\text{fm}$. The curve $V_0(R)$ falls as R grows: many wells share one binding energy. Overlay the mean-square radius for each solution and observe that radius and depth move together. That plot is the geometric content of Lecture 7's underdetermination statement.

Next, estimate the asymptotic decay constant $\gamma = \sqrt{2\mu B_d}/\hbar \approx 0.23\,\text{fm}^{-1}$ and the exterior scale $1/\gamma \approx 4.3\,\text{fm}$. Most of the probability density sits outside a 2 fm well, which is why low-energy scattering (Lecture 10) and peripheral transfer (Lecture 21) care about the asymptotic normalisation more than about the interior well shape. When Lecture 9 introduces the tensor force, interior physics returns for Q_d; until then, long-distance deuteron physics is deliberately simple.

Research frontier: EFT deuteron physics and student projects

The square-well deuteron of Lecture 7 teaches matching of interior and exterior wave functions and the consequences of weak binding. Modern few-body physics replaces that toy well by pionless and chiral EFTs that expand systematically in momenta over the breakdown scale, with quantified truncation errors [43, 44]. Lattice QCD is beginning to constrain low-energy constants that enter those EFTs, while precision deuteron observables (rms radius, quadrupole moment, photodisintegration, and electrodisintegration) benchmark the force. The pedagogical lesson survives: one binding energy cannot fix two potential parameters without additional data or a power-counting principle. Open research questions include the consistent inclusion of electromagnetic currents at the same chiral order as the Hamiltonian, the size of relativistic corrections, and how far pionless EFT can be pushed before explicit pions become mandatory. Connecting lattice-QCD phase shifts and deuteron properties to chiral LECs remains a multi-year programme rather than a finished calculation.

Students should also notice the difference between a potential that merely fits B_d and an EFT that predicts a tower of observables with truncation errors. Comparing a square-well deuteron to a published chiral calculation of the same radius and quadrupole moment is a concrete way to see what "systematic improvability" means in practice.

Student directions. (1) Computational: solve the square-well matching condition for several

R and plot the $V_0(R)$ curve. (2) Analytic limit: estimate the exterior probability $P_>$ as $B_d \to 0$ and discuss halo formation. (3) Literature reading: from a chiral-EFT review, list which orders first generate a tensor force and a three-nucleon force [43].

7.11 Problems with solutions

Problem 7.1 — Binding energy of the deuteron. Atomic masses: $m(^1\mathrm{H}) = 1.007825\,\mathrm{u}$, $m(n) = 1.008665\,\mathrm{u}$, $m(^2\mathrm{H}) = 2.014102\,\mathrm{u}$. With $c^2 = 931.5\,\mathrm{MeV/u}$, compute B_d and B_d/A.

Solution 7.1.

$$\begin{aligned} B_d &= [m(^1\mathrm{H}) + m(n) - m(^2\mathrm{H})]c^2 \\ &= (1.007825 + 1.008665 - 2.014102) \times 931.5\,\mathrm{MeV} \\ &= 0.002388 \times 931.5\,\mathrm{MeV} \approx 2.224\,\mathrm{MeV}. \end{aligned}$$

$$B_d/A \approx 1.11\,\mathrm{MeV},$$

much less than the mid-mass $\sim 8\,\mathrm{MeV}$ per nucleon: the deuteron is barely bound [6, 14].

Problem 7.2 — Solve the labelled square well. For $R = 2.00\,\mathrm{fm}$, $B_d = 2.2246\,\mathrm{MeV}$, and $\hbar^2/(2\mu) = 41.5\,\mathrm{MeV\,fm^2}$, solve $\xi\cot\xi = -\eta$, where $\eta = R\sqrt{B_d/[\hbar^2/(2\mu)]}$, on the first-state branch $\pi/2 < \xi < \pi$. Determine V_0.

Solution 7.2.

$$\eta = 2.00\sqrt{\frac{2.2246}{41.5}} = 0.463.$$

Numerical solution of $\xi\cot\xi = -0.463$ gives $\xi = 1.820$. Hence

$$V_0 = B_d + \frac{\hbar^2}{2\mu}\frac{\xi^2}{R^2} = 2.2246 + 41.5\frac{(1.820)^2}{(2.00)^2} \approx 36.6\,\mathrm{MeV}.$$

This value belongs to the explicitly labelled $R = 2.00\,\mathrm{fm}$ teaching model; B_d alone fixes a curve $V_0(R)$, not a unique potential [6].

Problem 7.3 — Bound-state condition V_0R^2. For an s-wave square well, at least one bound state requires $V_0R^2 > \pi^2\hbar^2/(8\mu) \approx 109\,\mathrm{MeV\,fm^2}$. (a) With $R = 1.73\,\mathrm{fm}$ and $V_0 = 46.7\,\mathrm{MeV}$ (triplet fit), compute V_0R^2 and compare with $109\,\mathrm{MeV\,fm^2}$. (b) The singlet channel has $V_0R^2 \sim 93.5\,\mathrm{MeV\,fm^2}$. What does that imply?

Solution 7.3. (a)

$$V_0R^2 = 46.7 \times (1.73)^2 \approx 46.7 \times 2.99 \approx 140\,\mathrm{MeV\,fm^2} > 109\,\mathrm{MeV\,fm^2}.$$

The triplet well is above threshold and binds the deuteron ($V_0R^2 \approx 140\,\mathrm{MeV\,fm^2}$ reproduces $B_d = 2.225\,\mathrm{MeV}$) [14].

(b) $93.5 < 109$: the singlet well is just *below* threshold. There is no bound ${}^1S_0$ pn state; low-energy singlet scattering shows a virtual state [14, 6].

Problem 7.4 — Normalisation and exterior probability. With $B_d = 2.224\,\mathrm{MeV}$ and $\hbar^2/m_N = 41.5\,\mathrm{MeV\,fm}^2$, compute $\gamma = \sqrt{2\mu B_d}/\hbar = \sqrt{m_N B_d}/\hbar$ and the decay length $1/\gamma$. For $R = 2.00\,\mathrm{fm}$ and $kR = 1.820$, use

$$A^{-2} = \frac{R}{2} - \frac{\sin(2kR)}{4k} + \frac{\sin^2(kR)}{2\gamma}, \qquad P_> = \frac{A^2\sin^2(kR)}{2\gamma}$$

to find the probability outside the well.

Solution 7.4.

$$\gamma = \sqrt{\frac{m_N B_d}{\hbar^2}} = \sqrt{\frac{B_d}{\hbar^2/m_N}} = \sqrt{\frac{2.224}{41.5}} \approx \sqrt{0.0536} \approx 0.231\,\mathrm{fm}^{-1}.$$

$$\frac{1}{\gamma} \approx 4.3\,\mathrm{fm} > R = 2.00\,\mathrm{fm}.$$

With $k = 1.820/R = 0.910\,\mathrm{fm}^{-1}$,

$$A^{-2} \approx 3.16\,\mathrm{fm}, \qquad P_> \approx 0.642.$$

About 64% of the radial probability lies outside the force range: the deuteron is a loosely bound, extended system [6, 14].

Problem 7.5 — Thermal deuterons. Estimate the temperature at which two deuterons can approach to $r \sim 1\,\mathrm{fm}$ against Coulomb repulsion. Use $k_B = 8.6\times10^{-5}\,\mathrm{eV/K}$ and $\alpha\,\hbar c \approx 1.44\,\mathrm{MeV\,fm}$.

Solution 7.5. In the rest frame of one deuteron the other approaches with relative velocity $2v$, so

$$4k_BT \sim \frac{\alpha\,\hbar c}{r} \approx \frac{1.44}{1} = 1.44\,\mathrm{MeV}.$$

$$T \sim \frac{1.44\times10^{6}\,\mathrm{eV}}{4\times8.6\times10^{-5}\,\mathrm{eV/K}} \sim 4\times10^{9}\,\mathrm{K}.$$

Nuclear contact among charged light nuclei requires stellar temperatures [3].

Problem 7.6 — Photodisintegration threshold. The $\gamma + {}^2\mathrm{H} \to p + n$ threshold is essentially B_d plus a small centre-of-mass correction. If a photon of energy $E_\gamma = 2.50\,\mathrm{MeV}$ breaks a deuteron at rest, what is the total kinetic energy of the $p+n$ system in the lab (neglect recoil of the compound system before breakup, to leading order)?

Solution 7.6. Energy conservation:

$$E_\gamma = B_d + T_p + T_n \quad\Rightarrow\quad T_p + T_n = 2.50 - 2.224 = 0.276\,\mathrm{MeV}.$$

The excess photon energy above threshold appears as kinetic energy of the two nucleons [6].

Problem 7.7 — Binding energy in MeV from mass defect. Using $m_d = 2.014102\,\mathrm{u}$, $m_p = 1.007825\,\mathrm{u}$, $m_n = 1.008665\,\mathrm{u}$, and $1\,\mathrm{u} = 931.494\,\mathrm{MeV}/c^2$, compute B_d.

Solution 7.7.

$$\Delta m = (m_p + m_n - m_d) = 0.002388\,\mathrm{u}, \quad B_d = \Delta m \times 931.494 \approx 2.224\,\mathrm{MeV},$$

the standard deuteron binding energy [6, 27].

Chapter 8

Deuteron Spin and Magnetic Moment

8.1 Nuclear spin and parity: careful definitions

The binding energy of the deuteron fixes the radial scale of a central NN well, under the assumption that the relative orbital angular momentum is $\ell = 0$. That assumption must now be tested with every other measured property of the ground state: the total angular momentum (nuclear spin), the parity, and the magnetic dipole moment [6, 2]. If those observables demanded $\ell \neq 0$, the simple central-force picture of Lectures 6–7 would have to be enlarged—and in fact the magnetic and quadrupole moments force a small D-wave component once the tensor force is admitted (Lectures 8–9).

8.1.1 What "nuclear spin" means

Each nucleon carries orbital angular momentum $\boldsymbol{\ell}$ and intrinsic spin $\boldsymbol{s} = \frac{1}{2}\hbar$ (in units where we often set $\hbar = 1$ for quantum numbers). Their sum is

$$\boldsymbol{j} = \boldsymbol{\ell} + \boldsymbol{s}. \tag{8.1}$$

In a nucleus the total angular momentum is the vector sum of all nucleon contributions. Nuclear physicists call this total $\boldsymbol{I}$ and name it the **nuclear spin**, even though it includes orbital motion as well as spin. The usual quantum rules apply:

$$\boldsymbol{I}^2 \to I(I+1)\hbar^2, \qquad I_z \to M_I\hbar, \quad M_I = -I, \dots, +I. \tag{8.2}$$

When two angular momenta j_1 and j_2 are coupled, the allowed total values follow the triangle rule

$$J = |j_1 - j_2|,\ |j_1 - j_2| + 1, \dots,\ j_1 + j_2. \tag{8.3}$$

This rule will be applied twice: first $\boldsymbol{S} = \boldsymbol{s}_p + \boldsymbol{s}_n$, then $\boldsymbol{I} = \boldsymbol{L} + \boldsymbol{S}$.

8.1.2 Parity

Parity is the behaviour of the spatial wave function under $\boldsymbol{r} \to -\boldsymbol{r}$. If $\psi(-\boldsymbol{r}) = +\psi(\boldsymbol{r})$ the parity is even ($\pi = +$); if $\psi(-\boldsymbol{r}) = -\psi(\boldsymbol{r})$ it is odd ($\pi = -$). A central-potential wave function separates as

$$\psi_{\ell m}(\boldsymbol{r}) = R_\ell(r) Y_{\ell m}(\theta, \phi). \tag{8.4}$$

Inversion leaves $r = |\boldsymbol{r}|$ unchanged but sends $(\theta, \phi) \to (\pi - \theta, \phi + \pi)$. Spherical harmonics obey

$$Y_{\ell m}(\pi - \theta, \phi + \pi) = (-1)^\ell Y_{\ell m}(\theta, \phi). \tag{8.5}$$

Therefore

$$\psi_{\ell m}(-\boldsymbol{r}) = (-1)^\ell \psi_{\ell m}(\boldsymbol{r}), \qquad \pi = (-1)^\ell. \tag{8.6}$$

Even ℓ gives even parity; odd ℓ gives odd parity.

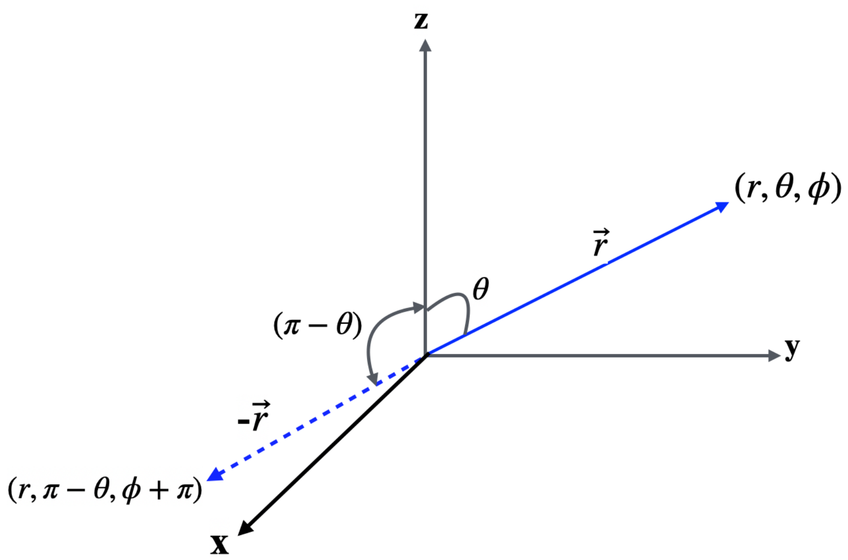


Figure 8.1: Inversion $\boldsymbol{r} \leftrightarrow -\boldsymbol{r}$. Nuclear states have definite parity when the Hamiltonian conserves parity.

8.1.3 Experimental assignment for the deuteron

Experiment gives

$$I^\pi(^2\mathrm{H}) = 1^+. \tag{8.7}$$

That single label will eliminate most of the angular-momentum possibilities we might have imagined.

8.2 Coupling two spins: four ways to make $I = 1$

The deuteron total angular momentum is

$$\boldsymbol{I} = \boldsymbol{s}_p + \boldsymbol{s}_n + \boldsymbol{L}, \tag{8.8}$$

where $\boldsymbol{L}$ is the relative orbital angular momentum. Each nucleon has $s = \frac{1}{2}$, so the total spin $\boldsymbol{S} = \boldsymbol{s}_p + \boldsymbol{s}_n$ obeys

$$S = \left|\tfrac{1}{2} - \tfrac{1}{2}\right|, \ldots, \tfrac{1}{2} + \tfrac{1}{2} = 0,\ 1. \tag{8.9}$$

For fixed S and L, the second application of Equation (8.3) gives

$$|L - S| \leq I \leq L + S. \tag{8.10}$$

We require $I = 1$:

- If $S = 0$, then $I = L$, so only $L = 1$ works.
- If $S = 1$, then $|L - 1| \leq 1 \leq L + 1$, which permits $L = 0, 1, 2$.

Thus there are exactly four couplings [6]:

(a) spins parallel ($S = 1$) with $L = 0$;

(b) spins antiparallel ($S = 0$) with $L = 1$;

(c) spins parallel ($S = 1$) with $L = 1$;

(d) spins parallel ($S = 1$) with $L = 2$.

Parity decides among them. The deuteron has *even* parity, and orbital parity is $(-1)^L$. Therefore L must be even: $L = 1$ is ruled out. Cases (b) and (c) drop. We are left with

$$^3S_1 \quad (L = 0,\ S = 1) \qquad \text{and} \qquad ^3D_1 \quad (L = 2,\ S = 1). \tag{8.11}$$

The bound state is **100% spin triplet**. There is no bound singlet pn state.

The notation $^{2S+1}L_I$ now follows directly. The superscript is the spin multiplicity $2S+1$, the letter denotes $L = 0, 1, 2, \ldots \to S, P, D, \ldots$, and the subscript is the total angular momentum I. Hence the two even-parity possibilities are $^3S_1$ and $^3D_1$.

For later expectation values, the stretched states also fix the Clebsch–Gordan convention explicitly:

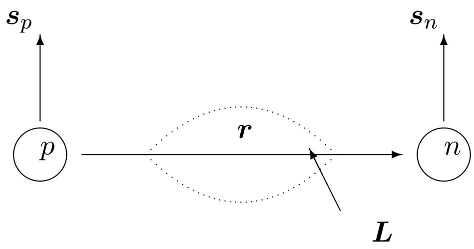
$$|^3S_1; 1, 1\rangle = |L{=}0, m_L{=}0\rangle|S{=}1, m_S{=}1\rangle, \tag{8.12}$$

$$|^3D_1; 1, 1\rangle = \sqrt{\frac{1}{10}}\,|2, 0\rangle|1, 1\rangle - \sqrt{\frac{3}{10}}\,|2, 1\rangle|1, 0\rangle + \sqrt{\frac{3}{5}}\,|2, 2\rangle|1, -1\rangle. \tag{8.13}$$

The squared coefficients sum to one. They give $\langle L_z\rangle = 3\hbar/2$ and $\langle S_z\rangle = -\hbar/2$, in agreement with the projection theorem below.

Key idea

Spin dependence of the nuclear force. Nature binds the triplet channel and not the singlet. The NN force depends on the relative orientation of the nucleon spins. A spin-independent central well would not explain that fact.



$I^\pi = 1^+ :\ ^3S_1$ dominant + small $^3D_1$

Figure 8.2: Schematic spins and orbital angular momentum in the NN system. The deuteron ground state is dominated by parallel spins and $L = 0$, with a small $L = 2$ admixture.

8.2.1 Triplet and singlet spin states

Start with the unique highest-projection product state,

$$|1,1\rangle = |{\uparrow\uparrow}\rangle. \tag{8.14}$$

Apply the total lowering operator $S_- = s_{p-} + s_{n-}$. Since $S_-|1,1\rangle = \hbar\sqrt{2}\,|1,0\rangle$,

$$S_-|{\uparrow\uparrow}\rangle = \hbar|{\downarrow\uparrow}\rangle + \hbar|{\uparrow\downarrow}\rangle, \tag{8.15}$$

$$\therefore \quad |1,0\rangle = \frac{1}{\sqrt{2}}\left(|{\uparrow\downarrow}\rangle + |{\downarrow\uparrow}\rangle\right). \tag{8.16}$$

A second lowering gives $|1,-1\rangle = |{\downarrow\downarrow}\rangle$. The three triplet states ($S = 1$) are therefore

$$|1,1\rangle = |{\uparrow\uparrow}\rangle, \qquad |1,0\rangle = \tfrac{1}{\sqrt{2}}\big(|{\uparrow\downarrow}\rangle + |{\downarrow\uparrow}\rangle\big), \qquad |1,-1\rangle = |{\downarrow\downarrow}\rangle. \tag{8.17}$$

The remaining normalized $M_S = 0$ combination must be orthogonal to $|1,0\rangle$; it is the singlet

$$|0,0\rangle = \tfrac{1}{\sqrt{2}}\big(|{\uparrow\downarrow}\rangle - |{\downarrow\uparrow}\rangle\big). \tag{8.18}$$

A useful operator that distinguishes them is $\boldsymbol{s}_p \cdot \boldsymbol{s}_n$. Square $\boldsymbol{S} = \boldsymbol{s}_p + \boldsymbol{s}_n$:

$$\boldsymbol{S}^2 = \boldsymbol{s}_p^2 + \boldsymbol{s}_n^2 + 2\boldsymbol{s}_p \cdot \boldsymbol{s}_n, \tag{8.19}$$

$$\therefore \quad \boldsymbol{s}_p \cdot \boldsymbol{s}_n = \frac{1}{2}\left(\boldsymbol{S}^2 - \boldsymbol{s}_p^2 - \boldsymbol{s}_n^2\right). \tag{8.20}$$

Because $s_p = s_n = \frac{1}{2}$, $\boldsymbol{s}_p^2 = \boldsymbol{s}_n^2 = \frac{3}{4}\hbar^2$. Thus

$$\langle \boldsymbol{s}_p \cdot \boldsymbol{s}_n \rangle = \frac{\hbar^2}{2}\left[S(S+1) - \frac{3}{4} - \frac{3}{4}\right] = \begin{cases} +\dfrac{\hbar^2}{4}, & S = 1, \\ -\dfrac{3\hbar^2}{4}, & S = 0. \end{cases} \tag{8.21}$$

A potential containing $\boldsymbol{\sigma}_p \cdot \boldsymbol{\sigma}_n$ can therefore deepen the triplet well relative to the singlet. That is why the square-well depth of Lecture 7 applies to the *triplet* channel; the singlet well is shallower and does not bind.

Caution

The singlet value is not zero. It is negative ($-3\hbar^2/4$). Saying the singlet product "vanishes" is a common slip; the correct eigenvalues matter when one writes $V_\sigma(r)$.

8.3 Magnetic moments: from classical orbits to g-factors

8.3.1 Nuclear magneton

Consider a charge e moving with speed v in a circular orbit of radius r. Its period, current, and loop area are

$$T = \frac{2\pi r}{v}, \qquad \mathcal{I} = \frac{e}{T} = \frac{ev}{2\pi r}, \qquad \mathcal{A} = \pi r^2. \tag{8.22}$$

The magnetic dipole moment is current times area,

$$\mu = \mathcal{I}\mathcal{A} = \frac{evr}{2}. \tag{8.23}$$

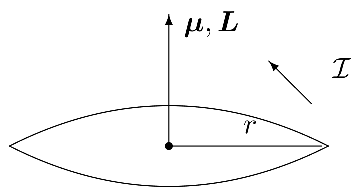
Since the orbital angular momentum is $L = mvr$, this becomes

$$\boldsymbol{\mu}_\ell = \frac{e}{2m}\boldsymbol{L}. \tag{8.24}$$

For a proton, set $m = m_p$ and factor out one quantum $\hbar$. This defines the **nuclear magneton**

$$\mu_N = \frac{e\hbar}{2m_p} \approx 3.15245 \times 10^{-8}\,\mathrm{eV/T}. \tag{8.25}$$

The displayed decimal is a rounded recommended-constant value; CODATA updates should be used when metrological precision matters. The lecture retains only the digits needed for the percent-level structure argument. It is smaller than the Bohr magneton by $m_e/m_p \approx 1/1836$. Ordinary magnetism of matter is electronic; nuclear magnetism is a delicate probe [6].



$\mu = \mathcal{I}\pi r^2 = eL/(2m)$

Figure 8.3: Classical orbital current: charge on a loop produces $\mu \propto qL/(2m)$. Neutrons have $q = 0$ and no orbital moment from net charge.

8.3.2 Orbital and spin contributions

We write

$$\mu_\ell = g_\ell\,\mu_N\,\frac{\ell_z}{\hbar}, \qquad \mu_s = g_s\,\mu_N\,\frac{s_z}{\hbar}. \tag{8.26}$$

For proton orbital motion $g_\ell = 1$; for the neutron $g_\ell = 0$. For free nucleon *spins*, measured magnetic moments, expressed in the CODATA-defined nuclear-magneton unit, give anomalous g-factors far from the Dirac value $g_s = 2$. Rounded for this calculation,

$$g_p = 5.5857, \tag{8.27}$$

$$g_n = -3.8261. \tag{8.28}$$

These are experimental nucleon properties, not predictions of CODATA; CODATA supplies the consistent constants and unit conversion. Equivalently, $\mu_p = +2.7928\,\mu_N$ and $\mu_n = -1.9130\,\mu_N$. The neutron's nonzero moment and the proton's large g_p are early evidence that nucleons are composite (meson clouds; today, quarks) [6, 2].

8.4 Deuteron magnetic moment: pure S-wave calculation

If the orbital angular momentum vanishes, there is no orbital magnetic moment: only the intrinsic spins of the proton and neutron contribute. With $I^\pi = 1^+$ and a pure ${}^3S_1$ configuration, the stretched state $M_I = 1$ has both spins aligned ($s_z = +\hbar/2$). The deuteron moment is then simply the sum of the free-nucleon moments [6],

$$\mu_d(L{=}0) = \mu_p + \mu_n = \frac{g_p + g_n}{2}\,\mu_N = \frac{5.5857 - 3.8261}{2}\,\mu_N = 0.8798\,\mu_N. \tag{8.29}$$

Experiment finds

$$\mu_d^{(\mathrm{exp})} = 0.8574376(4)\,\mu_N, \tag{8.30}$$

about $0.022\,\mu_N$ ($\sim 2.5\%$) below the pure S-wave value. The agreement is already striking: most of μ_d is indeed $\mu_p + \mu_n$. The residual deficit, however, is a genuine structural effect. Meson-exchange currents and relativistic corrections can shift the moment, but within the nonrelativistic NN picture the natural explanation is a small admixture of $L = 2$ in the ground-state wave function [6].

Key idea

Physics of the discrepancy. A pure s-wave deuteron would have $\mu_d = \mu_p + \mu_n$. The measured moment is slightly smaller, signalling a few-percent D-wave component and, with it, a non-central piece of the force.

8.5 D-wave admixture from μ_d

Write the ground state as a coherent mixture of even orbital waves allowed by $I^\pi = 1^+$,

$$|\psi\rangle = a_S\,|L{=}0\rangle + a_D\,|L{=}2\rangle, \qquad |a_S|^2 + |a_D|^2 = 1. \tag{8.31}$$

8.5.1 Expectation value of the magnetic-moment operator

The numerical D-state moment should not simply be quoted; it follows from an angular-momentum expectation value. In the centre-of-mass frame, and neglecting exchange currents, the z-component of the two-nucleon magnetic-moment operator is

$$\frac{\hat{\mu}_z}{\mu_N} = \frac{1}{2}\frac{\hat{L}_z}{\hbar} + g_p\frac{\hat{s}_{pz}}{\hbar} + g_n\frac{\hat{s}_{nz}}{\hbar}. \tag{8.32}$$

Only the proton contributes an orbital magnetic moment. Here $\boldsymbol{L} = \boldsymbol{r} \times \boldsymbol{p}$ always denotes the *relative* orbital angular momentum, not either particle's orbital momentum about an arbitrary laboratory origin. Its relation to the centre-of-mass orbital momenta can be obtained explicitly. Let $M = m_p + m_n$, $\boldsymbol{r} = \boldsymbol{r}_p - \boldsymbol{r}_n$, and let $\boldsymbol{p}$ be the relative momentum in the centre-of-mass frame. Then

$$\boldsymbol{r}_p = \frac{m_n}{M}\boldsymbol{r}, \qquad \boldsymbol{p}_p = \boldsymbol{p}, \tag{8.33}$$

so

$$\boldsymbol{L}_p = \boldsymbol{r}_p \times \boldsymbol{p}_p = \frac{m_n}{M}\,\boldsymbol{r} \times \boldsymbol{p} = \frac{m_n}{M}\boldsymbol{L} \simeq \frac{1}{2}\boldsymbol{L}. \tag{8.34}$$

Similarly $\boldsymbol{L}_n = (m_p/M)\boldsymbol{L}$, but the neutron has zero one-body orbital charge factor. Thus the exact coefficient of L_z is m_n/M, approximated by 1/2 in Equation (8.32). This states the CM orbital convention completely. Within the spin-triplet state the proton and neutron share the total spin symmetrically. Equivalently, the antisymmetric operator $\boldsymbol{s}_p - \boldsymbol{s}_n$ has zero diagonal matrix element between symmetric triplet spin states, and therefore

$$\langle s_{pz}\rangle = \langle s_{nz}\rangle = \frac{1}{2}\langle S_z\rangle. \tag{8.35}$$

The projection theorem itself follows in two steps. First, $\boldsymbol{J} = \boldsymbol{L} + \boldsymbol{S}$ implies

$$\boldsymbol{L}\cdot\boldsymbol{J} = \boldsymbol{L}\cdot(\boldsymbol{L}+\boldsymbol{S}) = \boldsymbol{L}^2 + \boldsymbol{L}\cdot\boldsymbol{S}, \tag{8.36}$$

$$\boldsymbol{J}^2 = \boldsymbol{L}^2 + \boldsymbol{S}^2 + 2\boldsymbol{L}\cdot\boldsymbol{S}. \tag{8.37}$$

Eliminating $\boldsymbol{L}\cdot\boldsymbol{S}$ gives

$$\boldsymbol{L}\cdot\boldsymbol{J} = \frac{1}{2}\left(\boldsymbol{J}^2 + \boldsymbol{L}^2 - \boldsymbol{S}^2\right). \tag{8.38}$$

In a state $|LSJM\rangle$, rotational symmetry says that the expectation value of $\boldsymbol{L}$ can point only along $\boldsymbol{J}$. Its z-projection is consequently

$$\langle L_z\rangle = \frac{\langle \boldsymbol{L}\cdot\boldsymbol{J}\rangle}{J(J+1)\hbar^2}\,\langle J_z\rangle. \tag{8.39}$$

Using $\langle J_z\rangle = M\hbar$ and the eigenvalues of the squared angular momenta in Equation (8.38) gives

$$\langle L_z\rangle = M\hbar\,\frac{J(J+1) + L(L+1) - S(S+1)}{2J(J+1)}, \tag{8.40}$$

$$\langle S_z\rangle = M\hbar\,\frac{J(J+1) + S(S+1) - L(L+1)}{2J(J+1)}. \tag{8.41}$$

The second line can be obtained either by interchanging L and S or from $\langle S_z\rangle = M\hbar - \langle L_z\rangle$. The observable deuteron moment is evaluated in the stretched state $J = M = 1$.

For $^3S_1$, $L = 0$ and $S = 1$, so

$$\frac{\langle L_z \rangle}{\hbar} = 1\,\frac{1(1+1) + 0(0+1) - 1(1+1)}{2(1)(1+1)} = \frac{2+0-2}{4} = 0, \tag{8.42}$$

$$\frac{\langle S_z \rangle}{\hbar} = 1\,\frac{1(1+1) + 1(1+1) - 0(0+1)}{2(1)(1+1)} = \frac{2+2-0}{4} = 1. \tag{8.43}$$

Equation (8.32) therefore reproduces

$$\begin{aligned} \mu(^3S_1) &= \left[\frac{1}{2}(0) + \frac{g_p + g_n}{2}(1)\right]\mu_N \\ &= \frac{g_p + g_n}{2}\,\mu_N = \mu_p + \mu_n = 0.8798\,\mu_N. \end{aligned} \tag{8.44}$$

For $^3D_1$, $L = 2$, $S = 1$, and $J = M = 1$. Equations (8.40)–(8.41) give, step by step,

$$\frac{\langle L_z \rangle}{\hbar} = 1\,\frac{1(1+1) + 2(2+1) - 1(1+1)}{2(1)(1+1)} = \frac{2+6-2}{4} = \frac{3}{2}, \tag{8.45}$$

$$\frac{\langle S_z \rangle}{\hbar} = 1\,\frac{1(1+1) + 1(1+1) - 2(2+1)}{2(1)(1+1)} = \frac{2+2-6}{4} = -\frac{1}{2}. \tag{8.46}$$

The orbital and spin angular momenta are therefore partly antiparallel in the D state. Substitution into (8.32) yields

$$\begin{aligned} \mu(^3D_1) &= \left[\frac{1}{2}\left(\frac{3}{2}\right) + \frac{g_p + g_n}{2}\left(-\frac{1}{2}\right)\right]\mu_N \\ &= \frac{1}{4}(3 - g_p - g_n)\,\mu_N \approx 0.310\,\mu_N. \end{aligned} \tag{8.47}$$

This is the expectation-value derivation of the smaller pure-D magnetic moment [6, 2].

In the one-body operator approximation, $\langle ^3S_1|\hat{\mu}_z|^3D_1\rangle = 0$: the angular functions with $L = 0$ and $L = 2$ are orthogonal, and the operator in (8.32) does not change L. The mixed-state expectation value can be expanded before the cross terms are discarded:

$$\begin{aligned} \langle \psi|\hat{\mu}_z|\psi\rangle &= |a_S|^2\langle ^3S_1|\hat{\mu}_z|^3S_1\rangle + |a_D|^2\langle ^3D_1|\hat{\mu}_z|^3D_1\rangle \\ &\quad + a_S^* a_D\langle ^3S_1|\hat{\mu}_z|^3D_1\rangle + a_D^* a_S\langle ^3D_1|\hat{\mu}_z|^3S_1\rangle \\ &= P_S\mu_S + P_D\mu_D. \end{aligned} \tag{8.48}$$

Therefore

$$\mu_d = P_S\,(0.880\,\mu_N) + P_D\,(0.310\,\mu_N), \qquad P_S + P_D = 1. \tag{8.49}$$

To obtain the commonly used form, substitute $P_S = 1 - P_D$ and $\mu_D = \frac{1}{4}(3 - g_p - g_n)\mu_N$:

$$\begin{aligned} \mu_d &= (1 - P_D)(\mu_p + \mu_n) + P_D\frac{3 - g_p - g_n}{4}\mu_N \\ &= (\mu_p + \mu_n) + P_D\left[\frac{3}{4}\mu_N - \frac{3}{2}(\mu_p + \mu_n)\right] \\ &= (\mu_p + \mu_n) - \frac{3}{2}\left[\mu_p + \mu_n - \frac{1}{2}\mu_N\right]P_D. \end{aligned} \tag{8.50}$$

Finally, requiring $\mu_d = 0.8574\,\mu_N$ and retaining the displayed rounded moments gives

$$
\begin{aligned}
0.8574 &= 0.8798(1 - P_D) + 0.310 P_D \\
&= 0.8798 - (0.8798 - 0.310) P_D, \qquad (8.51)\\
(0.8798 - 0.310) P_D &= 0.8798 - 0.8574, \\
P_D &= \frac{0.8798 - 0.8574}{0.8798 - 0.310} = 0.0393 \approx 0.039, \qquad P_S = 1 - P_D \approx 0.961. \qquad (8.52)
\end{aligned}
$$

This classroom estimate suggests a few-percent D-wave. It is *not* a model-independent measurement of P_D: meson-exchange currents and relativistic corrections also shift μ_d. Independent NN scattering analyses still find a few-percent D-state probability, so the qualitative conclusion is robust: the force cannot be purely central. Only a tensor interaction can mix $L = 0$ with $L = 2$ in a single eigenstate. The electric quadrupole moment $Q_d \neq 0$ (Lecture 9) establishes that mixing far more cleanly than the magnetic-moment deficit alone [6].

Caution

Do not promote $P_D \approx 4\%$ to an observable. Treat it as a pedagogical estimate that motivates looking for a tensor force; the decisive geometric evidence is the nonzero quadrupole moment.

Key idea

What Lectures 7–9 establish together. Weak binding fixes the depth and range of the residual force. Spin and parity fix the channel as mostly ${}^3S_1$. The magnetic-moment deficit *suggests* a small D-wave; the quadrupole moment *requires* one, and with it a tensor force.

8.6 Preview of heavier nuclei

In odd-A nuclei, pairing often cancels the moments of all but the last unpaired nucleon. Plotting μ versus I produces the Schmidt lines. Real nuclei fall between those lines because of configuration mixing, the many-body cousin of the deuteron's D-wave. The deuteron is the cleanest few-body example of the same idea: free moments first, then a small admixture [6, 1].

Takeaway

Takeaways (Lecture 8).

- $I^\pi = 1^+$: even parity forces even L; the bound state is spin triplet.
- Spin-dependent force: triplet binds, singlet does not.
- Pure S-wave $\mu_d = 0.880\,\mu_N$; experiment $0.857\,\mu_N$.
- Mixture $\sim 96\%\ S + \sim 4\%\ D$ is a classroom estimate of the magnetic deficit; quadrupole data (next lecture) establish the D-wave and the tensor force.

8.6.1 Nearly additive magnetic moment of the deuteron

To a first approximation $\mu_d \approx \mu_p + \mu_n \approx 0.880\,\mu_N$, as expected if orbital contributions vanish in a pure $^3S_1$ state. The measured $0.857\,\mu_N$ is close but not identical: the residual deficit is naturally attributed to a few-percent D-wave admixture (and, at a finer level, meson-exchange currents) [14, 6].

8.7 Deeper physics: magnetic moments beyond the deuteron

8.7.1 How μ is measured (Rabi and NMR)

Place a state of angular momentum I in a uniform field $\boldsymbol{B} = B\hat{\boldsymbol{z}}$. With $\hat{\mu}_z = g_I \mu_N \hat{I}_z/\hbar$, the Zeeman Hamiltonian is

$$\hat{H}_Z = -\hat{\boldsymbol{\mu}} \cdot \boldsymbol{B} = -g_I \mu_N B \frac{\hat{I}_z}{\hbar}. \tag{8.53}$$

Because $\hat{I}_z|I, M_I\rangle = M_I \hbar |I, M_I\rangle$,

$$E_{M_I} = -g_I \mu_N B M_I. \tag{8.54}$$

Adjacent substates therefore differ by

$$|\Delta E| = g_I \mu_N B = h\nu. \tag{8.55}$$

The Rabi molecular-beam method and nuclear magnetic resonance detect the frequency ν at which a radiofrequency field drives $\Delta M_I = \pm 1$ transitions. Since B is known, the resonance gives g_I, and the stretched-state moment is $\mu = g_I I \mu_N$. Free-nucleon moments $\mu_p = +2.7928\,\mu_N$ and $\mu_n = -1.9130\,\mu_N$ are among the most accurately known quantities in nuclear physics [2, 1].

8.7.2 Schmidt lines (preview of the shell model)

For odd-A nuclei the magnetic moment is often dominated by the last unpaired nucleon. Plotting μ versus I produces the Schmidt lines: two limiting curves for $j = \ell \pm \frac{1}{2}$. Real nuclei fall between the lines because of configuration mixing and meson-exchange currents. The deuteron is the few-body cousin of that story: free $\mu_p + \mu_n$ is the extreme independent-particle limit; a few-percent D-wave admixture is the simplest configuration mixing [6, 1]. The two Schmidt formulae are derived, rather than merely quoted, in Section 8.9.2.

8.7.3 Isospin language

Proton and neutron form an isospin doublet $t = \frac{1}{2}$. The deuteron has total isospin $T = 0$, and this assignment follows from exchange symmetry. For two nucleons, the spatial, spin, and isospin factors acquire under exchange the signs

$$(-1)^L, \qquad (-1)^{S+1}, \qquad (-1)^{T+1}, \tag{8.56}$$

respectively. Their product must be -1 for fermions, so

$$(-1)^{L+S+T} = -1. \tag{8.57}$$

The deuteron has even L and $S = 1$. Hence $L + S + T$ is odd only when $T = 0$. Explicitly, its isospin wave function is the antisymmetric combination

$$|T = 0, T_z = 0\rangle = \frac{1}{\sqrt{2}} \left(|pn\rangle - |np\rangle\right). \tag{8.58}$$

The $T = 1$, $S = 0$ pn partner of the virtual $^1S_0$ state is unbound, as are free nn and pp (after Coulomb). Charge independence of the strong force organises NN scattering lengths into a nearly universal pattern once Coulomb is removed. Lecture 10 develops this NN spin–isospin structure through phase shifts [4].

Remark

Quarks and anomalous moments. Dirac particles with charge e and mass m have $g = 2$. The proton's $g_p \approx 5.59$ and the neutron's nonzero moment are direct evidence of composite structure (charged quark currents and meson clouds), already anticipated in Lecture 7's residual-force discussion [2, 4].

8.7.4 Beyond the one-body magnetic operator

For the one-body operator derived in Section 8.5.1, the S–D cross term vanishes: $\hat{L}_z$, $\hat{s}_{pz}$, and $\hat{s}_{nz}$ do not change L, and the $L = 0$ and $L = 2$ angular functions are orthogonal. This gives the probability-weighted expectation value in Equation (8.50).

A precision calculation must add two-body meson-exchange currents, relativistic corrections, and small nucleon-structure effects. Those contributions alter the magnetic moment without being equivalent to a change in P_D. Modern potentials still find a few-percent D-state probability, so the classroom estimate is useful, but P_D itself is model dependent [6].

8.8 Magnetic moment deficit and D-wave probability

8.8.1 From free sum to measured μ_d

The independent-particle estimate $\mu_S \approx \mu_p + \mu_n = 0.8798\,\mu_N$ overshoots experiment ($\mu_d = 0.8574\,\mu_N$) by about 2.5%. The lecture attributes the deficit to a small $^3D_1$ admixture: orbital motion of the proton in $L = 2$ contributes $g_\ell^{(p)}\mu_N\langle L_z\rangle$, while the S–D interference shifts the spin projections. Keeping only the dominant S-wave spin term plus a D-wave orbital correction with probability P_D gives the schematic balance

$$\mu_d \approx (1 - P_D)(\mu_p + \mu_n) + P_D\, g_\ell^{(p)}\, \langle L_z\rangle_D, \tag{8.59}$$

where $\langle L_z\rangle_D$ is fixed by angular-momentum algebra in the stretched state $M_I = 1$.

8.8.2 A worked P_D estimate

Using $g_\ell^{(p)} = 1$ and typical shell-model couplings for $L = 2$, $S = 1$, $J = 1$, a few-percent D-wave probability suffices:

$$P_D \sim \frac{\mu_p + \mu_n - \mu_d}{(\mu_p + \mu_n) - g_\ell^{(p)} \langle L_z \rangle_D} \approx 0.04\text{–}0.06. \tag{8.60}$$

Inserting $P_D = 0.05$ into (8.59) lowers μ_d by $\sim 0.02\,\mu_N$, matching the observed deficit. The same P_D feeds the quadrupole moment (Lecture 9): μ_d and Q_d are correlated constraints on the tensor force, not independent knobs.

Key idea

Schmidt preview. Odd-A nuclei use the same logic at higher A: plot μ/μ_N versus I and compare with the Schmidt lines for $j = \ell \pm \frac{1}{2}$ (Section 8.9.2). The deuteron is the two-body limit where only one pair of nucleons contributes and configuration mixing is a few-percent D-wave rather than valence-shell fragmentation.

Remark

What P_D is not. Meson-exchange currents and relativistic corrections also shift μ_d at the percent level. Treating the entire deficit as orbital D-wave probability is a classroom estimate, not a unique inversion of data. Modern potentials quote P_D with model-dependent spread; the qualitative lesson (tensor force required) is robust [6, 2].

8.8.3 Isospin and the magnetic moment sum rule

In the $T = 0$ deuteron, the isospin wave function is antisymmetric under nucleon exchange (Section 8.7.3). The magnetic operator is isovector, so the $T = 0$ expectation value is not constrained to equal $\mu_p + \mu_n$ unless the spatial wave function is purely S-wave. The measured $\mu_d < \mu_p + \mu_n$ is therefore a joint statement about isospin symmetry and tensor-induced D-wave admixture. For comparison, the unbound $T = 1$, $S = 0$ virtual state has no stable magnetic moment, but its scattering length $a_s \approx -23.7\,\text{fm}$ (Lecture 10) shows how close the singlet channel comes to binding despite $\mu_p + \mu_n$ being irrelevant there.

8.8.4 Schmidt lines at $A = 2$ and $A = 15$

At $A = 2$ the "valence nucleon" picture is replaced by two active bodies, yet the logic survives: start from free moments, then reduce by orbital mixing. At $A = 15$, ^{15}N ($I = \frac{1}{2}^-$) and ^{15}O ($I = \frac{1}{2}^-$) provide mirror tests of Schmidt limits once the hole/particle sign is tracked. Lecture 14 plots the full Schmidt curves; the deuteron is the anchor point showing that even the simplest bound nucleus requires configuration mixing beyond free g-factors [1, 22].

Impulse versus consistent currents

State in one sentence the impulse approximation for μ_d and in a second sentence what "consistent two-body currents" means. That pair of sentences is the conceptual payload of Lecture 8 beyond the existence of a D-wave.

8.9 Further physics: magnetic moments and the nuclear magneton

To each angular momentum is associated a magnetic dipole

$$\boldsymbol{\mu} = g\,\mu_N \frac{\boldsymbol{j}}{\hbar}, \qquad \mu_N = \frac{e\hbar}{2m_p}. \tag{8.61}$$

For orbital motion of a proton, $g_\ell = 1$; for a neutron, $g_\ell = 0$. Spin g-factors of free nucleons are anomalous ($g_p \approx 5.59$, $g_n \approx -3.83$), a signal of internal structure [13, 6, 2, 5].

In odd-A nuclei a first approximation attributes μ to the last unpaired nucleon (Schmidt estimate). The deuteron is the clean two-body limit of the same idea: free moments first, then a small configuration admixture (D-wave) to match experiment. Exact g-factors of complex nuclei require interactions among many nucleons; the deuteron isolates the few-body physics.

8.9.1 Spin dependence from the deuteron alone

From $I^\pi = 1^+$ and $\mu_d \approx \mu_p + \mu_n$ one concludes that the ground state is mostly ${}^3S_1$. The absence of a bound ${}^1S_0$ pn state shows that the force is spin-dependent: both channels are attractive, but only the triplet is deep enough to bind [14, 6]. The small quadrupole moment and the μ_d deficit then require a few-percent ${}^3D_1$ admixture (tensor force; Lecture 9).

8.9.2 Schmidt lines for odd-A nuclei

For a single valence nucleon with orbital angular momentum ℓ and spin $s = \frac{1}{2}$, the one-body magnetic operator is

$$\frac{\hat{\mu}_z}{\mu_N} = g_\ell \frac{\hat{\ell}_z}{\hbar} + g_s \frac{\hat{s}_z}{\hbar}. \tag{8.62}$$

The measured moment is the expectation value in the stretched state $|j, m = j\rangle$. Since $j_z = \ell_z + s_z$, it is useful to write

$$\frac{\mu}{\mu_N} = g_\ell j + (g_s - g_\ell)\frac{\langle s_z\rangle}{\hbar}. \tag{8.63}$$

The projection theorem derived in Section 8.5.1, with $L \to \ell$, $S \to s$, and $M = j$, gives

$$\frac{\langle s_z\rangle}{\hbar} = j\,\frac{j(j+1) + s(s+1) - \ell(\ell+1)}{2j(j+1)}. \tag{8.64}$$

For aligned spin, $j = \ell + \frac{1}{2}$, so $\ell = j - \frac{1}{2}$ and $s(s+1) = \frac{3}{4}$. Substitution into Equation (8.64) gives

$$\begin{aligned}\frac{\langle s_z\rangle}{\hbar} &= \frac{j(j+1) + \frac{3}{4} - (j - \frac{1}{2})(j + \frac{1}{2})}{2(j+1)} \\ &= \frac{j^2 + j + \frac{3}{4} - (j^2 - \frac{1}{4})}{2(j+1)} = \frac{j+1}{2(j+1)} = \frac{1}{2}.\end{aligned} \tag{8.65}$$

Equation (8.63) then yields the upper Schmidt line,

$$\boxed{\frac{\mu}{\mu_N} = g_\ell j + \frac{1}{2}(g_s - g_\ell), \qquad j = \ell + \frac{1}{2}.} \tag{8.66}$$

Equivalently, $\mu/\mu_N = g_\ell \ell + g_s/2$: in the stretched state the orbital and spin projections are simply ℓ and $1/2$.

For anti-aligned spin, $j = \ell - \frac{1}{2}$, so $\ell = j + \frac{1}{2}$. Now

$$\begin{aligned} \frac{\langle s_z \rangle}{\hbar} &= \frac{j(j+1) + \frac{3}{4} - (j + \frac{1}{2})(j + \frac{3}{2})}{2(j+1)} \\ &= \frac{j^2 + j + \frac{3}{4} - (j^2 + 2j + \frac{3}{4})}{2(j+1)} = -\frac{j}{2(j+1)}. \end{aligned} \tag{8.67}$$

The lower Schmidt line is therefore

$$\boxed{\frac{\mu}{\mu_N} = g_\ell j - (g_s - g_\ell)\frac{j}{2(j+1)}, \qquad j = \ell - \frac{1}{2}.} \tag{8.68}$$

For an unpaired proton one inserts $g_\ell = 1$ and $g_s = g_p$; for an unpaired neutron, $g_\ell = 0$ and $g_s = g_n$. These are independent particle limits, not exact predictions. Experimental moments of odd-A nuclei generally lie between the Schmidt lines, shifted toward smaller $|\mu|$ by configuration mixing and meson-exchange currents [6, 2, 13]. The deuteron is the few-body prototype of that programme: start from free g-factors, then correct for orbital admixtures.

8.9.3 Nuclear magneton vs Bohr magneton

Because $m_p \approx 1836\, m_e$,

$$\begin{aligned} \mu_N &= \frac{m_e}{m_p}\, \mu_B \\ &= \frac{m_e}{m_p}\left(\frac{e\hbar}{2m_e}\right) = \frac{e\hbar}{2m_p} \\ &\approx \frac{1}{1836}\, \mu_B. \end{aligned} \tag{8.69}$$

Nuclear moments are therefore three orders of magnitude smaller than electronic moments, which is why nuclear magnetism is a precision hyperfine effect in atoms and a separate experimental industry (NMR, atomic beams, ion traps) [5, 2].

8.9.4 Projection theorem for $\langle s_z \rangle$

The Schmidt derivation (Section 8.9.2) rests on evaluating $\langle s_z \rangle$ in $|j, m = j\rangle$. For coupled $\boldsymbol{L} + \boldsymbol{S} = \boldsymbol{J}$, the Wigner–Eckart theorem gives

$$\frac{\langle s_z \rangle}{\hbar} = \frac{\langle \boldsymbol{J} \cdot \boldsymbol{S} \rangle}{j(j+1)\hbar^2}\, j = j\, \frac{j(j+1) + s(s+1) - \ell(\ell+1)}{2j(j+1)}, \tag{8.70}$$

which is (8.64). For the deuteron ($\ell \to L = 0$, $s \to S = 1$, $j \to I = 1$), $\langle S_z \rangle/\hbar = 0$ in the pure S-wave limit, so $\mu_S = \mu_p + \mu_n$ follows immediately. Only $L = 2$ admixture generates nonzero $\langle L_z \rangle$ and S–D cross terms in the full operator of Section 8.5.1.

8.9.5 Worked Schmidt comparison for ^{15}N

^{15}N ($Z = 7$, one proton hole on ^{16}O) has $I = \frac{1}{2}$ from the $1p_{1/2}$ orbit ($\ell = 1$, $j = \frac{1}{2}$). The lower Schmidt line with $g_\ell = 1$, $g_s = g_p$ gives

$$\frac{\mu}{\mu_N} = \frac{1}{2} + \frac{1}{2}(g_p - 1)\frac{1/2}{3/2} = \frac{1}{2} + \frac{1}{6}(g_p - 1) \approx 0.44, \tag{8.71}$$

so $\mu_{\text{Schmidt}} \approx 2.2\,\mu_N$. Experiment gives $\mu = -0.283\,\mu_N$: the hole sign and core polarisation reverse and reduce the moment, showing that Schmidt lines are bounds, not predictions, even near closed shells [6, 22]. The deuteron sits closer to the independent-particle limit because only two nucleons participate.

8.9.6 Link to Lecture 9 and 14

The same D-wave probability that lowers μ_d generates Q_d (Lecture 9) through (9.23). Lecture 14 generalises the Schmidt plot to all odd-A nuclei and introduces quenching of g_s. The deuteron is the minimal laboratory for all three ideas: anomalous free-nucleon g-factors, configuration mixing, and tensor-force induced orbital admixture [2].

8.9.7 Rabi resonance in the lab frame

In NMR one typically applies a transverse radiofrequency field $B_1 \cos\omega t$ near the Larmor frequency $\omega_0 = g\mu_N B/\hbar$. The Rabi condition $\hbar\omega = g\mu_N B$ is the same physics as the molecular-beam $\Delta M = \pm 1$ resonance, but in a condensed sample the signal is detected as a voltage induced in a pickup coil. For the deuteron, the small D-wave correction shifts ω_0 by parts per thousand relative to $\mu_p + \mu_n$, comparable to the shift used to infer P_D in precision beam experiments [2, 22].

8.9.8 Hyperfine structure in atomic deuterium

Replacing the proton in hydrogen by a deuteron changes the hyperfine splitting because $\mu_d \neq \mu_p$. The $1S$ hyperfine interval scales as $\mu_d I_d \cdot \langle \boldsymbol{I} \cdot \boldsymbol{J} \rangle$ for the electron coupling. The percent-level difference between μ_d and $\mu_p + \mu_n$ is therefore visible in atomic spectroscopy, independent of nuclear scattering experiments. Atomic and molecular data provide a third route to the same D-wave physics constrained by μ_d and Q_d (Lecture 9) [2, 6].

8.10 Deeper reading: μ_d, exchange currents, and Schmidt intuition

The deuteron magnetic moment lies close to but not exactly at $\mu_p + \mu_n$. A small D-wave admixture P_D shifts μ_d by an amount proportional to $(\mu_p + \mu_n - \frac{1}{2}\mu_N)$. Fitting μ_d alone does not fix P_D uniquely once meson-exchange currents are allowed: two-body currents can move μ_d at the same order as a change in P_D. Modern calculations therefore evaluate the current operator consistently with the Hamiltonian that generates the wave function.

Order of magnitude

If a pure S-wave estimate gives $\mu_S \approx 0.880\,\mu_N$ and the measured $\mu_d \approx 0.857\,\mu_N$, a few-percent P_D accounts for the difference in the impulse approximation. Exchange currents of relative size $\sim 1\%$ matter at the same level: that is why Lecture 8's message is qualitative structure ($I^\pi = 1^+$ requires D-wave) rather than a precision extraction of P_D from μ_d alone.

Schmidt lines in Lecture 14 reuse the same single-particle magnetic operators; quenching and two-body currents reappear there in heavier nuclei. The deuteron is the cleanest few-body laboratory in which those operator issues can be stated without a large valence space.

Relativistic and two-body current bookkeeping

Relativistic corrections to magnetic operators and explicit meson-exchange currents enter at similar numerical size for μ_d. Calculations that vary one while freezing the other can appear to "prefer" a particular P_D. The reliable approach is a single Hamiltonian with consistent currents. That standard, learned on the deuteron, reappears in medium-mass Gamow–Teller quenching (Lectures 14 and 19).

Spin and isospin structure

The deuteron is isoscalar ($T = 0$). Its magnetic moment therefore tests isoscalar combinations of nucleon moments and isoscalar exchange currents. Isovector magnetic properties appear more strongly in odd-mirror pairs and in $N = Z$ odd–odd nuclei. Keeping isoscalar versus isovector channels straight prevents mis-applying deuteron lessons to Schmidt deviations in heavy nuclei.

From μ_d to many-body operators

Write $\mu_d = \mu_S(1 - P_D) + \mu_D P_D$ with the textbook combinations of nucleon moments. Insert $P_D = 0.04$ and 0.06 and compare with experiment. The residual can be assigned either to a adjusted P_D or to an exchange-current shift of a few hundredths of a nuclear magneton. Without an independent constraint (quadrupole moment, form factors, or a consistent current calculation), that assignment is ambiguous.

This ambiguity is the few-body prototype of operator renormalisation in heavy nuclei. Lecture 14's Schmidt deviations and Lecture 19's Gamow–Teller quenching are the same story at larger mass: wave functions and operators must be improved together. Keep a short notebook table: observable, impulse-approximation prediction, size of two-body current correction, and whether the correction is isoscalar or isovector. The deuteron fills the first row of that table.

Magnetic moments diagnose operators. The deuteron teaches the lesson in a system small enough to solve; medium-mass nuclei reuse it under the names quenching and effective g-factors.

In practice, rework one numerical estimate from the lecture with a different parameter set and compare the shift to the quoted experimental uncertainty. If the shift exceeds the uncertainty, the estimate is teaching a scale, not delivering a fit. That distinction keeps theoretical cartoons honest when they sit beside evaluated data tables and modern many-body calculations. Along the course arc, write one sentence linking this chapter backward and one forward: the discipline of those two sentences is as important as any single formula. Examination problems rarely ask for a literature review, but they do ask for the scale argument that literature reviews presuppose.

Research frontier: currents, D-wave, and student projects

Lecture 8 showed that the deuteron magnetic moment μ_d and the assignment $I^\pi = 1^+$ require a few-percent D-wave admixture in the ground state. Contemporary calculations evaluate two-body meson-exchange currents and relativistic corrections consistently with the same chiral Hamiltonian used for the wave function, rather than adding ad hoc current operators after the fact [43]. Related electroweak currents underlie Gamow–Teller quenching and neutrinoless double-beta matrix elements in heavier nuclei: a landmark ab initio study showed that consistent two-body currents resolve much of the long-standing GT quenching puzzle [52]. Precision tables of magnetic moments remain essential benchmarks for both few-body and shell-model calculations [22]. What remains uncertain is how large residual quenching is in mid-mass and heavy nuclei once two-body currents and correlations are both included, and how those lessons translate into $0\nu\beta\beta$ operator uncertainties.

Polarisation observables and electromagnetic form factors of the deuteron provide additional constraints on the same D-wave physics. Modern analyses insist that the current operator and the wave function come from one consistent framework; otherwise apparent agreement with μ_d can hide cancelling mistakes. That consistency requirement is the bridge from the lecture toy model to contemporary few-body papers.

Those consistency checks also matter for polarisation transfer and for deuteron electrodisintegration, where different current recipes can shift extracted P_D values.

Student directions. (1) Analytic estimate: recompute μ_S and μ_D contributions for a given P_D. (2) Data project: using Stone's table, plot odd-proton Schmidt deviations for $20 \leq Z \leq 40$. (3) Short essay: explain why exchange currents matter more for magnetic moments than for the charge radius.

8.11 Problems with solutions

Problem 8.1 — Pure S-wave magnetic moment. Using $g_p = 5.5857$ and $g_n = -3.8261$, compute the pure $^3S_1$ prediction for μ_d and the fractional difference from the experimental value $0.8574\,\mu_N$.

Solution 8.1.

$$\mu_d(L{=}0) = \frac{g_p + g_n}{2}\,\mu_N = \frac{5.5857 - 3.8261}{2}\,\mu_N = 0.8798\,\mu_N.$$

$$\Delta\mu = 0.8798 - 0.8574 = 0.0224\,\mu_N, \qquad \frac{\Delta\mu}{\mu_d(L{=}0)} \approx 2.5\%.$$

Most of μ_d is $\mu_p + \mu_n$; the small deficit is structural [6].

Problem 8.2 — D-wave probability from μ_d. Assume

$$\mu_d = P_S\,(0.8798\,\mu_N) + P_D\,(0.310\,\mu_N), \quad P_S + P_D = 1,$$

and $\mu_d = 0.8574\,\mu_N$. Find P_D.

Solution 8.2.

$$0.8574 = 0.8798(1 - P_D) + 0.310\,P_D = 0.8798 - 0.5698\,P_D,$$

$$P_D = \frac{0.8798 - 0.8574}{0.5698} \approx 0.039 \approx 4\%, \quad P_S \approx 96\%.$$

This 4% is a one-body-operator, model-dependent classroom inference, not an observable: exchange currents, relativistic corrections, and the chosen NN potential shift the extracted P_D. The robust conclusion is only that a few-percent D-wave is natural [6, 14].

Problem 8.3 — Why $L = 1$ is excluded. List (S, L) combinations that can give $I = 1$ for pn and show which survive the experimental parity $\pi = +$.

Solution 8.3. Coupling the two spin-$\frac{1}{2}$ nucleons first gives

$$S = \left|\tfrac{1}{2} - \tfrac{1}{2}\right|, \ldots, \tfrac{1}{2} + \tfrac{1}{2} = 0,\ 1.$$

Angular-momentum addition then requires

$$|L - S| \le I \le L + S.$$

For $S = 0$, this reduces to $I = L$, so $I = 1$ requires $L = 1$. For $S = 1$, requiring $I = 1$ gives

$$|L - 1| \le 1 \le L + 1,$$

which is satisfied by $L = 0, 1, 2$. Thus the four possibilities are

$$(S, L) = (0, 1),\ (1, 0),\ (1, 1),\ (1, 2).$$

Parity is $\pi = (-1)^L$. The measured $\pi = +$ eliminates both $L = 1$ cases, leaving $(S, L) = (1, 0)$ and $(1, 2)$, or ${}^3S_1$ and ${}^3D_1$. Both have $S = 1$, so the bound state is entirely spin triplet even though it is not entirely S-wave [6].

Problem 8.4 — Nuclear magneton scale. Compute $\mu_N = e\hbar/(2m_p)$ in MeV/T using $e\hbar/(2m_e) = \mu_B = 5.788 \times 10^{-11}$ MeV/T and $m_p/m_e = 1836$. Why is nuclear magnetism hard to measure compared with electronic magnetism?

Solution 8.4.

$$\mu_N = \frac{m_e}{m_p}\,\mu_B = \frac{5.788\times 10^{-11}}{1836} \approx 3.15\times 10^{-14}\,\mathrm{MeV/T}.$$

Nuclear moments are ~ 2000 times smaller than electronic ones, so they appear only as tiny hyperfine shifts or require specialised methods (Rabi beams, NMR) [5, 2].

Problem 8.5 — Derive the $^3D_1$ magnetic moment. For a pure $^3D_1$ state, use the projection theorem to calculate $\langle L_z\rangle$ and $\langle S_z\rangle$ in the stretched state $J = M = 1$. Then evaluate

$$\frac{\hat{\mu}_z}{\mu_N} = \frac{1}{2}\frac{\hat{L}_z}{\hbar} + \frac{g_p+g_n}{2}\frac{\hat{S}_z}{\hbar}$$

and obtain the numerical moment using the g-factors of Problem 8.1.

Solution 8.5. For $L = 2$, $S = 1$, and $J = M = 1$,

$$\begin{aligned}
\frac{\langle L_z\rangle}{\hbar} &= M\frac{J(J+1)+L(L+1)-S(S+1)}{2J(J+1)}\\
&= \frac{2+6-2}{4} = \frac{3}{2},\\
\frac{\langle S_z\rangle}{\hbar} &= M\frac{J(J+1)+S(S+1)-L(L+1)}{2J(J+1)}\\
&= \frac{2+2-6}{4} = -\frac{1}{2}.
\end{aligned}$$

The magnetic moment is therefore

$$\begin{aligned}
\frac{\mu(^3D_1)}{\mu_N} &= \frac{1}{2}\left(\frac{3}{2}\right) + \frac{g_p+g_n}{2}\left(-\frac{1}{2}\right)\\
&= \frac{3}{4} - \frac{g_p+g_n}{4} = \frac{1}{4}(3-g_p-g_n).
\end{aligned}$$

Numerically,

$$\begin{aligned}
3-g_p-g_n &= 3-5.5857-(-3.8261)\\
&= 3-5.5857+3.8261 = 1.2404,\\
\therefore\qquad \mu(^3D_1) &= \frac{1.2404}{4}\,\mu_N = 0.3101\,\mu_N.
\end{aligned}$$

[6].

Problem 8.6 — Charge independence and no bound nn. Explain why charge independence of the strong force, together with the Pauli principle, is consistent with a bound pn ($s = 1$) deuteron and no bound nn or pp systems.

Solution 8.6. For two nucleons the exchange sign is

$$(-1)^L(-1)^{S+1}(-1)^{T+1} = (-1)^{L+S+T},$$

and Fermi statistics require this sign to be -1. In an S-wave, $L = 0$, so $S + T$ must be odd.

The deuteron channel has $S = 1$, hence $T = 0$. Its isospin state

$$\frac{1}{\sqrt{2}}(|pn\rangle - |np\rangle)$$

has no nn or pp member: two identical nucleons necessarily have $T = 1$. For nn or pp with $L = 0$ and $T = 1$, antisymmetry therefore forces $S = 0$. Charge independence relates that channel to the unbound $T = 1$, ${}^1S_0$ pn channel, apart from electromagnetic and small charge-dependent effects. Thus the bound $T = 0$, ${}^3S_1$ channel exists only for pn, while the allowed nn and pp S-wave channel does not bind [14, 6].

Problem 8.7 — Clebsch–Gordan check of the D state. Using Equation (8.13), verify normalisation and calculate $\langle L_z\rangle$ and $\langle S_z\rangle$ directly from the three product-state probabilities.

Solution 8.7. The probabilities are 1/10, 3/10, and 3/5, whose sum is one. Therefore

$$\frac{\langle L_z\rangle}{\hbar} = \frac{1}{10}(0) + \frac{3}{10}(1) + \frac{3}{5}(2) = \frac{3}{2},$$

and

$$\frac{\langle S_z\rangle}{\hbar} = \frac{1}{10}(1) + \frac{3}{10}(0) + \frac{3}{5}(-1) = -\frac{1}{2}.$$

Their sum is $M = 1$, as required.

Problem 8.8 — Pure S-wave magnetic moment. If the deuteron were a pure ${}^3S_1$ state, what would μ_d equal? Compare with the free-nucleon sum $\mu_p + \mu_n$.

Solution 8.8. For pure $L = 0$, orbital contributions vanish and $\mu_d = \mu_p + \mu_n \approx 2.793 - 1.913 = 0.880\,\mu_N$. The measured moment is slightly smaller, evidencing a few-percent D-wave admixture [6, 1].

Chapter 9

Quadrupole Moment and Tensor Force

"The mixing of ℓ values in the deuteron is the best evidence we have for the noncentral (tensor) character of the nuclear force."
K. S. Krane

9.1 Electric quadrupole moment from the multipole expansion

The magnetic moment of the deuteron is close to, but not equal to, the sum of the free nucleon moments. That small discrepancy already suggests a few percent D-wave admixture. A sharper geometric probe is the electric quadrupole moment: it reports the *shape* of the charge distribution. The definition used in nuclear physics follows from the multipole expansion of the electrostatic potential of a localised charge density, exactly as in the original lecture (Fig. 1 below) [6, 1, 2].

9.1.1 Potential of a nuclear charge distribution (Fig. 1)

Let $\rho(\boldsymbol{r}')$ be the charge density of the nucleus. The electric potential at an exterior point $\boldsymbol{r}$ is

$$\Phi(\boldsymbol{r}) = \frac{1}{4\pi\varepsilon_0} \int \frac{\rho(\boldsymbol{r}')}{|\boldsymbol{r} - \boldsymbol{r}'|} \, \mathrm{d}^3 r', \tag{9.1}$$

where $\boldsymbol{r}'$ labels a charge element inside the nucleus (Figure 9.1).

9.1.2 Multipole expansion far from the nucleus

Far from the nucleus one expands the Coulomb kernel in Legendre polynomials. Write

$$|\boldsymbol{r} - \boldsymbol{r}'| = r \left(1 - 2\frac{r'}{r} \cos\theta + \left(\frac{r'}{r}\right)^2 \right)^{1/2},$$

with θ the angle between $\boldsymbol{r}$ and $\boldsymbol{r}'$. For $r > r'$ the generating function of Legendre polynomials is

$$\frac{1}{|\boldsymbol{r} - \boldsymbol{r}'|} = \frac{1}{r} \sum_{\ell=0}^{\infty} \left(\frac{r'}{r}\right)^{\ell} P_\ell(\cos\theta). \tag{9.2}$$

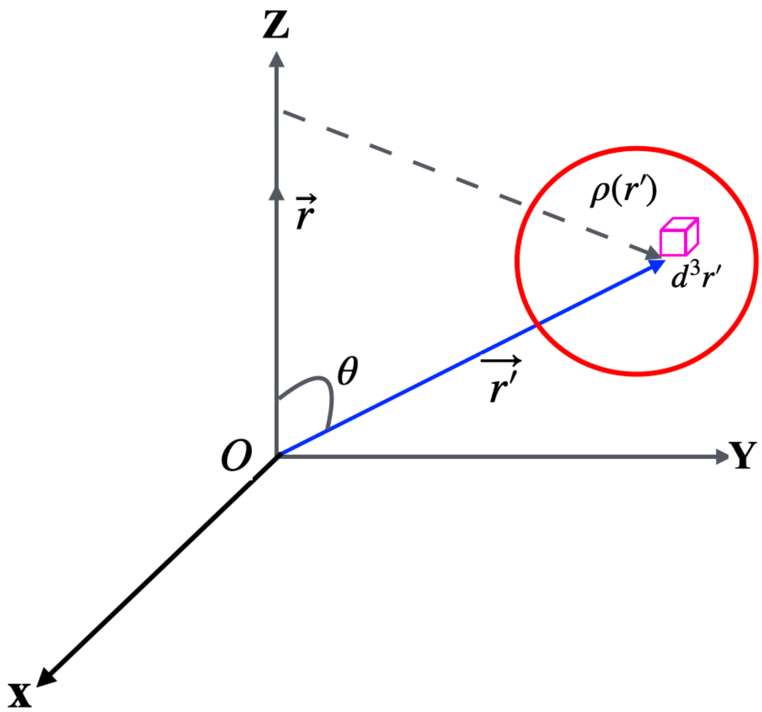


Figure 9.1: Charge distribution $\rho(\boldsymbol{r}')$ inside a nucleus (Fig. 1 of the lecture). An infinitesimal charge element occupies volume d^3r' at $\boldsymbol{r}'$. The field point $\boldsymbol{r}$ lies outside the distribution; θ is the angle between $\boldsymbol{r}$ and $\boldsymbol{r}'$.

The first three polynomials are $P_0 = 1$, $P_1(\cos\theta) = \cos\theta$, and $P_2(\cos\theta) = \frac{1}{2}(3\cos^2\theta - 1)$, so

$$\frac{1}{|\boldsymbol{r} - \boldsymbol{r}'|} = \frac{1}{r}\left(1 + \frac{r'}{r}\cos\theta + \frac{r'^2}{2r^2}(3\cos^2\theta - 1) + \cdots\right). \tag{9.3}$$

The quadrupole term is precisely the $\ell = 2$ piece of this expansion: it is of order $(r'/r)^2$ and must be retained if one wishes to discuss Q. (Omitting r'^2/r^2 would truncate before the quadrupole appears.)

Substitute (9.3) into (9.1):

$$\Phi(\boldsymbol{r}) = \frac{1}{4\pi\varepsilon_0}\int \rho(\boldsymbol{r}')\frac{1}{r}\left(1 + \frac{r'}{r}\cos\theta + \frac{r'^2}{2r^2}(3\cos^2\theta - 1) + \cdots\right)\mathrm{d}^3r'. \tag{9.4}$$

For an axially symmetric source whose symmetry axis is z, collecting powers of $1/r$ gives

$$\Phi(\boldsymbol{r}) = \frac{e}{4\pi\varepsilon_0 r}\left(Z + \frac{D}{r}\,P_1(\cos\theta) + \frac{Q_2}{r^2}\,P_2(\cos\theta) + \cdots\right), \tag{9.5}$$

where

- the **monopole** Z is the total charge (in units of e);
- D is the **electric dipole** moment;
- Q_2 is the Legendre-normalised quadrupole coefficient.

Nuclear ground states of definite parity have vanishing static electric dipole. The first multipole that can report a permanent deformation is therefore the quadrupole.

9.1.3 Definition of the quadrupole tensor

The Cartesian quadrupole tensor associated with the third term of the expansion is

$$Q_{ij} = \int \rho(\boldsymbol{r})\big(3x_i x_j - r^2\delta_{ij}\big)\,\mathrm{d}^3r, \tag{9.6}$$

with $i, j = x, y, z$ and δ_{ij} the Kronecker symbol. Examples:

$$Q_{xx} = \int \rho(\boldsymbol{r})\,(3x^2 - r^2)\,\mathrm{d}^3r, \tag{9.7}$$

$$Q_{xy} = \int \rho(\boldsymbol{r})\,(3xy)\,\mathrm{d}^3r. \tag{9.8}$$

There are nine components, but only five are independent (the tensor is symmetric and traceless). In nuclear spectroscopy the component measured in the stretched state is almost always

$$Q_{\text{spec}} \equiv \frac{Q_{zz}}{e} = \frac{1}{e}\int \rho(\boldsymbol{r})\,(3z^2 - r^2)\,\mathrm{d}^3r = \langle 3z^2 - r^2\rangle_{\text{charge}}. \tag{9.9}$$

Thus Q_{spec} has units of area and $eQ_{\text{spec}} = Q_{zz}$ has units of charge times area. Nuclear tables quote either Q_{spec} in barns or eQ_{spec} in $e \cdot \text{fm}^2$.

The coefficient in Equation (9.5) is

$$Q_2 = \frac{1}{e}\int \rho(\boldsymbol{r})\,r^2 P_2(\cos\theta)\,\mathrm{d}^3r = \frac{Q_{zz}}{2e} = \frac{Q_{\text{spec}}}{2}. \tag{9.10}$$

The factor of two follows from $r^2P_2(\cos\theta) = (3z^2 - r^2)/2$. This convention is used everywhere below: Q_2 belongs in the potential, while quoted nuclear moments are $Q_{\text{spec}} = Q_{zz}/e = 2Q_2$. Classically, $Q_{\text{spec}} > 0$ for a prolate (cigar-like) distribution along z, and $Q_{\text{spec}} < 0$ for an oblate (flattened) one.

9.1.4 Spherical symmetry forces vanishing quadrupole

If the charge density is spherically symmetric, $\rho = \rho(r)$ only, then by symmetry

$$\int z^2\rho\,\mathrm{d}^3r = \int x^2\rho\,\mathrm{d}^3r = \int y^2\rho\,\mathrm{d}^3r = \frac{1}{3}\int r^2\rho\,\mathrm{d}^3r. \tag{9.11}$$

Hence $3z^2 - r^2$ averages to zero and

$$Q_{zz} = \int \rho(\boldsymbol{r}')\,(3z^2 - r^2)\,\mathrm{d}^3r' = 0. \tag{9.12}$$

Equivalently, on a sphere $x^2 = y^2 = z^2$ so $r^2 = 3z^2$ and $3z^2 - r^2 = 0$ pointwise in the average sense of (9.12). **A spherical charge distribution has no electric quadrupole moment.**

9.2 Deuteron quadrupole moment

The deuteron's electric quadrupole moment was historically decisive: a nonzero Q_d cannot arise from a pure $L = 0$ wave function. Together with the magnetic-moment deficit (Lecture 8), it forces a tensor component in the NN force and a few-percent D-wave admixture. The argument below follows the multipole analysis of [6, 1].

9.2.1 Pure S-wave implies vanishing Q_d

Free protons and neutrons have vanishing electric quadrupole moments. Any nonzero Q of the deuteron must come from the *orbital* distribution of the proton's charge about the centre of mass.

If the NN potential were purely central, the ground state would be pure $L = 0$. The spatial wave function is then proportional to the spherical harmonic Y_{00}, which is angle-independent. The charge density is therefore spherically symmetric, and the multipole argument of Section 9.1.4 applies at once:

$$Q_d(L{=}0) = 0. \tag{9.13}$$

A pure s-wave deuteron would look electromagnetically spherical.

9.2.2 The measured value

Experiment finds a small but clearly nonzero moment for the bound deuteron [6]:

$$Q_d \equiv Q_{\text{spec}} = \frac{Q_{zz}}{e} = 2.88 \times 10^{-27}\,\text{cm}^2 = 0.00288\,\text{b}, \qquad eQ_d = 0.288\,\text{e}\cdot\text{fm}^2. \tag{9.14}$$

(The unit is area, as required by (9.9); one barn $= 10^{-28}\,\text{m}^2 = 100\,\text{fm}^2$.) The value is tiny compared with collective rare-earth rotors (several barns), but it is not zero. That single experimental fact rules out a pure $L = 0$ ground state and, with it, a purely central NN potential.

A geometric scale eR^2 with $R \sim 2\,\text{fm}$ is a few $e \cdot \text{fm}^2$; the measured $eQ_d \sim 0.3\,e \cdot \text{fm}^2$ is smaller, consistent with a mild deformation from a few-percent D-wave rather than a strongly deformed rotor.

9.2.3 Mixed wave function and the D-state

Using the mixed ground state of Lecture 8,

$$|\psi\rangle = a_S|S\rangle + a_D|D\rangle, \qquad |a_S|^2 + |a_D|^2 = 1,$$

the spectroscopic quadrupole moment in the stretched state is

$$Q_d = \left\langle \psi \left| \frac{\hat{Q}_{zz}}{e} \right| \psi \right\rangle = |a_S|^2 Q_{SS} + 2\,\text{Re}(a_S^* a_D)\, Q_{SD} + |a_D|^2 Q_{DD}. \tag{9.15}$$

The pure-S matrix element vanishes by spherical symmetry, $Q_{SS} = 0$. The interference and pure-D pieces may be written in terms of the radial functions $u(r) = rR_S(r)$ and $w(r) = rR_D(r)$

(standard Blatt–Weisskopf / Reid normalisation) as

$$Q_d = \frac{\sqrt{2}}{10}\int_0^\infty r^2\,(uw - \tfrac{w^2}{\sqrt{8}})\mathrm{d}r \quad \text{(units of area)}, \tag{9.16}$$

with eQ_d quoted in $e \cdot \mathrm{fm}^2$. Equivalently, the dominant contribution is the S–D interference term $\propto a_S a_D$; the pure D–D piece is smaller because P_D itself is small. Evaluating the radial integrals requires a model for $u(r)$ and $w(r)$. Realistic potentials fitted to NN scattering reproduce $Q_d \approx 0.286\,\mathrm{fm}^2$ with a few-percent D-state probability, consistent with the magnetic-moment estimate of Lecture 8 once exchange currents are acknowledged [6]. The positive sign of Q_d means a prolate (cigar-like) charge distribution along the spin axis.

Key idea

Two multipoles, one structure.

Observable	Pure S prediction	Experiment
μ_d/μ_N	0.880	0.857
Q_d	0	0.00288 b

Both point to a small D-wave. The quadrupole moment is an on/off signal of nonsphericity; the magnetic moment is a percent-level correction to an already large spin moment.

Caution

Extracting P_D from Q_d alone is model dependent because the D-state radial wave function is not measured directly. Potentials that produce a few-percent D-state also produce the observed Q_d, and scattering analyses agree. The qualitative need for a tensor force does not rest on a single fitted percentage [6].

9.3 Non-central force and the tensor operator

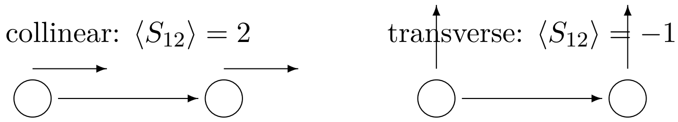
9.3.1 Why a central potential cannot mix L

In a purely central potential $V(r)$, orbital angular momentum is a constant of the motion: $[\boldsymbol{L}^2, H] = 0$. Energy eigenstates may therefore be chosen with definite L. A single eigenstate cannot be a coherent superposition of $L = 0$ and $L = 2$. Experiment, however, requires precisely that mixture: $Q_d \neq 0$ and the magnetic-moment deficit. The residual NN interaction therefore cannot be purely central; it must contain a piece that does not commute with $\boldsymbol{L}^2$.

Spin–spin forces of the form $V_\sigma(r)\,\boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2$ still depend only on the relative distance: they split triplet from singlet but do not mix different L. The operator that couples S and D must involve the orientation of the relative vector $\boldsymbol{r}$ with respect to the nucleon spins.

9.3.2 Magnetic-dipole analogy and the tensor operator

A familiar non-central interaction is the force between two magnetic dipoles. At fixed separation the energy depends on how the moments are oriented relative to $\hat{\boldsymbol{r}}$: collinear and sideways arrangements are not equivalent.



same separation r, different tensor energy

$$S_{12} = 3(\boldsymbol{\sigma}_1\cdot\hat{r})(\boldsymbol{\sigma}_2\cdot\hat{r}) - \boldsymbol{\sigma}_1\cdot\boldsymbol{\sigma}_2$$

Figure 9.2: Two dipoles at fixed r: collinear versus tilted. The force depends on orientation, so it is non-central.

Classically,

$$U \propto \frac{3(\boldsymbol{m}_1\cdot\hat{r})(\boldsymbol{m}_2\cdot\hat{r}) - \boldsymbol{m}_1\cdot\boldsymbol{m}_2}{r^3}.$$

Replacing magnetic moments by nucleon spin operators yields the **tensor operator**

$$S_{12} = 3(\boldsymbol{\sigma}_1\cdot\hat{r})(\boldsymbol{\sigma}_2\cdot\hat{r}) - \boldsymbol{\sigma}_1\cdot\boldsymbol{\sigma}_2, \tag{9.17}$$

and a potential piece $V_T(r)\,S_{12}$. In the coupled ${}^3S_1$–${}^3D_1$ subspace the angular matrix elements of S_{12} are standard:

$$\langle {}^3S_1|S_{12}|{}^3S_1\rangle = 0, \quad \langle {}^3S_1|S_{12}|{}^3D_1\rangle = \sqrt{8}, \quad \langle {}^3D_1|S_{12}|{}^3D_1\rangle = -2. \tag{9.18}$$

(For a spin-triplet product state polarised along an axis making angle ϑ with $\hat{r}$, $\langle S_{12}\rangle = 3\cos^2\vartheta - 1$: it is $+2$ for spins parallel to the separation and -1 for spins transverse to it. The interaction is therefore orientation dependent even at fixed r.)

The selection rule also follows from the operator structure: S_{12} is the scalar coupling of a rank-2 spherical harmonic $Y_2(\hat{r})$ to a rank-2 spin tensor. The orbital matrix element obeys

$$|L - L'| \le 2 \le L + L', \qquad (-1)^{L'} = (-1)^L,$$

so it connects $\Delta L = 0, \pm 2$ while conserving total J and parity. Starting from $L = 0$, the anisotropic part can therefore connect only $L' = 2$. This is the ${}^3S_1 \leftrightarrow {}^3D_1$ selection rule; the S-wave diagonal angular average vanishes. (The vanishing S–S diagonal shows that a pure central spin–spin force cannot mix L; only the off-diagonal tensor matrix element does.) Writing the radial wave as $\psi = u(r)Y_S + w(r)Y_D$, the coupled radial Schrödinger equations in the deuteron channel become

$$-\frac{\hbar^2}{m}\frac{\mathrm{d}^2u}{\mathrm{d}r^2} + V_C u + \sqrt{8}\,V_T w = Eu, \tag{9.19}$$

$$-\frac{\hbar^2}{m}\left(\frac{\mathrm{d}^2w}{\mathrm{d}r^2} - \frac{6w}{r^2}\right) + (V_C - 2V_T - 3V_{LS})w + \sqrt{8}\,V_T u = Ew, \tag{9.20}$$

where V_C collects central (including spin–spin) pieces and V_{LS} is the spin–orbit term. The off-diagonal $\sqrt{8}\,V_T$ is precisely what mixes u and w and generates $Q_d \neq 0$.

One-pion exchange produces exactly this tensor structure at long range, with radial factors $e^{-m_\pi r}/r$ and polynomials in $1/(m_\pi r)$ [6, 2, 4, 9]. In the deuteron channel V_T mixes ${}^3S_1$ with ${}^3D_1$, generates $Q_d \neq 0$, and shifts μ_d by a few percent. That mixing is the clearest bound-state evidence for the tensor character of the residual nuclear force [6].

Definition

Tensor force. The non-central NN potential contains $V_T(r)S_{12}$. In the deuteron it is responsible for S–D mixing, for $Q_d \neq 0$, and for the small correction to μ_d.

9.3.3 The full NN potential is richer

Phenomenological potentials include central, spin–spin, tensor, spin–orbit, and short-range terms, with isospin dependence. The deuteron bound state has taught us the first three. Scattering (all partial waves, both singlet and triplet, as functions of energy) fixes the rest. When the bound state runs out of information, the continuum (scattering) takes over.

9.4 Quadrupole moments of heavier nuclei (context)

Tables of quadrupole moments place the deuteron in perspective. Many nuclei have $|Q|$ of order eR^2 (single-particle scale). Rare-earth nuclei reach several barns: the whole proton distribution is collectively deformed. Closed-shell nuclei have $Q \approx 0$. For deformed heavy nuclei one distinguishes the **intrinsic** quadrupole moment Q_0 in the body-fixed frame from the **spectroscopic** moment measured in the laboratory; they differ by a projection factor depending on I. The deuteron is the few-body extreme: weak binding, small spectroscopic Q, tensor-driven rather than collective [6, 1].

9.5 Synthesis: Lectures 7–9

Lecture	Observable	Lesson
7	$B_d \approx 2.224\,\text{MeV}$	Range $\sim 2\,\text{fm}$, depth $\sim 35\,\text{MeV}$; barely bound
8	$I^\pi = 1^+$, $\mu_d = 0.857\,\mu_N$	Triplet; $\sim 96\%$ S + $\sim 4\%$ D; spin-dependent force
9	$Q_d = 0.00288\,\text{b}$	Nonspherical; tensor force $V_T S_{12}$

Takeaway

Physics of Lecture 9.

- $Q_{\text{spec}} = Q_{zz}/e = \langle 3z^2 - r^2\rangle_{\text{charge}}$: vanishes for spherical charge distributions.
- $Q_d = 0.00288\,\text{b} \neq 0$: the deuteron is not a pure S-state.
- Consistent few-percent D-wave from μ_d and Q_d.
- Tensor operator S_{12} encodes the non-central force; pion exchange is its long-range origin.

- Lectures 7–9 define the qualitative NN force; Lectures 10–11 develop scattering phase shifts and meson exchange for the course.

9.5.1 Nonzero Q_d and the tensor force

Free nucleons have vanishing electric quadrupole moments, so any $Q_d \neq 0$ must come from the orbital charge distribution. The measured deuteron quadrupole moment is small but nonzero ($Q_d \simeq 0.00288\,\mathrm{b}$); a pure $\ell = 0$ wave function cannot produce it. The tensor potential distinguishes geometries with $\boldsymbol{\sigma} \cdot \hat{r} \neq 0$ from those with $\boldsymbol{\sigma} \cdot \hat{r} = 0$, and thereby generates a permanent deformation [6, 14, 1].

9.6 Deeper physics: spectroscopic Q, shells, and one-pion tensor force

9.6.1 Intrinsic vs spectroscopic quadrupole moment

For deformed heavy nuclei one distinguishes the **intrinsic** quadrupole moment Q_0 in the body-fixed frame from the **spectroscopic** moment Q measured in the lab. They are related by a Clebsch–Gordan projection factor depending on I. For the deuteron $I = 1$ there is no collective rotor picture: Q_d is a few-body matrix element of $r^2 Y_{20}$ between S and D components. Standard values are

$$Q_d \approx +0.286\,\mathrm{fm}^2, \qquad eQ_d \approx +0.286\,\mathrm{e} \cdot \mathrm{fm}^2 \tag{9.21}$$

and stresses that free nucleons have $Q = 0$, so any nonzero deuteron Q is purely orbital / deformation physics [6].

9.6.2 Shell-model systematics of Q

The plot of quadrupole moments versus Z shows sign changes near magic numbers: closed shells are spherical ($Q \approx 0$); a valence proton hole just below a shell tends to give prolate $Q > 0$, while a particle just above tends toward oblate $Q < 0$ in the single-particle picture. Collective mid-shell nuclei have Q many times the single-particle estimate. The deuteron sits at the opposite extreme: no shells, weak binding, small Q generated by tensor-driven D-wave admixture [1].

9.6.3 One-pion exchange and S_{12}

The long-range NN force is one-pion exchange (Yukawa). The pseudoscalar pion coupling produces a potential containing the tensor operator S_{12} with radial dependence

$$V_T(r) \propto \frac{e^{-m_\pi r}}{r}\left(1 + \frac{3}{m_\pi r} + \frac{3}{(m_\pi r)^2}\right) \tag{9.22}$$

(up to spin–isospin factors). The angular matrix elements (9.18) then feed into the coupled equations (9.19)–(9.20), so pion exchange is the microscopic origin of the magnetic-dipole analogy in the lecture: it is non-central and mixes ${}^3S_1$–${}^3D_1$. Short-range repulsion, spin–orbit forces, and multi-pion / contact terms complete modern potentials (Argonne, CD-Bonn, chiral EFT). The deuteron Q_d and η_D (D/S asymptotic ratio) are primary constraints on V_T [6, 4, 2].

Key idea

Tensor force checklist.

- Operator: $S_{12} = 3(\boldsymbol{\sigma}_1 \cdot \hat{r})(\boldsymbol{\sigma}_2 \cdot \hat{r}) - \boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2$.
- Bound-state signals: $Q_d \neq 0$, $\mu_d \neq \mu_p + \mu_n$, few-percent D-state probability (model dependent).
- Continuum signals: mixing parameters in NN phase-shift analyses (ε_1, etc.).
- Microscopic origin: one-pion exchange at long range.

9.6.4 What remains for scattering

Bound-state multipoles fix the deuteron channel. Scattering states access all partial waves, both singlet and triplet, as functions of energy. Phase shifts $\delta_{^1S_0}(E)$, $\delta_{^3S_1}(E)$, P- and D-wave phase shifts, and mixing angles complete the NN interaction and feed into many-body nuclear structure. That programme is the natural continuation of Lectures 7–9 [2, 6].

9.7 Quadrupole moment and the tensor operator

9.7.1 Why $Q_d \neq 0$ requires S–D interference

A pure $^3S_1$ state has spherical symmetry and $Q_d = 0$. The measured $Q_d = +0.286\,\mathrm{fm}^2$ therefore proves an admixture of $L = 2$. In the coupled S–D basis the quadrupole operator $Q \propto r^2 Y_{20}$ has non-zero matrix elements between $L = 0$ and $L = 2$, so the expectation value scales as

$$Q_d \propto a_S a_D \, \langle r^2 \rangle_{SD} + a_D^2 \, \langle r^2 \rangle_{DD}, \tag{9.23}$$

with amplitudes a_S and a_D. The linear $a_S a_D$ term dominates when $a_D \ll 1$: Q_d is sensitive to the *phase* of the D-wave as well as its magnitude. The positive sign corresponds to prolate deformation (charge extended along the spin quantisation axis).

9.7.2 Angular structure of S_{12}

The tensor operator

$$S_{12} = 3(\boldsymbol{\sigma}_1 \cdot \hat{r})(\boldsymbol{\sigma}_2 \cdot \hat{r}) - \boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2 \tag{9.24}$$

couples spin to the separation vector. For nucleon spins aligned along $\hat{z}$, a short calculation gives $S_{12} = +2$; for spins perpendicular to $\hat{r}$ in the equatorial plane, $S_{12} = -1$. A central force averaged over a spherical s-wave vanishes on S_{12}, but the $^3S_1$–$^3D_1$ coupling does not. One-pion exchange generates (9.24) with Yukawa range $r_0 = \hbar/(m_\pi c)$ (Lecture 11), providing the microscopic link between pion physics and deuteron deformation.

Order-of-magnitude Q_d

With $r_{\rm rms} \approx 2.1\,\text{fm}$, a classical estimate $Q \sim er^2 \sim 0.4\,\text{e}\cdot\text{fm}^2$ sets the scale. The measured $eQ_d \approx 0.29\,\text{e}\cdot\text{fm}^2$ is smaller because only the few-percent D-wave contributes coherently. Requiring (9.23) to match experiment with $a_D^2 \sim 0.05$ is consistent with the P_D inferred from μ_d in Lecture 8.

Remark

Continuum counterpart. Bound-state $\eta_D = a_D/a_S$ at large r is mirrored in NN scattering by the ${}^3S_1$–${}^3D_1$ mixing parameter $\varepsilon_1(E)$ (Lecture 10). Potentials tuned to B_d, μ_d, and Q_d must simultaneously reproduce low-energy triplet phase shifts; failure in any one channel signals missing physics (three-body forces, relativistic effects, or inconsistent currents).

9.7.3 Electrostatic multipole expansion at $A = 2$

Outside the charge distribution the deuteron potential admits

$$\Phi(r,\theta) = \frac{e}{r} + \frac{1}{4\pi\varepsilon_0}\frac{1}{2r^3}\sum_{m=-2}^{2} Q_{2m}\,Y_{2m}(\theta,\phi) + \cdots \tag{9.25}$$

The monopole is the proton charge (neutron uncharged); the dipole vanishes for parity reasons; the quadrupole is the first deformation signal. Because $Q_d \propto a_S a_D$, measuring Q_d fixes the *product* of S and D amplitudes, while μ_d (Lecture 8) constrains their relative phase and spin content. No single deuteron observable alone yields $P_D = a_D^2$; the trio (B_d, μ_d, Q_d) is the minimal static set [6, 14].

9.7.4 Tensor force in heavier nuclei (preview)

The same S_{12} operator that mixes ${}^3S_1$ and ${}^3D_1$ in the deuteron contributes to spin–orbit splittings in the mean field (Lecture 12) and to the alignment of high-momentum pairs in short-range correlation studies. At $A = 2$ the tensor expectation is a few percent; in $A \sim 50$ nuclei it can drive shell reordering when combined with deformation (Lecture 15). The deuteron is the cleanest place to learn the operator before it is masked by many-body averaging [1, 4].

9.8 Further physics: multipoles beyond the quadrupole

Outside a localised charge distribution the electrostatic potential admits a multipole expansion: monopole (total charge), dipole (vanishes for definite-parity ground states), quadrupole, octupole, ... At large distance the lowest non-vanishing multipole dominates [13, 1, 6].

The same logic applies to magnetic multipoles and to radiation. In γ decay, multipole selection rules (angular momentum and parity of the photon field) organise transition rates (a topic for later lectures). For the deuteron ground state the essential static multipoles are already in hand: magnetic dipole μ_d and electric quadrupole Q_d. Together they require a tensor force and a few-percent D-wave, completing the qualitative NN picture of Lectures 7–9.

9.8.1 Deformation and quadrupole moments of heavier nuclei

Nuclei near closed (magic) shells are nearly spherical and have small Q. Nuclei far from magic numbers are often permanently deformed and can have large spectroscopic quadrupole moments; rotational bands appear in the spectrum [14, 6, 13]. The deuteron is the opposite extreme: few-body, weakly deformed, with Q_d generated by a small D-wave rather than collective rotation.

Static multipoles (monopole, dipole, quadrupole, ...) also organise electromagnetic radiation: γ transitions carry multipolarity $E1, M1, E2, \ldots$ with selection rules fixed by angular momentum and parity [13], the dynamical counterpart of the static moments used here.

9.8.2 Deuteron quadrupole moment in numbers

The free neutron and proton have vanishing electric quadrupole moments, so any $Q_d \neq 0$ must come from orbital structure. Experiment gives

$$Q_d = +0.00288\,\mathrm{b} = +0.286\,\mathrm{fm}^2, \tag{9.26}$$

small compared with heavy deformed nuclei but definitely nonzero [6]. A pure $^3S_1$ wave function would yield $Q_d = 0$. A mixed $S{+}D$ wave function produces

$$Q_d \propto a_S a_D\,\langle r^2\rangle_{SD} + a_D^2\,\langle r^2\rangle_{DD}, \tag{9.27}$$

so Q_d is linearly sensitive to the S–D interference. Realistic potentials that fit scattering and B_d reproduce Q_d with a_D^2 of a few percent, consistent with the magnetic-moment estimate [6, 14].

The positive sign of Q_d corresponds to a prolate (cigar-like) deformation: more charge along the symmetry axis than in the equatorial plane.

9.8.3 Single-particle vs collective quadrupole moments

For a single valence proton in an orbit of mean-square radius $\langle r^2\rangle \sim R^2 \sim r_0^2 A^{2/3}$, one expects

$$|Q| \lesssim e\,R^2 \sim 0.06\text{–}0.5\,e\cdot\mathrm{b} \tag{9.28}$$

from light to heavy nuclei. Many odd-A moments fall in that range, but rare-earth and actinide nuclei reach several $e \cdot \mathrm{b}$: the entire charged core is permanently deformed, so many protons contribute coherently [6, 2]. Even–even nuclei have vanishing spectroscopic Q in the laboratory frame if $I = 0$, but large *intrinsic* quadrupole moments show up in rotational spectra and in Coulomb excitation.

9.8.4 Tensor operator S_{12} and noncentral force

The simplest scalar operator that couples spin and relative coordinate is

$$S_{12} = 3(\boldsymbol{\sigma}_1 \cdot \hat{\boldsymbol{r}})(\boldsymbol{\sigma}_2 \cdot \hat{\boldsymbol{r}}) - \boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2. \tag{9.29}$$

A potential $V_T(r)\,S_{12}$ is the tensor force. It vanishes when averaged over a spherically symmetric s-wave alone, but it mixes $^3S_1$ and $^3D_1$ and generates Q_d. In heavier nuclei the same tensor force contributes to the spin-orbit splitting and to few-nucleon binding, though its expectation value is partially averaged by many-body motion [14, 6].

One-pion exchange naturally produces a tensor potential of Yukawa range $r_0 \approx 1.4\,\mathrm{fm}$; that is the long-range reason the deuteron is not a pure s-wave state [9, 14].

9.8.5 Parity and multipole selection

A static electric multipole of order ℓ has parity $(-1)^\ell$; a magnetic multipole has parity $(-1)^{\ell+1}$. Nuclear ground states of definite parity therefore forbid static electric dipoles: the lowest allowed electric deformation is the quadrupole. That is why Q is the first multipole beyond the monopole in the structural discussion of the deuteron and of deformed heavy nuclei [1, 6].

9.8.6 Evaluating S_{12} in the S–D basis

For two spin-$\frac{1}{2}$ particles with $\boldsymbol{S} = \boldsymbol{s}_1 + \boldsymbol{s}_2$, the tensor operator (9.24) has non-zero matrix elements only between states of the same total S and differing L by $\Delta L = 2$. In the ${}^3S_1$–${}^3D_1$ coupled channel, the off-diagonal element $\langle D|S_{12}|S\rangle$ is proportional to $\langle 2,0;1,0|1,0\rangle$ Clebsch–Gordan factors and sets the mixing rate. A radial integral over $V_T(r)\,u_S(r)\,u_D(r)$ converts the operator into an energy shift and a level coupling. One-pion exchange supplies $V_T(r) \propto e^{-m_\pi r}/r^3$ at long range (Lecture 11), so the D-wave admixture is tied directly to pion physics rather than to an arbitrary off-diagonal potential.

9.8.7 Consistency of μ_d, Q_d, and η_D

Modern NN fits require simultaneous agreement with:

1. B_d (Lecture 7 square-well threshold);
2. μ_d (Lecture 8 magnetic moment);
3. Q_d and η_D (this lecture);
4. triplet s-wave scattering length a_t (Lecture 10).

A potential that fits B_d but misses Q_d typically has incorrect tensor strength; one that fits Q_d but misses μ_d may have right tensor geometry but wrong current operators. The deuteron is therefore a *constraint bundle*, not a single-number benchmark.

Classical quadrupole vs experiment

A uniform charge distribution in a prolate spheroid with semi-axis $a = 2.0\,\mathrm{fm}$ and $b = 1.5\,\mathrm{fm}$ has $Q \sim \frac{2}{5}e(a^2 - b^2) \approx 0.5\,\mathrm{e}\cdot\mathrm{fm}^2$. The measured $eQ_d \approx 0.29\,\mathrm{e}\cdot\mathrm{fm}^2$ is smaller because only the D-wave fraction ($\sim 5\%$) contributes. Dividing the experimental value by the classical estimate gives a rough $a_D^2 \sim 0.06$, consistent with the P_D from Lecture 8.

9.8.8 Bridge to shell-model Q systematics

Heavy-nucleus quadrupole moments (Section 9.6.2) arise from collective deformation or valence-shell orbits, not from S–D mixing. The tensor force that shapes the deuteron reappears in the nuclear mean field as spin–orbit and deformation-driving terms (Lectures 12–13). The sign changes of Q near magic numbers and the large Q of rare-earth nuclei (Lecture 15) are many-body amplifications of the same operator structure seen at $A = 2$ [6, 1].

9.8.9 Asymptotic D/S ratio η_D

At large r the S and D components behave as $u_S \sim e^{-\gamma r}$ and $u_D \sim \eta_D e^{-\gamma r}$ with the same γ (Lecture 7). Electron scattering and tensor-polarisation observables constrain η_D independently of the interior potential. Because $Q_d \propto a_S a_D$ and μ_d depends on the same product with different angular weighting, (Q_d, η_D, μ_d) overconstrain fits if the current operator is held fixed. Modern analyses treat η_D as a standard deuteron observable alongside B_d and r_{rms} [6, 14].

9.9 Deeper reading: Q_d, S_{12}, and high-momentum pairs

Without the tensor operator S_{12}, a pure S-wave deuteron has vanishing quadrupole moment. The measured $Q_d > 0$ is therefore direct evidence for tensor-driven S–D mixing. One-pion exchange supplies the long-range tensor force; short-range cutoffs regulate the contact pieces. In many-body systems the same tensor correlations produce high-momentum neutron–proton pairs observed as short-range correlations and drive shell evolution away from stability (Lecture 12).

A classical estimate $Q \sim r^2$ with $r \sim 2\,\text{fm}$ gives the correct mega-femtometre-squared scale, but the quantum matrix element depends on the relative phase of S and D waves. When reading SRC ratios, keep them distinct from mean-field single-particle energy shifts: both are tensor-related, yet they probe different momentum domains. Lectures 10–11 then embed this force piece into scattering and meson exchange more broadly.

Operator identities and expectation values

Evaluate S_{12} for spin-aligned and spin-antialigned configurations to recover the classic factors $+2$ and -1. Those factors explain why the tensor force is attractive in the deuteron channel and repulsive in others. In momentum space, the tensor force mixes high relative momenta, feeding the SRC plateau observed in exclusive electron scattering.

Shell-evolution sketch

A schematic tensor force shifts single-particle energies of $j_>$ and $j_<$ partners differently as protons or neutrons fill spin–orbit partners. That differential shift is a leading explanation for magic-number migration (Lecture 12). Classroom discussions should separate this mean-field tensor effect from high-momentum SRC fractions: same operator family, different probes.

Tensor force across scales

At long distance, one-pion exchange generates a tensor potential $\propto S_{12}(e^{-m_\pi r}/r)$ times a radial function. At short distance, contacts and heavier-meson exchanges cut off the singularity. The deuteron D-wave lives primarily in the intermediate region where one-pion exchange is trustworthy, which is why Q_d is a relatively clean tensor observable compared with high-momentum SRC ratios.

In a nucleus, the same operator has a mean-field avatar (single-particle energy shifts) and a short-range avatar (high-momentum pairs). A student literature note should cite one example of each and refuse to conflate their percentages. Bridging to Lecture 12, sketch how filling a $j_>$ proton orbital shifts neutron $j_<$ energies via the tensor force: that cartoon is sufficient preparation for modern shell-evolution discussions without requiring a full G-matrix derivation.

The tensor force is the structural thread from Q_d to shell evolution and SRC. Keeping long-distance and short-distance tensor manifestations separate is essential for clear reasoning.

In practice, rework one numerical estimate from the lecture with a different parameter set and compare the shift to the quoted experimental uncertainty. If the shift exceeds the uncertainty, the estimate is teaching a scale, not delivering a fit. That distinction keeps theoretical cartoons honest when they sit beside evaluated data tables and modern many-body calculations. Along the course arc, write one sentence linking this chapter backward and one forward: the discipline of those two sentences is as important as any single formula. Examination problems rarely ask for a literature review, but they do ask for the scale argument that literature reviews presuppose.

Research frontier: tensor force, SRC, and student projects

The deuteron quadrupole moment of Lecture 9 is the cleanest static signature of the tensor force: without S_{12}, a pure S-wave deuteron would have $Q_d = 0$. In many-body systems the same tensor component drives shell evolution in exotic nuclei (migration of single-particle energies and islands of inversion) and correlates high-momentum nucleon–nucleon pairs observed as short-range correlations (SRC) in exclusive electron scattering [45, 43]. Chiral EFT organises the tensor force from one-pion exchange upward, while SRC studies connect laboratory high-momentum fractions to the dense-matter equation of state and to the neutron-proton dominance of correlated pairs [44]. Open questions include the quantitative link between measured SRC ratios and the high-density symmetry energy, and how much of shell evolution is tensor-driven versus continuum- or central-force-driven in specific isotopic chains.

Experimentally, SRC studies use exclusive $(e, e'pN)$ reactions to isolate high-relative-momentum pairs, while structure studies infer tensor-driven shell evolution from energy gaps and spectroscopic factors. A student literature survey that keeps those two communities' observables distinct avoids mixing correlation fractions with mean-field single-particle shifts. Keeping SRC ratios and mean-field shell gaps as distinct observables prevents double counting when students survey the tensor-force literature.

Student directions. (1) Analytic exercise: evaluate S_{12} for aligned and perpendicular spin orientations and recover the lecture factors $+2$ and -1. (2) Scale estimate: estimate the classical quadrupole scale $Q \sim r^2$ for $r \sim 2\,\text{fm}$ and compare with Q_d. (3) Literature note: summarise one experimental SRC ratio (e.g. *np* dominance) and its implication for neutron-rich matter.

9.10 Problems with solutions

Problem 9.1 — $Q = 0$ for a sphere. Show that $Q_{zz} = \int \rho(3z^2 - r^2)\,\mathrm{d}^3r$ vanishes for any spherically symmetric $\rho(r)$.

Solution 9.1. Spherical symmetry $\Rightarrow \int x^2\rho = \int y^2\rho = \int z^2\rho = \frac{1}{3}\int r^2\rho$. Hence

$$\int (3z^2 - r^2)\rho = 3 \cdot \tfrac{1}{3}\int r^2\rho - \int r^2\rho = 0.$$

A pure $L = 0$ deuteron would have $Q_d = 0$ [6].

Problem 9.2 — Units of Q_d. The deuteron quadrupole moment is $Q_d = 0.00288\,\mathrm{b}$ and also $2.88 \times 10^{-27}\,\mathrm{cm}^2$. Convert the barn value to $\mathrm{e}\cdot\mathrm{fm}^2$ ($1\,\mathrm{b} = 100\,\mathrm{fm}^2$) and verify consistency with $10^{-27}\,\mathrm{cm}^2$.

Solution 9.2.

$$0.00288\,\mathrm{b} = 0.00288 \times 100 = 0.288\,\mathrm{fm}^2, \qquad eQ_d = 0.288\,\mathrm{e}\cdot\mathrm{fm}^2$$

(the first is $Q_d = Q_{zz}/e$; the second restores the charge unit). Also $1\,\mathrm{b} = 10^{-28}\,\mathrm{m}^2 = 10^{-24}\,\mathrm{cm}^2$, so

$$0.00288\,\mathrm{b} = 0.00288 \times 10^{-24}\,\mathrm{cm}^2 = 2.88 \times 10^{-27}\,\mathrm{cm}^2,$$

matching the lecture value. Compared with eR^2 for $R \sim 2\,\mathrm{fm}$ ($\sim 4\,e\cdot\mathrm{fm}^2$), the deformation is mild [6].

Problem 9.3 — Valid far-field expansion and the factor of two. Starting from the exact identity

$$\frac{1}{|\boldsymbol{r} - \boldsymbol{r}'|} = \frac{1}{r}\left(1 - 2\epsilon\cos\theta + \epsilon^2\right)^{-1/2} \qquad \left(\epsilon \equiv \frac{r'}{r} < 1\right),$$

expand through $O(\epsilon^2)$. For an axial charge distribution show that the potential coefficient $Q_2 = e^{-1}\int \rho r'^2 P_2(\cos\theta')\,\mathrm{d}^3r'$ satisfies $Q_2 = Q_{zz}/(2e) = Q_{\mathrm{spec}}/2$.

Solution 9.3. Put $x = -2\epsilon\cos\theta + \epsilon^2$ in $(1+x)^{-1/2} = 1 - \frac{1}{2}x + \frac{3}{8}x^2 + \cdots$. Keeping all terms through ϵ^2,

$$\frac{1}{|\boldsymbol{r} - \boldsymbol{r}'|} = \frac{1}{r}\left[1 + \epsilon\cos\theta + \frac{\epsilon^2}{2}(3\cos^2\theta - 1) + \cdots\right]$$

The ϵ^2 term is exactly $(r'^2/r^2)P_2(\cos\theta)$. Along the symmetry axis,

$$eQ_2 = \int \rho r'^2 P_2(\cos\theta')\,\mathrm{d}^3r' = \frac{1}{2}\int \rho(3z'^2 - r'^2)\,\mathrm{d}^3r' = \frac{1}{2}Q_{zz}.$$

Hence $Q_2 = Q_{\mathrm{spec}}/2$, which is the required convention check [6, 1].

Problem 9.4 — Why a tensor force is required. Given $I^\pi = 1^+$, $\mu_d \neq \mu_p + \mu_n$, and $Q_d \neq 0$, explain why a purely central NN potential is ruled out.

Solution 9.4. A central potential conserves L, so each energy eigenstate has definite L. Even parity and $I = 1$ allow $L = 0$ and $L = 2$ separately, but not a coherent mixture in one eigenstate. Nonzero Q_d requires nonsphericity (D-wave), and the μ_d deficit requires the same mixture. The tensor force $V_T(r)S_{12}$ is the non-central operator that mixes ${}^3S_1$ and ${}^3D_1$ [6, 14].

Problem 9.5 — Tensor operator on a pure S-wave. The tensor operator is $S_{12} = 3(\boldsymbol{\sigma}_1 \cdot \hat{r})(\boldsymbol{\sigma}_2 \cdot \hat{r}) - \boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2$. Argue why $\langle S_{12} \rangle = 0$ in a pure ${}^3S_1$ state with spherical average, yet S_{12} can still mix S and D waves.

Solution 9.5. In a pure s-wave the orbital density is isotropic. Averaging $(\boldsymbol{\sigma}_1 \cdot \hat{r})(\boldsymbol{\sigma}_2 \cdot \hat{r})$ over angles reduces it to $\frac{1}{3}\boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2$, so $\langle S_{12} \rangle = 0$. Off-diagonal matrix elements between $L = 0$ and $L = 2$ need not vanish: S_{12} transforms as a rank-2 tensor and connects $\Delta L = 2$. That is how a small D-wave is generated even though the pure-S diagonal expectation is zero [14, 6].

Problem 9.6 — Geometric scale of Q. Estimate the single-particle quadrupole scale eR^2 for $R = r_0 A^{1/3}$ with $r_0 = 1.2\,\mathrm{fm}$ at $A = 2$ and at $A = 160$. Compare with $Q_d = 0.288\,e \cdot \mathrm{fm}^2$ and with rare-earth moments of several $e \cdot \mathrm{b}$.

Solution 9.6.

$$R(A{=}2) \approx 1.2 \times 1.26 \approx 1.5\,\mathrm{fm}, \quad eR^2 \approx 2.3\,e \cdot \mathrm{fm}^2;$$

$$R(A{=}160) \approx 1.2 \times 5.4 \approx 6.5\,\mathrm{fm}, \quad eR^2 \approx 42\,e \cdot \mathrm{fm}^2 = 0.42\,e \cdot \mathrm{b}.$$

Q_d is $\ll eR^2(A{=}2)$ (weak D-wave deformation). Collective rare-earth $Q \sim$ few b exceed the single-particle scale by an order of magnitude (coherent core deformation) [6, 2].

Problem 9.7 — Tensor orientation and selection rule. For a triplet product state with both spins polarised along z, evaluate $\langle S_{12} \rangle$ when $\hat{r} \parallel \hat{z}$ and when $\hat{r} \perp \hat{z}$. Then use the rank and parity of S_{12} to explain why it couples $L = 0$ to $L = 2$, but not to $L = 1$.

Solution 9.7. For the aligned triplet, $\langle \boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2 \rangle = 1$. If $\hat{r} \parallel \hat{z}$, each projected Pauli operator has eigenvalue $+1$, so $\langle S_{12} \rangle = 3 - 1 = 2$. If $\hat{r} \perp \hat{z}$, each one-spin transverse expectation vanishes, so $\langle S_{12} \rangle = -1$. The interaction is orientation dependent.

The orbital part is rank 2 and even under parity. The triangle rule permits $|L - L'| \leq 2 \leq L + L'$, while parity requires L and L' to have the same parity. From $L = 0$, only $L' = 2$ survives; $L' = 1$ has the wrong parity. Thus the tensor force produces the observed S–D mixing.

Problem 9.8 — Pure S wave and Q_d. Show that a pure $\ell = 0$ deuteron wave function cannot produce a nonzero electric quadrupole moment. What does $Q_d \neq 0$ therefore imply?

Solution 9.8. An $\ell = 0$ spatial wave function is isotropic (Y_{00}); the quadrupole operator averages to zero for a spherically symmetric charge distribution. Measured $Q_d \neq 0$ therefore requires a D-wave ($\ell = 2$) component and a tensor force [6, 1].

Chapter 10

Scattering of Nucleons

"Although the study of the deuteron gives us a number of clues about the nucleon–nucleon interaction, the total amount of information available is limited... To study the nucleon–nucleon interaction in different configurations, we can perform nucleon–nucleon scattering experiments."

K. S. Krane

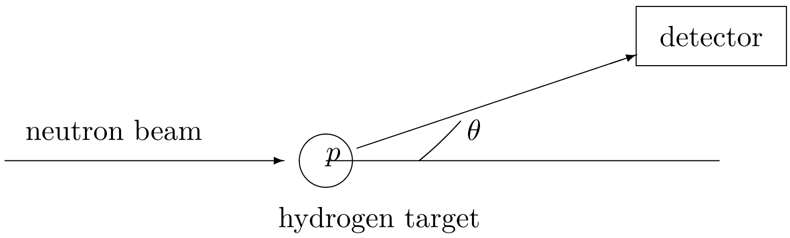
The deuteron (Lectures 7–9) is the only bound two-nucleon system. It samples the residual NN force in a single channel, mostly ${}^3S_1$ with a few-percent D-wave, at a fixed average separation fixed by weak binding. That is not enough to reconstruct the full force. There are no bound excited states to explore other partial waves or spin orientations, and the continuum is invisible from multipole moments alone [6].

Scattering supplies what the bound state cannot: controllable energy, access to $\ell = 0, 1, 2, \ldots$, and both spin channels. In this lecture the experimental definition of the cross section is tied to partial-wave analysis. Phase shifts δ_ℓ become the language in which theory and experiment meet for a short-range central potential [6, 1, 2, 14].

10.1 Why scatter neutrons on protons?

A scattering experiment uses a monoenergetic **projectile** beam and a **target** with many scattering centres. For *n–p* scattering the projectiles are neutrons of chosen kinetic energy and the target is hydrogen: each scattering centre is a free proton (bound only in the atom, irrelevant at nuclear energies). A detector at polar angle θ relative to the beam axis counts scattered particles. For a central force the yield does not depend on the azimuthal angle ϕ.

Hydrogen is essential for a clean NN measurement. On a heavy nucleus an incident nucleon would scatter from several target nucleons within the force range, so the observed yield would mix many-body effects. On hydrogen each encounter is a single *np* collision; multiple scattering from different atoms is rare if the target is thin [6]. The same philosophy motivates electron scattering on thin foils when one wants the elementary charge form factor (Lecture 1): isolate one interaction per projectile.



centre-of-mass angle; thin target isolates one np collision

Figure 10.1: Schematic n–p scattering setup: monoenergetic projectile beam, hydrogen target, and detector at angle θ.

10.2 Differential and total cross sections

The number $\mathrm{d}N$ of particles recorded in time t into solid angle $\mathrm{d}\Omega$ grows with exposure time, detector acceptance, beam intensity I (particles per unit area per unit time), and the number of target centres N_t (or areal density n_a times the illuminated area). These experimental knobs are factored out by defining the **differential scattering cross section** $\mathrm{d}\sigma/\mathrm{d}\Omega$ through

$$\mathrm{d}N = \frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}\, t\, \mathrm{d}\Omega\, I\, N_t. \tag{10.1}$$

Equivalently,

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} = \frac{\mathrm{d}N}{\mathrm{d}\Omega\, n_a\, N_p}, \qquad N_p = tIA, \tag{10.2}$$

with A the beam area. The cross section has dimensions of area; the historical unit is the barn,

$$1\,\mathrm{b} = 10^{-28}\,\mathrm{m}^2 = 100\,\mathrm{fm}^2.$$

Nuclear NN cross sections at low energy are tens of barns, huge compared with the geometric size of a nucleon ($\sim 1\,\mathrm{fm}^2$), a quantum enhancement characteristic of low-energy s-wave scattering [6, 1].

Physically, $\mathrm{d}\sigma/\mathrm{d}\Omega$ is the effective area that the target presents for scattering into $\mathrm{d}\Omega$. It packages all of the unknown dynamics into a measurable quantity: once $\mathrm{d}\sigma/\mathrm{d}\Omega$ is known, one can compare theories without redoing the experiment for every proposed potential.

Angles and energies may be quoted in the laboratory or centre-of-mass (CM) frame. Partial-wave formulae below are written in the CM frame; converting to laboratory angles for equal-mass np scattering is a standard kinematic exercise. The **total** elastic cross section is the angular integral

$$\sigma_{\mathrm{tot}} = \int \frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}\, \mathrm{d}\Omega. \tag{10.3}$$

For pure s-wave scattering, $\mathrm{d}\sigma/\mathrm{d}\Omega$ is isotropic in the CM frame and $\sigma_{\mathrm{tot}} = 4\pi\,(\mathrm{d}\sigma/\mathrm{d}\Omega)$.

10.3 Central potential, plane waves, and partial waves

A **central** potential $V(r)$ depends only on the nucleon–nucleon separation. Angular momentum about the force centre is then conserved, so eigenstates of definite orbital angular momentum ℓ

are useful. In the absence of interaction the incident beam is a plane wave along z,

$$\psi_{\text{inc}} = e^{ikz}, \qquad k = \frac{p}{\hbar} = \frac{\sqrt{2\mu E}}{\hbar}, \tag{10.4}$$

where $\mu \approx m_N/2$ is the reduced mass of the NN system and E is the centre-of-mass kinetic energy. Multiplying by $e^{-iEt/\hbar}$ makes the wave travel in the $+z$ direction: toward the target for $z < 0$ and away from it for $z > 0$ [6].

A plane wave has definite energy and momentum along z, but it is *not* an eigenstate of $\boldsymbol{L}^2$. It is a coherent superposition of all orbital angular momenta. Because of axial symmetry about the beam, only projections $m = 0$ appear. The expansion in spherical waves (partial waves) reads

$$e^{ikz} = \sum_{\ell=0}^{\infty} i^{\ell}(2\ell+1)\, j_\ell(kr)\, P_\ell(\cos\theta), \tag{10.5}$$

where j_ℓ are spherical Bessel functions and P_ℓ Legendre polynomials. Each term $\ell = 0, 1, 2, \ldots$ is an s, p, d, … wave with a definite impact-parameter character in the semiclassical picture.

10.4 Phase shifts: what a short-range potential can do

Detectors sit centimetres from the target, while the nuclear force acts only for $r \lesssim$ a few fm. Outside the force range the wave is free again; the potential cannot change its wavelength there. What it *can* do is shift the phase of each outgoing partial wave relative to the free solution.

Free spherical waves have the asymptotic form

$$j_\ell(kr) \sim \frac{\sin(kr - \ell\pi/2)}{kr} \qquad (kr \to \infty). \tag{10.6}$$

In the presence of a short-range central potential the radial function $u_\ell(r) = rR_\ell(r)$ for r larger than the force range becomes

$$u_\ell(r) \sim \sin(kr - \ell\pi/2 + \delta_\ell), \tag{10.7}$$

where $\delta_\ell(E)$ is the **phase shift** in channel ℓ [6, 2]. Attractive potentials typically pull nodes of u_ℓ toward the origin (positive δ_ℓ); repulsive potentials push nodes outward (negative δ_ℓ).

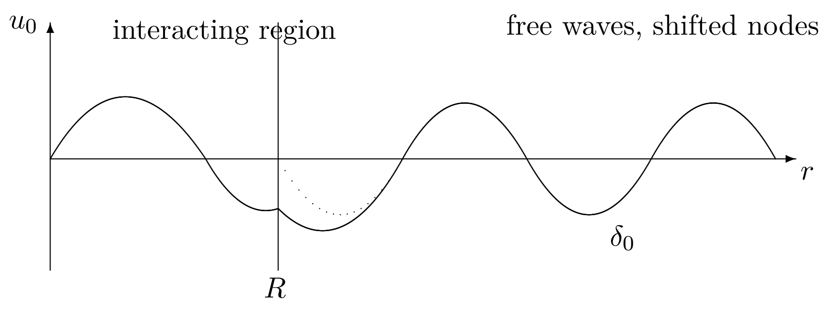


Figure 10.2: Schematic s-wave radial wave $u(r)$. Inside the force range the potential distorts the wave; outside, the free oscillation is shifted by the phase δ_0.

All information about $V(r)$ that elastic scattering at energy E can provide is encoded in the set $\{\delta_\ell(E)\}$. Solving the radial Schrödinger equation for a model potential and reading δ_ℓ is the theoretical route; measuring $\mathrm{d}\sigma/\mathrm{d}\Omega$ and fitting δ_ℓ is the experimental route. Phase shifts themselves are not directly observed: they are a convenient parameterisation of the measured cross sections (and polarisations). The two routes meet in (10.12) below.

For $\ell = 0$ one may write the exterior wave as a combination of incoming and outgoing spherical waves $e^{\pm ikr}/r$. Scattering does not create or destroy particles in elastic scattering, so it cannot change the *amplitudes* of those waves; it can only change the relative phase of the outgoing piece. That single phase is δ_0 (up to a conventional factor of two in intermediate steps of the algebra) [6].

10.4.1 Matching any partial wave at the force boundary

Let the potential vanish for $r > R$, and define the logarithmic derivative of the regular interior solution by

$$\mathcal{L}_\ell(E) = \left.\frac{u_\ell'}{u_\ell}\right|_R .$$

With Riccati–Bessel functions $\hat{j}_\ell(x) = xj_\ell(x)$ and $\hat{n}_\ell(x) = xn_\ell(x)$, write

$$u_\ell(r) = C_\ell\left[\hat{j}_\ell(kr)\cos\delta_\ell - \hat{n}_\ell(kr)\sin\delta_\ell\right].$$

Continuity of u_ℓ and u_ℓ' at R gives

$$\tan\delta_\ell = \frac{k\hat{j}_\ell'(kR) - \mathcal{L}_\ell\hat{j}_\ell(kR)}{k\hat{n}_\ell'(kR) - \mathcal{L}_\ell\hat{n}_\ell(kR)}. \tag{10.8}$$

Thus the interior radial equation supplies one logarithmic derivative, which fixes δ_ℓ, then S_ℓ, then the cross section.

10.5 Cross section from phase shifts

Matching the asymptotic partial-wave solution to an incident plane wave plus outgoing scattered spherical wave produces the scattering amplitude $f(\theta)$. Outside the force range the radial wave in channel ℓ may be written as a superposition of incoming and outgoing spherical waves,

$$u_\ell(r) \propto e^{-i(kr-\ell\pi/2)} - S_\ell\, e^{i(kr-\ell\pi/2)}, \tag{10.9}$$

with the unitary S-matrix element $S_\ell = e^{2i\delta_\ell}$ for a real central potential (elastic scattering). The coefficient of the outgoing scattered wave relative to the free case defines the partial-wave amplitude

$$f_\ell = \frac{S_\ell - 1}{2ik} = \frac{e^{2i\delta_\ell} - 1}{2ik} = \frac{e^{i\delta_\ell}\sin\delta_\ell}{k}. \tag{10.10}$$

The incoming and outgoing radial probability currents are proportional to their squared amplitudes. Elastic flux conservation therefore requires $|S_\ell| = 1$, so $S_\ell = e^{2i\delta_\ell}$ with real δ_ℓ. If inelastic channels open, one instead writes $S_\ell = \eta_\ell e^{2i\delta_\ell}$, $0 \le \eta_\ell < 1$; all formulae below use the elastic limit $\eta_\ell = 1$. Summing over partial waves with the Legendre weights of the plane-wave expansion gives

$$f(\theta) = \sum_{\ell=0}^{\infty} (2\ell+1)\, f_\ell\, P_\ell(\cos\theta) = \frac{1}{k} \sum_{\ell=0}^{\infty} (2\ell+1)\, e^{i\delta_\ell} \sin\delta_\ell\, P_\ell(\cos\theta). \tag{10.11}$$

The CM differential cross section is therefore

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} = |f(\theta)|^2 = \frac{1}{k^2} \left| \sum_{\ell=0}^{\infty} (2\ell+1)\, e^{i\delta_\ell} \sin\delta_\ell\, P_\ell(\cos\theta) \right|^2. \tag{10.12}$$

(The overall phase convention for $e^{i\delta_\ell}$ varies among sources; the measurable $|f|^2$ is fixed by the δ_ℓ.) The factor $(2\ell+1)$ counts magnetic substates of a given ℓ; the Legendre polynomials build the angular pattern of each partial wave.

When only $\ell = 0$ contributes, $P_0 = 1$ and

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} = \frac{\sin^2\delta_0}{k^2}, \tag{10.13}$$

independent of angle in the CM frame. For an s wave alone,

$$\sigma_{\text{tot}} = \int \frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}\, \mathrm{d}\Omega = \frac{\sin^2\delta_0}{k^2} \int \mathrm{d}\Omega = \frac{4\pi}{k^2} \sin^2\delta_0.$$

For a general partial-wave sum the different ℓ interfere in $\mathrm{d}\sigma/\mathrm{d}\Omega$, but after angular integration the cross terms vanish by orthogonality $\int P_\ell P_{\ell'}\, \mathrm{d}\Omega = 4\pi\delta_{\ell\ell'}/(2\ell+1)$. Each multipole then contributes separately:

$$\sigma_{\text{tot}} = \frac{4\pi}{k^2} \sum_{\ell=0}^{\infty} (2\ell+1) \sin^2\delta_\ell = \frac{4\pi}{k^2} \sin^2\delta_0 \quad (s\text{-wave only}). \tag{10.14}$$

Thus a single number $\delta_0(E)$ determines the entire low-energy elastic yield. The unitarity bound $\sin^2\delta_0 \le 1$ implies $\sigma_{\text{tot}} \le 4\pi/k^2$: at very low energy the quantum cross section can greatly exceed the geometric size of the potential well [6, 14].

Caution

Factor-of-four check. The physical outgoing-to-incoming phase is $S_\ell = e^{2i\delta_\ell}$, not $e^{i\delta_\ell}$. Hence

$$\left| \frac{S_\ell - 1}{2ik} \right|^2 = \frac{|2ie^{i\delta_\ell} \sin\delta_\ell|^2}{4k^2} = \frac{\sin^2\delta_\ell}{k^2}.$$

For an s wave this gives $\sigma_0 = 4\pi \sin^2\delta_0/k^2$. Dropping the factor 2 in $S - 1 = 2ie^{i\delta} \sin\delta$ produces the erroneous factor-of-four cross section.

10.5.1 Optical theorem from the same partial waves

Unitarity links the total cross section to the forward scattering amplitude. Once partial-wave unitarity is written as $S_\ell = e^{2i\delta_\ell}$, the optical theorem is automatic: every absorbed or elastically redirected flux appears as $\operatorname{Im} f(0)$. Measuring only the small-angle interference with Coulomb scattering can therefore constrain σ_{tot} [6, 14].

At $\theta = 0$, $P_\ell(1) = 1$, and therefore

$$\operatorname{Im} f(0) = \frac{1}{k}\sum_{\ell=0}^{\infty}(2\ell+1)\sin^2\delta_\ell.$$

Comparison with Equation (10.14) yields

$$\boxed{\sigma_{\text{tot}} = \frac{4\pi}{k}\operatorname{Im} f(0)}. \tag{10.15}$$

The optical theorem is therefore the forward-angle statement of partial-wave unitarity.

For equal-mass np scattering the laboratory and CM scattering angles are related by $\theta_{\text{lab}} = \theta_{\text{CM}}/2$, and

$$\left(\frac{d\sigma}{d\Omega}\right)_{\text{lab}} = 4\cos\theta_{\text{lab}}\left(\frac{d\sigma}{d\Omega}\right)_{\text{CM}}. \tag{10.16}$$

Partial-wave formulae in this lecture are always in the CM frame unless stated otherwise.

Key idea

Phase shifts δ_ℓ are the interface between potential and data. Theory: solve the radial equation for $V(r)$ and extract δ_ℓ. Experiment: measure $d\sigma/d\Omega$ and determine δ_ℓ.

10.6 Low energy: only s-wave scattering

Semiclassically, a projectile of momentum $p = \sqrt{2\mu E}$ that grazes the force range r_0 carries angular momentum $\ell\hbar \sim pr_0$. For nuclear $r_0 \sim 2\,\text{fm}$ and $E \sim 1\,\text{MeV}$ one finds $pr_0/\hbar \ll 1$: the projectile cannot bring $\ell \geq 1$ inside the nuclear force range without flying past outside the well. Only $\ell = 0$ scatters. At $E \sim 10\,\text{MeV}$, $\ell = 1$ begins to contribute [6].

A sharper estimate uses the free kinetic energy scale associated with confinement in a well of size R,

$$T \sim \frac{\ell(\ell+1)\hbar^2}{2\mu R^2}.$$

For $\ell = 1$, $R \sim 1\,\text{fm}$, and $\mu c^2 \sim 500\,\text{MeV}$, T is tens of MeV. Incident energies far below that scale justify the s-wave truncation [6].

This is why low-energy np data are so powerful and so simple: a single phase $\delta_0(E)$ (actually one for each spin channel) carries the dynamics, and $\sigma(\theta)$ is nearly isotropic.

10.7 Square well in the continuum: matching for $E > 0$

The same square well used for the deuteron bound state can be solved for positive energy. For $\ell = 0$,

$$V(r) = \begin{cases} -V_0 & r < R, \\ 0 & r > R, \end{cases} \qquad E > 0. \tag{10.17}$$

Inside, $u(r) = A\sin(k_1 r)$ with $k_1 = \sqrt{2\mu(V_0+E)}/\hbar$ (regularity at the origin again kills the cosine). Outside,

$$u(r) = C\sin(kr+\delta_0), \qquad k = \frac{\sqrt{2\mu E}}{\hbar}. \tag{10.18}$$

Continuity of u and u' at $r = R$ yields

$$k\cot(kR+\delta_0) = k_1\cot(k_1 R). \tag{10.19}$$

Solving for the phase shift,

$$\delta_0 = \arctan\!\left(\frac{k}{k_1}\tan(k_1 R)\right) - kR \pmod{\pi}. \tag{10.20}$$

Given E, V_0, and R, one thus obtains $\delta_0(E)$ and then predicts σ_{tot} from (10.14) [6].

In the zero-energy limit $k \to 0$, $k_1 \to K \equiv \sqrt{2\mu V_0}/\hbar$ and the scattering length extracted from $a = -\lim(\tan\delta_0/k)$ becomes

$$a = R - \frac{\tan(KR)}{K}. \tag{10.21}$$

(If $KR = \pi/2$, $a \to \infty$: the zero-energy bound-state threshold of Lecture 7.) Then $\sigma_{\text{tot}} \to 4\pi a^2$.

As $V_0 \to 0$, $\delta_0 \to 0$ and there is no scattering. For the attractive well fitted to the deuteron ($V_0 \approx 35\,\text{MeV}$, $R \sim 2\,\text{fm}$), low-energy δ_0 is large and σ_{tot} is tens of barns. Experiment finds the zero-energy unpolarised np cross section near 20 b, not the $\sim$ 4–5 b that a pure triplet estimate would suggest. The resolution is that free protons and neutrons scatter in a statistical mixture of triplet and singlet spins: three triplet states and one singlet, so the spin weights are 3/4 and 1/4. Writing

$$\sigma = \tfrac{3}{4}\sigma_t + \tfrac{1}{4}\sigma_s, \tag{10.22}$$

and taking $\sigma_t \approx 4\pi a_t^2 \sim 4.6\,\text{b}$ from the deuteron (triplet) channel leaves $\sigma_s \sim 70\,\text{b}$: the singlet interaction is strong but just unbound, producing a large negative scattering length [6, 14].

10.8 Scattering length and effective range

At vanishing energy, $\delta_0 \to 0$ in such a way that

$$a = -\lim_{k\to 0}\frac{\tan\delta_0}{k} \tag{10.23}$$

defines the **scattering length** a. At low energy $\delta_0 \to 0$ with $\tan\delta_0 \sim \delta_0 \sim \sin\delta_0$, so

$$\tan\delta_0 \approx -ka \quad\Rightarrow\quad \sin\delta_0 \approx -ka$$

(to leading order in k). Insert into the s-wave total cross section:

$$\sigma_{\text{tot}} = \frac{4\pi}{k^2}\sin^2\delta_0 \approx \frac{4\pi}{k^2}(ka)^2 = 4\pi a^2. \tag{10.24}$$

Thus as $k \to 0$,

$$\sigma_{\text{tot}} \to 4\pi a^2. \tag{10.25}$$

The sign of a carries physics: a large *positive* scattering length is characteristic of a barely bound state (triplet, deuteron); a large *negative* scattering length signals a virtual state just above threshold (singlet) [6, 14].

Checked low-energy np numbers

Standard values are $a_t \approx +5.4\,\text{fm}$ (triplet) and $a_s \approx -23.7\,\text{fm}$ (singlet). The unpolarised zero-energy cross section is then

$$\begin{aligned}\sigma &= 4\pi\left(\tfrac{3}{4}a_t^2 + \tfrac{1}{4}a_s^2\right)\\ &= 4\pi\left(\tfrac{3}{4}(5.4)^2 + \tfrac{1}{4}(23.7)^2\right) \approx 4\pi(21.9 + 140.4) \approx 20\,\text{b},\end{aligned}$$

matching experiment. The huge singlet piece dominates because $|a_s|$ is so large: the singlet potential nearly binds [6, 14].

A more accurate low-energy expansion follows from matching the exterior wave to an effective-range expansion of the logarithmic derivative. Writing

$$k\cot\delta_0 = -\frac{1}{a} + \frac{1}{2}r_0k^2 + \cdots, \tag{10.26}$$

defines the effective range r_0 of fermi size. The first two terms already reproduce low-energy $\delta_0(E)$ in each spin channel without committing to a full potential at every energy [6, 14]. Fitting a and r_0 is how modern analyses encode low-energy NN physics.

The s-wave amplitude itself is

$$f_0(k) = \frac{1}{k\cot\delta_0 - ik} = \frac{1}{-1/a + \frac{1}{2}r_0k^2 - ik + \cdots}. \tag{10.27}$$

This form exposes both unitarity (the $-ik$ term) and the pole structure. A shallow bound state is a pole at $k = i\kappa$, $\kappa > 0$. Setting the denominator to zero,

$$-\frac{1}{a} + \frac{1}{2}r_0(-\kappa^2) - i(i\kappa) = 0 \quad \Rightarrow \quad -\frac{1}{a} - \frac{1}{2}r_0\kappa^2 + \kappa = 0, \tag{10.28}$$

so

$$\frac{1}{a} = \kappa - \frac{1}{2}r_0\kappa^2 + \cdots. \tag{10.29}$$

At leading order $a \simeq 1/\kappa > 0$, as in the triplet deuteron channel. A pole on the negative imaginary axis is a virtual state and corresponds at leading order to a large negative a, as in the singlet channel. For the square well, $KR = \pi/2$ makes $a = R - \tan(KR)/K$ diverge: the pole reaches zero energy at exactly the bound-state threshold found in Lecture 7.

Key idea

The deuteron fixes the *triplet* well near threshold. Scattering reveals a strong *singlet* interaction that fails to bind, and it maps how $\delta_\ell(E)$ evolve as higher partial waves open with energy.

10.8.1 Scattering lengths, Coulomb interference, and unitarity

Low-energy np data separate into triplet (positive scattering length, bound-state related) and singlet (negative scattering length, virtual state) s waves. In pp scattering the Coulomb and nuclear amplitudes interfere, so the observed cross section is not the pure nuclear value; phase-shift analyses still extract $\delta_0(E)$ and thence scattering lengths after Coulomb modifications. The optical theorem $\sigma_{\text{tot}} = (4\pi/k)\,\text{Im}\,f(0)$ follows from forward unitarity of the same partial-wave expansion [6, 14, 2].

10.9 Deeper physics: partial waves, scattering length, and unitarity

10.9.1 From plane wave to partial-wave cross section

The lecture expands a plane wave in partial waves (Equation (10.5)) and shows that a short-range potential can only shift the phase of each outgoing channel (Equation (10.7)). The differential cross section then becomes a sum over ℓ:

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} = \frac{1}{k^2}\left|\sum_{\ell=0}^{\infty}(2\ell+1)\,e^{i\delta_\ell}\,\sin\delta_\ell\,P_\ell(\cos\theta)\right|^2, \tag{10.30}$$

which is (10.12) in the main text. At low energy only $\ell = 0$ contributes (the centrifugal barrier $\sim \ell(\ell+1)/r^2$ suppresses higher waves), so the angular distribution is isotropic and fixing $\delta_0(E)$ fixes the entire cross section.

10.9.2 Scattering length and effective range

Expanding $\delta_0(E) \approx -ka + \mathcal{O}(k^3)$ defines the **scattering length** a (Equation (10.23)). For the np triplet channel, $a_t \approx +5.4\,\text{fm}$: the interaction is weakly attractive at threshold, consistent with a shallow bound state (the deuteron, Lecture 7). The next term in the low-energy expansion defines the **effective range** r_e (Equation (10.26)), which sets the curvature of $\delta_0(E)$ and is linked to the interior shape of $V(r)$. Together (a, r_e) are the minimal low-energy data for each spin channel; they cannot be read from B_d alone.

Geometric cross section at nuclear size

For a black-disk target of radius $R \approx 1.2\, A^{1/3}$ fm,

$$\sigma_{\text{geo}} = \pi R^2 \approx \begin{cases} 45\,\text{mb} & A = 12, \\ 1.3\,\text{b} & A = 208, \end{cases} \tag{10.31}$$

using $1\,\text{b} = 100\,\text{mb}$. Measured total reaction cross sections at tens of MeV often approach this geometric scale, while elastic NN cross sections at low energy are set instead by a^2 because the force range is only a few fm [6, 2].

10.9.3 Optical theorem as a consistency check

The optical theorem (Equation (10.15)) ties the forward amplitude to the total cross section:

$$\sigma_{\text{tot}} = \frac{4\pi}{k}\, \Im f(0). \tag{10.32}$$

In partial-wave language, $\sigma_{\text{tot}} = (4\pi/k) \sum_\ell (2\ell + 1) \sin^2 \delta_\ell$. Each δ_ℓ must be real for elastic scattering, so the sum is automatically positive: unitarity is built in. When imaginary optical potentials are introduced for heavy-ion reactions (absorption into other channels), δ_ℓ acquires an imaginary part and the same relation remains the bookkeeping check that flux removed from the elastic channel equals flux into all inelastic channels.

10.9.4 Levinson's theorem and the deuteron pole

For a short-range central potential with one s-wave bound state, Levinson's theorem states $\delta_0(E = 0) = +\pi/2$. The scattering length is then positive and large when the bound state is weakly bound ($a \gg R$). The deuteron pole in the T-matrix at $E = -B_d$ is the analytic continuation of this phase-shift behaviour: scattering and bound-state data are linked, not independent fits [6, 2]. Lecture 7's square well sits at the same threshold; Lecture 11's Yukawa tail sets the slope of $\delta_0(E)$ at higher energy through r_e.

10.9.5 When partial waves beyond $\ell = 0$ matter

At centre-of-mass energy $E \sim 50\,\text{MeV}$, the p-wave ($\ell = 1$) channel opens with centrifugal barrier $\hbar^2 \ell(\ell + 1)/(2\mu r^2)$ suppressing low-r penetration. Triplet p waves and tensor mixing of s and d waves become comparable in importance. At low energy ($E \lesssim 10\,\text{MeV}$), however, the story collapses to real δ_0 and the isotropic cross section (10.33), which is why hydrogen-target neutron scattering at thermal energies isolates the singlet/triplet spin structure introduced in Lecture 7.

10.10 Further physics: coherent scattering and spin channels

At very low energy the de Broglie wavelength of a neutron can exceed the size of a hydrogen molecule. Scattering from the two protons then interferes coherently, and the cross sections of ortho- and parahydrogen differ because the total proton spin of the molecule weights triplet and singlet *np* amplitudes differently. That difference is direct evidence that $\sigma_t \neq \sigma_s$ [6].

The optical analogy remains useful: an attractive well of nuclear size diffracts the incident wave much as an obstacle of size R diffracts light of wavelength $\lambda \sim 2\pi/k$. Partial-wave analysis is the quantum bookkeeping of that diffraction pattern, partial wave by partial wave.

10.10.1 Source-map limitation

No text file named `Lecture10_raw.txt` was present in the supplied course archive when this chapter was remediated. Consequently, statements cannot be mapped line-by-line to an original Lecture 10 raw transcript. The derivations follow the cited treatments in keys, and every redrawn figure has an explicit record in the relevant `SOURCES.txt`. This documents the gap rather than inventing a raw provenance.

10.10.2 Low-energy s-wave cross section

When only $\ell = 0$ contributes, (10.12) reduces to

$$\sigma_{\text{tot}} = \frac{4\pi}{k^2}\sin^2\delta_0 \approx 4\pi a^2 \qquad (k \to 0), \tag{10.33}$$

using $\delta_0 \approx -ka$. For *np* scattering in the triplet channel, $a_t \approx +5.4\,\text{fm}$ gives $\sigma_t \approx 4\pi a_t^2 \approx 0.37\,\text{b}$, while the singlet channel with $a_s \approx -23.7\,\text{fm}$ yields $\sigma_s \approx 7\,\text{b}$. The dramatic difference between spin channels is direct evidence of the spin-dependent force introduced phenomenologically in Lecture 7's two-well picture [6, 14].

10.10.3 Effective range and the interior of $V(r)$

The effective-range expansion (Equation (10.26))

$$k\cot\delta_0 = -\frac{1}{a} + \frac{1}{2}r_e k^2 + \cdots \tag{10.34}$$

connects the curvature of $\delta_0(E)$ at threshold to the rms extent of the interaction region. For the triplet channel, $r_{e,t} \approx 1.7\,\text{fm}$, comparable to the deuteron size. The deuteron binding energy fixes the *slope* of $k\cot\delta_0$ near $E = 0$ (Levinson's theorem links one bound state to a $\pi/2$ phase shift at threshold), but r_e carries independent information about the potential shape inside $r \lesssim 2\,\text{fm}$. Fitting both a and r_e breaks the (R, V_0) degeneracy of the square well (Lecture 7 further material).

Optical theorem at $E = 50\,\text{MeV}$

For *np* scattering near 50 MeV, several partial waves contribute and $\sigma_{\text{tot}} \sim 50$–$100\,\text{mb}$. The optical theorem (Equation (10.15)) requires that the forward scattering amplitude's imaginary part reproduce this total. At higher energy, the geometric scale πR^2 with $R \sim 1\,\text{fm}$ gives $\sim 10\,\text{mb}$ per nucleon–nucleon collision, while heavy-ion reaction cross sections grow with $A^{2/3}$ (Lecture 10 main text, Equation (10.3)).

10.10.4 Bridge to Lecture 11 and the deuteron

Yukawa potentials with range $\hbar/(m_\pi c)$ generate phase shifts that can be fitted to (a, r_e) in each spin channel (Lecture 11). The deuteron (Lectures 7–9) is the bound-state pole in the triplet s-wave amplitude (Equation (10.29)): scattering and bound-state data are two views of the same T-matrix. Tensor and spin–orbit terms mix partial waves at higher ℓ, but at low energy the story reduces to real δ_0 and the unitarity sum in (10.33) [2, 6].

10.10.5 Coherent scattering from parahydrogen

When the neutron wavelength exceeds the H_2 bond length, scattering from the two protons adds coherently. Parahydrogen ($I_{\text{mol}} = 0$) weights the singlet np channel; orthohydrogen ($I_{\text{mol}} = 1$) weights the triplet. The measured difference in total cross sections at $E \lesssim 0.1\,\text{eV}$ directly confirms $\sigma_t \neq \sigma_s$ without resolving individual phase shifts. That experiment is historically important because it established spin-dependent NN forces before direct accelerator phase-shift analyses [6, 14].

10.10.6 Partial-wave unitarity and the $\ell = 1$ turn-on

As energy increases, the $\ell = 1$ partial wave turns on when $kR_{\text{eff}} \sim 1$. For $A = 12$ targets at $E \sim 20\,\text{MeV}$, several partial waves contribute and diffraction minima appear in $\mathrm{d}\sigma/\mathrm{d}\Omega$. The optical theorem (Equation (10.15)) still holds channel by channel: the imaginary part of each partial-wave amplitude fixes the contribution to σ_{tot}. For NN scattering at low energy the unitarity sum collapses to $\sin^2\delta_0$, linking the deuteron pole (Lecture 7) to the threshold cross sections quoted in (10.33) [2, 6].

10.11 Deeper reading: unitarity, optical theorem, and optical potentials

At unitarity, an S-wave cross section saturates $\sigma = 4\pi/k^2$. Low-energy np scattering is large because the deuteron is near threshold, not because the force is singular in a naive sense. The optical theorem links $\operatorname{Im} f(0)$ to the total cross section and thereby constrains absorptive (imaginary) potentials once inelastic channels open.

$$\sigma_{\text{tot}} = \frac{4\pi}{k}\operatorname{Im} f(\theta = 0). \tag{10.35}$$

Optical potentials used in Lecture 21 reactions are the many-body descendants of this idea: a complex mean field that reproduces elastic scattering while accounting for absorption into other channels. Uncertainty quantification of those potentials is now part of extracting spectroscopic factors. Partial-wave analyses of NN scattering remain the calibration point for the chiral forces of Lecture 11 and for the few-body calculations that start from the deuteron (Lectures 7–9).

Coulomb versus nuclear phases

In charged-particle scattering, Coulomb phases multiply nuclear partial wave amplitudes. Extracting nuclear phase shifts therefore requires a careful Coulomb subtraction, especially at low energy and high Z. Neutron scattering avoids that complication and historically fixed much of the low-energy NN database.

Optical potential uncertainties

When an optical potential is fitted to elastic angular distributions, imaginary depths are constrained by absorption, real depths by refractive patterns and rainbows. Propagating the fit covariance into a transfer calculation is now expected practice. A spectroscopic factor quoted to three digits without an optical-model error bar is incomplete.

Worked unitarity and effective-range numbers

At $E_{\text{lab}} = 1\,\text{keV}$ for np, k is tiny and $\sigma \approx 4\pi a^2$ with a of several femtometres yields tens of barns: huge compared with πR^2 for a nucleus. That enhancement is threshold physics, not a failure of geometry. Adding the effective-range term $-\frac{1}{2}r_e k^2$ in $k\cot\delta_0$ curves the cross section upward in energy; fitting both a and r_e is the minimal data-driven model before introducing a potential shape.

For charged projectiles, replace the discussion with Coulomb-modified scattering lengths. The important conceptual transfer to Lecture 21 is that reaction theory inherits these low-energy constants when it uses NN potentials or optical models constrained by elastic data. Always ask which elastic database and which energy range constrained the interaction behind a published transfer calculation.

Scattering theory supplies the boundary conditions of nuclear reactions. Partial waves, scattering lengths, and optical potentials are not a side topic; they are the entrance channel to Lectures 21–22.

In practice, rework one numerical estimate from the lecture with a different parameter set and compare the shift to the quoted experimental uncertainty. If the shift exceeds the uncertainty, the estimate is teaching a scale, not delivering a fit. That distinction keeps theoretical cartoons honest when they sit beside evaluated data tables and modern many-body calculations. Along the course arc, write one sentence linking this chapter backward and one forward: the discipline of those two sentences is as important as any single formula. Examination problems rarely ask for a literature review, but they do ask for the scale argument that literature reviews presuppose.

Synthesis problem

Write a half-page note that (i) restates the chapter's central scale or selection rule in one equation, (ii) evaluates it numerically for one concrete nucleus or energy, and (iii) names the neighbouring lecture that uses the result. Keep the note free of research-frontier speculation; the goal is fluency with the lecture physics itself. If an assumption fails (Born approximation, spherical mean field, allowed transition, one-dimensional barrier), state the failure mode in a final sentence.

That habit converts passive reading into examination-ready understanding and makes the later research-frontier box read as an extension rather than a replacement.

Research frontier: scattering with uncertainty quantification and student projects

Partial-wave analysis and optical potentials of Lecture 10 are being rebuilt with quantified uncertainties. Bayesian fits of nucleon–nucleon phase shifts, optical-model parameters with posterior predictive distributions, and EFT-based interactions embedded in reaction frameworks replace older point estimates that quoted only best-fit χ^2 [44, 43]. Rare-isotope beams demand reaction theories that treat breakup, continuum coupling, and target excitation on equal footing with structure, because spectroscopic factors extracted without those couplings can be misleading [53]. R-matrix methods remain central for resonance analyses in light systems, but modern implementations emphasise consistent channel truncations and covariance of resonance parameters. Uncertainties that still limit many applications include the imaginary (absorptive) part of optical potentials for exotic projectiles and the propagation of interaction uncertainties into transfer and knockout cross sections.

For radioactive beams, the optical potential is often underconstrained because elastic data are sparse. Uncertainty quantification then becomes part of the structure extraction: a spectroscopic factor quoted without an optical-model error bar is incomplete. Learning to read covariance bands on angular distributions is therefore a modern experimental-theory skill.

Reporting a spectroscopic factor together with an optical-model error band is now expected practice in rare-isotope reaction papers.

That habit of quoting covariances alongside central values is the main cultural shift relative to older optical-model papers.

Student directions. (1) Data exercise: from a published S-wave phase shift, reconstruct the scattering length and effective range. (2) Scale estimate: compute the geometric $\sigma = \pi R^2$ scale for $A = 12$ and $A = 208$ and compare with measured reaction cross sections at one energy. (3) Methods outline: explain how an optical-model imaginary depth is constrained by elastic angular distributions.

10.12 Problems with solutions

Problem 10.1 — Cross section dimensions. From $\mathrm{d}N = \sigma\, t\, \mathrm{d}\Omega\, I\, N_t$, show that σ has dimensions of area.

Solution 10.1. $\mathrm{d}N$ is dimensionless (a count). tI has dimensions of (time)$\times$(area^{-1} time^{-1})=area^{-1}; N_t is a number; $\mathrm{d}\Omega$ is dimensionless. Hence σ must supply an area so that the product is dimensionless.

Problem 10.2 — Low-energy total cross section. At low energy only δ_0 is nonzero. Show that $\sigma_{\text{tot}} = 4\pi k^{-2} \sin^2 \delta_0$. If $\delta_0 = \pi/2$, what is σ_{tot}?

Solution 10.2. From $f = (e^{i\delta_0} \sin \delta_0)/k$ one has $\mathrm{d}\sigma/\mathrm{d}\Omega = |f|^2 = \sin^2 \delta_0/k^2$, isotropic in the CM frame. Integrate:

$$\sigma_{\text{tot}} = \int \frac{\sin^2 \delta_0}{k^2} \, \mathrm{d}\Omega = \frac{4\pi}{k^2} \sin^2 \delta_0.$$

At resonance-like $\delta_0 = \pi/2$, $\sigma_{\text{tot}} = 4\pi/k^2$ (unitarity limit for s-wave) [6].

Problem 10.3 — Why only s-wave at 1 MeV. Estimate $\ell_{\max} \sim pr_0/\hbar$ for $E = 1\,\text{MeV}$ (lab-scale NN), $r_0 = 2\,\text{fm}$, $\hbar c = 200\,\text{MeV fm}$, $\mu c^2 \approx 500\,\text{MeV}$. Argue that $\ell = 0$ dominates.

Solution 10.3.

$$pc = \sqrt{2\mu c^2 \, E} \approx \sqrt{1000\,\text{MeV}^2} \approx 32\,\text{MeV}, \qquad \frac{pr_0}{\hbar} = \frac{(pc)\, r_0}{\hbar c} \approx \frac{32 \times 2}{200} \approx 0.3.$$

Values $\ell \geq 1$ require order-1 or larger; only $\ell = 0$ fits inside the nuclear range at this energy [6].

Problem 10.4 — Square-well matching and threshold. For the s-wave square well, match $u_{\text{in}} = A \sin(k_1 r)$ to $u_{\text{out}} = C \sin(kr + \delta_0)$ at R. Derive $k \cot(kR + \delta_0) = k_1 \cot(k_1 R)$, obtain the zero-energy scattering length, and identify the first bound-state threshold.

Solution 10.4. Continuity of u and u', followed by division, gives

$$k_1 \cot(k_1 R) = k \cot(kR + \delta_0).$$

As $k \to 0$, the exterior zero-energy form is proportional to $r - a$. Matching its logarithmic derivative to the interior one, with $K = \sqrt{2\mu V_0}/\hbar$, yields

$$a = R - \frac{\tan(KR)}{K}.$$

The first divergence occurs at $KR = \pi/2$, or $V_0 R^2 = \pi^2 \hbar^2/(8\mu)$, exactly the first bound-state threshold.

Problem 10.5 — Unitarity and optical theorem. Starting with $S_\ell = e^{2i\delta_\ell}$, show that $f_\ell = (S_\ell - 1)/(2ik) = e^{i\delta_\ell} \sin \delta_\ell / k$. Then derive the optical theorem from the partial-wave expansion.

Solution 10.5. The identity $e^{2i\delta} - 1 = 2ie^{i\delta} \sin \delta$ gives the stated amplitude and ensures the correct factor of four in $\sigma_\ell = 4\pi(2\ell + 1) \sin^2 \delta_\ell / k^2$. At zero angle,

$$\operatorname{Im} f(0) = \frac{1}{k} \sum_\ell (2\ell + 1) \sin^2 \delta_\ell.$$

Multiplication by $4\pi/k$ reproduces the summed total cross section:

$$\sigma_{\text{tot}} = \frac{4\pi}{k} \operatorname{Im} f(0).$$

Problem 10.6 — Effective-range pole check. Insert $k = i\kappa$ into

$$f_0(k) = \frac{1}{-1/a + \frac{1}{2}r_0 k^2 - ik}$$

and derive the shallow-bound-state pole condition. Explain the signs of the triplet and singlet scattering lengths.

Solution 10.6. At $k = i\kappa$, the denominator vanishes when

$$-\frac{1}{a} - \frac{1}{2}r_0\kappa^2 + \kappa = 0, \qquad \frac{1}{a} = \kappa - \frac{1}{2}r_0\kappa^2.$$

Thus a shallow physical bound state has $a > 0$ at leading order, matching the triplet deuteron channel. A virtual-state pole on the negative imaginary axis gives a large negative a, matching the unbound singlet channel.

Problem 10.7 — Optical theorem check. For pure s-wave scattering with phase δ_0, write σ_{tot} from partial waves and $\operatorname{Im} f(0)$. Verify that they satisfy $\sigma_{\text{tot}} = (4\pi/k) \operatorname{Im} f(0)$.

Solution 10.7.

$$\sigma_{\text{tot}} = \frac{4\pi}{k^2}\sin^2\delta_0, \qquad f(0) = \frac{e^{2i\delta_0} - 1}{2ik} \quad \Rightarrow \quad \operatorname{Im} f(0) = \frac{\sin^2\delta_0}{k}.$$

Hence $(4\pi/k) \operatorname{Im} f(0) = 4\pi \sin^2\delta_0/k^2 = \sigma_{\text{tot}}$ [6, 14].

Chapter 11

Theories of Nuclear Forces

"Yukawa noticed that the range of nuclear forces $r_0 \simeq 1.4\,\mathrm{fm}$ *corresponds to a particle of mass* $m_\pi c^2 \simeq 140\,\mathrm{MeV}$*."*

paraphrasing H. Yukawa (1935)

Lectures 7–10 constrain the residual NN force from two sides: the deuteron multipoles and low-energy phase shifts. Both approaches are phenomenological: they fix a potential (or its phase shifts) to data. This lecture asks a deeper question: *what carries the force?* In modern quantum field theory, forces between particles arise from the exchange of virtual quanta. For electromagnetism the quantum is the massless photon and the potential is Coulomb. For the residual strong force the lightest quantum is the massive pion and the potential is Yukawa [9, 14, 5, 15, 2].

11.1 Three layers of description: from Coulomb to QED

Electromagnetism shows how the same interaction can be told at increasing depth.

Action at a distance. Two charges q_1 and q_2 exert equal and opposite forces

$$F = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r^2} \tag{11.1}$$

instantaneously across empty space. The formula works at low speeds, but instantaneous action conflicts with relativity: no influence can travel faster than light.

Classical field. Charge q_1 does not "reach out" to q_2. It produces an electric field filling space; q_2 feels a force because it sits in that field. The field is the intermediary. Changes in the field propagate at finite speed c.

Quantum field (QED). The electromagnetic field is quantized. Energy and momentum are exchanged in discrete packets (photons). The Coulomb force is the low-energy limit of exchanging *virtual* photons: off-shell quanta that mediate the interaction without appearing as free radiation. Virtual photons share the quantum numbers of real photons (massless, chargeless, spin 1); only their four-momentum differs from the on-shell condition $q^2 = 0$ [5, 15].

The same ladder applies to nucleons N_1 and N_2. Classically one might imagine a potential $V(r)$. At the quantum level the interaction is the exchange of virtual mesons. Unlike the photon, those mesons may carry mass and electric charge, and that changes the range and the selection rules of the force.

11.2 Range of a force from the uncertainty principle

A useful *heuristic* estimates the range of a force mediated by a massive quantum. Creating a virtual particle of rest energy mc^2 requires an energy fluctuation of order $\Delta E \sim mc^2$. The uncertainty principle then suggests a lifetime

$$\Delta t \lesssim \frac{\hbar}{\Delta E} \sim \frac{\hbar}{mc^2}. \tag{11.2}$$

Even if the quantum travels near the speed of light, the farthest it can go before being reabsorbed is of order

$$R \lesssim c\,\Delta t \sim \frac{\hbar c}{mc^2} = \frac{\hbar}{mc}. \tag{11.3}$$

This Compton wavelength of the mediator sets the force range. Two limits are immediate:

- **Massless mediator** ($m = 0$): $R \to \infty$. The force is long-ranged (Coulomb, gravity in the Newtonian limit).
- **Massive mediator**: R is finite. The force is short-ranged; beyond a few Compton wavelengths the exchange is exponentially suppressed.

This argument is pedagogical, not a derivation of field theory: it does not claim that energy conservation is violated. The rigorous statement is that a massive propagator $1/(q^2 + m^2c^2)$ Fourier-transforms to an exponentially screened potential, derived next from the Klein–Gordon equation [9, 14, 5, 2].

That single scale explains why nuclear forces die within a few fm while electrostatics still acts across atomic distances.

11.2.1 Predicting the pion mass from the nuclear range

Electron scattering and low-energy NN data fix the range of the residual strong force at $R \sim 1$–$2\,\text{fm}$. Taking $R \approx 1.4$–$2\,\text{fm}$ and $\hbar c \approx 197\,\text{MeV fm}$,

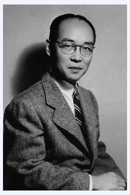
$$mc^2 \sim \frac{\hbar c}{R} \approx \frac{197}{1.4\text{–}2} \approx 100\text{–}140\,\text{MeV}. \tag{11.4}$$

Yukawa made this prediction in 1935, years before the pion was discovered. The observed mesons have

$$m_{\pi^\pm}c^2 \approx 139.6\,\text{MeV}, \qquad m_{\pi^0}c^2 \approx 135\,\text{MeV},$$

exactly the scale required by the nuclear force range [9, 14, 2].

Historical note: Hideki Yukawa (1907–1981)



Hideki Yukawa, working in Osaka in 1934–1935, proposed that the short-range nuclear force is mediated by the exchange of a massive boson. From the empirical range $r_0 \sim 1$–$2\,\mathrm{fm}$ he estimated a mediator mass of order $100\,\mathrm{MeV}/c^2$. The particle, the pion, was later found with $m_\pi c^2 \approx 140\,\mathrm{MeV}$, and the associated potential falls exponentially with range fixed by m_π. Yukawa received the 1949 Nobel Prize in Physics, the first Nobel awarded to a Japanese scientist. His insight is the microscopic origin of the residual NN force developed in this lecture [9, 14, 2].

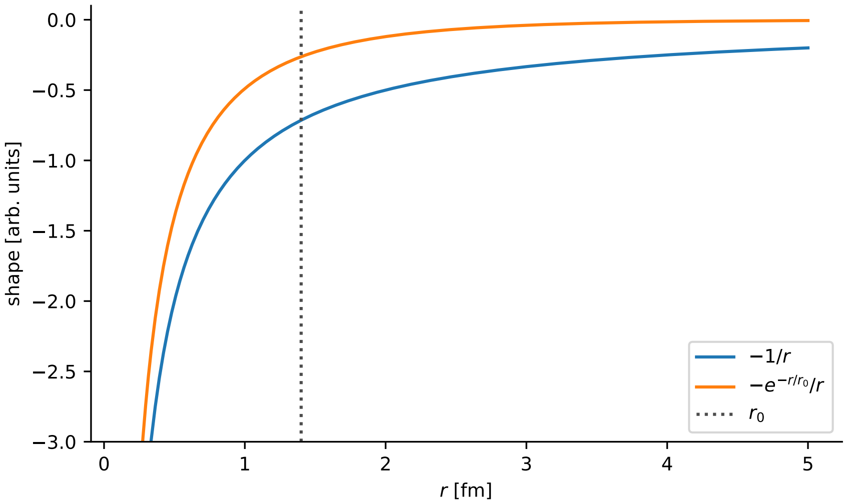


Figure 11.1: Coulomb ($-1/r$, massless exchange) versus Yukawa ($-e^{-r/r_0}/r$, massive exchange). The exponential kills the force beyond $r_0 \sim \hbar/(m_\pi c)$.

Key idea

Finite nuclear range $\Leftrightarrow$ massive exchange quantum. The pion mass and the fermi-scale range of the residual NN force are two faces of $R \sim \hbar/(m_\pi c)$.

11.3 From the Klein–Gordon equation to the Yukawa potential

The form of the potential follows from the relativistic wave equation for a spinless field of mass m. We use $\boldsymbol{r}$ in fm, $\boldsymbol{q}$ and $\mu = mc/\hbar$ in fm^{-1}, and energies in MeV; g is dimensionless, so $\hbar c$ is displayed explicitly. The free Klein–Gordon equation is

$$\frac{1}{c^2}\frac{\partial^2\psi}{\partial t^2} - \nabla^2\psi + \mu^2\psi = 0, \qquad \mu = \frac{mc}{\hbar}. \tag{11.5}$$

For $m = 0$ this reduces to the wave equation for electromagnetic potentials. A static point source for the massless case yields Coulomb's law. To keep source strength, test-particle coupling, and sign distinct, let a positive source g generate a scalar field ϕ through

$$(\nabla^2 - \mu^2)\phi(r) = -g\hbar c\,\delta^3(\boldsymbol{r}), \qquad V(r) = -g\phi(r). \tag{11.6}$$

The last relation chooses the attractive scalar channel; changing the coupling or operator channel can change the sign of the physical potential. For $r \neq 0$ the radial equation is

$$\frac{1}{r}\frac{\mathrm{d}^2}{\mathrm{d}r^2}(r\phi) - \mu^2\phi = 0,$$

so $r\phi = Ae^{-\mu r} + Be^{+\mu r}$. Boundedness at infinity forces $B = 0$, hence $\phi(r) = Ae^{-\mu r}/r$. To fix A, integrate (11.6) over a ball of radius $\varepsilon \to 0$:

$$\int_{r<\varepsilon}(\nabla^2 - \mu^2)\phi\,\mathrm{d}^3r = \oint_{\partial B}\nabla\phi\cdot\mathrm{d}\boldsymbol{S} - \mu^2\int\phi\,\mathrm{d}^3r = -g\hbar c. \tag{11.7}$$

The volume integral of $\phi \sim 1/r$ vanishes as $\varepsilon \to 0$, while $\nabla\phi \sim -A\hat{r}/r^2$ gives $\oint\nabla\phi\cdot\mathrm{d}\boldsymbol{S} = -4\pi A$. Therefore $A = g\hbar c/(4\pi)$, and the chosen test coupling gives

$$V(r) = -\frac{g^2}{4\pi}\frac{\hbar c}{r}\,e^{-\mu r} = -\frac{g^2}{4\pi}\frac{\hbar c}{r}\,e^{-r/r_0}, \qquad r_0 = \frac{\hbar}{mc} \approx 1.4\,\mathrm{fm} \quad (\text{for } m = m_\pi). \tag{11.8}$$

This is a pedagogical scalar Yukawa well. Physical one-pion exchange has spin–isospin operators, not a universal central attraction.

Fourier-transform route. The same potential is the Fourier transform of the massive propagator. In the static Born approximation,

$$\tilde{V}(\boldsymbol{q}) = -\frac{g^2\hbar c}{q^2 + \mu^2}, \tag{11.9}$$

and

$$\begin{aligned} V(r) &= \int\frac{\mathrm{d}^3q}{(2\pi)^3}\,\tilde{V}(\boldsymbol{q})\,e^{i\boldsymbol{q}\cdot\boldsymbol{r}} = -\frac{g^2\hbar c}{2\pi^2}\int_0^\infty\frac{q^2\,\mathrm{d}q}{q^2+\mu^2}\frac{\sin(qr)}{qr} \\ &= -\frac{g^2\hbar c}{4\pi}\frac{e^{-\mu r}}{r}, \end{aligned} \tag{11.10}$$

recovering (11.8). In the limit $m \to 0$ ($\mu \to 0$) one recovers the Coulomb shape $1/r$; for finite m the exponential cuts off the force beyond r_0 [14, 5].

In quantum field theory the same potential is the Fourier transform of the propagator $1/(q^2 + m^2c^2)$ of the exchanged meson in the static Born approximation [14, 5].

11.4 Pions as mediators of the residual strong force

Real pions are copiously produced in accelerators and in cosmic-ray showers in the atmosphere. Virtual pions of the same mass and quantum numbers can be exchanged between nucleons; the associated range is their Compton wavelength $\hbar/(m_\pi c) \sim 1.4\,\mathrm{fm}$.

It is crucial to distinguish the **fundamental** strong force from the **residual** force between nucleons. Inside a nucleon, quarks exchange gluons and the interaction is extremely strong and confining. A proton or neutron as a whole is a colour singlet (“white”). When two colour-neutral nucleons approach within a fermi, residual colour forces remain, analogous to van der Waals forces between neutral atoms. Pion exchange is the longest-range piece of that residual interaction

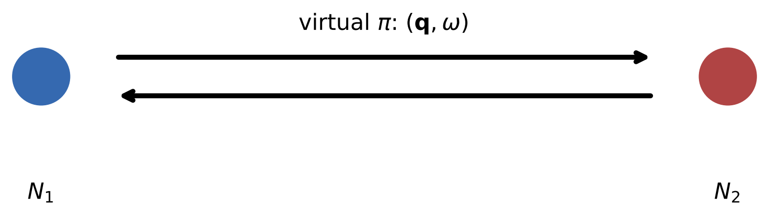


Figure 11.2: Schematic meson exchange between two nucleons. The virtual pion mediates a short-range residual force of Compton wavelength $\hbar/(m_\pi c)$.

[15, 4, 14].

11.4.1 Charge exchange: *pp* versus *np*

Neutral pion exchange preserves nucleon identity:

$$p \to p + \pi^0, \qquad n \to n + \pi^0. \tag{11.11}$$

Charged pion exchange converts neutron $\leftrightarrow$ proton:

$$n \to p + \pi^-, \qquad p + \pi^- \to n, \tag{11.12}$$
$$p \to n + \pi^+, \qquad n + \pi^+ \to p. \tag{11.13}$$

Consequently:

- In *pp* (or *nn*) scattering, if both nucleons keep their charges, only π^0 exchange is allowed at the one-pion level.
- In *np* scattering, π^0, π^+, and π^- can all contribute.

Because $m_{\pi^\pm}$ and m_{π^0} differ by only a few MeV, the *pp* and *np* forces are nearly equal once the Coulomb repulsion between protons is removed. That near-equality is the charge independence of the residual strong force, already used when arguing that only the *pn* triplet binds (Lecture 8) [14, 6].

11.5 One-pion exchange, tensor force, and the short-range core

Single-pion exchange (OPE) produces not only the radial Yukawa factor (11.8) but also spin-dependent and tensor structures. Put $\mu_\pi = m_\pi c/\hbar$, so both q and μ_π have dimensions of inverse length. In one common static pseudovector convention,

$$\widetilde{V}_\pi(\boldsymbol{q}) = -\frac{f_{\pi NN}^2}{\mu_\pi^2}\,(\boldsymbol{\tau}_1 \cdot \boldsymbol{\tau}_2)\frac{(\boldsymbol{\sigma}_1 \cdot \boldsymbol{q})(\boldsymbol{\sigma}_2 \cdot \boldsymbol{q})}{q^2 + \mu_\pi^2}\,\hbar c, \tag{11.14}$$

where $f_{\pi NN}$ is dimensionless. Since multiplication by $q_i q_j$ becomes differentiation of $Y(r) = e^{-\mu_\pi r}/(4\pi r)$, decompose the two derivatives using

$$S_{12}(\hat{r}) = 3(\boldsymbol{\sigma}_1 \cdot \hat{r})(\boldsymbol{\sigma}_2 \cdot \hat{r}) - \boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2. \tag{11.15}$$

The coordinate-space result can then be written

$$V_\pi(\boldsymbol{r}) = C_\pi(\boldsymbol{\tau}_1 \cdot \boldsymbol{\tau}_2)\left[(\boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2)W_S(r) + S_{12}(\hat{r})W_T(r)\right], \tag{11.16}$$

where the overall C_π depends on the coupling convention, while the distributional radial factors are

$$W_S(r) = \frac{\mu_\pi^2}{3}\frac{e^{-\mu_\pi r}}{r} - \frac{4\pi}{3}\delta^3(\boldsymbol{r}), \tag{11.17}$$

$$W_T(r) = \frac{\mu_\pi^2}{3}\frac{e^{-\mu_\pi r}}{r}\left(1 + \frac{3}{\mu_\pi r} + \frac{3}{(\mu_\pi r)^2}\right). \tag{11.18}$$

The delta-function is the zero-range part generated by differentiating $Y(r)$. A point-nucleon OPE formula is not trustworthy at $r = 0$: phenomenological potentials regulate this term, while effective field theory absorbs its short-distance content into contact operators. The finite-range spin–spin and tensor tails are the model-independent lesson. The potential piece $V_T(r)S_{12}$ is attractive in the deuteron channel and is the long-range origin of the S–D mixing, the quadrupole moment $Q_d \neq 0$, and the small correction to μ_d discussed in Lecture 9 [14, 6]. In multi-nucleon systems the tensor force largely averages out; in the two-body problem it is essential for binding.

A pure attractive Yukawa well is not enough for nuclear saturation. Phenomenological and meson-exchange potentials add a strong **repulsive core** at $r \lesssim 0.5\,\mathrm{fm}$, often modelled as another Yukawa with a heavier mass (or as ω and multi-pion exchange). Heavier mediators have shorter Compton wavelengths, so they act only at short distance, exactly where the hard core is needed to keep nucleons from collapsing into a single blob of pion-range size [14, 4].

A realistic NN potential therefore has a hierarchy of ranges [14]:

Distance	Dominant physics	Mediators / language
$r \gtrsim 1.4\,\mathrm{fm}$	long-range attraction, tensor	π (OPEP)
$r \sim 0.7$–$1.4\,\mathrm{fm}$	intermediate attraction	multi-π, ρ
$r \lesssim 0.5\,\mathrm{fm}$	short-range repulsion	ω, quark Pauli / contact terms in EFT

Modern chiral effective field theory organises the same hierarchy as an expansion in soft momenta: one-pion exchange at long range, multi-pion exchange at intermediate range, and short-range contact operators fitted to phase shifts. For this course it is enough that OPEP explains the long-range tensor force while shorter-range physics is constrained by scattering data (Lecture 10), not invented from the deuteron alone.

11.5.1 Why two-nucleon forces are not the whole Hamiltonian

Integrating out unresolved pion and nucleon excitations also generates irreducible three-nucleon (3N) operators. The classic mechanism is two-pion exchange in which one nucleon is virtually excited while interacting with two neighbours. Modern chiral EFT therefore contains consistent NN, 3N, and higher-body terms in a controlled hierarchy. 3N forces are smaller than the leading NN interaction but are needed quantitatively for few-body binding, nuclear saturation, shell evolution, and neutron-rich matter. They must not be mimicked by simply retuning the long-range OPE tail [6, 14].

Empirically, the need for many-body forces appears already in light nuclei: potentials fitted only to NN data underbind or overbind ^{3}H and ^{4}He unless a 3N piece is added. The same operators matter in neutron-star matter, where β equilibrium and a small proton fraction amplify density-dependent 3N contributions beyond what laboratory few-body systems probe [14, 4].

11.5.2 Phenomenology and mechanism

Historically, NN potentials were first fitted to deuteron properties and scattering phase shifts with adjustable strengths and ranges (phenomenology). Yukawa's insight and modern one-boson-exchange models derive the long-range form from a field-theoretic mechanism. The two approaches meet in practice: phase shifts (Lecture 10) and the deuteron multipoles (Lectures 7–9) remain the data that any mechanism-based potential must reproduce [6, 14].

Observable (L7–10)	Inferred operator	Mechanism (this lecture)
B_d, no excited states	depth/range of attraction	residual force scale
$I^\pi = 1^+$, triplet binds	spin dependence	$\boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2$
$Q_d \neq 0$, μ_d deficit	tensor force S_{12}	OPEP tensor
$a_t > 0$, $a_s < 0$, $\delta_\ell(E)$	channel-dependent V	OPE + shorter range

Weak interactions provide a contrast. Exchange of the heavy Z^0 ($m_Z c^2 \approx 91\,\text{GeV}$) yields a Yukawa range $\hbar c/m_Z c^2 \sim 2\times10^{-3}\,\text{fm}$ and an effective $V_0 R^2$ five orders of magnitude smaller than the strong residual force, negligible for nuclear binding though essential for neutrino scattering [14]. Within the strong sector alone one also layers masses: the light pion sets the longest NN range ($\sim 1.4\,\text{fm}$); heavier mesons (ρ, ω, ...) supply shorter-range attraction and repulsion; a repulsive core at $r \lesssim 0.5\,\text{fm}$ is required by high-energy phase shifts and keeps nuclei from collapsing under OPE attraction alone [9, 14, 6].

Key idea

Lectures 7–11 in one line. The deuteron and phase shifts map the residual NN potential; pion exchange explains why that potential is short-ranged, spin- and tensor-dependent, and charge-independent at long distance.

11.5.3 Klein–Gordon origin of the Yukawa potential

Static solutions of the Klein–Gordon equation with a point source yield the Yukawa form $V \propto e^{-\mu r}/r$; the massless limit recovers Coulomb. Feynman-diagram language makes the same point: pion exchange builds the long-range NN force, photon exchange the electromagnetic piece, while Z^0 exchange is negligible for NN binding but dominates neutrino–nucleon scattering. Modern EFT organises these ranges as a controlled expansion about the pion scale [9, 14, 44, 6].

11.6 Deeper physics: Yukawa range and the force hierarchy

11.6.1 Compton wavelength as force range

The lecture estimates the range of an interaction mediated by a particle of mass m as $R \sim \hbar/(mc)$ (Equation (11.3)). For the pion,

$$r_\pi = \frac{\hbar}{m_\pi c} \approx \frac{197\,\mathrm{MeV\,fm}}{140\,\mathrm{MeV}} \approx 1.4\,\mathrm{fm}, \tag{11.19}$$

matching the NN force range inferred from phase shifts and from the deuteron rms radius (Lecture 7). Yukawa's 1935 insight was to invert this logic: a force of nuclear range implies an exchanged quantum of pion mass [9, 14].

The Yukawa potential (attractive scalar channel; same convention as (11.8) in the lecture body)

$$V_Y(r) = -\frac{g^2}{4\pi}\,\frac{\hbar c}{r}\,e^{-r/r_\pi} \tag{11.20}$$

Fourier-transforms to a propagator $\propto 1/(q^2 + m_\pi^2 c^2)$. At momenta $q \ll m_\pi c$ the potential looks Coulomb-like ($\propto 1/r$); at $qr_\pi \gg 1$ it is exponentially suppressed. Nuclear saturation (Lecture 2) requires that attraction turn to repulsion at shorter distance, so one-pion exchange is necessary but not sufficient for a realistic NN potential.

11.6.2 Three rungs on the force ladder

Key idea

Force hierarchy (order of magnitude).

1. **QCD binding** inside a nucleon: $E \sim m_\pi c^2 \sim 140\,\mathrm{MeV}$ confinement scale.
2. **Residual NN force:** $B_d \sim 2\,\mathrm{MeV}$, range $\sim$ 1–2 fm.
3. **Electromagnetic:** Coulomb between protons, $\alpha\hbar c/r$, long ranged but $\alpha \approx 1/137$.

The residual force is $\sim 10^{-2}$ of the strong scale because nucleons are colour singlets: gluon exchange is suppressed, leaving pion exchange as the leading colour-neutral interaction [15, 4].

Weak force range for comparison

Replacing m_π by $m_W \approx 80\,\mathrm{GeV}$ in (11.19) gives a range $\sim 10^{-3}$ fm, explaining why weak decays appear as point interactions at nuclear scales while pion exchange still shapes the NN potential [5].

Remark

Beyond one pion. Heavy mesons (ρ, ω, σ-like states) contribute at $r \lesssim 1\,\mathrm{fm}$ and generate the repulsive core seen in phase shift analyses (Lecture 10). Chiral EFT organises these contributions by momentum scale, linking the long-range pion physics of this lecture to the three-nucleon forces needed for $A > 2$ binding (Lecture 12 further material).

11.6.3 Virtual pions and the residual coupling

The pions exchanged between nucleons are *virtual*: their four-momentum satisfies $q^2 \neq m_\pi^2 c^2$, so the exchange duration is $\Delta t \sim \hbar/(m_\pi c^2)$ as in Equation (11.2). The coupling $g_{\pi NN}$ is larger than α, but the Yukawa factor e^{-r/r_π} suppresses the force at distance. That combination explains why the residual force binds nuclei ($B/A \sim 8\,\mathrm{MeV}$) yet remains feeble compared with quark confinement inside each nucleon [15, 14, 5].

11.6.4 Charge independence and pion exchange

One-pion exchange respects approximate charge independence: π^0 and $\pi^\pm$ exchanges relate *pp*, *nn*, and *pn* channels. Coulomb breaks the symmetry for *pp*, which is why only the *np* deuteron binds (Lecture 7). The range estimate (11.19) is tied to the same physics that organises triplet and singlet scattering lengths in Lecture 10 [14, 9].

11.6.5 Electromagnetic vs strong hierarchy at $1\,\mathrm{fm}$

At $r = 1\,\mathrm{fm}$, Coulomb repulsion between two protons is $\sim e^2/(4\pi\varepsilon_0 r) \approx 1.4\,\mathrm{MeV}$, comparable to a fraction of the NN well depth but far below the strong scale inside each nucleon. Pion exchange provides $\sim 10\,\mathrm{MeV}$ attraction at the same distance, overwhelming Coulomb for *np* pairs. The hierarchy table in this lecture is therefore quantitative: charge independence works for *np* and *nn* scattering because the residual strong force dominates Coulomb at nuclear separation [14, 5, 15].

Physics-depth close

If you remember only one equation from Lecture 11, let it be $R \approx \hbar/(m_\pi c)$. Everything else in modern force models is an organised correction around that long-distance scale.

11.7 Further physics: residual force and colour neutrality

Nucleons are colour singlets. The fundamental strong interaction binds quarks via gluon exchange and confines colour. The force between two nucleons is a residual, colour-van-der-Waals effect: short-ranged, saturating, and much weaker than the force inside a single nucleon [15, 4, 14]. Pion exchange is the longest piece of that residual force; heavier mesons and quark substructure dominate only at sub-fermi distances where the repulsive core appears.

11.7.1 Yukawa range: a numerical derivation

Inserting $m_\pi c^2 = 140\,\mathrm{MeV}$ and $\hbar c = 197\,\mathrm{MeV\,fm}$,

$$r_\pi = \frac{\hbar}{m_\pi c} = \frac{197}{140}\,\mathrm{fm} \approx 1.41\,\mathrm{fm}. \tag{11.21}$$

The same estimate applied to the ρ meson ($m_\rho c^2 \approx 770\,\text{MeV}$) gives $r \sim 0.26\,\text{fm}$: heavier mediators are shorter ranged and dominate the inner part of the NN potential where the repulsive core appears in phase-shift analyses (Lecture 10). The pion sets the *outer* shape; multiple exchanges build the inner repulsion [14, 9].

11.7.2 Force hierarchy in numbers

Compare energy scales at $r \sim 1\,\text{fm}$:

Interaction	Scale at 1 fm	Range
Strong (QCD)	$\sim 100\,\text{MeV}$ binding	confinement
Residual NN	$\sim 10\,\text{MeV}$ (well depth)	$\sim 1.4\,\text{fm}$
Coulomb (two protons)	$\sim 1.4\,\text{MeV}$	long
Weak	$\ll 1\,\text{keV}$	$\sim 10^{-3}\,\text{fm}$

The residual force is weak because colour-singlet nucleons exchange colour-neutral mesons (Lecture 11 main text). Coulomb is long ranged but multiply charged by $\alpha \approx 1/137$. The deuteron binding energy $B_d \sim 2\,\text{MeV}$ (Lecture 7) sits naturally on the residual row, not the QCD row [15, 5].

Pion mass from range (Yukawa 1935)

If nuclear forces vanish beyond $R = 2\,\text{fm}$, then $mc^2 \sim \hbar c/R \approx 100\,\text{MeV}$, bracketing the pion mass before it was discovered experimentally. The factor-of-two agreement is fortuitous (the force does not cut off sharply), but the *order* of magnitude established meson exchange as the carrier of the residual strong force [9, 14].

11.7.3 Bridge to deuteron and saturation

One-pion exchange generates the tensor force of Lecture 9 and contributes to the spin–orbit splittings of Lecture 12. Without short-range repulsion (from heavier exchanges), however, nuclear matter would collapse: the saturation density of Lecture 2 requires a balance between attraction at $r \sim 1$–$2\,\text{fm}$ and repulsion at $r \lesssim 1\,\text{fm}$. Three-nucleon forces (Lecture 11 research frontier) provide additional repulsion needed for correct $A > 2$ binding [43, 44].

11.7.4 Two-pion vs one-pion exchange

At $r \lesssim 1\,\text{fm}$, two-pion exchange and heavier meson (ρ, ω) diagrams contribute comparable strength to one-pion exchange. Their short range produces the repulsive core required for saturation (Lecture 2) without altering the long-range a_t fixed by one-pion physics. Chiral EFT orders these contributions by momentum scale, so the Yukawa range of this lecture is the *leading* term in a controlled expansion rather than the complete potential [14, 4].

11.8 Deeper reading: range, hierarchy, and three-nucleon forces

Yukawa's relation $R \sim \hbar/(mc)$ with $m = m_\pi$ yields $\sim 1.4\,\mathrm{fm}$, matching the internucleon spacing of Lecture 2. Heavier mesons generate shorter ranges; contact terms absorb unresolved short-distance physics in an EFT organisation. Three-nucleon forces appear at a definite chiral order and are required for saturation: fitting only NN data typically misplaces binding beyond $A \sim 3$.

Students should estimate $m_\pi c^2 = 140\,\mathrm{MeV}$ and $\hbar c = 197\,\mathrm{MeV\,fm}$ once by hand to own the range formula. Similarity renormalisation group evolutions later soften the interaction for many-body methods (Lecture 12), but they induce many-body operators that must be tracked. The lecture arc is: deuteron phenomenology (Lectures 7–9), scattering constraints (Lecture 10), and meson-exchange organisation (Lecture 11) before mean-field shells.

Power counting intuition

Leading one-pion exchange is fixed by chiral symmetry and the pion-nucleon coupling; contacts absorb short-range physics order by order. Truncation errors can be estimated by comparing successive orders rather than by a single χ^2. That culture of uncertainty is the modern replacement for quoting one preferred hard-core potential.

Softening and induced forces

SRG evolution decouples high-momentum matrix elements, improving many-body convergence, but induces three- and four-body operators. Discarding induced many-body pieces can reintroduce saturation errors. Lecture 11's range argument still sets the physical long-distance scale those softened interactions must preserve.

Mass–range table

Compute $\hbar/(mc)$ for $m = m_\pi, m_\rho, m_N$. The progression $1.4\,\mathrm{fm} \to 0.3\,\mathrm{fm} \to 0.2\,\mathrm{fm}$ organises long-range tensor physics, short-range repulsion, and unresolved contact structure. Chiral EFT keeps the pion explicit and buries the rest in contacts ordered by momentum. That is a bookkeeping revolution relative to listing every meson by name, but Yukawa's range formula remains the intuition check.

For three-nucleon forces, compare published binding energies of ^{3}H and ^{4}He with and without $3N$ terms in a given Hamiltonian family (literature values suffice). The shift is mega-electronvolt-scale, comparable to the entire deuteron binding: bulk saturation cannot ignore it. Lecture 2's Fermi-gas estimate assumed an effective interaction; Lecture 11 finally says where that interaction comes from.

Meson exchange organises the residual force by range. Yukawa's formula remains the intuition, while chiral EFT supplies the modern bookkeeping for contacts and three-nucleon forces.

In practice, rework one numerical estimate from the lecture with a different parameter set and compare the shift to the quoted experimental uncertainty. If the shift exceeds the uncertainty, the estimate is teaching a scale, not delivering a fit. That distinction keeps theoretical cartoons honest when they sit beside evaluated data tables and modern many-body calculations. Along the course arc, write one sentence linking this chapter backward and one forward: the discipline of those two sentences is as important as any single formula. Examination problems rarely ask for a literature review, but they do ask for the scale argument that literature reviews presuppose.

Synthesis problem

Write a half-page note that (i) restates the chapter's central scale or selection rule in one equation, (ii) evaluates it numerically for one concrete nucleus or energy, and (iii) names the neighbouring lecture that uses the result. Keep the note free of research-frontier speculation; the goal is fluency with the lecture physics itself. If an assumption fails (Born approximation, spherical mean field, allowed transition, one-dimensional barrier), state the failure mode in a final sentence. That habit converts passive reading into examination-ready understanding and makes the later research-frontier box read as an extension rather than a replacement.

Research frontier: three-nucleon forces, SRG, and student projects

Yukawa's one-pion exchange (Lecture 11) is the longest-range piece of the residual NN force and still sets the scale $\hbar/(m_\pi c) \sim 1.4\,\mathrm{fm}$. Chiral EFT shows that consistent three-nucleon forces appear at a definite order and are required for saturation and for spectra of light nuclei; fitting only two-nucleon data typically misplaces binding beyond $A \sim 3$ [43, 44]. Similarity renormalization group (SRG) and in-medium SRG methods soften interactions for many-body calculations while tracking induced many-body operators that must not be discarded silently [46]. Connecting lattice-QCD low-energy constants to these EFTs is a long-term goal of the field and a central recommendation of community roadmaps [54]. Remaining uncertainties include the chiral-order truncation error of three-nucleon forces in medium-mass nuclei and the regulator dependence of SRG-evolved Hamiltonians when induced four-body forces are omitted.

A practical computational direction is to compare binding energies of $A = 3, 4$ systems with and without an explicit $3N$ term in a published Hamiltonian family. Even a literature-only comparison of tabulated results shows why the pion-exchange intuition of the lecture must be completed by many-body forces before saturation is trustworthy.

Community roadmaps treat that few-body to many-body pipeline as unfinished business rather than a solved derivation.

Student directions. (1) Analytic estimate: evaluate the Yukawa range $\hbar/(m_\pi c)$ numerically. (2) Conceptual note: explain why a pure two-nucleon force fitted to NN scattering underbinds or overbinds nuclei beyond $A \sim 3$ without $3N$ forces. (3) Methods sketch: outline an SRG flow that decouples high-momentum modes [46].

11.9 Problems with solutions

Problem 11.1 — Pion mass from range. If the nuclear force range is $R = 1.4\,\mathrm{fm}$ and $\hbar c = 197\,\mathrm{MeV\,fm}$, estimate the mediator mass $mc^2 \approx \hbar c/R$. Compare with $m_\pi c^2 \approx 140\,\mathrm{MeV}$.

Solution 11.1. The Compton relation $R = \hbar/(mc)$ rearranges to

$$mc^2 = \frac{\hbar c}{R} \approx \frac{197\,\mathrm{MeV\,fm}}{1.4\,\mathrm{fm}} \approx 141\,\mathrm{MeV},$$

in excellent agreement with $m_\pi c^2 \approx 140\,\mathrm{MeV}$ [9, 14].

Problem 11.2 — Photon vs pion range. Why does electromagnetism have infinite range while the residual strong force does not, in the exchange picture?

Solution 11.2. The photon is massless, so Δt can be arbitrarily long and $R \sim c\Delta t$ unbounded. The pion is massive, so $\Delta t \lesssim \hbar/(m_\pi c^2)$ and $R \sim \hbar/(m_\pi c)$ is a few fm [5, 9].

Problem 11.3 — Which pions in *pp* vs *np*?. Which pion species can mediate *pp* scattering if both protons remain protons? Which can mediate *np*?

Solution 11.3. *pp*: only π^0 (charged emission would turn a proton into a neutron). *np*: π^0, π^+, and π^- all allowed via charge-exchange and neutral channels [14].

Problem 11.4 — Yukawa vs Coulomb. The static Yukawa potential may be written $V(r) = -(g^2/4\pi)(\hbar c/r)e^{-\mu r}$. Show that $\mu \to 0$ recovers a Coulomb-like $1/r$ law. What physical limit does $\mu \to 0$ correspond to?

Solution 11.4. As $\mu \to 0$, $e^{-\mu r} \to 1$, so $V \to -(g^2/4\pi)(\hbar c/r)$, the massless-exchange (infinite-range) Coulomb form. That is the photon limit of meson exchange [9, 14].

Problem 11.5 — Range from a heavier meson. Estimate the Compton range $\hbar/(mc)$ for a meson with $mc^2 = 780\,\mathrm{MeV}$ (ω/ρ scale). Compare with the pion range 1.4 fm.

Solution 11.5.

$$R = \frac{\hbar c}{mc^2} \approx \frac{197\,\mathrm{MeV\,fm}}{780\,\mathrm{MeV}} \approx 0.25\,\mathrm{fm}.$$

Heavier mesons mediate much shorter-range NN forces than the pion; they contribute attraction/repulsion inside $\sim 1\,\mathrm{fm}$ [14, 6].

Problem 11.6 — Why a repulsive core is needed. One-pion exchange is attractive in the deuteron channel. What experimental fact at short distance requires an additional repulsive core, and what would happen to nuclei without it?

Solution 11.6. High-energy NN phase shifts and the hard behaviour of the short-range wave function require a repulsive core for $r \lesssim 0.5\,\mathrm{fm}$. Without it, OPE attraction alone would collapse nuclear matter to unphysically high density [6, 14].

Chapter 12

The Nuclear Shell Model

"The fundamental assumption of the shell model [is that] the motion of a single nucleon is governed by a potential representing the average interaction with all the other nucleons."
K. S. Krane

The liquid-drop SEMF (Lectures 4–5) describes bulk binding but cannot explain why certain proton or neutron numbers are especially stable. Just as atomic chemistry is organised by electronic shells, nuclear data show **magic numbers**

$$N, Z = 2, 8, 20, 28, 50, 82, 126 \tag{12.1}$$

at which nuclei are more tightly bound, more abundant, and more spherical than the SEMF alone predicts [6, 14, 1]. The nuclear shell model attributes those gaps to single-particle orbitals in a mean field, completed by a strong spin–orbit force.

12.1 Atomic shells as a guide

In atoms, major shells and subshells organise the periodic table. Closed shells (noble gases) are chemically inert: ionisation energies peak, radii are compact, and valence chemistry is nearly absent. Alkali metals, with a single electron outside a closed shell, share chemical behaviour because that valence electron dominates bonding.

The same logic applies to nuclei, with two important differences. First, both protons and neutrons fill independent sequences of levels (two interpenetrating Fermi seas). Second, the mean field is short-ranged and saturates, so the single-particle spectrum is not hydrogenic. Still, when N or Z reaches a magic number, separation energies jump, radii and abundances show discontinuities, and residual shell effects appear on top of the smooth SEMF [14, 6]. Historically, these "magic" irregularities were recognised experimentally long before Mayer and Jensen introduced the nuclear spin–orbit force that rearranges the harmonic-oscillator levels into the observed sequence $2, 8, 20, 28, 50, 82, 126$ [6].

12.2 Empirical evidence for magic numbers

Several independent observables mark shell closures.

Binding-energy residuals. The difference between measured B/A and the SEMF shows positive peaks near magic N or Z (extra binding of closed shells). Pronounced humps appear near $N = 28$, $N = 50$, $N = 82$ and $Z = 28, 50, 82$, among others (Fig. 12.1).

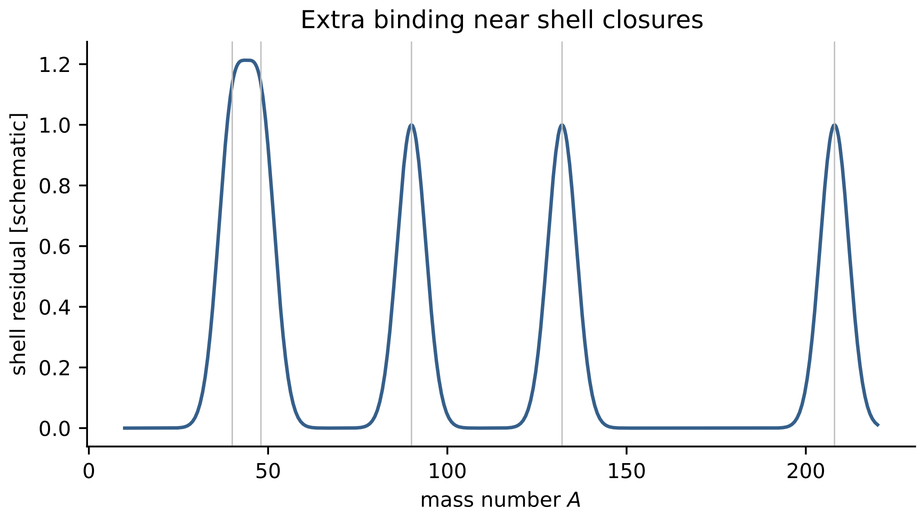


Figure 12.1: Schematic extra-binding residual near shell closures. This vector redraw shows the qualitative signature, not digitized data [1, 6].

Nuclear radii. The ratio of the measured change in nuclear radius to a smooth estimate, when N increases by two, fluctuates near $N = 20, 28, 50, 82$ (Fig. 12.2).

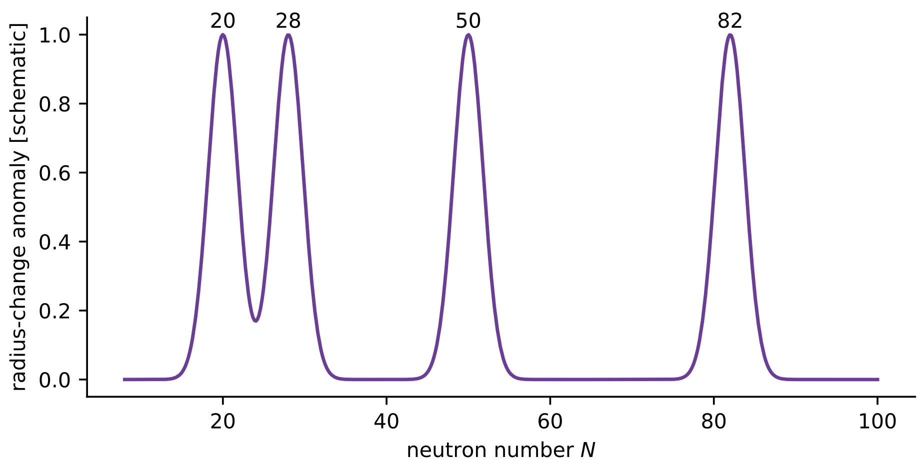


Figure 12.2: Schematic radius-change anomalies near magic neutron numbers; the vector redraw is qualitative rather than a measured dataset [1].

Stable isotones. The number of stable isotones peaks at magic neutron numbers (Fig. 12.3): closed neutron shells favour more proton partners along the valley of stability.

Separation energies and first 2^+. Approaching a closure from below, the last occupied orbit becomes more strongly bound; crossing it, the next nucleon must enter the higher shell. Thus $S_n(N, Z)$ shows a pronounced *downward* step from magic N to $N + 1$, and S_p behaves

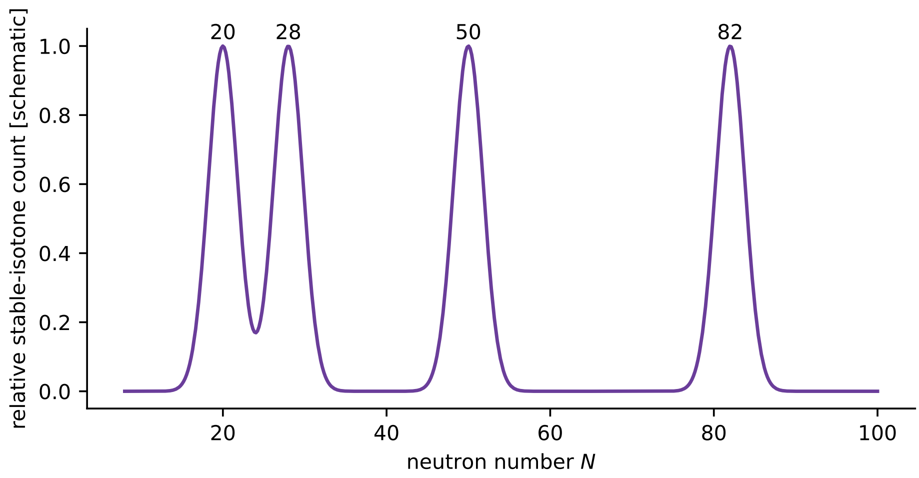


Figure 12.3: Schematic enhancement of stable-isotone counts at magic N. The vector redraw is qualitative rather than a measured dataset [1].

analogously across magic Z. Equivalently, removing a nucleon from the closed-shell nucleus costs unusually high separation energy. Even–even nuclei near magic numbers also have higher-lying first 2^+ states (more rigid, spherical cores).

The experimentally identified closures are

$$2,\ 8,\ 20,\ 28,\ 50,\ 82,\ 126.$$

Beyond 126, laboratory superheavy nuclei still show evidence for further shell gaps [6, 1].

Key idea

Magic numbers are experimental facts. The shell model is the mean-field theory that predicts them once a realistic single-particle potential and spin–orbit coupling are chosen.

12.3 Mean field and single-particle motion

The translationally invariant intrinsic Hamiltonian of A nucleons may be written schematically

$$H_{\text{int}} = \sum_{i=1}^{A} \frac{\boldsymbol{p}_i^2}{2m} - \frac{\boldsymbol{P}_{\text{cm}}^2}{2Am} + \sum_{i<j} v_{ij} + \sum_{i<j<k} V_{ijk}, \tag{12.2}$$

where $\boldsymbol{P}_{\text{cm}} = \sum_i \boldsymbol{p}_i$, v_{ij} contains the strong interaction, Coulomb repulsion among protons, and weaker electromagnetic corrections, and V_{ijk} denotes genuine three-nucleon forces discussed in Lecture 11. Subtracting the centre-of-mass (CM) kinetic energy prevents spurious translation of the whole nucleus from being counted as intrinsic excitation. Solving (12.2) exactly is impossible for medium and heavy nuclei. The independent-particle approximation replaces the two-body sum by a one-body mean field $U(\boldsymbol{r})$ that represents the average interaction of one nucleon with all others:

$$H \simeq \sum_{i=1}^{A} \left[\frac{\boldsymbol{p}_i^2}{2m} + U(\boldsymbol{r}_i) \right] + V_{\text{res}}. \tag{12.3}$$

Here V_{res} collects residual two- and three-body effects not absorbed into U. Equation (12.3) also assumes that CM contamination has been removed or is negligible in the chosen intrinsic basis. Pairing and collective correlations live in V_{res} and become essential away from closed shells (Lectures 13–15).

The mean field is not an externally imposed force. It is generated by the nucleons themselves, so a microscopic calculation determines the occupied orbitals and U self-consistently: the orbitals produce a density, the density produces a new field, and the cycle is repeated until both agree. The Pauli principle is essential in this picture. It forces identical nucleons to occupy different quantum states and builds separate proton and neutron Fermi seas. Nuclear saturation then explains why the depth and interior density of the mean field remain roughly independent of A, while its radius grows approximately as $A^{1/3}$.

For spherical nuclei one takes a central potential $U(\boldsymbol{r}) = V(r)$. Spherical symmetry implies that $\boldsymbol{L}^2$ and L_z commute with the single-particle Hamiltonian, so the orbital wave function separates as

$$\psi_{n\ell m}(\boldsymbol{r}) = \frac{u_{n\ell}(r)}{r}\, Y_{\ell m}(\hat{r}). \tag{12.4}$$

Substitute into the single-particle Schrödinger equation

$$-\frac{\hbar^2}{2m}\nabla^2\psi + V(r)\psi = E\psi.$$

The Laplacian in spherical coordinates yields the radial equation for $u(r)$:

$$-\frac{\hbar^2}{2m}\frac{\mathrm{d}^2u}{\mathrm{d}r^2} + \left[V(r) + \frac{\ell(\ell+1)\hbar^2}{2mr^2}\right]u = E\,u. \tag{12.5}$$

Boundary conditions are

$$u(0) = 0 \qquad \text{(regularity at the origin)} \tag{12.6}$$

and, for a bound state in a finite well,

$$u(r) \to 0 \quad \text{as} \quad r \to \infty. \tag{12.7}$$

Protons and neutrons fill independent sequences of solutions of (12.5) (two Fermi seas). Each spatial orbital of angular momentum ℓ admits $2(2\ell+1)$ identical nucleons once spin $s = \frac{1}{2}$ is included, before spin–orbit coupling is turned on. Different choices of $V(r)$ change level spacings but, without spin–orbit coupling, fail to produce the full set of magic numbers [6, 1, 13].

12.4 Infinite spherical well

Take $V = 0$ for $r < R$ and $V = \infty$ for $r \geq R$. Inside the well the potential term vanishes and (12.5) becomes

$$-\frac{\hbar^2}{2m}\frac{\mathrm{d}^2u}{\mathrm{d}r^2} + \frac{\ell(\ell+1)\hbar^2}{2mr^2}u = E\,u. \tag{12.8}$$

Define the wave number

$$k = \sqrt{\frac{2mE}{\hbar^2}}, \tag{12.9}$$

so that $E = \hbar^2 k^2/(2m)$. The regular solution of (12.8) is proportional to a spherical Bessel function,

$$u(r) = A\, r\, j_\ell(kr), \tag{12.10}$$

because $j_\ell(\rho)$ satisfies the free radial equation with $\rho = kr$. The infinite wall requires $u(R) = 0$, hence

$$j_\ell(kR) = 0. \tag{12.11}$$

Let $x_{\ell,n}$ denote the nth positive zero of j_ℓ. Then

$$k_{n\ell} = \frac{x_{\ell,n}}{R}, \qquad E_{n\ell} = \frac{\hbar^2 x_{\ell,n}^2}{2mR^2}. \tag{12.12}$$

For $\ell = 0$,

$$j_0(\rho) = \frac{\sin\rho}{\rho},$$

so the zeros are $x_{0,n} = n\pi$ with $n = 1, 2, 3, \ldots$. Higher ℓ require tabulated zeros. Ordering the resulting energies produces the sequence

$$1s,\ 1p,\ 1d,\ 2s,\ 1f,\ 2p,\ 1g,\ \ldots$$

(with $1d$ and $2s$ nearly degenerate in some realisations). Unlike the Coulomb atom, there is no prohibition on $1p$, $1d$, $1f$: the nuclear mean field is not $1/r$.

Each spatial orbit of orbital angular momentum ℓ holds $2(2\ell + 1)$ identical nucleons once spin is included: $2\ell + 1$ orbital projections and two spin states. Cumulative filling therefore closes shells at

$$\begin{gathered} 1s \to 2, \\ 1s + 1p \to 2 + 6 = 8, \\ 1s + 1p + 1d + 2s \to 8 + 10 + 2 = 20, \end{gathered}$$

but not at 28, 50, 82, 126 [6, 1].

12.5 Isotropic harmonic oscillator

A second standard potential is the three-dimensional isotropic oscillator

$$V(r) = \tfrac{1}{2} m\omega^2 r^2. \tag{12.13}$$

In Cartesian coordinates the Hamiltonian is a sum of three independent one-dimensional oscillators,

$$H = \sum_{i=x,y,z} \left(\frac{p_i^2}{2m} + \tfrac{1}{2} m\omega^2 x_i^2 \right), \tag{12.14}$$

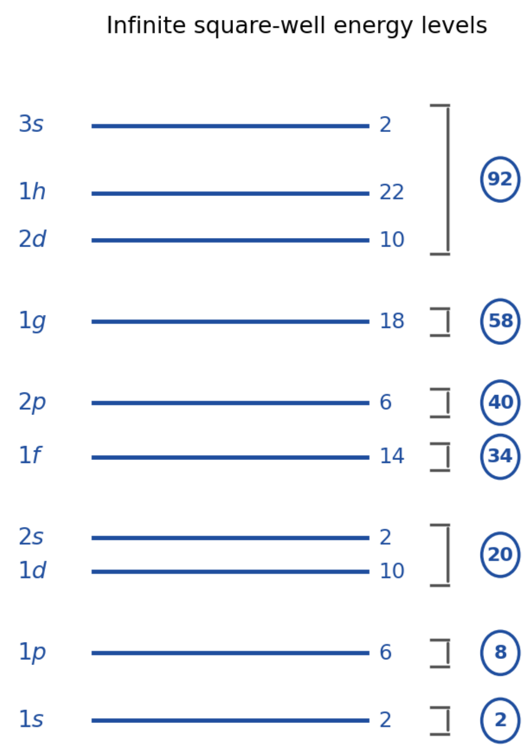


Figure 12.4: Infinite square-well single-particle levels with degeneracies and cumulative shell closures (original lecture figure).

with eigenvalues

$$E = (n_x + n_y + n_z + \tfrac{3}{2})\hbar\omega, \qquad n_x, n_y, n_z = 0, 1, 2, \ldots. \tag{12.15}$$

Define the principal oscillator quantum number

$$N = n_x + n_y + n_z = 0, 1, 2, \ldots. \tag{12.16}$$

Then

$$E_N = \left(N + \tfrac{3}{2}\right)\hbar\omega. \tag{12.17}$$

Equivalently, in spherical coordinates one writes $N = 2n_r + \ell$ with radial quantum number $n_r = 0, 1, 2, \ldots$ and orbital angular momentum ℓ. Nuclear spectroscopic labels conventionally shift the radial count so that the lowest s orbit is called $1s$ (corresponding to $n_r = 0$).

All states with the same N are degenerate. The number of Cartesian solutions with $n_x + n_y + n_z = N$ is

$$\binom{N+2}{2} = \frac{(N+1)(N+2)}{2}. \tag{12.18}$$

Including two spin projections multiplies by 2, so the occupancy of major shell N is

$$g_N = (N+1)(N+2). \tag{12.19}$$

Explicitly:

N	Orbitals (nuclear labels)	Occupancy g_N	Cumulative
0	$1s$	2	2
1	$1p$	6	8
2	$1d, 2s$	12	20
3	$1f, 2p$	20	40
4	$1g, 2d, 3s$	30	70

Cumulative magic numbers from pure oscillator filling are therefore $2, 8, 20, 40, 70, \ldots$. Again the empirical sequence beyond 20 is missed [6, 14, 13].

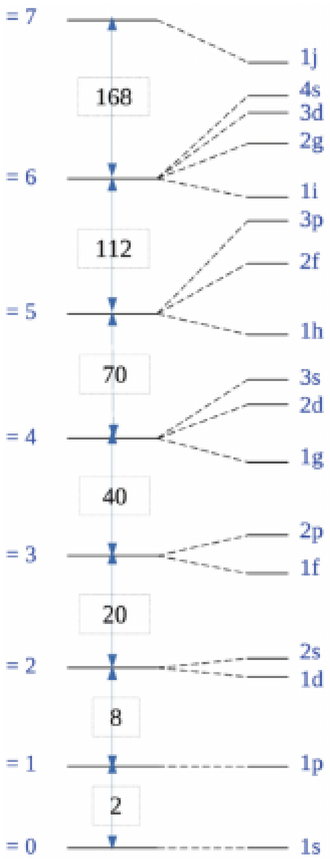


Figure 12.5: Harmonic-oscillator major shells with cumulative occupancies (original lecture figure).

Remark

Both the infinite well and the oscillator are unphysical as absolute walls: a particle cannot be removed by any finite energy. Real nuclei have finite separation energies $S_n, S_p \sim$ a few to tens of MeV. Finite wells (Lecture 13) fix that defect without, by themselves, fixing the magic numbers past 20.

12.6 Toward a realistic potential: Woods–Saxon shape

A finite square well allows a nucleon to have finite removal (separation) energy but retains a discontinuous edge. Nuclear density falls smoothly over a 10%–90% surface thickness $t = 4a \ln 3$, typically 2–3 fm. The most successful central mean field is the Woods–Saxon form

$$V(r) = -\frac{V_0}{1 + \exp\big[(r - R)/a\big]}, \tag{12.20}$$

with $R = r_0 A^{1/3}$ and diffuseness $a \sim 0.5\,\mathrm{fm}$ (Fig. 12.6). Even so, Woods–Saxon alone does not rearrange the magic numbers past 20; the essential missing piece is spin–orbit coupling [10, 6].

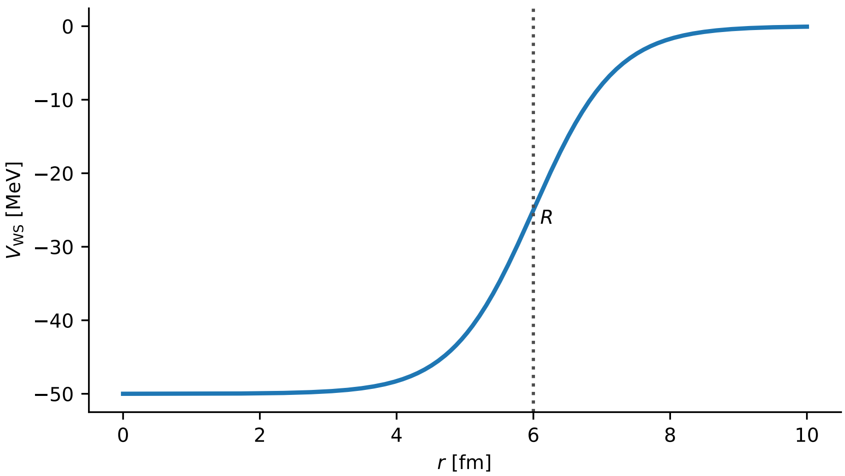


Figure 12.6: Woods–Saxon single-particle potential: flat attractive interior, diffuse surface, and finite depth (original vector evaluation of the analytic form; Woods and Saxon [10]).

12.7 Spin–orbit coupling and the correct magic numbers

Historical note: Mayer and Jensen (1949)

Maria Goeppert Mayer and, independently, **J. Hans D. Jensen** (with Haxel and Suess) showed that a strong nuclear spin–orbit force $V_{\mathrm{so}}(r)\,\boldsymbol{\ell}\cdot\boldsymbol{s}$ splits each $\ell > 0$ level into $j = \ell \pm \frac{1}{2}$ and rearranges the spectrum so that the empirical magic numbers appear. They shared the 1963 Nobel Prize in Physics with Wigner. The nuclear spin–orbit force is much stronger than its atomic electromagnetic counterpart and is a residual effect of the underlying NN interaction [6, 14].

Without spin–orbit coupling the single-particle Hamiltonian commutes with ℓ_z and s_z. A dimensionally explicit phenomenological choice is

$$H_{\mathrm{so}} = C_{\mathrm{so}}\,\frac{1}{r}\frac{\mathrm{d}V_{\mathrm{WS}}}{\mathrm{d}r}\,\frac{\boldsymbol{\ell}\cdot\boldsymbol{s}}{\hbar^2}, \tag{12.21}$$

where C_{so} has dimensions of length squared; hence $(1/r)\mathrm{d}V/\mathrm{d}r$ times C_{so} has dimensions of energy. Because V_{WS} changes most rapidly near $r \simeq R$, the nuclear spin–orbit interaction is surface peaked. High-ℓ orbits are especially sensitive: their centrifugal barrier pushes their probability density toward the surface, precisely where the derivative is largest. The constant and its sign are fitted so that the $j = \ell + \frac{1}{2}$ partner is the lower one. With this term, the good quantum numbers change: H no longer commutes with ℓ_z or s_z, but still commutes with ℓ^2, s^2, $\boldsymbol{j}^2$ and j_z, where

$$\boldsymbol{j} = \boldsymbol{\ell} + \boldsymbol{s}, \qquad j = \ell \pm \tfrac{1}{2} \quad (\ell \geq 1). \tag{12.22}$$

States are therefore labelled $n\ell_j$ (e.g. $1d_{5/2}$, $1d_{3/2}$).

12.7.1 Expectation value of the spin–orbit operator

Start from the operator identity obtained by expanding $\boldsymbol{j}^2 = (\boldsymbol{\ell} + \boldsymbol{s})^2$:

$$\boldsymbol{j}^2 = \boldsymbol{\ell}^2 + \boldsymbol{s}^2 + 2\,\boldsymbol{\ell} \cdot \boldsymbol{s}, \tag{12.23}$$

and therefore

$$\boldsymbol{\ell} \cdot \boldsymbol{s} = \frac{1}{2}(\boldsymbol{j}^2 - \boldsymbol{\ell}^2 - \boldsymbol{s}^2). \tag{12.24}$$

On a simultaneous eigenstate of $\boldsymbol{j}^2$, $\boldsymbol{\ell}^2$ and $\boldsymbol{s}^2$ with eigenvalues $j(j+1)\hbar^2$, $\ell(\ell+1)\hbar^2$ and $s(s+1)\hbar^2$,

$$\langle \boldsymbol{\ell} \cdot \boldsymbol{s} \rangle = \frac{\hbar^2}{2}\big[j(j+1) - \ell(\ell+1) - s(s+1)\big]. \tag{12.25}$$

For nucleons $s = \frac{1}{2}$, so $s(s+1) = \frac{3}{4}$.

Partner $j = \ell + \frac{1}{2}$.

$$j(j+1) = \left(\ell + \tfrac{1}{2}\right)\left(\ell + \tfrac{3}{2}\right) = \ell^2 + 2\ell + \tfrac{3}{4},$$

hence

$$j(j+1) - \ell(\ell+1) - s(s+1) = (\ell^2 + 2\ell + \tfrac{3}{4}) - (\ell^2 + \ell) - \tfrac{3}{4} = \ell,$$

and

$$\langle \boldsymbol{\ell} \cdot \boldsymbol{s} \rangle_{j=\ell+1/2} = \frac{\hbar^2}{2}\,\ell. \tag{12.26}$$

Partner $j = \ell - \frac{1}{2}$.

$$j(j+1) = \left(\ell - \tfrac{1}{2}\right)\left(\ell + \tfrac{1}{2}\right) = \ell^2 - \tfrac{1}{4},$$

hence

$$j(j+1) - \ell(\ell+1) - s(s+1) = (\ell^2 - \tfrac{1}{4}) - (\ell^2 + \ell) - \tfrac{3}{4} = -(\ell+1),$$

and

$$\langle \boldsymbol{\ell} \cdot \boldsymbol{s} \rangle_{j=\ell-1/2} = -\frac{\hbar^2}{2}\,(\ell+1). \tag{12.27}$$

Treating H_{so} as a first-order perturbation to a central-field radial state $u_{n\ell}$, normalized by $\int_0^\infty |u_{n\ell}(r)|^2 \mathrm{d}r = 1$, gives

$$\Delta E^{(1)}_{n\ell j} = C_{\text{so}} \int_0^\infty |u_{n\ell}(r)|^2 \frac{1}{r}\frac{\mathrm{d}V_{\text{WS}}}{\mathrm{d}r}\,\mathrm{d}r\,\frac{\langle \boldsymbol{\ell} \cdot \boldsymbol{s} \rangle_j}{\hbar^2}. \tag{12.28}$$

Defining the energy coefficient multiplying $\langle \boldsymbol{\ell} \cdot \boldsymbol{s} \rangle$ as $\langle f \rangle$, the nuclear ordering corresponds to $\langle f \rangle < 0$, so the $j = \ell + \frac{1}{2}$ partner is lowered and the $j = \ell - \frac{1}{2}$ partner is raised. Their splitting is

$$\Delta E_{\text{so}} = E_{j=\ell-1/2} - E_{j=\ell+1/2} = -\langle f \rangle \, \frac{\hbar^2}{2} \, (2\ell + 1). \tag{12.29}$$

Large-ℓ orbits therefore split most; s states ($\ell = 0$) do not split at all.

That reordering is decisive. The high-j members of each major shell ($1f_{7/2}$, $1g_{9/2}$, $1h_{11/2}$, $1i_{13/2}$) drop into lower major shells and open gaps at the observed magic numbers [16, 17, 6, 14]:

- $1f_{7/2}$ (occupancy $2j + 1 = 8$) alone closes a shell at $20 + 8 = 28$;
- after 28, filling $2p_{3/2}(4) + 1f_{5/2}(6) + 2p_{1/2}(2) + 1g_{9/2}(10) = 22$ yields $28 + 22 = 50$;
- the next major regrouping, including $1h_{11/2}$ (occupancy 12), yields 82;
- including $1i_{13/2}$ (occupancy 14) yields 126.

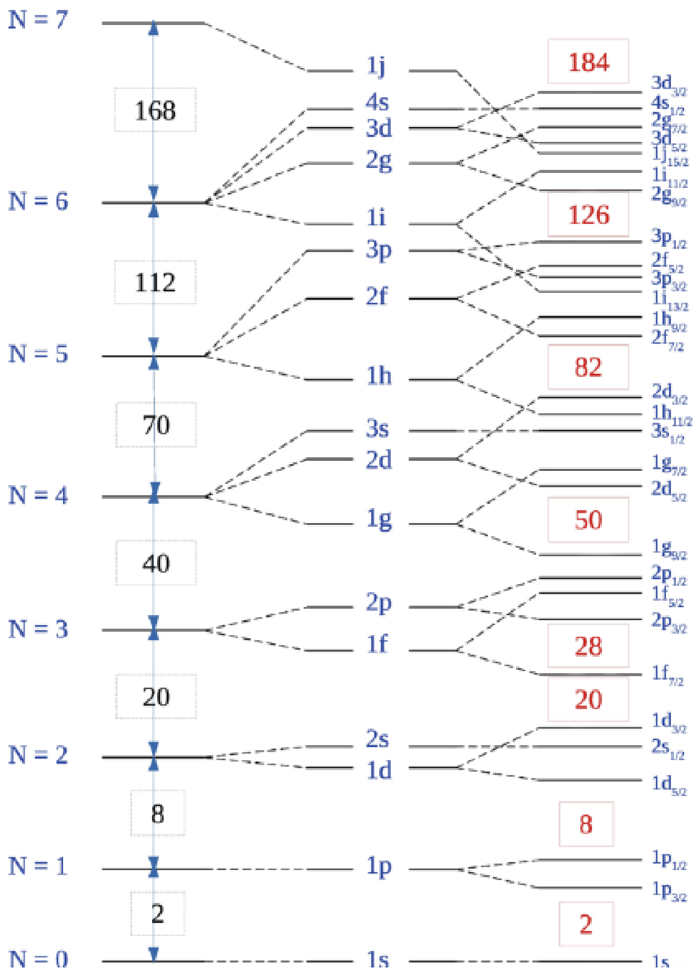


Figure 12.7: Harmonic-oscillator levels with ℓ^2 and spin–orbit splitting. Empirical magic numbers are highlighted (original lecture figure; cf. Mayer [16] and Haxel, Jensen and Suess [17]).

Each j subshell holds $2j + 1$ identical nucleons (protons or neutrons separately). Cumulative filling:

Newly added subshell(s)	Total occupancy N or Z
$1s_{1/2}$	2
$+1p_{3/2}, 1p_{1/2}$	8
$+1d_{5/2}, 2s_{1/2}, 1d_{3/2}$	20
$+1f_{7/2}$	28
$+2p_{3/2}, 1f_{5/2}, 2p_{1/2}, 1g_{9/2}$	50
$+1g_{7/2}, 2d_{5/2}, 2d_{3/2}, 3s_{1/2}, 1h_{11/2}$	82
$+1h_{9/2}, 2f_{7/2}, \ldots, 1i_{13/2}$	126

Key idea

Central potentials alone give $2, 8, 20$. Spin–orbit coupling is required for $28, 50, 82, 126$. Remember the mechanism, not only the list: surface-peaked spin–orbit coupling lowers the high-j ($j = \ell + \frac{1}{2}$) orbit, that orbit carries $2j + 1$ nucleons, and its large occupancy creates the next shell gap.

Caution

Nuclear spin–orbit ordering is opposite to the atomic fine-structure pattern for electrons: in nuclei $j = \ell + \frac{1}{2}$ lies *below* $j = \ell - \frac{1}{2}$. The strength is also orders of magnitude larger (MeV scale vs. fractions of an eV in atoms), because the force is residual strong interaction, not electromagnetism.

12.8 Why the potential shape alone is not enough

Finite square wells, wells with rounded edges, and the Woods–Saxon form all improve separation energies and level densities relative to infinite walls. They do *not*, by themselves, produce the full magic sequence: the level order remains close to that of the oscillator or infinite well until $\boldsymbol{\ell} \cdot \boldsymbol{s}$ is added. Conversely, once a strong spin–orbit term is present, magic numbers appear for a wide class of reasonable central shapes. The empirical gaps are therefore a robust signature of spin–orbit physics, not a fine-tuned accident of one potential [6, 1].

12.8.1 Magic numbers near stability and far from it

Closed shells at $2, 8, 20, 28, 50, 82, 126$ organise separation energies, radii, and abundances near the valley. Far from stability, radioactive-beam spectroscopy shows that some of these gaps quench while new ones appear: the one-body spin–orbit "caricature" must be understood as emerging from more complicated nuclear forces that evolve with occupation [6, 1, 14, 45].

12.9 Deeper physics: harmonic-oscillator shells and spin–orbit splitting

12.9.1 HO quantum numbers and naive closures

In a three-dimensional harmonic oscillator,

$$E_N = \hbar\omega\left(N + \frac{3}{2}\right), \qquad N = 2n_r + \ell, \tag{12.30}$$

with radial quantum number n_r and orbital ℓ. Degenerate shells at $N = 0, 1, 2, 3, \ldots$ would give closures at 2, 6, 12, 20 nucleons if all ℓ values up to N were filled with their full degeneracies $(2\ell + 1) \times 2$ (spin). That pattern matches the first two magic numbers (2 and 8) but fails at 20: experiment demands 20, 28, 50, 82, 126 [6, 1].

12.9.2 Spin–orbit splitting in one formula

A nucleon with velocity $\boldsymbol{v}$ in orbit ℓ sees a magnetic field from the nuclear centre, and Thomas precession fixes the single-particle spin–orbit potential to the form

$$\hat{H}_{\text{so}} = -f(r)\, \boldsymbol{L} \cdot \boldsymbol{S}, \tag{12.31}$$

with $f(r) > 0$ in the nuclear interior. States with $j = \ell + \frac{1}{2}$ ($\boldsymbol{L}$ and $\boldsymbol{S}$ aligned) are pushed *down*; states with $j = \ell - \frac{1}{2}$ are pushed *up*. The $1f_{7/2}$ orbit dropping below $2p_{3/2}$ at $N = 28$ is the canonical example: without (12.31), the shell at $N = 20$ would close too early.

Splitting scale estimate

Take $\hbar\omega \approx 41\, A^{-1/3}$ MeV and a typical $f(r) \sim 10$ MeV acting on $\langle \boldsymbol{L} \cdot \boldsymbol{S} \rangle \sim \ell/2$ for $j = \ell + \frac{1}{2}$. The resulting ~ 5–10 MeV splittings between $j = \ell \pm \frac{1}{2}$ partners are comparable to the HO spacing and suffice to reorder levels across major shells. Magic numbers are therefore not a property of HO alone but of HO plus a strong spin–orbit force (Mayer and Jensen, 1949) [6, 14].

Remark

Limits. The HO picture ignores deformation (Lecture 15), pairing gaps, and configuration mixing. Far from stability, traditional closures migrate (islands of inversion). The filling diagram in the main text is a mean-field map: residual interactions spread single-particle strength and shift separation energies by hundreds of keV even near closed shells.

12.9.3 Magic numbers as discontinuities in observables

Each magic number appears simultaneously in:

- a jump in two-nucleon separation energy S_{2n} (Section 12.10);
- a peak in the first 2^+ energy in even–even neighbours (Lecture 15, Figure 15.2);
- a reduction in collectivity (small $B(E2)$ near closed cores).

The HO plus spin–orbit model predicts *where* those discontinuities occur; residual interactions set their *size*. Filling the WS diagram (Lecture 13) is the operational step that turns magic numbers into I^π and Schmidt assignments (Lecture 14) [1, 6].

12.9.4 Pairing and even–even 0^+ ground states

Even–even nuclei have $I^\pi = 0^+$ because like nucleons pair to $J = 0$ (Section 13.6). Pairing gaps of ~ 1 MeV smooth odd–even staggering in separation energies but do not remove magic-number jumps at major shells. The shell model therefore uses a mean field (this lecture) plus pairing (later refinements) before configuration mixing and collectivity (Lecture 15) enter [14, 6].

12.9.5 Harmonic-oscillator frequency $\hbar\omega$

The empirical rule $\hbar\omega \approx 41\,A^{-1/3}$ MeV sets the spacing between major HO shells and therefore the scale of separation-energy jumps at magic numbers. For ^{208}Pb, $\hbar\omega \approx 7$ MeV, comparable to the spin–orbit splitting of the $1i_{13/2}$ and $2g_{9/2}$ orbitals near the neutron drip line. Residual interactions of similar magnitude mix those orbits in exotic nuclei, which is why the independent-particle picture weakens far from stability even when magic numbers persist in lighter isotopes [1, 6].

12.10 Further physics: separation energies and double magic nuclei

Doubly magic nuclei (Z and N both magic), such as ^{4}He, ^{16}O, ^{40}Ca, ^{48}Ca, and ^{208}Pb, combine large S_n and S_p, spherical shape, and simple particle/hole spectra when one nucleon is added or removed [14, 6, 20]. They are the benchmarks against which residual interactions and ab initio methods are tested.

The one-nucleon separation energies

$$S_n(A,Z) = B(A,Z) - B(A-1,Z), \tag{12.32}$$

$$S_p(A,Z) = B(A,Z) - B(A-1,Z-1) \tag{12.33}$$

jump when a shell closes, in direct analogy with atomic ionisation energies. Two-nucleon separation energies S_{2n} and S_{2p} are often smoother against odd–even staggering and therefore especially useful for mapping shell evolution far from stability.

Magic numbers need not be universal. Far from the valley of β stability, especially in neutron-rich light and medium-mass nuclei, experimental spectroscopy shows that traditional closures such as $N = 20$ or $N = 28$ can weaken or migrate (“islands of inversion”). The next predicted closures beyond $N = 126$ (e.g. $N = 184$, $Z = 114$ or nearby) underlie the search for superheavy islands of stability [14, 1].

12.10.1 HO filling through $N = 40$ and where it fails

In the pure HO model (Equation (12.17)), each major shell N contains orbits with $\ell = N, N-2, \ldots$ up to parity constraints. Counting degeneracies $(2\ell + 1) \times 2$ and accumulating:

N	Orbits	Cumulative nucleons	Experiment
0	$1s_{1/2}$	2	2
1	$1p_{3/2}, 1p_{1/2}$	6	6
2	$1d_{5/2}, 2s_{1/2}, 1d_{3/2}$	12	—
3	$1f_{7/2}, \ldots$	20 (HO)	20
with spin–orbit: $1f_{7/2}$ drops $\to$ magic 28			

The failure at $N = 2$ (HO predicts 12, magic is 8) is cured by spin–orbit pushing $1p_{3/2}$ down from the $N = 2$ shell. The failure at 20 without spin–orbit is cured by $1f_{7/2}$ dropping to open the gap at 28 [6, 1].

12.10.2 Spin–orbit splittings in the WS spectrum

In a Woods–Saxon plus spin–orbit potential (Lecture 13), the splitting between $j = \ell + \frac{1}{2}$ and $j = \ell - \frac{1}{2}$ partners is largest for high ℓ near the surface, where $f(r)$ peaks. The $1f_{7/2}$ orbital ($\ell = 3$) experiences a $\sim 7\,\text{MeV}$ downward shift relative to its $j = \ell - \frac{1}{2}$ partner, enough to reorder the $N = 3$ shell. Without this term, magic numbers would stop at 20 and the periodic table of nuclei would not exist in its observed form [6, 14].

Shell gap vs $\hbar\omega$

For ^{40}Ca, the proton separation energy jump across $Z = 20$ is $\Delta S_p \sim 15\,\text{MeV}$, while typical single-particle spacings within a major shell are ~ 1–$3\,\text{MeV}$. The gap is therefore an order of magnitude larger than the residual interaction scale, justifying the independent-particle picture for one nucleon outside ^{40}Ca (Lecture 13).

12.10.3 Bridge to Lectures 13–15

The filling diagram derived here supplies the orbit labels for I^π assignments (Lecture 13) and Schmidt moments (Lecture 14). Mid-shell even–even nuclei violate the independent-particle picture for $E(2^+)$ (Lecture 15): the same shell gaps that produce magic numbers also control when collectivity turns on. Separation-energy jumps at magic N (Section 12.10) are the experimental signature that the HO plus spin–orbit map captures the mean field [1, 20].

12.10.4 Two-nucleon separation energies near ^{40}Ca

The quantity $S_{2n}(A, Z) = B(A, Z) - B(A - 2, Z)$ smooths odd–even pairing staggering and highlights shell closures. For the calcium chain, S_{2n} shows pronounced steps at $N = 20$ and $N = 28$, matching the WS filling diagram (Figure 13.2). Mid-shell between closures, S_{2n} falls and $E(2^+)$ in even–even neighbours drops (Lecture 15), marking the onset of collectivity. Magic numbers are therefore visible simultaneously in separation energies, radii (Figure 12.2), and excited-state systematics [6, 1].

12.11 Deeper reading: magic numbers, spin–orbit, and migration

The harmonic-oscillator magic numbers $2, 8, 20, 40, \ldots$ fail above ^{40}Ca without a strong spin–orbit splitting that produces $28, 50, 82, 126$. In neutron-rich nuclei those closures can migrate or disappear (islands of inversion), while new closures appear. The tensor force of Lecture 9 is one microscopic driver; continuum coupling near drip lines is another (Lecture 13).

Ab initio valence-space methods now derive effective interactions from chiral Hamiltonians rather than fitting only two-body matrix elements to spectra. When comparing a textbook USD-type interaction with an ab initio valence Hamiltonian, ask which observables were used in the fit or validation and how uncertainties are quoted. Magic numbers remain the organisational map; they are no longer immutable integers.

Spin–orbit energy scale

A spin–orbit splitting of several mega-electronvolts near the Fermi surface reshuffles oscillator shells into the empirical magic sequence. Without it, ^{48}Ca and ^{208}Pb would not be doubly magic in the textbook sense. Measuring single-particle energies with transfer reactions (Lecture 21) is how those splittings are tracked into exotic isotopes.

Effective interactions

Phenomenological valence interactions encode core polarisation in fitted two-body matrix elements. Ab initio valence-space methods compute those effects from a parent Hamiltonian. When spectra agree but charge radii disagree, the operator (effective charge) is usually the culprit, not the energies alone (Lecture 14–15 themes).

Filling diagram audit

Reproduce the oscillator filling through $N = 40$ and then insert spin–orbit splitting to recover $28, 50, 82$. Mark which empirical magic numbers require the spin–orbit step. Next, list three nuclei often cited as islands of inversion and write one sentence each on the spectroscopic evidence (low $E(2^+)$, large $B(E2)$, or intruder configurations). That audit turns magic numbers from memorised integers into falsifiable claims.

When reading an ab initio spectrum for a medium-mass nucleus, record the chiral order, many-body method, and model-space truncation in a three-line note. Without those, a predicted gap cannot be compared fairly to experiment or to a phenomenological shell-model fit. Lectures 13–15 then test the resulting single-particle and collective consequences.

Magic numbers are experimental claims with theoretical mechanisms. Spin–orbit coupling, tensor forces, and continuum effects are the mechanisms this course emphasises.

In practice, rework one numerical estimate from the lecture with a different parameter set and compare the shift to the quoted experimental uncertainty. If the shift exceeds the uncertainty, the estimate is teaching a scale, not delivering a fit. That distinction keeps theoretical cartoons honest when they sit beside evaluated data tables and modern many-body calculations. Along the course arc, write one sentence linking this chapter backward and one forward: the discipline of those two sentences is as important as any single formula. Examination problems rarely ask for a literature review, but they do ask for the scale argument that literature reviews presuppose.

Synthesis problem

Write a half-page note that (i) restates the chapter's central scale or selection rule in one equation, (ii) evaluates it numerically for one concrete nucleus or energy, and (iii) names the neighbouring lecture that uses the result. Keep the note free of research-frontier speculation; the goal is fluency with the lecture physics itself. If an assumption fails (Born approximation, spherical mean field, allowed transition, one-dimensional barrier), state the failure mode in a final sentence. That habit converts passive reading into examination-ready understanding and makes the later research-frontier box read as an extension rather than a replacement.

Research frontier: ab initio shells and student projects

Magic numbers of Lecture 12 are no longer viewed as immutable integers carved into the chart of nuclides. In neutron-rich nuclei, traditional closures migrate or disappear ("islands of inversion"), while new closures appear; the tensor force and continuum coupling are key microscopic drivers [45]. Ab initio methods (coupled cluster, IMSRG, and valence-space IMSRG) now compute medium-mass nuclei from chiral Hamiltonians and derive effective shell-model interactions microscopically rather than by purely phenomenological fits [46, 53]. What remains uncertain is the quantitative accuracy of ab initio spectra near open-shell nuclei where collective correlations are strong, and how to assign meaningful theory-uncertainty bars when truncating the many-body expansion. Facility programmes at rare-isotope laboratories are designed to test these predictions with transfer, knockout, and γ-ray spectroscopy far from stability [54].

Students reading ab initio papers should look for three reported items: the chiral order of the Hamiltonian, the many-body truncation, and a quantified uncertainty estimate. Without those, a predicted magic gap is difficult to compare with experiment. The lecture filling diagram remains the map against which those predictions are judged.

Comparing two published IMSRG or coupled-cluster gap predictions for the same nucleus, with their stated truncations, is a useful student literature exercise.

Facility proposals increasingly ask for such theory-uncertainty statements before claiming a new closure has been established.

Student directions. (1) Table exercise: reproduce the oscillator filling table through $N = 40$ and mark where experiment disagrees. (2) Case study: pick one island-of-inversion nucleus (e.g. ^{32}Mg) and summarise the spectroscopic evidence for a weakened $N = 20$ gap. (3) Short essay: contrast phenomenological USD-type interactions with a valence-space IMSRG Hamiltonian.

12.12 Problems with solutions

Problem 12.1 — Occupancy of a j subshell. How many identical nucleons can occupy a $1g_{9/2}$ orbit? What cumulative magic number is completed when $1g_{9/2}$ is filled after the $N = 28$

core and the $2p_{3/2}, 1f_{5/2}, 2p_{1/2}$ subshells?

Solution 12.1. A j subshell holds $2j+1$ identical nucleons, so

$$2j + 1 = 2 \cdot \tfrac{9}{2} + 1 = 10.$$

After the $N = 28$ core the next orbits fill as

$$\begin{aligned} 2p_{3/2} &: \ 4, \\ 1f_{5/2} &: \ 6, \\ 2p_{1/2} &: \ 2, \\ 1g_{9/2} &: \ 10, \end{aligned}$$

for a total of $4 + 6 + 2 + 10 = 22$. The cumulative closure is $28 + 22 = 50$ [6, 16].

Problem 12.2 — Why HO misses magic 28. The isotropic oscillator produces closures at $2, 8, 20, 40, \ldots$. Explain in one or two sentences how spin–orbit coupling creates a gap at 28.

Solution 12.2. In the $N = 3$ oscillator shell the spin–orbit term lowers $1f_{7/2}$ ($j = \ell + \frac{1}{2}$, $\ell = 3$) below the remaining $2p$ and $1f_{5/2}$ partners. The isolated $1f_{7/2}$ subshell holds $2j + 1 = 8$ nucleons, so a new gap opens at $20 + 8 = 28$ [6, 14, 17].

Problem 12.3 — $\boldsymbol{\ell} \cdot \boldsymbol{s}$ expectation values. Using $\boldsymbol{\ell} \cdot \boldsymbol{s} = \frac{1}{2}(\boldsymbol{j}^2 - \boldsymbol{\ell}^2 - \boldsymbol{s}^2)$, show that $\langle \boldsymbol{\ell} \cdot \boldsymbol{s} \rangle = \frac{1}{2}\ell\hbar^2$ for $j = \ell + \frac{1}{2}$ and $-\frac{1}{2}(\ell+1)\hbar^2$ for $j = \ell - \frac{1}{2}$. Then show that the partner splitting is proportional to $2\ell + 1$.

Solution 12.3. With $s = \frac{1}{2}$, $s(s+1) = \frac{3}{4}$. For $j = \ell + \frac{1}{2}$,

$$j(j+1) = \left(\ell + \tfrac{1}{2}\right)\left(\ell + \tfrac{3}{2}\right) = \ell^2 + 2\ell + \tfrac{3}{4},$$

so

$$j(j+1) - \ell(\ell+1) - s(s+1) = \ell$$

and $\langle \boldsymbol{\ell} \cdot \boldsymbol{s} \rangle = \frac{1}{2}\ell\hbar^2$. For $j = \ell - \frac{1}{2}$,

$$j(j+1) - \ell(\ell+1) - s(s+1) = -(\ell+1),$$

hence $-\frac{1}{2}(\ell+1)\hbar^2$. The difference of the two expectation values is $\frac{1}{2}(2\ell+1)\hbar^2$, so the energy splitting scales as $2\ell + 1$ once $\langle f(r) \rangle$ is factored out [6, 13].

Problem 12.4 — Oscillator degeneracy. Derive the occupancy $g_N = (N+1)(N+2)$ of major shell N in the isotropic three-dimensional oscillator, including spin.

Solution 12.4. The number of non-negative integer solutions of $n_x + n_y + n_z = N$ is $\binom{N+2}{2} = (N+1)(N+2)/2$. Each spatial state admits two spin projections, so

$$g_N = 2 \cdot \frac{(N+1)(N+2)}{2} = (N+1)(N+2).$$

For $N = 0, 1, 2, 3$ one recovers $2, 6, 12, 20$, and the cumulative closures $2, 8, 20, 40$ [6, 1].

Problem 12.5 — CM subtraction and separation step. Why is $\boldsymbol{P}_{\mathrm{cm}}^2/(2Am)$ subtracted from the many-body Hamiltonian? Also state the expected direction of the one-neutron separation energy step when going from a magic-N isotope to its $N+1$ neighbour.

Solution 12.5. The subtraction removes kinetic energy of translating the nucleus as a whole, leaving only intrinsic motion; otherwise spurious CM excitations can pollute the shell spectrum. The first neutron above a closed shell occupies a higher, less-bound orbital, so S_n drops from magic N to $N+1$. Conversely, removing a neutron from the closed-shell nucleus requires an unusually large S_n [6, 1].

Problem 12.6 — Empirical magic signatures. List three independent experimental signatures of a magic number and explain briefly why each appears at shell closures [6, 1].

Solution 12.6. (i) Extra binding and jumps in S_n or S_p (closed shells are harder to remove nucleons from). (ii) Kinks or jumps in charge radii / abundance systematics. (iii) Spherical ground states and high first 2^+ energies in even–even neighbours of magic cores (collectivity is quenched). All follow from a large single-particle gap at the Fermi surface [6, 1].

Chapter 13

Shell Model: Woods–Saxon Potential and Ground-State J^π

“For realistic calculations, more accurate potentials are preferred… Among the various models, the Woods–Saxon potential is particularly effective.”
lecture notes, PH5001

Lecture 12 showed that magic numbers beyond 20 require spin–orbit coupling, and that the choice of infinite well versus harmonic oscillator does not change those magic numbers once $\boldsymbol{\ell} \cdot \boldsymbol{s}$ is strong enough. Realistic calculations still need a finite, diffuse mean field: the **Woods–Saxon** potential. With that spectrum one reads ground-state spin and parity from the extreme single-particle filling rules [6, 1, 10].

13.1 Why infinite potentials fail as absolute models

An infinite wall forbids nucleon removal: separation energies would be infinite. Real nuclei have finite S_n and S_p. A finite square well allows separation but retains a discontinuous edge. Nuclear density and the mean field fall smoothly over a surface thickness of order 2–3 fm (Lecture 2). The potential should follow that density. Phenomenologically, depths and radii chosen to match optical-model scattering and separation energies already point toward $V_0 \sim 50\,\mathrm{MeV}$ and $R \sim 1.25\,A^{1/3}\,\mathrm{fm}$; the Woods–Saxon form below implements that diffuse edge [10, 6, 1].

13.2 Woods–Saxon single-particle potential

Woods and Saxon introduced this radial shape as the real part of a diffuse *optical-model* potential for nucleon–nucleus scattering [10]. Its finite depth and smooth surface also make it a useful phenomenological bound-state mean field; that later use is not a first-principles derivation. The standard central form is

$$V_{\mathrm{WS}}(r) = -\frac{V_0}{1 + \exp\big[(r - R)/a\big]}, \tag{13.1}$$

with $R = r_0 A^{1/3}$, $r_0 \approx 1.25\,\mathrm{fm}$, diffuseness $a \approx 0.5$–$0.7\,\mathrm{fm}$, and depth $V_0 \sim 50\,\mathrm{MeV}$ chosen to match separation energies [10, 6, 1]. The potential is nearly flat and attractive in the nuclear interior, then rises smoothly through the surface to zero outside. At $r = R$ one has $V = -V_0/2$. The 10%–90% surface thickness follows directly from the Woods–Saxon shape. Define

$$f(r) = \frac{-V_{\rm WS}(r)}{V_0} = \frac{1}{1 + \exp[(r-R)/a]}.$$

Solving $f(r) = p$ gives

$$\frac{1}{p} - 1 = e^{(r-R)/a}, \qquad r(p) = R + a\ln\left(\frac{1-p}{p}\right). \tag{13.2}$$

Thus the inner 90% point and outer 10% point are

$$r(0.9) = R - a\ln 9, \qquad r(0.1) = R + a\ln 9. \tag{13.3}$$

Their radial separation is

$$t = r(0.1) - r(0.9) = 2a\ln 9 = 4a\ln 3 \approx 4.4\,a, \tag{13.4}$$

of order 2–3 fm for realistic a.

The radial Schrödinger equation for the reduced wave function $u(r)$ is

$$-\frac{\hbar^2}{2m}\frac{\mathrm{d}^2u}{\mathrm{d}r^2} + \left[V_{\rm WS}(r) + \frac{\ell(\ell+1)\hbar^2}{2mr^2}\right]u = E\,u, \tag{13.5}$$

with $u(0) = 0$. Outside the nucleus, $V \to 0$ for a neutron. It is useful to combine the potential and centrifugal terms into

$$V_{\rm eff}(r) = V_{\rm WS}(r) + \frac{\ell(\ell+1)\hbar^2}{2mr^2}. \tag{13.6}$$

For $\ell > 0$, the positive centrifugal barrier suppresses the wave function near the origin and shifts probability toward the nuclear surface. Far outside, both terms approach zero. The radial equation then reduces to

$$\frac{\mathrm{d}^2u}{\mathrm{d}r^2} = \frac{2m|E|}{\hbar^2}\,u = \kappa^2 u \qquad (E < 0), \tag{13.7}$$

whose two solutions are $e^{+\kappa r}$ and $e^{-\kappa r}$. Normalisability eliminates the growing solution, leaving

$$u(r) \underset{r\to\infty}{\longrightarrow} C\,e^{-\kappa r}, \qquad \kappa = \sqrt{\frac{2m|E|}{\hbar^2}}. \tag{13.8}$$

Matching the interior numerical solution to this exterior exponential quantises the allowed energies. There is therefore a finite number of bound orbits, and the least-bound nucleon has a realistic separation energy of a few to tens of MeV.

The decay length $1/\kappa \propto 1/\sqrt{|E|}$ is important physics: a weakly bound neutron has a long exterior tail. Near the neutron drip line this produces diffuse skins and, for low ℓ where the centrifugal barrier is small, halo structure. A bound proton also decays, but its asymptotic tail is modified by the repulsive Coulomb potential rather than being a pure exponential at finite radius.

For protons one must add the Coulomb potential of the remaining charge distribution. A simple estimate for a uniformly charged sphere of radius R is

$$V_{\mathrm{C}}(r) = \begin{cases} \dfrac{(Z-1)e^2}{8\pi\epsilon_0 R}(3 - r^2/R^2), & r \le R, \\ \dfrac{(Z-1)e^2}{4\pi\epsilon_0 r}, & r > R. \end{cases} \tag{13.9}$$

Consequently the effective proton potential is raised and is less binding than the neutron potential at the same A. It is misleading to call the Woods–Saxon geometry itself "wider": the Coulomb tail is long-ranged, while the spatial extent of an orbital depends on its binding and angular momentum. Coulomb repulsion is one reason heavy stable nuclei require $N > Z$ [6, 1].

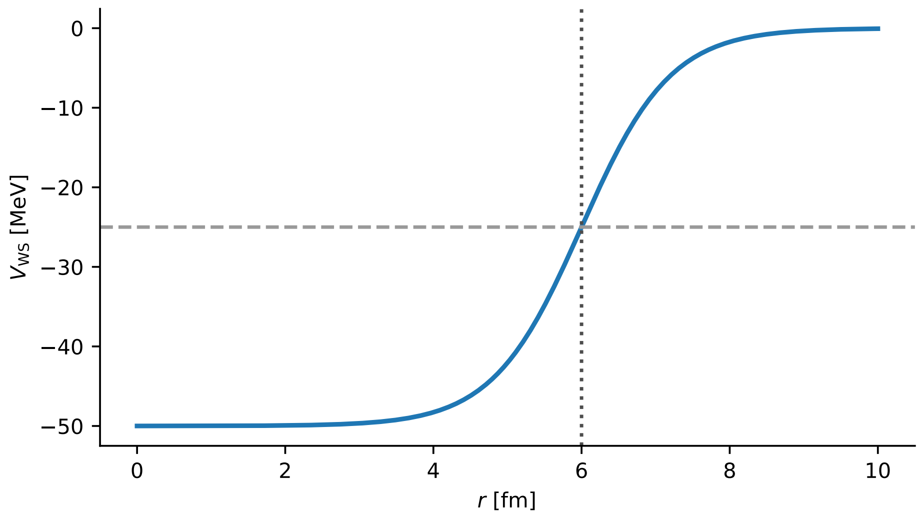


Figure 13.1: Woods–Saxon potential $V(r)$ used as the basic single-particle nuclear mean field. The red marker indicates the half-depth radius R (original vector evaluation of the analytic form; Woods and Saxon [10]).

13.3 From oscillator to Woods–Saxon to spin–orbit

Compared with the oscillator, a finite diffuse well lowers high-ℓ orbitals preferentially (they live nearer the surface) and lifts accidental degeneracies such as $1d$–$2s$ and $1f$–$2p$. Typical trends when passing from oscillator (HO) to Woods–Saxon (WS) before spin–orbit:

- $1f$ drops more than $2p$;
- $1g$ drops strongly, approaching the $1f, 2p$ region;
- $1d$ and $2s$ split, with $1d$ usually lower;
- $2d$ and $3s$ descend; the uppermost level becomes $1h$.

Adding $\boldsymbol{\ell} \cdot \boldsymbol{s}$ then splits each $\ell > 0$ level: $1p \to p_{3/2}, p_{1/2}$; $1d \to d_{5/2}, d_{3/2}$; $1f \to f_{7/2}, f_{5/2}$; $1g \to g_{9/2}, g_{7/2}$; $1h \to h_{11/2}, h_{9/2}$. The magic numbers remain $2, 8, 20, 28, 50, 82, 126$, but the detailed spacings become realistic (Fig. 13.2).

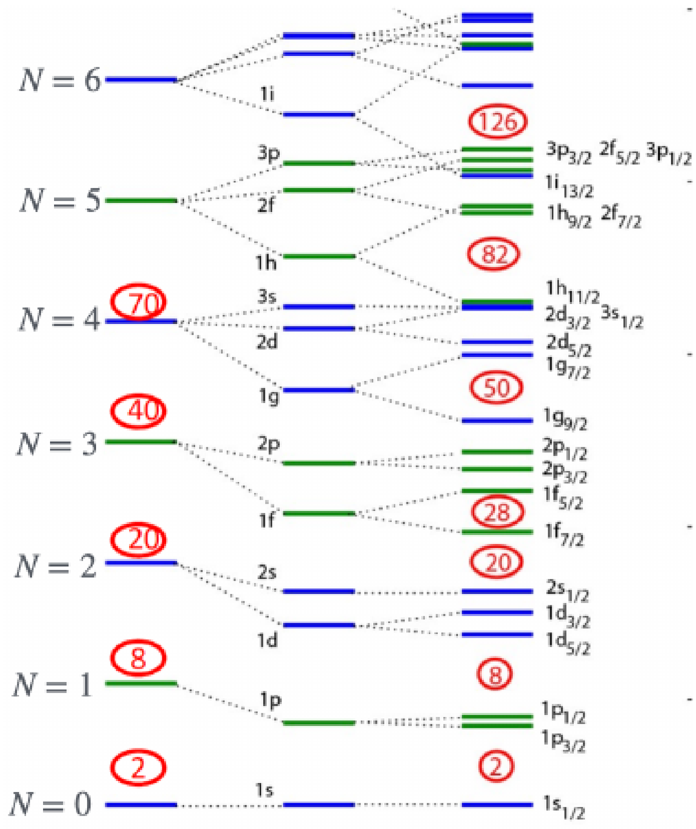


Figure 13.2: Single-particle spectrum: harmonic oscillator (HO), Woods–Saxon (WS), and Woods–Saxon plus spin–orbit (WS+LS). Red circles mark shell closures; see [1, 6]).

Key idea

Magic numbers are robust once spin–orbit is present. Energy values and intra-shell ordering depend on the central potential; Woods–Saxon is the standard realistic choice. The finite depth controls separation energies, the diffuseness controls surface physics, the centrifugal term separates different ℓ, and spin–orbit coupling separates $j = \ell \pm \frac{1}{2}$.

13.4 Extreme single-particle model

In the **extreme single-particle** (or independent-particle) limit one assumes:

1. nucleons fill the lowest available single-particle orbits;
2. residual interactions are dominated by pairing of like nucleons into $J = 0$ pairs;
3. ground-state properties of odd-A nuclei (spin, parity, magnetic moment) are carried by the last unpaired nucleon.

Even–even nuclei then have all nucleons paired. The model is most reliable near closed shells; mid-shell, configuration mixing and collective motion become essential (Lecture 15). Empirically it assigns the correct I^π for a large majority of odd-A ground states near magic cores, and it is somewhat less successful for excited spectra once several valence nucleons share an open shell [6, 1]. The magnetic moment (Lecture 14) then provides an independent test of whether the valence orbit really is pure.

13.5 Spin and parity: notation

Nuclear "spin" means total angular momentum $\boldsymbol{I}$ (often written J). For a single nucleon,

$$\boldsymbol{j} = \boldsymbol{\ell} + \boldsymbol{s}, \qquad |j| = \sqrt{j(j+1)}\,\hbar, \qquad \pi = (-1)^\ell. \tag{13.10}$$

The spectroscopic label is I^π (or J^π). Example: a nucleon in $d_{5/2}$ ($\ell = 2$, $j = \frac{5}{2}$) has $I^\pi = \frac{5}{2}^+$.

13.6 Even–even ground states and paired 0^+ structure

In the paired mean-field limit, like nucleons occupy time-reversed states and couple predominantly to total angular momentum zero. Consider two identical nucleons in the same $n\ell_j$ orbit. The Pauli principle restricts the allowed total angular momenta of the pair to the even values

$$I = 0, 2, \ldots, 2j - 1 \tag{13.11}$$

(odd multipoles are excluded for two identical fermions in one j shell). Among these, the residual short-range attraction favours the maximum spatial overlap, which occurs for the $I = 0$ pair. That pair has positive parity regardless of whether ℓ is even or odd, because

$$\pi_{\text{pair}} = (-1)^\ell (-1)^\ell = +1. \tag{13.12}$$

If the even–even ground state is dominated by this paired structure,

$$I^\pi = 0^+, \tag{13.13}$$

independent of which orbits are filled. Observed even–even nuclear ground states are 0^+, but that empirical rule is not proved merely by particle counting: deformation and configuration mixing can make the wave function highly correlated while preserving 0^+ [6, 1, 20]. The magnetic moment vanishes as well (Lecture 14).

13.7 Odd-A nuclei: last nucleon rules

Either Z or N is odd. All but one nucleon form $J = 0$ pairs; the unpaired nucleon occupies the last partially filled orbit $n\ell_j$ and determines

$$I = j, \qquad \pi = (-1)^\ell. \tag{13.14}$$

This extreme single-particle assignment is the first thing to check against a measured ground-state J^π [6, 1]. It works best near closed shells and for light-to-medium nuclei; far from magic numbers, residual interactions mix nearby j orbits and can rearrange which configuration lies lowest. Even then the odd-nucleon picture usually predicts the correct *parity* and a short list of candidate spins, which is why it remains a workhorse for first interpretations of radioactive beams and isomer searches.

13.7.1 Example: ^{15}N

Nitrogen-15 has $Z = 7$, $N = 8$. Neutrons fill a closed shell through $1p_{1/2}$. The proton occupation is

$$(1s_{1/2})^2(1p_{3/2})^4(1p_{1/2})^1. \tag{13.15}$$

The unpaired proton is in $1p_{1/2}$ ($\ell = 1$, $j = \frac{1}{2}$), so

$$I^\pi = \tfrac{1}{2}^-, \tag{13.16}$$

in agreement with experiment. The same last-proton assignment also predicts a Schmidt magnetic moment near the $j = \ell - 1/2$ line (Lecture 14), so spin, parity, and μ form a consistent checklist for this simple hole state.

13.7.2 Nearby configurations and mixing: ^{61}Ni

Nickel-61 has $Z = 28$, $N = 33$. The protons fill the closed $Z = 28$ shell. Above the $N = 28$ neutron core one must place five valence neutrons in the $2p_{3/2}$, $1f_{5/2}$ and $2p_{1/2}$ sequence. Two limiting basis configurations illustrate the role of residual interactions:

$$(2p_{3/2})^4(1f_{5/2})^1 \quad \Rightarrow \quad I^\pi = \tfrac{5}{2}^-, \tag{13.17}$$

$$(2p_{3/2})^3(1f_{5/2})^2 \quad \Rightarrow \quad I^\pi = \tfrac{3}{2}^-. \tag{13.18}$$

In the second basis configuration the two $1f_{5/2}$ neutrons can form a $J = 0$ pair, leaving the odd neutron in $2p_{3/2}$. Experiment finds $I^\pi = \frac{3}{2}^-$ for the ground state of ^{61}Ni [20, 21, 6]. This fixes the total quantum numbers, not a unique occupation: the physical state is a shell-model superposition of nearby configurations with $I^\pi = 3/2^-$, including core-coupled pieces.

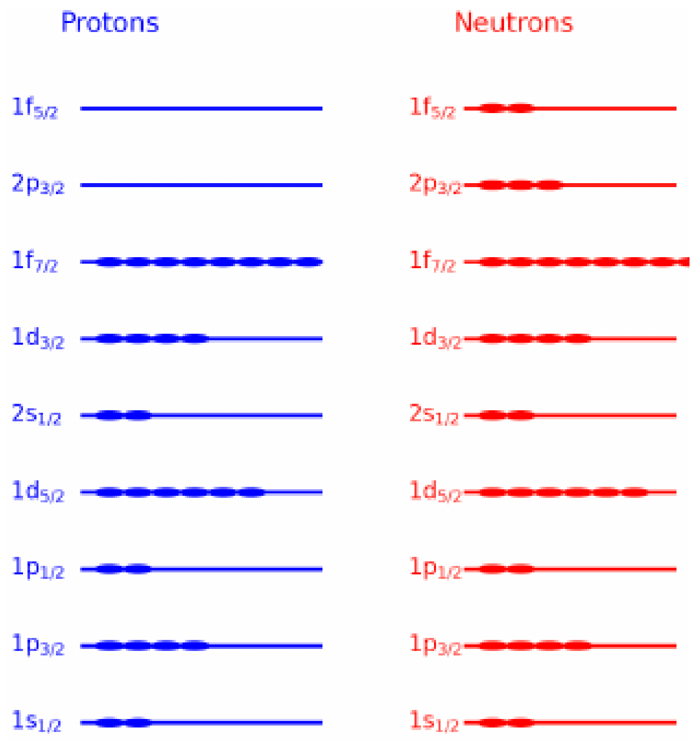


Figure 13.3: One limiting ^{61}Ni valence configuration, $(2p_{3/2})^3(1f_{5/2})^2$, compatible with a $3/2^-$ odd neutron. The evaluated ground-state spin is $3/2^-$, but the physical wave function mixes this and other $3/2^-$ configurations [20, 21].

The lesson is not a universal rule that "pairing is always stronger in high-ℓ orbits", but rather that residual pairing and configuration mixing can reorganise nearby basis states when single-particle spacings are small. The extreme model supplies useful labels, while spectroscopic factors and a full diagonalisation test their purity [6, 1].

13.7.3 Example: ^{17}O

Oxygen-17 ($Z = 8$, $N = 9$) has closed proton shells and neutrons $(1s_{1/2})^2(1p_{3/2})^4(1p_{1/2})^2(1d_{5/2})^1$ (Fig. 13.4). The valence neutron is $1d_{5/2}$, so

$$I^\pi = \tfrac{5}{2}^+. \tag{13.19}$$

Evaluated data confirm the ground state $5/2^+$ and a first excited $1/2^+$ state at 871 keV, corresponding to promotion of the valence neutron to $2s_{1/2}$ [20, 21]. Negative-parity states involve p-shell holes. Excited states of ^{17}O are treated systematically in Lecture 15.

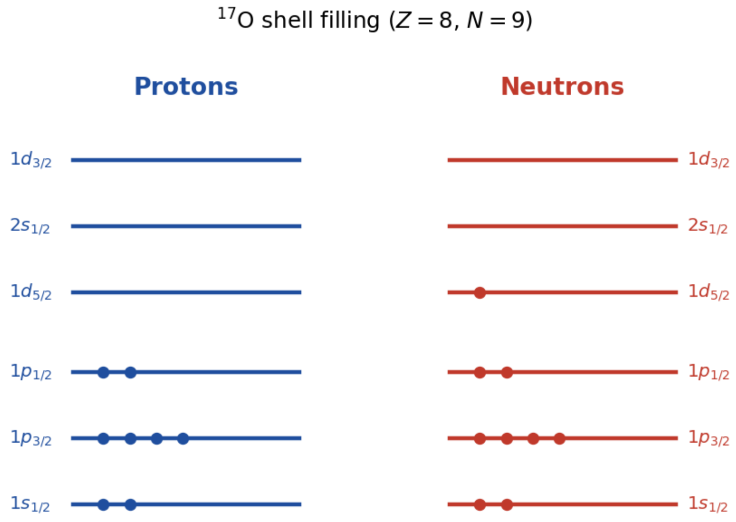


Figure 13.4: Shell-model level occupation for ^{17}O: closed proton shell and unpaired $1d_{5/2}$ neutron (original lecture figure).

13.8 Odd–odd nuclei

Both an unpaired proton and an unpaired neutron contribute. Their angular momenta $\boldsymbol{j}_p$ and $\boldsymbol{j}_n$ couple to a multiplet of total angular momentum

$$|j_p - j_n| \le I \le j_p + j_n, \tag{13.20}$$

in integer steps, with parity

$$\pi = (-1)^{\ell_p + \ell_n}. \tag{13.21}$$

The triangle rule alone does *not* select the ground-state member of the multiplet: residual proton–neutron interactions decide the ordering. Nordheim's empirical rules provide a first guide [19, 6], but they are not rigorous theorems. Roughly, when the odd proton and odd neutron occupy orbits whose intrinsic spins are parallel (or antiparallel) in a prescribed way relative to their ℓ, the lowest multiplet member tends to be the extreme $I = |j_p \pm j_n|$ rather than a mid-range value; modern residual interactions in shell-model spaces supersede the simple rule wherever spectroscopic data exist. Few odd–odd nuclei are stable (Lecture 5); ^{14}N ($I^\pi = 1^+$) is a classic light example [20].

Key idea

Even–even: 0^+. Odd A: I^π of the last unpaired nucleon (subject to pairing reordering near closely spaced levels). Odd–odd: an allowed multiplet $|j_p - j_n| \le I \le j_p + j_n$, with the ground member fixed by residual pn interactions.

13.9 Pairing and the SEMF

Two nucleons in the same j orbit prefer $I_{\text{pair}} = 0$. That pairing energy is the microscopic origin of the SEMF pairing term δ (Lecture 4) and of the even–even preference for stability. Breaking a pair costs $\sim$ 1–2 MeV and is the first step toward many low-lying even–even excitations (two-quasiparticle states), while odd-A spectra often show the pure single-particle sequence until core excitations set in [6].

13.9.1 ESP assignments and neutron versus proton surfaces

Textbook ESP examples remain the cleanest pedagogy: unpaired $p_{1/2}$ in ^{15}O and $d_{5/2}$ in ^{17}O match the observed spins and parities. Woods–Saxon densities measured for charge and mass show that neutron and proton surfaces need not coincide in heavy nuclei, which is why mean-field radii for neutrons and protons are fitted separately in modern optical and Skyrme analyses [6, 1].

13.10 Deeper physics: Woods–Saxon orbits and I^π assignments

13.10.1 Woods–Saxon parameters and level ordering

The Woods–Saxon well (Equation (13.1))

$$V(r) = -\frac{V_0}{1 + \exp[(r - R)/a]} \tag{13.22}$$

interpolates between $-V_0$ in the interior and zero outside, with surface diffuseness $a \approx 0.65$ fm. Solving the radial Schrödinger equation (Equation (13.5)) with a spin–orbit term yields the single-particle spectrum of Figure 13.2. The centrifugal barrier $\ell(\ell+1)\hbar^2/(2mr^2)$ raises p waves relative to s waves; spin–orbit further splits each ℓ into $j = \ell \pm \frac{1}{2}$. Near closed shells the extreme model (Lecture 13) assigns ground-state I^π from the last unpaired nucleon.

13.10.2 Worked I^π assignments

Consider ^{17}O ($N = 9$, $Z = 8$). The closed ^{16}O core ($N = 8$ magic) leaves one neutron outside. Filling the levels in Figure 13.2 places the valence neutron in the $1d_{5/2}$ orbit ($\ell = 2$, $j = \frac{5}{2}$). Parity is $(-1)^\ell = +$, so

$$I^\pi(^{17}\text{O}) = \tfrac{5}{2}^+, \tag{13.23}$$

matching ENSDF [20, 6]. For ^{15}N ($N = 7$, one proton hole on ^{16}O), the last unpaired proton occupies $1p_{1/2}$ ($\ell = 1$, $j = \frac{1}{2}$), giving $I^\pi = \frac{1}{2}^-$.

Key idea

Assignment checklist.

1. Identify the closed core (magic N or Z).
2. Locate the last partially filled orbit in the WS diagram.
3. Read $I = j$ and $\pi = (-1)^\ell$ for a single unpaired nucleon.

4. Flag mid-shell nuclei (^{61}Ni, etc.) where configuration mixing invalidates step 3.

Remark

Protons vs neutrons. Coulomb repulsion raises proton levels relative to neutron levels at the same A, so mirror nuclei are similar but not identical. Transfer reactions measure proton and neutron spectroscopic factors separately (Lecture 15 further material). Woods–Saxon is a starting map, not the final word for drip-line nuclei where the continuum must be included.

13.10.3 Centrifugal barrier and orbit ordering

The radial equation (Equation (13.5)) includes $\ell(\ell+1)\hbar^2/(2mr^2)$, which raises p-wave orbits above s-waves and d-waves above p-waves at the same n. Combined with spin–orbit splitting, this produces the familiar ordering $1s_{1/2}, 1p_{3/2}, 1p_{1/2}, 1d_{5/2}, \ldots$ of Figure 13.2. Assigning I^π is then a bookkeeping exercise *provided* the last nucleon sits in a orbit separated from neighbours by more than the residual interaction scale ($\sim$ 1–3 MeV near closed shells, larger mid-shell) [6, 20].

13.10.4 From orbits to transfer-reaction quantum numbers

One-nucleon transfer reactions (Lecture 15) populate the same orbits with angular-momentum transfer $\Delta L = \ell$ fixed by the reaction mechanism. If ^{17}O is truly $1d_{5/2}$, a (d,p) cross section on ^{16}O should peak at the angular momentum matched to $\ell = 2$. Mismatch between angular distributions and the WS assignment signals configuration mixing (^{61}Ni is the standard cautionary example) [1, 6].

13.10.5 Parity and allowed I^π combinations

For a single nucleon, $\pi = (-1)^\ell$. Even–even ground states are 0^+ (Section 13.6). Odd–odd nuclei require coupling of two unpaired nucleons and can have I^π values not accessible to a single orbit (for example 0^- from $p_{1/2} \otimes p_{1/2}$). The extreme model's last-nucleon rule applies only to odd-A nuclei; attempting to assign odd–odd ground states from one orbit alone is a common student error [6, 20].

Surface thickness

For diffuseness $a \simeq 0.6$ fm, the 10%–90% skin thickness is $t = 4a \ln 3 \simeq 2.6$ fm, matching Lecture 1's empirical surface. Changing a at fixed central depth moves single-particle orbitals differently for low-ℓ and high-ℓ states because of their radial weighting at the surface.

Configuration mixing in mid-shell nuclei fragments the single-particle strength: the spectroscopic factor to a given final state can be $\ll 1$ even when a naive filling diagram suggests a pure orbital. Always treat lecture I^π rules as the first guess, then look for nearby levels of the same parity that signal mixing.

Depth-extra reinforcement for continuum orbitals

A positive-energy "bound-state" root in a truncated basis is not a resonance. Record energy and width from a continuum calculation before comparing with knockout data. This remark protects Lecture 13 assignments near drip lines from a common misreading of numerical output.

Physics-depth close

The Woods–Saxon model is the usable mean field for I^π assignments. Its parameters encode saturation scales from Lecture 2 and its failures encode correlation physics from Lectures 14–15. Treat every assigned I^π as a hypothesis to check against ENSDF, not as a theorem.

Parameter correlations and drip-line caution

Depth, radius, and diffuseness of the Woods–Saxon well are correlated when only one eigenvalue is constrained. Fitting several isotopic chains and both neutron and proton orbitals reduces that ambiguity. Spin–orbit depths are further constrained by magic-number splittings from Lecture 12. Near drip lines, a 5% depth change can unbind a weakly held orbital, so continuum methods replace naive bound-state roots. Mid-shell nuclei fragment single-particle strength: spectroscopic factors from transfer (Lecture 21) then replace the extreme last-nucleon rule. Keep a short checklist: closed core? last orbit? $I = j$ and $\pi = (-1)^\ell$? nearby same-parity levels signalling mixing? That checklist is Lecture 13's practical payload.

Surface weighting

High-ℓ orbitals weight the surface more than low-ℓ orbitals. Changing diffuseness at fixed central density therefore reorders levels even when the rms radius from Lecture 1 is held fixed. Surface physics and shell ordering are inseparable.

13.11 Further physics: proton versus neutron potentials

Because protons feel Coulomb repulsion in addition to the nuclear mean field, the proton single-particle spectrum is not identical to the neutron spectrum at the same A. Relative to neutrons, proton orbits are pushed upward, and the least-bound proton typically has a smaller separation energy near the proton drip line than the analogous neutron near the neutron drip line [6, 1].

In transfer and knockout reactions one therefore maps proton and neutron spectroscopic factors separately. Near closed shells the extreme single-particle assignments for I^π remain robust; far from stability the same residual interactions that reorder ^{61}Ni can rearrange whole sequences of levels and even dissolve traditional magic numbers. Modern shell-model and density-functional calculations keep the Woods–Saxon (or Hartree–Fock) mean field as the starting point and diagonalise a residual interaction in a truncated valence space.

13.11.1 Surface thickness of a Woods–Saxon potential

For $V(r) = -V_0/[1+\exp((r-R)/a)]$, the density falls from 90% to 10% of its central value over a distance

$$t \approx 4a \ln 3 \approx 4.4a. \tag{13.24}$$

With $a = 0.65\,\mathrm{fm}$, $t \approx 2.9\,\mathrm{fm}$, comparable to the nuclear skin thickness inferred from electron scattering (Lecture 1). The radius $R \approx 1.25\,A^{1/3}\,\mathrm{fm}$ sets the bulk size; a sets the diffuseness. Deeply bound orbits ($1s$, $1p$) probe the interior; weakly bound valence orbits ($1g$, $1h$) probe the surface region where $f(r)$ in the spin–orbit term peaks [6, 10].

13.11.2 Configuration mixing beyond the extreme model

^{61}Ni ($N = 33$, one neutron outside ^{60}Ni) is often quoted as $I^\pi = \frac{3}{2}^-$ from the $1f_{5/2}$ orbit, but the full spectrum shows multiple low-lying states within hundreds of keV. Residual interactions mix $1f_{5/2}$, $2p_{3/2}$, and $2p_{1/2}$ configurations, so the last-nucleon picture is a zeroth-order guide only. The extreme model is reliable when the shell gap exceeds the residual interaction ($\Delta\varepsilon \gg V_{\mathrm{res}}$); it fails in mid-shell where $E(2^+)$ drops to tens of keV (Lecture 15) [6, 1, 20].

Mirror pair ^{17}O / ^{17}F

Both have one nucleon outside ^{16}O. Coulomb shifts the proton levels upward, so ^{17}F ($I = \frac{5}{2}^+$, $1d_{5/2}$ proton) mirrors ^{17}O ($I = \frac{5}{2}^+$, $1d_{5/2}$ neutron) in I^π but with different separation energies. The magnetic moments differ because the valence nucleon charge differs (Lecture 14): Schmidt lines apply separately to protons and neutrons [6, 20].

13.11.3 Bridge to Lectures 14–15

I^π from the last nucleon (Section 13.7) is tested by magnetic moments (Lecture 14) and by transfer reactions that measure spectroscopic factors (Lecture 15). A large spectroscopic factor confirms a pure single-particle assignment; fragmentation of strength signals mixing. Near closed shells both tests usually pass; mid-shell they fail together, signalling the onset of collectivity [1].

13.11.4 Woods–Saxon parameters in numbers

Typical values for medium-mass nuclei are $V_0 \approx 50\,\mathrm{MeV}$, $R \approx 1.25\,A^{1/3}\,\mathrm{fm}$, $a \approx 0.65\,\mathrm{fm}$, and a spin–orbit strength that splits $1f_{7/2}$ and $1f_{5/2}$ by several MeV. For ^{208}Pb, $R \approx 6.5\,\mathrm{fm}$ and the least-bound valence neutrons ($1i_{13/2}$, $2g_{9/2}$) sit only $\sim 5\,\mathrm{MeV}$ below continuum threshold, foreshadowing the drip-line physics discussed in the research frontier below [6, 10, 20].

13.12 Deeper reading: I^π, spectroscopic factors, and the continuum

The extreme single-particle picture assigns I^π from the last odd nucleon's orbital in a Woods–Saxon potential. It works near closed shells and fails when configuration mixing or deformation dominates (Lecture 15). Spectroscopic factors from transfer or knockout measure fragmentation of that single-particle strength; quenching relative to independent-particle estimates is common and partly a reaction-theory issue (Lecture 21).

Near drip lines, separation energies of a few hundred kilo-electronvolts make continuum coupling mandatory: resonances replace bound orbitals. Halo candidates revisit Lecture 7's exterior-probability theme. Always cross-check lecture I^π assignments against evaluated data for a few odd-A neighbours to see where the simple rule still holds.

Depth, radius, and diffuseness trade-offs

Woods–Saxon parameters are correlated: a deeper, narrower well can mimic a shallower, wider one for a single eigenvalue. Fitting several isotopic chains and both neutron and proton orbitals reduces that ambiguity. Spin–orbit depth is additionally constrained by splitting systematics from Lecture 12.

Resonance widths

Unbound orbitals acquire widths Γ. A level listed with $E > 0$ in a naive bound-state solver is not a physical state until a continuum method assigns a resonance energy and width. Halo candidates combine small S_n with large radii: revisit Lecture 7's exterior probability for intuition.

Parameter sensitivity

Vary the Woods–Saxon depth by 5% at fixed radius and watch a weakly bound orbital's separation energy move by hundreds of kilo-electronvolts. Near the drip line that shift can unbind the state entirely. Conversely, a deeply bound orbital barely moves. This sensitivity explains why continuum methods are mandatory for halo candidates yet optional for deeply bound closed-shell orbitals.

Assign I^π for ^{17}O, ^{41}Ca, and ^{209}Pb using a filling diagram, then check evaluated data. Where the assignment works, the extreme single-particle picture is validated; where multiple nearby states share strength, Lecture 15 collectivity or configuration mixing is already leaking into the ground state. Keep both outcomes as expected, not as embarrassments.

The Woods–Saxon model is the usable mean field for I^π assignments. Its parameters encode saturation scales from Lecture 2 and its failures encode correlation physics from Lectures 14–15.

In practice, rework one numerical estimate from the lecture with a different parameter set and compare the shift to the quoted experimental uncertainty. If the shift exceeds the uncertainty, the estimate is teaching a scale, not delivering a fit. That distinction keeps theoretical cartoons honest when they sit beside evaluated data tables and modern many-body calculations. Along the course arc, write one sentence linking this chapter backward and one forward: the discipline of those two sentences is as important as any single formula. Examination problems rarely ask for a literature review, but they do ask for the scale argument that literature reviews presuppose.

Synthesis problem

Write a half-page note that (i) restates the chapter's central scale or selection rule in one equation, (ii) evaluates it numerically for one concrete nucleus or energy, and (iii) names the neighbouring lecture that uses the result. Keep the note free of research-frontier speculation; the goal is fluency with the lecture physics itself. If an assumption fails (Born approximation, spherical mean field, allowed transition, one-dimensional barrier), state the failure mode in a final sentence. That habit converts passive reading into examination-ready understanding and makes the later research-frontier box read as an extension rather than a replacement.

Research frontier: continuum shell structure and student projects

The Woods–Saxon extreme single-particle model of Lecture 13 remains the first interpretation of odd-A ground-state I^π, but near drip lines the continuum and particle decay widths must be included. Halo nuclei and unbound resonances require open quantum-system methods; spectroscopic factors extracted from knockout and transfer are often quenched relative to independent-particle estimates [45, 53]. Ab initio and density-functional approaches aim to predict drip lines and S_n surfaces before experiment reaches them [46]. Uncertainties that still matter include the continuum discretisation in theoretical models, the reaction-theory corrections needed to extract spectroscopic factors, and incomplete mass and lifetime data along the most exotic isotopic chains. The lecture's single-particle language is therefore a starting map, not a finished atlas, for nuclei with small separation energies.

Near the drip line, even the definition of a single-particle energy becomes subtle because resonances have widths. Open-system formulations replace bound orbitals with Gamow or Berggren bases. Keeping that distinction clear prevents over-interpreting a Woods–Saxon level diagram for nuclei with S_n of only a few hundred keV.

Distinguishing narrow resonances from true bound orbitals is essential before assigning lecture-style I^π labels near the drip line.

Student directions. (1) Potential exercise: for a Woods–Saxon profile, plot $V(r)$ and estimate the 10%–90% surface thickness $t = 4a \ln 3$. (2) Data check: assign I^π for ^{17}O and ^{15}N and verify against ENSDF [20]. (3) Discussion: explain why ^{61}Ni needs configuration mixing even though a simple last-neutron rule exists.

13.13 Problems with solutions

Problem 13.1 — Woods–Saxon radius. For ^{208}Pb with $r_0 = 1.25\,\text{fm}$, compute $R = r_0 A^{1/3}$. If $a = 0.65\,\text{fm}$, estimate the 10%–90% surface thickness $t = 4a \ln 3$.

Solution 13.1.

$$A^{1/3} = 208^{1/3} \approx 5.924, \qquad R = 1.25 \times 5.924 \approx 7.40\,\text{fm}.$$

The surface thickness is

$$t = 4a \ln 3 = 4 \times 0.65 \times \ln 3 \approx 2.86\,\text{fm},$$

which is the Woods–Saxon 10%–90% surface thickness. It is not a neutron-skin thickness (the neutron–proton radius difference) [1, 6, 10].

Problem 13.2 — Ground-state J^π. Predict I^π for ^{17}O and for ^{40}Ca in the extreme single-particle model.

Solution 13.2. ^{17}O: unpaired neutron in $1d_{5/2}$ ($\ell = 2$, $j = \frac{5}{2}$) $\Rightarrow \frac{5}{2}^+$. ^{40}Ca: doubly magic even–even $\Rightarrow 0^+$ [6, 20].

Problem 13.3 — ^{15}N configuration. Write the proton and neutron occupations of ^{15}N and deduce I^π.

Solution 13.3. Neutrons: closed through $1p_{1/2}$ ($N = 8$). Protons: $(1s_{1/2})^2(1p_{3/2})^4(1p_{1/2})^1$. Unpaired proton in $1p_{1/2} \Rightarrow I^\pi = \frac{1}{2}^-$ [6, 20].

Problem 13.4 — Odd–odd multiplet. An unpaired $d_{5/2}$ proton and an unpaired $p_{1/2}$ neutron form an odd–odd configuration. List the allowed I^π values and state what additional physics is needed to choose the ground member.

Solution 13.4.

$$|j_p - j_n| = 2, \qquad j_p + j_n = 3,$$

so $I = 2, 3$. Parity $\pi = (-1)^{2+1} = -$, hence 2^- and 3^-. Residual proton–neutron interactions (guided empirically by Nordheim rules) select which of these is the ground state [19, 6].

Problem 13.5 — Even–even ground state. Why does the extreme single-particle model assign $I^\pi = 0^+$ to all even–even ground states? What residual interaction enforces that?

Solution 13.5. Like nucleons pair into $J = 0$ time-reversed couples, so an even number of protons and of neutrons leave no unpaired angular momentum: $I = 0$ and positive parity. The residual pairing force (SEMF δ; shell-model pairing matrix elements) selects 0^+ as the lowest configuration [6, 1].

Problem 13.6 — Hole vs particle near magic cores. ^{39}K has one proton hole relative to ^{40}Ca. What I^π does the extreme single-particle model predict, and why is a hole in j assigned the same j^π as a particle in j?

Solution 13.6. The last occupied proton orbit below $Z = 20$ is $1d_{3/2}$, so a proton hole gives $I^\pi = \frac{3}{2}^+$. A closed-shell-minus-one hole transforms like a particle in the same j under angular momentum, hence the same Schmidt/ESP assignment [6, 20].

Chapter 14

Schmidt Magnetic Moments

"In the extreme single-particle model the magnetic moment of an odd-A nucleus is attributed to the last unpaired nucleon."

T. Schmidt; extreme single-particle shell model

Lecture 8 treated the deuteron as a two-body magnetic moment. For heavier odd-A nuclei the extreme shell model makes a parallel claim: paired nucleons cancel, and μ is that of the valence nucleon in orbit $n\ell_j$. Plotting the resulting $\mu(j)$ gives the **Schmidt lines** [6, 1, 2].

14.1 Even–even nuclei: vanishing moment

In an even–even ground state every nucleon is paired to $I = 0$. Orbital and spin magnetism cancel within each pair, so

$$\mu = 0 \qquad (I^\pi = 0^+). \tag{14.1}$$

Experiment confirms that even–even ground states carry no static magnetic dipole (consistent with $I = 0$). Excited 2^+ states can have nonzero moments, but those probe collective currents rather than a single unpaired nucleon. The vanishing ground-state moment is why Schmidt-line systematics are discussed for odd-A nuclei [6, 1].

14.2 Nuclear magneton and g-factors

Magnetic moments of nuclei are measured in units of the nuclear magneton

$$\mu_N = \frac{e\hbar}{2m_p} \approx 5.05 \times 10^{-27}\,\mathrm{J\,T^{-1}}, \tag{14.2}$$

about 1/1836 of the Bohr magneton $\mu_B = e\hbar/(2m_e)$ because $m_p \gg m_e$. The z-component of the magnetic-moment operator for a single nucleon is

$$\mu_z = \left(g_\ell\,\ell_z + g_s\,s_z\right)\frac{\mu_N}{\hbar}, \tag{14.3}$$

with free-nucleon g-factors

$$\begin{aligned} g_\ell^{(p)} &= 1, \quad g_s^{(p)} = 5.5857, \\ g_\ell^{(n)} &= 0, \quad g_s^{(n)} = -3.8260. \end{aligned} \tag{14.4}$$

A useful way to remember the orbital factors is to begin with the magnetic moment of a particle of charge q and mass m:

$$\boldsymbol{\mu}_\ell = \frac{q}{2m}\,\boldsymbol{\ell} = \frac{q}{e}\frac{m_p}{m}\,\frac{\mu_N}{\hbar}\,\boldsymbol{\ell}. \tag{14.5}$$

For a proton, $q = e$ and $m \simeq m_p$, giving $g_\ell^{(p)} = 1$; for a neutron, $q = 0$, giving $g_\ell^{(n)} = 0$ in the leading free one-body operator. In a nucleus, exchange currents and core polarisation can generate effective orbital contributions, so these are baseline coefficients rather than exact in-medium constants. The spin factors follow from the measured intrinsic moments:

$$\mu_p = g_s^{(p)}\frac{\mu_N}{\hbar}\left(\frac{\hbar}{2}\right) = 2.7928\,\mu_N, \qquad \mu_n = \frac{g_s^{(n)}}{2}\mu_N = -1.9130\,\mu_N. \tag{14.6}$$

A pointlike charged Dirac fermion has $g_s = 2$ at tree level; radiative corrections modify that value. The large anomalous moments of the proton and neutron reflect their quark substructure: the proton (uud) and neutron (udd) carry internal charge currents even though the neutron is neutral overall. The neutron therefore has no orbital g_ℓ from net charge, yet a large spin moment [6, 5].

Because of the strong spin–orbit force, the good single-particle angular momentum is $\boldsymbol{j} = \boldsymbol{\ell}+\boldsymbol{s}$. The spectroscopic moment is the expectation value of μ_z in the stretched state $m_j = j$:

$$\mu = \langle j, m_j = j|\, g_\ell \ell_z + g_s s_z\, |j, m_j = j\rangle \frac{\mu_N}{\hbar}. \tag{14.7}$$

14.3 Case $j = \ell + \frac{1}{2}$: unique m_ℓ, m_s

For $j = \ell + \frac{1}{2}$ and $m_j = j = \ell + \frac{1}{2}$, the only combination consistent with $m_\ell + m_s = m_j$ is

$$m_\ell = \ell, \qquad m_s = +\tfrac{1}{2}. \tag{14.8}$$

The coupled state is therefore a single uncoupled basis vector,

$$|j = \ell + \tfrac{1}{2}, m_j = j\rangle = |\ell, m_\ell = \ell\rangle|\tfrac{1}{2}, m_s = \tfrac{1}{2}\rangle. \tag{14.9}$$

Consequently

$$\langle \ell_z \rangle = \ell\hbar, \qquad \langle s_z \rangle = \tfrac{1}{2}\hbar, \tag{14.10}$$

$$\begin{aligned}\frac{\mu}{\mu_N} &= \frac{1}{\hbar}\left(g_\ell \ell\hbar + g_s\frac{\hbar}{2}\right)\\ &= g_\ell \ell + \tfrac{1}{2}g_s.\end{aligned} \tag{14.11}$$

Using $\ell = j - \frac{1}{2}$ gives, one algebraic step at a time,

$$\frac{\mu}{\mu_N} = g_\ell\left(j - \tfrac{1}{2}\right) + \tfrac{1}{2}g_s = j\,g_\ell + \tfrac{1}{2}(g_s - g_\ell). \tag{14.12}$$

Specialising to free g-factors gives the first Schmidt formula

$$\frac{\mu}{\mu_N} = \begin{cases} j + 2.293 & \text{odd proton } (g_\ell = 1,\ g_s = 5.586), \\ -1.913 & \text{odd neutron } (g_\ell = 0,\ g_s = -3.826). \end{cases} \tag{14.13}$$

For a $d_{5/2}$ proton ($\ell = 2$, $j = \frac{5}{2}$), $\mu \approx 4.79\,\mu_N$ on the Schmidt line. For any odd neutron with $j = \ell + \frac{1}{2}$, the Schmidt value equals the free neutron's intrinsic moment, $-1.913\,\mu_N = g_s^{(n)}\mu_N/2$, independent of j, because the leading orbital term vanishes.

14.4 Case $j = \ell - \frac{1}{2}$: two mixed components

For $j = \ell - \frac{1}{2}$ and $m_j = j$, two (m_ℓ, m_s) pairs contribute:

$$\begin{aligned} |1\rangle &= |m_\ell = \ell,\ m_s = -\tfrac{1}{2}\rangle, \\ |2\rangle &= |m_\ell = \ell - 1,\ m_s = +\tfrac{1}{2}\rangle. \end{aligned} \tag{14.14}$$

The stretched state is the normalised combination

$$|j, m_j = j\rangle = a|1\rangle + b|2\rangle. \tag{14.15}$$

14.4.1 Coefficients from $J_+|j, j\rangle = 0$

The raising operator $J_+ = L_+ + S_+$ annihilates the highest-weight state. The two ladder operators act according to

$$L_+|\ell, m_\ell\rangle = \hbar\sqrt{\ell(\ell+1) - m_\ell(m_\ell+1)}\,|\ell, m_\ell + 1\rangle, \tag{14.16}$$

$$S_+|\tfrac{1}{2}, m_s\rangle = \hbar\sqrt{\tfrac{1}{2}(\tfrac{1}{2}+1) - m_s(m_s+1)}\,|\tfrac{1}{2}, m_s + 1\rangle. \tag{14.17}$$

Therefore

$$J_+|1\rangle = (L_+ + S_+)|\ell, -\tfrac{1}{2}\rangle = \hbar|\ell, +\tfrac{1}{2}\rangle, \tag{14.18}$$

$$J_+|2\rangle = (L_+ + S_+)|\ell - 1, +\tfrac{1}{2}\rangle = \hbar\sqrt{2\ell}\,|\ell, +\tfrac{1}{2}\rangle. \tag{14.19}$$

Here the compact ket on the right denotes $|m_\ell = \ell, m_s = +\frac{1}{2}\rangle$. Acting on the mixture now gives

$$\begin{aligned} 0 &= J_+|j, j\rangle = aJ_+|1\rangle + bJ_+|2\rangle \\ &= \hbar(a + b\sqrt{2\ell})|\ell, +\tfrac{1}{2}\rangle. \end{aligned} \tag{14.20}$$

Since the ket and $\hbar$ are nonzero, the coefficient must vanish:

$$a + b\sqrt{2\ell} = 0 \qquad \Rightarrow \qquad a = -b\sqrt{2\ell}, \tag{14.21}$$

together with normalisation $|a|^2 + |b|^2 = 1$. Therefore

$$|a|^2 = \frac{2\ell}{2\ell + 1}, \qquad |b|^2 = \frac{1}{2\ell + 1}. \tag{14.22}$$

Choosing an irrelevant overall phase so that $b > 0$, the normalized state can be written explicitly as

$$|j = \ell - \tfrac{1}{2}, j\rangle = -\sqrt{\frac{2\ell}{2\ell+1}}\,|1\rangle + \sqrt{\frac{1}{2\ell+1}}\,|2\rangle. \tag{14.23}$$

These are precisely the Clebsch–Gordan coefficients for coupling ℓ and $s = \frac{1}{2}$ to $j = \ell - \frac{1}{2}$.

14.4.2 Expectation value of μ_z

Diagonal matrix elements of the magnetic operator are

$$\begin{aligned}\langle 1|\mu_z/\mu_N|1\rangle &= g_\ell \ell - \tfrac{1}{2} g_s,\\ \langle 2|\mu_z/\mu_N|2\rangle &= g_\ell(\ell-1) + \tfrac{1}{2} g_s,\end{aligned} \tag{14.24}$$

and cross terms vanish by orthogonality of the $|m_\ell m_s\rangle$ basis. More explicitly, ℓ_z and s_z are diagonal in this basis, so

$$\langle 1|\ell_z|2\rangle = \langle 1|s_z|2\rangle = 0. \tag{14.25}$$

This diagonality, not orthogonality alone for an arbitrary operator, is the reason the interference terms vanish. Expanding the full matrix element,

$$\begin{aligned}\frac{\mu}{\mu_N} &= (a^*\langle 1| + b^*\langle 2|)\frac{g_\ell \ell_z + g_s s_z}{\hbar}(a|1\rangle + b|2\rangle)\\ &= |a|^2\langle 1|\mu_z/\mu_N|1\rangle + |b|^2\langle 2|\mu_z/\mu_N|2\rangle\\ &\quad + a^* b\langle 1|\mu_z/\mu_N|2\rangle + b^* a\langle 2|\mu_z/\mu_N|1\rangle.\end{aligned} \tag{14.26}$$

The last line is zero, and the weighted average is therefore

$$\frac{\mu}{\mu_N} = |a|^2(g_\ell \ell - \tfrac{1}{2} g_s) + |b|^2(g_\ell(\ell-1) + \tfrac{1}{2} g_s). \tag{14.27}$$

Insert $|a|^2 = 2\ell/(2\ell+1)$ and $|b|^2 = 1/(2\ell+1)$:

$$\begin{aligned}\frac{\mu}{\mu_N} &= \frac{2\ell}{2\ell+1}(g_\ell \ell - \tfrac{1}{2} g_s) + \frac{1}{2\ell+1}(g_\ell(\ell-1) + \tfrac{1}{2} g_s)\\ &= \frac{1}{2\ell+1}\Big[2\ell^2 g_\ell - \ell g_s + (\ell-1) g_\ell + \tfrac{1}{2} g_s\Big]\\ &= \frac{1}{2\ell+1}\Big[(2\ell^2 + \ell - 1) g_\ell - (\ell - \tfrac{1}{2}) g_s\Big].\end{aligned} \tag{14.28}$$

Now set $j = \ell - \frac{1}{2}$, so $\ell = j + \frac{1}{2}$ and $2\ell + 1 = 2j + 2$. The two coefficients in the numerator become

$$2\ell^2 + \ell - 1 = 2(j + \tfrac{1}{2})^2 + (j + \tfrac{1}{2}) - 1 = 2j^2 + 3j = j(2j+3), \tag{14.29}$$

$$\ell - \tfrac{1}{2} = j. \tag{14.30}$$

Thus

$$\frac{\mu}{\mu_N} = \frac{j(2j+3)g_\ell - jg_s}{2(j+1)} = \frac{j}{2(j+1)}\big[(2j+3)g_\ell - g_s\big]. \tag{14.31}$$

Adding and subtracting $jg_\ell/[2(j+1)]$ in the last expression gives the equivalent standard Schmidt form

$$\frac{\mu}{\mu_N} = \frac{j}{2(j+1)}\big[(2j+3)g_\ell - g_s\big] = j\,g_\ell - \frac{j}{2(j+1)}(g_s - g_\ell). \tag{14.32}$$

Specialising to free g-factors:

$$\frac{\mu}{\mu_N} = \begin{cases} \dfrac{j}{j+1}\,(j - \frac{1}{2}g_s^{(p)} + \frac{3}{2}) = \dfrac{j(j-1.293)}{j+1} & \text{odd proton,} \\ -\dfrac{j}{2(j+1)}\,g_s^{(n)} = +1.913\,\dfrac{j}{j+1} & \text{odd neutron.} \end{cases} \tag{14.33}$$

For an odd neutron in $p_{1/2}$ ($j = \frac{1}{2}$), $\mu/\mu_N = 1.913 \times \frac{1}{2}/\frac{3}{2} = 0.638$. These limiting curves are the Schmidt lines first discussed by Schmidt in 1937 [18, 6, 1].

14.4.3 Independent check: the projection theorem

The same result follows without writing Clebsch–Gordan coefficients. Rotational symmetry requires the expectation value of any vector operator in $|jm_j\rangle$ to point along $\boldsymbol{j}$. Hence

$$\langle \ell_z \rangle = \frac{\langle \boldsymbol{\ell}\cdot\boldsymbol{j}\rangle}{j(j+1)\hbar^2}\,m_j\hbar, \qquad \langle s_z \rangle = \frac{\langle \boldsymbol{s}\cdot\boldsymbol{j}\rangle}{j(j+1)\hbar^2}\,m_j\hbar. \tag{14.34}$$

Using $\boldsymbol{j} = \boldsymbol{\ell} + \boldsymbol{s}$,

$$2\boldsymbol{\ell}\cdot\boldsymbol{j} = \boldsymbol{j}^2 + \boldsymbol{\ell}^2 - \boldsymbol{s}^2, \tag{14.35}$$

$$2\boldsymbol{s}\cdot\boldsymbol{j} = \boldsymbol{j}^2 - \boldsymbol{\ell}^2 + \boldsymbol{s}^2. \tag{14.36}$$

For the spectroscopic state $m_j = j$, division by $\hbar$ gives

$$\frac{\langle \ell_z\rangle}{\hbar} = \frac{j(j+1) + \ell(\ell+1) - s(s+1)}{2(j+1)}, \tag{14.37}$$

$$\frac{\langle s_z\rangle}{\hbar} = \frac{j(j+1) - \ell(\ell+1) + s(s+1)}{2(j+1)}. \tag{14.38}$$

Substitution into $\mu/\mu_N = g_\ell\langle \ell_z\rangle/\hbar + g_s\langle s_z\rangle/\hbar$, with $s = \frac{1}{2}$, reproduces Eqs. (14.13) and (14.32). This is the same projection theorem used for the deuteron in Lecture 8; it is a valuable check because it keeps every orbital and spin term visible.

14.4.4 Why a one-hole state has the same Schmidt value

A filled j subshell has zero total moment because the contributions from m and $-m$ cancel. To construct a hole state with total $M = +j$, remove the particle with $m = -j$. The remaining shell therefore has

$$\mu_{\text{hole}}(M = +j) = 0 - \mu_{\text{particle}}(m = -j) = \mu_{\text{particle}}(m = +j), \tag{14.39}$$

because a vector moment changes sign under $m \to -m$. Thus a particle and a hole in the same j orbit have the same Schmidt moment in the extreme model, even though their many-body occupations are complementary.

14.5 Schmidt lines and experiment

Plotting μ against j for odd-proton nuclei gives two rising lines ($j = \ell \pm \frac{1}{2}$). For odd-neutron nuclei the $j = \ell + \frac{1}{2}$ line is flat at $-1.913\,\mu_N$ and the $j = \ell - \frac{1}{2}$ line is positive and slowly rising. Experimental ground-state moments of odd-A nuclei typically fall *between* the two Schmidt lines for the assigned j (Fig. 14.1) [18, 6, 2, 22]. The Schmidt curves are theoretical single-particle limits, not rigorous mathematical bounds; they organise the data and diagnose configuration purity.

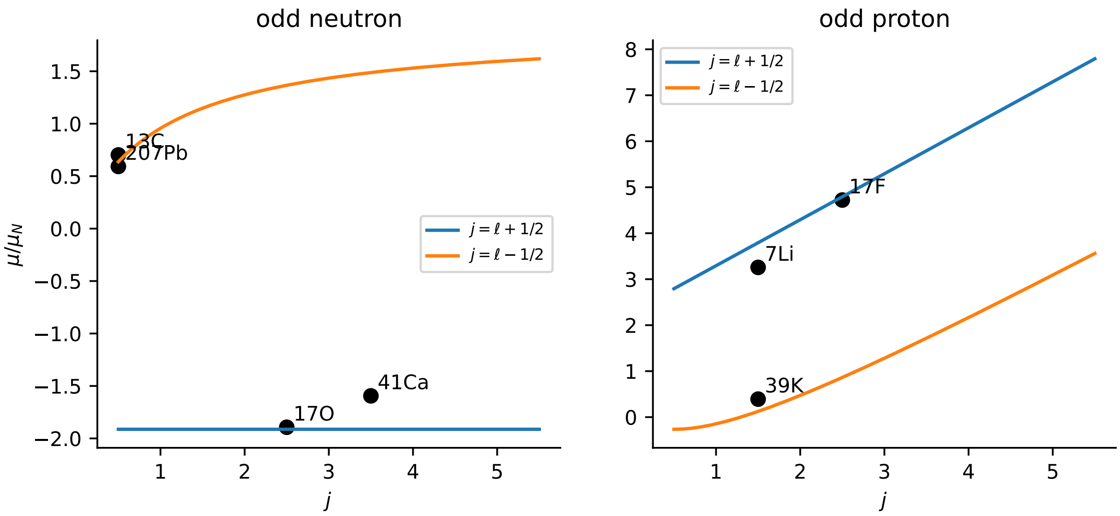


Figure 14.1: Free-nucleon Schmidt curves with selected measured ground-state magnetic moments from the Stone/NNDC compilation [22, 20]. Curves are single-particle estimates, not bounds.

14.6 Why real nuclei lie between the Schmidt lines

The extreme single-particle picture is too simple:

- configuration mixing admixes other orbitals with the same I^π;
- meson-exchange currents modify the free one-body magnetic operator inside the nucleus;
- collective core polarisation contributes for soft nuclei far from closed shells;
- restricted shell-model spaces often use effective spin g-factors with $|g_s|$ reduced from the free value. A rough sd- or pf-shell starting estimate is $g_s^{\text{eff}} \sim (0.7\text{–}0.8)g_s^{\text{free}}$, but the factor depends

on model space, interaction, mass region, and operator; it is neither a universal property of nuclear matter nor interchangeable with axial-current quenching in weak transitions [6, 1].

A historical remark clarifies the g_s point: free-nucleon magnetic moments already deviate from the Dirac values $g_s = 2$ and 0 because of the meson cloud around each nucleon. Inside a nucleus that cloud is distorted by neighbouring nucleons, so it is unsurprising that the in-medium effective g_s differs from the free one [6, 18]. The net effect is that measured $|\mu|$ values are usually smaller than the free Schmidt prediction for the assigned j, with scatter set by configuration purity. Near doubly magic cores (e.g. ^{17}O, ^{41}Ca) agreement with Schmidt values is better; mid-shell nuclei deviate more [6, 2, 1, 22]. When a measured μ lies close to one Schmidt line, the valence assignment $j = \ell \pm \frac{1}{2}$ is strongly supported; intermediate values tag mixing.

Key idea

Schmidt lines are the shell-model baseline for μ. They fix the single-particle g-factors and $j = \ell \pm \frac{1}{2}$ assignment; deviations measure correlations beyond the extreme model.

14.6.1 Schmidt systematics and vanishing odd multipoles

Measured odd-A moments fall between the Schmidt lines with substantial scatter; agreement is best near doubly magic cores. Static electric dipole (and other odd-parity) moments vanish by symmetry for eigenstates of good parity, so the spectroscopic electromagnetic characterisation of ground states is carried by μ and Q [18, 6, 1].

14.7 Deeper physics: Schmidt lines, g-factors, and quenching

14.7.1 Schmidt limits as independent-particle bounds

For an odd-A nucleus with one unpaired nucleon in orbit ℓ, the magnetic operator (14.3) evaluated in the stretched state $|j, m = j\rangle$ yields the Schmidt formulae (Equation (8.66), Equation (8.68)). For an unpaired proton,

$$\frac{\mu}{\mu_N} = \begin{cases} j + \frac{1}{2}(g_p - 1) = \ell + 1 + \frac{1}{2}(g_p - 1), & j = \ell + \frac{1}{2}, \\ j - \frac{j}{2(j+1)}(g_p - 1), & j = \ell - \frac{1}{2}, \end{cases} \tag{14.40}$$

with free $g_p = 5.5857$. Real nuclei fall *between* the two curves because core polarisation, configuration mixing, and meson-exchange currents reduce the effective spin response.

14.7.2 Quenching as a compact summary

When a measured moment lies below the upper Schmidt line, one often writes an **effective** spin g-factor $g_s^{\text{eff}} < g_s^{\text{free}}$. This is a phenomenological compression of several mechanisms, not a fundamental constant. Near ^{16}O or ^{40}Ca, odd neighbours such as ^{17}O ($\mu = +1.89\,\mu_N$) sit close to the $d_{5/2}$ Schmidt prediction ($\sim 1.91\,\mu_N$), while mid-shell nuclei can deviate by tens of percent [6, 22].

Schmidt deviation for ^{15}N

^{15}N has an unpaired proton in $1p_{1/2}$ ($I = \frac{1}{2}$). The Schmidt formula with $g_\ell = 1$, $g_s = g_p$, $j = \ell - \frac{1}{2}$ gives $\mu/\mu_N = \frac{1}{2} + \frac{1}{6}(g_p - 1) \approx 0.44$, i.e. $\mu_{\text{Schmidt}} \approx +2.2\,\mu_N$. Experiment gives $\mu = -0.283\,\mu_N$: the hole character and core polarisation reverse and quench the moment. Even near ^{16}O, Schmidt lines are guides, not precision predictions [6, 22].

Remark

Link to electroweak observables. Quenching of spin operators also appears in Gamow–Teller β decay and in neutrinoless double-beta matrix elements. Modern calculations renormalise *both* the Hamiltonian and the operator consistently rather than applying an ad hoc g_A^{eff} after the fact (Lecture 14 further material).

14.7.3 Even–even neighbours as Schmidt context

Schmidt lines apply only to odd-A nuclei. The adjacent even–even core has $\mu = 0$ (Section 14.1), and its first 2^+ energy (Lecture 15) indicates whether the core is rigid (high $E(2^+)$, Schmidt works well) or soft/collective (low $E(2^+)$, Schmidt fails). Plotting μ against $E(2^+)$ of the $A - 1$ neighbour separates closed-shell odd nuclei from mid-shell ones without invoking a full shell-model calculation [6, 22].

14.7.4 Sign of μ and hole vs particle orbits

A valence *proton* and a valence *proton hole* on the same orbit produce opposite Schmidt limits once the hole sign $(-1)^{N_\text{h}}$ is applied. ^{15}N (p hole on ^{16}O) and ^{17}F (p particle, $1d_{5/2}$) illustrate mirror magnetic behaviour tied to the same WS filling diagram (Lecture 13). Confusing particle and hole signs is a common source of error when overlaying measured moments on Schmidt plots [2, 6].

14.7.5 Landé g-factor for odd–even nuclei

When several nucleons are unpaired (odd–odd nuclei), the Landé formula generalises the Schmidt limit. For odd-A nuclei with one valence nucleon, the effective g-factor reduces to the Schmidt value in the stretched state. Measuring μ and I separately determines an empirical g_I, which can be compared with shell-model calculations that include core polarisation. Near closed shells the agreement is often within 10%; mid-shell nuclei require full diagonalisation in a truncated space [6, 22].

Remark

Isoscalar versus spin contributions. Orbital g_ℓ factors are close to bare values; spin g_s factors show larger medium renormalisations. Magnetic moments therefore diagnose spin correlations more sharply than electric quadrupole moments diagnose charge deformation (Lecture 15), although both feel configuration mixing.

A practical student plot: Schmidt lines for odd protons with five measured mid-shell moments overlaid. The pattern of deviations is the entire qualitative message of this chapter.

14.8 Further physics: magnetic moments as configuration tags

Measured magnetic moments are among the most sensitive tests of the valence configuration near closed shells. If μ lies close to one Schmidt line, the extreme single-particle assignment $j = \ell \pm \frac{1}{2}$ is strongly supported. Intermediate values signal configuration mixing or collective admixtures [6, 2, 22].

Effective g_s factors reduced from the free nucleon values are a compact way to summarise medium modifications, but they absorb several distinct mechanisms: core polarisation, meson-exchange currents, and truncated model spaces. For that reason one should not treat a single quenching factor as fundamental. Near doubly magic cores such as ^{16}O and ^{40}Ca, moments of neighbouring odd-A nuclei remain comparatively close to Schmidt predictions; far from magic numbers, collective contributions dominate and the Schmidt picture is only a reference baseline.

14.8.1 Plotting Schmidt lines: odd-proton example

For an unpaired proton, (14.40) gives two curves in the $(I, \mu/\mu_N)$ plane. At $I = \frac{1}{2}$ (from $p_{1/2}$ or $f_{7/2}$), the upper line yields $\mu/\mu_N = 0.44$ and the lower line gives $\mu/\mu_N = -0.25$. At $I = \frac{7}{2}$ (from $f_{7/2}$ aligned), $\mu/\mu_N = 4.71$ (upper) and 3.79 (lower). Measured moments of odd-Z nuclei between $20 \leq Z \leq 40$ cluster between the curves, with deviations growing toward mid-shell where collective admixtures appear [22, 6].

14.8.2 Quenching: one number, several mechanisms

Writing $\mu_{\text{exp}} = g_s^{\text{eff}} \mu_{\text{Schmidt}}$ compresses:

- **Configuration mixing:** valence spin is shared among several j-shells (Lecture 13).
- **Core polarisation:** paired nucleons are excited by the valence nucleon and contribute opposite magnetism.
- **Meson-exchange currents:** two-body operators (pp, pn currents) add to the one-body Schmidt operator (Lecture 8 further material).

Separating these requires a model, not just a table of moments. The deuteron deficit is the minimal case where only mechanism (iii) and a few-percent D-wave compete [2, 52].

Gamow–Teller quenching scale

Historically, shell-model β decay rates required $g_A^{\text{eff}}/g_A \approx 0.75$. Modern ab initio calculations with two-body currents recover much of that quenching without an ad hoc factor, paralleling the magnetic-moment story. The same operator renormalisation feeds $0\nu\beta\beta$ matrix elements (Lecture 14 research frontier) [52, 51].

14.8.3 Bridge to Lecture 15 spectroscopic factors

A nucleus whose moment lies on a Schmidt line usually has a large spectroscopic factor for adding or removing the valence nucleon (Lecture 15). When both μ and S deviate, configuration mixing is confirmed from two independent probes. Even–even neighbours with low $E(2^+)$ (Lecture 15) signal that the odd-A Schmidt picture is only a local approximation near closed shells [6, 1].

14.8.4 Stone table as experimental anchor

Evaluated magnetic moments (Stone tables) remain the benchmark for shell-model and ab initio calculations. Near ^{16}O and ^{40}Ca, odd neighbours often lie within $\sim 20\%$ of Schmidt lines; in the $A \sim 100$ region deviations exceed 50%, signalling collective admixtures and strong configuration mixing. Any student overlay of Schmidt curves should quote uncertainties on μ and note whether the state is isomeric or ground-state [22, 6, 2].

14.8.5 Core polarisation as a picture for quenching

When a valence nucleon moves, it polarises closed-shell pairs in the core: particle–hole excitations of the core contribute magnetism opposite to the valence spin. In a truncated shell-model space this appears as configuration mixing; in an ab initio calculation it is part of the correlated wave function. Either way, the measured moment lies between the Schmidt lines. The deuteron has no inert core, so its quenching is dominated by D-wave mixing and exchange currents rather than core polarisation, simplifying the interpretation at $A = 2$ [6, 52, 2].

14.9 Deeper reading: g-factors, quenching, and operators

Schmidt lines bound single-particle magnetic moments for odd-A nuclei near closed shells. Mid-shell moments fall inside the lines because of configuration mixing. An effective g_s quenching was long used phenomenologically; modern work attributes much of Gamow–Teller and magnetic quenching to two-body currents treated consistently with correlations, a lesson already previewed in the deuteron (Lecture 8).

For $0\nu\beta\beta$ matrix elements (Lecture 19), the same operator renormalisation issues return at higher stakes: an inconsistent g_A^{eff} stacked on top of an already-renormalised operator double counts physics. Evaluated moment tables remain the benchmark against which both shell-model and ab initio calculations are judged.

Magnetic moments as operator diagnostics

Because magnetic moments weight spin strongly, they expose quenching and meson-exchange currents more clearly than binding energies do. A model that fits spectra with effective charges may still fail moments unless operators are renormalised consistently. Build a habit of checking moments whenever a new effective interaction is proposed.

Mirror magnetic moments

Differences between mirror magnetic moments isolate isovector operator pieces and Coulomb-induced configuration changes. They are a precision testbed complementary to isoscalar deuteron lessons (Lecture 8).

Schmidt diagram exercise

Draw Schmidt lines for odd protons and odd neutrons. Overlay ^{17}F, ^{17}O, a mid-shell odd-A nucleus, and a deformed rare-earth moment. The closed-shell neighbours hug the lines; the mid-shell point falls inside; the deformed example may require collective g_R contributions. Write one sentence for each point naming the dominant physics.

Then estimate a naive quenching factor from one Gamow–Teller $\log ft$ compared with an extreme single-particle matrix element. Discuss why that single number should not be transplanted into a $0\nu\beta\beta$ calculation that already includes two-body currents. The exercise links Lectures 8, 14, and 19 into one operator story.

Schmidt lines are baselines, not predictions for mid-shell nuclei. Deviations teach configuration mixing and operator renormalisation, preparing the weak-interaction chapters.

In practice, rework one numerical estimate from the lecture with a different parameter set and compare the shift to the quoted experimental uncertainty. If the shift exceeds the uncertainty, the estimate is teaching a scale, not delivering a fit. That distinction keeps theoretical cartoons honest when they sit beside evaluated data tables and modern many-body calculations. Along the course arc, write one sentence linking this chapter backward and one forward: the discipline of those two sentences is as important as any single formula. Examination problems rarely ask for a literature review, but they do ask for the scale argument that literature reviews presuppose.

Synthesis problem

Write a half-page note that (i) restates the chapter's central scale or selection rule in one equation, (ii) evaluates it numerically for one concrete nucleus or energy, and (iii) names the neighbouring lecture that uses the result. Keep the note free of research-frontier speculation; the goal is fluency with the lecture physics itself. If an assumption fails (Born approximation, spherical mean field, allowed transition, one-dimensional barrier), state the failure mode in a final sentence. That habit converts passive reading into examination-ready understanding and makes the later research-frontier box read as an extension rather than a replacement.

Research frontier: electroweak operators and student projects

Schmidt lines (Lecture 14) still organise magnetic moments of odd-A nuclei near closed shells, but modern theory computes effective g-factors and Gamow–Teller operators from correlated wave functions and two-body currents. A landmark ab initio study showed that consistent two-body currents resolve much of the long-standing GT quenching, with direct implications for β-decay rates and for neutrinoless double-beta nuclear matrix elements [52, 51]. Evaluated moment tables remain the experimental reference against which shell-model and ab initio calculations are judged [22]. Open questions include how large residual quenching is in heavy nuclei, how operator renormalisation should be matched to the Hamiltonian truncation, and how those lessons propagate into $0\nu\beta\beta$ matrix-element uncertainties that limit neutrino-mass sensitivity claims [50]. Treating Hamiltonian correlations and electroweak operators inconsistently remains a common source of theoretical bias.

A concrete way to see operator renormalisation is to compare a bare GT matrix element in a small valence space with a published calculation that includes two-body currents. The fractional change is often comparable to the historical quenching factor, which is why modern papers discourage applying an extra ad hoc g_A^{eff} on top of already-renormalised operators.

The same consistency requirement appears in calculations of magnetic moments far from Schmidt lines.

Student directions. (1) Data plot: draw Schmidt lines for odd protons and overlay five measured moments. (2) Estimate: extract a single phenomenological quenching factor g_A^{eff}/g_A from one mid-shell GT transition and discuss its limitations. (3) Essay: explain why operator renormalisation and Hamiltonian correlations must be treated consistently for $0\nu\beta\beta$ matrix elements.

14.10 Problems with solutions

Problem 14.1 — Schmidt value for $d_{5/2}$ proton. Compute the Schmidt moment for an odd proton in $j = \ell + \frac{1}{2} = \frac{5}{2}$.

Solution 14.1. For $j = \ell + \frac{1}{2}$ and free proton g-factors,

$$\frac{\mu}{\mu_N} = j\, g_\ell + \tfrac{1}{2}(g_s - g_\ell) = \tfrac{5}{2} \cdot 1 + \tfrac{1}{2}(5.586 - 1) = 2.5 + 2.293 = 4.793.$$

Equivalently, $\mu/\mu_N = j + 2.293$ [18, 6].

Problem 14.2 — Neutron $p_{1/2}$ Schmidt moment. For an odd neutron with $j = \ell - \frac{1}{2} = \frac{1}{2}$ ($\ell = 1$), compute the Schmidt moment in nuclear magnetons.

Solution 14.2. For an odd neutron with $j = \ell - \frac{1}{2}$,

$$\frac{\mu}{\mu_N} = +1.913\, \frac{j}{j+1} = 1.913 \times \frac{1/2}{3/2} = \frac{1.913}{3} \approx 0.638.$$

Equivalently, $\mu/\mu_N = -g_s^{(n)} j/(2(j+1))$ with $g_s^{(n)} = -3.826$ yields the same result [6, 18].

Problem 14.3 — Neutron $j = \ell + \frac{1}{2}$. Show that any odd neutron with $j = \ell + \frac{1}{2}$ has Schmidt moment $\mu = -1.913\, \mu_N$, independent of j.

Solution 14.3. With $g_\ell^{(n)} = 0$ and $g_s^{(n)} = -3.826$, the $j = \ell + \frac{1}{2}$ formula collapses to

$$\frac{\mu}{\mu_N} = \tfrac{1}{2} g_s = -1.913,$$

for every such orbit. Only the spin moment contributes [6, 1].

Problem 14.4 — Coefficients a and b. For $j = \ell - \frac{1}{2}$ and $m_j = j$, derive $|a|^2 = 2\ell/(2\ell+1)$ and $|b|^2 = 1/(2\ell+1)$ from $J_+|j,j\rangle = 0$.

Solution 14.4. Write $|j,j\rangle = a|1\rangle + b|2\rangle$ with $|1\rangle = |\ell, -\frac{1}{2}\rangle$ and $|2\rangle = |\ell-1, +\frac{1}{2}\rangle$. Acting with $J_+ = L_+ + S_+$ and using $L_+|\ell,m\rangle = \hbar\sqrt{\ell(\ell+1) - m(m+1)}\,|\ell, m+1\rangle$ gives the condition $a + b\sqrt{2\ell} = 0$. Together with $|a|^2 + |b|^2 = 1$ one obtains

$$|a|^2 = \frac{2\ell}{2\ell+1}, \qquad |b|^2 = \frac{1}{2\ell+1},$$

as in the main text [6].

Problem 14.5 — Even–even magnetic moment. Why is the ground-state magnetic moment of an even–even nucleus zero in the extreme single-particle (paired) picture?

Solution 14.5. All nucleons are paired to total $I = 0$. With $\boldsymbol{\mu}$ a vector operator, $\langle I = 0|\mu_z|I = 0\rangle = 0$. Experimentally even–even ground states have vanishing moments within measurement precision [6, 1].

Problem 14.6 — Reading a Schmidt diagram. An odd-proton ground state has measured μ close to the $j = \ell + \frac{1}{2}$ Schmidt line and far from $j = \ell - \frac{1}{2}$. What assignment does this support, and what would an intermediate μ suggest?

Solution 14.6. Closeness to one Schmidt line supports a nearly pure single-j valence proton with that ℓ, j. Intermediate values indicate configuration mixing, core polarisation, and/or quenched effective g_s [18, 6, 22].

Chapter 15

Shell-Model Excited States and Collective Motion

"Experimentally, it is observed that nuclei can have several excited states... The nuclear shell model is instrumental in explaining these excited states."
lecture notes, PH5001

Ground-state I^π and Schmidt moments test the filling of the mean field. Excited states test whether the same orbitals, with particle promotions and pair breaking, organise the low-lying spectrum. Near closed shells the extreme shell model works well (classic example: ^{17}O). In mid-shell even–even nuclei the first 2^+ state is collective, and vibrational or rotational models are required [6, 1, 14].

15.1 Ground state of ^{17}O

Oxygen-17 has $Z = 8$, $N = 9$. Protons fill the closed shell through $1p_{1/2}$ (magic $Z = 8$) and contribute $I = 0$. Neutrons fill

$$(1s_{1/2})^2\,(1p_{3/2})^4\,(1p_{1/2})^2\,(1d_{5/2})^1. \tag{15.1}$$

The unpaired neutron is in $1d_{5/2}$ ($\ell = 2$, $j = \frac{5}{2}$), so

$$I^\pi = \tfrac{5}{2}^+, \tag{15.2}$$

matching experiment. The closed proton shell and nearly closed neutron p shell make ^{17}O a textbook single-particle nucleus: the core is ^{16}O and the valence neutron is weakly coupled to it.

15.2 Single-particle excitations of ^{17}O

Promoting the valence neutron to a higher empty orbit changes I^π without breaking the ^{16}O core:

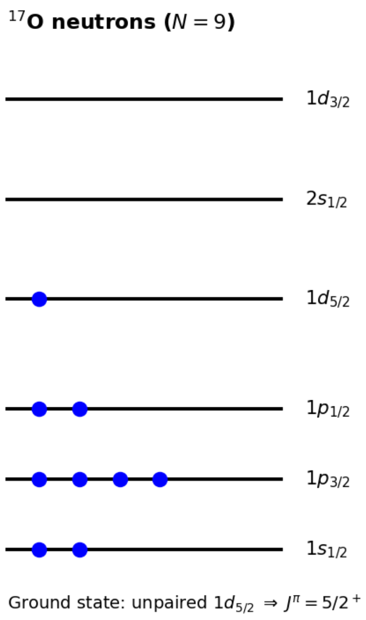


Figure 15.1: Neutron filling for ^{17}O. The ground state is fixed by the unpaired $1d_{5/2}$ neutron.

Configuration (valence n)	I^π	Character
$1d_{5/2}$	$\frac{5}{2}^+$	ground state
$2s_{1/2}$	$\frac{1}{2}^+$	first s-wave excitation
$1d_{3/2}$	$\frac{3}{2}^+$	spin–orbit partner of $d_{5/2}$

These positive-parity states appear in the low-lying spectrum and carry large single-particle strength in one-neutron transfer. Evaluated levels place the $1/2^+$ state at 871 keV and show additional positive- and negative-parity excitations consistent with sd and p-shell rearrangements [20, 21, 6].

15.3 Particle–hole excitations in ^{17}O

Higher states involve exciting a nucleon from a filled orbit, leaving a hole.

Hole in $1p_{3/2}$. Promote one neutron from $1p_{3/2}$ into $1d_{5/2}$. After the transition one has three neutrons in $1p_{3/2}$ (odd occupancy) and two in $1d_{5/2}$. The two $d_{5/2}$ particles can couple to $J_d = 0, 2, 4$, and this pair angular momentum must then couple to the $p_{3/2}$ hole:

$$I = |J_d - \tfrac{3}{2}|, \ldots, J_d + \tfrac{3}{2}, \qquad \pi = -. \tag{15.3}$$

Only the $J_d = 0$ component gives a pure $I^\pi = 3/2^-$ state; the full particle–hole configuration generates a negative-parity multiplet.

Hole in $1p_{1/2}$. Promote from $1p_{1/2}$ into $1d_{5/2}$: one unpaired neutron remains in $1p_{1/2}$. Coupling the $d^2_{5/2}$ pair $J_d = 0, 2, 4$ to the $p_{1/2}$ hole gives $I = |J_d - \frac{1}{2}|, \ldots, J_d + \frac{1}{2}$, all with negative parity. The $J_d = 0$ component gives $I^\pi = 1/2^-$, but it is not the only member.

Multi-particle couplings. When three unpaired neutrons sit in different orbits (e.g. $1p_{1/2}$, $1d_{5/2}$, $2s_{1/2}$), their angular momenta couple to several total I. Parity of a multi-particle configuration is the product of single-particle parities,

$$\pi = \prod_i (-1)^{\ell_i}. \tag{15.4}$$

A state such as $\frac{5}{2}^-$ can arise from coupling three unequal j values with overall negative parity. The extreme model still provides the language of orbits and parities even when residual interactions fix the detailed multiplet ordering.

15.4 Even–even example: ^{38}Ar

Argon-38 has $Z = 18$, $N = 20$. Neutrons fill the magic $N = 20$ shell; protons sit two holes below $Z = 20$. The ground state is even–even, so

$$I^\pi = 0^+. \tag{15.5}$$

Because $N = 20$ is closed, low-lying excitations often rearrange protons.

Proton $1d_{3/2} \to 1f_{7/2}$. In a restricted two-active-proton picture, paired spectator holes are assumed to carry $J = 0$. The remaining $d_{3/2}$ and $f_{7/2}$ angular momenta then run from $|7/2 - 3/2| = 2$ to $7/2 + 3/2 = 5$:

$$I = 2, 3, 4, 5, \qquad \pi = (-1)^2(-1)^3 = -. \tag{15.6}$$

States $2^-, 3^-, 4^-, 5^-$ are therefore allowed basis states. Residual interactions mix these with other cross-shell configurations and set their ordering; the triangle rule alone does not identify any evaluated level [6, 20, 21].

Proton $2s_{1/2} \to 1d_{3/2}$. With all other active particles paired to $J = 0$, the $s_{1/2}^{-1} d_{3/2}$ coupling allows $I^\pi = 1^+, 2^+$. This is a basis multiplet, not a prediction of its energy. Evaluated data instead provide the benchmark: ^{38}Ar has a first 2^+ level at 2167.6 keV, while a second 0^+ level occurs at 3377.5 keV and another 2^+ at about 3937 keV [20, 21]. Their wave functions require configuration mixing; none should be assigned from counting alone.

15.5 Systematics of the lowest 2^+ in even–even nuclei

In the great majority of even–even nuclei the lowest excited state is 2^+. Important exceptions occur near doubly closed shells, where a 0^+ or 3^- state may intervene; globally, however, a low-lying 2^+ cannot be explained by single-particle counting alone: the same pattern appears for hundreds of nuclei over a wide range of A (Fig. 15.2).

Empirical systematics of the lowest $E(2^+)$ show three regimes [6, 1, 14]:

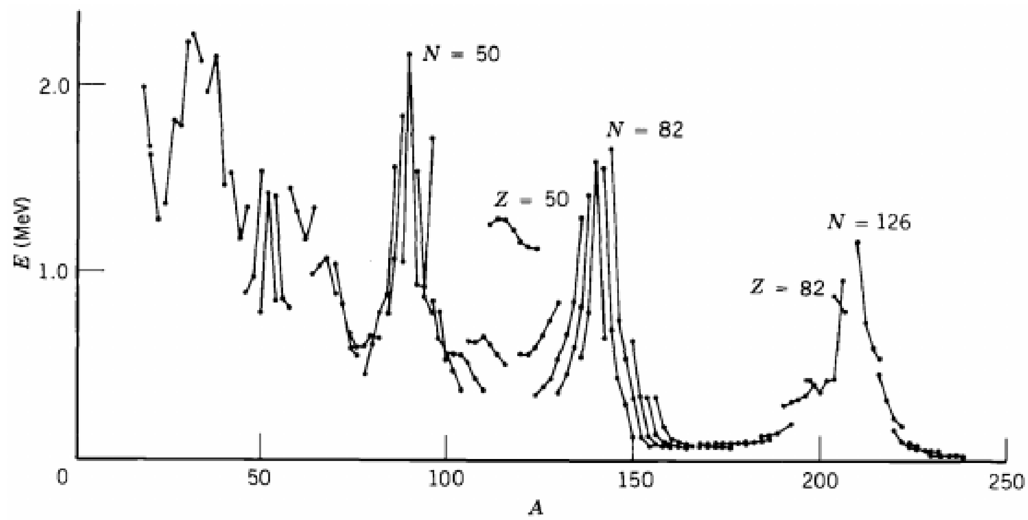


Figure 15.2: Energies of the lowest 2^+ states of even-Z, even-N nuclei versus A. Peaks occur at shell closures; mid-shell regions show much lower $E(2^+)$ ([6]).

1. **Magic and near-magic nuclei:** $E(2^+)$ is large (several MeV), reflecting rigid spherical cores.
2. **Spherical and weakly collective regions:** typical $E(2^+)$ values are of order 1 MeV. An approximately harmonic quadrupole vibrator has $E(4^+)/E(2^+) \approx 2$, while transitional nuclei need not obey this ideal ratio.
3. **Deformed regions $A \sim$ 150–190 and $A \gtrsim 230$:** $E(2^+)$ drops to tens of keV, with $E(4^+)/E(2^+) \approx 10/3 \approx 3.33$, the rigid-rotor ratio.

Peaks of $E(2^+)$ at magic N or Z reconnect these collective trends to shell structure: closed shells stiffen the nucleus against deformation.

Key idea

Odd-A neighbours of closed shells: single-particle and particle–hole excitations. Even–even mid-shell nuclei: collective 2^+ (vibration or rotation). The shell model alone is incomplete for the latter.

15.6 Vibrational model

For spherical even–even nuclei ($A \lesssim 150$) the equilibrium shape is spherical. Small excess energy can excite volume-conserving surface oscillations. Just as several valence pairs outside a closed shell can cooperate in microscopic quadrupole vibrations, many valence pairs far from magic numbers can permanently distort the core into a deformed equilibrium shape (next section): vibration and rotation are the soft and hard ends of that same cooperative tendency [6, 1].

The nuclear radius is expanded in spherical harmonics,

$$R(\theta, \phi, t) = \bar{R}\left[1 + \sum_{\lambda=0}^{\infty} \sum_{\mu=-\lambda}^{\lambda} \alpha_{\lambda\mu}(t)\, Y_{\lambda\mu}(\theta, \phi)\right], \tag{15.7}$$

with multipolarity λ and deformation amplitudes $\alpha_{\lambda\mu}(t)$. Because R is real,

$$\alpha_{\lambda,-\mu} = (-1)^\mu \alpha^*_{\lambda\mu}. \tag{15.8}$$

Thus the $2\lambda+1$ complex-looking coefficients contain only $2\lambda+1$ independent real coordinates; quadrupole motion has five, not ten, real degrees of freedom. To derive the volume constraint, write the volume enclosed by the surface as

$$\mathcal{V} = \frac{1}{3}\int R^3(\theta,\phi,t)\,\mathrm{d}\Omega. \tag{15.9}$$

For small amplitudes, $(1+\epsilon)^3 = 1+3\epsilon+\mathcal{O}(\epsilon^2)$, so Eq. (15.7) gives

$$\begin{aligned}\mathcal{V} &= \frac{\bar{R}^3}{3}\int\left[1+3\sum_{\lambda\mu}\alpha_{\lambda\mu}Y_{\lambda\mu}\right]\mathrm{d}\Omega + \mathcal{O}(\alpha^2)\\ &= \frac{4\pi\bar{R}^3}{3}\left(1+\frac{3\alpha_{00}}{\sqrt{4\pi}}\right)+\mathcal{O}(\alpha^2),\end{aligned} \tag{15.10}$$

where $\int Y_{\lambda\mu}\mathrm{d}\Omega = \sqrt{4\pi}\,\delta_{\lambda 0}\delta_{\mu 0}$. Keeping $\mathcal{V} = 4\pi\bar{R}^3/3$ therefore requires

$$\alpha_{00} = 0 \qquad \text{to first order.} \tag{15.11}$$

At second order a small monopole correction compensates the volume change from $|\alpha_{\lambda\mu}|^2$. The intrinsic parity of a one-phonon excitation of multipolarity λ is $(-1)^\lambda$, because $Y_{\lambda\mu}(-\hat{r}) = (-1)^\lambda Y_{\lambda\mu}(\hat{r})$.

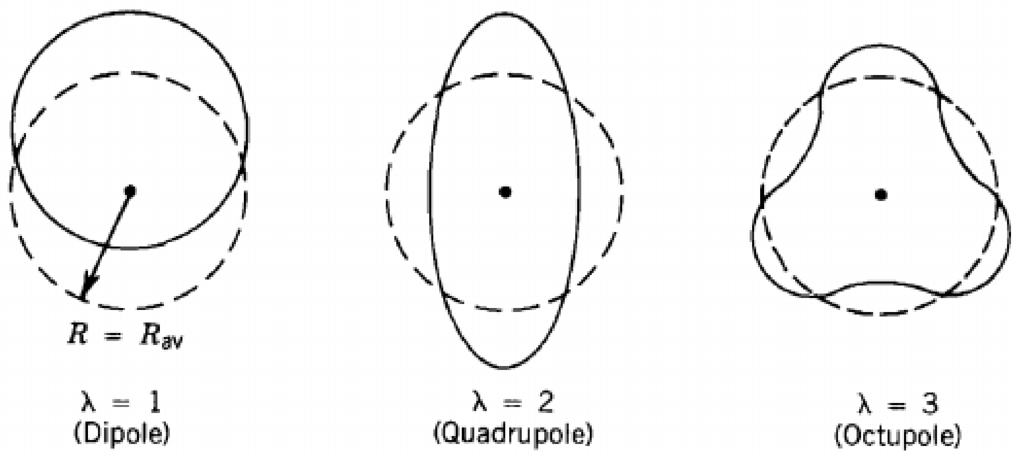


Figure 15.3: Lowest vibrational modes of a nucleus: dipole ($\lambda = 1$), quadrupole ($\lambda = 2$), octupole ($\lambda = 3$) ([6]).

Breathing mode ($\lambda = 0$). With $Y_{00} = 1/\sqrt{4\pi}$,

$$R(\theta,\phi,t) = \bar{R}\left(1+\frac{\alpha_{00}(t)}{\sqrt{4\pi}}\right). \tag{15.12}$$

The nucleus expands and contracts uniformly without changing shape. It is *not* an incompressible surface vibration: strict volume conservation sets the independent first-order α_{00} to zero. The physical isoscalar giant-monopole (breathing) resonance is a compressional density mode, lies at high excitation energy, and is outside the low-lying incompressible shape spectrum considered here.

Dipole mode ($\lambda = 1$). A pure $\lambda = 1$ surface deformation is equivalent to a displacement of the nuclear centre of mass. To see this explicitly, translate a sphere by a small vector $\boldsymbol{d}$. Along direction $\hat{r}$, its new surface is

$$R'(\hat{r}) = \bar{R} + \boldsymbol{d} \cdot \hat{r} + \mathcal{O}(d^2/\bar{R}). \tag{15.13}$$

The angular function $\boldsymbol{d} \cdot \hat{r}$ is a linear combination of the three $Y_{1\mu}$. Thus a $\lambda = 1$ deformation merely moves the whole sphere and can be removed by choosing the centre-of-mass origin. It is not an intrinsic excitation. Low-lying shape phonons begin at quadrupole order. (Giant dipole resonances are a different, high-energy isovector phenomenon in which protons and neutrons oscillate against each other.)

Quadrupole mode ($\lambda = 2$). The nucleus oscillates between prolate and oblate ellipsoidal shapes. This mode produces the dominant low-lying vibrational spectrum of spherical even–even nuclei.

Octupole mode ($\lambda = 3$). More complex pear-shaped surface oscillations; the one-phonon state is 3^-.

15.6.1 Phonons and the vibrational spectrum

After imposing the reality condition, each of the $2\lambda + 1$ independent real normal coordinates behaves as a harmonic oscillator. For one such coordinate the Hamiltonian has the form

$$H_{\lambda\mu} = \frac{\pi_{\lambda\mu}^2}{2B_\lambda} + \frac{1}{2} C_\lambda \alpha_{\lambda\mu}^2, \qquad \omega_\lambda = \sqrt{\frac{C_\lambda}{B_\lambda}}, \tag{15.14}$$

where B_λ is the collective inertia and C_λ the restoring-force constant. Canonical quantisation gives

$$E_{n_\lambda} = \left(n_\lambda + \tfrac{1}{2}\right) \hbar\omega_\lambda. \tag{15.15}$$

For a degenerate multipole there are $2\lambda + 1$ coordinates, so its vacuum zero-point energy is $(2\lambda + 1)\hbar\omega_\lambda/2$; for quadrupole motion it is $5\hbar\omega_2/2$, not $\hbar\omega_2/2$. Only excitation energies are relevant here, so that common offset is subtracted and n_λ phonons cost $n_\lambda \hbar\omega_\lambda$. These quanta are called **phonons**. The lowest vibrational excitation is one quadrupole phonon:

$$E = \hbar\omega_2, \qquad I^\pi = 2^+, \tag{15.16}$$

because the phonon carries angular momentum $\lambda = 2$ and parity $(-1)^\lambda = +$. For even–even nuclei the ground state is 0^+ and this 2^+ is the first excited state of the vibrational model. In real spherical nuclei the one-phonon energy is typically $\hbar\omega_2 \sim 0.5$–1 MeV; anharmonicity and mixing with two-quasiparticle excitations keep the two-phonon multiplet from being perfectly degenerate, but the $0^+, 2^+, 4^+$ clustering near $2\hbar\omega_2$ remains a standard spectroscopic fingerprint [1, 6].

Two quadrupole phonons have energy $2\hbar\omega_2$. Coupling two identical $I = 2$ bosons allows only the totally symmetric states $I = 0, 2, 4$ (the odd values $1, 3$ are forbidden for identical phonons), all with positive parity: the two-phonon triplet $0^+, 2^+, 4^+$.

Three quadrupole phonons give energy $3\hbar\omega_2$ and allowed $I^\pi = 0^+, 2^+, 3^+, 4^+, 6^+$. One octupole phonon gives

$$E = \hbar\omega_3, \qquad I^\pi = 3^-. \tag{15.17}$$

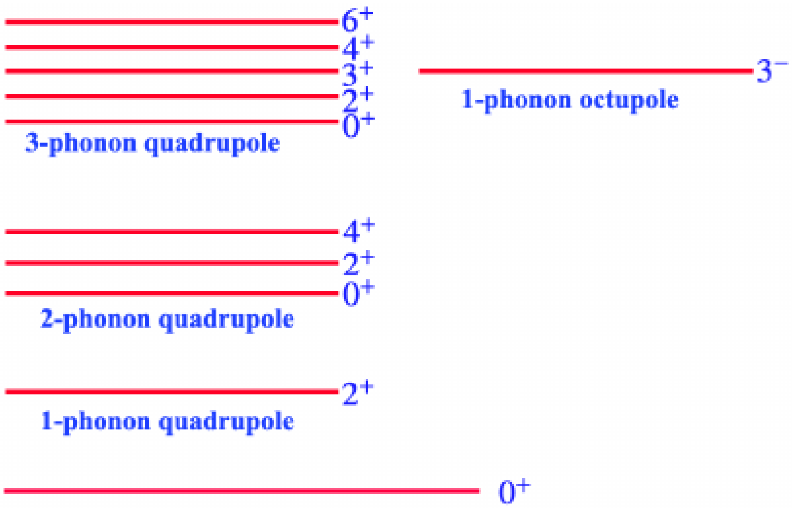


Figure 15.4: Schematic vibrational spectrum of an even–even nucleus: one-, two- and three-phonon quadrupole multiplets and the one-phonon octupole 3^-.

In the harmonic limit the two-phonon 4^+ and the one-phonon 2^+ satisfy

$$\frac{E(4^+)}{E(2^+)} = \frac{2\hbar\omega_2}{\hbar\omega_2} = 2, \tag{15.18}$$

matching the empirical ratio for many nuclei with $A \sim 30$–150. The equality is an ideal harmonic limit, not an exact universal rule: anharmonicities split the two-phonon multiplet and shift the ratio away from 2. For an incompressible liquid drop with surface tension σ and density ρ, the classical surface-mode frequencies scale as

$$\omega_\lambda = \sqrt{\frac{\sigma(\lambda-1)(\lambda+2)\lambda}{\rho\,\bar{R}^3}}, \tag{15.19}$$

so higher multipoles are stiffer and lie higher in energy [6, 1, 14].

15.7 Rotational model

Nuclei with large static quadrupole moments (notably $A \sim 150$–190 and $A \gtrsim 230$) are permanently deformed (Fig. 15.5). For an axially symmetric rotor the intrinsic deformation may be parametrised by a quadrupole parameter β_2 through

$$R(\theta) = \bar{R}(1 + \beta_2 Y_{20}(\theta) + \cdots). \tag{15.20}$$

Rotation about an axis perpendicular to the symmetry axis produces a quantized rotational band.

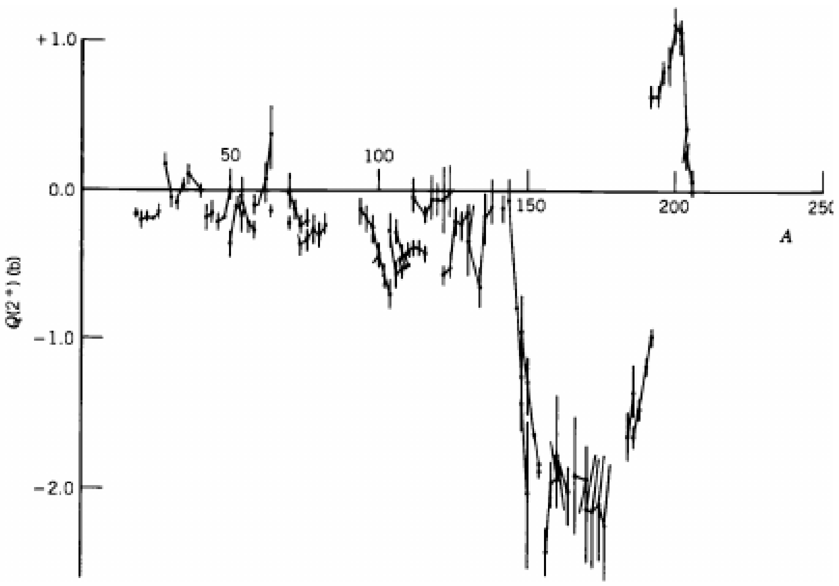


Figure 15.5: Electric quadrupole moments of the lowest 2^+ states of even–even nuclei. Large $|Q|$ in deformed regions signals permanent deformation ([6]).

Classically $E = \frac{1}{2}\mathcal{I}\omega^2$ and $L = \mathcal{I}\omega$, so $E = L^2/(2\mathcal{I})$. In quantum mechanics the squared angular momentum of the rotor is replaced by $I(I+1)\hbar^2$, so

$$E_I = \frac{\hbar^2}{2\mathcal{I}} I(I+1). \tag{15.21}$$

For an axially symmetric intrinsic state, K is the projection of $\boldsymbol{I}$ on the symmetry axis. The sequence below assumes an even–even, reflection-symmetric ground configuration with $K = 0$, a single rigid moment of inertia, and invariance under a rotation by π about an axis perpendicular to the symmetry axis. Under these assumptions only even I with positive parity occur:

$$I^\pi = 0^+, 2^+, 4^+, 6^+, \ldots \tag{15.22}$$

(odd-I members are excluded). The first few energies are therefore

$$E(0^+) = 0, \tag{15.23}$$

$$E(2^+) = \frac{\hbar^2}{2\mathcal{I}}\, 2 \cdot 3 = \frac{6\hbar^2}{2\mathcal{I}}, \tag{15.24}$$

$$E(4^+) = \frac{\hbar^2}{2\mathcal{I}}\, 4 \cdot 5 = \frac{20\hbar^2}{2\mathcal{I}}, \tag{15.25}$$

and their ratio is

$$\frac{E(4^+)}{E(2^+)} = \frac{20}{6} = \frac{10}{3} \approx 3.33, \tag{15.26}$$

the classic rotor ratio. Rotational 2^+ energies are typically of order 10–100 keV, much lower than vibrational ~ 1 MeV levels [6, 14, 1].

The same deformation that produces the rotational band also generates a large intrinsic quadrupole moment Q_0. For a uniformly charged, slightly deformed nucleus,

$$Q_0 \simeq \frac{3}{\sqrt{5\pi}} Ze\bar{R}^2\beta_2 \qquad \text{(to first order in } \beta_2\text{)}. \tag{15.27}$$

The laboratory spectroscopic quadrupole moment of a band member is

$$Q_s(I,K) = \frac{3K^2 - I(I+1)}{(I+1)(2I+3)}\, Q_0. \tag{15.28}$$

For the $K = 0$ ground-state band this is negative when $Q_0 > 0$: a prolate intrinsic shape does not imply a positive laboratory Q_s. The reduced transition probability is

$$B(E2; IK \to I-2, K) = \frac{5}{16\pi} Q_0^2 \left|\langle IK\, 2\, 0 | I-2, K\rangle\right|^2. \tag{15.29}$$

Thus Q_s is linear in Q_0, whereas $B(E2)$ is quadratic in Q_0. Enhanced $B(E2)$ values (many Weisskopf units) are a hallmark of collective rotation, in contrast to single-particle estimates near closed shells [6, 1].

15.8 Limitations of the extreme shell model

The extreme single-particle picture fails when:

- several valence nucleons occupy an open shell (need configuration mixing and residual two-body matrix elements);
- the nucleus is soft or deformed far from magic numbers (collective rotational/vibrational degrees of freedom);
- continuum and cluster structures appear near drip lines.

Modern shell-model calculations keep the mean-field orbitals but diagonalise a residual interaction in a truncated valence space. Collective models build on the same single-particle structure: magic numbers still mark where sphericity is preferred. A practical diagnostic is the energy ratio $E(4^+)/E(2^+)$: values near 2 favour vibration, near 10/3 favour rotation, and intermediate values signal transitional nuclei. Large $B(E2)$ enhancements relative to Weisskopf units reinforce a collective assignment [6, 1, 14].

Key idea

Near closed shells, excited states are particle promotions and particle–hole excitations on top of a magic core. ^{17}O is the cleanest example. Mid-shell even–even spectra demand collective vibration ($E(4^+)/E(2^+) \approx 2$) or rotation ($E(4^+)/E(2^+) \approx 3.33$).

15.8.1 Single-particle, vibrational, and rotational spectra

A single atlas of spectra contrasts regimes: ^{17}O shows single-particle promotions on a magic core; mid-shell ^{106}Pd shows vibrational phonon spacing; actinide ^{242}Pu shows a long rotational sequence $0^+, 2^+, \ldots$. Moments of inertia approach rigid-body values only far from magic numbers, confirming collective rather than single-particle rotation [14, 6, 1].

15.9 Deeper physics: $E(2^+)$ systematics and collective ratios

15.9.1 Three regimes of the first 2^+ state

The lowest 2^+ excitation in even–even nuclei (Section 15.5) separates three physical regimes. Near doubly magic nuclei, $E(2^+) \sim 1$–$4\,\mathrm{MeV}$: the core is spherical and particle–hole excitations cost shell-gap energy. In transitional nuclei, $E(2^+) \sim 0.5$–$1\,\mathrm{MeV}$: quadrupole vibrations of a nearly spherical surface (Section 15.6) become competitive. In well-deformed rotors ($A \sim 150$–190, actinides), $E(2^+) \sim 10$–$100\,\mathrm{keV}$: the entire nucleus rotates collectively (Section 15.7).

15.9.2 Vibration vs rotation: the energy ratio test

A harmonic quadrupole vibrator has equally spaced phonon levels, so

$$\frac{E(4^+)}{E(2^+)} \approx 2 \qquad \text{(vibrator)}, \tag{15.30}$$

while a rigid asymmetric rotor has

$$\frac{E(4^+)}{E(2^+)} = \frac{4(4+1) - 2(2+1)}{2(2+1) - 0(0+1)} = \frac{10}{3} \approx 3.33 \qquad \text{(rigid rotor)}. \tag{15.31}$$

These ratios are the quickest experimental discriminator between surface vibration and permanent deformation [6, 1].

^{152}Sm: a textbook rotor

^{152}Sm has $E(2^+) = 121.8\,\mathrm{keV}$ and $E(4^+) = 366.5\,\mathrm{keV}$, giving $E(4^+)/E(2^+) \approx 3.01$, close to $10/3$. The same nucleus in the samarium chain sits in the deformed rare-earth region where $B(E2; 0^+ \to 2^+)$ values are large (many collective nucleons). By contrast, ^{114}Cd near the $Z = 50$ shell shows $E(4^+)/E(2^+) \approx 2.0$–$2.4$, consistent with vibrational motion of a spherical core.

Remark

Shell-model incompleteness. The extreme single-particle model (Lecture 13) cannot produce hundreds of keV 2^+ states in mid-shell even–even nuclei: promoting one nucleon across a shell gap costs $\sim \hbar\omega \sim 10\,\mathrm{MeV}$. Collectivity requires coherent motion of many nucleons. Peaks of $E(2^+)$ at magic N or Z (Figure 15.2) reconnect collective physics to the shell closures of Lecture 12.

15.9.3 $B(E2)$ as the dynamical partner of $E(2^+)$

The energy $E(2^+)$ locates the band head; the reduced transition probability $B(E2; 0^+ \to 2^+)$ measures collective quadrupole strength. Vibrators have moderate $B(E2)$; well-deformed rotors reach many Weisskopf units. A nucleus can have low $E(2^+)$ but small $B(E2)$ if the state is a two-phonon excitation rather than a rotor band head. Using energy ratios (15.30)–(15.31) together with $B(E2)$ disambiguates models that ratios alone leave degenerate [6, 14].

15.9.4 Phonons and the five quadrupole coordinates

The expansion (15.7) contains five independent real quadrupole amplitudes ($\lambda = 2$). Quantising these as harmonic oscillators gives a phonon spectrum with (15.30); rotors orient deformation in space and produce (15.31). The shell model of Lectures 12–14 supplies single-particle energies but not these collective coordinates, which is why vibration and rotation enter as complementary pictures [1, 6].

15.9.5 Critical-point analogy for shape transitions

The drop of $E(2^+)$ toward mid-shell even–even nuclei resembles a softening mode as the nucleus approaches a deformation instability. The ratio $E(4^+)/E(2^+)$ sliding from ~ 2 (vibrator) toward 10/3 (rotor) is an empirical order parameter for that transition. Microscopic theories (beyond-mean-field DFT, Monte Carlo shell model) aim to predict *where* on the chart the transition occurs without fitting $E(2^+)$ by hand. The shell closures of Lecture 12 mark the boundaries where the order parameter returns to large values (stiff spherical cores) [14, 45, 6].

Collective $B(E2)$ values can exceed Weisskopf units by orders of magnitude in rotors, while seniority-dominated near-spherical nuclei stay closer to single-particle estimates (Lecture 20). Shape coexistence in regions such as Ni or Hg isotopes shows low-lying 0^+ states whose charge radii and monopole strengths reveal deformed intruders. Use more than one observable before assigning a geometric label.

Ratio plus lifetime

Never assign a rotor on R_{42} alone; demand a lifetime or $B(E2)$. Conversely, a large $B(E2)$ with $R_{42} \sim 2.2$ may signal a transitional or coexistence regime rather than a textbook vibrator.

Physics-depth close

Collective models earn their keep when they predict ratios and trends, not when they decorate a single level energy with a geometric adjective. Demand $B(E2)$, monopole data, or radii before accepting a shape assignment.

15.10 Further physics: spectroscopic factors

One-nucleon transfer reactions such as (d, p) or (p, d) measure how much of a pure single-particle orbital is present in a nuclear state. The spectroscopic factor S quantifies that overlap: $S = 1$

for an ideal single-particle state on an inert closed core, while $S < 1$ signals fragmentation of the single-particle strength among several eigenstates [6, 1].

Near doubly magic cores, low-lying odd-A states often carry large spectroscopic factors, consistent with the extreme shell model. As one moves away from closed shells, residual interactions spread the strength and S decreases. Sum rules constrain the total strength available in a given orbit: the summed spectroscopic factors for adding or removing a nucleon from a closed shell cannot exceed the orbit's degeneracy. Fragmentation of spectroscopic strength is therefore direct experimental evidence for the limitations of the extreme independent-particle picture and for the need for configuration mixing or collective degrees of freedom.

15.10.1 Energy ratios as collective diagnostic

Collective models predict level spacing ratios that depend on the symmetry of the motion. For a harmonic quadrupole vibrator (Equation (15.30)),

$$E(0^+) = 0, \quad E(2^+) = \hbar\omega, \quad E(4^+) = 2\hbar\omega, \quad E(6^+) = 3\hbar\omega, \tag{15.32}$$

so $E(4^+)/E(2^+) = 2$ and $E(6^+)/E(2^+) = 3$. A rigid rotor (Equation (15.31)) gives $E(4^+)/E(2^+) = 10/3$ and $E(6^+)/E(2^+) = 7/3 \approx 2.33$. Transitional nuclei often fall between these limits, signalling γ-soft or shape-coexistence physics rather than pure vibration or rotation [6, 14].

15.10.2 Worked ratio: ^{152}Sm vs ^{114}Cd

^{152}Sm: $E(2^+) = 121.8\,\text{keV}$, $E(4^+) = 366.5\,\text{keV}$, ratio $= 3.01 \approx 10/3$ (rotor). ^{114}Cd: $E(2^+) \approx 558\,\text{keV}$, $E(4^+) \approx 1158\,\text{keV}$, ratio ≈ 2.08 (vibrator). The two nuclei sit on opposite sides of the $A \sim 150$ deformed region: Sm is mid-shell rare earth; Cd is near the $Z = 50$ shell closure where $E(2^+)$ rises (Figure 15.2) and collectivity weakens [6, 20].

15.10.3 Spectroscopic factors and the independent-particle limit

For (d,p) on ^{16}O populating the ^{17}O $d_{5/2}$ state, spectroscopic factors $S \approx 0.8$–1.0 confirm the extreme shell-model assignment (Lecture 13). In mid-shell nuclei, $S \sim 0.3$–0.5 is common: the same residual interactions that lower $E(2^+)$ fragment single-particle strength. Sum rules guarantee that total strength for a given orbit is conserved, but it spreads across many eigenstates [6, 1].

Single-particle vs collective $E(2^+)$ scale

Promoting one nucleon across a shell gap costs $\Delta\varepsilon \sim \hbar\omega \sim 10\,\text{MeV}$ (Lecture 12). Observed $E(2^+) \sim 100\,\text{keV}$ in deformed nuclei therefore requires $\sim 10^2$ nucleons contributing coherently to the quadrupole operator, not one valence nucleon. That 10^4 scale separation is the central reason collectivity lies outside the shell model of Lectures 12–14 [14].

15.10.4 Bridge to shell closures

Peaks in $E(2^+)$ at magic N or Z (Figure 15.2) mark where the shell model regains validity: closed shells resist deformation, so the first 2^+ is a particle–hole excitation, not a collective band head. The vibration/rotation ratio test is therefore most informative *between* magic numbers, where the filling diagram of Lecture 12 predicts mid-shell softness [1, 6].

15.10.5 Sum rule for spectroscopic strength

For a given single-particle orbit α, the sum of spectroscopic factors over all final states populated in one-nucleon transfer cannot exceed the degeneracy of α. When residual interactions spread strength, individual S values fall but the sum is preserved (up to reaction mechanism factors). Mid-shell nuclei with fragmented strength and low $E(2^+)$ therefore tell a consistent story: the independent-particle picture fails for *both* static moments (Lecture 14) and transfer strength at the same time [6, 1].

15.11 Deeper reading: $E(2^+)$, $B(E2)$, and shape coexistence

Vibrational nuclei have $E(4^+)/E(2^+) \simeq 2$; rigid rotors approach 10/3. Real nuclei interpolate, and some exhibit shape coexistence: low-lying deformed and spherical configurations compete. Monopole transition strengths and lifetime patterns distinguish scenarios that a single energy ratio cannot.

Large γ-ray arrays map these systematics across the chart, especially with radioactive beams. Microscopic theories aim to predict the onset of collectivity without fitted geometric parameters, connecting back to shell evolution (Lecture 12) and forward to isomer spectroscopy (Lecture 20). Plot $E(2^+)$ versus N for an isotopic chain as a standard student exercise: shell closures appear as peaks, deformation onsets as valleys.

Phase transitions in finite nuclei

The spherical–deformed transition can be discussed with control parameters such as valence nucleon number. Order parameters include β_2 deformation and $B(E2; 0^+ \to 2^+)$. Finite-system "phase transitions" are smoothed, yet energy-ratio trajectories still reveal rapid structural change. Pair that discussion with charge-radius kinks from Lecture 3.

Triaxiality

Rigid triaxial and soft γ-unstable models can yield similar low spin spectra. Higher spin levels, odd–even staggering of γ bands, and Coulomb excitation relative phases help discriminate. Do not assign triaxiality from a single $E(2^+)$ value.

Energy-ratio trajectory

For an isotopic chain (e.g. even–even Sm or Zr), plot $R_{42} = E(4^+)/E(2^+)$ versus N. Mark the vibrational locus near 2 and the rotational locus near 10/3. Identify where the chain crosses from one regime toward the other and whether a shape coexistence region is claimed in the literature. Add $B(E2)$ values if available: energies alone can mislead.

Connect to Lecture 3 by checking whether charge radii kink at the same neutron number where R_{42} drops. A simultaneous radius and collectivity change is strong evidence for deformation onset. Connect to Lecture 20 by asking how multipolarity assignments and lifetimes confirmed the 2^+ and 4^+ spin sequence used in the ratio.

Collectivity is organised by energy ratios, transition strengths, and radii together. No single observable deserves a geometric label on its own.

In practice, rework one numerical estimate from the lecture with a different parameter set and compare the shift to the quoted experimental uncertainty. If the shift exceeds the uncertainty, the estimate is teaching a scale, not delivering a fit. That distinction keeps theoretical cartoons honest when they sit beside evaluated data tables and modern many-body calculations. Along the course arc, write one sentence linking this chapter backward and one forward: the discipline of those two sentences is as important as any single formula. Examination problems rarely ask for a literature review, but they do ask for the scale argument that literature reviews presuppose.

Synthesis problem

Write a half-page note that (i) restates the chapter's central scale or selection rule in one equation, (ii) evaluates it numerically for one concrete nucleus or energy, and (iii) names the neighbouring lecture that uses the result. Keep the note free of research-frontier speculation; the goal is fluency with the lecture physics itself. If an assumption fails (Born approximation, spherical mean field, allowed transition, one-dimensional barrier), state the failure mode in a final sentence. That habit converts passive reading into examination-ready understanding and makes the later research-frontier box read as an extension rather than a replacement.

Research frontier: shapes, collectivity, and student projects

Vibrational and rotational models of Lecture 15 are the entry point to modern collective physics: shape coexistence, triaxiality, and quantum phase transitions between spherical, deformed, and γ-soft nuclei [45]. Large γ-ray tracking arrays and radioactive beams map $E(2^+)$, $B(E2)$, and monopole transition strengths across the chart, revealing where geometric models succeed and where microscopic configuration mixing is mandatory [53, 54]. Microscopic generators (beyond-mean-field density-functional theory, Monte Carlo shell model, and ab initio methods) aim to predict where collectivity onsets without fitted geometric parameters. Uncertainties that remain active research topics include the soft-γ versus rigid-triaxial distinction in transitional nuclei, the role of pairing in shape coexistence, and the quantitative accuracy of predicted $B(E2)$ values far from stability. The lecture's geometric language remains useful precisely when students learn its domain of validity.

For transitional nuclei, competing interpretations (shape coexistence versus a soft γ degree of freedom) can fit the same low-lying spectrum yet disagree on monopole strengths and lifetime patterns. Student projects that gather $E(0^+)$, $\rho^2(E0)$, and $B(E2)$ together learn how modern spectroscopy discriminates among collective scenarios.

Collecting $E(0^+)$, $\rho^2(E0)$, and lifetimes together is the modern way to test coexistence scenarios beyond a single $E(2^+)$ ratio.
Student directions. (1) Data project: from NuDat, plot $E(2^+)$ versus A for even-even Cd or Sm isotopes [21]. (2) Analytic check: compute the rigid-rotor ratio $E(4^+)/E(2^+)$ for one deformed nucleus and compare with 10/3. (3) Case study: identify one textbook case of shape coexistence and list the key experimental signatures.

15.12 Problems with solutions

Problem 15.1 — First excited state of ^{17}O. In the extreme single-particle model, what is I^π if the valence neutron of ^{17}O is promoted from $1d_{5/2}$ to $2s_{1/2}$? Compare with the evaluated excitation energy.

Solution 15.1. $2s_{1/2}$ has $\ell = 0$, $j = \frac{1}{2}$, so $I^\pi = \frac{1}{2}^+$. ENSDF places this state at 871 keV, matching the observed first excited state [20, 21, 6].

Problem 15.2 — ^{38}Ar multiplet. A proton is excited from $1d_{3/2}$ to $1f_{7/2}$ in ^{38}Ar. Assuming all spectator particles and holes couple to $J = 0$, list the allowed I^π values for the two active proton angular momenta. Explain why this does not determine the observed level ordering.

Solution 15.2. $j_1 = \frac{3}{2}$, $j_2 = \frac{7}{2}$ give

$$|j_1 - j_2| \le I \le j_1 + j_2 \quad \Rightarrow \quad I = 2, 3, 4, 5.$$

Parity $(-1)^{2+3} = -$, so $2^-, 3^-, 4^-, 5^-$ are allowed. Residual interactions and mixing with other cross-shell configurations determine their energies and strengths, so the triangle rule alone cannot assign an evaluated ^{38}Ar level [6, 20, 21].

Problem 15.3 — Rotor vs vibrator ratio. Compute $E(4^+)/E(2^+)$ for a pure axial rotor with $E(I) \propto I(I+1)$. Compare with the ideal harmonic vibrator value.

Solution 15.3. For the rotor,

$$E(2) \propto 2 \cdot 3 = 6, \qquad E(4) \propto 4 \cdot 5 = 20,$$

so the ratio is $20/6 = 10/3 \approx 3.33$. For a harmonic quadrupole vibrator, $E(4^+) = 2\hbar\omega_2$ while $E(2^+) = \hbar\omega_2$, so the ratio is 2 [6, 14].

Problem 15.4 — Why $\lambda = 1$ is not a shape phonon. Explain why a pure $\lambda = 1$ term in the surface expansion does not generate an intrinsic low-lying vibrational excitation.

Solution 15.4. A $\lambda = 1$ deformation displaces the nuclear surface in a way that is equivalent to a shift of the centre of mass. Transforming to the CM frame removes the amplitude, so there is no intrinsic shape excitation. Low-lying surface phonons therefore begin at quadrupole order ($\lambda = 2$) [6, 1].

Problem 15.5 — Extracting a rotor moment from $B(E2)$. For a $K = 0$ band,

$$B(E2; 2^+ \to 0^+) = \frac{Q_0^2}{16\pi}.$$

If $B(E2; 2^+ \to 0^+) = 0.50\, e^2\mathrm{b}^2$, find Q_0. Then give the spectroscopic moment $Q_s(2, 0) = -2Q_0/7$.

Solution 15.5.

$$Q_0 = \sqrt{16\pi B(E2)} = \sqrt{8\pi}\, e\mathrm{b} \approx 5.01\, e\mathrm{b}, \qquad Q_s(2, 0) = -\frac{2}{7}Q_0 \approx -1.43\, e\mathrm{b}.$$

The negative laboratory Q_s is consistent with a positive intrinsic Q_0 for a prolate $K = 0$ rotor [6].

Problem 15.6 — Rotor prediction from one level. An even–even $K = 0$ band has $E(2^+) = 80\,\mathrm{keV}$. In the rigid-rotor limit, predict $E(4^+)$ and $E(6^+)$.

Solution 15.6. Since $E(I) = AI(I + 1)$, $A = 80/6\,\mathrm{keV}$. Therefore

$$E(4^+) = \frac{20}{6}(80) = 267\,\mathrm{keV}, \qquad E(6^+) = \frac{42}{6}(80) = 560\,\mathrm{keV}.$$

Agreement with these ratios and enhanced in-band $B(E2)$ values are both needed before identifying a rotational band [6, 14].

Problem 15.7 — Two-phonon vibrational multiplet. In the harmonic quadrupole vibrator, what I^π values appear at $2\hbar\omega_2$? Why are 1^+ and 3^+ absent?

Solution 15.7. Two identical quadrupole phonons couple symmetrically to $I^\pi = 0^+, 2^+, 4^+$ only. Odd spins $(1^+, 3^+)$ are antisymmetrised under boson exchange and are forbidden for identical phonons [1, 6].

Chapter 16

Radioactivity and Alpha Decay

"A factor of 2 in energy means a factor of 10^{24} *in half-life! The theoretical explanation of this Geiger–Nuttall rule in 1928 was one of the first triumphs of quantum mechanics."*
K. S. Krane

The semi-empirical mass formula and the β-stability parabolas (earlier chapters) predict which nuclei are energetically favoured against nucleon rearrangement. Experiment organises itself differently: on the chart of nuclides, every known species is plotted by neutron number N and proton number Z, and the dominant decay mode in each region is read directly from the map [6, 1, 14, 2]. Heavy nuclei predominantly emit α particles; nuclei far from the valley of β stability adjust N/Z by β decay. This chapter develops the phenomenology of radioactivity and α decay: the chart, the decay Q-value, two-body kinematics, the exponential decay law, the Geiger–Nuttall systematics, and the potential-barrier picture that Gamow used to explain the extreme sensitivity of the half-life to Q.

16.1 Chart of nuclides and where α decay lives

Figure 16.1 shows the chart of nuclides: each point is a known nuclide, with Z on the horizontal axis and N on the vertical axis. Stable species form a narrow band called the *valley of stability*. Away from that band, characteristic decay modes dominate: β^- emission on the neutron-rich side, β^+ emission and/or electron capture on the proton-rich side, and α decay in the upper-right corner where both Z and N are large. The map is not decorative: once Q_α opens and competes with β, the Geiger–Nuttall sensitivity of the α lifetime often decides which channel is observed first [6, 1].

Why heavy nuclei emit α particles. The red-shaded region in Fig. 16.1 corresponds to very heavy nuclei, typically $Z \gtrsim 82$ (lead, bismuth, and the actinides). An α particle is a ^{4}He nucleus ($Z = 2$, $N = 2$). Emitting it reduces both Z and N by 2, moving the system diagonally down and to the left on the chart toward more stable configurations [6, 1].

Three physical reasons combine to favour α emission in this region:

(i) **Coulomb repulsion.** The SEMF Coulomb term grows as $Z(Z-1)/A^{1/3}$. For large Z, the electrostatic energy of packing many protons into a finite volume makes the parent less bound than lighter fragments would be.

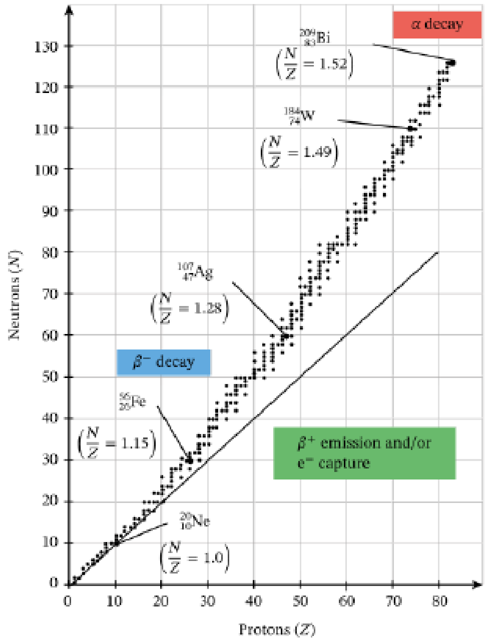


Figure 16.1: Chart of nuclides showing regions of dominant decay mode: α decay, β^- decay, and β^+/electron capture.

(ii) **Declining B/A.** Beyond the iron peak, the average binding energy per nucleon decreases for very heavy nuclei, so releasing a tightly bound cluster can lower the total rest mass.

(iii) **The anomalously large binding of ^{4}He.** Helium-4 is doubly magic ($Z = N = 2$) with experimental binding energy $B_\alpha \approx 28.3\,\text{MeV}$ (about 7 MeV per nucleon). That large B_α enters the Q-value with a positive sign and makes α emission energetically preferred over most other light clusters [6, 1, 14].

Stable versus long-lived nuclei. Around $A \sim 200$ one finds the heaviest stable nuclides; ^{208}Pb is the heaviest stable nucleus. ^{209}Bi, long treated as stable, is now known to undergo extremely slow α decay. Beyond bismuth no stable species exist; heavy nuclides decay by α and/or β chains toward lead. Many nuclides with $150 < A < 210$ are not strictly stable but are *long-lived*: their α half-lives are long on geological scales; some exceed the age of the Earth, while the primordial uranium isotopes remain because a non-negligible fraction survives (^{238}U, ^{235}U, and ^{232}Th). The chart therefore distinguishes three categories: stable nuclei in the valley, long-lived radioactive nuclei with human-scale or geological half-lives, and short-lived nuclei studied in the laboratory.

Key idea

On the chart of nuclides, α decay occupies the upper-right corner ($Z \gtrsim 82$). Each emission step moves $(A, Z) \to (A - 4, Z - 2)$, equivalently $(N, Z) \to (N - 2, Z - 2)$. ^{208}Pb is the heaviest stable nucleus; the neighboring ^{209}Bi is radioactive but extraordinarily long-lived.

16.2 Particle emission versus fission

When a heavy nucleus breaks apart, the process is classified by the size of the fragments:

- **Particle emission:** one fragment is much smaller than the other (proton, neutron, deuteron, α, or a slightly heavier cluster).
- **Fission:** the nucleus splits into two fragments of comparable mass (treated later in the course).

Among light ejectiles, α emission is usually dominant because $B_\alpha \approx 28.3\,\mathrm{MeV}$ is exceptionally large. For ^{235}U, emission of a neutron, proton, or ^{3}He has $Q < 0$, while α emission has $Q \approx +4.7\,\mathrm{MeV}$ [1]. Heavier clusters such as ^{12}C or ^{14}C can be emitted in rare instances, but the thicker Coulomb barrier and small preformation probability suppress those channels by many orders of magnitude relative to α decay [6].

16.3 The α decay scheme and Q-value

Alpha decay is the emission of a tightly bound ^{4}He cluster from a heavy parent. The large binding of the α ($B_\alpha \approx 28.3\,\mathrm{MeV}$) makes the channel energetically open for many nuclei with $A \gtrsim 150$, yet lifetimes span many decades because of Coulomb-barrier tunneling (Lecture 17). Spectroscopy of α lines still maps low-lying daughter states with keV precision [6, 1].

The basic process is

$$ {}^{A}_{Z}X \;\rightarrow\; {}^{A-4}_{Z-2}Y + {}^{4}_{2}\mathrm{He} + Q. \tag{16.1}$$

The decay is spontaneous only if $Q > 0$: the total rest mass of the products is less than that of the parent, and the excess appears as kinetic energy of the daughter and the α particle.

16.3.1 Q from nuclear masses

Care is required with atomic versus nuclear masses: evaluated tables quote atomic masses, and the α entry is a neutral helium atom, not a bare $^4\mathrm{He}^{2+}$ ion. Electron bookkeeping cancels for α decay when consistent atomic entries are used, but using a nuclear α mass against atomic parents introduces a spurious shift of order $2m_ec^2$ [6, 27].

Let m_N denote nuclear masses (masses of the bare nuclei, without electrons). Then

$$ Q = \Big[m_\mathrm{N}({}^{A}_{Z}X) - m_\mathrm{N}({}^{A-4}_{Z-2}Y) - m_\mathrm{N}({}^{4}\mathrm{He})\Big]c^2. \tag{16.2}$$

A positive Q means the parent nucleus is heavier than the sum of the daughter and α masses.

16.3.2 Q from atomic masses: cancellation of electrons

In the laboratory one uses atomic masses m_atom, which include Z electrons for a neutral atom. Write the relation between atomic and nuclear masses as

$$ m_\mathrm{atom}({}^{A}_{Z}X) = m_\mathrm{N}({}^{A}_{Z}X) + Zm_e - \frac{B_e(Z)}{c^2}, \tag{16.3}$$

where m_e is the electron mass and $B_e(Z)$ is the total electronic binding energy of the neutral atom (a few keV per electron, negligible on the MeV scale of nuclear decays).

For the three bodies in Eq. (16.1):

$$m_{\text{atom}}({}^{A}_{Z}X) = m_{\text{N}}({}^{A}_{Z}X) + Zm_e - \frac{B_e(Z)}{c^2}, \tag{16.4}$$

$$m_{\text{atom}}({}^{A-4}_{Z-2}Y) = m_{\text{N}}({}^{A-4}_{Z-2}Y) + (Z-2)m_e - \frac{B_e(Z-2)}{c^2}, \tag{16.5}$$

$$m_{\text{atom}}({}^{4}\text{He}) = m_{\text{N}}({}^{4}\text{He}) + 2m_e - \frac{B_e(2)}{c^2}. \tag{16.6}$$

Subtract Eq. (16.5) and Eq. (16.6) from Eq. (16.4):

$$\begin{aligned} m_{\text{atom}}({}^{A}_{Z}X) &- m_{\text{atom}}({}^{A-4}_{Z-2}Y) - m_{\text{atom}}({}^{4}\text{He}) \\ &= \Big[m_{\text{N}}({}^{A}_{Z}X) - m_{\text{N}}({}^{A-4}_{Z-2}Y) - m_{\text{N}}({}^{4}\text{He})\Big] \\ &\quad + \underbrace{\big[Zm_e - (Z-2)m_e - 2m_e\big]}_{=0} \\ &\quad - \frac{B_e(Z) - B_e(Z-2) - B_e(2)}{c^2}. \end{aligned} \tag{16.7}$$

The electron-mass term vanishes identically because the parent atom has Z electrons, the daughter has $Z-2$, and the helium atom has 2, so $Z = (Z-2)+2$. The remaining electronic binding-energy difference is of order 10 keV and is neglected in nuclear energetics [1, 6]. Therefore

$$Q \approx \Big[m_{\text{atom}}({}^{A}_{Z}X) - m_{\text{atom}}({}^{A-4}_{Z-2}Y) - m_{\text{atom}}({}^{4}\text{He})\Big]c^2. \tag{16.8}$$

Caution

In Eq. (16.8), $m_{\text{atom}}({}^{4}\text{He})$ is the mass of a *neutral helium atom*, not the mass of the bare α particle. Using the wrong mass tabulation entry is a common source of error of about $2m_ec^2$, with smaller electronic-binding corrections.

16.3.3 Q from binding energies

The binding energy of a nucleus is defined by $m_{\text{N}}c^2 = Zm_pc^2 + Nm_nc^2 - B$. For the decay (16.1), the mass difference can be rewritten entirely in terms of binding energies:

$$\begin{aligned} Q &= \Big[m_{\text{N}}(X) - m_{\text{N}}(Y) - m_{\text{N}}(\alpha)\Big]c^2 \\ &= \bigg[\Big(Zm_p + Nm_n - \frac{B_X}{c^2}\Big) - \Big((Z-2)m_p + (N-2)m_n - \frac{B_Y}{c^2}\Big) - \Big(2m_p + 2m_n - \frac{B_\alpha}{c^2}\Big)\bigg]c^2. \end{aligned} \tag{16.9}$$

The nucleon mass terms cancel because $A = (A-4)+4$ and $Z = (Z-2)+2$:

$$Q = B_Y + B_\alpha - B_X. \tag{16.10}$$

Thus α decay is exothermic when the combined binding of the daughter plus the α exceeds the binding of the parent.

Numerical value of B_α

Evaluated data give $B_\alpha = B(^4\text{He}) \approx 28.296\,\text{MeV}$ [20, 21, 1]. This value should be taken from experiment: the SEMF significantly underbinds the doubly magic ^{4}He and must not be used for B_α.

16.3.4 SEMF estimates and the $A \sim 150$ puzzle

Parent and daughter binding energies can be estimated with the semi-empirical mass formula (pairing term δ often omitted for a first trend):

$$B = a_v A - a_s A^{2/3} - a_c \frac{Z(Z-1)}{A^{1/3}} - a_{\text{sym}} \frac{(N-Z)^2}{A} + \delta. \tag{16.11}$$

Along the β-stable line for fixed A, the atomic number is approximately

$$Z \approx \frac{A/2}{1 + 0.0078\,A^{2/3}}. \tag{16.12}$$

Insert Eqs. (16.11) and (16.12) into Eq. (16.10) with the experimental $B_\alpha = 28.3\,\text{MeV}$. At fixed A, the parent has (A, Z) and the daughter $(A-4, Z-2)$, so

$$\begin{aligned} Q_\alpha^{(\text{SEMF})} &= B(A-4, Z-2) + B_\alpha - B(A, Z) \\ &= a_v\big[(A-4) - A\big] - a_s\big[(A-4)^{2/3} - A^{2/3}\big] \\ &\quad - a_c\left[\frac{(Z-2)(Z-3)}{(A-4)^{1/3}} - \frac{Z(Z-1)}{A^{1/3}}\right] \\ &\quad - a_{\text{sym}}\left[\frac{(N'-Z')^2}{A-4} - \frac{(N-Z)^2}{A}\right] + B_\alpha, \end{aligned} \tag{16.13}$$

with $N = A - Z$ and $N' = A - 4 - (Z-2) = N - 2$. Expanding for large A with Z on the β-stable line (16.12), the volume piece alone contributes $-4a_v$ while the Coulomb release and the fixed B_α win for heavy A. Numerically this estimate crosses $Q_\alpha > 0$ near $A \gtrsim 145$–150.

Yet the chart shows many stable or long-lived nuclides through the lead–bismuth region: ^{208}Pb is the heaviest stable nucleus, whereas ^{209}Bi is an extremely long-lived α emitter. Beyond bismuth one finds no stable species. Two complementary explanations resolve the apparent contradiction:

(i) **Shell and pairing effects.** The SEMF is a smooth average. Individual nuclides can have Q_α shifted by several MeV by shell closures (notably $N = 126$) and odd-even pairing, so some isotopes with SEMF $Q > 0$ are actually sub-threshold or have very small Q [6, 1].

(ii) **$Q > 0$ is necessary but not sufficient.** Even when $Q_\alpha > 0$, the half-life depends exponentially on Q through Coulomb-barrier tunneling (Lecture 17). For $Q \lesssim 4\,\text{MeV}$, lifetimes can exceed $10^{17}\,\text{s}$ and the decay is effectively unobservable in ordinary laboratory times [1, 6]. A nucleus can therefore have positive α decay energy yet appear stable on human timescales.

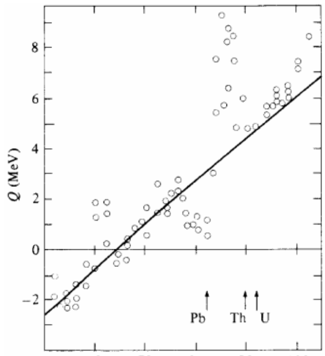


Figure 16.2: Experimental α-decay Q-values (circles) versus proton number Z, compared with the SEMF estimate (solid line). Multiple circles at the same Z correspond to different neutron numbers. Adapted from Cottingham and Greenwood [23].

Figure 16.2 shows that Q becomes positive for heavy Z near the actinides, with scatter at fixed Z from different N. Adding neutrons to a heavy isotope generally lowers Q_α and lengthens the half-life.

16.4 Kinematics: sharing of Q between α and daughter

The Q-value is the total kinetic energy released. If the parent nucleus is initially at rest, momentum conservation requires the daughter and α to recoil in opposite directions with equal magnitude of linear momentum.

Step 1: Momentum conservation.

$$\boldsymbol{p}_d + \boldsymbol{p}_\alpha = 0 \quad \Rightarrow \quad |\boldsymbol{p}_d| = |\boldsymbol{p}_\alpha| = p. \tag{16.14}$$

Step 2: Kinetic energies in terms of p.

$$K_d = \frac{p^2}{2M_d}, \qquad K_\alpha = \frac{p^2}{2M_\alpha}, \tag{16.15}$$

where M_d and M_α are the nuclear masses of the daughter and the α particle.

Step 3: Energy conservation.

$$K_d + K_\alpha = Q. \tag{16.16}$$

Step 4: Eliminate p. Substitute the kinetic-energy expressions:

$$\frac{p^2}{2M_d} + \frac{p^2}{2M_\alpha} = Q. \tag{16.17}$$

Factor out $p^2/2$:

$$\frac{p^2}{2}\left(\frac{1}{M_d}+\frac{1}{M_\alpha}\right)=Q. \tag{16.18}$$

Combine the fractions:

$$\frac{1}{M_d}+\frac{1}{M_\alpha}=\frac{M_d+M_\alpha}{M_d M_\alpha}. \tag{16.19}$$

Hence

$$p^2=\frac{2QM_dM_\alpha}{M_d+M_\alpha}. \tag{16.20}$$

Step 5: Solve for K_α and K_d.

$$K_\alpha=\frac{p^2}{2M_\alpha}=\frac{2QM_dM_\alpha}{M_d+M_\alpha}\cdot\frac{1}{2M_\alpha}=Q\,\frac{M_d}{M_d+M_\alpha}. \tag{16.21}$$

Similarly,

$$K_d=Q\,\frac{M_\alpha}{M_d+M_\alpha}. \tag{16.22}$$

Step 6: Approximate for heavy nuclei. For $A\gg 4$, $M_d\approx(A-4)u$ and $M_\alpha\approx 4u$, so $M_d+M_\alpha\approx Au$. Therefore

$$K_\alpha\approx Q\left(1-\frac{4}{A}\right),\qquad K_d\approx Q\,\frac{4}{A}. \tag{16.23}$$

For $A=200$, $K_\alpha/Q=1-4/200=0.98$: about 98% of the energy is carried by the α particle and about 2% by the recoiling daughter [6, 1]. Detectors measure K_α; from it one reconstructs $Q=K_\alpha(M_d+M_\alpha)/M_d$.

Remark

The recoiling daughter ($\sim 100\,\mathrm{keV}$ for $Q\sim 5\,\mathrm{MeV}$) has enough energy to break chemical bonds, so α sources must be thin coatings if one wishes to avoid recoil losses from the active layer [6].

16.5 Exponential decay law, activity, mean life, and half-life

Radioactive decay is a Poisson process: for a single nucleus that has not yet decayed at time t, the probability to decay in the next interval $\mathrm{d}t$ is $\lambda\,\mathrm{d}t$, where λ is the decay constant (probability per unit time).

Step 1: Differential equation for $N(t)$. For a large sample, the rate of change of the number of undecayed nuclei is proportional to the number present:

$$\frac{\mathrm{d}N}{\mathrm{d}t}=-\lambda N(t). \tag{16.24}$$

Step 2: Solve the equation. Separate variables and integrate from $t=0$ (when $N=N_0$):

$$\int_{N_0}^{N(t)}\frac{\mathrm{d}N}{N}=-\lambda\int_0^t\mathrm{d}t \quad\Rightarrow\quad \ln\frac{N(t)}{N_0}=-\lambda t \quad\Rightarrow\quad N(t)=N_0\,e^{-\lambda t}. \tag{16.25}$$

Step 3: Mean life. The mean (average) lifetime of a nucleus in the sample is

$$\tau = \int_0^\infty t\,\lambda e^{-\lambda t}\,\mathrm{d}t = \frac{1}{\lambda}. \tag{16.26}$$

(One integration by parts gives $\tau = 1/\lambda$.)

Step 4: Half-life. The half-life $t_{1/2}$ is defined by $N(t_{1/2}) = N_0/2$:

$$\frac{N_0}{2} = N_0\,e^{-\lambda t_{1/2}} \quad \Rightarrow \quad e^{-\lambda t_{1/2}} = \frac{1}{2} \quad \Rightarrow \quad \lambda t_{1/2} = \ln 2. \tag{16.27}$$

Therefore

$$t_{1/2} = \frac{\ln 2}{\lambda} = \frac{0.693}{\lambda}, \qquad \tau = \frac{t_{1/2}}{\ln 2} \approx 1.443\,t_{1/2}. \tag{16.28}$$

Step 5: Activity. The activity (decay rate) is

$$\mathcal{A}(t) = -\frac{\mathrm{d}N}{\mathrm{d}t} = \lambda N(t) = \lambda N_0\,e^{-\lambda t}. \tag{16.29}$$

Activity also falls by a factor of two in one half-life. In SI units, $\mathcal{A}$ is measured in becquerels (Bq = decays per second); the curie ($1\,\mathrm{Ci} = 3.7 \times 10^{10}\,\mathrm{Bq}$) remains common in nuclear medicine [13, 6].

Key idea

Two laboratory observables characterise α decay: Q (from K_α and kinematics) and $t_{1/2}$ (from the time dependence of activity). They are related by the Geiger–Nuttall rule, explained by Gamow tunneling in Lecture 17.

16.6 Geiger–Nuttall rule

In 1911, Geiger and Nuttall discovered empirically that α emitters with large disintegration energy have short half-lives, and conversely [6, 2]. For known α emitters, Q spans roughly 4–9.5 MeV while $t_{1/2}$ spans from microseconds to $\sim 10^9$ years. The empirical relation is

$$\log_{10}(t_{1/2}/\mathrm{s}) = a\,Q^{-1/2} + b, \tag{16.30}$$

where a and b depend primarily on Z. Different isotopic chains (e.g. thorium, uranium) form parallel lines on the plot.

Even–even versus odd-A nuclei. When all α emitters are plotted together, there is scatter about the average trend. Restricting to fixed Z and even–even (Z, N) yields smooth curves. Odd-A and odd–odd nuclei follow the same inverse trend but lie *above* the even–even lines: their half-lives are longer by factors of 2 to 10^3 at comparable Q [6]. The prominent example is ^{235}U (even Z, odd N): its long half-life ($t_{1/2} \approx 7 \times 10^8$ y) allows natural uranium to persist and thermal-neutron fission reactors to operate.

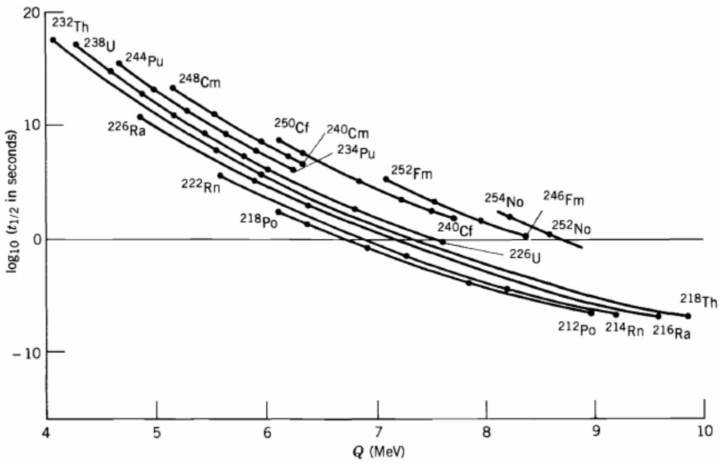


Figure 16.3: Geiger–Nuttall systematics: $\log_{10}(t_{1/2}/\mathrm{s})$ versus α-decay energy Q (MeV). Even–even nuclei form smooth curves for fixed Z [24, 6].

Orders of magnitude. ^{232}Th ($Q \approx 4.08\,\mathrm{MeV}$, $t_{1/2} \approx 1.4 \times 10^{10}\,\mathrm{y}$) and ^{218}Th ($Q \approx 9.85\,\mathrm{MeV}$, $t_{1/2} \approx 10^{-7}\,\mathrm{s}$) illustrate that a factor of about 2 in Q corresponds to roughly 10^{24} in $t_{1/2}$ [1, 6]. This extreme sensitivity is the central puzzle solved by Gamow's tunneling theory.

16.7 Toward Gamow's theory: preformation and the potential

Historical note: Gamow, Gurney, and Condon (1928)

George Gamow (1904–1968), together with Gurney and Condon, explained Geiger–Nuttall systematics by quantum tunneling of a preformed α through the Coulomb barrier [25, 26, 6, 14, 2]. The success of the one-body model does not prove that α clusters exist permanently inside heavy nuclei; it shows that decay rates behave *as if* an α were preformed at the nuclear surface.

In the Gamow model one assumes that two protons and two neutrons are already correlated as an α cluster inside the parent nucleus. The cluster moves in a potential energy $V(r)$ due to the residual nucleus, where r is the distance between their centres:

- **Nuclear well ($r < R$).** The strong force dominates. The potential is approximated by a square well of depth $V_0 \sim 30$–$40\,\mathrm{MeV}$. The Coulomb repulsion is small inside the nuclear volume and is neglected in this first approximation.

- **Coulomb barrier ($r > R$).** The nuclear force has short range; outside the surface the repulsive Coulomb potential remains:

$$V_{\mathrm{C}}(r) = \frac{(Z-2)\cdot 2e^2}{4\pi\varepsilon_0\, r} = \frac{Z_1 Z_2 e^2}{4\pi\varepsilon_0\, r}, \tag{16.31}$$

  with $Z_1 = Z - 2$ (charge of the daughter) and $Z_2 = 2$ (charge of the α).

For $\ell = 0$ the centrifugal term vanishes; $\ell \neq 0$ adds $\hbar^2\ell(\ell+1)/(2\mu r^2)$ to the barrier (Lecture 17).

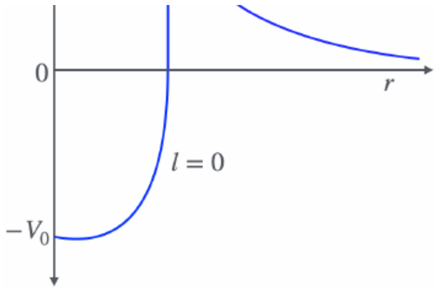


Figure 16.4: Nuclear square well joined to the Coulomb barrier $V_C(r)$ for $\ell = 0$. The horizontal line marks the available energy $E = Q$. The classically forbidden region lies between the inner radius R and the outer turning point b where $V_C(b) = Q$.

Barrier height: numerical example. The Coulomb barrier height is the value of V_C at the effective contact distance. For $Z = 72$ and $A \approx 216$, the daughter radius is

$$R_d \approx 1.25\, A^{1/3}\,\text{fm} \approx 1.25 \times 216^{1/3}\,\text{fm} \approx 1.25 \times 6\,\text{fm} \approx 7.5\,\text{fm}. \tag{16.32}$$

Including the α-particle radius $R_\alpha \approx 1.5\,\text{fm}$, the centre-to-centre distance at contact is

$$R \approx R_d + R_\alpha \approx 9\,\text{fm}. \tag{16.33}$$

At this separation,

$$V_C \approx \frac{2 \times (Z-2) \times e^2/(4\pi\varepsilon_0)}{R} \approx \frac{2 \times 70 \times 1.44\,\text{MeV fm}}{9\,\text{fm}} \approx 22.4\,\text{MeV}. \tag{16.34}$$

Typical α-decay Q values are only 4–9 MeV, so $Q \ll V_C(R)$: classically the α cannot escape. Quantum mechanics permits tunneling through the barrier between R and b; the calculation of the penetration probability is the subject of Lecture 17.

Takeaway

α decay in heavy nuclei is favoured by large B_α and Coulomb energy considerations ($Q > 0$ from roughly $A \sim 150$ onward in the SEMF). Observed stability through ^{208}Pb, and the extraordinary longevity of ^{209}Bi, require tunneling: $Q > 0$ is necessary but not sufficient. The Geiger–Nuttall correlation and the potential-barrier picture set the stage for the Gamow factor derived in the next chapter.

16.7.1 Why heavy nuclei prefer α emission

Alpha emission is favoured among heavy spontaneous ejecta because the α is an exceptionally tightly bound cluster: the mass reduction for emitting ^{4}He often exceeds that for emitting a free nucleon or other light fragment. Coulomb disruption grows as Z^2 while volume binding grows only as A, so α radioactivity concentrates in the upper right of the chart [6, 1, 13].

16.8 Deeper physics: Q_α and Geiger–Nuttall slopes

16.8.1 Evaluating Q_α from masses

The α-decay energy release is fixed by mass conservation. For ${}^{A}_{Z}X \to {}^{A-4}_{Z-2}Y + \alpha$,

$$Q_\alpha = [M(X) - M(Y) - M({}^4\mathrm{He})]c^2, \tag{16.35}$$

using atomic masses so that the two emitted electrons of the neutral daughter cancel with the parent atomic electrons [6, 1]. The same Q appears as the α kinetic energy in the centre-of-mass frame when recoil is neglected ($A \gg 4$). Shell and pairing corrections appear as local deviations from a smooth SEMF trend; precision work requires evaluated tables rather than the liquid-drop estimate alone [20, 21].

Numerical Q_α for ^{212}Po

Using neutral-atom masses from evaluated data, $M({}^{212}\mathrm{Po}) - M({}^{208}\mathrm{Pb}) - M({}^4\mathrm{He})$ gives $Q_\alpha \approx 8.95\,\mathrm{MeV}$ and $t_{1/2} \approx 3 \times 10^{-7}\,\mathrm{s}$ [6, 20]. Compare with ^{238}U: $Q_\alpha \approx 4.27\,\mathrm{MeV}$ and $t_{1/2} \approx 4.5 \times 10^9\,\mathrm{y}$. The higher-$Q$ isotope is shorter lived by roughly twenty-four orders of magnitude, the central puzzle of the Geiger–Nuttall plot [24, 2].

16.8.2 Reading the Geiger–Nuttall slope

Empirically, even–even α emitters of fixed Z obey (16.30). The slope a is positive: each increment in $Q^{-1/2}$ shortens the half-life. A rough Gamow estimate (Lecture 17) shows that the tunneling exponent scales as $\sqrt{\mu/Q}$ through the Coulomb turning point $b \propto 1/Q$, so $\log t_{1/2}$ varies nearly linearly with $Q^{-1/2}$ over the observed 4–9 MeV window [25, 6]. The intercept b depends primarily on Z because the barrier height grows with daughter charge. Different isotopic chains therefore form parallel lines on a $\log t_{1/2}$ versus Q plot.

The physical content is extreme sensitivity: the forbidden region width $b - R$ shrinks only as $Q^{-1/2}$ while the penetration probability falls exponentially in that width. A factor of two in Q therefore changes $t_{1/2}$ by many orders of magnitude, which is why uranium persists in nature while polonium isotopes with $Q \sim 9\,\mathrm{MeV}$ decay in microseconds [1, 2]. The SEMF predicts a smooth $Q_\alpha(A)$ trend; shell closures near $N = 126$ and $Z = 82$ produce kinks that appear as vertical shifts on the Geiger–Nuttall plot for fixed Z [6, 20].

Remark

The Geiger–Nuttall rule is a *correlation*, not a substitute for microscopic calculation. Odd-A and odd–odd parents typically lie above the even–even curves (longer lifetimes at the same Q) because angular momentum and hindrance reduce the preformation overlap. Treating all nuclei with one line overpredicts decay rates for ^{235}U by orders of magnitude [6].

Key idea

Two observables, one exponential law. Laboratory α spectroscopy measures Q from the kinetic energy K_α; activity measurements give $t_{1/2}$. Their extreme correlation across twenty-plus decades of lifetime is explained only when quantum tunneling through the Coulomb barrier is included. Lecture 17 derives the Gamow factor that turns this empirical slope into a calculable penetration probability [14, 25].

16.8.3 Assumptions and limits of the Q_α analysis

Equation (16.35) assumes the decay proceeds to the ground state of the daughter and that recoil corrections are negligible. Decay to excited daughter states reduces the α energy by the excitation energy and splits the Geiger–Nuttall systematics into fine-structure branches [6]. Atomic-electron rearrangement after recoil contributes at the eV level and is ignored in nuclear Q tabulations [1]. For very heavy nuclei, Q_α alone does not determine the dominant decay mode: spontaneous fission competes when the fissility parameter Z^2/A exceeds the critical value of Lecture 23 [6, 56]. The Geiger–Nuttall plot remains the first diagnostic because it compresses energetics and lifetime into one figure before any microscopic model is invoked.

16.9 Further physics from the literature

Even–even systematics and hindrance. Geiger–Nuttall plots restricted to fixed Z and even–even (Z, N) form smooth curves, while odd-A and odd–odd nuclei lie systematically above those curves (longer lifetimes at the same Q) [6]. The deviation is quantified by the *hindrance factor* $F = \lambda_{\text{theory}}/\lambda_{\text{exp}}$: values near unity indicate a normal Gamow transition; $F \gg 1$ signals poor parent-daughter overlap or selection-rule suppression.

Neutron number and shell effects. Adding neutrons to a heavy isotope generally lowers Q_α and lengthens $t_{1/2}$. A pronounced discontinuity near $N = 126$ appears in $Q(A)$ plots because the daughter from α emission often sits near that closed neutron shell [6, 1]. The SEMF captures the average trend but not these local kinks; evaluated masses from [20, 21] are required for quantitative work.

Competition with other channels. For very heavy nuclei, spontaneous fission competes with α decay; the dominant mode depends on Z, A, and shell structure. Cluster radioactivity (^{14}C, ^{24}Ne, and heavier ejectiles) and proton radioactivity near the proton drip line are rare variants of the same barrier penetration physics with different charge, mass, and preformation factors [6, 1, 14].

Historical context. Rutherford's 1911 interpretation of Geiger–Marsden scattering established the nuclear atom; the same α particles from natural sources later provided discrete energy spectra that confirmed energy quantisation in nuclear decays [2, 14]. Modern silicon detectors resolve α groups to keV precision, enabling branching-ratio studies (Lecture 17).

16.9.1 From Q_α systematics to tunneling (Lecture 17)

The Geiger–Nuttall plot compresses two independent measurements into one empirical law. To move from correlation to mechanism, combine the Q-value with the Coulomb turning point $b = Z_1 Z_2 e^2/(4\pi\varepsilon_0 Q)$. For fixed Z, increasing Q shrinks the classically forbidden width $b - R$ and lowers the Gamow exponent G. A back-of-the-envelope estimate for even–even nuclei uses $\log_{10} t_{1/2} \propto G/\ln 10 \propto Z_1 Z_2/\sqrt{Q}$, matching the observed $Q^{-1/2}$ slope of (16.30) [25, 6].

Slope ratio between thorium and uranium chains

On a $\log t_{1/2}$ versus $Q^{-1/2}$ plot, thorium ($Z = 90$) and uranium ($Z = 92$) even–even chains are nearly parallel but offset vertically. The offset reflects the Z-dependence of the barrier height at fixed A: higher daughter charge raises $V_{\mathrm{C}}(R)$ and lengthens lifetimes at the same Q. A 2 MeV increase in Q_α typically shortens $t_{1/2}$ by ten to fifteen orders of magnitude for actinides [6, 1]. This is the quantitative content behind the qualitative Geiger–Nuttall figure in the main lecture.

16.9.2 Competition with fission and the decay chain

For $A \gtrsim 230$, spontaneous fission competes with α emission. The fissility parameter Z^2/A of Lecture 23 sets the fission barrier scale; when B_f drops below a few MeV, fission half-lives can fall below α half-lives regardless of a favourable Q_α [6, 56]. Superheavy synthesis chains terminate when neither α nor fission allows further Z increase: the same barrier-penetration language governs both channels. Cluster radioactivity (^{14}C emission from ^{223}Ra) is a rare third variant with higher Coulomb barrier and smaller preformation factor [14, 1].

Key idea

Arc through the course. α decay (Lectures 16–17) supplies the Gamow exponent reused in fusion (Lectures 25–26). Fission (Lectures 23–24) competes when Z^2/A is large. Reading a Geiger–Nuttall point together with Q_α from AME2020 and a fissility estimate is a compact summary of heavy-nucleus stability [27, 6].

16.9.3 Worked link: from Q_α to half-life orders of magnitude

Take two even–even polonium isotopes on the same $Z = 84$ chain. If $Q_1/Q_2 \approx 4$ (a factor of two in $Q^{-1/2}$), the Gamow exponent changes by roughly $\sqrt{Q_1/Q_2} - 1 \approx 100\%$ in the leading $1/\sqrt{Q}$ term, and $\log_{10}(t_{1/2})$ shifts by $\sim$ 10–15 units for typical actinide barriers [6, 25]. This is the quantitative reason that ^{218}Th ($Q \approx 9.85$ MeV, $t_{1/2} \sim 10^{-7}$ s) and ^{232}Th ($Q \approx 4.08$ MeV, $t_{1/2} \sim 10^{10}$ y) coexist on one smooth Geiger–Nuttall line. When moving to reactions, the same Q bookkeeping sets thresholds for (α, n) and (α, p) channels on light targets (Lecture 22) [14, 1].

16.9.4 Hindrance and the odd-A example

Odd-A α emitters such as ^{235}U lie above the even–even Geiger–Nuttall curves because angular momentum and poor parent-daughter overlap reduce the preformation factor [6, 1]. The hindrance factor $F = \lambda_{\text{theory}}/\lambda_{\text{exp}}$ quantifies the deviation: $F \approx 1$ for normal even–even transitions, $F \gg 1$ for suppressed ones. This is why natural uranium persists for 10^9 years despite a modest $Q_\alpha \approx 4.27\,\mathrm{MeV}$. The same overlap language reappears in transfer spectroscopic factors (Lecture 21) and in β-decay $\log ft$ values (Lecture 19) [6, 14].

16.10 Deeper reading: Q_α, Geiger–Nuttall, and preformation

Geiger–Nuttall plots correlate $\log T_{1/2}$ with $1/\sqrt{Q_\alpha}$ because the Gamow penetrability dominates the lifetime. Preformation (the many-body overlap that prepares an α particle) multiplies the attempt frequency but usually varies far less than the exponential tunneling factor. Fine structure to excited daughter states tests that overlap beyond a single ground-state branch.

Superheavy α chains (Lecture 25) use the same Q_α arithmetic with AME masses where available. Competition with fission (Lectures 23–24) terminates many chains. Compute Q_α for ^{212}Po from mass excesses once by hand: the exercise ties Lecture 4 binding differences to Lecture 17 tunneling.

Reduced widths and dimensionless hindrance

Beyond Geiger–Nuttall systematics, reduced widths compare α emission probabilities after removing the known penetrability. Small reduced widths signal poor preformation or structural mismatch with the daughter. Fine-structure branching ratios to rotational bands in deformed emitters further constrain the parent wave function.

Cluster radioactivity

Heavier cluster emission (^{14}C, etc.) is rarer but conceptually identical: Q value plus penetrability plus preformation. It reinforces that α decay is one member of a cluster family rather than an isolated miracle of $A = 4$.

Geiger–Nuttall construction

From AME masses, compute Q_α for a sequence of even–even polonium or radon isotopes and plot $\log T_{1/2}$ versus $1/\sqrt{Q_\alpha}$. Fit a straight line and note outliers. Outliers often signal hindered transitions, wrong spin assignments, or isomer contamination. The plot makes Lecture 17's exponential penetrability visible before any WKB integral is written.

Discuss preformation qualitatively: nuclei with clear α-cluster correlations may sit systematically below the average line (faster decay) after penetrability is removed. That residual is the structural content beyond Gamow's barrier factor and the bridge to microscopic cluster models used in superheavy spectroscopy.

Alpha decay links mass differences to tunneling. Geiger–Nuttall plots make the exponential visible; microscopic preformation explains residual scatter about the trend line.

In practice, rework one numerical estimate from the lecture with a different parameter set and compare the shift to the quoted experimental uncertainty. If the shift exceeds the uncertainty, the estimate is teaching a scale, not delivering a fit. That distinction keeps theoretical cartoons honest when they sit beside evaluated data tables and modern many-body calculations. Along the course arc, write one sentence linking this chapter backward and one forward: the discipline of those two sentences is as important as any single formula. Examination problems rarely ask for a literature review, but they do ask for the scale argument that literature reviews presuppose.

Synthesis problem

Write a half-page note that (i) restates the chapter's central scale or selection rule in one equation, (ii) evaluates it numerically for one concrete nucleus or energy, and (iii) names the neighbouring lecture that uses the result. Keep the note free of research-frontier speculation; the goal is fluency with the lecture physics itself. If an assumption fails (Born approximation, spherical mean field, allowed transition, one-dimensional barrier), state the failure mode in a final sentence. That habit converts passive reading into examination-ready understanding and makes the later research-frontier box read as an extension rather than a replacement.

Research frontier: α decay, clusters, and student projects

Geiger–Nuttall systematics (Lecture 16) remain useful for quick half-life estimates, but microscopic preformation factors, cluster radioactivity, and α chains from superheavy elements are active frontiers [55]. Precision Q_α values from AME2020 and decay spectroscopy at SHE factories map shell effects near the predicted island of stability around $Z \sim 114$–126 and $N \sim 184$ [27, 53]. Competition between α emission and spontaneous fission terminates many heavy decay chains and sets practical limits on how far synthesis can be pushed. What remains uncertain includes absolute preformation probabilities, the location of the next closed shells beyond oganesson, and the reliability of extrapolated Q_α values when mass models disagree. Gamow's exponential barrier penetration still dominates the lifetime, but the many-body overlap that prepares the α particle is no longer treated as a universal constant.

Cluster radioactivity and fine-structure α branching to excited daughter states provide additional microscopic tests of preformation. When a decay chain populates both ground and excited states, the branching ratios constrain the parent wave function beyond a single Geiger–Nuttall point. That is a natural extension of the lecture systematics.

Student directions. (1) Data exercise: from AME2020, compute Q_α for ^{212}Po and ^{252}Cf and relate $\log T_{1/2}$ to $1/\sqrt{Q}$. (2) Conceptual note: sketch why a large Q_α alone does not guarantee a short half-life if the preformation factor is tiny. (3) Literature summary: summarise one recent SHE α-decay chain and its Z, N assignments [55].

16.11 Problems with solutions

Problem 16.1 — K_α fraction for $A = 220$. For α decay of a nucleus with $A = 220$, estimate the fraction of Q carried by the α particle using Eq. (16.23).

Solution 16.1.

Step 1. Write the kinematic ratio from momentum conservation:

$$\frac{K_\alpha}{Q} = \frac{M_d}{M_d + M_\alpha}.$$

Step 2. Approximate masses by nucleon count: $M_d \approx (A-4)u$ and $M_\alpha \approx 4u$, so $M_d+M_\alpha \approx Au$.

Step 3. Substitute $A = 220$:

$$\frac{K_\alpha}{Q} \approx 1 - \frac{4}{A} = 1 - \frac{4}{220} = 1 - 0.01818 = 0.982.$$

The α particle carries about 98.2% of Q; the daughter recoil is about 1.8% [6, 1].

Problem 16.2 — Half-life and mean life from λ. A sample has decay constant $\lambda = 1.0 \times 10^{-6}\,\mathrm{s}^{-1}$. Find (a) the half-life $t_{1/2}$ in days, and (b) the mean life τ.

Solution 16.2.

Step 1 (half-life). From Eq. (16.28):

$$t_{1/2} = \frac{\ln 2}{\lambda} = \frac{0.693}{1.0 \times 10^{-6}\,\mathrm{s}^{-1}} = 6.93 \times 10^5\,\mathrm{s}.$$

Step 2 (convert to days).

$$t_{1/2} = \frac{6.93 \times 10^5}{86400\,\mathrm{s/day}} \approx 8.0\,\mathrm{days}.$$

Step 3 (mean life).

$$\tau = \frac{1}{\lambda} = \frac{1}{1.0 \times 10^{-6}} = 1.0 \times 10^6\,\mathrm{s} \approx 11.6\,\mathrm{days}.$$

Note $\tau = t_{1/2}/\ln 2 \approx 1.443\,t_{1/2}$.

Problem 16.3 — Activity decay over three half-lives. At $t = 0$ a source has activity $\mathcal{A}_0 = 3.7 \times 10^4\,\mathrm{Bq}$ (1 mCi). What is the activity after $3\,t_{1/2}$?

Solution 16.3.

Step 1. Activity follows Eq. (16.29): $\mathcal{A}(t) = \mathcal{A}_0\, e^{-\lambda t}$.

Step 2. After one half-life, $\mathcal{A}$ falls by a factor of 2. After three half-lives:

$$\mathcal{A}(3t_{1/2}) = \frac{\mathcal{A}_0}{2^3} = \frac{\mathcal{A}_0}{8}.$$

Step 3. Numerical value:

$$\mathcal{A}(3t_{1/2}) = \frac{3.7 \times 10^4}{8} = 4.6 \times 10^3\,\mathrm{Bq}.$$

Problem 16.4 — Coulomb barrier height for $Z = 92$. Estimate the Coulomb barrier height for α decay of ^{238}U ($Z = 92$, $A = 238$) using $R_d = 1.25A^{1/3}\,\text{fm}$ and $R_\alpha = 1.5\,\text{fm}$. Compare with typical $Q \approx 4.3\,\text{MeV}$.

Solution 16.4.

Step 1 (daughter radius).

$$R_d = 1.25 \times 238^{1/3}\,\text{fm} \approx 1.25 \times 6.20\,\text{fm} \approx 7.75\,\text{fm}.$$

Step 2 (contact distance).

$$R = R_d + R_\alpha \approx 7.75 + 1.5 = 9.25\,\text{fm}.$$

Step 3 (barrier height). With $Z_1 = Z - 2 = 90$:

$$V_\text{C}(R) = \frac{2 \times 90 \times 1.44\,\text{MeV fm}}{9.25\,\text{fm}} \approx 28\,\text{MeV}.$$

Step 4 (comparison). $Q \approx 4.3\,\text{MeV} \ll 28\,\text{MeV}$, so the decay proceeds only by quantum tunneling. The ratio $Q/V_\text{C} \sim 0.15$ explains the enormous half-life ($\sim 4.5 \times 10^9\,\text{y}$) [6, 20].

Problem 16.5 — Geiger–Nuttall estimate. Two even–even isotopes of the same element have α energies $Q_1 = 5.0\,\text{MeV}$ and $Q_2 = 6.0\,\text{MeV}$. If the first has $t_{1/2}^{(1)} = 10^6\,\text{s}$, estimate $t_{1/2}^{(2)}$ using $\log_{10} t_{1/2} = aQ^{-1/2} + b$ with the same a, b.

Solution 16.5.

Step 1. Subtract the two Geiger–Nuttall equations:

$$\log_{10} t_{1/2}^{(2)} - \log_{10} t_{1/2}^{(1)} = a\left(Q_2^{-1/2} - Q_1^{-1/2}\right).$$

Step 2. A Geiger–Nuttall plot of $\log_{10} t_{1/2}$ against $Q^{-1/2}$ has a *positive* slope: larger $Q^{-1/2}$ means a lower decay energy and a longer half-life. For a representative heavy isotopic chain, take $a \approx +140\,\text{MeV}^{1/2}$ (the fitted value depends on Z).

$$Q_2^{-1/2} - Q_1^{-1/2} = \frac{1}{\sqrt{6.0}} - \frac{1}{\sqrt{5.0}} = 0.4082 - 0.4472 = -0.039.$$

$$\log_{10} \frac{t_{1/2}^{(2)}}{t_{1/2}^{(1)}} \approx +140 \times (-0.039) \approx -5.46.$$

Step 3.

$$t_{1/2}^{(2)} \approx 10^6 \times 10^{-5.46} \approx 3.5\,\text{s}.$$

A 20% increase in Q therefore shortens the estimated half-life by about 3×10^5. The numerical value is only illustrative because a must be fitted for the isotopic chain, but the sign and exponential sensitivity are unambiguous [6, 1].

Problem 16.6 — Geiger–Nuttall reading. Two even–even α emitters with the same Z differ by $\Delta Q_\alpha = 0.5\,\mathrm{MeV}$ near $Q_\alpha \sim 5\,\mathrm{MeV}$. Qualitatively, which has the shorter half-life, and by roughly how many orders of magnitude can λ change for such ΔQ?

Solution 16.6. Larger Q means a thinner Coulomb barrier and larger Gamow penetration, hence shorter $t_{1/2}$. Geiger–Nuttall systematics show that few-hundred-keV changes in Q can shift half-lives by many orders of magnitude (often $\sim 10^2$–10^4 or more, depending on Z) [24, 6].

Chapter 17

Alpha Decay: Tunneling and the Gamow Factor

"The central feature of this one-body model is that the α particle is preformed inside the parent nucleus... the theory works quite well, especially for even–even nuclei."

K. S. Krane

Lecture 16 established the observables Q and $t_{1/2}$, the Geiger–Nuttall correlation, and the nuclear-well plus Coulomb-barrier potential. This chapter derives the Gamow tunneling factor in full: rectangular-barrier warm-up with the *reduced mass* μ, the Coulomb-barrier integral evaluated without skipped steps, the assault frequency, the decay rate $\lambda = S_\alpha\, f\, P$, and the fine structure of α spectra from angular-momentum and parity selection rules [6, 1, 14, 2].

17.1 Classically forbidden region

Quantum tunneling through that barrier turns a classically stable α+daughter configuration into a slow decay. The transmission probability falls exponentially with barrier height and width, which is why α lifetimes span many orders of magnitude while Q values change by only a few MeV [6, 1].

The available energy $E = Q$ of an emitted α particle is typically 4–9 MeV, much less than the Coulomb barrier height $B \sim 20$–30 MeV at the nuclear surface. Three regions of the radial potential are distinguished (Fig. 17.1):

(i) **Well ($r < R$).** The α moves inside the attractive nuclear potential (depth $V_0 \sim 35$ MeV) with kinetic energy approximately $Q + V_0$.

(ii) **Barrier ($R < r < b$).** The Coulomb potential exceeds the total energy Q. Classically the particle cannot enter this region: its kinetic energy would have to be negative.

(iii) **Asymptotic ($r > b$).** The Coulomb potential falls below Q; the α escapes with kinetic energy Q.

The outer *classical turning point* b is defined by

$$V_C(b) = Q, \qquad b = \frac{Z_1 Z_2 e^2}{4\pi\varepsilon_0\, Q}, \tag{17.1}$$

with $Z_1 = Z - 2$ (daughter charge) and $Z_2 = 2$ (α charge).

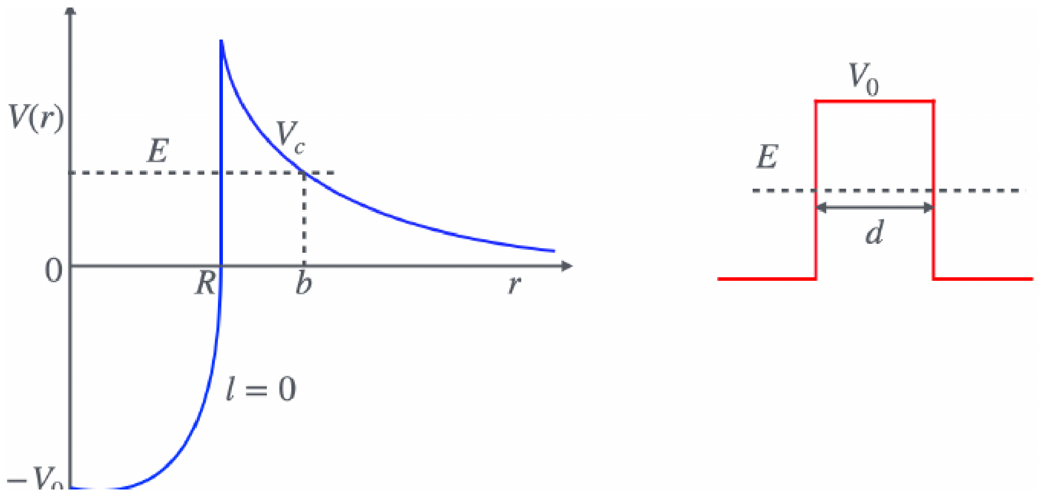


Figure 17.1: Left: nuclear well plus Coulomb barrier with total energy $E = Q$, inner radius R, and outer turning point b. Right: rectangular barrier of height V_0 and width d used as a warm-up for tunneling theory.

Quantum mechanics permits a non-zero probability for the α to traverse the classically forbidden interval $R \le r \le b$. One may express the lifetime as $\tau = \tau_0/P$, where P is the escape probability and τ_0 is a formation time scale; the course uses the equivalent form $\lambda = S_\alpha\, f\, P$ with assault frequency f and preformation factor S_α [1, 6].

17.2 Rectangular barrier and the reduced mass

For a one-dimensional rectangular barrier of height $V_0 > E$ and width d, the Schrödinger equation in the forbidden region gives an evanescent wavefunction $\psi \propto e^{-\gamma r}$ with

$$\gamma = \frac{\sqrt{2\mu(V_0 - E)}}{\hbar}, \tag{17.2}$$

where μ is the *reduced mass* of the α-daughter system:

$$\mu = \frac{m_\alpha\, M_d}{m_\alpha + M_d}. \tag{17.3}$$

For heavy nuclei $M_d \gg m_\alpha$, so $\mu \approx m_\alpha$ to good approximation, but all barrier formulas must be written with μ for consistency with the two-body Coulomb problem [1].

Matching the wavefunction and its derivative at the barrier boundaries yields the standard WKB result for the transmission probability:

$$P \approx e^{-2\gamma d} = \exp\left[-\frac{2d}{\hbar}\sqrt{2\mu(V_0 - E)}\right]. \tag{17.4}$$

The factor $(V_0 - E)$ is the amount by which the potential exceeds the particle energy inside the barrier.

17.3 Coulomb barrier and the Gamow integral

For a non-rectangular barrier, one divides the interval into slices of width dr and multiplies the elementary penetration factors. This WKB approximation gives

$$P \approx \exp\left[-2\int_R^b \gamma(r)\,\mathrm{d}r\right] = e^{-2G}, \tag{17.5}$$

where

$$\gamma(r) = \frac{\sqrt{2\mu[V(r)-Q]}}{\hbar}, \qquad G = \int_R^b \gamma(r)\,\mathrm{d}r. \tag{17.6}$$

Barrier excess measured from the turning point. For $R \le r \le b$, the quantity that enters the WKB wave number is

$$V_{\mathrm{C}}(r) - Q = \frac{Z_1 Z_2 e^2}{4\pi\varepsilon_0}\left(\frac{1}{r} - \frac{1}{b}\right), \tag{17.7}$$

which vanishes at b, since $Q = Z_1 Z_2 e^2/(4\pi\varepsilon_0 b)$. Substituting into Eq. (17.6):

$$G = \frac{\sqrt{2\mu}}{\hbar}\sqrt{\frac{Z_1 Z_2 e^2}{4\pi\varepsilon_0}}\int_R^b \sqrt{\frac{1}{r} - \frac{1}{b}}\,\mathrm{d}r. \tag{17.8}$$

17.4 Evaluating the integral: every step

Define

$$I = \int_R^b \sqrt{\frac{1}{r} - \frac{1}{b}}\,\mathrm{d}r = \int_R^b \sqrt{\frac{b-r}{br}}\,\mathrm{d}r. \tag{17.9}$$

Step 1: Trigonometric substitution. Set

$$r = b\sin^2\theta. \tag{17.10}$$

Differentiate:

$$\mathrm{d}r = 2b\sin\theta\cos\theta\,\mathrm{d}\theta. \tag{17.11}$$

Express $1/r$:

$$\frac{1}{r} = \frac{1}{b\sin^2\theta}. \tag{17.12}$$

Form the combination under the square root:

$$\frac{1}{r} - \frac{1}{b} = \frac{1}{b\sin^2\theta} - \frac{1}{b} = \frac{1-\sin^2\theta}{b\sin^2\theta} = \frac{\cos^2\theta}{b\sin^2\theta}. \tag{17.13}$$

Take the square root (positive in the interval of interest):

$$\sqrt{\frac{1}{r} - \frac{1}{b}} = \sqrt{\frac{\cos^2\theta}{b\sin^2\theta}} = \frac{\cos\theta}{\sqrt{b}\,\sin\theta} = \frac{\cot\theta}{\sqrt{b}}. \tag{17.14}$$

Step 2: Rewrite the integral in θ.

$$I = \int \frac{\cot\theta}{\sqrt{b}} \cdot 2b\sin\theta\cos\theta\,\mathrm{d}\theta. \tag{17.15}$$

Simplify the integrand:

$$\frac{\cot\theta}{\sqrt{b}} \cdot 2b\sin\theta\cos\theta = \frac{\cos\theta}{\sin\theta\sqrt{b}} \cdot 2b\sin\theta\cos\theta = \frac{2b\cos^2\theta}{\sqrt{b}} = 2\sqrt{b}\cos^2\theta. \tag{17.16}$$

Hence

$$I = 2\sqrt{b}\int \cos^2\theta\,\mathrm{d}\theta. \tag{17.17}$$

Step 3: Integrate $\cos^2\theta$. Use $\cos^2\theta = \frac{1}{2}(1+\cos 2\theta)$:

$$I = 2\sqrt{b}\int \frac{1+\cos 2\theta}{2}\,\mathrm{d}\theta = \sqrt{b}\int (1+\cos 2\theta)\,\mathrm{d}\theta = \sqrt{b}\left(\theta + \frac{\sin 2\theta}{2}\right). \tag{17.18}$$

Step 4: Limits of integration. When $r = R$:

$$\sin^2\theta_R = \frac{R}{b} \quad \Rightarrow \quad \theta_R = \arcsin\sqrt{\frac{R}{b}}. \tag{17.19}$$

When $r = b$:

$$\sin^2\theta_b = 1 \quad \Rightarrow \quad \theta_b = \frac{\pi}{2}. \tag{17.20}$$

Step 5: Evaluate at the limits.

$$I = \sqrt{b}\left[\frac{\pi}{2} + \frac{\sin\pi}{2}\right] - \sqrt{b}\left[\arcsin\sqrt{\frac{R}{b}} + \frac{\sin(2\theta_R)}{2}\right]. \tag{17.21}$$

Since $\sin\pi = 0$:

$$I = \sqrt{b}\left[\frac{\pi}{2} - \arcsin\sqrt{\frac{R}{b}} - \frac{\sin(2\theta_R)}{2}\right]. \tag{17.22}$$

Step 6: Simplify $\sin(2\theta_R)$. Use $\sin(2\theta_R) = 2\sin\theta_R\cos\theta_R$ with $\sin\theta_R = \sqrt{R/b}$ and $\cos\theta_R = \sqrt{1 - R/b}$:

$$\sin(2\theta_R) = 2\sqrt{\frac{R}{b}}\sqrt{1-\frac{R}{b}} = 2\sqrt{\frac{R}{b}\left(1-\frac{R}{b}\right)}. \tag{17.23}$$

Therefore

$$\frac{\sin(2\theta_R)}{2} = \sqrt{\frac{R}{b}\left(1-\frac{R}{b}\right)}. \tag{17.24}$$

Step 7: Use $\arccos$ form. Since $\arccos x + \arcsin x = \pi/2$,

$$\frac{\pi}{2} - \arcsin\sqrt{\frac{R}{b}} = \arccos\sqrt{\frac{R}{b}}. \tag{17.25}$$

The final result is

$$I = \sqrt{b}\left[\arccos\sqrt{\frac{R}{b}} - \sqrt{\frac{R}{b}\left(1-\frac{R}{b}\right)}\right]. \tag{17.26}$$

Step 8: Assemble G and P. Substituting Eq. (17.26) into Eq. (17.8):

$$G = \frac{b}{\hbar}\sqrt{2\mu Q}\left[\arccos\sqrt{\frac{R}{b}} - \sqrt{\frac{R}{b}\left(1-\frac{R}{b}\right)}\right], \tag{17.27}$$

where we used $Q = Z_1Z_2e^2/(4\pi\varepsilon_0 b)$, so the two factors of $\sqrt{b}$ combine to b. This form is dimensionless; an extra $\sqrt{b}$ would make G dimensionful. The tunneling probability is

$$P = e^{-2G}. \tag{17.28}$$

17.5 Barrier height B and the small Q/B limit

The Coulomb barrier height at $r = R$ is

$$B = \frac{Z_1Z_2e^2}{4\pi\varepsilon_0\, R}, \qquad \frac{R}{b} = \frac{Q}{B}. \tag{17.29}$$

In terms of B and Q, Eq. (17.27) becomes

$$G = \frac{Z_1Z_2e^2}{4\pi\varepsilon_0\,\hbar}\sqrt{\frac{2\mu}{Q}}\left[\arccos\sqrt{\frac{Q}{B}} - \sqrt{\frac{Q}{B}\left(1-\frac{Q}{B}\right)}\right]. \tag{17.30}$$

Small Q/B expansion. For typical α decays, $Q/B \sim 0.1$–0.2. Expand for $Q/B \ll 1$:

$$\arccos\sqrt{\frac{Q}{B}} \approx \frac{\pi}{2} - \sqrt{\frac{Q}{B}}, \qquad \sqrt{\frac{Q}{B}\left(1-\frac{Q}{B}\right)} \approx \sqrt{\frac{Q}{B}}. \tag{17.31}$$

The bracket becomes $\pi/2 - 2\sqrt{Q/B}$. Keeping the leading term $\pi/2$ and using $Q = \frac{1}{2}\mu v^2$ for the asymptotic α speed v:

$$G \approx \frac{\pi Z_1Z_2e^2}{4\pi\varepsilon_0\,\hbar v} = \frac{Z_1Z_2e^2}{4\varepsilon_0\,\hbar v}, \qquad v = \sqrt{\frac{2Q}{\mu}}. \tag{17.32}$$

Thus $P \sim \exp(-2G)$ depends exponentially on $1/v \propto Q^{-1/2}$, which is the Geiger–Nuttall rule [6, 1, 14].

Key idea

Because $P \sim \exp(-c/\sqrt{Q})$, a small increase in Q reduces G and multiplies the decay rate by many orders of magnitude. The Geiger–Nuttall rule is a direct consequence of Coulomb-barrier tunneling.

17.6 Assault frequency, preformation, and $\lambda = S_\alpha f P$

The tunneling probability P is the escape probability *per attempt*. The α bounces inside the well, presenting itself at the barrier with characteristic frequency f. The decay constant is

$$\lambda = S_\alpha \, f \, P, \tag{17.33}$$

where S_α is the *preformation factor* (probability that nucleons are clustered as an α at the barrier surface). It is a model-dependent overlap factor, not a universal observable; setting $S_\alpha = 1$ is only a crude single-particle upper-normalisation. Extracted values depend on the barrier radius and nuclear-structure model, and hindered transitions can have $S_\alpha \ll 1$ [6].

Assault frequency. If the α travels back and forth across diameter $2R$ at interior speed v_0, the time for one round trip is $2R/v_0$, so

$$f = \frac{v_0}{2R}. \tag{17.34}$$

Inside the well the kinetic energy is $Q + V_0$ (where V_0 is the well depth, typically $\sim 35\,\mathrm{MeV}$):

$$\frac{1}{2}\mu v_0^2 = Q + V_0 \quad \Rightarrow \quad v_0 = \sqrt{\frac{2(Q+V_0)}{\mu}}. \tag{17.35}$$

Hence

$$f = \frac{1}{2R}\sqrt{\frac{2(Q+V_0)}{\mu}} = \sqrt{\frac{Q+V_0}{2\mu R^2}}. \tag{17.36}$$

Combining with Eq. (17.28):

$$\lambda = S_\alpha \sqrt{\frac{Q+V_0}{2\mu R^2}}\, e^{-2G}, \qquad t_{1/2} = \frac{\ln 2}{\lambda}. \tag{17.37}$$

Caution

Absolute half-life calculations are extremely sensitive to the assumed radius: A 4% change in R can change calculated $t_{1/2}$ by a factor of about five. Measured lifetimes are therefore sometimes used to infer an effective sum of nuclear and α radii [6].

17.7 Fine structure, branching, and selection rules

When the daughter is left in an excited state, each branch has its own $Q_i = Q_{\mathrm{gs}} - E_i^*$ and its own required orbital angular momentum ℓ carried by the α. The centrifugal barrier adds $\hbar^2\ell(\ell+1)/(2\mu r^2)$ to $V(r)$, raising G and suppressing high-ℓ branches [6, 1].

Selection rules. The α particle has $J^\pi = 0^+$: intrinsic spin 0 and positive intrinsic parity. Angular momentum and parity conservation therefore require

$$\boldsymbol{J}_i = \boldsymbol{J}_f + \boldsymbol{\ell}, \qquad \pi_i = \pi_f \times (-1)^\ell. \tag{17.38}$$

Allowed ℓ values satisfy $|J_i - J_f| \le \ell \le J_i + J_f$ with the parity condition. In practice, the lowest allowed ℓ dominates because it minimises the barrier.

17.7.1 Even–even example: ^{244}Cm $\to$ ^{240}Pu

Both parent and daughter ground states are 0^+. The daughter rotational band begins at 2^+ (42.824 keV), 4^+ (141.690 keV), and 6^+ (294.319 keV) [20, 21]. DDEP/LNHB evaluated absolute α-emission probabilities are [20, 21, 28]:

Daughter state	E^* (keV)	ℓ	Branch (%)
0^+ (gs)	0	0	76.7(4)
2^+	42.824	2	23.3(4)
4^+	141.690	4	0.0204(15)
6^+	294.319	6	0.00352(18)

The ground-state branch ($\ell = 0$) dominates. The 2^+ branch has both a smaller Q_i and an $\ell = 2$ centrifugal barrier. Higher members of the rotational band are strongly suppressed; note especially that the 4^+ and 6^+ probabilities are hundredths and thousandths of a percent, respectively [6, 28].

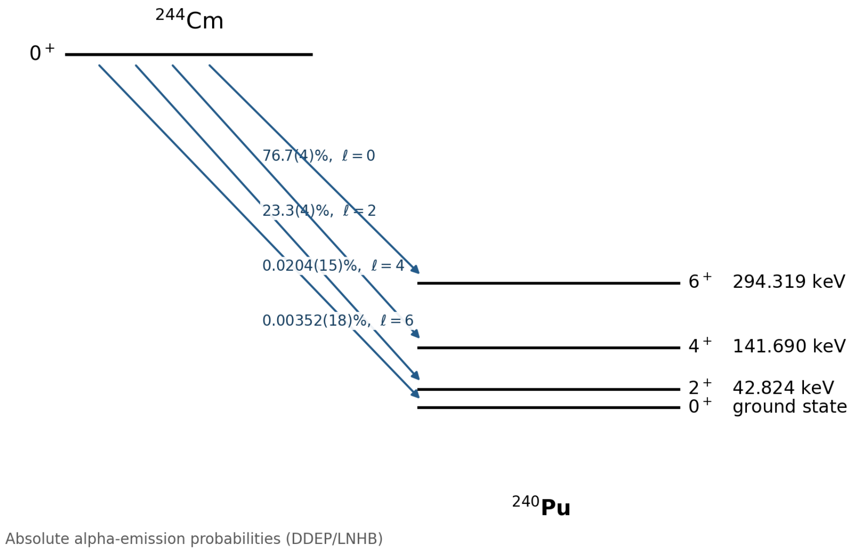


Figure 17.2: Branching ratios for α decay of ^{244}Cm to levels of ^{240}Pu, using DDEP/LNHB evaluated absolute α-emission probabilities [20, 21, 28].

17.7.2 Odd-A example: ^{243}Am $\to$ ^{239}Np

Remark

The original lecture slides incorrectly labelled this case as ^{249}Am. The correct parent is ^{243}Am ($J^\pi = \frac{5}{2}^-$), which α-decays to ^{239}Np ($J^\pi = \frac{5}{2}^+$ ground state) [20, 21, 6].

Ground-state branch. Parent $\frac{5}{2}^-$ to daughter $\frac{5}{2}^+$ requires a change of parity, so ℓ must be odd. The lowest allowed value is $\ell = 1$. The centrifugal barrier makes this branch highly hindered: evaluated data give a branching ratio of only 0.240(3)% [20, 21, 28].

Dominant branch to $\frac{5}{2}^-$ at 74.660 keV. The daughter level at $E^* = 74.660\,\text{keV}$ has $J^\pi = \frac{5}{2}^-$, matching the parent parity. An $\ell = 0$ transition is therefore allowed. Although Q_i is reduced, the much thinner $\ell = 0$ barrier dominates, and this branch carries 86.74(5)% of the α intensity [20, 21, 28, 6].

Other branches. The 31.130 keV, $\frac{7}{2}^+$ level receives 0.192(3)% and requires odd ℓ, with $\ell_{\text{min}} = 1$. The 117.84 keV, $\frac{7}{2}^-$ and 173.02 keV, $\frac{9}{2}^-$ levels receive 11.46(5)% and 1.383(7)%, respectively; parity is unchanged and $\ell_{\text{min}} = 2$ for both. Figure 17.3 shows these principal DDEP branches.

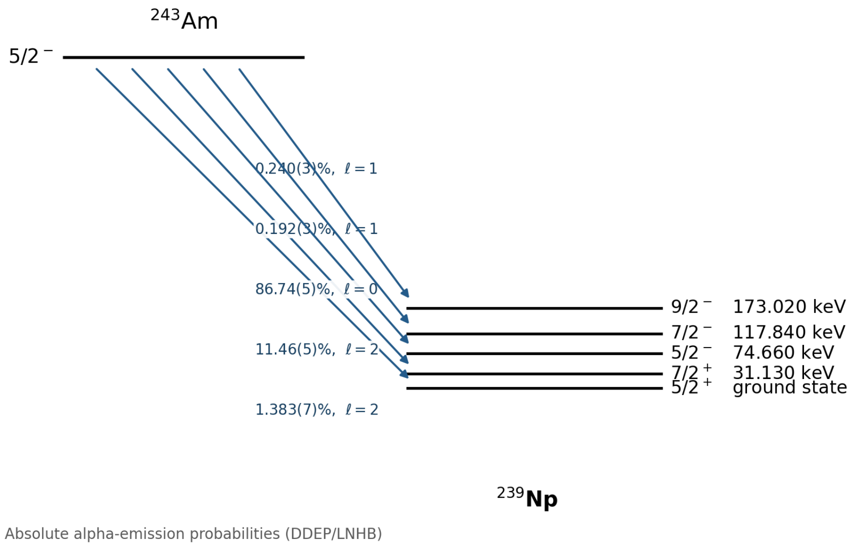


Figure 17.3: Branching ratios for α decay of ^{243}Am to levels of ^{239}Np, using DDEP evaluated absolute α-emission probabilities; the isotope is corrected from ^{249}Am to ^{243}Am [20, 21, 28].

Hindrance factors. The hindrance factor $F = \lambda_{\text{theory}}/\lambda_{\text{exp}}$ (or equivalently the ratio of predicted to observed branch intensity) quantifies deviations from the Gamow estimate. Large F indicates poor parent-daughter overlap or selection-rule suppression beyond the centrifugal barrier [6].

17.8 Why many nuclei with $Q > 0$ still appear stable

The SEMF gives $Q_\alpha > 0$ already near $A \sim 150$, yet many nuclides up to ^{208}Pb are stable or long-lived. If G is large, e^{-2G} is extraordinarily small and $t_{1/2}$ can exceed 10^{17} s ($\sim 10^{10}$ years). Only when Q rises into the actinide range do laboratory half-lives become short. This is precisely the Geiger–Nuttall message encoded in $P = e^{-2G}$: $Q > 0$ is necessary for spontaneous α decay, but the half-life is fixed by tunneling.

Takeaway

The Gamow theory explains (i) why $Q > 0$ does not imply short lifetimes, (ii) the Geiger–Nuttall $Q^{-1/2}$ correlation, and (iii) the fine structure of α spectra through ℓ-dependent barriers and selection rules. Use reduced mass μ throughout, and remember $\lambda = S_\alpha f P$.

17.8.1 Exponential Gamow penetration

The transmission probability through the Coulomb barrier is of the Gamow form $P \sim e^{-2G}$ with G the barrier integral. Because G depends strongly on Q and Z, half-lives change by many orders of magnitude while Q_α changes by only a factor of two—the quantitative content of the Geiger–Nuttall correlation [25, 24, 6, 14].

17.9 Deeper physics: the Gamow factor and WKB tunneling

17.9.1 The Gamow integral for a Coulomb barrier

For $\ell = 0$ and energy $E = Q$, the WKB penetration exponent for a Coulomb barrier is the Gamow integral

$$G = \int_R^b \kappa(r)\,\mathrm{d}r, \qquad \kappa(r) = \frac{\sqrt{2\mu[V_\mathrm{C}(r) - Q]}}{\hbar}, \tag{17.39}$$

with inner radius R and outer turning point $b = Z_1Z_2e^2/(4\pi\varepsilon_0 Q)$ from (17.1) [6, 1]. Evaluating the integral for a pure Coulomb tail gives the standard form

$$G \approx \frac{Z_1Z_2e^2}{4\pi\varepsilon_0\hbar}\sqrt{\frac{2\mu}{Q}}\left[\arccos\sqrt{\frac{Q}{V_\mathrm{C}(R)}} - \sqrt{\frac{Q}{V_\mathrm{C}(R)}\left(1 - \frac{Q}{V_\mathrm{C}(R)}\right)}\right], \tag{17.40}$$

and the escape probability is $P \approx e^{-2G}$. The reduced mass $\mu = m_\alpha M_d/(m_\alpha + M_d)$ enters every WKB formula; for heavy parents $\mu \simeq m_\alpha$ to within a few percent [1].

Order-of-magnitude Gamow exponent

For ^{212}Po ($Z = 84$, $Q \approx 8.95\,\mathrm{MeV}$), take $R \approx 1.2\,A^{1/3}\,\mathrm{fm} \approx 7\,\mathrm{fm}$ and $V_\mathrm{C}(R) \approx 25\,\mathrm{MeV}$. Then $2G \sim 30$–40, so $P \sim e^{-2G} \sim 10^{-14}$ and, with an assault frequency $f \sim 10^{21}\,\mathrm{s}^{-1}$, $\lambda \sim 10^7\,\mathrm{s}^{-1}$, consistent with the microsecond half-life [6, 14]. A 4% increase in effective R lengthens $t_{1/2}$ by roughly a factor of five because G is so steep [6].

17.9.2 Rectangular barrier as a WKB warm-up

Before the Coulomb integral, a rectangular barrier of height $V_0 > E$ and width d gives (17.4). The exponent is linear in d and in $\sqrt{V_0 - E}$, with μ in the denominator of γ. This model captures the essential physics: classically forbidden regions admit exponentially small transmission, and heavier ejectiles (larger μ) or lower Q (thicker barrier) suppress the rate dramatically. The full Gamow result replaces the constant height by the $1/r$ Coulomb tail, which is why $\log t_{1/2}$ tracks $Q^{-1/2}$ rather than a simple exponential in Q [25, 2].

Key idea

Assault frequency and preformation. The decay rate is $\lambda = S_\alpha f P$ with preformation factor S_α and assault frequency $f \sim v/(2R)$. Gamow theory explains the Q dependence through $P = e^{-2G}$; shell effects and hindrance enter S_α. Even–even nuclei with $F \approx 1$ validate the minimal model; large hindrance factors signal poor cluster overlap at the surface [6, 14].

17.9.3 Centrifugal barrier and angular momentum

For $\ell \neq 0$ the effective barrier acquires a centrifugal term $\hbar^2\ell(\ell+1)/(2\mu r^2)$ that raises G and lengthens $t_{1/2}$. Fine-structure α groups populating excited daughter states with spin $I > 0$ therefore require a larger Gamow exponent than ground-state $\ell = 0$ decay [6, 1]. In fusion, the same centrifugal term suppresses low-energy cross sections for partial waves with $\ell > 0$, filtering which angular-momentum channels contribute at sub-barrier energies (Lecture 25) [14]. The rectangular-barrier warm-up (17.4) is the constant-height limit; the Coulomb tail makes G depend on $\log Q$ as well as $1/\sqrt{Q}$, which is why the Geiger–Nuttall slope is not a simple exponential in Q alone.

17.9.4 Deformed nuclei and effective barrier radii

For deformed heavy nuclei ($A \gtrsim 230$) the Coulomb barrier is not spherically symmetric at the surface. The α cluster exits preferentially where the barrier is lowest, effectively reducing the barrier width compared with a spherical average [6, 1]. Practitioners sometimes quote an *effective* radius R_{eff} fitted to measured half-lives; a 4% change in R_{eff} can shift $t_{1/2}$ by about a factor of five because of the exponential in G [6]. This sensitivity is the same reason that sub-barrier fusion cross sections (Lecture 25) are difficult to predict from a single contact radius: orientation and deformation matter as much as the spherical estimate [55]. Fine-structure studies that compare ground-state and excited-state α branches constrain both R_{eff} and the hindrance factor S_α [14].

Key idea

Exponential leverage. If $2G \sim 60$, then $\Delta G = 1$ changes P by a factor $e^2 \sim 7$. Barrier modeling at the mega-electronvolt level is therefore mandatory for order-of-magnitude half-lives, which is why Lecture 16's Geiger–Nuttall correlation is powerful yet not a substitute for a full WKB evaluation when Q or Z changes substantially.

17.10 Further physics from the literature

Radius sensitivity and effective sizes. Absolute Gamow half-lives depend exponentially on R through both G and f. A 4% change in the radius parameter can shift calculated $t_{1/2}$ by a factor of about five; practitioners sometimes invert the problem and infer an effective R-sum from measured lifetimes [6]. Deformed nuclei ($A \gtrsim 230$) further modify the barrier because the Coulomb potential is not spherically symmetric at the surface.

Preformation and cluster decay. The preformation factor S_α accounts for the probability that four nucleons are correlated as an α at the nuclear surface. Heavier cluster emissions (^{14}C, ^{24}Ne, and others) use the same $P = e^{-2G}$ framework but with larger reduced mass, higher Coulomb barrier, and vastly smaller S_α [6, 1, 14]. The first observed ^{14}C branch from ^{223}Ra had relative intensity $\sim 10^{-9}$ compared with α decay.

Proton radioactivity. Near the proton drip line, one-proton emission becomes the $Z_2 = 1$ analogue of Gamow theory when $Q_p > 0$. The same WKB integral applies with $Z_1 = Z - 1$, $Z_2 = 1$, and μ replaced by the proton-daughter reduced mass [6, 1].

Alignment experiments. When neither parent nor daughter has spin 0, several ℓ values may contribute to the same transition. Measuring the angular distribution of α particles relative to a spin-aligned source disentangles the ℓ-components; an aligned analysis for ^{253}Es is given in [6].

17.10.1 Time-reversed Gamow factor and fusion (Lectures 25–26)

Alpha decay and fusion differ only in boundary conditions on the same Coulomb barrier. Outgoing α particles tunnel from $r = R$ to $r = b$; incoming projectiles tunnel inward. The WKB exponent (17.6) is identical in form, with μ now the reduced mass of the colliding pair. For $^{12}\text{C} +^{12}\text{C}$ at $E_{\text{CM}} = 1\,\text{MeV}$, $2G$ is still large ($\sim$ 15–20), so the fusion cross section remains tiny despite $E > 0$ [6, 14]. Stellar rates average this tunneling probability over a Maxwell–Boltzmann distribution (Lecture 26).

Reduced mass in α decay versus $p + p$ fusion

For ^{212}Po α decay, $\mu \approx m_\alpha$ to 0.5%. For $p + p$ fusion, $\mu = m_p/2$. The Gamow energy parameter $E_G = 2\mu(\pi\alpha\hbar c)^2$ for $p + p$ is orders of magnitude smaller than for heavy-ion fusion because $Z_1 Z_2 = 1$. That is why hydrogen burning proceeds at keV Gamow-peak energies while superheavy synthesis requires MeV bombarding energies [36, 55].

17.10.2 From cluster decay to reaction cross sections

Cluster radioactivity generalises Gamow theory: the ejectile carries larger μ and higher $Z_1 Z_2$, so G grows and lifetimes stretch to 10^9 y or more for ^{14}C branches [6, 14]. The same penetration integral appears in heavy-ion fusion when the compound nucleus decays by α emission after capture. Direct-reaction stripping (Lecture 21) probes surface overlap that preformation factors encode in decay; compound-nucleus evaporation (Lecture 22) is the inelastic counterpart of sequential α and γ cascades [6, 39].

Fine-structure α decay to excited daughter states splits one Geiger–Nuttall point into several branches with different Q values. Each branch tests a different radial overlap because the daughter wave function must match a specific final-state configuration. Hindrance factors for odd-A parents play the same role as suppressed spectroscopic factors in transfer reactions: the barrier integral is unchanged, but the nuclear overlap is small [6, 1]. Proton radioactivity near the drip line is the $Z_2 = 1$ limit of the same WKB framework.

Remark

The assault frequency $f \sim v/(2R) \sim 10^{21}\,\mathrm{s}^{-1}$ sets the attempt rate but not the steep Q dependence: changing f by an order of magnitude shifts $\log t_{1/2}$ by only one unit, whereas a modest Q change moves $\log t_{1/2}$ by ten or more. This hierarchy justifies treating G as the primary variable when reading Geiger–Nuttall data, and treating S_α as a secondary correction except for strongly hindered transitions [6, 14].

17.10.3 Cluster decay as a test of the Gamow framework

The first observed ^{14}C branch from ^{223}Ra had relative intensity $\sim 10^{-9}$ compared with α decay [6, 14]. The larger ejectile mass and higher Coulomb charge raise G so much that the preformation factor must also be tiny. Cluster decay is therefore a sensitive probe of both barrier penetration and surface clustering. In the reaction context, light-ion transfer and knockout reactions on radioactive beams test similar overlap integrals at the nuclear surface (Lecture 21) [53]. The course arc from Geiger–Nuttall (Lecture 16) through G (Lecture 17) to sub-barrier fusion (Lecture 25) is one continuous story about exponential sensitivity to barrier geometry.

17.11 Deeper reading: WKB sensitivity and assault frequency

Because $P \sim e^{-2G}$, a 1 MeV shift in barrier height or a change in the inner radius of a few tenths of a femtometre can move half-lives by orders of magnitude. That exponential sensitivity is why microscopic fission theory invests so much effort in inertias and potential-energy surfaces (Lecture 23), not only in a one-dimensional Coulomb integral.

The assault frequency ν in $\lambda = \nu P$ is only an order-of-magnitude prefactor in the pedagogical model; modern calculations replace it with collective inertia along a fission or cluster path. Reduced-mass corrections $\mu \approx m_\alpha(A-4)/A$ matter at the few-percent level for light emitters and less for heavy ones, but they should be written explicitly when comparing textbook estimates to data.

Rectangular versus Coulomb barriers

A rectangular barrier of height V_0 and width L yields $G \approx L\sqrt{2\mu(V_0 - E)}/\hbar$, exposing exponential dependence algebraically. Replacing it with a Coulomb integral changes the Q-dependence into the Geiger–Nuttall form but preserves the exponential message. Students should evaluate both once.

Angular momentum barriers

Emitting α particles with $\ell > 0$ adds a centrifugal term that increases G and suppresses high-ℓ branches. That is why ground-state to ground-state $\ell = 0$ transitions often dominate when spin allows, connecting to selection themes in Lecture 20.

WKB numerical sketch

Approximate a Coulomb barrier as a triangle or use the analytic Gamow integral for ^{212}Po with $Q_\alpha \approx 8.95\,\text{MeV}$. Show that $2G$ is tens of units, so e^{-2G} is astronomically small, yet multiplied by a nuclear assault frequency $\sim 10^{20}\,\text{s}^{-1}$ it can yield a macroscopic half-life. Change Q by $0.5\,\text{MeV}$ and recompute to feel the exponential lever arm.

Relate this sensitivity to fission (Lecture 23): replacing the α Coulomb barrier with a many-dimensional fission barrier preserves the exponential, not the detailed prefactor. Students who master one WKB estimate here will not be surprised by order-of-magnitude fission lifetime uncertainties later.

WKB tunneling is the quantitative engine behind α lifetimes and a prototype for fission and fusion barrier penetration in later lectures.

In practice, rework one numerical estimate from the lecture with a different parameter set and compare the shift to the quoted experimental uncertainty. If the shift exceeds the uncertainty, the estimate is teaching a scale, not delivering a fit. That distinction keeps theoretical cartoons honest when they sit beside evaluated data tables and modern many-body calculations. Along the course arc, write one sentence linking this chapter backward and one forward: the discipline of those two sentences is as important as any single formula. Examination problems rarely ask for a literature review, but they do ask for the scale argument that literature reviews presuppose.

Synthesis problem

Write a half-page note that (i) restates the chapter's central scale or selection rule in one equation, (ii) evaluates it numerically for one concrete nucleus or energy, and (iii) names the neighbouring lecture that uses the result. Keep the note free of research-frontier speculation; the goal is fluency with the lecture physics itself. If an assumption fails (Born approximation, spherical mean field, allowed transition, one-dimensional barrier), state the failure mode in a final sentence. That habit converts passive reading into examination-ready understanding and makes the later research-frontier box read as an extension rather than a replacement.

Research frontier: microscopic tunneling and student projects

The Gamow tunneling picture of Lecture 17 (WKB barrier penetration for α emission) is the conceptual ancestor of modern fission and cluster decay calculations. Today's theory evaluates multidimensional potential-energy surfaces with collective inertias, rather than a one-dimensional Coulomb barrier with an ad hoc assault frequency [56, 55]. Preformation is reframed as the overlap of the parent ground state with a cluster-plus-daughter configuration. Gamow's original insight (exponential sensitivity to the barrier integral) survives, but quantitative half-lives demand controlled many-body input [25]. Remaining uncertainties of even $1\,\text{MeV}$ in barrier height can change spontaneous-fission or cluster-decay lifetimes by orders of magnitude, which is why microscopic inertia and dissipation modelling is as important as the barrier shape itself. The lecture's rectangular-barrier toy model remains the right first step for seeing that exponential dependence analytically.

In fission, the analogue of the assault frequency is a collective inertia along the fission path; different microscopic prescriptions can change lifetimes dramatically even when barriers look similar. Student reading should therefore separate barrier-height uncertainty from inertia uncertainty when comparing published spontaneous-fission half-lives.
Separating barrier-height errors from inertia errors is the practical checklist when comparing published spontaneous-fission half-lives.
Those two error sources rarely scale the same way with mass number, which students can illustrate with a simple WKB model.
Student directions. (1) Analytic estimate: evaluate a rectangular-barrier Gamow factor and show the exponential dependence on width and height. (2) Kinematics check: compare reduced-mass corrections $\mu \approx m_\alpha(A-4)/A$ for $A = 50$ and $A = 250$. (3) Literature reading: from a modern fission-theory review, list two sources of order-of-magnitude uncertainty in WKB half-lives [56].

17.12 Problems with solutions

Problem 17.1 — Turning point b. For α decay with $Z_1 = 90$, $Z_2 = 2$, and $Q = 5.0\,\text{MeV}$, calculate the classical turning point b. Use $e^2/(4\pi\varepsilon_0) = 1.44\,\text{MeV fm}$.

Solution 17.1.

Step 1. From Eq. (17.1):

$$b = \frac{Z_1 Z_2 e^2/(4\pi\varepsilon_0)}{Q}.$$

Step 2. Numerator:

$$Z_1 Z_2 \times \frac{e^2}{4\pi\varepsilon_0} = 90 \times 2 \times 1.44\,\text{MeV fm} = 259.2\,\text{MeV fm}.$$

Step 3.

$$b = \frac{259.2\,\text{MeV fm}}{5.0\,\text{MeV}} = 51.8\,\text{fm} \approx 52\,\text{fm}.$$

The α is classically forbidden for $R \lesssim r \lesssim 52\,\text{fm}$ [1, 6].

Problem 17.2 — Reduced mass and γ. For α decay of a heavy nucleus ($A = 220$), compute μ in u and estimate γ for a rectangular barrier with $V_0 - E = 20\,\text{MeV}$ and $\hbar c = 197\,\text{MeV fm}$.

Solution 17.2.

Step 1 (masses). $m_\alpha \approx 4\,\text{u}$, $M_d \approx 216\,\text{u}$.

Step 2 (reduced mass).

$$\mu = \frac{m_\alpha M_d}{m_\alpha + M_d} = \frac{4 \times 216}{220}\,\mathrm{u} = 3.927\,\mathrm{u}.$$

Since $M_d \gg m_\alpha$, μ is close to m_α.

Step 3 (convert to energy units). $1\,\mathrm{u}\,c^2 = 931.5\,\mathrm{MeV}$, so $\mu c^2 = 3.927 \times 931.5 \approx 3658\,\mathrm{MeV}$.

Step 4 (γ).

$$\gamma = \frac{\sqrt{2\mu(V_0 - E)}}{\hbar} = \frac{\sqrt{2 \times 3658 \times 20}}{197\,\mathrm{MeV\,fm}} = \frac{\sqrt{146320}}{197} \approx 1.94\,\mathrm{fm}^{-1}.$$

For a 10 fm wide rectangular barrier, $P \approx e^{-2\gamma d} \approx e^{-38.8} \approx 10^{-17}$ [1].

Problem 17.3 — Gamow factor sensitivity. If $2G = 60$ at one Q and a small increase in Q reduces $2G$ by 5% (from 60 to 57), by what factor does $P = e^{-2G}$ increase? What is the corresponding change in half-life?

Solution 17.3.

Step 1 (probability ratio).

$$\frac{P'}{P} = \frac{e^{-57}}{e^{-60}} = e^3 = 20.1.$$

Step 2 (half-life ratio). Since $t_{1/2} \propto 1/P$:

$$\frac{t'_{1/2}}{t_{1/2}} = \frac{P}{P'} = e^{-3} = 0.050.$$

The half-life drops by a factor of about 20. A few percent change in G (from a modest change in Q) can change lifetimes by orders of magnitude, explaining Geiger–Nuttall systematics [6, 1].

Problem 17.4 — Small Q/B limit for G. For $Z_1 = 90$, $Z_2 = 2$, $Q = 5.0\,\mathrm{MeV}$, and barrier height $B = 25\,\mathrm{MeV}$, evaluate the approximate Gamow factor $G \approx \pi Z_1 Z_2 e^2/(4\pi\varepsilon_0 \hbar v)$ with $v = \sqrt{2Q/\mu}$ and $\mu \approx 4\,\mathrm{u}$.

Solution 17.4.

Step 1 (μ in SI-consistent units). Use $\mu c^2 \approx 3727\,\mathrm{MeV}$ (since $\mu \approx 4\,\mathrm{u}$).

Step 2 (speed).

$$v = c\sqrt{\frac{2Q}{\mu c^2}} = c\sqrt{\frac{10}{3727}} = 0.052c \approx 1.56 \times 10^7\,\mathrm{m/s}.$$

Step 3 (G). Using $\hbar c = 197\,\text{MeV fm}$ and $Z_1 Z_2 e^2/(4\pi\varepsilon_0) = 259.2\,\text{MeV fm}$:

$$G \approx \frac{\pi \times 259.2}{197 \times (v/c)} = \frac{\pi \times 259.2}{197 \times 0.0518} \approx 79.8.$$

Thus the leading small-Q/B estimate gives $P \approx e^{-2G} = e^{-159.6} \approx 5 \times 10^{-70}$.

Step 4 (finite-radius check). Here $Q/B = 0.20$, so the exact bracket in Eq. (17.30) is

$$\arccos\sqrt{0.20} - \sqrt{0.20(0.80)} = 0.707.$$

The exact finite-radius result is

$$G_{\text{exact}} = \frac{259.2}{197}\sqrt{\frac{2(3727)}{5.0}}\,(0.707) \approx 35.9, \qquad P_{\text{exact}} \approx e^{-71.8} \approx 7 \times 10^{-32}.$$

The leading value $G \approx 79.8$ deliberately drops the surface correction $-2\sqrt{Q/B}$; at $Q/B = 0.20$ that correction is not small numerically. Both results show the exponential suppression, while the comparison demonstrates why quantitative half-lives require the finite nuclear radius [6, 14].

Problem 17.5 — Selection rules for ^{243}Am. The parent ^{243}Am has $J^\pi = \frac{5}{2}^-$. For daughter ^{239}Np levels with $J^\pi = \frac{5}{2}^+$ (gs), $\frac{7}{2}^+$, and $\frac{5}{2}^-$ (75 keV), list the lowest allowed ℓ for each and predict which branch dominates.

Solution 17.5.

Step 1 (parity rule). $\pi_i = \pi_f(-1)^\ell$. With $\pi_i = -$:

- $\frac{5}{2}^+$: need $(-1)^\ell = -1$, so ℓ odd. Lowest: $\ell = 1$.
- $\frac{7}{2}^+$: parity changes, so ℓ is odd. Angular coupling $|5/2 - 7/2| \le \ell \le 5/2 + 7/2$ gives $\ell \ge 1$; hence the lowest value is $\ell = 1$.
- $\frac{5}{2}^-$: parity match, ℓ even. $|5/2 - 5/2| = 0$, so $\ell = 0$ is allowed.

Step 2 (dominant branch). The $\ell = 0$ branch to $\frac{5}{2}^-$ at 74.660 keV has the thinnest barrier despite the reduced Q_i. DDEP gives 86.74(5)% intensity to this state versus 0.240(3)% to the ground state [20, 21, 28, 6].

Problem 17.6 — Assault frequency estimate. For $Q = 5\,\text{MeV}$, $V_0 = 35\,\text{MeV}$, $R = 7.5\,\text{fm}$, and $\mu \approx 4\,\text{u}$, estimate the assault frequency $f = v_0/(2R)$.

Solution 17.6.

Step 1 (interior speed).

$$v_0 = \sqrt{\frac{2(Q+V_0)}{\mu}} = \sqrt{\frac{2 \times 40\,\text{MeV}}{\mu}}.$$

With $\mu c^2 \approx 3727\,\text{MeV}$:

$$v_0 = c\sqrt{\frac{80}{3727}} \approx 0.147c \approx 4.4 \times 10^7\ \text{m/s}.$$

Step 2 (frequency).

$$f = \frac{v_0}{2R} = \frac{4.4 \times 10^7\,\text{m/s}}{2 \times 7.5 \times 10^{-15}\,\text{m}} = 2.9 \times 10^{21}\,\text{s}^{-1}.$$

Step 3 (interpretation). The α presents itself at the barrier $\sim 10^{21}$ times per second, but escapes only with probability $P = e^{-2G} \ll 1$, yielding a much smaller $\lambda = S_\alpha f P$ [6, 1].

Problem 17.7 — Turning point from Q. For $Q_\alpha = 5.0\,\text{MeV}$ and daughter $Z_d = 90$, estimate the outer classical turning point $b = Z_d Z_\alpha e^2/(4\pi\varepsilon_0 Q)$ using $e^2/(4\pi\varepsilon_0) = 1.44\,\text{MeV fm}$.

Solution 17.7.

$$b = \frac{90 \times 2 \times 1.44}{5.0} \approx 52\,\text{fm}.$$

Much larger than the nuclear radius $\sim 7\,\text{fm}$: the α must tunnel a thick Coulomb barrier [6, 1].

Chapter 18

Beta Decay: Modes, Q-Values, and the Neutrino

"I have done something very bad today by proposing a particle that cannot be detected; it is something no theorist should ever do."

W. Pauli (1930), on the neutrino

Beta decay rearranges N and Z at fixed A through the weak interaction. When the neutron-to-proton ratio N/Z deviates from the value required for stability on the chart of nuclides, the nucleus can lower its mass by converting a neutron into a proton (β^- decay) or a proton into a neutron (β^+ decay or electron capture). This lecture develops the three decay modes with their full reaction equations (including neutrinos), derives Q-values first from nuclear masses and then from tabulated atomic masses (showing every electron term until it cancels), compares the energetics of β^+ and electron capture, and explains the continuous β spectrum that forced Pauli's neutrino hypothesis and Fermi's theory of β decay [6, 1, 2, 5].

18.1 When the N/Z ratio is wrong

A stable nucleus requires a proper balance of neutrons and protons. For light nuclei the stable N/Z ratio is near unity; for heavy nuclei it rises toward ~ 1.5 because the Coulomb repulsion among protons must be compensated by extra neutrons [6, 1]. Lectures 4–5 already located that valley with the SEMF mass parabola; β decay is the weak process that moves a nucleus along an isobar toward the valley when the strong and electromagnetic binding leave a neighbouring charge lower in mass.

Two generic failures illustrate the logic:

1. **Excess of neutrons.** Consider $Z = 10$ with $N = 14$ on the stability line. If N increases to 16 while Z stays fixed, the N/Z ratio moves away from stability. A neutron inside the nucleus can convert to a proton, increasing Z by one and decreasing N by one. This is β^- decay.
2. **Excess of protons.** With the same $Z = 10$, $N = 14$ reference, suppose Z increases to 12 while N remains 14. Even though neutrons still outnumber protons, the proton count exceeds the stable value for this mass number. A proton converts to a neutron, decreasing Z by one. This is β^+ decay or electron capture.

Both conversions are mediated by the weak interaction. At the quark level the fundamental processes are

$$n \to p + e^- + \bar{\nu}_e, \qquad p \to n + e^+ + \nu_e, \qquad p + e^- \to n + \nu_e. \tag{18.1}$$

Electrons and positrons are not pre-existing constituents of the nucleus; they are created at the instant of decay, in contrast with α particles in α decay [6, 2]. When both β^+ and electron capture are energetically open, capture often dominates at low Q because it avoids the $2m_ec^2$ threshold of positron emission (derived below) and uses atomic electrons already present at the nucleus [1, 6].

18.2 Three modes of beta decay

For a parent nucleus ${}^A_Z X$ the three observed modes are:

Beta minus (β^-).

$$ {}^A_Z X \;\to\; {}^{A}_{Z+1} Y + e^- + \bar{\nu}_e. \tag{18.2}$$

The proton number increases by one; a neutron becomes a proton. The free neutron decays by this mode with mean lifetime $\tau_n \approx 879\,\mathrm{s}$, not half-life; the corresponding half-life is $t_{1/2} = \tau_n \ln 2 \approx 609\,\mathrm{s}$. Its energy release is $Q \approx 0.782\,\mathrm{MeV}$ [6, 20].

Beta plus (β^+).

$$ {}^A_Z X \;\to\; {}^{A}_{Z-1} Y + e^+ + \nu_e. \tag{18.3}$$

The proton number decreases by one. A *free* proton cannot undergo β^+ decay because the three-body final rest mass $m_n + m_e + m_\nu$ exceeds m_p. Inside a proton-rich nucleus, however, the change in nuclear binding energy can make the conversion exothermic. This does not imply decay of an isolated proton. Examples include ${}^{11}\mathrm{C}$ and ${}^{18}\mathrm{F}$, used in positron-emission tomography [6, 1, 21].

Electron capture (EC).

$$ {}^A_Z X + e^- \;\to\; {}^{A}_{Z-1} Y + \nu_e. \tag{18.4}$$

The nucleus captures an atomic electron (usually from the K shell), converting a proton to a neutron. The daughter atom is left with a vacancy that fills by characteristic X-rays or Auger electrons. EC and β^+ both reduce Z by one and convert a proton to a neutron [6, 1].

Key idea

β^-: $Z \to Z+1$, antineutrino $\bar{\nu}_e$ emitted. β^+ and EC: $Z \to Z-1$, neutrino ν_e emitted. Free neutron decays; free proton does not. All three modes require a light, neutral, spin-$\frac{1}{2}$ neutrino (or antineutrino) to conserve energy, momentum, and angular momentum [6, 2].

Historical note

In December 1930 Wolfgang Pauli proposed an undetected neutral particle to save energy and angular momentum in β decay. Enrico Fermi named it the *neutrino* and built the first quantitative theory in 1934 [6, 2, 5]. Direct detection came only in 1956 (Reines and Cowan); today we distinguish ν_e from $\bar{\nu}_e$ in each decay mode as written above.

18.3 Q-value for β^-: nuclear masses, then atomic masses

Consider reaction (18.2). The Q-value is the difference between initial and final rest mass energies (excluding the negligible nuclear recoil):

$$Q_{\beta^-} = \Big[m'(^A_Z X) - m'(^{A}_{Z+1}Y) - m_e - m(\bar{\nu}_e)\Big]c^2, \tag{18.5}$$

where $m'(\cdot)$ denotes *nuclear* mass (nucleons only). The antineutrino mass is of order eV and is neglected: $m(\bar{\nu}_e) \approx 0$.

Step 1: express nuclear masses through atomic masses. Tables list *atomic* masses $m(\cdot)$ of neutral atoms. A neutral atom $^A_Z X$ contains the nucleus plus Z orbital electrons. Exactly, $m' = m - Zm_e + B_{\text{at}}/c^2$, where B_{at} is the positive total electronic binding energy. Neglecting the small B_{at}/c^2 mass correction for the present derivation,

$$m'(^A_Z X) = m(^A_Z X) - Zm_e. \tag{18.6}$$

For the daughter $^{A}_{Z+1}Y$, which has $Z+1$ electrons in its neutral atom,

$$m'(^{A}_{Z+1}Y) = m(^{A}_{Z+1}Y) - (Z+1)m_e. \tag{18.7}$$

Step 2: substitute into (18.5).

$$\begin{aligned} Q_{\beta^-} &= \Big[(m(X) - Zm_e) - (m(Y) - (Z+1)m_e) - m_e\Big]c^2 \\ &= \Big[m(X) - Zm_e - m(Y) + (Z+1)m_e - m_e\Big]c^2. \end{aligned} \tag{18.8}$$

Step 3: collect electron terms. The electron mass terms are

$$-Zm_e + (Z+1)m_e - m_e = -Zm_e + Zm_e + m_e - m_e = 0. \tag{18.9}$$

Every explicit electron mass cancels. Therefore

$$Q_{\beta^-} = [m(^A_Z X) - m(^{A}_{Z+1}Y)]c^2, \tag{18.10}$$

using tabulated atomic masses directly [6, 1]. The energy Q_{β^-} is shared by the electron, antineutrino, and (negligibly) the recoiling daughter:

$$Q_{\beta^-} = K_e + E_\nu, \qquad K_{\max} \simeq Q_{\beta^-} \quad \text{(massless-neutrino, infinite-daughter-mass limit)}. \tag{18.11}$$

For a parent at rest, the exact endpoint is obtained from relativistic energy–momentum conservation. If M_i and M_f are the consistently defined initial and daughter masses, then

$$K_{\max} = \frac{M_i^2 + m_e^2 - (M_f + m_\nu)^2}{2M_i} c^2 - m_e c^2. \tag{18.12}$$

For $M_f \gg m_e$, this is $K_{\max} \approx Q - m_\nu c^2 - p_{e,\max}^2/(2M_f)$. Thus the endpoint is slightly below Q even for $m_\nu = 0$, because the daughter recoils.

Remark

Small differences in electronic binding between parent and daughter atoms (typically keV) are omitted above. Precision work sometimes retains them; for course estimates they are negligible compared with $Q \sim 1$–$3\,\text{MeV}$ [6].

18.4 Q-value for β^+: the $2m_ec^2$ threshold

For reaction (18.3), with nuclear masses and $m(\nu_e) \approx 0$,

$$Q_{\beta^+} = \Big[m'(X) - m'(Y) - m_e\Big]c^2, \tag{18.13}$$

where m_e is the positron mass ($m_{e^+} = m_{e^-}$). Using (18.6) with $m'(Y) = m(Y) - (Z-1)m_e$,

$$\begin{aligned} Q_{\beta^+} &= \Big[(m(X) - Zm_e) - (m(Y) - (Z-1)m_e) - m_e\Big]c^2 \\ &= \Big[m(X) - Zm_e - m(Y) + (Z-1)m_e - m_e\Big]c^2. \end{aligned} \tag{18.14}$$

Collecting electron terms:

$$-Zm_e + (Z-1)m_e - m_e = -Zm_e + Zm_e - m_e - m_e = -2m_e. \tag{18.15}$$

Hence

$$Q_{\beta^+} = [m(X) - m(Y)]c^2 - 2m_ec^2. \tag{18.16}$$

Positron emission requires not only $m(X) > m(Y)$ but

$$m(X) - m(Y) > 2m_e \quad \Longleftrightarrow \quad Q_{\beta^+} > 0, \tag{18.17}$$

with threshold $2m_ec^2 = 1.022\,\text{MeV}$. Typical β Q-values are only a few MeV, so this threshold is a substantial fraction of the available energy [6, 1].

18.5 Q-value for electron capture

With neutral-atom masses, the *atomic* EC Q-value is

$$Q_{\text{EC}}^{\text{atom}} = [m(X) - m(Y)]c^2. \tag{18.18}$$

This quantity is a property of the neutral parent and daughter ground states; one must not subtract a shell binding energy from its definition. For capture from shell i to a daughter nuclear level of excitation E_f^*, energy conservation instead gives the channel neutrino energy

$$E_{\nu,i} = Q_{\mathrm{EC}}^{\mathrm{atom}} - E_f^* - B_i - E_{\mathrm{rec}}, \tag{18.19}$$

up to small differences associated with the final atomic configuration. Here B_i is the positive binding energy of the captured electron and $E_{\mathrm{rec}} \simeq p_\nu^2/(2M_f)$. The daughter vacancy subsequently releases X rays and/or Auger electrons. Thus B_i lowers a particular neutrino-line energy, while the total energy release including atomic relaxation remains $Q_{\mathrm{EC}}^{\mathrm{atom}} - E_f^*$.

Caution

B_i is *not* universally a few eV. For K-shell capture in medium and heavy elements it can be tens of keV; for iodine, $B_K \approx 33.17\,\mathrm{keV}$. Always use the binding energy of the shell from which capture occurs [6, 1].

Comparison of β^+ and EC. Subtracting (18.16) from the atomic EC Q-value,

$$Q_{\mathrm{EC}}^{\mathrm{atom}} - Q_{\beta^+} = 2m_e c^2 = 1.022\,\mathrm{MeV}. \tag{18.20}$$

Whenever

$$0 < [m(X) - m(Y)]c^2 < 2m_e c^2, \tag{18.21}$$

electron capture is energetically allowed ($Q_{\mathrm{EC}}^{\mathrm{atom}} > 0$) for at least one accessible shell and daughter state, but positron emission is forbidden ($Q_{\beta^+} < 0$). This situation is common for moderately proton-rich nuclei [6, 1, 20].

Key idea

Atomic-mass formulas: $Q_{\beta^-} = [m(X) - m(Y)]c^2$, $Q_{\beta^+} = [m(X) - m(Y) - 2m_e]c^2$, $Q_{\mathrm{EC}}^{\mathrm{atom}} = [m(X) - m(Y)]c^2$. EC is favored over β^+ by exactly $2m_e c^2$ at the atomic Q-value level. For capture from shell i, however, $E_{\nu,i} = Q_{\mathrm{EC}}^{\mathrm{atom}} - E_f^* - B_i - E_{\mathrm{rec}}$.

18.6 Continuous spectrum: ^{64}Cu and the missing energy

The Q-value appears as kinetic energy shared among the light decay products. The recoiling daughter nucleus carries negligible energy ($\sim$ keV or less). For β^- decay the electron and antineutrino share Q_{β^-}; neither body is at rest in general, so the electron kinetic energy K_e is *continuous* from near zero up to $K_{\max} \approx Q$.

Copper-64 is a remarkable laboratory example: it decays by both β^- and β^+ (and EC), so both spectra in Figs. 18.1 and 18.2 can be measured on one isotope [1, 21].

In the 1920s, before the neutrino was known, a *discrete* electron line was expected. If only the electron and heavy daughter emerged, two-body kinematics would fix the electron energy (as in α decay). Calorimetric experiments showed that the continuous distribution is intrinsic to the decay, not caused by energy loss in the source [6, 2]. Electrons were often observed with

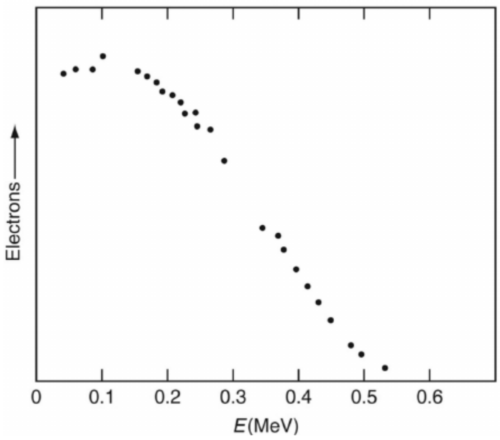


Figure 18.1: Energy spectrum of electrons from β^- decay of ^{64}Cu to ^{64}Zn. The endpoint near 0.58 MeV equals Q_{β^-} for the ground-state transition. Data from Langer, Moffat, and Price [29], as after [1, 6].

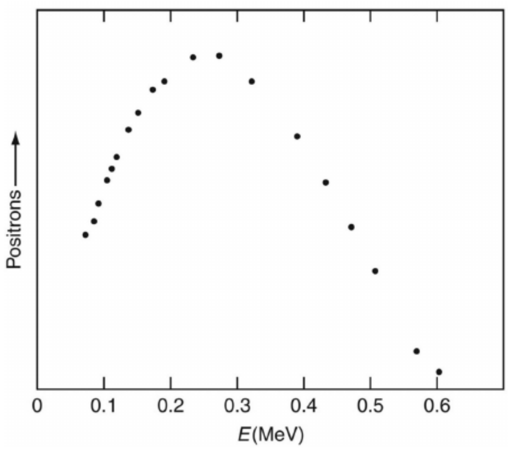


Figure 18.2: Energy spectrum of positrons from β^+ decay of ^{64}Cu to ^{64}Ni. Endpoint near 0.65 MeV. Same experimental reference as Fig. 18.1 [29, 1, 6].

kinetic energy well below Q: where was the missing energy?

Pauli's solution (1930): a second, electrically neutral, highly penetrating particle carries away the "missing" energy and spin. Fermi (1934) incorporated the neutrino into a quantitative theory. The continuous spectrum is therefore a *three-body* phase-space effect, not an experimental artefact [6, 2, 5].

Caution

The continuous spectrum is three-body kinematics, not instrumental smearing. The endpoint energy measures Q (and, in precision experiments near the endpoint, constrains the neutrino mass; see Lecture 19). Internal conversion and atomic effects can add low-energy structure, but the broad envelope is phase space [6, 14].

18.7 Toward Fermi theory: golden rule and plane-wave leptons

Fermi treated the weak interaction as a small perturbation on nuclear states. The transition rate from initial state ψ_i to final state ψ_f is Fermi's golden rule [6, 1, 14]:

$$\lambda = \frac{2\pi}{\hbar}\,|\langle\psi_f|H_{\text{int}}|\psi_i\rangle|^2\,\frac{\mathrm{d}n}{\mathrm{d}E_f}, \tag{18.22}$$

where $\mathrm{d}n/\mathrm{d}E_f$ is the density of final states. For β^- decay,

$$\psi_i = \psi_p, \qquad \psi_f = \psi_d\,\psi_e\,\psi_\nu, \tag{18.23}$$

with ψ_p and ψ_d the parent and daughter nuclear wave functions.

The electron and antineutrino are treated as free plane waves normalized in volume V:

$$\psi_e(\boldsymbol{r}) = \frac{1}{\sqrt{V}}e^{i\boldsymbol{p}_e\cdot\boldsymbol{r}/\hbar} = \frac{1}{\sqrt{V}}\left(1 + \frac{i\boldsymbol{p}_e\cdot\boldsymbol{r}}{\hbar} + \cdots\right), \tag{18.24}$$

and similarly for ψ_ν. For typical β momenta ($p \sim 1\,\text{MeV}/c$) and nuclear radius $R \sim 5\,\text{fm}$, the dimensionless parameter $pR/\hbar \sim 0.03 \ll 1$. The *allowed* approximation keeps only the constant term ($\ell = 0$ for the lepton pair). The nuclear physics is then absorbed into a matrix element M_{fi} nearly independent of lepton energy, and the spectrum shape comes entirely from the lepton phase space developed in Lecture 19 [6, 1].

Key idea

Students should remember:

- Wrong $N/Z \Rightarrow \beta^-$ (neutron rich) or β^+/EC (proton rich).
- Free neutron decays; free proton does not ($m_n > m_p$).
- Atomic-mass $Q_{\beta^-} = [m(X) - m(Y)]c^2$; all electron masses cancel.
- Q_{β^+} has an extra $-2m_ec^2$; EC is $\approx 1.022\,\text{MeV}$ more favorable.
- Continuous spectrum $\Rightarrow$ three bodies $\Rightarrow$ neutrino.

- Golden rule: rate $\propto |M_{fi}|^2 \times$ (phase space).

18.7.1 Electron capture and double β

Electron capture probes s-electron density at the nucleus and therefore competes with β^+ once Q_{EC} is open. Double β decay, though second-order weak, is observed (or tightly limited) in even–even systems where single β is pairing-blocked; it is a bridge between nuclear structure and lepton-number conservation tests [6, 1, 14].

18.8 Deeper physics: Q_β, the continuous spectrum, and the neutrino

18.8.1 Energy release in β^- decay

For β^- decay $X \to Y + e^- + \bar{\nu}_e$, the nuclear Q-value follows the same mass-difference pattern as α decay,

$$Q_{\beta^-} = [M(X) - M(Y)]c^2, \tag{18.25}$$

using atomic masses of neutral atoms [6, 1]. The electron and antineutrino share this energy continuously because the three-body final state has a continuum of momentum partitions. The electron kinetic energy K therefore spans $0 \le K \le Q$ (minus small recoil and atomic binding corrections). A discrete line at $K = Q$ would imply a two-body final state; its absence led Pauli to postulate the neutrino [2, 14].

Free neutron decay

The simplest case is $n \to p + e^- + \bar{\nu}_e$ with $Q_{\beta^-} \approx 0.782\,\text{MeV}$ [6, 20]. The measured mean lifetime $\tau_n \approx 879\,\text{s}$ maps to $t_{1/2} \approx 609\,\text{s}$. Proton recoil lowers the electron endpoint by only $\sim 0.1\,\text{keV}$, negligible on the MeV scale. This decay is the laboratory standard for testing the V-A structure of the weak interaction [2].

18.8.2 Why the spectrum is continuous

In a two-body decay (for example α emission) energy and momentum fix the final kinetic energies uniquely. In β decay there are three light final particles. For fixed electron momentum p_e, the antineutrino can carry any momentum consistent with energy conservation, so many final configurations share the same K but differ in neutrino energy. Fermi's golden rule (Lecture 19) counts these configurations: the available phase volume grows with K and with $(Q - K)$ when $m_\nu = 0$, producing the characteristic rising-then-falling spectrum [6, 14].

Remark

Electron capture (EC) competes with β^+ emission when $Q_{\text{EC}} > 0$. The observable Q for EC is reduced by the binding energy of the captured atomic electron, typically tens of keV. For heavy nuclei the continuous β spectrum shape is unchanged in principle, but atomic and chemical effects modify the low-energy end [1, 6].

Key idea

From spectrum shape to Fermi theory. The continuous β spectrum is not an experimental artefact: calorimetric measurements on ^{210}Bi confirmed that the electrons themselves carry a continuum of energies [1]. Lecture 19 converts this observation into a quantitative phase-space formula and Kurie plots that separate nuclear matrix elements from kinematics [6].

18.8.3 Limits of the two-body analogy

Unlike α decay, β spectra depend on the weak interaction structure (V-A) and on atomic Coulomb corrections $F(Z, K)$. The endpoint $K_{\text{max}} \approx Q$ is robust, but the shape near $K = 0$ is modified for high-Z emitters because the Coulomb field favours low-energy electrons in β^- decay [6, 1]. Neutrino mass $m_\nu \neq 0$ replaces $(Q - K)^2$ by $(Q - K)\sqrt{(Q - K)^2 - m_\nu^2 c^4}$ and shifts the endpoint to $Q - m_\nu c^2$; this effect is searched at the sub-eV scale in tritium decay [2, 14]. Double β decay bypasses the continuous spectrum when single β is blocked by mass parabola considerations (Lecture 5) [6, 5].

18.8.4 Angular correlation and the V-A structure

The Wu experiment on polarised ^{60}Co showed that β decay violates parity: electrons emerge preferentially opposite to the nuclear spin [2, 6]. This selects the V-A form of the weak current and explains why only left-handed neutrinos appear at low energy. The angular distribution modifies the spectrum shape beyond pure phase space but does not remove the continuum: even with parity violation, three-body kinematics allows a range of electron energies at fixed nuclear orientation [14, 5]. Positron emitters used in PET (^{11}C, ^{18}F) and β^- sources in radiotherapy rely on the same Q_β bookkeeping developed in this lecture [6, 1].

Electron capture competes with β^+ when Q_{EC} is small and binding energies favour capture. In stellar environments, continuum electron capture alters effective rates: a reminder that laboratory Q values are necessary but not always sufficient for astrophysical weak rates. Delayed-neutron branching ratios add another applied data need for reactor and r-process networks.

18.9 Further physics from the literature

Free neutron decay. The simplest β^- example is free neutron decay, $n \to p + e^- + \bar{\nu}_e$, with energy release about 0.782 MeV; proton recoil makes the exact electron endpoint slightly lower. The measured neutron mean lifetime is $\tau_n \approx 879$ s, corresponding to $t_{1/2} \approx 609$ s. These observables provide precise tests of the V-A structure of the weak interaction [6, 2, 20].

Double beta decay. When single β decay is blocked (for example, when the intermediate nucleus would have higher mass), some even–even nuclei can decay by double β decay. Two-neutrino double β ($2\nu\beta\beta$) has been observed; neutrinoless double β ($0\nu\beta\beta$), if found, would prove that neutrinos are Majorana particles and would constrain the absolute neutrino mass scale [6, 1, 14, 5].

Parity violation. The Wu experiment (1957) showed that β decay does not conserve parity: electrons are preferentially emitted opposite to the nuclear spin in polarised ^{60}Co decay. This established the V-A form of the weak current and explained why only left-handed neutrinos and right-handed antineutrinos appear in β decay at low energy [2, 5, 6].

Medical and geochemical applications. Positron emitters (^{11}C, ^{18}F) and β^- sources are used in PET and radiotherapy. The β^- decay of ^{14}C provides radiocarbon dating. EC of ^{40}K contributes to the Earth's internal heat [6, 1, 13].

From the literature

Calorimetric measurements on ^{210}Bi proved the continuous spectrum is a property of the decay electrons themselves, not energy loss before detection [1]. Representative β^-, β^+, and EC processes with Q and half-lives for order-of-magnitude estimates are tabulated in [6].

18.9.1 From the continuous spectrum to Fermi phase space (Lecture 19)

Pauli's neutrino hypothesis explains energy and momentum conservation in three-body decay. Fermi theory makes the spectrum quantitative: for each electron kinetic energy K, the available neutrino energy is $Q - K$ (when $m_\nu = 0$), and the phase-space volume grows as $K^2(Q-K)^2$ in the non-relativistic limit [6, 14]. The rising low-energy part of the spectrum reflects increasing electron phase space; the fall near $K = Q$ reflects shrinking neutrino phase space. Coulomb correction $F(Z, K)$ distorts the shape for high-Z emitters but does not remove the continuum [1].

Endpoint shift from a finite neutrino mass

If $m_\nu c^2 = 1\,\text{eV}$, the electron endpoint moves from $K_{\max} = Q$ to $K_{\max} \approx Q - m_\nu c^2$. For tritium with $Q \approx 18.6\,\text{keV}$, a $1\,\text{eV}$ neutrino mass shifts the endpoint by $\sim 0.005\%$, requiring precision spectroscopy to resolve [2, 6]. The Kurie plot remains linear in K only when the phase-space factor $(Q-K)\sqrt{(Q-K)^2 - m_\nu^2 c^4}$ is used instead of $(Q-K)^2$.

18.9.2 β cascades toward reactions and reactors

Each fission fragment β-decays toward stability (Lecture 24), producing reactor antineutrinos and decay heat. The same continuous spectrum shapes reactor $\bar{\nu}_e$ flux calculations used in oscillation experiments [14, 2]. β-delayed neutron emission from rare isotopes links weak decay to (n, γ) reaction networks in the r-process (Lecture 22) [53]. Forbidden transitions and atomic effects can mimic sterile-neutrino distortions if not modelled before invoking new physics [50].

Key idea

Arc through the course. Continuous β spectra (Lecture 18) become Kurie plots and $\log ft$ values (Lecture 19), then γ cascades (Lecture 20), then compound-nucleus β-delayed processes in reactions and fission (Lectures 21–24). The weak interaction is the slow step in the solar pp chain (Lecture 26) [36, 6].

18.9.3 Reactor antineutrinos and the β-decay network

Each fission event produces neutron-rich fragments that β-decay toward stability with a broad distribution of Q_β values. The cumulative $\bar{\nu}_e$ flux is therefore a weighted sum of many continuous spectra, not a single Kurie line [14, 2]. Precision reactor oscillation experiments require forbidden transitions and quasi-degenerate fragment chains to be modelled at the few-percent level. β-delayed neutron emission from exotic waiting-point nuclei links the same weak process to (n, γ) reaction networks in the r-process (Lecture 22) [53]. The neutrino hypothesis of Lecture 18 is therefore not historical background alone: it is the bookkeeping tool that makes three-body phase space calculable.

18.9.4 Medical and geochemical β sources

The β^- decay of ^{14}C with $Q_\beta \approx 156\,\mathrm{keV}$ provides radiocarbon dating; electron capture in ^{40}K contributes to the Earth's internal heat budget [6, 1]. These applications use the integrated decay rate $\lambda = \ln 2/t_{1/2}$, not the spectrum shape, but the continuous spectrum confirms that each decay carries a variable electron energy. In reactor physics, fragment β chains after fission (Lecture 24) produce both decay heat and antineutrinos; the cumulative spectrum is a superposition of thousands of channels with different Q values and $\log ft$ strengths [35, 14]. Understanding the allowed spectrum of Lecture 18 is prerequisite to Fermi theory (Lecture 19) and to all precision weak-interaction work that follows.

18.10 Deeper reading: endpoints, neutrinos, and delayed neutrons

The continuous β spectrum and the neutrino hypothesis of Lecture 18 underwrite endpoint searches for the absolute neutrino mass and for sterile-neutrino distortions. Atomic final-state distributions in tritium and pile-up systematics in calorimetric ^{163}Ho experiments are modern versions of the same spectral-shape problem.

Beta-delayed neutron emission appears when Q_β exceeds S_n of the daughter: a direct link between Lecture 5 masses and Lecture 24 reactor delayed-neutron fractions. Always compute Q from atomic masses with the correct electron-mass bookkeeping before discussing spectral endpoints.

Kurie plots and endpoints

A Kurie plot linearises an allowed spectrum so the intercept reads Q. Detector response, screening, and forbidden shape factors curve that line. Endpoint neutrino-mass searches are battles against those systematics at the electronvolt level. Nuclear physicists contribute clean Q values from traps; atomic and detector physicists own much of the shape.

Reactor antineutrino connection

Reactor spectra fold fission yields (Lecture 24) with β branching ratios and spectral shapes from this chapter. Uncertainties in forbidden transitions near endpoints feed sterile-neutrino anomaly discussions. The continuous spectrum is therefore both fundamental and applied.

Endpoint arithmetic

Using atomic masses, compute Q_{β^-} for ^{14}C and Q_{EC} for a small-Q case such as ^{7}Be. Sketch the allowed spectrum shape $\propto pE(Q-E)^2$ for the former and discuss why capture dominates when positron emission is energetically closed or suppressed. Estimate the Kurie intercept shift for a 1 eV neutrino mass relative to a kilo-electronvolt-scale Q: the fractional effect is tiny, which is why endpoint experiments are heroic.

Tie delayed neutrons to Lecture 5: when $Q_\beta > S_n$(daughter), a branch can emit neutrons after β decay. That branch is nuclear structure plus mass arithmetic, and it feeds reactor control data in Lecture 24.

The continuous β spectrum forced the neutrino's invention and still anchors endpoint mass experiments. Nuclear Q values are the indispensable calibration.

In practice, rework one numerical estimate from the lecture with a different parameter set and compare the shift to the quoted experimental uncertainty. If the shift exceeds the uncertainty, the estimate is teaching a scale, not delivering a fit. That distinction keeps theoretical cartoons honest when they sit beside evaluated data tables and modern many-body calculations. Along the course arc, write one sentence linking this chapter backward and one forward: the discipline of those two sentences is as important as any single formula. Examination problems rarely ask for a literature review, but they do ask for the scale argument that literature reviews presuppose.

Synthesis problem

Write a half-page note that (i) restates the chapter's central scale or selection rule in one equation, (ii) evaluates it numerically for one concrete nucleus or energy, and (iii) names the neighbouring lecture that uses the result. Keep the note free of research-frontier speculation; the goal is fluency with the lecture physics itself. If an assumption fails (Born approximation, spherical mean field, allowed transition, one-dimensional barrier), state the failure mode in a final sentence. That habit converts passive reading into examination-ready understanding and makes the later research-frontier box read as an extension rather than a replacement.

Research frontier: precision β decay and student projects

The continuous β spectrum and neutrino hypothesis of Lecture 18 underpin today's searches for the absolute neutrino-mass scale (tritium endpoint spectroscopy and ^{163}Ho calorimetry) and for sterile-neutrino spectral distortions. Rare-isotope beams enable β-delayed neutron spectroscopy that constrains r-process paths and reactor antineutrino inventories, while ultra-sensitive $0\nu\beta\beta$ experiments use the same weak-interaction language at a deeper level [50, 51, 53]. Atomic-mass Q values from traps remain the cleanest absolute energy calibrations for these programmes. Open issues include atomic final-state distributions in tritium, incomplete β-delayed neutron branching ratios for exotic precursors, and the nuclear-matrix-element bottleneck for interpreting a future $0\nu\beta\beta$ signal as a Majorana mass. Community roadmaps rank ton-scale double-beta experiments among the highest-priority construction projects [54].

Reactor antineutrino spectra and sterile-neutrino searches add applied motivation for precision β shapes. Forbidden transitions and atomic effects can mimic sterile-neutrino distortions if not modelled carefully. The lecture Kurie analysis is the right starting point for understanding why those systematics matter.

Atomic final-state effects and forbidden shape factors must be controlled before a spectral distortion is attributed to new neutrino physics.

Connecting those applied spectra back to the lecture continuous beta-spectrum derivation keeps the physics organised.

Student directions. (1) Data exercise: compute Q_{β^-} from atomic masses for ^{14}C and ^{37}Ar electron capture. (2) Sensitivity estimate: estimate the Kurie endpoint shift corresponding to a 0.5 eV neutrino mass in tritium. (3) Literature note: summarise one β-delayed neutron measurement relevant to reactor antineutrino anomalies or r-process waiting points.

18.11 Problems with solutions

Problem 18.1 — Derive Q_{β^-} from atomic masses. Show step by step that for ${}^A_Z X \to {}^{A}_{Z+1}Y + e^- + \bar{\nu}_e$,

$$Q_{\beta^-} = \left[m({}^A_Z X) - m({}^{A}_{Z+1}Y)\right]c^2, \tag{18.26}$$

starting from nuclear masses and neglecting $m(\bar{\nu}_e)$ and electronic binding [6].

Solution 18.1. **Step 1.** Write the nuclear-mass Q-value:

$$Q_{\beta^-} = \left[m'(X) - m'(Y) - m_e\right]c^2.$$

Step 2. Relate nuclear and atomic masses:

$$m'(X) = m(X) - Zm_e, \qquad m'(Y) = m(Y) - (Z+1)m_e.$$

Step 3. Substitute:

$$Q_{\beta^-} = \left[m(X) - Zm_e - m(Y) + (Z+1)m_e - m_e\right]c^2.$$

Step 4. Simplify the electron terms:

$$-Zm_e + (Z+1)m_e - m_e = (-Z + Z + 1 - 1)m_e = 0.$$

Step 5. Conclude $Q_{\beta^-} = [m(X) - m(Y)]c^2$. All three electron masses (one in the daughter atom, one emitted, one subtracted explicitly) cancel identically [6, 1].

Problem 18.2 — β^+ threshold and EC window. The atomic mass excess of a parent exceeds that of the daughter by $\Delta m\, c^2 = 1.50\,\text{MeV}$ (parent heavier). Can the nucleus decay by β^+? By EC to the daughter ground state, first neglecting shell binding and recoil? What if $\Delta m\, c^2 = 0.80\,\text{MeV}$?

Solution 18.2. Case A: $\Delta m\, c^2 = 1.50\,\text{MeV}$.

$Q_{\beta^+} = 1.50 - 1.022 = 0.48\,\text{MeV} > 0$: positron emission is allowed.

$Q_{\text{EC}}^{\text{atom}} = 1.50\,\text{MeV} > 0$: EC is also allowed. Neglecting recoil, capture from shell i gives $E_{\nu,i} \approx 1.50\,\text{MeV} - B_i$. Both modes compete; the atomic EC Q-value exceeds Q_{β^+} by $2m_ec^2$ [6, 1].

Case B: $\Delta m\, c^2 = 0.80\,\text{MeV}$.

$Q_{\beta^+} = 0.80 - 1.022 = -0.22\,\text{MeV} < 0$: positron emission is forbidden.

$Q_{\text{EC}}^{\text{atom}} = 0.80\,\text{MeV} > 0$: EC is open provided an accessible shell and daughter level satisfy $B_i + E_f^* + E_{\text{rec}} < 0.80\,\text{MeV}$. This illustrates the window $0 < \Delta m\, c^2 < 2m_ec^2$ where EC alone occurs [1, 20].

Problem 18.3 — ^{64}Cu endpoint. Using atomic mass tables, estimate Q_{β^-} for $^{64}\text{Cu} \to {}^{64}\text{Zn} + e^- + \bar{\nu}_e$ and compare with the endpoint in Fig. 18.1.

Solution 18.3. From ENSDF/NuDat [20, 21], the ground-state Q_{β^-} for ^{64}Cu is 0.579 MeV. The tabulated value is 0.579 MeV from atomic masses, in agreement with the Langer *et al.* spectrum endpoint near 0.58 MeV (Fig. 18.1) [1, 6]. The equality $K_{\max} = Q$ holds only in the infinite-daughter-mass, massless-neutrino limit. In reality, $K_{\max} \approx Q - p_{e,\max}^2/(2M_f) - m_\nu c^2$, so recoil lowers the endpoint slightly. The spectrum falls to zero there because the neutrino phase space vanishes (derived fully in Lecture 19).

Problem 18.4 — Why is the spectrum continuous?. Explain in three sentences why α decay gives discrete α energies but β^- decay gives a continuous electron spectrum, even for a ground-state to ground-state transition.

Solution 18.4. α decay is a two-body process (α + daughter nucleus). Energy and momentum conservation fix the α kinetic energy uniquely (up to small recoil corrections). β^- decay is a three-body process (electron + antineutrino + daughter). One continuous parameter (for example the antineutrino energy) remains undetermined, so the electron kinetic energy ranges continuously from 0 to Q [6, 2].

Problem 18.5 — Neutrino vs antineutrino labels. In β^-, β^+, and electron capture, which light fermions are emitted or absorbed? Why does this matter for lepton number?

Solution 18.5. β^- emits $e^- + \bar{\nu}_e$; β^+ emits $e^+ + \nu_e$; EC absorbs e^- and emits ν_e. Lepton number is conserved in each case ($L_e = +1$ for e^-, ν_e and $L_e = -1$ for $e^+, \bar{\nu}_e$) [6, 1].

Problem 18.6 — Comparing Q_{EC} and $Q_{\beta+}$. Derive the relation $Q_{\beta+} = Q_{\mathrm{EC}} - 2m_e c^2$ (neglecting binding and recoil) and state the window where only EC is open.

Solution 18.6. Atomic-mass bookkeeping gives $Q_{\mathrm{EC}} = [m(X) - m(Y)]c^2$ while positron emission must create e^+ and leave one fewer atomic electron, costing $2m_e c^2$. Thus $Q_{\beta+} = Q_{\mathrm{EC}} - 2m_e c^2$. When $0 < Q_{\mathrm{EC}} < 2m_e c^2$, only EC (and not β^+) is open [6, 1].

Chapter 19

Fermi Theory of Beta Decay

"Fermi's theory of β decay is the prototype of all weak-interaction calculations: a contact coupling times phase space."
after E. Fermi (1934)

Lecture 18 introduced the continuous β spectrum and Fermi's golden rule. This lecture completes the original course derivation: plane-wave leptons and the allowed approximation, phase-space counting for electron and neutrino (with explicit volume cancellation), conversion to the electron kinetic energy K with the *correct* momentum–energy factor structure, the Coulomb Fermi function $F(Z, K)$, Fermi–Kurie plots, and the classification of allowed Fermi and Gamow–Teller transitions versus forbidden decays [6, 1, 14, 2].

19.1 Matrix element and plane-wave leptons

The partial decay rate depends on the matrix element of the weak interaction H_{int} between initial and final states:

$$|M_{if}|^2 = |\langle \psi_{\text{d}}\, \psi_e\, \psi_\nu | H_{\text{int}} | \psi_{\text{p}} \rangle|^2 \,. \tag{19.1}$$

In Fermi's original contact form the leptons are created at a point, so the nuclear matrix element factorises from the lepton wave-function normalisation once plane waves (or Coulomb-distorted continuum waves) are chosen. That factorisation is what lets the spectrum shape be fixed largely by phase space and $F(Z, K)$, with nuclear structure entering mainly through an overall allowed matrix element [6, 14]. For β^- decay, ψ_{p} is the parent nucleus, ψ_{d} the daughter, and ψ_e, ψ_ν the electron and antineutrino.

Free leptons are modelled as plane waves normalised in a large volume V:

$$\psi_e(\boldsymbol{r}) = \frac{1}{\sqrt{V}} e^{i\boldsymbol{p}_e \cdot \boldsymbol{r}/\hbar}, \qquad \psi_\nu(\boldsymbol{r}) = \frac{1}{\sqrt{V}} e^{i\boldsymbol{p}_\nu \cdot \boldsymbol{r}/\hbar}. \tag{19.2}$$

Expanding the electron factor,

$$\frac{1}{\sqrt{V}} e^{i\boldsymbol{p}_e \cdot \boldsymbol{r}/\hbar} = \frac{1}{\sqrt{V}} \left(1 + \frac{i\boldsymbol{p}_e \cdot \boldsymbol{r}}{\hbar} + \frac{(i\boldsymbol{p}_e \cdot \boldsymbol{r})^2}{2!\,\hbar^2} + \cdots \right). \tag{19.3}$$

The **allowed approximation** keeps only the leading constant term, equivalent to evaluating the lepton wave function at the origin ($\boldsymbol{r} = \mathbf{0}$). This corresponds to orbital angular momentum $\ell = 0$ for the lepton pair. The nuclear matrix element

$$M_{if} = \int \psi_{\mathrm{d}}^{*}(\boldsymbol{r})\, H_{\mathrm{int}}\, \psi_{\mathrm{p}}(\boldsymbol{r})\, \mathrm{d}^3 r \tag{19.4}$$

is then nearly constant over the β spectrum, and the energy dependence comes from phase space [6, 1].

Squaring the transition amplitude, each lepton contributes a factor $1/V$:

$$|\langle \psi_f | H_{\mathrm{int}} | \psi_i \rangle|^2 = \frac{|M_{if}|^2}{V^2}. \tag{19.5}$$

This $1/V^2$ will cancel against the V^2 from lepton phase space (Section 19.2).

19.2 Phase-space density and volume cancellation

Fermi's golden rule for β decay contains the square of a weak matrix element times the density of final lepton states. For plane-wave leptons in a normalisation volume V, each wave function contributes a factor $V^{-1/2}$, so $|M|^2 \propto V^{-2}$. The momentum-space density of states contributes a compensating V^2. The cancellation is not a bookkeeping trick: it guarantees that rates are independent of the fictitious box used to quantise continuum states [6].

19.2.1 Single-particle density of states

Consider a free particle of momentum p confined to a cube of side L, volume $V = L^3$. Standing-wave quantisation gives momentum components $p_x = n_x \pi\hbar/L$ with positive integers n_x. The spacing in each direction is $\pi\hbar/L$, so the volume per quantum state in momentum space is

$$\left(\frac{\pi\hbar}{L}\right)^3 = \left(\frac{h}{2L}\right)^3 = \frac{h^3}{8V}.$$

The factor 8 is essential: $h = 2\pi\hbar$.

The number of states with momentum magnitude between p and $p + \mathrm{d}p$ equals the number of lattice points in a spherical shell. Only the positive octant ($p_x, p_y, p_z > 0$) contributes, giving shell volume $\pi p^2 \mathrm{d}p/2$. Dividing by the volume per state,

$$\mathrm{d}n = \frac{\pi p^2\, \mathrm{d}p/2}{h^3/(8V)} = \frac{4\pi p^2\, \mathrm{d}p\, V}{h^3}. \tag{19.6}$$

Equivalently, the uncertainty-principle argument assigns one state per cell

$$\Delta x\, \Delta y\, \Delta z\, \Delta p_x\, \Delta p_y\, \Delta p_z \sim h^3$$

in phase space, yielding the same result [6, 1].

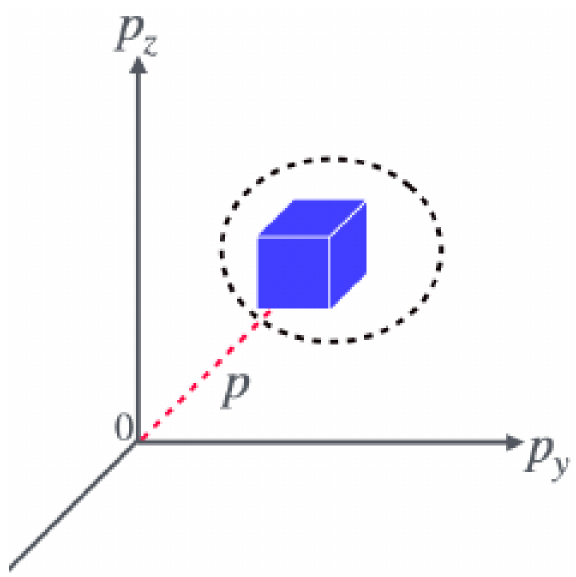


Figure 19.1: Momentum-space shell used to count lepton final states. Each quantum standing-wave state in the positive octant occupies $h^3/(8V)$ in momentum space; the factor 8 gives $\mathrm{d}n = 4\pi p^2\,\mathrm{d}p\,V/h^3$. Spin multiplicities, when summed explicitly, multiply this orbital count and may instead be absorbed into the overall weak-interaction constant.

19.2.2 Two leptons: product and cancellation

For β^- decay there are two free leptons. The combined number of final states in intervals $\mathrm{d}p_e$ and $\mathrm{d}p_\nu$ is the product:

$$\mathrm{d}n = \frac{4\pi p_e^2\,\mathrm{d}p_e\,V}{h^3} \cdot \frac{4\pi p_\nu^2\,\mathrm{d}p_\nu\,V}{h^3} = \frac{(4\pi)^2V^2}{h^6}\,p_e^2\,p_\nu^2\,\mathrm{d}p_e\,\mathrm{d}p_\nu. \tag{19.7}$$

From Fermi's golden rule (Lecture 18), the partial transition rate into this lepton momentum interval is

$$\mathrm{d}\lambda \propto \frac{|M_{if}|^2}{V^2} \cdot \frac{(4\pi)^2V^2}{h^6}\,p_e^2\,p_\nu^2\,\mathrm{d}p_e\,\mathrm{d}p_\nu = \frac{|M_{if}|^2}{h^6}\,p_e^2\,p_\nu^2\,\mathrm{d}p_e\,\mathrm{d}p_\nu. \tag{19.8}$$

The normalisation volume V cancels completely:

$$\frac{1}{V^2} \times V^2 = 1. \tag{19.9}$$

The decay rate is independent of the arbitrary box size and depends only on $|M_{if}|^2$ and lepton momenta [6, 1, 14].

19.3 Spectrum in kinetic energy: correct momentum–energy factors

We now express $\mathrm{d}\lambda$ in terms of electron kinetic energy K.

Energy conservation. Neglecting nuclear recoil and neutrino mass,

$$Q = K + E_\nu, \qquad p_\nu c = E_\nu = Q - K, \qquad p_\nu^2 = \frac{(Q-K)^2}{c^2}. \tag{19.10}$$

Relativistic electron kinematics. Total electron energy $E_e = K + m_e c^2$. The invariant relation $E_e^2 = p_e^2 c^2 + m_e^2 c^4$ gives

$$(K + m_e c^2)^2 = p_e^2 c^2 + m_e^2 c^4, \tag{19.11}$$

so

$$p_e^2 = \frac{K^2 + 2K m_e c^2}{c^2} = \frac{K(K + 2m_e c^2)}{c^2}, \qquad p_e = \frac{\sqrt{K(K+2m_e c^2)}}{c}. \tag{19.12}$$

Differentiating p_e with respect to K. From $p_e^2 c^2 = K^2 + 2K m_e c^2$, differentiate both sides:

$$2p_e c^2 \, \mathrm{d}p_e = (2K + 2m_e c^2)\, \mathrm{d}K = 2(K + m_e c^2)\, \mathrm{d}K. \tag{19.13}$$

Hence

$$\mathrm{d}p_e = \frac{K + m_e c^2}{p_e c^2}\, \mathrm{d}K = \frac{K + m_e c^2}{\sqrt{K(K+2m_e c^2)}} \frac{\mathrm{d}K}{c}. \tag{19.14}$$

Neutrino phase-space factor at fixed K. At fixed electron kinetic energy K, the final-state energy constraint fixes p_ν. For a massless neutrino, $\mathrm{d}E_\nu/\mathrm{d}p_\nu = c$, so integrating the energy-conserving delta function contributes $1/c$.

Assembling the spectrum. The differential rate in $\mathrm{d}K$ is proportional to $p_e^2\, p_\nu^2\, \mathrm{d}p_e/c$. Substituting (19.11), (19.10), and (19.14):

$$\begin{aligned}
\frac{p_e^2\, p_\nu^2}{c}\, \mathrm{d}p_e &\propto \frac{K(K+2m_e c^2)}{c^2} \cdot \frac{(Q-K)^2}{c^2} \cdot \frac{K + m_e c^2}{\sqrt{K(K+2m_e c^2)}} \frac{\mathrm{d}K}{c} \cdot \frac{1}{c} \\
&= \frac{(K + m_e c^2)\, K(K+2m_e c^2)\, (Q-K)^2}{c^6 \sqrt{K(K+2m_e c^2)}}\, \mathrm{d}K \\
&= \frac{\sqrt{K(K+2m_e c^2)}\, (K + m_e c^2)\, (Q-K)^2}{c^6}\, \mathrm{d}K.
\end{aligned} \tag{19.15}$$

Absorbing c^{-6} and other energy-independent constants into C, the **correct allowed spectrum** is

$$N(K)\, \mathrm{d}K = C\, |M_{if}|^2\, \underbrace{\sqrt{K(K+2m_e c^2)}}_{p_e c}\, \underbrace{(K + m_e c^2)}_{E_e}\, (Q-K)^2\, \mathrm{d}K. \tag{19.16}$$

Key idea

Correct factor structure (memorise this form).

$$N(K)\, \mathrm{d}K \propto p_e\, E_e\, (Q-K)^2\, \mathrm{d}K, \tag{19.17}$$

$$N(K) \propto \sqrt{K(K+2m_e c^2)}\, (K + m_e c^2)\, (Q-K)^2. \tag{19.18}$$

Here $p_e c = \sqrt{K(K + 2m_e c^2)}$ and $E_e = K + m_e c^2$. The factor $(Q-K)^2$ comes from the neutrino phase space. This is *not* the same as $K\sqrt{K+2m_e c^2}\,(K+m_e c^2)(Q-K)^2$: the momentum factor is $p_e c = \sqrt{K(K+2m_e c^2)}$, not $K\sqrt{K+2m_e c^2}$. The erroneous extra factor $\sqrt{K}$ arises if one writes $p_e^2 \propto K(K+2m_e c^2)$ but replaces $p_e\,\mathrm{d}p_e$ inconsistently [6, 1].

Ref. [1] writes the same result as $(T_e^2 + 2T_e m_e c^2)^{1/2}(Q - T_e)^2(T_e + m_e c^2)$, which is identical to (19.18) with $T_e = K$ [1, 6].

Momentum form. In terms of electron momentum p_e, with $K = (p_e c)^2/(2m_e c^2) = p_e^2/(2m_e)$ only in the *non-relativistic* limit and more generally from (19.12),

$$N(p_e)\,\mathrm{d}p_e \propto p_e^2\,(Q-K)^2\,\mathrm{d}p_e, \qquad K = K(p_e). \tag{19.19}$$

Both (19.17) and (19.18) must be used consistently when changing variables.

Non-relativistic and ultrarelativistic limits. For $K \ll m_e c^2$, $p_e c \simeq \sqrt{2m_e c^2 K}$ and $E_e \simeq m_e c^2$, so

$$N_{\mathrm{NR}}(K) \propto \sqrt{K}\,(Q-K)^2.$$

For $K \gg m_e c^2$, $p_e c \simeq E_e \simeq K$, so

$$N_{\mathrm{UR}}(K) \propto K^2(Q-K)^2.$$

These are limiting shapes, not interchangeable approximations; when $F(Z,K)$ is neglected they peak at $Q/5$ and $Q/2$, respectively.

19.4 Fermi function $F(Z,K)$

The plane-wave approximation ignores the Coulomb interaction between the emitted β particle and the daughter nucleus of charge $+Ze$. For β^- decay the interaction is attractive; for β^+ it is repulsive. Quantum mechanically, this modifies the electron/positron wave function near the nucleus and multiplies the spectrum by the **Fermi function**

$$F(Z,K) = \frac{2\pi\eta}{1 - e^{-2\pi\eta}}, \tag{19.20}$$

with

$$\eta = \pm\frac{Ze^2}{4\pi\varepsilon_0\,\hbar v}. \tag{19.21}$$

The plus sign applies to β^- (electron attracted); the minus sign to β^+ (positron repelled). Here Z is the *daughter* atomic number and v is the lepton speed [6, 1].

Relativistic velocity. The relativistic relation is

$$v = \frac{p_e c^2}{E_e} = \frac{\sqrt{K(K+2m_e c^2)}}{K + m_e c^2}\,c. \tag{19.22}$$

A *non-relativistic* estimate $v \approx \sqrt{2K/m_e}$ is sometimes used for low K, but MeV-scale β particles require (19.22) [6, 1, 14].

Using $e^2/(4\pi\varepsilon_0) = \hbar c/137 \approx 1.44\,\text{MeV fm}$ and $\hbar c \approx 197\,\text{MeV fm}$,

$$\eta = \pm\frac{Z \times 1.44\,\text{MeV fm}\,c}{197\,\text{MeV fm}\,v} = \pm\frac{Z \times 1.44}{197}\,\frac{c}{v}. \tag{19.23}$$

The full spectrum including Coulomb corrections is

$$N(K)\,\mathrm{d}K \propto F(Z,K)\,\sqrt{K(K+2m_ec^2)}\,(K+m_ec^2)\,(Q-K)^2\,\mathrm{d}K. \tag{19.24}$$

$F > 1$ enhances low-energy β^- electrons and suppresses low-energy positrons, explaining why the ^{64}Cu β^- and β^+ spectra in Lecture 18 differ in shape even though the bare phase space (19.16) is the same [1, 6].

Remark

The Fermi function (19.20) is most important at low K, where v is small and η is large. Its plane-wave limit is $F \to 1$ as $\eta \to 0$ (weak Coulomb field, $Z\alpha c/v \to 0$). Merely taking $K \to \infty$ gives $v \to c$ and $\eta \to \pm Z\alpha$, so F need not equal unity for a high-Z daughter. Fully relativistic Coulomb wave functions (Fermi–Longmire) are used in precision work [6, 1].

19.5 Fermi–Kurie plot

To test (19.24) against data, one linearises the spectrum. From (19.19) and (19.24), the momentum-space intensity is

$$I(p_e) \propto p_e^2\,(Q-K)^2\,F(Z,K). \tag{19.25}$$

Therefore

$$\sqrt{\frac{N(p_e)}{p_e^2\,F(Z,K)}} \propto (Q-K). \tag{19.26}$$

A plot of $\sqrt{N/(p^2F)}$ versus K (or total electron energy) is a **Fermi–Kurie plot**. For allowed transitions with constant $|M_{if}|^2$, the plot is a straight line intercepting $K = Q$ at zero intensity [6, 1].

^{66}Ga: a hindered Fermi caution, not a clean benchmark. $^{66}\text{Ga}(0^+) \to {}^{66}\text{Zn}(0^+) + \beta^+ + \nu_e$: $\Delta I = 0$ and no parity change permit an allowed Fermi transition, but the ground-state branch is strongly suppressed by isospin. Camp and Langer found $\log ft \simeq 7.9$ and reported an energy-dependent shape, making this branch sensitive to isospin mixing and higher-order terms. It must therefore not be presented as generic evidence that every allowed $0^+ \to 0^+$ Kurie plot is perfectly linear [31, 6].

^{91}Y: forbidden example. $^{91}\text{Y}(\frac{1}{2}^-) \to {}^{91}\text{Zr}(\frac{5}{2}^+) + e^- + \bar{\nu}_e$: $|\Delta I| = 2$ with parity change. The lepton pair must carry angular momentum; the $\ell = 0$ approximation fails. The Kurie

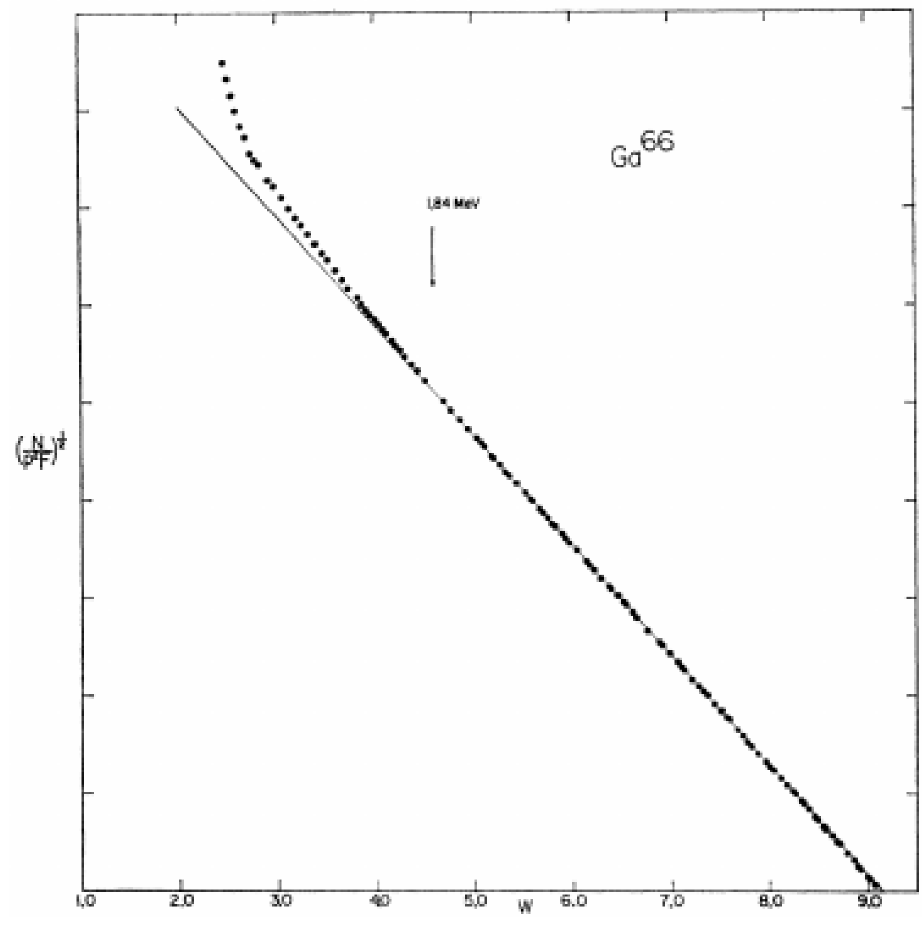


Figure 19.2: Fermi–Kurie plot for $^{66}\mathrm{Ga} \xrightarrow{\beta^+} {}^{66}\mathrm{Zn}$, ground-state $0^+ \to 0^+$ branch. This branch is allowed by spin and parity but strongly isospin forbidden ($\log ft \simeq 7.9$); Camp and Langer reported a non-statistical shape [31]; see also [6, 1].

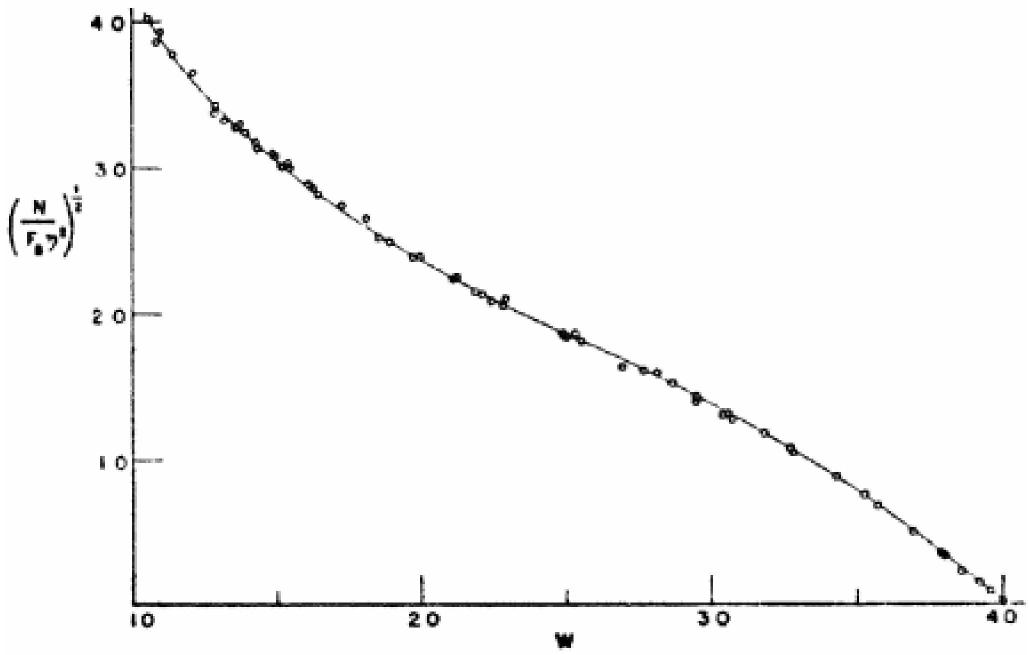


Figure 19.3: Fermi–Kurie plot for ${}^{91}\mathrm{Y} \xrightarrow{\beta^-} {}^{91}\mathrm{Zr}$: first-forbidden-unique transition. The conventional allowed reduction is curved, while division by the leading $p^2 + q^2$ shape factor restores a line [30]; see also [6, 1].

plot (Fig. 19.3) bends upward (convex): a shape factor $S(K)$ multiplies the allowed spectrum [30, 6, 1].

19.6 Allowed Fermi and Gamow–Teller transitions

The lepton wave function expansion (19.2) links orbital angular momentum to the power of $\boldsymbol{p} \cdot \boldsymbol{r}/\hbar$. Keeping only the constant term is the *allowed* ($\ell = 0$) approximation.

Electron and neutrino each have spin $s = \frac{1}{2}$. The total leptonic spin can be

$$S = 0 \quad \text{(singlet, spins antiparallel)} \qquad \text{or} \qquad S = 1 \quad \text{(triplet, spins parallel)}. \tag{19.27}$$

With $\ell = 0$, the total leptonic angular momentum is $J = \ell + S = S$.

Fermi (vector) transitions. $S = 0$, $J = 0$. Selection rules:

$$\Delta I = 0, \qquad \Delta\pi = 0 \quad \text{(no parity change)}. \tag{19.28}$$

Pure Fermi includes $0^+ \to 0^+$. In the allowed approximation the dimensionless nuclear matrix element is

$$M_F = \left\langle f \left| \sum_{k=1}^{A} \tau_k^{\pm} \right| i \right\rangle, \tag{19.29}$$

where $\tau_k^{\pm}$ converts a neutron to a proton or vice versa, according to the decay mode [6, 14].

Gamow–Teller (axial) transitions. $S = 1$, $J = 1$. Selection rules:

$$\Delta I = 0,\ \pm 1 \quad (\text{but not } 0^+ \to 0^+), \qquad \Delta\pi = 0. \tag{19.30}$$

The corresponding vector matrix element is

$$\boldsymbol{M}_{GT} = \left\langle f \left| \sum_{k=1}^{A} \boldsymbol{\sigma}_k \tau_k^{\pm} \right| i \right\rangle. \tag{19.31}$$

After the appropriate spin sums, an allowed mixed transition has nuclear strength

$$\mathcal{B}_{\text{allowed}} = g_V^2 |M_F|^2 + g_A^2 |M_{GT}|^2, \tag{19.32}$$

with vector and axial-vector couplings g_V and g_A. Their normalization must be stated consistently with the weak constant G_β; nuclear-medium calculations often use an effective g_A, so an extracted M_{GT} is convention dependent [6, 14, 2]. The pure Fermi limit has $M_{GT} = 0$; the pure Gamow–Teller limit has $M_F = 0$. For an unpolarized allowed rate the two contributions add as in (19.32), without an $M_F M_{GT}$ interference term.

Key idea

Allowed ($\ell = 0$): Fermi ($S = 0$, $\Delta I = 0$, no parity change) vs Gamow–Teller ($S = 1$, $\Delta I = 0, \pm 1$, not $0 \to 0$, no parity change). Both use the spectrum (19.18); the difference is in $|M_{if}|^2$, not in the leading phase space.

19.7 Forbidden transitions, shape factors, and ft values

If $\ell = 0$ transitions are suppressed by angular momentum or parity selection rules, the next term in the plane-wave expansion $(\boldsymbol{p} \cdot \boldsymbol{r}/\hbar)$ contributes. Transitions with leptonic orbital angular momentum $\ell \geq 1$ are called **forbidden**; the rate decreases rapidly as ℓ increases.

Order of forbiddenness. $\ell = 1$: first forbidden; $\ell = 2$: second forbidden; and so on [6, 1].

First-forbidden selection rules (summary). For $\ell = 1$, parity changes: $\Delta\pi = (-1)^\ell = -1$.

- First-forbidden Fermi ($\ell = 1$, $S = 0$): $\Delta I = 0, \pm 1$, $\Delta\pi = -1$.
- First-forbidden Gamow–Teller ($\ell = 1$, $S = 1$): $\Delta I = 0, \pm 1, \pm 2$, $\Delta\pi = -1$.

This orbital-and-spin coupling summary is only a mnemonic. Actual first-forbidden rates contain several vector and axial operators; "unique" transitions with $|\Delta I| = 2$ and parity change have a simpler leading shape factor, whereas non-unique transitions mix multiple matrix elements. The $^{91}\mathrm{Y}(\tfrac{1}{2}^-) \to {}^{91}\mathrm{Zr}(\tfrac{5}{2}^+)$ ground-state branch is first-forbidden unique [30, 6, 1].

Shape factor. Forbidden transitions multiply the allowed spectrum by an energy-dependent **shape factor** $S(K)$:

$$N(K)\,\mathrm{d}K = S(K)\,N_0(K)\,\mathrm{d}K, \tag{19.33}$$

where $N_0(K)\,\mathrm{d}K$ is the allowed form (19.24). When $S(K)$ is accounted for, the Kurie plot can be straightened [6, 1].

Dimensionless statistical rate function. Introduce electron total energy, momentum, and endpoint in electron-mass units:

$$W = \frac{E_e}{m_e c^2}, \qquad p = \frac{p_e c}{m_e c^2} = \sqrt{W^2 - 1}, \qquad W_0 = 1 + \frac{Q}{m_e c^2}.$$

For an allowed branch with a massless neutrino, the dimensionless integrated phase-space factor is

$$f(Z, W_0) = \int_1^{W_0} F(Z, W)\, pW\, (W_0 - W)^2\, \mathrm{d}W. \tag{19.34}$$

Finite-size, screening, radiative, recoil, and shape-factor corrections can be included in the integrand for precision work.

Partial half-life and the ft constant. With the coupling convention used in Eq. (19.32), the allowed partial rate is

$$\lambda_{\text{branch}} = \frac{G_\beta^2 m_e^5 c^4}{2\pi^3 \hbar^7}\, f(Z, W_0)\, \left[g_V^2 |M_F|^2 + g_A^2 |M_{GT}|^2 \right]. \tag{19.35}$$

If the branch fraction is b, its partial half-life is $t_{\text{branch}} = t_{1/2}^{\text{total}}/b$, not the total half-life. Since $\lambda_{\text{branch}} = \ln 2 / t_{\text{branch}}$, rearrangement gives

$$f\, t_{\text{branch}} = \frac{K}{g_V^2 |M_F|^2 + g_A^2 |M_{GT}|^2}, \qquad K = \frac{2\pi^3 \hbar^7 \ln 2}{G_\beta^2 m_e^5 c^4} \simeq 6.1 \times 10^3\,\mathrm{s}, \tag{19.36}$$

where the quoted numerical constant depends on whether g_V and CKM factors are included in G_β. Thus ft removes the leading phase-space dependence but still reflects the nuclear matrix elements and coupling convention. Values of $\log_{10}(ft/\mathrm{s})$ classify transitions:

$\log_{10} ft$	Classification
3–3.5	Superallowed Fermi
~ 5	Allowed
6–10	First forbidden
> 10	Higher forbidden

Superallowed $0^+ \to 0^+$ Fermi decays determine the vector coupling G_V and, combined with muon decay, the CKM element V_{ud} [6, 1, 20].

Caution

"Forbidden" does not mean impossible: ^{91}Y β-decays with $t_{1/2} \approx 58\,\mathrm{d}$. It means the transition is suppressed relative to allowed decays with similar Q [6].

Key idea

Students should remember:

- State counting: the positive-octant standing-wave cell is $h^3/(8V)$, hence $\mathrm{d}n = 4\pi p^2\,\mathrm{d}p\,V/h^3$; two leptons give $p_e^2 p_\nu^2$; V^2 cancels $1/V^2$ from $|M|^2$.
- Correct spectrum: $N(K) \propto p_e E_e (Q-K)^2$, equivalently $\sqrt{K(K+2m_ec^2)}(K+m_ec^2)(Q-K)^2$.
- Fermi function: $F = 2\pi\eta/(1-e^{-2\pi\eta})$, $\eta = \pm Ze^2/(4\pi\varepsilon_0\hbar v)$, $v = p_ec^2/E_e$.
- Kurie plot: $\sqrt{N/(p^2F)}$ vs K linear for allowed decays.
- Fermi: $S=0$, $\Delta I = 0$; GT: $S=1$, $\Delta I = 0, \pm 1$, not $0 \to 0$, with operators $\sum_k \tau_k^{\pm}$ and $\sum_k \boldsymbol{\sigma}_k \tau_k^{\pm}$. Forbidden transitions require operator-dependent shape factors.
- Comparative half-life: $f = \int_1^{W_0} FpW(W_0-W)^2\,\mathrm{d}W$ and $ft_{\text{branch}} = K/(g_V^2|M_F|^2 + g_A^2|M_{GT}|^2)$.

19.7.1 Linear Kurie plots and forbidden shape factors

For allowed decays, plotting $\sqrt{N(p)/(p^2F)}$ versus kinetic energy yields a straight line that extrapolates to Q. First-forbidden spectra need an additional shape factor $S(p,q)$ before linearity is restored, as in classic ^{91}Y analyses. Finite source thickness curves the low-energy end of Kurie plots and must be controlled experimentally [6, 1].

19.8 Deeper physics: Fermi phase space and Kurie linearity

19.8.1 Allowed phase space in electron energy

For allowed β^- decay with $m_\nu = 0$, Fermi theory gives the partial spectrum (up to the Coulomb factor $F(Z,K)$)

$$\frac{\mathrm{d}N}{\mathrm{d}K} \propto |M_{if}|^2\,F(Z,K)\,K\,(Q-K)^2\,p_e, \tag{19.37}$$

where p_e is the electron momentum and $(Q-K)^2$ is the neutrino phase-space factor [6, 1]. The nuclear matrix element $|M_{if}|^2$ is nearly constant across the spectrum in the allowed approximation (lepton wave functions evaluated at the origin). All energy dependence then comes from lepton phase space and the Coulomb correction $F(Z,K)$, which enhances low-energy electrons in β^- decay of high-Z nuclei [6, 14].

Kurie plot for ^{64}Cu

Define the Kurie function $\sqrt{K(Q-K)^2F(Z,K)^{-1}\,\mathrm{d}N/\mathrm{d}K}$. For an allowed transition with constant $|M_{if}|^2$, this quantity is linear in K and extrapolates to zero at $K=Q$ [6, 1]. Historical Kurie plots for ^{64}Cu show straight lines whose intercepts fix Q to better than 10 keV. Curvature signals forbidden transitions, neutrino mass, or incorrect $F(Z,K)$ [2].

19.8.2 The comparative half-life ft

Integrating (19.37) over K defines the phase-space factor $f(Z, Q)$. The product

$$ft = \frac{\ln 2}{f(Z, Q)\,|M_{if}|^2} \tag{19.38}$$

removes kinematics so that $\log_{10} ft$ reflects only the nuclear matrix element [6]. Superallowed $0^+ \to 0^+$ Fermi transitions cluster near $\log ft \approx 3.0$, setting the strength of the weak interaction. Forbidden decays have much larger $\log ft$ because additional lepton momentum factors suppress the rate [1, 14].

Key idea

Limits of the allowed approximation. The expansion of the electron plane wave at the origin fails when leptons carry orbital angular momentum ($\ell > 0$ for the lepton pair). Forbidden decays then acquire extra factors of K or $(Q - K)$ in the spectrum and curved Kurie plots. The allowed framework remains the baseline from which all higher-forbidden corrections are measured [6, 2].

19.8.3 Integrating the spectrum: the f factor

The total decay rate is obtained by integrating (19.37) from $K = 0$ to $K = Q$. The result defines the integrated phase-space factor $f(Z, Q)$, which depends on Z through $F(Z, K)$ and grows rapidly with Q. For superallowed $0^+ \to 0^+$ transitions, $\log ft \approx 3.0$; for first-forbidden decays, $\log ft$ can exceed 8 [6, 20]. The f-factor tabulations remove the kinematic Q dependence so that nuclear structure enters only through $|M_{if}|^2$. This separation is what allows measured half-lives from one isotope to calibrate weak-interaction strength for entirely different nuclei [1, 14].

19.8.4 Forbidden transitions and Kurie curvature

First-forbidden decays introduce additional lepton momentum factors (linear or quadratic in K) that curve the Kurie plot [6, 1]. The classification (allowed, first forbidden, second forbidden) is determined by nuclear spin and parity change together with the lepton orbital angular momentum required [14, 2]. Reactor antineutrino anomalies at short baseline have been discussed in terms of both sterile neutrinos and forbidden β branches of fission fragments; disentangling the two requires accurate shape factors [50]. The allowed Kurie linearity derived in this lecture is therefore not only a classroom exercise but a quality-control step in precision weak-interaction physics.

Comparative half-lives ft classify transitions: superallowed Fermi, allowed Gamow–Teller, and various forbidden degrees. When quoting $\log ft$, state whether screening and radiative corrections were applied. For double-β matrix elements, compare at least two many-body methods before converting a half-life limit into a mass limit.

19.9 Further physics from the literature

Neutrino mass and the endpoint. All expressions above assume $m_\nu = 0$, so $p_\nu c = E_\nu = Q - K$ and $(Q-K)^2$ appears in the spectrum. If $m_\nu \neq 0$, near the endpoint $(Q-K)$ is the neutrino total energy (neglecting recoil), and

$$p_\nu c = \sqrt{(Q-K)^2 - m_\nu^2 c^4}.$$

The massless factor $(Q-K)^2$ is therefore replaced by

$$(Q-K)\sqrt{(Q-K)^2 - m_\nu^2 c^4}\,\Theta(Q-K-m_\nu c^2).$$

The electron endpoint moves to $K_{\text{max}} \simeq Q - m_\nu c^2 - E_{\text{rec}}$, rather than remaining at Q. Precision β-spectroscopy near the tritium endpoint constrains m_{ν_e} [6, 14, 2].

Comparative half-lives and weak coupling. The product ft in (19.36) removes the calculated phase-space integral $f(Z,Q)$ so that $\log_{10} ft$ reflects only nuclear matrix elements. Superallowed $0^+ \to 0^+$ Fermi transitions (for example ^{14}O, ^{10}C) give the most precise value of G_V. Combined with muon decay, this determines the Cabibbo angle and V_{ud} in the quark mixing matrix [6, 1, 20].

From Fermi contact to W boson. Fermi's 1934 point interaction is the low-energy limit of charged-current exchange mediated by $W^\pm$ bosons ($m_W \approx 80\,\text{GeV}/c^2$). At nuclear β energies the propagator momentum is so large that the interaction looks local, with strength set by G_F [14, 5, 2].

Experimental databases. Half-lives, Q-values, and branching ratios used in problems are tabulated in ENSDF and NuDat 3 [20, 21]. Kurie-plot data for ^{64}Cu, ^{66}Ga, and ^{91}Y appear in the classic papers by Langer *et al.* (1949) and Camp and Langer (1963) [6, 1].

From the literature

The β spectrum is the earliest example of a three-body decay treated with relativistic phase space; the same methods reappear in muon and heavy-lepton decays [14]. Surveys also discusses modern tritium experiments and the KATRIN limit on m_{ν_e} [2].

Applications. The β spectrum shape enters dosimetry (medical isotopes), reactor antineutrino flux calculations, and detector calibration. NMR spin polarisation relies on β–γ angular correlations in allowed decays [13, 6].

19.9.1 Deriving Kurie linearity from phase space

For allowed β^- decay with constant $|M_{if}|^2$ and $m_\nu = 0$, divide (19.37) by $K(Q-K)^2F(Z,K)$ and take the square root:

$$\sqrt{\frac{\mathrm{d}N/\mathrm{d}K}{K(Q-K)^2F(Z,K)}} \propto |M_{if}|, \tag{19.39}$$

which is constant in K [6, 1]. Plotting the left-hand side versus K therefore gives a straight line intercepting $K = Q$. Curvature signals forbidden transitions (extra powers of K or $Q - K$), incorrect $F(Z, K)$, or neutrino mass near the endpoint [2]. This is why Kurie plots remain the standard diagnostic in precision β spectroscopy.

Superallowed $0^+ \to 0^+$ calibration

Superallowed Fermi transitions have $|M_{if}|^2 = T = 1$ isospin lowering with no angular-momentum barrier. Measured $\log ft$ values cluster near 3.0, fixing the vector coupling G_V [6, 20]. A 10% error in the integrated phase-space factor $f(Z, Q)$ propagates directly into G_V. Kurie-linearity checks on each transition verify that the allowed approximation holds before using the rate for fundamental-constant work [1, 14].

19.9.2 From $\log ft$ to γ selection rules and reactions

After β decay, daughter levels de-excite by γ emission governed by multipole selection rules (Lecture 20). A large $\log ft$ for the β branch implies a hindered weak matrix element; the subsequent γ cascade may include hindered $E1$ or $M1$ transitions with large internal conversion [6, 34]. In reactions, Gamow–Teller strength probed by charge-exchange experiments is the isospin analogue of allowed β decay (Lecture 21) [14, 53]. The phase-space factor f is the weak-interaction counterpart of the E_γ^{2L+1} phase space for photon emission.

Remark

Forbidden β decays feed reactor antineutrino systematics because their shape factors modify the spectrum at the few-percent level. Treating all fission fragments as allowed emitters biases oscillation fits. The Kurie formalism is the tool for separating those contributions [50, 6].

19.9.3 Charge-exchange reactions as β strength in reverse

Charge-exchange reactions such as (p, n) or $(^3\text{He}, t)$ on stable targets populate the same Gamow–Teller strength that β decay would connect if isospin were reversed [14, 53]. The cross section at zero momentum transfer is proportional to $B(GT)$, the same reduced matrix element that sets $\log ft$ in β decay. This link explains why nuclear-structure calculations that quench $B(GT)$ in the shell model affect both reactor antineutrino fluxes and $0\nu\beta\beta$ matrix elements [52, 50]. Kurie linearity is the low-energy laboratory check that the phase-space factorisation is correct before importing $\log ft$ systematics into astrophysical networks.

19.9.4 Double beta decay and lepton-number violation

Two-neutrino double β decay ($2\nu\beta\beta$) is an allowed second-order weak process within the Standard Model; neutrinoless double β ($0\nu\beta\beta$) would prove that neutrinos are Majorana particles [6, 5, 51]. The phase-space treatment of Lecture 19 extends to $0\nu\beta\beta$ with a different lepton factor but the same nuclear matrix-element bottleneck [52, 50]. The $\log ft$ systematics for single β decay calibrate many-body methods that are then applied to $0\nu\beta\beta$ matrix elements. Ton-scale experiments searching for $0\nu\beta\beta$ are among the highest-priority construction projects in nuclear science [54, 51].

19.10 Deeper reading: phase space, ft values, and matrix elements

Fermi theory organises allowed spectra through phase space f and a nuclear matrix element. A linear Kurie plot is the signature of an allowed vector/axial transition with approximately constant $|M_{if}|^2$. Forbidden decays acquire shape factors; quenching of Gamow–Teller strength connects to Lecture 14 and to two-body currents.

Neutrinoless double-β decay reuses the same weak language at second order in lepton-number violation. Half-life limits translate into Majorana mass limits only after choosing a nuclear matrix element set: state that choice explicitly. Superallowed $0^+ \to 0^+$ Fermi decays remain the precision electroweak calibration points of nuclear β decay.

Vector versus axial probes

Pure Fermi transitions test the vector current and CKM unitarity inputs; Gamow–Teller transitions test axial strength and nuclear spin correlations. Mixing ratios in mixed transitions require correlation measurements. Keeping those channels distinct clarifies why quenching is primarily an axial-current story in medium-mass nuclei.

Phase-space sensitivity

Because f rises steeply with Q, a 50 keV mass error near a small endpoint strongly affects extracted $|M_{if}|^2$ from a measured half-life. Mass precision and β spectroscopy are coupled experimental programmes, not sequential afterthoughts.

Comparative half-life drill

Take a superallowed Fermi transition with known $t_{1/2}$ and Q, look up or approximate the phase-space factor f, and form ft. Compare with a free-neutron lifetime scale to see why nuclear matrix elements of order unity are special. Then take a strongly hindered forbidden decay and note the jump in $\log ft$.

Write a one-paragraph explanation of why $2\nu\beta\beta$ is allowed in the Standard Model while $0\nu\beta\beta$ is not, emphasising lepton number and the role of nuclear matrix elements in the rate. State that operator quenching lessons from Lectures 8 and 14 feed directly into those matrix elements. This paragraph is the conceptual endpoint of the weak-interaction arc that began with continuous spectra in Lecture 18.

Fermi theory converts Q values and matrix elements into rates. Comparative half-lives classify transitions; neutrinoless double-β decay reuses the same language at a deeper level.

In practice, rework one numerical estimate from the lecture with a different parameter set and compare the shift to the quoted experimental uncertainty. If the shift exceeds the uncertainty, the estimate is teaching a scale, not delivering a fit. That distinction keeps theoretical cartoons honest when they sit beside evaluated data tables and modern many-body calculations. Along the course arc, write one sentence linking this chapter backward and one forward: the discipline of those two sentences is as important as any single formula. Examination problems rarely ask for a literature review, but they do ask for the scale argument that literature reviews presuppose.

Synthesis problem

Write a half-page note that (i) restates the chapter's central scale or selection rule in one equation, (ii) evaluates it numerically for one concrete nucleus or energy, and (iii) names the neighbouring lecture that uses the result. Keep the note free of research-frontier speculation; the goal is fluency with the lecture physics itself. If an assumption fails (Born approximation, spherical mean field, allowed transition, one-dimensional barrier), state the failure mode in a final sentence. That habit converts passive reading into examination-ready understanding and makes the later research-frontier box read as an extension rather than a replacement.

Research frontier: Fermi theory, quenching, and $0\nu\beta\beta$

Fermi's golden-rule treatment (Lecture 19) still organises allowed spectra and ft values, but nuclear matrix elements for forbidden decays and for neutrinoless double-β decay require correlated many-body theory. Consistent two-body currents reduce GT quenching and revise $0\nu\beta\beta$ matrix-element estimates [52]. Ton-scale $0\nu\beta\beta$ experiments are a highest-priority construction recommendation of the 2023 nuclear-science long-range plan [54, 51, 50]. What remains uncertain is not whether lepton-number violation would be revolutionary if observed, but how large the nuclear matrix element is, and therefore how a half-life limit translates into a Majorana-mass limit. Forbidden β decays in nuclei relevant to reactor antineutrino spectra add a second applied frontier where shape factors and quenching must be controlled. The lecture's phase-space and selection-rule language remains the correct entry point to those advanced topics.

When comparing $0\nu\beta\beta$ half-life limits across isotopes, students should convert to an effective Majorana mass only after stating which matrix element set was assumed. Different many-body methods still disagree at the factor-of-two level in some cases, which dominates the particle-physics interpretation even when the experimental exposure is excellent.

Stating the matrix-element set used when converting a half-life limit into an effective Majorana mass is mandatory for an honest comparison across isotopes.

Student directions. (1) Derivation check: show why an allowed Kurie plot is linear for constant $|M_{if}|^2$. (2) Data estimate: extract $\log ft$ from a measured half-life and phase-space factor for one superallowed transition. (3) Short report: contrast $2\nu\beta\beta$ (Standard Model) with $0\nu\beta\beta$ (lepton-number violation) and the role of nuclear matrix elements.

19.11 Problems with solutions

Problem 19.1 — Derive the spectrum from phase space. Starting from $\mathrm{d}\lambda \propto p_e^2 p_\nu^2\,\mathrm{d}p_e/c$, where $1/c$ comes from the massless-neutrino energy delta function, with $p_\nu c = Q - K$ and relativistic electron kinematics, derive

$$N(K)\,\mathrm{d}K \propto \sqrt{K(K+2m_ec^2)}\,(K+m_ec^2)\,(Q-K)^2\,\mathrm{d}K. \tag{19.40}$$

Show every algebraic step [1, 6].

Solution 19.1. **Step 1: Neutrino factor.** $p_\nu c = E_\nu = Q - K$, so $p_\nu^2 = (Q-K)^2/c^2$.

Step 2: Electron momentum squared. From $E_e = K + m_ec^2$ and $E_e^2 = p_e^2c^2 + m_e^2c^4$:

$$p_e^2c^2 = K^2 + 2Km_ec^2 \quad \Rightarrow \quad p_e^2 = \frac{K(K+2m_ec^2)}{c^2}.$$

Step 3: Differentiate for $\mathrm{d}p_e$. From $p_e^2c^2 = K^2 + 2Km_ec^2$:

$$2p_ec^2\,\mathrm{d}p_e = 2(K+m_ec^2)\,\mathrm{d}K \quad \Rightarrow \quad \mathrm{d}p_e = \frac{K+m_ec^2}{p_ec^2}\,\mathrm{d}K.$$

Step 4: Assemble.

$$\frac{p_e^2p_\nu^2}{c}\,\mathrm{d}p_e = \frac{K(K+2m_ec^2)}{c^2}\cdot\frac{(Q-K)^2}{c^2}\cdot\frac{K+m_ec^2}{p_ec^2}\,\mathrm{d}K\cdot\frac{1}{c}.$$

Step 5: Substitute p_e. Use $p_e = \sqrt{K(K+2m_ec^2)}/c$:

$$\frac{p_e^2p_\nu^2}{c}\,\mathrm{d}p_e = \frac{K(K+2m_ec^2)(Q-K)^2(K+m_ec^2)}{c^6\sqrt{K(K+2m_ec^2)}}\,\mathrm{d}K.$$

Step 6: Simplify. Cancel one factor $\sqrt{K(K+2m_ec^2)}$ from numerator and denominator:

$$\frac{p_e^2p_\nu^2}{c}\,\mathrm{d}p_e = \frac{\sqrt{K(K+2m_ec^2)}\,(K+m_ec^2)\,(Q-K)^2}{c^6}\,\mathrm{d}K.$$

Step 7: Identify p_ec and E_e.

$$\sqrt{K(K+2m_ec^2)} = p_ec, \qquad K+m_ec^2 = E_e,$$

so $N(K)\,\mathrm{d}K \propto p_eE_e(Q-K)^2\,\mathrm{d}K$, as required [1, 6].

Problem 19.2 — Endpoint and maximum of $N(K)$. For allowed β^- decay with $Q = 1.0\,\mathrm{MeV}$ and $m_ec^2 = 0.511\,\mathrm{MeV}$, show that $N(K) = 0$ at $K = Q$ and estimate $K_{\max}$ where $N(K)$ peaks (numerically or graphically).

Solution 19.2. At $K = Q$: the factor $(Q-K)^2 = 0$, so $N(Q) = 0$. This is the spectrum endpoint.

For the maximum, set $dN/dK = 0$ on the allowed form (ignoring F):

$$N(K) \propto \sqrt{K(K+2m_ec^2)}\,(K+m_ec^2)\,(Q-K)^2.$$

It is convenient to differentiate $\ln N$:

$$\frac{K+m_ec^2}{K(K+2m_ec^2)}+\frac{1}{K+m_ec^2}-\frac{2}{Q-K}=0.$$

With $Q=1.0\,\text{MeV}$ and $m_ec^2=0.511\,\text{MeV}$, the root is

$$K_{\text{peak}}=0.338\,\text{MeV}.$$

This lies between the non-relativistic limiting estimate $Q/5$ and the ultrarelativistic estimate $Q/2$. The spectrum peaks well below Q because the $(Q-K)^2$ factor favors energy sharing with the neutrino [6, 1].

Problem 19.3 — Fermi parameter η. For β^- decay to a daughter with $Z=30$, find η and comment on $F(Z,K)$ at $K=0.10\,\text{MeV}$ and $K=1.0\,\text{MeV}$. Use $v=p_ec^2/E_e$.

Solution 19.3. At $K=0.10\,\text{MeV}$:

$$p_ec=\sqrt{0.10\times1.122}=0.335\,\text{MeV},\quad E_e=0.611\,\text{MeV},$$

$$v=\frac{p_ec^2}{E_e}=0.548c,\qquad \eta=+\frac{30\times1.44}{197\times0.548}\approx0.40.$$

Then $F=2\pi\eta/(1-e^{-2\pi\eta})\approx2.73$: Coulomb enhancement is large.

At $K=1.0\,\text{MeV}$: $p_ec=\sqrt{1.0\times2.022}=1.422\,\text{MeV}$, $E_e=1.511\,\text{MeV}$, $v=0.941c$, $\eta\approx0.233$, and $F\approx1.90$. Enhancement decreases at higher K [6, 1].

Problem 19.4 — Kurie plot intercept. A Fermi–Kurie plot of $\sqrt{N/(p^2F)}$ versus K for an allowed β^- decay is a straight line with slope $-0.50\,\text{MeV}^{-1}$ and intercept at $K=2.50\,\text{MeV}$ on the horizontal axis. What is Q?

Solution 19.4. For allowed decays, $\sqrt{N/(p^2F)}\propto(Q-K)$. The intercept where intensity vanishes is $K=Q$. Therefore $Q=2.50\,\text{MeV}$ [6, 1].

Problem 19.5 — Classify ^{66}Ga and ^{91}Y. State whether each decay is allowed or forbidden, Fermi or Gamow–Teller, and predict the Kurie-plot shape.

Solution 19.5. $^{66}\text{Ga}(0^+)\to{}^{66}\text{Zn}(0^+)$**:** $\Delta I=0$, $\Delta\pi=0$, so it is allowed Fermi by spin and parity. It is nevertheless strongly isospin forbidden ($\log ft\simeq7.9$); the Camp–Langer spectrum was reported as non-statistical, so it is not a clean generic allowed benchmark [31, 6].

$^{91}\text{Y}(\frac{1}{2}^-)\to{}^{91}\text{Zr}(\frac{5}{2}^+)$**:** $|\Delta I|=2$, parity changes: this is first-forbidden unique (minimum $\ell=1$). The conventional allowed Kurie plot is non-linear until the first-forbidden-unique shape factor is included [30, 6, 1].

Problem 19.6 — Common error check. A student writes $N(K)\propto K\sqrt{K+2m_ec^2}(K+m_ec^2)(Q-K)^2$. Explain why this differs from the correct formula and identify the extra factor.

Solution 19.6. The correct momentum factor is $p_e c = \sqrt{K(K + 2m_e c^2)}$, not $K\sqrt{K + 2m_e c^2}$. The student expression has an extra factor of $\sqrt{K}$ relative to the correct form $\sqrt{K(K + 2m_e c^2)}(K + m_e c^2)(Q - K)^2$. This error typically arises from using $p_e^2 \propto K(K + 2m_e c^2)$ but failing to include the $\mathrm{d}p_e$ Jacobian correctly, or from confusing $p_e c$ with $K\sqrt{K + 2m_e c^2}$ [1, 6].

Problem 19.7 — Kurie plot of an allowed spectrum. Define the Kurie (Fermi–Kurie) function for an allowed β spectrum. What shape do you expect versus electron kinetic energy K if $F(Z, K)$ is correctly included and $m_\nu = 0$?

Solution 19.7.

$$K(K) = \sqrt{\frac{N(K)}{p_e E_e F(Z, K)}} \propto (Q - K)$$

for allowed transitions with massless neutrinos. A linear plot versus K extrapolates to the endpoint Q [6, 1].

Chapter 20

Gamma Decay

"Gamma radiation is the electromagnetic de-excitation of the nucleus: same species, lower energy, multipole selection rules, and competition from internal conversion."
K. S. Krane

Alpha and beta decay change the nuclear identity (Z and/or A). **Gamma decay** does not: the same nucleus drops from an excited state to a lower level by emitting a photon. Excited states are populated most often by preceding β or α decay (or by nuclear reactions). The transition energy is typically tens of keV to a few MeV. This chapter develops multipole classification (EL, ML), the long-wavelength expansion, angular-momentum and parity selection rules, transition strengths and widths, angular correlations, and competition with internal conversion [6, 1, 14, 2, 32].

20.1 De-excitation energy and three standard sources

For a transition between two states of the same nuclide,

$$^{A}_{Z}X^{*}(J_i^{\pi_i}) \longrightarrow {}^{A}_{Z}X(J_f^{\pi_f}) + \gamma, \tag{20.1}$$

neither A nor Z changes. The level spacing $E_0 = E_i - E_f$ is shared by the photon and the recoil of the final nucleus, so E_γ is slightly smaller than E_0. Excited states may be fed by alpha or beta decay, electron capture, inelastic scattering, or a capture reaction; the subsequent photon cascade reveals the daughter level scheme [6, 1].

20.1.1 ^{137}Cs: a line accompanied by conversion electrons

About 94.6% of ^{137}Cs beta decays feed the 661.7 keV, $11/2^-$ isomer of ^{137}Ba, whose half-life is 2.55 min. Its decay to the $3/2^+$ ground state has $\Delta J = 4$ and changes parity, so the leading photon multipole is $M4$ (the next allowed order is $E5$). The high order explains the isomerism and also makes internal conversion appreciable. A BrIcc calculation for $Z = 56$, $E_{\mathrm{tr}} = 661.7$ keV, pure $M4$, gives an illustrative total coefficient $\alpha_T \simeq 0.11$, hence $f_\gamma = 1/(1 + \alpha_T) \simeq 0.90$. The exact coefficient used in an analysis must state the BrIcc atomic-vacancy convention and be checked against the evaluated transition record [6, 20, 21, 34].

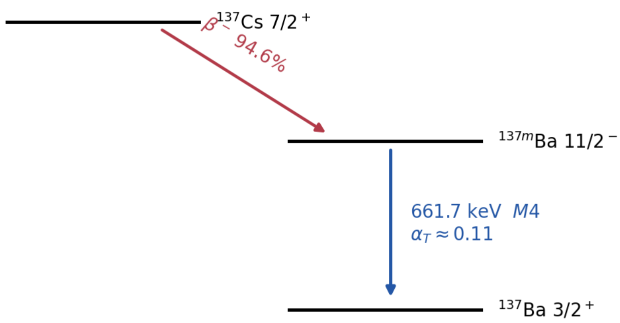


Figure 20.1: The ^{137}Cs decay scheme, showing the dominant beta feeding of the 661.7 keV ^{137}Ba isomer. Evaluated values should be taken from ENSDF/NuDat rather than read from this schematic [20, 21].

20.1.2 ^{60}Co: an ordered two-photon cascade

^{60}Co beta decay strongly populates the 2.505 MeV, 4^+ state of ^{60}Ni. Energy subtraction fixes the order of the cascade:

$$2.505\,\text{MeV}\,(4^+) \xrightarrow[E_\gamma=2.505-1.332]{1.173\,\text{MeV}\,(E2)} 1.332\,\text{MeV}\,(2^+) \xrightarrow[E_\gamma=1.332-0]{1.332\,\text{MeV}\,(E2)} 0\,(0^+). \tag{20.2}$$

The raw lecture prose interchanged these two photon assignments; the level differences above give the corrected ordering. The $2^+ \to 0^+$ leg is forced to be pure $E2$. The $4^+ \to 2^+$ leg permits $E2, M3, E4, \ldots$; it is conventionally treated as overwhelmingly $E2$, but "pure" is an experimental qualification, not a consequence of $\Delta J = 2$ alone. In particular this cascade is not the standard adjacent-order $M1 + E2$ mixing case. Its directional correlation is useful in coincidence experiments [6, 1].

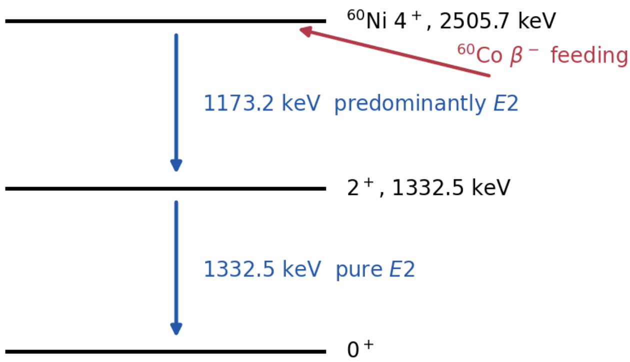


Figure 20.2: Partial ^{60}Co → ^{60}Ni decay scheme. The printed labels are 1.1732 and 1.3325 MeV; rounded values are used in the text.

20.1.3 ^{22}Na: nuclear and annihilation photons

^{22}Na decays mainly by positron emission, with a smaller electron-capture branch, to the 1.274 MeV state of ^{22}Ne. Its de-excitation photon is therefore observed together with two 511 keV annihilation photons from the positron. The 1.274 MeV line is nuclear; the 511 keV lines are annihilation radiation [6, 20].

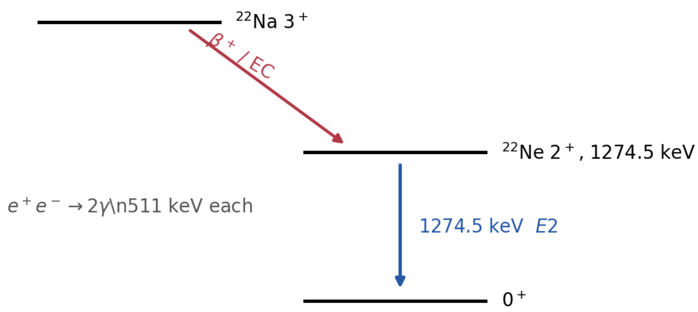


Figure 20.3: The ^{22}Na decay scheme. Branch percentages are retained as lecture-level rounded values; ENSDF is the authority for evaluated intensities [20].

Gamma ray or X-ray? The names identify origin, not a non-overlapping energy range. A gamma ray comes from nuclear levels; an X-ray comes from an electronic-shell vacancy. Consequently a low-energy gamma ray may have less energy than a hard X-ray [1, 2].

20.2 From a localized source to electromagnetic multipoles

Let the transition frequency be $\omega = E_\gamma/\hbar$, the photon wave number $k = \omega/c$, and the nuclear radius $R \simeq 1.2A^{1/3}$ fm. The expansion parameter is

$$kR = \frac{E_\gamma R}{\hbar c} \simeq 6.08 \times 10^{-3} A^{1/3} \left(\frac{E_\gamma}{\mathrm{MeV}}\right). \tag{20.3}$$

For $A = 100$ and $E_\gamma = 1\,\mathrm{MeV}$, $kR \simeq 0.028$. Thus the wavelength is long compared with the source, and the plane-wave factor can be ordered as

$$e^{i\mathbf{k}\cdot\mathbf{r}} = 1 + i\mathbf{k}\cdot\mathbf{r} - \frac{(\mathbf{k}\cdot\mathbf{r})^2}{2!} + \cdots . \tag{20.4}$$

After separating the angular dependence into scalar and vector spherical harmonics, successive terms are electric or magnetic multipoles. Their amplitudes contain powers of kR, and their rates contain E_γ^{2L+1}; this is the dynamical origin of the strong suppression of high L [6, 1, 2].

20.2.1 Long-wavelength operators

With the standard spherical-harmonic normalization, the leading electric operator is

$$\mathcal{M}(EL, M) = \int d^3r\, \rho(\mathbf{r})\, r^L Y_{LM}(\hat{\mathbf{r}}) \simeq e \sum_{p=1}^{Z} r_p^L Y_{LM}(\hat{\mathbf{r}}_p). \tag{20.5}$$

Only protons appear in the one-body charge form, although collective motion and effective charges can make the matrix element a property of many nucleons. A convenient current-density form of the magnetic operator is

$$\mathcal{M}(ML, M) = -\frac{i}{c}\sqrt{\frac{L}{L+1}} \int d^3r\, r^L\, \mathbf{Y}_{LLM}(\hat{\mathbf{r}}) \cdot \mathbf{j}(\mathbf{r}), \tag{20.6}$$

up to the conventional overall phase. For $M1$, its familiar one-body limit is

$$\mathcal{M}(M1, M) = \sqrt{\frac{3}{4\pi}}\, \mu_N \sum_a \left(g_{\ell,a}\boldsymbol{\ell}_a + g_{s,a}\mathbf{s}_a\right)_M. \tag{20.7}$$

Equations (20.5) and (20.6) also make the parity transparent: ρ is a scalar, $\mathbf{j}$ is a polar vector, and $\mathbf{r} \times \mathbf{j}$ is axial.

Why no single-photon $E0$ radiation? For $L = 0$, $\mathcal{M}(E0)$ begins with the conserved total charge. A radial pulsation may alter the charge distribution inside the nucleus, but outside it the monopole field remains a static Coulomb field; no transverse $1/r$ radiation field results. There is likewise no $M0$ moment in ordinary electrodynamics. A nuclear $E0$ transition can nevertheless couple directly to an atomic electron or, above threshold, create an e^-e^+ pair [6, 14].

20.3 Angular momentum, parity, and multipole mixing

A radiation multipole of rank L transfers angular momentum $L\hbar$. The Wigner–Eckart theorem gives the same triangle condition as vector addition:

$$|J_i - J_f| \le L \le J_i + J_f, \qquad L = 1, 2, 3, \ldots. \tag{20.8}$$

The lower bound is therefore

$$L_{\min} = \max(1, |J_i - J_f|) \tag{20.9}$$

provided $J_i + J_f \ge 1$. It is incorrect to replace (20.8) by the oversimplified statement $\Delta J = L$: for example, a $2 \to 2$ transition may contain $L = 1, 2, 3, 4$. If either endpoint has $J = 0$, however, the triangle collapses to a single value $L = J_{\text{nonzero}}$.

For $J_i = J_f = 0$, the triangle permits only $L = 0$, while a real single photon requires $L \ge 1$:

$$0 \longrightarrow 0 \quad \Rightarrow \quad \text{single-photon emission is forbidden.} \tag{20.10}$$

This statement applies to any $0 \to 0$ pair of parities, not just $0^+ \to 0^+$.

The parity carried by the two families is

$$EL: \quad \pi_f = \pi_i(-1)^L, \tag{20.11}$$

$$ML: \quad \pi_f = \pi_i(-1)^{L+1}. \tag{20.12}$$

Thus parity changes for $E1, M2, E3, M4, \ldots$, and is unchanged for $M1, E2, M3, E4, \ldots$. For each allowed L, parity selects one of EL or ML. Adjacent allowed orders may both occur: a same-parity $\Delta J = 1$ transition can be $M1+E2$, whereas an opposite-parity one can be $E1+M2$. Such a transition is not "half electric and half magnetic"; its amplitudes must be measured.

Remark

Assignment algorithm: (1) list every integer L satisfying (20.8), excluding $L = 0$; (2) apply (20.11)–(20.12); (3) compare the lowest one or two surviving multipoles using rates, angular correlations, polarization, or conversion coefficients. "Allowed" does not mean "equally probable" [6, 1].

20.4 Transition strengths, Weisskopf units, lifetimes, and widths

For an operator $\mathcal{M}(\sigma L)$, $\sigma = E, M$, define the reduced transition probability

$$B(\sigma L; J_i \to J_f) = \frac{1}{2J_i + 1} \left|\langle J_f \left\| \mathcal{M}(\sigma L) \right\| J_i\rangle\right|^2 . \tag{20.13}$$

For electric radiation, if $B(EL)$ is quoted numerically in $e^2\mathrm{fm}^{2L}$, the radiative decay constant is

$$\lambda_\gamma(EL) = \frac{8\pi(L+1)}{L[(2L+1)!!]^2} \left(\frac{E_\gamma}{\hbar c}\right)^{2L+1} \frac{e^2}{4\pi\epsilon_0\hbar} \frac{B(EL)}{e^2}. \tag{20.14}$$

The dimensions are explicit: $(E_\gamma/\hbar c)^{2L+1}$ contributes $\mathrm{fm}^{-(2L+1)}$, $(e^2/4\pi\epsilon_0)(B/e^2)$ contributes $\mathrm{MeV\,fm}^{2L+1}$, and division by $\hbar$ leaves s^{-1}. Magnetic rates have the same E_γ^{2L+1} dependence when $B(ML)$ is defined with the magnetic operator; magnetic strengths are conventionally reported in $\mu_N^2\mathrm{fm}^{2L-2}$ [6, 1].

20.4.1 The Weisskopf benchmark

The single-particle estimate replaces detailed wave functions by a uniform radial amplitude inside $R = 1.2A^{1/3}$ fm. The radial matrix element of r^L against a constant density is

$$\frac{\displaystyle\int_0^R r^L\, r^2\, \mathrm{d}r}{\displaystyle\int_0^R r^2\, \mathrm{d}r} = \frac{\displaystyle\int_0^R r^{L+2}\, \mathrm{d}r}{R^3/3} = \frac{3}{L+3}\, R^L. \tag{20.15}$$

Squaring and attaching the conventional angular-average factor $1/(4\pi)$ produces the electric Weisskopf unit. For magnetic radiation the single-nucleon current supplies a moment of order μ_N, while a rank L operator supplies R^{L-1}. In the conventional Weisskopf prescription the reduced single-particle estimate is

$$|\langle \mathcal{M}(ML)\rangle_{\mathrm{W}}| = \sqrt{\frac{10}{\pi}}\, \frac{3}{L+3} R^{L-1} \mu_N.$$

Squaring it (with the reduced-probability spin average understood) produces the magnetic expression below. Thus the conventional Weisskopf units are

$$B_{\mathrm{W}}(EL) = \frac{1}{4\pi} \left(\frac{3}{L+3}\right)^2 R^{2L}\, e^2, \tag{20.16}$$

$$B_{\mathrm{W}}(ML) = \frac{10}{\pi} \left(\frac{3}{L+3}\right)^2 R^{2L-2}\, \mu_N^2. \tag{20.17}$$

A measured strength is then stated as B/B_{W} in Weisskopf units (W.u.). This is a scale, not a precision prediction: configuration changes can hinder a transition, while coherent collective motion can enhance $E2$ strengths far above one W.u. [6, 1, 14].

Substitute $B_{\mathrm{W}}(EL)$ into the multipole rate (20.14) (with $R = r_0 A^{1/3}$, $r_0 = 1.2\,\mathrm{fm}$) and restore conventional units for E_γ in MeV. For the leading electric dipole case $L = 1$,

$$\lambda_{\mathrm{W}}(E1) \propto E_\gamma^3\, B_{\mathrm{W}}(E1) \propto E_\gamma^3\, R^2 \propto A^{2/3} E_\gamma^3, \tag{20.18}$$

and the numerical prefactor evaluates to $1.02 \times 10^{14}\, A^{2/3} E_\gamma^3$ (in s^{-1}). Higher multipoles bring higher powers of E_γ and of R (hence of A). The resulting internally consistent set of rough rates, with E_γ in MeV and λ_{W} in s^{-1}, is:

L	$\lambda_{\mathrm{W}}(EL)$	$\lambda_{\mathrm{W}}(ML)$
1	$1.02 \times 10^{14} A^{2/3} E_\gamma^3$	$3.15 \times 10^{13} E_\gamma^3$
2	$7.28 \times 10^{7} A^{4/3} E_\gamma^5$	$2.24 \times 10^{7} A^{2/3} E_\gamma^5$
3	$33.9 A^2 E_\gamma^7$	$10.4 A^{4/3} E_\gamma^7$
4	$1.07 \times 10^{-5} A^{8/3} E_\gamma^9$	$3.27 \times 10^{-6} A^2 E_\gamma^9$

At fixed A and E_γ, each increase in L is usually a severe penalty; in medium and heavy nuclei an EL Weisskopf rate is often roughly two orders of magnitude above the corresponding ML rate. The memorable 10^{-5} and 10^2 rules are therefore only order-of-magnitude summaries, not constants independent of A, E_γ, or structure [6, 1].

20.4.2 From a rate to a line width

For a level with independent exit channels c,

$$\lambda_{\mathrm{tot}} = \sum_c \lambda_c, \qquad \tau = \lambda_{\mathrm{tot}}^{-1}, \qquad t_{1/2} = \tau \ln 2, \qquad \Gamma = \hbar \lambda_{\mathrm{tot}} = \frac{\hbar}{\tau}. \tag{20.19}$$

Here Γ is the full natural width in energy units and $\hbar = 6.5821196 \times 10^{-16}\,\mathrm{eV\,s}$. A branching fraction is $b_c = \lambda_c/\lambda_{\mathrm{tot}} = \Gamma_c/\Gamma$. Long lifetime means narrow natural width. High-L, low-energy, or structurally hindered transitions may produce metastable nuclear isomers [6, 1, 14].

20.5 Angular distributions and gamma–gamma correlations

An ensemble with equally populated magnetic substates has no preferred axis, so its single-photon distribution is isotropic even though each multipole has nontrivial angular structure. An anisotropy appears when the initial spins are aligned, or when detecting a first photon γ_1 in a cascade selects an unequal population of substates in the intermediate level. Relative to the γ_1 direction, the second photon has

$$W(\theta) = 1 + \sum_{\substack{k=2 \\ k\ \mathrm{even}}}^{k_{\max}} A_k P_k(\cos\theta). \tag{20.20}$$

The coefficients depend on the three spins, the multipole orders, detector acceptance, and any attenuation before the second decay. Odd Legendre terms are absent for the usual unpolarized electromagnetic cascade [6, 2].

For adjacent allowed multipoles, such as $M1+E2$, the reduced matrix elements in Eq. (20.13) cannot be divided directly: they carry different electromagnetic dimensions. Define instead the dimensionless *radiation-amplitude* mixing ratio

$$\delta = \frac{\mathcal{A}(E2; J_i \to J_f)}{\mathcal{A}(M1; J_i \to J_f)}, \tag{20.21}$$

where each $\mathcal{A}(\sigma L)$ includes the multipole-dependent radiation normalization and its power of $E_\gamma/\hbar c$. Equivalently, $|\delta|^2 = \lambda_\gamma(E2)/\lambda_\gamma(M1)$ when interference has been removed by angular integration. The sign depends on the adopted phase convention and must be reported with it. After angular integration the idealized fractions are $f_{E2} = \delta^2/(1+\delta^2)$ and $f_{M1} = 1/(1+\delta^2)$, while the sign of δ remains visible through interference in A_k. Angular correlations determine L and mixing; linear polarization is needed to distinguish electric from magnetic character when angular distributions alone are degenerate [6].

20.6 Worked multipole assignments

20.6.1 ^{72}Ge: $0^+ \to 0^+$ and $2^+ \to 0^+$

Germanium-72 has a 0^+ ground state, a 0^+ first excited state at 691 keV ($t_{1/2} = 0.42\,\mu$s), and a 2^+ state at 834 keV ($t_{1/2} = 3.1$ ps).

$0^+ \to 0^+$ at 691 keV. Here $J_i = J_f = 0$, so (20.8) forces $L = 0$. No photon multipole exists. Gamma emission is forbidden. The state still decays by **internal conversion** (Section 20.7): the nuclear transition energy is transferred to an atomic electron [6, 1].

$2^+ \to 0^+$ at 834 keV.

$$|2-0| \le L \le 2+0 \quad \Rightarrow \quad L = 2. \tag{20.22}$$

Both states have positive parity, so $\pi_f = \pi_i$. For $L = 2$, electric radiation satisfies $\pi_f = \pi_i(-1)^2 = \pi_i$. The transition is pure $E2$. The short half-life is consistent with a low multipole order [6].

20.6.2 ^{117}In: large ΔJ

An excited state of ^{117}In at 315 keV has $J^\pi = \frac{1}{2}^-$; the ground state is $\frac{9}{2}^+$. Then

$$\left|\tfrac{1}{2} - \tfrac{9}{2}\right| \le L \le \tfrac{1}{2} + \tfrac{9}{2} \quad \Rightarrow \quad L = 4 \text{ or } 5. \tag{20.23}$$

Parity *changes* ($- \to +$).

- For $L = 4$: magnetic parity rule $\pi_f = \pi_i(-1)^5 = -\pi_i$ matches a change. Allowed multipole: $M4$.

- For $L = 5$: electric rule $\pi_f = \pi_i(-1)^5 = -\pi_i$ also matches. Allowed multipole: $E5$.

The extra power of $E_\gamma R/\hbar c$ makes $E5$ much slower than $M4$ on the Weisskopf scale despite the usual electric-over-magnetic advantage at equal L. The leading assignment is therefore $M4$; its high order explains the isomeric lifetime [6, 1].

Assign multipolarity for $3^- \to 0^+$

Possible L: $|3-0| \leq L \leq 3$, so $L = 3$. Parity changes. Electric $E3$ has $\pi_f = \pi_i(-1)^3 = -\pi_i$. Magnetic $M3$ would not change parity. Result: pure $E3$.

20.7 Internal conversion

Internal conversion competes with γ emission whenever atomic electrons overlap the nuclear transition field. It is especially important for low-energy, high-multipolarity transitions (where radiative rates are suppressed) and for $0^+ \to 0^+$ cases where a single photon is forbidden. Conversion-electron spectroscopy therefore complements γ spectroscopy in assigning spins and multipolarities [6, 1].

20.7.1 Mechanism

In **internal conversion** (IC) the nuclear transition field ejects an electron already present in the atom. This is a single quantum process, not gamma emission followed by photoelectric absorption. Inner-shell wave functions have the greatest nuclear overlap, so K, L, and M conversion are especially important. For conversion from shell s,

$$K_{e,s} = E_{\text{tr}} - B_s - E_{\text{rec},e} \simeq E_{\text{tr}} - B_s, \tag{20.24}$$

provided $E_{\text{tr}} > B_s$. Different shell binding energies therefore produce discrete conversion-electron lines. The vacancy subsequently gives characteristic X-rays or Auger electrons; these atomic emissions should not be confused with the nuclear transition itself [6, 1, 2].

20.7.2 Conversion coefficient

Define partial decay constants λ_γ and λ_{IC} for photon emission and internal conversion. The **internal conversion coefficient** is

$$\alpha \equiv \frac{\lambda_{\text{IC}}}{\lambda_\gamma} = \frac{N_e}{N_\gamma}, \tag{20.25}$$

with N_e and N_γ the numbers of conversion electrons and photons in the same transition after efficiency and fluorescence corrections. One also defines shell coefficients α_K, α_L, $\ldots$ with $\alpha = \alpha_K + \alpha_L + \cdots$. The total electromagnetic width is

$$\lambda_{\text{em}} = \lambda_\gamma + \lambda_{\text{IC}} = \lambda_\gamma(1+\alpha), \qquad \Gamma_{\text{em}} = \hbar\lambda_{\text{em}}. \tag{20.26}$$

Consequently

$$f_\gamma = \frac{1}{1+\alpha}, \qquad f_{\mathrm{IC}} = \frac{\alpha}{1+\alpha}, \qquad \lambda_\gamma = \frac{\ln 2}{t_{1/2}(1+\alpha)} \tag{20.27}$$

for a level whose only decay is this one electromagnetic transition. With several branches j, one instead sums $\lambda_{\mathrm{level}} = \sum_j (1+\alpha_j)\lambda_{\gamma j}$. This correction is essential before inferring $B(\sigma L)$ from a measured level lifetime. Conversion grows with Z, falls rapidly with transition energy, and generally becomes more important at high multipole order [6, 1, 14].

20.7.3 $0^+ \to 0^+$: conversion only

For $0^+ \to 0^+$, $\lambda_\gamma = 0$ for single-photon emission. Below the pair threshold, the ordinary electromagnetic channel is $E0$ internal conversion. Internal pair formation opens at $E_{\mathrm{tr}} > 2m_e c^2 = 1.022\,\mathrm{MeV}$. Because the reference photon rate vanishes, the usual ratio $\alpha = \lambda_{\mathrm{IC}}/\lambda_\gamma$ is not a useful finite quantity for a pure $E0$ transition; $E0$ strengths are instead described by electronic factors and a nuclear monopole strength. The 691 keV 0^+ state of ^{72}Ge is the lecture's below-threshold conversion example [6, 14].

Caution

Do not use α without specifying the transition. It depends on Z, E_{tr}, shell, and multipolarity. Nor should an observed level half-life be identified with a photon partial half-life until all conversion and competing branches have been removed [6].

20.8 Nuclear recoil and resonant absorption (outline)

If the free nucleus is initially at rest and emits a photon of energy E_γ, momentum conservation requires nuclear recoil momentum $p = E_\gamma/c$. Non-relativistically,

$$E_R = \frac{p^2}{2M} = \frac{E_\gamma^2}{2Mc^2}, \tag{20.28}$$

so energy conservation gives

$$E_\gamma = E_0 - E_R \approx E_0 - \frac{E_0^2}{2Mc^2}, \tag{20.29}$$

where E_0 is the level difference. For absorption, the photon must supply $E_0 + E_R$. Free-nucleus emission and absorption lines are therefore separated by $\sim 2E_R$ (often eV scale), which can block resonance unless Doppler or natural widths overlap.

In a solid the recoil momentum can be taken by the crystal as a whole. The zero-phonon probability is finite and depends on temperature, gamma energy, and the lattice vibrational spectrum; it is not a sharp $E_R < \hbar\omega_{\mathrm{phonon}}$ threshold. Recoilless emission and absorption constitute the **Mössbauer effect**, used for ultra-high-resolution nuclear spectroscopy [6, 14, 2].

Key idea

Students should remember:

- γ decay: same nucleus; $X^* \to X + \gamma$.
- Multipoles EL, ML; no $E0$ photon; no magnetic monopole.
- $|J_i - J_f| \le L \le J_i + J_f$, $L \ge 1$; $0 \to 0$ γ forbidden.
- Parity: EL has $\pi_f = \pi_i(-1)^L$; ML has $\pi_f = \pi_i(-1)^{L+1}$.
- $\lambda_\gamma \propto E_\gamma^{2L+1} B(\sigma L)$; Weisskopf units are benchmarks, not precision predictions.
- $\tau = 1/\lambda_{\text{tot}}$, $t_{1/2} = \tau \ln 2$, and $\Gamma = \hbar/\tau$.
- $\alpha = \lambda_{\text{IC}}/\lambda_\gamma$ and $f_\gamma = 1/(1+\alpha)$; a pure $E0$ transition needs separate notation.

20.8.1 Multipoles, Weisskopf units, and conversion

Parity and angular-momentum conservation restrict which multipoles EL and ML can connect two levels; odd static multipoles vanish in states of good parity. Weisskopf single-particle estimates set the baseline; enhancements signal collectivity, while $0^+ \to 0^+$ cases force internal conversion or pair creation instead of a single photon [6, 1, 2].

20.9 Deeper physics: multipoles, Weisskopf units, and conversion

20.9.1 Electric and magnetic multipole rates

An excited nuclear level can de-excite by emitting a photon of multipolarity σL ($\sigma = E, M$). In the long-wavelength limit the partial width scales as

$$\lambda_{\sigma L} \propto \frac{E_\gamma^{2L+1}}{(2L+1)!!}\,\frac{L+1}{L}\,|B(\sigma L)|^2, \tag{20.30}$$

where $B(EL)$ and $B(ML)$ are reduced transition probabilities [6, 1]. Selection rules follow from angular-momentum addition: an EL transition changes parity by $(-1)^L$, an ML transition by $(-1)^{L+1}$. The Weisskopf estimate replaces $|B(\sigma L)|^2$ by a single-particle value to define one "Weisskopf unit" (W.u.) [14].

Weisskopf estimate for an $E2$ transition

For a single-particle $E2$ transition with $E_\gamma = 0.5\,\text{MeV}$ and $R \approx 5\,\text{fm}$, the Weisskopf estimate gives a partial half-life of order $10^{-12}\,\text{s}$ [6]. Measured $B(E2)$ values for collective rotational bands can exceed the Weisskopf value by tens to hundreds, signalling coherent motion of many nucleons. Conversely, $B(E2) \ll 1$ W.u. marks a hindered transition with poor radial overlap [14, 1].

20.9.2 Internal conversion as a competing channel

The same nuclear matrix element drives γ emission and internal conversion: an atomic electron carries away the transition energy instead of a photon. The total de-excitation rate includes a conversion coefficient α,

$$\lambda_{\text{tot}} = \lambda_\gamma(1+\alpha), \tag{20.31}$$

so the observed photon intensity is reduced by $1/(1+\alpha)$ [6, 1]. Conversion dominates for low E_γ and high Z because the atomic electron can match the nuclear transition momentum near the nucleus. For a 100 keV $E2$ transition in $Z = 90$, α can exceed unity and most decays proceed without an observed γ [6, 34].

Key idea

From spectroscopy to reactions. Multipole selection rules constrain γ cascades following β or α decay and compound-nucleus evaporation (Lecture 21). A measured $B(E2)$ quantifies collectivity; a large α at low energy warns that photon intensities alone undercount the de-excitation rate [6, 20].

20.9.3 Angular correlations and mixed multipoles

When two successive γ rays are emitted from aligned nuclei, their angular correlation constrains the multipolarities and the mixing ratio δ between competing EL and ML amplitudes [6, 1]. Internal conversion competes with the same nuclear matrix element, so a γ-γ correlation analysis must include conversion branches that remove flux from the photon channel [34]. In compound-nucleus decay, the first emitted γ often carries the highest multipolarity allowed by angular-momentum selection, setting the pattern for subsequent statistical cascades analysed with Hauser–Feshbach theory (Lecture 21) [6, 14].

20.9.4 Collective $E2$ strength and rotational bands

Rotational bands in even–even nuclei exhibit a sequence of $E2$ transitions with similar $B(E2)$ values, often tens of W.u., signalling coherent quadrupole motion [14, 6]. The Weisskopf single-particle estimate is deliberately crude: it assumes one proton or neutron moving in a single orbit. Collective enhancement occurs when many nucleons contribute in phase. In reactions, Coulomb excitation of deformed targets populates the same $E2$ strength that γ spectroscopy measures in decay cascades (Lecture 21) [6, 2]. Conversion coefficients must still be applied when inferring $B(E2)$ from photon intensities alone.

Angular correlation coefficients and linear polarisation measurements assign multipolarities when conversion data are incomplete. Isomer traps and storage rings extend lifetime measurements into exotic regions where electronic timing is difficult. The nuclear clock theme shows that "γ decay" spans from mega-electronvolts down to electronvolts.

Physics-depth close

Selection rules, Weisskopf estimates, and conversion coefficients are the minimal toolbox for γ-ray spectroscopy. Isomers and the nuclear clock show that the same toolbox spans mega-electronvolts to electronvolts.

20.10 Further physics from the literature

20.10.1 What a measured lifetime actually determines

Suppose a level has gamma branches j, branching photon intensities $I_{\gamma j}$, and conversion coefficients α_j. The level half-life determines only the sum

$$\frac{\ln 2}{t_{1/2}} = \sum_j (1+\alpha_j)\lambda_{\gamma j}.$$

To extract $B(\sigma L)$ for one branch, the experimental analysis must (i) normalize the photon intensities, (ii) restore the unobserved conversion strength by multiplying branch j by $1+\alpha_j$, (iii) form its partial rate, and only then (iv) divide out the phase-space factor E_γ^{2L+1}. Confusing a level half-life with a photon partial half-life can produce a large error for a low-energy, high-Z transition [6, 1].

20.10.2 What one Weisskopf unit does and does not mean

One W.u. is the strength generated by a deliberately crude single-particle radial estimate. It provides a common scale across nuclei and energies, but it is not the statement that every one-particle transition has exactly that strength. A value far below one W.u. signals hindrance from configuration, isospin, or radial mismatch; a large $E2$ value often signals coherent collective motion. Collective quadrupole motion can make $E2$ transitions compete strongly even where a naive hierarchy would suggest otherwise [6, 1, 14].

20.10.3 A spectroscopy decision chain

No single observable usually supplies a complete assignment.

1. Level energies and coincidences establish the cascade.
2. Spin information restricts L through the triangle inequality.
3. Known parities select EL or ML for each L.
4. Angular correlations constrain L and the mixing ratio δ.
5. Linear polarization distinguishes electric from magnetic character.
6. Shell-resolved conversion coefficients $\alpha_K, \alpha_L, \ldots$ provide an independent multipolarity test.
7. A conversion-corrected partial lifetime yields $B(\sigma L)$, which can then be compared with a nuclear model.

This chain explains why selection rules by themselves identify possibilities, not measured amplitudes [6, 2, 20].

20.10.4 Internal conversion calculations

The nuclear matrix element is essentially common to gamma emission and conversion, so their ratio depends mainly on atomic structure, Z, transition energy, and multipolarity. Accurate coefficients require relativistic electron wave functions and finite nuclear size. Evaluated level schemes tabulate experimental or calculated shell coefficients; the rough Z^3 and inverse-energy trends are valuable intuition but should not replace those tables [6, 1, 20, 21].

20.10.5 Recoil, line width, and the Mössbauer limit

The free-nucleus recoil shift can exceed the natural width by many orders of magnitude even though it is tiny compared with E_γ. A solid can absorb momentum without creating a phonon, giving a zero-phonon line. Scanning source and absorber through resonance then resolves isomer shifts, electric quadrupole splittings, and magnetic hyperfine splittings. The 14.4 keV transition of ^{57}Fe is the standard example [6, 14, 2].

20.10.6 Weisskopf estimates and conversion-corrected lifetimes

The total de-excitation rate of a level sums all γ and conversion channels. For branch j with multipolarity σL,

$$\frac{\ln 2}{t_{1/2}} = \sum_j (1+\alpha_j)\,\lambda_{\gamma j}, \tag{20.32}$$

where $\lambda_{\gamma j} \propto E_{\gamma j}^{2L+1} |B(\sigma L)_j|^2$ [6, 1]. A Weisskopf estimate supplies a single-particle $|B(\sigma L)|^2$ for order-of-magnitude checks. If $\alpha_j \gg 1$, the level half-life is dominated by conversion even when the γ branch is invisible in a spectrum [34, 20].

Conversion correction for a $Z = 82$ transition

Consider an $M1$ transition at $E_\gamma = 100\,\text{keV}$ in $Z = 82$. Tabulated α_K can exceed 10^3 for this combination [6, 34]. The measured level half-life then exceeds the photon partial half-life by the factor $1+\alpha \approx 10^3$. Quoting $B(M1)$ from the level lifetime without conversion correction would overestimate the nuclear matrix element by thirty orders of magnitude in rate.

20.10.7 From γ cascades to compound-nucleus decay

Compound-nucleus evaporation (Lecture 21) is a sequence of statistical γ and particle emissions. Each step obeys the same multipole and Hauser–Feshbach logic as discrete spectroscopy, but level densities replace resolved states [6, 1]. Fission fragments emit prompt γ rays before β decay (Lecture 24); stellar fusion chains terminate in γ emission from excited ^{4}He and CNO intermediates (Lecture 26) [36, 14]. Internal conversion competes most strongly at low E_γ and high Z, exactly where fission-fragment cascades are rich in low-energy transitions.

Key idea

Arc through the course. Multipole rules connect β-delayed γ spectroscopy (Lectures 18–19) to reaction-residue γ detection (Lecture 21) and fission-fragment decay heat (Lecture 24). Weisskopf units provide the first sanity check on any quoted $B(\sigma L)$ before invoking collectivity or hindrance [6, 14].

20.10.8 Mössbauer spectroscopy as the zero-recoil limit

When a nucleus in a solid emits a γ ray without recoil energy loss to lattice phonons, the natural line width is not Doppler-broadened by the free-nucleus recoil [6, 14]. The 14.4 keV transition in ^{57}Fe is the standard example: isomer shifts probe chemical environments, and quadrupole splitting maps local electric-field gradients. This is the same $E2/M1$ multipole language applied to condensed matter rather than to discrete level schemes in a decay chain. Recoil energy $E_{\rm rec} = E_\gamma^2/(2Mc^2)$ is tiny compared with E_γ but enormous compared with eV-scale nuclear widths, which is why the Mössbauer condition requires a rigid lattice [2].

20.10.9 Isomer spectroscopy and K forbiddenness

Long-lived isomers in heavy and exotic nuclei often decay by hindered multipole transitions or by internal conversion when E_γ is small [6, 34]. K-isomers in deformed nuclei violate the selection rules for fast $E1$ decay because the projection quantum number K is not conserved in the available final states. Tracking arrays that record γ-γ coincidences apply the lecture selection rules event by event, filtering multipolarity before a level scheme is built [20, 53]. The ^{229}Th nuclear clock isomer at ~ 8 eV is an extreme case where conversion and radiative decay compete in a domain usually dominated by atomic physics [59].

20.11 Deeper reading: Weisskopf estimates, conversion, and isomers

Weisskopf single-particle estimates give reference lifetimes for EL and ML multipoles. Collective enhancement of $E2$ and K-forbidden hindrance of isomers are measured against those benchmarks. Internal conversion coefficients rise rapidly at low energy and high Z, so intensity balances in heavy nuclei require conversion tables, not γ counts alone.

The ^{229}Th nuclear clock isomer at ~ 8 eV pushes the same multipole ideas into the optical regime. Tracking arrays turn Lecture 20 selection rules into coincidence filters for constructing level schemes far from stability, linking back to collective structure (Lecture 15) and forward to reaction γ tagging (Lecture 21).

Weisskopf unit arithmetic

Compute a Weisskopf $E2$ lifetime for a 100 keV transition in a medium-mass nucleus and compare with a measured collective lifetime: the ratio is the enhancement factor. Repeat for an $M1$ single-particle estimate versus a hindered isomer. Those two ratios teach more than memorising selection-rule tables.

Internal conversion as a diagnostic

Conversion coefficient ratios α_K/α_L help assign multipolarity when angular distributions are unavailable. At very low energy, conversion can dominate emission entirely, which is why some isomers are detected via electrons or photons from subsequent atomic cascades.

Multipolarity decision tree

Given a transition energy, ΔI, and $\Delta\pi$, list the lowest allowed multipoles. Estimate Weisskopf lifetimes and decide whether the transition is likely prompt or isomeric. If Z is high and E is low, check conversion coefficients before predicting a γ branching ratio. This decision tree is exactly what analysis pipelines automate in tracking-array studies.

For the nuclear clock, note that an optical-energy nuclear transition blurs the line between atomic and nuclear spectroscopy: the multipole is still $M1$ (to leading order), but the engineering is laser physics. Lecture 20's selection rules remain the correct language even at $8\,\mathrm{eV}$.

Gamma spectroscopy turns selection rules into event filters. Weisskopf estimates and conversion coefficients remain the first tools for reading modern level schemes.

In practice, rework one numerical estimate from the lecture with a different parameter set and compare the shift to the quoted experimental uncertainty. If the shift exceeds the uncertainty, the estimate is teaching a scale, not delivering a fit. That distinction keeps theoretical cartoons honest when they sit beside evaluated data tables and modern many-body calculations. Along the course arc, write one sentence linking this chapter backward and one forward: the discipline of those two sentences is as important as any single formula. Examination problems rarely ask for a literature review, but they do ask for the scale argument that literature reviews presuppose.

Synthesis problem

Write a half-page note that (i) restates the chapter's central scale or selection rule in one equation, (ii) evaluates it numerically for one concrete nucleus or energy, and (iii) names the neighbouring lecture that uses the result. Keep the note free of research-frontier speculation; the goal is fluency with the lecture physics itself. If an assumption fails (Born approximation, spherical mean field, allowed transition, one-dimensional barrier), state the failure mode in a final sentence. That habit converts passive reading into examination-ready understanding and makes the later research-frontier box read as an extension rather than a replacement.

Research frontier: γ spectroscopy, isomers, and the nuclear clock

Multipole selection rules and internal conversion (Lecture 20) are everyday tools of modern γ-ray tracking arrays. Isomer spectroscopy of exotic and superheavy nuclei constrains deformation and K-forbiddenness, while conversion-coefficient evaluations (BrIcc) remain essential for intensity balances in complex decay schemes [34, 20]. A striking frontier is the ^{229}Th nuclear clock isomer: radiative decay of the $\sim 8\,\mathrm{eV}$ state has been observed, opening a path to a nucleus-based optical frequency standard and to tests of fundamental-constant drift [59]. Uncertainties that still matter include precise isomer energies and branching ratios in exotic nuclei, conversion-coefficient systematics at low energy, and the engineering path from a laboratory isomer observation to a closed-loop nuclear clock. The lecture's Weisskopf estimates and multipolarity diagnostics remain the first tools students use when reading modern γ-spectroscopy papers.

Beyond the thorium clock, isomer research with radioactive beams maps K-isomer lifetimes and hindered transitions that diagnose axial asymmetry and pairing. Tracking arrays that record γ-γ coincidences turn the lecture selection rules into event-by-event multipolarity filters, which is why modern papers still begin with angular-correlation and conversion arguments.

Student directions. (1) Weisskopf exercise: apply single-particle estimates to an $E2$ and an $M1$ transition of given energy and compare lifetimes. (2) Data lookup: using BrIcc or tables, find α_K for a 100 keV $E2$ in $Z = 50$ versus $Z = 90$. (3) Literature summary: summarise the ^{229}Th isomer energy measurement and why a nuclear clock could complement atomic clocks [59].

20.12 Problems with solutions

Problem 20.1: Multipole for $2^+ \to 0^+$. A nuclear transition proceeds from $J_i^\pi = 2^+$ to $J_f^\pi = 0^+$. List all multipoles allowed by angular momentum and parity, and identify the dominant one [6].

Solution 20.1. Step 1: Angular momentum.

$$|2 - 0| \le L \le 2 + 0 \quad \Rightarrow \quad L = 2.$$

Only quadrupole radiation is allowed.

Step 2: Parity. Both states have $\pi = +1$, so parity does not change. For $L = 2$,

$$EL: \ \pi_f = \pi_i(-1)^2 = \pi_i \quad \text{(allowed)}, \qquad ML: \ \pi_f = \pi_i(-1)^3 = -\pi_i \quad \text{(forbidden)}.$$

Step 3: Dominant multipole. The only allowed multipole is $E2$.

Problem 20.2: ^{117}In $1/2^- \to 9/2^+$. For $J_i^\pi = \frac{1}{2}^-$ to $J_f^\pi = \frac{9}{2}^+$, find allowed L values and the corresponding EL/ML assignments. Which multipole dominates? [6, 1]

Solution 20.2. Step 1:

$$\left|\tfrac{1}{2} - \tfrac{9}{2}\right| \le L \le \tfrac{1}{2} + \tfrac{9}{2} \quad \Rightarrow \quad L = 4, 5.$$

Step 2: Parity changes $(- \to +)$.

- $L = 4$: ML has $\pi_f = \pi_i(-1)^5 = -\pi_i \Rightarrow M4$. EL would preserve parity $\Rightarrow$ forbidden.
- $L = 5$: EL has $\pi_f = \pi_i(-1)^5 = -\pi_i \Rightarrow E5$. ML would preserve parity $\Rightarrow$ forbidden.

Step 3: Compare orders, not only electric versus magnetic character. The $E5$ amplitude contains one additional power of the small parameter kR, so its rate contains roughly an additional $(kR)^2$ suppression. That penalty overwhelms the usual enhancement of an electric multipole over a magnetic multipole of the same order. The dominant assignment is $M4$.

Problem 20.3: Why $0^+ \to 0^+$ cannot emit a photon. Show from the triangle inequality that a $0^+ \to 0^+$ transition cannot proceed by single-photon emission. What is the alternative electromagnetic channel? [6]

Solution 20.3. For $J_i = J_f = 0$,

$$|0-0| \le L \le 0+0 \quad \Rightarrow \quad L = 0.$$

A real photon multipole requires $L \ge 1$. Hence $\lambda_\gamma = 0$ for single-photon emission. The transition proceeds by internal conversion (electric monopole coupling to an atomic electron), or by e^+e^- pair creation if $E_{\mathrm{tr}} > 1.022\,\mathrm{MeV}$.

Problem 20.4: Conversion coefficient. A transition emits $N_\gamma = 8.0 \times 10^6$ photons and $N_e = 2.0 \times 10^6$ conversion electrons (all shells). Find α and the fraction of decays that proceed by photon emission [1].

Solution 20.4.

$$\alpha = \frac{N_e}{N_\gamma} = \frac{2.0 \times 10^6}{8.0 \times 10^6} = 0.25.$$

Total electromagnetic decays $N_{\mathrm{tot}} = N_\gamma + N_e = 1.0 \times 10^7$. Photon fraction:

$$\frac{N_\gamma}{N_{\mathrm{tot}}} = \frac{1}{1+\alpha} = \frac{1}{1.25} = 0.80 \quad (80\%).$$

Problem 20.5: Recoil energy. Estimate the nuclear recoil energy $E_R = E_\gamma^2/(2Mc^2)$ for the 662 keV γ ray of ^{137}Ba ($Mc^2 \approx 137 \times 931\,\mathrm{MeV}$). Is E_R large compared with eV-scale chemical shifts? [6]

Solution 20.5.

$$Mc^2 \approx 1.275 \times 10^5\,\mathrm{MeV}, \qquad E_\gamma = 0.662\,\mathrm{MeV}.$$

$$E_R = \frac{(0.662)^2}{2 \times 1.275 \times 10^5}\,\mathrm{MeV} = \frac{0.438}{2.55 \times 10^5}\,\mathrm{MeV} \approx 1.72 \times 10^{-6}\,\mathrm{MeV} = 1.72\,\mathrm{eV}.$$

Yes: E_R is of order eV, comparable to or larger than many chemical and hyperfine shifts, which is why free-atom resonant absorption is hindered and the Mössbauer (recoilless) effect is powerful in solids.

Problem 20.6: Assign $3^- \to 2^+$. List every allowed multipole for $3^- \to 2^+$, and identify the one normally expected to dominate [6, 1].

Solution 20.6. Angular momentum:

$$|3-2| \le L \le 3+2 \quad \Rightarrow \quad L = 1, 2, 3, 4, 5.$$

Parity changes. Lowest $L = 1$:

$$E1: \ \pi_f = \pi_i(-1)^1 = -\pi_i \quad \text{(allowed)}, \qquad M1: \ \text{parity unchanged (forbidden)}.$$

Applying the parity test to every L gives

$$E1, \ M2, \ E3, \ M4, \ E5.$$

The lowest member, $E1$, normally dominates because each higher order brings additional powers of kR. Structural hindrance can alter quantitative branching, so "dominant $E1$" is a Weisskopf-scale expectation, not an extra selection rule.

Problem 20.7: Isomeric hindrance. Explain qualitatively why an $M4$ isomer can have a half-life of minutes while a typical $E2$ state lives for picoseconds [6, 14].

Solution 20.7. Step 1: Why high L is costly. In the long-wavelength expansion a rank-L amplitude carries powers of $kR = E_\gamma R/\hbar c \ll 1$. Increasing L therefore strongly suppresses the squared amplitude, in addition to double-factorial factors.

Step 2: Compare energy dependences. The Weisskopf table gives $\lambda(M4) \propto E_\gamma^9$, whereas $\lambda(E2) \propto E_\gamma^5$, and magnetic radiation has an additional penalty at equal L. A low-energy $M4$ can therefore be slower by many orders of magnitude.

Step 3: Interpret the lifetime. Selection rules that force $M4$ can make the state isomeric. The exact lifetime cannot follow from a fixed "penalty per order" alone: A, E_γ, internal conversion, and the nuclear matrix element must also be specified.

Problem 20.8: Long-wavelength parameter. For $A = 125$ and $E_\gamma = 1.00\,\text{MeV}$, calculate R, kR, and the leading squared-amplitude penalty $(kR)^2$ for raising the multipole order by one. Explain why the complete Weisskopf ratio is not exactly $(kR)^2$ [6, 2].

Solution 20.8. Step 1: Radius.

$$R = 1.2A^{1/3}\,\text{fm} = 1.2(125)^{1/3}\,\text{fm} = 6.00\,\text{fm}.$$

Step 2: Expansion parameter.

$$kR = \frac{E_\gamma R}{\hbar c} = \frac{(1.00\,\text{MeV})(6.00\,\text{fm})}{197.327\,\text{MeV fm}} = 3.04 \times 10^{-2}.$$

Step 3: Rate scale.

$$(kR)^2 = 9.25 \times 10^{-4}.$$

The full ratio also contains different $(2L+1)!!$ factors, L-dependent coefficients, radial matrix elements, and electric or magnetic scales.

Problem 20.9: Weisskopf rate, lifetime, and width. For a 1.00 MeV $E2$ transition with $A = 100$, use

$$\lambda_{\mathrm{W}}(E2) = 7.28 \times 10^7 A^{4/3} E_\gamma^5 \ \mathrm{s}^{-1}$$

to calculate the Weisskopf mean life, half-life, and natural width. If the measured partial half-life is 2.00 ps and conversion is negligible, estimate $B(E2)$ in W.u. [1].

Solution 20.9. Step 1: Weisskopf rate.

$$A^{4/3} = 100^{4/3} = 464.2, \qquad \lambda_{\mathrm{W}} = 3.38 \times 10^{10}\,\mathrm{s}^{-1}.$$

Step 2: Time scales.

$$\tau_{\mathrm{W}} = \lambda_{\mathrm{W}}^{-1} = 2.96 \times 10^{-11}\,\mathrm{s}, \qquad t_{1/2,\mathrm{W}} = \tau_{\mathrm{W}} \ln 2 = 2.05 \times 10^{-11}\,\mathrm{s}.$$

Step 3: Width.

$$\Gamma_{\mathrm{W}} = \hbar\lambda_{\mathrm{W}} = (6.5821 \times 10^{-16}\,\mathrm{eV\,s})(3.38 \times 10^{10}\,\mathrm{s}^{-1}) = 2.22 \times 10^{-5}\,\mathrm{eV}.$$

Step 4: Strength in W.u. At fixed A, E_γ, L, the rate is proportional to $B(E2)$:

$$\frac{B(E2)}{B_{\mathrm{W}}(E2)} = \frac{\ln 2/(2.00 \times 10^{-12}\,\mathrm{s})}{3.38 \times 10^{10}\,\mathrm{s}^{-1}} = 10.3 \text{ W.u.}$$

Problem 20.10: Conversion-corrected partial rate. A level decays through one electromagnetic transition with $t_{1/2} = 10.0\,\mathrm{ns}$ and $\alpha = 3.00$. Find the total, photon, and conversion decay constants, the photon and electron fractions, and the total and photon partial widths [6, 1].

Solution 20.10. Step 1: Total rate.

$$\lambda_{\mathrm{tot}} = \frac{\ln 2}{t_{1/2}} = 6.93 \times 10^7\,\mathrm{s}^{-1}.$$

Step 2: Separate the channels.

$$\lambda_\gamma = \frac{\lambda_{\mathrm{tot}}}{1+\alpha} = 1.73 \times 10^7\,\mathrm{s}^{-1}, \qquad \lambda_{\mathrm{IC}} = \alpha\lambda_\gamma = 5.20 \times 10^7\,\mathrm{s}^{-1}.$$

Step 3: Fractions.

$$f_\gamma = \frac{1}{4} = 0.250, \qquad f_{\mathrm{IC}} = \frac{3}{4} = 0.750.$$

Step 4: Widths.

$$\Gamma_{\mathrm{tot}} = 4.56 \times 10^{-8}\,\mathrm{eV}, \qquad \Gamma_\gamma = 1.14 \times 10^{-8}\,\mathrm{eV}.$$

Using the level half-life directly as the photon partial half-life would overestimate the photon strength by $1 + \alpha = 4$.

Problem 20.11: Reading an angular correlation. A two-photon cascade has

$$W(\theta) = 1 + 0.20P_2(\cos\theta) - 0.05P_4(\cos\theta).$$

(a) Why is an unconditioned gamma ray from the same unpolarized source isotropic? (b) Evaluate $W(0°)$ and $W(90°)$. (c) What additional measurement distinguishes electric from magnetic character? [6]

Solution 20.11. Step 1: Origin of the axis. Before detecting the first photon, all magnetic substates are equally populated and no direction is preferred. Detection of γ_1 defines an axis and aligns the intermediate ensemble.

Step 2: Evaluate the Legendre polynomials. Since $P_2(1) = P_4(1) = 1$,

$$W(0°) = 1 + 0.20 - 0.05 = 1.15.$$

Since $P_2(0) = -1/2$ and $P_4(0) = 3/8$,

$$W(90°) = 1 - 0.10 - 0.05\left(\frac{3}{8}\right) = 0.88125.$$

Step 3: Electric or magnetic? Angular correlations constrain multipole order and mixing. A linear polarization measurement, commonly using polarization-dependent Compton scattering, distinguishes electric from magnetic character.

Problem 20.12: Dimensionless mixing ratio. An $M1 + E2$ transition has measured radiation-amplitude mixing ratio $\delta = -0.30$. Find the angle-integrated $M1$ and $E2$ fractions and explain what information is lost if only those fractions are reported.

Solution 20.12.

$$f_{E2} = \frac{\delta^2}{1+\delta^2} = \frac{0.09}{1.09} = 0.0826, \qquad f_{M1} = \frac{1}{1+\delta^2} = 0.917.$$

The fractions contain δ^2, so they lose the negative relative phase. Angular-correlation or polarization interference terms retain the sign. A ratio of the bare $E2$ and $M1$ reduced matrix elements would not be a dimensionless replacement for δ.

Problem 20.13: BrIcc correction for $^{137\mathrm{m}}$Ba. Treat the 661.7 keV, $11/2^- \to 3/2^+$ transition as pure $M4$, and use the illustrative BrIcc value $\alpha_T = 0.11$. (a) Verify the leading multipole. (b) Find photon and conversion fractions. (c) If 10^6 isomers decay through this branch, estimate both yields [20, 21, 34].

Solution 20.13. $\Delta J = 4$ and parity changes; $M4$ is allowed and is lower order than the also-allowed $E5$. Therefore

$$f_\gamma = \frac{1}{1.11} = 0.901, \qquad f_{\mathrm{IC}} = \frac{0.11}{1.11} = 0.0991.$$

The expected yields are 9.01×10^5 photons and 9.91×10^4 conversion electrons before detector efficiencies. A precision result must use the shell sum and vacancy convention associated with the selected BrIcc record.

Chapter 21

Nuclear Reactions: Concepts and Cross Sections

"A nuclear reaction is a rearrangement of nucleons under the strong interaction. The cross section converts a microscopic probability into a measurable counting rate."
K. S. Krane

Radioactive decay is spontaneous. **Nuclear reactions** are induced: a projectile strikes a target and nucleons are redistributed. This chapter introduces the notation $a + A \to b + B$, the conservation laws that decide whether a channel is possible, and a dimensionally careful definition of differential and total cross sections. It then develops flux and luminosity, laboratory versus centre-of-mass (CM) energies, reaction types, and direct, pre-equilibrium, compound, and statistical mechanisms with their characteristic timescales and Bohr's independence hypothesis [6, 1, 14, 2].

21.1 General form of a nuclear reaction

A two-body reaction is written

$$a + A \longrightarrow b + B, \tag{21.1}$$

or in compact spectroscopic notation

$$A(a, b)B. \tag{21.2}$$

Here a is the **projectile**, A the **target**, b the detected light product, and B the **residual** (often left in the target). Examples: $^{14}\mathrm{N}(\alpha, p)^{17}\mathrm{O}$, $^{238}\mathrm{U}(n, \gamma)^{239}\mathrm{U}$. If several light particles emerge one writes $A(a, np)B$, and so on [6, 1].

21.1.1 Conservation laws as channel filters

Every proposed channel must conserve total energy, linear momentum, angular momentum, electric charge, and baryon (nucleon) number. For strong and electromagnetic reactions parity is also conserved; isospin is approximate because Coulomb and charge-dependent nuclear terms break the symmetry. Thus

$$Z_a + Z_A = Z_b + Z_B, \qquad A_a + A_A = A_b + A_B, \tag{21.3}$$
$$P_a^\mu + P_A^\mu = P_b^\mu + P_B^\mu, \qquad \mathbf{J}_i = \mathbf{J}_f. \tag{21.4}$$

The four-momentum equation contains both energy and three-momentum conservation and is valid in every inertial frame. Charge and nucleon number are quick bookkeeping tests; four-momentum conservation supplies the Q-value and the kinematic restrictions developed in the next lecture. Lepton number must additionally be conserved when leptons occur in a weak reaction [6, 1, 14].

Decay versus reaction. In α or β decay a single parent rearranges spontaneously. In a reaction two nuclei interact because the experimenter supplies a beam (or a neutron flux). Gamma decay changes neither Z nor A; reactions generally do.

21.2 Experimental setup

A typical arrangement is:

(i) an accelerator (or radioactive source, or reactor) that produces a beam of projectiles a with known kinetic energy;

(ii) a thin target containing nuclei A, in vacuum to avoid air scattering;

(iii) one or more detectors that count product particles b in a solid angle $d\Omega$ at laboratory angle θ, and measure their kinetic energy.

The residual B often stops in the target. The raw data are counts versus energy at each angle: the ingredients of a differential cross section [6, 1].

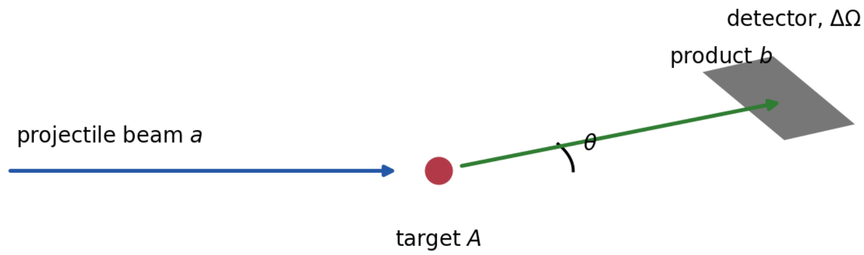


Figure 21.1: Schematic of a fixed-target scattering/reaction experiment: beam, target, and detector at angle θ subtending solid angle $d\Omega$.

Heavy-ion beams open additional reaction mechanisms; this chapter focuses on light projectiles $(n, p, d, \alpha, \dots)$ [14, 2].

21.3 Definition of the cross section

A nuclear reaction cross section is an effective area: it converts an incident flux into a counting rate without requiring a microscopic impact parameter. Laboratory beams are typically far larger than a single nucleus, so the cross section absorbs the quantum dynamics (Coulomb, strong interaction, open channels) into a number with units of area. Barns ($1\,\mathrm{b} = 10^{-28}\,\mathrm{m}^2$) match nuclear scales; thermal capture or resonant fission can be many barns, while rare processes may be microbarns or smaller [6, 1].

21.3.1 Flux, luminosity, and counting rate

Let Φ be the incident **flux density**,

$$\Phi = \frac{\mathrm{d}N_a}{\mathrm{d}A\,\mathrm{d}t}, \tag{21.5}$$

in projectiles per unit area per unit time. If N_t target nuclei are illuminated uniformly, the differential reaction rate defines the **differential cross section**:

$$\mathrm{d}\dot{N} = \Phi N_t \frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}\,\mathrm{d}\Omega, \qquad \frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} = \frac{\mathrm{d}\dot{N}}{\Phi N_t\,\mathrm{d}\Omega}. \tag{21.6}$$

The partial cross section for channel c, and the sum over all nonelastic channels, are

$$\sigma_c = \int_{4\pi} \frac{\mathrm{d}\sigma_c}{\mathrm{d}\Omega}\,\mathrm{d}\Omega, \qquad \sigma_{\mathrm{reac}} = \sum_{c\neq\mathrm{elastic}} \sigma_c. \tag{21.7}$$

An inclusive cross section sums over unobserved final states; an exclusive measurement tags a specific channel. Always name which one is intended when quoting a number from an evaluation. cross section specifies them. These definitions prevent the common mistake of calling a count, a rate, and a cross section the same quantity.

Physical meaning. The cross section is an effective area: if every projectile that hits a disk of area σ centred on a target nucleus reacted, the observed rate would match experiment. It is not generally the geometrical area of a nucleus: interference and resonances can make it much smaller or much larger [6, 1, 14].

21.3.2 Areal density, luminosity, and attenuation

If a target has volume density n, thickness t, and illuminated area A', then

$$N_t = nA't = n_a A', \qquad n_a \equiv nt. \tag{21.8}$$

The beam rate is $\dot{N}_a = \Phi A'$. Their product gives the fixed-target **luminosity**,

$$\mathcal{L} = \dot{N}_a n_a, \qquad [\mathcal{L}] = \mathrm{area}^{-1}\mathrm{time}^{-1}, \tag{21.9}$$

and hence

$$N_{\mathrm{obs}} = T \int_{\Delta\Omega} \mathcal{L}\,\epsilon(\Omega) \frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}\,\mathrm{d}\Omega. \tag{21.10}$$

Here T is live time and ϵ includes intrinsic efficiency, dead-time correction, and analysis acceptance. For nonuniform profiles, $\mathcal{L} = \int \Phi(x,y) n_a(x,y)\, dA$.

For a thin target $P_{\text{int}} \simeq n_a \sigma_{\text{tot}}$. The exact attenuation law is

$$I(t) = I(0)e^{-n\sigma_{\text{tot}}t}, \qquad P_{\text{int}} = 1 - e^{-n\sigma_{\text{tot}}t}. \tag{21.11}$$

For thick targets one must also integrate the energy-dependent cross section as the beam loses energy in the material [6, 1, 2].

21.3.3 Units: the barn

Because a rate equals luminosity times cross section, σ has units of **area**. Steradians are dimensionless, but "per sr" is retained to display the angular normalization of $d\sigma/d\Omega$. The practical nuclear unit is the barn:

$$1\,\text{b} = 10^{-24}\,\text{cm}^2 = 10^{-28}\,\text{m}^2. \tag{21.12}$$

Typical reaction cross sections range from microbarns to thousands of barns (resonant neutron capture) [6, 1].

Remark

Rutherford scattering already used the same idea: $d\sigma/d\Omega$ is the quantity that multiplies flux and target density to give a counting rate. Nuclear reactions generalise that language to all exit channels [2, 14].

21.4 Laboratory and centre-of-mass frames

Theory is simplest in the **centre-of-mass (CM)** frame, where total momentum vanishes. In the laboratory the target is usually at rest. For non-relativistic two-body kinematics with projectile mass m_a, target mass m_A, and laboratory kinetic energy E_{lab},

$$E_{\text{cm}} = \frac{m_A}{m_a + m_A}\, E_{\text{lab}} = \frac{\mu}{m_a}\, E_{\text{lab}}, \tag{21.13}$$

where $\mu = m_a m_A/(m_a + m_A)$ is the reduced mass. Derivation: the CM velocity in the lab is $v_{\text{cm}} = m_a v_1/(m_a + m_A)$. In the CM frame the relative speed remains v_1, and

$$E_{\text{cm}} = \tfrac{1}{2}\mu v_1^2 = \frac{m_A}{m_a + m_A}\left(\tfrac{1}{2}m_a v_1^2\right) = \frac{m_A}{m_a + m_A}\, E_{\text{lab}}. \tag{21.14}$$

Only E_{cm} is available for excitation and reaction Q-value balance; the remainder is locked in CM motion of the system as a whole [6, 1].

The laboratory scattering angle θ_{lab} and CM angle θ_{cm} are related (elastic equal-mass special cases aside) by

$$\tan\theta_{\text{lab}} = \frac{\sin\theta_{\text{cm}}}{\cos\theta_{\text{cm}} + m_a/m_A} \tag{21.15}$$

for elastic scattering of a on A (standard textbook form). Angular distributions should always state the frame [6, 14].

21.5 Coulomb barrier

For charged projectiles the long-range force is Coulomb repulsion. At separation r,

$$V_C(r) = \frac{1}{4\pi\varepsilon_0}\frac{Z_a Z_A e^2}{r}. \tag{21.16}$$

Near the nuclear surface $r \approx R_a + R_A \approx r_0(A_a^{1/3} + A_A^{1/3})$ the Coulomb barrier height is of order several MeV for light ions on medium targets, and tens of MeV for heavy systems. Below the barrier, elastic Coulomb (Rutherford) scattering dominates; nuclear reactions require penetration of the barrier (tunnelling) or energies above it. Neutrons feel no Coulomb barrier and can react at thermal energies—which is why thermal neutron capture and fission play an outsized role in reactors and in laboratory neutron spectroscopy [6, 1, 2].

At short distance the attractive nuclear potential takes over, producing a pocket that supports compound-nucleus resonances (next chapter and fission lectures). The competition between barrier penetration and nuclear absorption also underlies the energy dependence of capture and fission cross sections (Lecture 24): once a slow neutron is captured, the available excitation is the neutron separation energy, often several MeV above the ground state of the compound system.

21.6 Types of processes

21.6.1 Scattering versus reaction

If $a = b$ and $A = B$ as nuclear species, one speaks of **scattering**. If identities change ($a \neq b$ or $A \neq B$), one speaks of a **reaction** in the narrow sense. Both are nuclear reactions in the broad sense [6].

21.6.2 Elastic and inelastic scattering

Elastic scattering leaves A in its ground state. Total kinetic energy is conserved in the CM frame (rest masses unchanged):

$$K_a + K_A = K'_a + K'_A. \tag{21.17}$$

Inelastic scattering excites A (or a) by energy E^*:

$$K_a + K_A = K'_a + K'_A + E^*. \tag{21.18}$$

The detected spectrum shows discrete peaks at kinetic energies reduced by E^* (recoil corrected) [6, 1].

21.6.3 Transfer: stripping and pickup

In a **transfer reaction** one or more nucleons move between projectile and target.

- **Stripping:** the projectile loses a nucleon to the target. Example: (d,p) deposits a neutron; the proton is detected.
- **Pickup:** the projectile gains a nucleon from the target. Example: (p,d) picks up a neutron.

Transfer angular distributions are sensitive to the orbital angular momentum of the transferred nucleon and are a primary tool of nuclear spectroscopy [6, 14].

21.6.4 Capture

In a **capture reaction** the projectile is absorbed and the residual de-excites, often by γ emission: $A(n,\gamma)B$, $A(p,\gamma)B$. Neutron capture on heavy nuclei is central to reactors and nucleosynthesis [6, 1].

21.7 Direct, pre-equilibrium, and compound mechanisms

21.7.1 Timescales

Two limiting mechanisms organise most data:

(1) **Direct reactions.** The projectile interacts with one or a few surface nucleons and exits promptly. The characteristic time is the transit time across the nucleus,

$$t_{\text{direct}} \sim \frac{2R}{v} \sim 10^{-22}\,\text{s} \tag{21.19}$$

for $\sim 20\,\text{MeV}$ nucleons ($v \sim 0.2c$, $R \sim 10\,\text{fm}$). Angular distributions are **forward peaked**: memory of the beam direction survives.

(2) **Compound-nucleus (CN) reactions.** The projectile is absorbed; its energy is shared among many nucleons until a quasi-equilibrated excited nucleus C^* forms. A resolved compound level of total width Γ has mean life

$$\tau_{\text{CN}} = \frac{\hbar}{\Gamma}. \tag{21.20}$$

Thus widths from $1\,\text{MeV}$ down to $10^{-6}\,\text{eV}$ correspond to $\tau \simeq 6.6 \times 10^{-22}$ up to $6.6 \times 10^{-10}\,\text{s}$. There is no universal 10^{-16}–$10^{-15}\,\text{s}$ compound lifetime: broad, overlapping resonances can be much shorter and isolated neutron resonances much longer. What distinguishes an equilibrated compound mechanism is the separation from the transit/equilibration time and the loss of detailed entrance-channel memory. Emission resembles evaporation; in the CM frame the angular distribution is nearly **symmetric** (often nearly isotropic for high level density) [6, 1, 14].

Real reactions can interpolate between these limits. In a **pre-equilibrium** reaction the projectile has undergone several collisions and shared part, but not all, of its energy before emission. Such spectra retain some forward memory while extending toward the softer, evaporation-like compound spectrum. Once equilibration is reached, a **statistical** description replaces individual many-body amplitudes by level densities and channel transmission coefficients. The classic experimental test of entrance-channel amnesia is Ghoshal's comparison of the same compound nucleus formed by different entrance channels: when the compound picture holds, branching to a given exit channel depends on excitation energy, not on how the compound system was assembled [39, 6]. The Hauser–Feshbach cross section has the schematic factorisation

$$\sigma_{aA\to bB} \sim \sum_{J\pi} \sigma^{J\pi}_{\text{form}}(aA)\, \frac{T^{J\pi}_{bB}}{\sum_c T^{J\pi}_c}, \tag{21.21}$$

with angular-momentum and parity conservation imposed in each J^π class. The numerator forms the compound state and the ratio is its branching probability. This is the quantitative statistical implementation of Bohr independence. The factorisation is valid after averaging over enough compound levels that formation and decay amplitudes decorrelate. Near thresholds, for low level density, for doorway states, or when direct and compound amplitudes interfere, a simple Hauser–Feshbach average can fail. Width-fluctuation corrections are also needed when entrance and exit channels remain correlated [6, 1].

Symbolically,

$$a + A \;\longrightarrow\; C^* \;\longrightarrow\; b + B. \tag{21.22}$$

21.7.2 de Broglie wavelength and energy regime

The projectile de Broglie wavelength (often quoted as the reduced wavelength $\bar{\lambda} = \hbar/p$) sets the spatial scale of the interaction:

$$\bar{\lambda} = \frac{\hbar}{p} = \frac{\hbar c}{\sqrt{2m_a c^2\, E}}. \tag{21.23}$$

Using $\hbar c \approx 197\,\text{MeV}\cdot\text{fm}$ and $m_p c^2 \approx 938\,\text{MeV}$, a $10\,\text{MeV}$ proton has

$$\bar{\lambda} \approx \frac{197}{\sqrt{2\times 938\times 10}} \approx \frac{197}{137} \approx 1.4\,\text{fm}, \tag{21.24}$$

while the full wavelength $h/p = 2\pi\bar{\lambda}$ is a few fm, comparable to a medium nucleus and favourable to CN formation. At $E \sim 40\,\text{MeV}$, $\bar{\lambda} \sim 0.7\,\text{fm}$ approaches nucleon size and direct processes gain weight [6, 1].

21.7.3 Bohr independence hypothesis

Bohr's **independence hypothesis** states that once C^* is formed at a given excitation energy, spin, and parity, it forgets the entrance channel. Decay branching ratios depend only on the properties of C^*, not on whether C^* was made by $p + A$ or $\alpha + A'$.

Classic test: Ghoshal's experiment comparing

$$p + {}^{63}\text{Cu} \to {}^{64}\text{Zn}^* \to \text{various exit channels} \tag{21.25}$$

with

$$\alpha + {}^{60}\text{Ni} \to {}^{64}\text{Zn}^* \to \text{same exit channels}. \tag{21.26}$$

The comparison requires more than matching a laboratory energy: the two entrances must form the same ${}^{64}\text{Zn}^*$ excitation interval and must populate comparable J^π ensembles (or the data must be averaged so that differences are removed). One then compares *decay branching patterns* after separating entrance-channel formation factors. Ghoshal found closely corresponding excitation-function structures for common exit channels, supporting compound-nucleus independence within those conditions; the absolute cross sections need not coincide [39, 6, 1].

Matching excitation energies. Absorption of a proton on stationary ^{63}Cu forms ^{64}Zn* with excitation

$$\Delta E = (m_p + m_{\mathrm{Cu}} - m_{\mathrm{Zn}})c^2 + \frac{m_{\mathrm{Cu}}}{m_p + m_{\mathrm{Cu}}} K_p, \tag{21.27}$$

because only the CM kinetic energy is available internally. Therefore the laboratory energy needed for a chosen ΔE is

$$K_p = \left[\Delta E - (m_p + m_{\mathrm{Cu}} - m_{\mathrm{Zn}})c^2\right]\left(1 + \frac{m_p}{m_{\mathrm{Cu}}}\right). \tag{21.28}$$

An analogous formula with $(m_\alpha, m_{\mathrm{Ni}})$ replaces (m_p, m_{Cu}) for the α channel. Different mass combinations imply different laboratory energy scales for the same ΔE, which is why Ghoshal plots use two shifted energy axes [39, 6].

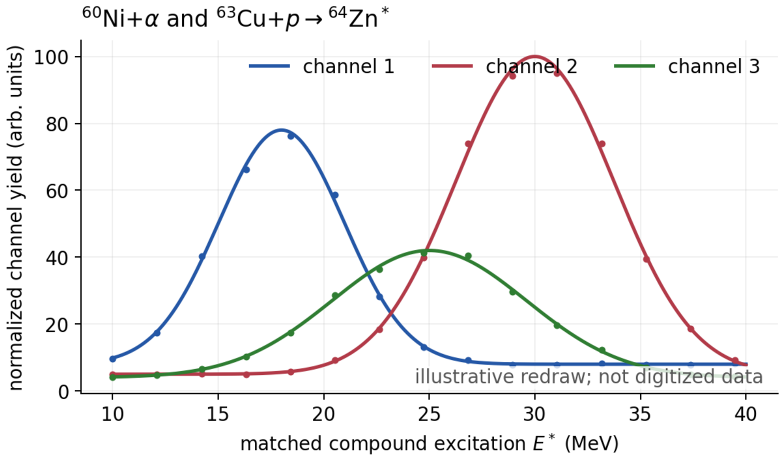


Figure 21.2: The same ^{64}Zn* compound system formed through ^{60}Ni $+ \alpha$ and ^{63}Cu $+ p$. Corresponding decay structures are compared only after matching compound excitation and accounting for the populated J^π distributions and formation normalizations. Redrawn schematic after Ghoshal; points are illustrative, not digitized data [39].

Bohr compound nucleus (1936)

Niels Bohr proposed that low-energy nuclear reactions proceed through a long-lived intermediate nucleus in statistical equilibrium. The picture explains dense resonances, nearly isotropic evaporation spectra, and entrance-channel independence. Direct reactions (Butler, Austern, ...) emerged later as the complementary short-timescale limit [6, 14, 2].

Key idea

Students should remember:

- Reaction: $a + A \to b + B$ or $A(a, b)B$.
- $\mathrm{d}\dot{N} = \Phi N_t (\mathrm{d}\sigma/\mathrm{d}\Omega)\mathrm{d}\Omega$, or $\dot{N} = \mathcal{L}\sigma$; counts require live time and efficiency.

- σ has units of area; $1\,\mathrm{b} = 10^{-24}\,\mathrm{cm}^2$.
- $E_{\mathrm{cm}} = m_A E_{\mathrm{lab}}/(m_a + m_A)$.
- Elastic vs inelastic; stripping/pickup; capture; Coulomb barrier for charged beams.
- Direct $\sim 10^{-22}\,\mathrm{s}$, forward peaked; pre-equilibrium interpolates; for a resolved CN level $\tau = \hbar/\Gamma$, with a broad range set by its width; statistical decay is more symmetric in the CM.

21.7.4 Direct memory versus compound amnesia

Direct processes remember the beam direction (forward peaking); equilibrated compound decay forgets it (near-isotropic evaporation in the CM). Ghoshal-type comparisons of the same compound nucleus formed in different entrance channels remain the textbook test of that amnesia. Stripping reactions such as (d, p) map single-particle strength on the target [39, 6, 1].

21.8 Deeper physics: direct versus compound reactions and geometric cross sections

21.8.1 Geometric cross section as a scale

When two nuclei collide, a useful order-of-magnitude target area is the geometric cross section

$$\sigma_{\mathrm{geom}} \approx \pi(R_1 + R_2)^2, \qquad R \approx r_0 A^{1/3}, \tag{21.29}$$

with $r_0 \approx 1.2\,\mathrm{fm}$ [6, 14]. Elastic and reaction cross sections at tens of MeV per nucleon often approach this scale when absorption is strong. The *reaction* cross section $\sigma_{\mathrm{reaction}}$ measures flux removed from the elastic channel; at high energy $\sigma_{\mathrm{reaction}} \lesssim \sigma_{\mathrm{geom}}$, while at low energy Coulomb repulsion suppresses both [6, 1].

Geometric estimate for $^{12}\mathrm{C} + ^{12}\mathrm{C}$

With $R \approx 1.2 \times 12^{1/3}\,\mathrm{fm} \approx 2.6\,\mathrm{fm}$ for each nucleus, $R_1 + R_2 \approx 5.2\,\mathrm{fm}$ and $\sigma_{\mathrm{geom}} \approx \pi(5.2\,\mathrm{fm})^2 \approx 85\,\mathrm{fm}^2 \approx 850\,\mathrm{mb}$ [6]. Measured fusion cross sections at barrier energies are smaller because tunneling and angular-momentum filtering reduce the effective area (Lecture 25).

21.8.2 Direct versus compound mechanisms

Direct reactions (stripping, pickup, knockout) transfer one or a few nucleons in a single encounter. Characteristic times are $\sim 10^{-22}\,\mathrm{s}$ (the transit time across a nuclear diameter). Angular distributions are strongly forward peaked and carry information about transferred ℓ [6, 14]. **Compound-nucleus (CN) reactions** form a long-lived equilibrated intermediate ($\sim 10^{-21}$ to $10^{-15}\,\mathrm{s}$). Decay is statistical: angular distributions are nearly isotropic in the CM frame, and outgoing energies follow an evaporation spectrum [6, 1].

Remark

The labels "direct" and "compound" are dynamical limits, not exclusive channels. The same final state can receive contributions from both mechanisms. Ghoshal's classic comparison of (p, p) and (p, n) on the same target illustrates how compound-nucleus averaging smooths excitation functions while direct channels retain angular structure [39, 6].

Key idea

Link to decay and fission. Direct reactions probe single-particle overlap (as in β and γ matrix elements). Compound-nucleus decay is the inelastic analogue of the statistical evaporation that follows fission or fusion (Lectures 22–24). The geometric cross section sets the flux scale; the mechanism determines how that flux is distributed in angle and energy [14, 2].

21.8.3 Lifetime scales and the Hauser–Feshbach limit

Direct reactions complete in $\sim R/v \sim 10^{-22}$ s. Compound-nucleus lifetimes range from 10^{-21} s for light nuclei to 10^{-15} s or longer for heavy systems near the fission barrier [6, 14]. The Weisskopf estimate for a single γ step is $\sim 10^{-13}$ s for a MeV transition; a compound nucleus typically emits many γ rays before fission or particle emission, so the total CN lifetime is the sum of many statistical steps. Pre-equilibrium emission fills the gap between direct and compound time scales and produces forward-peaked continuum particles that are neither pure direct nor pure evaporation [6, 1].

21.8.4 Optical model and absorption

The complex optical potential used for elastic scattering has an imaginary part that removes flux from the elastic channel into reaction channels [6, 14]. The optical theorem relates the forward elastic amplitude to the total cross section, so a good optical-model fit to elastic data constrains the reaction cross section as well. DWBA calculations for transfer reactions use the same optical potentials to generate distorted waves; spectroscopic factors extracted from (d, p) cross sections therefore inherit optical-model uncertainties that modern analyses quote as covariance bands [53, 39]. This is the reaction-theory counterpart of hindrance factors in decay: the barrier or potential is known approximately, but the overlap integral is model dependent.

Geometric scale

For $R = 1.2(A_t^{1/3} + A_p^{1/3})$ fm with $A_t = 12$, $A_p = 2$, $\pi R^2 \sim 400$ mb, a typical nuclear scale. Measured reaction cross sections can be larger (refractive/optical effects) or smaller (strong absorption and Q-value windows). Absolute normalisation is therefore a physics measurement, not a bookkeeping afterthought.

21.9 Further physics from the literature

Optical model and elastic scattering. Elastic scattering of nucleons is described by a complex optical potential whose imaginary part accounts for absorption into non-elastic channels. The resulting diffraction patterns and reaction cross sections connect to the total cross section via the optical theorem [6, 14, 2].

Optical theorem and unitarity. For wave number k, the forward elastic amplitude fixes the total loss of incident flux,

$$\sigma_{\mathrm{tot}} = \frac{4\pi}{k}\,\mathrm{Im}\, f(\theta = 0).$$

This is not a geometrical identity; it follows from probability conservation of the scattering matrix. The imaginary optical potential transfers flux from the explicitly treated elastic channel into all unresolved reaction channels.

DWBA for transfer. Direct stripping and pickup are analysed with the distorted-wave Born approximation (DWBA). Angular distributions determine the transferred orbital angular momentum ℓ; spectroscopic factors measure single-particle occupation relative to a pure shell-model configuration [6, 14].

Hauser–Feshbach theory. Statistical CN decay is computed with Hauser–Feshbach theory: transmission coefficients for all open channels, spin and parity conservation, and level densities. Bohr independence is built in by construction [6, 1]. Its predictions inherit uncertainties from optical potentials, level densities, gamma-strength functions, discrete-level schemes, and width-fluctuation corrections. Varying only one ingredient while holding the others fixed does not provide a complete model uncertainty; yield comparisons should distinguish experimental normalization errors from these correlated theory systematics.

What the mechanism labels do and do not mean. "Direct", "pre-equilibrium", and "compound" are dynamical limits, not three mutually exclusive final-state labels. The same measured channel can receive coherent direct and resonant amplitudes. A smooth forward component, an evaporation-like energy spectrum, and narrow fluctuations in an excitation function are therefore complementary evidence, rather than infallible one-feature classifiers. This distinction is standard throughout nuclear-reaction analyses [32, 6, 1].

Heavy-ion reactions. Heavy projectiles open fusion, deep-inelastic scattering, and multi-nucleon transfer. Timescales and angular distributions again separate quasi-elastic direct processes from long-lived fused systems [14, 2].

21.9.1 Quantitative direct versus compound discrimination

Direct stripping (d, p) on a target of mass A at beam energy $E_d \sim 10$–$20\,\mathrm{MeV}$ typically peaks in cross section near $\theta \sim 0^\circ$ with a characteristic diffraction pattern set by transferred ℓ. The differential cross section at the peak is often 0.1–10 mb/sr, while compound-nucleus background at the same angle may be $\sim 0.01\,\mathrm{mb/sr}$ [6, 39]. The ratio of forward peak to isotropic background is a practical discriminator: direct reactions carry angular information; compound reactions average it away [14, 2].

Geometric versus measured reaction cross section

For $^{12}\text{C} + ^{12}\text{C}$ at $E_{\text{CM}} \approx 5\,\text{MeV}$, $\sigma_{\text{geom}} \approx 850\,\text{mb}$ but measured fusion cross sections are $\sim$ 1–10 mb because of Coulomb tunneling and angular-momentum filtering [6]. At neutron energies near 1 MeV on ^{238}U, σ_{reaction} can approach hundreds of barns, comparable to πR^2 for $R \approx 7\,\text{fm}$, because the Coulomb barrier is absent [35, 6].

21.9.2 From direct reactions to kinematics and fission

Transfer reactions determine Q-values and excitation energies through two-body kinematics (Lecture 22). Stripping on fissile targets populates states that decay by fission when $E^* > B_f$ (Lecture 23). Knockout reactions on radioactive beams test the same single-particle overlap that preformation factors encode in α decay (Lectures 16–17) [53, 6]. Hauser–Feshbach compound decay feeds γ cascades analysed with multipole tools from Lecture 20.

Remark

The optical theorem links forward elastic scattering to the total cross section. Absorptive imaginary potentials that describe compound-nucleus formation therefore appear already in elastic channel analyses, connecting reaction cross sections to the same flux-conservation logic used in decay widths [6, 14].

21.9.3 From stripping to fissile breeding

Neutron transfer and capture reactions on fertile ^{238}U populate ^{239}U, which β-decays to fissile ^{239}Pu (Lecture 24) [6, 35]. The (d, p) reaction on actinide targets is used to map single-particle states that also control α-decay preformation (Lectures 16–17) [53]. Geometric cross sections set the scale for neutron absorption in reactors; direct and compound mechanisms determine how much of that flux goes into useful channels versus background [1, 39].

21.9.4 Charge exchange and β strength in reactions

Charge-exchange reactions probe Gamow–Teller strength complementary to β decay (Lecture 19) [14, 53]. Transfer reactions on radioactive beams map single-particle occupancies that also control α-decay preformation (Lectures 16–17). The geometric cross section sets the overall flux; direct versus compound mechanisms determine whether angular distributions carry structural information or are washed out by statistical decay [6, 39]. Hauser–Feshbach theory connects compound cross sections to γ-cascade multipolarities (Lecture 20) through level densities and transmission coefficients [1, 6].

21.9.5 Hauser–Feshbach and the compound-nucleus limit

Hauser–Feshbach theory computes compound-nucleus decay as a statistical balance among open channels weighted by transmission coefficients and level densities [6, 1]. Bohr independence (decay independent of formation) is built into the theory by construction. Modern applications inherit uncertainties from optical potentials, γ-strength functions, and level-density models [53, 39]. The lecture distinction between direct (peripheral, angular-structure-rich) and compound (near-isotropic, energy-averaged) mechanisms remains the organising principle for interpreting rare-isotope reaction data at FRIB and comparable facilities [54, 6].

21.10 Deeper reading: direct versus compound and geometric scales

Direct reactions are peripheral and angular-structure-rich; compound reactions are energy-averaged and closer to isotropic after many intermediate-state scramble. Ghoshal-type tests illustrate compound formation; DWBA analyses of (d, p) illustrate direct transfer. Geometric cross sections πR^2 set the scale, but absolute spectroscopic factors inherit optical-potential uncertainties.

Rare-isotope beams amplify those issues: elastic data for exotic projectiles are sparse, so Bayesian optical potentials and continuum coupling matter. Classify any published angular distribution before trusting a quoted spectroscopic factor. Lectures 22's kinematics then tell you where to place detectors; Lecture 10's scattering theory tells you what the distorted waves mean.

entrangling structure and reaction

A measured differential cross section equals a structure factor times a reaction factor (schematically). Changing the optical potential changes the reaction factor and thus the extracted structure. Publish both or publish neither as a precision claim. Coupled-channels analyses that include target excitation blur the simple direct/compound dichotomy but still begin from Lecture 21's classification.

Charge exchange

Charge-exchange reactions map Gamow–Teller strength relevant to β decay and to astrophysical electron capture. Their interpretation reuses Lectures 14 and 19 operator themes inside a reaction framework from this chapter.

Classification drill

Find one published (d, p) angular distribution on a closed-shell target and one compound-nucleus excitation function. Write three bullets for each: angular shape, energy dependence, and what structure quantity (if any) was extracted. Then estimate πR^2 and compare with the magnitude of the reported cross section.

Discuss how continuum coupling and core excitation modify DWBA for exotic beams. The goal is not to run a coupled-channels code in this course, but to read a modern reaction paper without confusing a fitted spectroscopic factor with a pure shell-model occupation. Lectures 10 and 22 supply the scattering and kinematic prerequisites for that reading.

Reactions entangle structure with mechanism. Direct versus compound is the first cut; optical-model uncertainty is the modern second cut.

In practice, rework one numerical estimate from the lecture with a different parameter set and compare the shift to the quoted experimental uncertainty. If the shift exceeds the uncertainty, the estimate is teaching a scale, not delivering a fit. That distinction keeps theoretical cartoons honest when they sit beside evaluated data tables and modern many-body calculations. Along the course arc, write one sentence linking this chapter backward and one forward: the discipline of those two sentences is as important as any single formula. Examination problems rarely ask for a literature review, but they do ask for the scale argument that literature reviews presuppose.

Synthesis problem

Write a half-page note that (i) restates the chapter's central scale or selection rule in one equation, (ii) evaluates it numerically for one concrete nucleus or energy, and (iii) names the neighbouring lecture that uses the result. Keep the note free of research-frontier speculation; the goal is fluency with the lecture physics itself. If an assumption fails (Born approximation, spherical mean field, allowed transition, one-dimensional barrier), state the failure mode in a final sentence. That habit converts passive reading into examination-ready understanding and makes the later research-frontier box read as an extension rather than a replacement.

Research frontier: rare-isotope reactions and student projects

Direct and compound reactions of Lecture 21 are the workhorses of structure studies with radioactive beams. Transfer, knockout, and charge-exchange reactions at FRIB, FAIR, RIKEN, and TRIUMF map spectroscopic factors and Gamow–Teller strength far from stability; theory must treat continuum and core excitation to interpret the cross sections [53, 54]. Classic compound-nucleus tests such as Ghoshal's still illustrate the mechanism, but modern analyses include uncertainty quantification of optical potentials and level densities [39]. Open questions include the quenching of spectroscopic factors relative to independent-particle estimates, the validity of sudden approximations in knockout, and how to propagate optical-model uncertainties into extracted structure quantities. The lecture distinction between direct (peripheral, angular-structure-rich) and compound (near-isotropic, energy-averaged) mechanisms remains the organising principle for reading modern papers.

A recurring pitfall is to treat a DWBA spectroscopic factor as a pure structure number. Modern analyses vary optical potentials within uncertainty bands and report the induced spread. Student projects that repeat a published angular-distribution fit with two optical-potential families learn that lesson quantitatively.

The lecture direct-versus-compound classification remains the first cut before those uncertainty bands are interpreted.

Facility white papers list transfer, knockout, and charge exchange among the first experiments for new beam species for exactly this experimental reason.

Student directions. (1) Classification exercise: classify a published (d, p) study as direct or compound using angular distributions. (2) Scale estimate: compute a geometric cross section and compare with a measured reaction cross section at Coulomb-barrier energy. (3) Methods project: outline extracting a spectroscopic factor from published angular-distribution data with a DWBA code.

21.11 Problems with solutions

Problem 21.1: Differential cross section from counts. A beam of flux density $\Phi = 2.0 \times 10^{10}\,\mathrm{cm^{-2}\,s^{-1}}$ illuminates $N_t = 1.5 \times 10^{19}$ target nuclei. In $\Delta t = 100\,\mathrm{s}$, a detector with $\Delta\Omega = 5.0 \times 10^{-3}\,\mathrm{sr}$ records $N = 3.0 \times 10^4$ product particles (assume $\mathrm{d}\sigma/\mathrm{d}\Omega$ constant over the detector). Find $\mathrm{d}\sigma/\mathrm{d}\Omega$ in mb/sr [6, 1].

Solution 21.1. From $\mathrm{d}\dot{N} = \Phi N_t(\mathrm{d}\sigma/\mathrm{d}\Omega)\mathrm{d}\Omega$, integrated over the run,

$$N = \Phi N_t \frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}\Delta\Omega\,\Delta t.$$

$$\begin{aligned}
\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} &= \frac{N}{\Phi N_t\,\Delta\Omega\,\Delta t} \\
&= \frac{3.0 \times 10^4}{(2.0 \times 10^{10})(1.5 \times 10^{19})(5.0 \times 10^{-3})(100)} \\
&= \frac{3.0 \times 10^4}{1.5 \times 10^{30}} \\
&= 2.0 \times 10^{-26}\,\mathrm{cm^2/sr}.
\end{aligned}$$

Since $1\,\mathrm{b} = 10^{-24}\,\mathrm{cm}^2$,

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} = 0.020\,\mathrm{b/sr} = 20\,\mathrm{mb/sr}.$$

Problem 21.2: Areal density. A gold foil ($A = 197$, density $\rho = 19.3\,\mathrm{g\,cm^{-3}}$) has thickness $t = 2.0\,\mu\mathrm{m}$. Estimate the areal density $n_a = nt$ in nuclei per cm^2. Use $N_A = 6.02 \times 10^{23}\,\mathrm{mol}^{-1}$ [1].

Solution 21.2. Number density

$$n = \frac{\rho N_A}{A} = \frac{19.3 \times 6.02 \times 10^{23}}{197} = 5.90 \times 10^{22}\,\mathrm{cm^{-3}}.$$

$$t = 2.0 \times 10^{-4}\,\mathrm{cm}, \qquad n_a = nt = 1.18 \times 10^{19}\,\mathrm{cm^{-2}}.$$

Problem 21.3: Lab to CM energy. A 12.0 MeV proton beam strikes a stationary $^{12}\mathrm{C}$ target. Find E_{cm}. What fraction of E_{lab} is unavailable for excitation? [6]

Solution 21.3.

$$E_{\mathrm{cm}} = \frac{m_C}{m_p + m_C}E_{\mathrm{lab}} \approx \frac{12}{13} \times 12.0\,\mathrm{MeV} = 11.08\,\mathrm{MeV}.$$

Unavailable fraction (CM motion):

$$\frac{m_p}{m_p + m_C} = \frac{1}{13} \approx 7.7\%.$$

Problem 21.4: Coulomb barrier estimate. Estimate V_C at contact for $\alpha + {}^{208}\text{Pb}$ using $R = 1.2(A_\alpha^{1/3} + A_{\text{Pb}}^{1/3})\,\text{fm}$ and $e^2/(4\pi\varepsilon_0) = 1.44\,\text{MeV}\cdot\text{fm}$ [6, 1].

Solution 21.4.

$$R = 1.2(4^{1/3} + 208^{1/3}) = 1.2(1.59 + 5.92) = 9.01\,\text{fm}.$$

$$V_C = \frac{(2)(82)(1.44)}{9.01} = \frac{236.2}{9.01} \approx 26.2\,\text{MeV}.$$

An α beam well below $\sim 26\,\text{MeV}$ (lab, with CM correction) interacts mainly by Coulomb scattering unless it tunnels.

Problem 21.5: Direct transit time. Estimate the transit time of a 20 MeV proton across a nucleus of diameter 10 fm. Use non-relativistic kinematics with $m_pc^2 \approx 938\,\text{MeV}$ [6].

Solution 21.5.

$$\tfrac{1}{2}mv^2 = 20\,\text{MeV} \quad \Rightarrow \quad \frac{v}{c} = \sqrt{\frac{2K}{mc^2}} = \sqrt{\frac{40}{938}} \approx 0.206.$$

$$t = \frac{10\,\text{fm}}{0.206\,c} = \frac{10 \times 10^{-15}\,\text{m}}{0.206 \times 3.00 \times 10^8\,\text{m}\,\text{s}^{-1}} \approx 1.6 \times 10^{-22}\,\text{s}.$$

This matches the direct-reaction timescale $\sim 10^{-22}\,\text{s}$.

Problem 21.6: Classify the process. Classify each as elastic, inelastic, stripping, pickup, or capture [6, 1]:

(a) ${}^{40}\text{Ca}(p,p){}^{40}\text{Ca}$ (g.s. to g.s.);

(b) ${}^{40}\text{Ca}(p,p'){}^{40}\text{Ca}^*$;

(c) ${}^{40}\text{Ca}(d,p){}^{41}\text{Ca}$;

(d) ${}^{40}\text{Ca}(p,d){}^{39}\text{Ca}$;

(e) ${}^{40}\text{Ca}(n,\gamma){}^{41}\text{Ca}$.

Solution 21.6. (a) Elastic scattering.
(b) Inelastic scattering.
(c) Stripping (deuteron loses a neutron).
(d) Pickup (proton gains a neutron).
(e) Radiative capture.

Problem 21.7: Bohr independence. Two entrance channels form the same compound nucleus C^* at the same excitation energy and spin. What does Bohr independence predict for the relative yields of two exit channels $b_1 + B_1$ and $b_2 + B_2$? Why must laboratory energies differ for $p + A$ versus $\alpha + A'$? [6]

Solution 21.7. Independence predicts that branching ratios (relative partial widths) of C^* are the same regardless of entrance channel. Absolute cross sections still contain an entrance formation factor, but excitation functions for a given exit channel should match when plotted versus excitation energy of C^*.

Laboratory energies differ because the Q-value for forming C and the mass ratios in $E_{\text{cm}} = m_A K_a/(m_a + m_A)$ differ for (p, A) and (α, A'). Matching ΔE requires different K_a scales (Ghoshal plot with two energy axes).

Problem 21.8: Luminosity, efficiency, and target attenuation. A $5.0 \times 10^8\,\text{s}^{-1}$ beam crosses a foil of areal density $n_a = 2.0 \times 10^{18}\,\text{cm}^{-2}$. A channel has $\sigma_c = 80\,\text{mb}$, and the apparatus accepts 2.0% of its products with 65% net efficiency. (a) Find the luminosity. (b) Find the observed rate. (c) If the total interaction cross section is 1.5 b, check the thin-target approximation [6, 1].

Solution 21.8. Step 1: luminosity.

$$\mathcal{L} = \dot{N}_a n_a = (5.0 \times 10^8)(2.0 \times 10^{18}) = 1.0 \times 10^{27}\,\text{cm}^{-2}\,\text{s}^{-1}.$$

Step 2: produced and observed rates. Since $80\,\text{mb} = 8.0 \times 10^{-26}\,\text{cm}^2$,

$$\dot{N}_c = \mathcal{L}\sigma_c = 80\,\text{s}^{-1}, \qquad \dot{N}_{\text{obs}} = 80(0.020)(0.65) = 1.04\,\text{s}^{-1}.$$

Step 3: attenuation test.

$$\tau = n_a \sigma_{\text{tot}} = (2.0 \times 10^{18})(1.5 \times 10^{-24}) = 3.0 \times 10^{-6}, \qquad P_{\text{int}} = 1 - e^{-\tau} \simeq \tau.$$

The quadratic correction is of relative order $\tau/2$, so attenuation is negligible and the thin-target yield formula is justified.

Problem 21.9: Yield with propagated uncertainty. A thin-target run has $\dot{N}_a = (5.00 \pm 0.10) \times 10^8\,\text{s}^{-1}$, $n_a = (2.00 \pm 0.06) \times 10^{18}\,\text{cm}^{-2}$, efficiency $\epsilon = 0.650 \pm 0.020$, live time $T = 1000\,\text{s}$, and $\sigma = 80.0\,\text{mb}$. Find the expected count and its uncertainty, assuming independent inputs and negligible counting fluctuation [6, 1].

Solution 21.9.

$$N = \dot{N}_a n_a \sigma \epsilon T = (5.00 \times 10^8)(2.00 \times 10^{18})(8.00 \times 10^{-26})(0.650)(1000) = 5.20 \times 10^4.$$

For a product of independent quantities,

$$\left(\frac{u_N}{N}\right)^2 = \left(\frac{0.10}{5.00}\right)^2 + \left(\frac{0.06}{2.00}\right)^2 + \left(\frac{0.020}{0.650}\right)^2 = 0.00225.$$

Thus $u_N/N = 0.0474$, $u_N = 2.47 \times 10^3$, and $N = (5.20 \pm 0.25) \times 10^4$. A real extraction would also include background, solid-angle, dead-time, and cross-section-model systematics.

Problem 21.10: When Hauser–Feshbach is unreliable. For each case state whether a basic Hauser–Feshbach average is well motivated: (a) hundreds of overlapping compound levels in each J^π bin; (b) one isolated near-threshold resonance; (c) a strong forward direct-transfer amplitude in the same exit channel. Name the missing physics where needed [6, 1].

Solution 21.10. (a) is the intended statistical limit, provided all important transmission coefficients and level densities are included. (b) requires an explicit Breit–Wigner or R-matrix treatment and may need width-fluctuation corrections. (c) requires a direct-reaction amplitude and, where coherent, its interference with the compound amplitude. The labels describe dynamical limits, not mutually exclusive measured channels.

Chapter 22

Nuclear Reaction Kinematics and Spectroscopy

"The energy of the outgoing particle is a spectrometer for the excitation of the residual nucleus, once Q-value kinematics are under control."
K. S. Krane

The preceding lecture classified reaction mechanisms. This chapter connects measured kinetic energies and angles to nuclear excitation energies: two-body momentum and energy conservation, the Q-value, the full non-relativistic relation among K_a, K_b, θ, and masses, thresholds for endothermic reactions, neutron capture on ^{238}U as a worked example, and the idea of resonant energy dependence of the cross section [6, 1, 14, 2].

22.1 Experimental geometry and what is measured

Consider the binary reaction

$$X(a,b)Y \qquad \text{or} \qquad a + X \to b + Y \tag{22.1}$$

in the laboratory frame with target X at rest. Projectile a has kinetic energy K_a and momentum $\mathbf{p}_a$. Product b is detected at laboratory angle θ with kinetic energy K_b and momentum $\mathbf{p}_b$. Residual Y recoils at angle ϕ with kinetic energy K_Y and momentum $\mathbf{p}_Y$. The residual is often unobserved: its recoil is inferred from two-body conservation once masses and Q are known. In practice a magnetic spectrometer or solid-state detector measures K_b versus θ; discrete peaks map excited states of Y, while continuum and broadening reveal unresolved levels or three-body contributions [6, 1].

Measurable quantities are typically K_a (set by the accelerator), θ, and K_b (from the detector). The residual kinetic energy K_Y is usually not measured directly. Nuclear data tables supply rest masses. The unknown of spectroscopic interest is often the excitation energy ΔE of Y [6, 1].

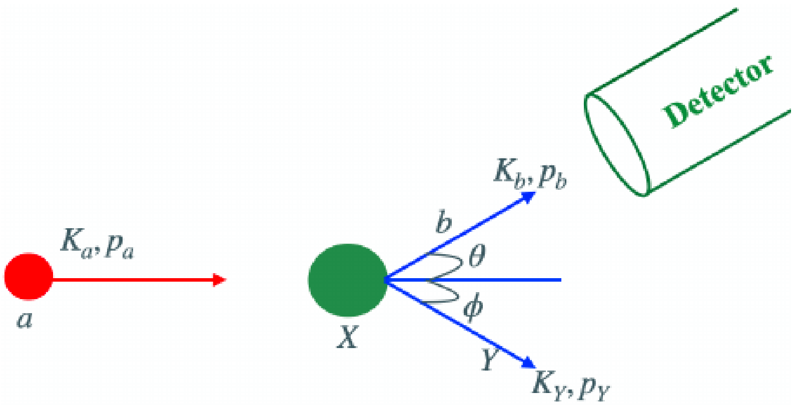


Figure 22.1: Laboratory setup for a two-body reaction $X(a,b)Y$: projectile a, target X, light product b at angle θ, residual Y at angle ϕ, and a detector that measures K_b (lecture figure).

22.2 Conservation laws

Two-body nuclear reactions are fixed, once masses and ΔE are known, by energy and three-momentum conservation. Angular distributions then carry the dynamical information (partial waves, resonances, direct transfer). The equations below are laboratory statements for a target at rest; CM kinematics follow by a Galilean (or relativistic) boost and are preferred when comparing to partial-wave theory [6, 1].

22.2.1 Momentum

With the beam along $+x$ and the reaction in the xy plane,

$$p_a = p_b \cos\theta + p_Y \cos\phi, \tag{22.2}$$

$$0 = p_b \sin\theta - p_Y \sin\phi. \tag{22.3}$$

(The target is at rest, so initial $p_y = 0$.) These two equations determine the unobserved recoil direction ϕ once p_b and θ are measured, which is the practical lever for missing-mass spectroscopy.

22.2.2 Energy and Q-value

Including rest energies and allowing residual excitation ΔE,

$$m_a c^2 + m_X c^2 + K_a = m_b c^2 + m_Y c^2 + K_b + K_Y + \Delta E. \tag{22.4}$$

Define the reaction **Q-value** from ground-state rest masses,

$$Q \equiv (m_a + m_X - m_b - m_Y)c^2. \tag{22.5}$$

Then (22.4) becomes

$$Q + K_a = K_b + K_Y + \Delta E. \tag{22.6}$$

- $Q > 0$: **exoergic** (exothermic); energy is released.

- $Q < 0$: **endoergic** (endothermic); a minimum beam energy is required (Section 22.5).

If Y is left in its ground state, set $\Delta E = 0$. Excited states appear as discrete peaks at lower K_b for fixed θ [6, 1, 14].

22.2.3 Invariant definition and exact CM energy

The mass-difference definition of Q is frame independent. Introduce the Mandelstam invariant s, with units of energy squared,

$$s = (E_a + E_X)^2 - c^2|\mathbf{p}_a + \mathbf{p}_X|^2. \tag{22.7}$$

For a target at rest, $E_X = m_X c^2$ and

$$s = (m_a c^2)^2 + (m_X c^2)^2 + 2m_X c^2(m_a c^2 + K_a). \tag{22.8}$$

Thus the exact energy available in the CM is $\sqrt{s}$. If the residual is excited, its effective rest mass is

$$m_Y^* = m_Y + \frac{\Delta E}{c^2}, \qquad Q^* = Q - \Delta E. \tag{22.9}$$

The final CM momentum follows without choosing a laboratory angle:

$$p_f^* c = \frac{\sqrt{\lambda(s, (m_b c^2)^2, (m_Y^* c^2)^2)}}{2\sqrt{s}}, \qquad \lambda(x, y, z) = x^2 + y^2 + z^2 - 2xy - 2xz - 2yz. \tag{22.10}$$

The channel is open only when the square root is real. Equations (22.8)–(22.10) are the exact relativistic two-body result; the following laboratory formula is their low-energy reduction [6, 14].

Physical meaning of ΔE. Each bound level of Y produces a kinematic locus $K_b(\theta)$. Measuring K_b at known θ and K_a is nuclear level spectroscopy via reaction kinematics.

22.3 Eliminating the unobserved residual

Nuclear kinetic energies of a few MeV on mass $\sim A \times 931$ MeV justify the non-relativistic relations

$$K_Y = \frac{p_Y^2}{2m_Y}, \qquad p_a^2 = 2m_a K_a, \qquad p_b^2 = 2m_b K_b. \tag{22.11}$$

From (22.2)–(22.3),

$$p_Y \cos\phi = p_a - p_b \cos\theta, \qquad p_Y \sin\phi = p_b \sin\theta. \tag{22.12}$$

Squaring and adding eliminates ϕ:

$$p_Y^2 = (p_a - p_b\cos\theta)^2 + (p_b \sin\theta)^2 = p_a^2 + p_b^2 - 2p_a p_b \cos\theta. \tag{22.13}$$

Therefore

$$K_Y = \frac{p_Y^2}{2m_Y} = \frac{m_a}{m_Y}K_a + \frac{m_b}{m_Y}K_b - \frac{2}{m_Y}\sqrt{m_a m_b K_a K_b}\,\cos\theta. \tag{22.14}$$

Substitute (22.14) into (22.6):

$$Q + K_a = K_b + \frac{m_a}{m_Y} K_a + \frac{m_b}{m_Y} K_b - \frac{2}{m_Y}\sqrt{m_a m_b K_a K_b}\,\cos\theta + \Delta E. \tag{22.15}$$

Collecting terms,

$$Q + K_a\Big(1 - \frac{m_a}{m_Y}\Big) = K_b\Big(1 + \frac{m_b}{m_Y}\Big) - \frac{2}{m_Y}\sqrt{m_a m_b K_a K_b}\,\cos\theta + \Delta E. \tag{22.16}$$

Solving for the excitation energy,

$$\Delta E = Q - K_b\Big(1 + \frac{m_b}{m_Y}\Big) + K_a\Big(1 - \frac{m_a}{m_Y}\Big) + \frac{2}{m_Y}\sqrt{m_a m_b K_a K_b}\,\cos\theta. \tag{22.17}$$

All quantities on the right are either tabulated (Q, masses) or measured (K_a, K_b, θ). Equation (22.17) is the working formula of two-body reaction spectroscopy in the non-relativistic limit [6, 1].

22.3.1 From CM observables to laboratory observables

For a non-relativistic binary exit channel, let V be the CM speed in the laboratory and u_b the speed of b in the CM. Galilean velocity addition gives

$$\mathbf{v}_{b,\text{lab}} = V\hat{\mathbf{x}} + u_b(\cos\theta_{\text{cm}}\hat{\mathbf{x}} + \sin\theta_{\text{cm}}\hat{\mathbf{y}}). \tag{22.18}$$

Therefore, with $\gamma_k \equiv V/u_b$,

$$\tan\theta_{\text{lab}} = \frac{\sin\theta_{\text{cm}}}{\gamma_k + \cos\theta_{\text{cm}}}, \tag{22.19}$$

$$K_{b,\text{lab}} = \frac{1}{2} m_b \left(V^2 + u_b^2 + 2Vu_b\cos\theta_{\text{cm}}\right). \tag{22.20}$$

Solid angles also transform. Away from kinematic turning points, $\mathrm{d}\Omega = 2\pi\,\mathrm{d}(\cos\theta)$. Write

$$\cos\theta_{\text{lab}} = \frac{\gamma_k + \cos\theta_{\text{cm}}}{\sqrt{1 + 2\gamma_k\cos\theta_{\text{cm}} + \gamma_k^2}}$$

and differentiate with respect to $\mu \equiv \cos\theta_{\text{cm}}$. With $N = \gamma_k + \mu$ and $D = \sqrt{1 + 2\gamma_k\mu + \gamma_k^2}$,

$$\frac{\mathrm{d}(\cos\theta_{\text{lab}})}{\mathrm{d}\mu} = \frac{D - N\cdot(\gamma_k/D)}{D^2} = \frac{D^2 - N\gamma_k}{D^3} = \frac{1 + \gamma_k\mu}{D^3}, \tag{22.21}$$

so

$$\left|\frac{\mathrm{d}\Omega_{\text{lab}}}{\mathrm{d}\Omega_{\text{cm}}}\right| = \frac{|1 + \gamma_k\cos\theta_{\text{cm}}|}{(1 + 2\gamma_k\cos\theta_{\text{cm}} + \gamma_k^2)^{3/2}}, \quad \left(\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}\right)_{\text{lab}} = \left(\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}\right)_{\text{cm}}\left|\frac{\mathrm{d}\Omega_{\text{cm}}}{\mathrm{d}\Omega_{\text{lab}}}\right|. \tag{22.22}$$

One CM angle can map to a sharply focused laboratory cone; in some endothermic reactions two kinematic branches can occur at one forward laboratory angle. This is why an angular distribution must name its frame [6, 1].

Caution

Masses in (22.17) must be consistent (nuclear or atomic with the same convention on both sides of Q). Evaluated atomic masses from AME2020 are appropriate when electron numbers cancel. For precise work one uses relativistic kinematics; the structure of the argument is unchanged [6, 14, 27].

22.4 Quadratic equation for K_b and kinematic loci

For fixed K_a, θ, and ΔE, equation (22.15) is quadratic in $\sqrt{K_b}$ (or in K_b after clearing the square root). Writing $x = \sqrt{K_b}$,

$$\Big(1+\frac{m_b}{m_Y}\Big)x^2 - \frac{2\cos\theta}{m_Y}\sqrt{m_a m_b K_a}\;x + \Big[K_a\Big(\frac{m_a}{m_Y}-1\Big) - Q + \Delta E\Big] = 0. \tag{22.23}$$

A physical root needs $K_b > 0$ and a non-negative discriminant. This restricts the allowed beam energy, angle, and excitation, and underlies the threshold formulae [6, 1].

22.5 Threshold for endothermic reactions

At threshold the products have no relative momentum in the CM; in the laboratory they move together, because setting both final laboratory momenta to zero would violate momentum conservation. With $M_f^* = m_b + m_Y + \Delta E/c^2$, the invariant condition is $s_{\text{th}} = (M_f^* c^2)^2$. Combining it with (22.8) gives the **exact** laboratory threshold

$$K_a^{\text{th}} = \frac{(M_f^*)^2 - (m_a+m_X)^2}{2m_X}\,c^2. \tag{22.24}$$

Since $Q^* = Q - \Delta E = (m_a + m_X - M_f^*)c^2$, equivalently

$$K_a^{\text{th}} = -Q^*\left(1+\frac{m_a}{m_X}\right) + \frac{(Q^*)^2}{2m_X c^2}. \tag{22.25}$$

The final term is the small relativistic mass correction. Dropping it gives the standard non-relativistic result

$$K_a^{\text{th}} = (-Q+\Delta E)\frac{m_a+m_X}{m_X} = (-Q+\Delta E)\Big(1+\frac{m_a}{m_X}\Big). \tag{22.26}$$

The factor $(1 + m_a/m_X) > 1$ appears because part of the beam energy is tied up in CM motion. For $Q^* < 0$ the threshold is positive. Even if the ground-state reaction is exoergic, a level with $\Delta E > Q$ has $Q^* < 0$ and therefore a threshold [6, 1, 14].

Exoergic case. If $Q^* > 0$, the channel has zero mass-difference threshold (neglecting angular-momentum constraints). Charged-particle cross sections can still be strongly suppressed at low energy; tunnelling makes the Coulomb barrier a suppression, not a sharp kinematic threshold. At $Q^* = 0$ the channel opens at zero energy with vanishing final relative momentum.

22.6 Example: neutron capture on ^{238}U

Consider radiative capture

$$n + {}^{238}\mathrm{U} \longrightarrow {}^{239}\mathrm{U}^* \longrightarrow {}^{239}\mathrm{U} + \gamma. \tag{22.27}$$

Using evaluated atomic masses, the ground-state Q-value is

$$Q = \big(m_n + m({}^{238}\mathrm{U}) - m({}^{239}\mathrm{U})\big)c^2 \approx 4.806\,\mathrm{MeV} \qquad \text{(evaluated value)}. \tag{22.28}$$

In the laboratory the target is at rest. Momentum conservation requires the compound nucleus to carry the neutron's momentum p_n. The residual kinetic energy is

$$K_U = \frac{p_n^2}{2m_U} = K_n \frac{m_n}{m_U}. \tag{22.29}$$

Energy balance for forming ^{239}U at excitation ΔE reads

$$K_n + Q = K_U + \Delta E = K_n \frac{m_n}{m_U} + \Delta E, \tag{22.30}$$

so

$$\Delta E = Q + K_n\Big(1 - \frac{m_n}{m_U}\Big) \approx Q + K_n \qquad (m_n \ll m_U). \tag{22.31}$$

For a **thermal** neutron ($K_n \approx 0$),

$$\Delta E \approx Q \approx 4.806\,\mathrm{MeV}. \tag{22.32}$$

Capture therefore populates levels of ^{239}U near 4.806 MeV of excitation. Discrete levels above that energy are reached by increasing K_n. When K_n matches a compound-nucleus resonance, the capture cross section rises sharply. The energy dependence of $\sigma(K_n)$ shows a series of peaks: **neutron resonances**. Levels below S_n are not populated by capture of free neutrons, because even $K_n = 0$ already deposits $\sim Q$ [6, 1, 2, 27, 20, 21].

Remark

The same $Q \approx 4.806\,\mathrm{MeV}$ sets the neutron separation energy of ^{239}U: $S_n({}^{239}\mathrm{U}) = Q$. Resonances just above S_n dominate low-energy neutron physics of ^{238}U and enter reactor design [6, 1, 35, 33].

22.7 Cross section versus energy: resonances (concept)

Plotting σ against incident energy for a fixed channel often reveals narrow peaks. Each peak corresponds to a quasi-stationary state of the compound system whose energy matches the available CM energy. For slow neutrons on heavy targets the level density is high and widths are often narrow (eV scale), producing the rich resonance forests of evaluated libraries such as ENDF; charged-particle reactions usually show broader, overlapping structure once Coulomb

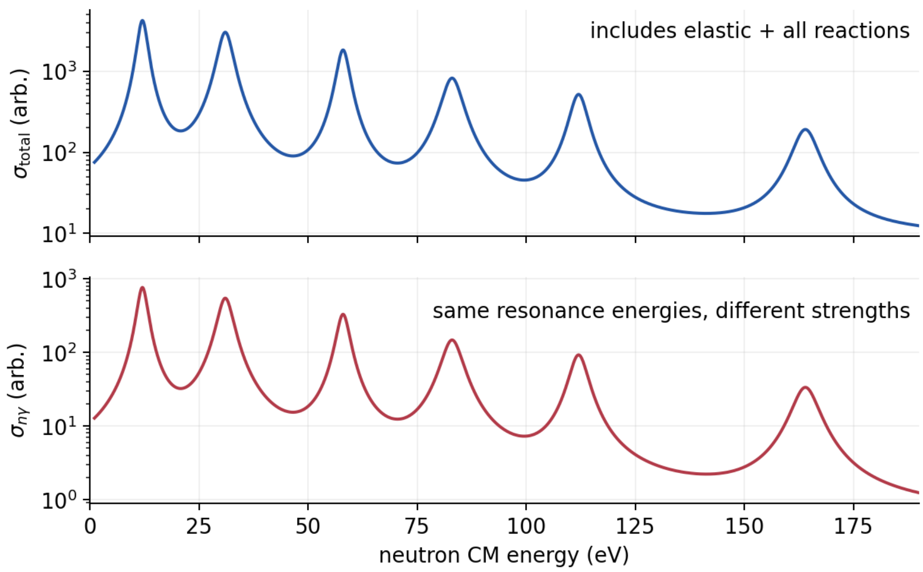


Figure 22.2: Schematic distinction between the evaluated *total* neutron cross section and the (n, γ) partial cross section for ^{238}U. Resonance energies can coincide while peak heights and backgrounds differ because the total includes elastic and every open reaction channel. Curves are redrawn for instruction and are not evaluated numerical data [35].

barriers raise the relevant excitation [6, 1, 35]. Qualitatively:

- on resonance, the reaction probability is large;
- off resonance, other processes (potential scattering, direct channels) may dominate.

The line shape follows by writing a resonant amplitude with pole $E_R - i\Gamma/2$. An entrance wave in partial wave J supplies the geometrical area π/k_i^2 and spin fraction $g_J = (2J+1)/[(2j_a+1)(2j_A+1)]$. Formation contributes the entrance branching amplitude $\sqrt{\Gamma_i}$, decay contributes $\sqrt{\Gamma_f}$, and propagation contributes $(E - E_R + i\Gamma/2)^{-1}$. Squaring this product gives, for an isolated level of spin J, entrance channel $i = a + A$, and exit channel $f = b + B$, the single-level **Breit–Wigner** cross section is

$$\sigma_{i\to f}(E) = \frac{\pi}{k_i^2}\, g_J \, \frac{\Gamma_i \Gamma_f}{(E - E_R)^2 + (\Gamma/2)^2}, \qquad g_J = \frac{2J+1}{(2j_a+1)(2j_A+1)}, \tag{22.33}$$

If the integrated final-state cross section counts an unordered pair of identical particles once, multiply the right-hand side by $1/S_f$, where $S_f = 1 + \delta_{bB}$. The entrance flux has no such division; this convention must be used consistently in the inverse reaction. Here $k_i = p_i/\hbar$, E_R is the CM resonance energy, Γ_i and Γ_f are partial widths, and $\Gamma = \sum_c \Gamma_c$. At the peak, $\sigma_{i\to f}(E_R) = (4\pi/k_i^2) g_J (\Gamma_i/\Gamma)(\Gamma_f/\Gamma)$: formation and decay branching both matter. The lifetime is

$$\tau \sim \frac{\hbar}{\Gamma}: \tag{22.34}$$

narrow resonances live longer. Energy-dependent penetrabilities can make $\Gamma_c(E)$ vary across a peak, while overlapping levels and a direct amplitude produce interference beyond the isolated-resonance approximation. The kinematic message is that discrete ΔE values map to discrete K_b peaks and incident formation energies [6, 14, 2].

22.8 Inverse reactions and detailed balance

The inverse of $a + A \to b + B$ is $b + B \to a + A$, and

$$Q_{\text{inverse}} = -Q_{\text{forward}}. \tag{22.35}$$

Time-reversal invariance does *not* say that the two cross sections are numerically equal: incident flux, phase space, and spin multiplicities differ. For unpolarized massive two-body channels at the same total CM energy, define $g_i = (2j_a + 1)(2j_A + 1)$, $g_f = (2j_b + 1)(2j_B + 1)$, and $S_i = 1 + \delta_{aA}$, $S_f = 1 + \delta_{bB}$. If each identical final pair is counted once, reciprocity gives

$$g_i\, p_i^2\, S_f\, \sigma_{aA\to bB} = g_f\, p_f^2\, S_i\, \sigma_{bB\to aA}. \tag{22.36}$$

Both cross sections must correspond to the same nuclear states and invariant energy.

Photon channel. A real photon has two transverse polarization states, not $2j_\gamma + 1 = 3$. For $a + A \leftrightarrow B + \gamma$, with distinct a, A and unpolarized beams/targets, the separately averaged capture and photodisintegration cross sections obey

$$(2j_a + 1)(2j_A + 1)\, p_{aA}^2\, \sigma_{aA\to B\gamma} = 2(2J_B + 1)\, p_\gamma^2\, \sigma_{\gamma B\to aA}, \tag{22.37}$$

where $p_\gamma = E_\gamma/c$. If a photon cross section is quoted for one fixed polarization rather than averaged over both, the factor 2 must be changed accordingly; identical massive particles require the S factors of Eq. (22.36). This polarization formula, rather than a formal substitution $j_\gamma = 1$ into the massive-particle expression, is used to connect capture and photodisintegration [6, 14].

Key idea

Students should remember:

- $Q = (m_a + m_X - m_b - m_Y)c^2$; $Q + K_a = K_b + K_Y + \Delta E$.
- Eliminate K_Y with $p_Y^2 = p_a^2 + p_b^2 - 2p_a p_b \cos\theta$.
- ΔE from measured K_b, θ, K_a via (22.17).
- Endothermic threshold: $K_a^{\text{th}} \simeq (-Q + \Delta E)(1 + m_a/m_X)$; use (22.24) for the exact result.
- $n + {}^{238}\text{U}$: $Q \simeq 4.806\,\text{MeV}$ sets capture excitation for slow neutrons; resonances map levels above S_n.
- Cross section peaks at compound resonances (Breit–Wigner shape).
- Inverse reactions have opposite Q-values; detailed balance also includes spin weights and CM momenta.

22.8.1 Breit–Wigner resonances in evaluated data

Isolated compound resonances follow Breit–Wigner line shapes set by partial widths into entrance and exit channels. Slow-neutron data on heavy targets display forests of narrow resonances in evaluated libraries; the total cross section includes elastic and all open reaction channels, so partial (n, γ) peaks need not match total-cross-section heights [6, 1, 35].

22.9 Deeper physics: Q-value kinematics and reaction thresholds

22.9.1 Two-body Q and energy release

For a reaction $X(a, b)Y$, the Q-value is

$$Q = [M(X) + M(a) - M(b) - M(Y)]c^2, \tag{22.38}$$

using atomic masses when the participants are neutral atoms [6, 1]. If $Q > 0$ the reaction is exothermic: rest mass converts to kinetic energy of the products. If $Q < 0$ the reaction is endothermic and requires a minimum projectile energy (threshold) before it can occur. The two-body kinematic relations in the lecture connect Q to the measured kinetic energies of a and b in the laboratory frame [6, 14].

Threshold for $^{13}\mathrm{C}(p, \gamma)^{14}\mathrm{N}$

The ground-state capture has $Q \approx 7.55\,\mathrm{MeV}$ [6, 20]. Because the proton and target are both massive, the threshold laboratory energy E_{th} exceeds $|Q|$ when $Q < 0$; for exothermic capture, $Q > 0$ and the reaction is possible at zero projectile energy in principle. In practice, Coulomb barrier penetration requires $E_p \gtrsim 0.5\,\mathrm{MeV}$ for this channel in stellar hydrogen burning (Lecture 26) [36].

22.9.2 Threshold energy for endothermic reactions

When $Q < 0$, define $|Q|$ as the energy deficit. In the centre-of-mass frame the minimum kinetic energy that conserves momentum is

$$E_{\mathrm{th}} = |Q| \left(1 + \frac{m_a}{M_X}\right), \tag{22.39}$$

for a target at rest [6, 1]. The extra factor m_a/M_X supplies the recoil energy of the target: not all projectile energy is available to create the heavier product. For neutron-induced reactions ($m_a \ll M_X$) the threshold is nearly $|Q|$; for proton beams on heavy targets the correction can be tens of percent [14].

Key idea

From thresholds to fission and fusion. Endothermic thresholds govern which channels open as beam energy rises: the first excited state of Y adds $|Q|$ to the threshold. In fission, $Q > 0$ for actinides but a saddle barrier must still be surmounted (Lecture 23). In fusion, even exothermic reactions require tunneling through the Coulomb barrier (Lecture 25). Two-body kinematics is the common language linking all three domains [6, 2].

22.9.3 Centre-of-mass versus laboratory observables

Reaction cross sections and angular distributions are simplest in the CM frame, where momentum conservation is symmetric. Laboratory angles must be transformed when comparing data from inverse kinematics, where the heavy nucleus is the projectile [6, 1]. The Q-value is Lorentz invariant, but the kinetic-energy partition between products is not: the same $Q > 0$ capture reaction deposits more recoil energy in the target when the projectile is light (neutron capture) than when it is heavy (inverse kinematics with a radioactive beam) [14, 53]. Evaluated masses from AME2020 fix Q to keV precision; threshold calculations should use those values rather than SEMF estimates [27, 20].

22.9.4 Recoil and the light-particle limit

When the projectile is much lighter than the target ($m_a \ll M_X$), the centre-of-mass frame is nearly at rest in the laboratory and the target recoil is negligible. Neutron-induced reactions are the standard example: the Q-value appears almost entirely as product kinetic energy [6, 35]. When the projectile is comparable to or heavier than the target (inverse kinematics), laboratory-frame observables must be transformed carefully before extracting Q or excitation energies [53, 1]. Alpha-decay recoil (Lecture 16) is the single-body limit of the same momentum bookkeeping.

Jacobian singularities near kinematic extrema amplify cross-section uncertainties when transforming between frames. In inverse kinematics, a 1 mrad angular error near zero degrees can dominate the excitation-energy budget. Document angle calibration with elastic scattering peaks before trusting missing-mass spectra.

22.10 Further physics from the literature

Relativistic two-body kinematics. At beam energies of tens to hundreds of MeV per nucleon the non-relativistic reduction $p^2 = 2mK$ fails. Exact Lorentz-invariant s-channel kinematics replace (22.17); the Källén function in (22.10) supplies the exact two-body momentum. Modern analysis codes implement the full transformation between laboratory and CM frames [6, 14].

Missing-mass spectroscopy. When only particle b is detected, the missing mass of Y is reconstructed from four-momentum conservation. Peaks in the missing-mass spectrum are excited states of Y. The method is standard at intermediate-energy facilities [6, 1]. Explicitly,

$$M_{\text{miss}}^2 c^4 = (E_a + m_X c^2 - E_b)^2 - c^2|\mathbf{p}_a - \mathbf{p}_b|^2, \qquad \Delta E = (M_{\text{miss}} - m_Y)c^2.$$

This invariant form avoids assigning a non-relativistic recoil correction outside its range of validity.

R-matrix and Breit–Wigner parameters. Isolated resonances are parameterised by E_R, partial widths Γ_c, and channel radii in R-matrix theory. Overlapping resonances and direct–resonance interference require more elaborate treatments [6, 14, 2].

Reactor and astrophysical capture. The ^{238}U(n,γ) resonances control neutron economy in reactors. In astrophysics, (n,γ) rates on unstable nuclei set the s- and r-process paths; many rates are still extrapolated from statistical models constrained by stable-target data [6, 1, 2, 35].

Evaluated-data cross-check. The lecture's rounded 4.86 MeV capture energy has been corrected to $S_n(^{239}\mathrm{U}) \simeq 4.806\,\mathrm{MeV}$. Evaluated masses should be used for numerical Q-values, while evaluated reaction libraries should be used for resonance parameters; a plotted total cross section is not automatically the (n,γ) partial cross section [33, 35].

22.10.1 Relativistic Q-values and missing mass

When beam energies exceed tens of MeV per nucleon, the non-relativistic relation $p^2 = 2mK$ fails. The invariant mass squared of the initial state,

$$s = \left(E_a + M_X c^2\right)^2 - |\mathbf{p}_a|^2 c^2, \tag{22.40}$$

fixes the centre-of-mass energy available for reaction [6, 14]. Missing-mass spectroscopy reconstructs the excitation energy of Y when only particle b is detected:

$$M_{\text{miss}}^2 c^4 = (E_a + M_X c^2 - E_b)^2 - c^2 |\mathbf{p}_a - \mathbf{p}_b|^2. \tag{22.41}$$

Peaks in M_{miss} map excited states of Y without detecting the recoil [1, 6].

Threshold for ^{238}U(n,f)

Neutron-induced fission of ^{238}U has $Q \approx 6.5\,\mathrm{MeV}$ but requires $E_n \gtrsim 1\,\mathrm{MeV}$ in practice because the fission barrier $B_f \approx 6\,\mathrm{MeV}$ must be surmounted (Lecture 23) [6, 35]. The reaction threshold in (22.39) applies to endothermic channels; fission adds a saddle barrier on top of the Q-value bookkeeping. Fast-reactor spectra extend above this threshold, converting fertile ^{238}U to fissile ^{239}Pu via (n,γ) followed by β^- decay (Lecture 24) [1].

22.10.2 From kinematics to astrophysical rates

Radiative-capture Q-values set the energy of emitted γ rays and determine whether a channel is exothermic at stellar temperatures. The ^{12}C$(p,\gamma)^{13}$N reaction in the CNO cycle and ^{3}He$(^3\mathrm{He}, 2p)^4$He in the pp chain (Lecture 26) are exothermic with $Q > 0$ but still require tunneling through the Coulomb barrier (Lecture 25) [36, 6]. Inverse kinematics with radioactive beams reverses the roles of projectile and target but uses the same invariant-mass algebra [53, 54].

Key idea

Arc through the course. Two-body kinematics links α-decay recoil (Lecture 16), β endpoints (Lecture 18), direct-reaction energy sharing (Lecture 21), fission thresholds (Lectures 23–24), and Gamow-peak integrals (Lecture 26). Evaluated masses from AME2020 anchor every Q-value in the chain [27, 6].

22.10.3 Resonances and the Q scale in capture reactions

When a compound level sits near the threshold energy, the cross section is enhanced by a Breit–Wigner factor (Lecture 21) [6, 14]. The ^{238}U(n, γ) resonances that control reactor neutron economy are sharp because the neutron width is small compared with level spacing [35, 6]. In astrophysics, radiative capture on unstable nuclei often relies on statistical estimates anchored by stable-target data; the Q-value from AME2020 sets the absolute energy scale for those extrapolations [27, 53]. Missing-mass spectroscopy with radioactive beams is the modern implementation of the same two-body kinematics in inverse geometry [14, 2].

22.10.4 Inverse kinematics and radioactive beams

When the heavy nucleus is the projectile, laboratory angles are compressed into a forward cone and energy loss in the target smears the excitation-energy resolution [53, 1]. Missing-mass reconstruction remains the standard tool because detecting the heavy recoil is often impractical. Active-target time-projection chambers place the reaction vertex inside an extended gas volume, so vertex reconstruction and energy-loss corrections become part of the kinematics chain [54, 6]. The ideal two-body formulas of the main lecture are therefore the first step in an analysis notebook that lists corrections explicitly rather than treating them as afterthoughts.

22.11 Deeper reading: Q values, thresholds, and missing mass

Two-body kinematics converts measured angles and momenta into excitation energy once Q is known from masses. Thresholds for populating excited states follow by adding E_x to the ground-state endothermicity. Inverse kinematics with radioactive beams boosts forward focusing and demands careful angular-resolution budgets near zero degrees.

Missing-mass reconstruction uses the undetected recoil's inferred four-momentum; invariant-mass methods use decay products of unbound states. Active-target TPCs add vertex and energy-loss corrections on top of the ideal two-body map. Always compute Q from AME masses before simulating a spectrometer acceptance.

Inclusive versus exclusive

Inclusive measurements integrate over unobserved particles and need careful efficiency modelling; exclusive two-body measurements lean on this chapter's algebra. Missing-mass resolution is limited by beam energy spread, target thickness, and angle. Quote the resolution budget when claiming a new excited state.

Threshold cusps

Opening a new channel redistributes flux and can produce cusps or rounded steps in excitation functions. Interpreting those features requires the Q-value map of Lecture 22 together with continuum coupling ideas from Lectures 10 and 21.

Kinematic lab exercise

Pick $^{12}\mathrm{C}(p, p')^{12}\mathrm{C}^*$ to the first 2^+ state. Compute Q for elastic and inelastic channels, the threshold beam energy, and the laboratory scattering angle corresponding to a given centre-of-mass angle at one beam energy. Propagate a 0.1° angle error into E_x near the kinematic limit.

Repeat in inverse kinematics for a hypothetical radioactive beam on a proton target. Observe the forward focusing and the heightened angle sensitivity. This pair of calculations is the everyday toolkit for spectrometer design and for understanding why missing-mass resolution budgets are dominated by angles and energy loss rather than by algebra errors in Q.

Kinematics is the map between detector coordinates and nuclear excitation. Inverse kinematics and missing mass are applications of the same two-body algebra.

In practice, rework one numerical estimate from the lecture with a different parameter set and compare the shift to the quoted experimental uncertainty. If the shift exceeds the uncertainty, the estimate is teaching a scale, not delivering a fit. That distinction keeps theoretical cartoons honest when they sit beside evaluated data tables and modern many-body calculations. Along the course arc, write one sentence linking this chapter backward and one forward: the discipline of those two sentences is as important as any single formula. Examination problems rarely ask for a literature review, but they do ask for the scale argument that literature reviews presuppose.

Synthesis problem

Write a half-page note that (i) restates the chapter's central scale or selection rule in one equation, (ii) evaluates it numerically for one concrete nucleus or energy, and (iii) names the neighbouring lecture that uses the result. Keep the note free of research-frontier speculation; the goal is fluency with the lecture physics itself. If an assumption fails (Born approximation, spherical mean field, allowed transition, one-dimensional barrier), state the failure mode in a final sentence. That habit converts passive reading into examination-ready understanding and makes the later research-frontier box read as an extension rather than a replacement.

Research frontier: inverse kinematics and missing-mass methods

Two-body kinematics (Lecture 22) is indispensable for inverse-kinematics experiments with radioactive beams, active-target time-projection chambers, and storage-ring reactions. Missing-mass spectroscopy reconstructs excitation energies when the recoil is undetected; thresholds and Q values from AME2020 set the absolute energy scale [27, 53]. Facility upgrades emphasise high-rigidity spectrometers and storage rings precisely because kinematic acceptance and luminosity limit the reach to the most exotic beams [54]. Uncertainties that students should watch for in published analyses include angular-resolution propagation near zero degrees, target thickness and energy-loss corrections, and the difference between missing-mass and invariant-mass reconstructions for unbound states. The lecture's two-body algebra is therefore not merely homework: it is the everyday language of radioactive-beam proposals and spectrometer design.

Active-target TPCs blur the lecture's sharp "thin-target" idealisation because the reaction vertex sits inside an extended gas volume. Reconstructing the vertex and correcting energy loss become part of the kinematics chain. Understanding two-body relations first makes those corrections intelligible rather than mysterious.
Vertex reconstruction and energy-loss corrections turn the ideal two-body formulas into the real analysis chain of an active-target experiment.
Writing the ideal kinematic map first, then listing corrections, is the recommended analysis notebook structure.
Spectrometer design documents for FRIB and FAIR make the same kinematic trade-offs explicit when choosing angular coverage and rigidity.
Student directions. (1) Kinematics exercise: for a given $X(a,b)Y$ reaction, compute Q and the threshold for the first excited state of Y. (2) Error propagation: show how a 1 mrad angular error propagates into K_b near zero degrees in inverse kinematics. (3) Methods comparison: contrast missing-mass and invariant-mass methods for unbound resonant states.

22.12 Problems with solutions

Problem 22.1: Q-value from masses. Using rest energies $m_ac^2 = 938.3\,\mathrm{MeV}$, $m_Xc^2 = 11174.9\,\mathrm{MeV}$, $m_bc^2 = 1875.6\,\mathrm{MeV}$, $m_Yc^2 = 10252.5\,\mathrm{MeV}$, compute Q for $a + X \to b + Y$. Is the reaction exoergic or endoergic? [6]

Solution 22.1.

$$Q = (938.3 + 11174.9) - (1875.6 + 10252.5) = 12113.2 - 12128.1 = -14.9\,\mathrm{MeV}.$$

Endoergic ($Q < 0$).

Problem 22.2: Threshold energy. For a reaction with $Q = -14.9\,\mathrm{MeV}$, $\Delta E = 0$, projectile mass number $A_a = 1$, target $A_X = 12$, find the laboratory threshold K_a^{th} [6, 1].

Solution 22.2.

$$K_a^{\mathrm{th}} = (-Q)\Big(1 + \frac{m_a}{m_X}\Big) = 14.9\Big(1 + \frac{1}{12}\Big) = 14.9 \times \frac{13}{12} = 16.1\,\mathrm{MeV}.$$

Problem 22.3: Residual momentum. In a reaction at $\theta = 30^\circ$, $p_a = 150\,\mathrm{MeV}/c$, $p_b = 120\,\mathrm{MeV}/c$. Find p_Y from momentum conservation [1].

Solution 22.3.

$$p_Y^2 = p_a^2 + p_b^2 - 2p_ap_b\cos\theta = 150^2 + 120^2 - 2(150)(120)\cos 30^\circ.$$

$$\cos 30^\circ = \sqrt{3}/2 \approx 0.866, \qquad 2(150)(120)(0.866) = 31176.$$

$$p_Y^2 = 22500 + 14400 - 31176 = 5724, \qquad p_Y = 75.7\,\mathrm{MeV}/c.$$

Problem 22.4: Excitation from K_b (forward angle). For $n + {}^{12}\mathrm{C} \to n' + {}^{12}\mathrm{C}^*$ treat the inelastic neutron as $a = b = n$, $X = Y = {}^{12}\mathrm{C}$, $Q = 0$. At $\theta = 0$ and $K_a = 20.0\,\mathrm{MeV}$, a peak is found at $K_b = 15.5\,\mathrm{MeV}$. Estimate ΔE using $\Delta E \approx K_a - K_b$ (heavy residual, forward kinematics), then correct with the leading recoil terms of (22.17) using $m_n/m_C = 1/12$ [6].

Solution 22.4. Crude estimate:

$$\Delta E \approx 20.0 - 15.5 = 4.5\,\mathrm{MeV}.$$

With $Q = 0$, $m_a = m_b = m_n$, $m_Y = m_C$, $\theta = 0$, $\cos\theta = 1$,

$$\begin{aligned}\Delta E &= -K_b\Big(1 + \frac{1}{12}\Big) + K_a\Big(1 - \frac{1}{12}\Big) + \frac{2}{12}\sqrt{K_a K_b}\\ &= -15.5 \times \frac{13}{12} + 20.0 \times \frac{11}{12} + \frac{1}{6}\sqrt{20 \times 15.5}.\end{aligned}$$

$$\sqrt{310} \approx 17.61, \quad \frac{1}{6} \times 17.61 \approx 2.94,$$

$$\Delta E = -16.79 + 18.33 + 2.94 = 4.48\,\mathrm{MeV}.$$

The residual is excited by about 4.5 MeV.

Problem 22.5: Thermal capture on ^{238}U. Show that a neutron with $K_n = 0.025\,\mathrm{eV}$ on ^{238}U produces essentially the same excitation as a zero-energy neutron if $Q = 4.806\,\mathrm{MeV}$. Compute ΔE from (22.31) [6, 1].

Solution 22.5.

$$\frac{m_n}{m_U} \approx \frac{1}{239} \approx 4.2 \times 10^{-3}, \qquad K_n = 2.5 \times 10^{-8}\,\mathrm{MeV}.$$

$$\Delta E = Q + K_n\Big(1 - \frac{m_n}{m_U}\Big) = 4.806\,\mathrm{MeV} + 2.5 \times 10^{-8}\,\mathrm{MeV} \times 0.996 \approx 4.806\,\mathrm{MeV}.$$

Thermal kinetic energy is negligible next to Q.

Problem 22.6: Resonance and lifetime. A neutron resonance has total width $\Gamma = 0.10\,\mathrm{eV}$. Estimate the compound-state lifetime $\tau = \hbar/\Gamma$. Use $\hbar = 6.58 \times 10^{-16}\,\mathrm{eV}\cdot\mathrm{s}$ [6, 14].

Solution 22.6.

$$\tau = \frac{6.58 \times 10^{-16}\,\mathrm{eV}\cdot\mathrm{s}}{0.10\,\mathrm{eV}} = 6.6 \times 10^{-15}\,\mathrm{s}.$$

This is a long-lived compound resonance, far longer than a direct transit time $\sim 10^{-22}\,\mathrm{s}$. Compound lifetimes span a broad range because the widths of individual levels span a broad range.

Problem 22.7: Why excited residuals lower K_b. From $Q + K_a = K_b + K_Y + \Delta E$, explain why peaks corresponding to higher ΔE appear at lower K_b in a fixed-angle spectrum for fixed K_a [6, 1].

Solution 22.7. For fixed entrance energy K_a and fixed Q, the sum $K_b + K_Y + \Delta E$ is fixed. Increasing ΔE reduces the kinetic energy available to share between b and Y. At a fixed angle the kinematic solution for K_b therefore decreases: excited-state peaks sit at lower detected energy than the ground-state peak. That is the basis of missing-mass / reaction spectroscopy.

Problem 22.8: Exact versus non-relativistic threshold. For $a + X \to b + Y$, take $Q = -20.0\,\text{MeV}$, $m_a c^2 = 938.3\,\text{MeV}$, $m_X c^2 = 11174.9\,\text{MeV}$, and $\Delta E = 2.0\,\text{MeV}$. Find (a) the non-relativistic threshold, (b) the exact correction, and (c) their fractional difference [6, 14].

Solution 22.8. Step 1: effective Q-value.

$$Q^* = Q - \Delta E = -22.0\,\text{MeV}.$$

Step 2: leading threshold.

$$K_{\text{th}}^{\text{NR}} = -Q^*\left(1 + \frac{m_a}{m_X}\right) = 22.0\left(1 + \frac{938.3}{11174.9}\right) = 23.85\,\text{MeV}.$$

Step 3: exact mass correction.

$$\delta K_{\text{rel}} = \frac{(Q^*)^2}{2m_X c^2} = \frac{(22.0)^2}{2(11174.9)} = 0.0217\,\text{MeV}.$$

Hence

$$K_{\text{th}}^{\text{exact}} = 23.87\,\text{MeV}, \qquad \frac{\delta K_{\text{rel}}}{K_{\text{th}}^{\text{NR}}} = 9.1 \times 10^{-4}.$$

The correction is small here, but the invariant derivation identifies it and remains valid when the non-relativistic approximation does not.

Problem 22.9: Reciprocity of an inverse reaction. At one CM energy, suppose a and A have spins 1/2 and 0, while b and B have spins 0 and 1. If $p_i = 60\,\text{MeV}/c$, $p_f = 90\,\text{MeV}/c$, and $\sigma_{aA\to bB} = 12\,\text{mb}$, find $\sigma_{bB\to aA}$. Assume no identical-particle factors [6].

Solution 22.9. Step 1: spin statistical weights.

$$g_i = (2\tfrac{1}{2} + 1)(2\cdot 0 + 1) = 2, \qquad g_f = (2\cdot 0 + 1)(2\cdot 1 + 1) = 3.$$

Step 2: apply detailed balance.

$$g_i p_i^2 \sigma_{i\to f} = g_f p_f^2 \sigma_{f\to i},$$

so

$$\sigma_{f\to i} = \frac{g_i}{g_f}\frac{p_i^2}{p_f^2}\sigma_{i\to f} = \frac{2}{3}\left(\frac{60}{90}\right)^2(12\,\mathrm{mb}) = 3.56\,\mathrm{mb}.$$

The inverse cross section is smaller even though the transition matrix element is constrained by time reversal, because flux, phase space, and spin multiplicities differ.

Problem 22.10: Breit–Wigner peak and area. An isolated resonance has $g_J = 1$, $k = 0.50\,\mathrm{fm}^{-1}$, $\Gamma_i = 0.20\,\mathrm{eV}$, $\Gamma_f = 0.30\,\mathrm{eV}$, and $\Gamma = 0.80\,\mathrm{eV}$. Find (a) the peak cross section and (b) $\int \sigma(E)\,\mathrm{d}E$, treating k and widths as constant across the line [6, 14].

Solution 22.10.

$$\sigma(E_R) = \frac{4\pi}{k^2}\frac{\Gamma_i}{\Gamma}\frac{\Gamma_f}{\Gamma} = \frac{4\pi}{0.25}(0.25)(0.375) = 4.71\,\mathrm{fm}^2 = 47.1\,\mathrm{mb}.$$

Using $\int_{-\infty}^{\infty} \mathrm{d}x/(x^2+a^2) = \pi/a$,

$$\int \sigma(E)\,\mathrm{d}E = \frac{2\pi^2}{k^2}\frac{\Gamma_i\Gamma_f}{\Gamma} = 5.92\,\mathrm{fm}^2\,\mathrm{eV} = 59.2\,\mathrm{mb\,eV}.$$

Peak height depends on branching ratios, while the resonance area also carries the absolute width scale.

Problem 22.11: Capture–photodisintegration balance. For $a(1/2) + A(0) \leftrightarrow B(1/2) + \gamma$, let $p_{aA} = 20\,\mathrm{MeV}/c$, $p_\gamma = 5\,\mathrm{MeV}/c$, and $\sigma_{aA\to B\gamma} = 10\,\mu\mathrm{b}$. Find the unpolarized inverse cross section, assuming distinct massive particles [6, 14].

Solution 22.11. The massive entrance degeneracy is $(2j_a+1)(2j_A+1) = 2$. The photon–nucleus channel has $2(2J_B+1) = 4$, where 2 counts transverse photon polarizations. Hence

$$\sigma_{\gamma B\to aA} = \frac{2(20)^2}{4(5)^2}(10\,\mu\mathrm{b}) = 80\,\mu\mathrm{b}.$$

Using $2j_\gamma + 1 = 3$ would be incorrect for a real photon.

Chapter 23

Nuclear Fission I

"The discovery of fission, and its interpretation as a liquid-drop splitting, transformed nuclear physics from a laboratory science into a source of industrial power."

L. Meitner and O. R. Frisch (1939)

Nuclear fission is the splitting of a heavy nucleus into two fragments of comparable mass. Every commercial nuclear power reactor extracts energy from this process. In this chapter we define fission, recount the 1938–1939 discovery, derive the Q-value for symmetric fission from the semi-empirical mass formula (SEMF), obtain the energetic condition $Z^2/A > 17.6$, and explain why nuclei that are energetically free to fission still do not spontaneously split: the *fission barrier* of the Bohr–Wheeler liquid-drop picture. Neutron-induced fission and the compound-nucleus mechanism close the discussion and prepare the following lecture, *Nuclear Fission II*, [6, 1, 14, 2].

23.1 Definition and energy scale

Fission is the nuclear reaction in which a heavy parent X splits into two roughly equal fragments Y and Z,

$$X \;\to\; Y + Z \qquad (A \approx A_1 + A_2,\ Z \approx Z_1 + Z_2). \tag{23.1}$$

"Roughly equal" means the two fragments have comparable mass numbers; the extreme opposite is light-particle emission (α, proton, neutron), already treated in earlier chapters. In practice a few free neutrons are also emitted, so a more complete schematic is $X \to Y + Z + \nu n$ with $\nu \sim 2$–3. The essential physics is still the binary split of a heavy system into two medium-mass nuclei [6, 1].

Key idea

Fission: a heavy nucleus divides into two fragments of comparable mass ($A_1 \sim A_2$). Light-particle emission is *not* fission. Typical energy release for $A \sim 240$ is of order 200 MeV per event.

Why so much energy? The binding-energy-per-nucleon curve B/A versus A rises from light nuclei to a broad maximum near iron and nickel ($A \sim 56$–62, $B/A \sim 8.7$–8.8 MeV) and then *declines* slowly for heavy nuclei. For uranium-region nuclei ($A \sim 230$–240) one has $B/A \sim 7.6$ MeV; for fragments near $A \sim 100$–140 one finds $B/A \sim 8.5$ MeV. Moving nucleons from a low-B/A parent into higher-B/A daughters releases energy of order

$$Q \sim \left[(B/A)_{\text{frag}} - (B/A)_{\text{parent}}\right] A \sim (8.5 - 7.6)\,\text{MeV} \times 240 \approx 216\,\text{MeV}. \tag{23.2}$$

That is the characteristic fission energy scale. A full SEMF derivation of when $Q > 0$ occupies Section 23.4.

23.2 Historical path: neutrons, barium, and the liquid drop

Historical note

1932–1938: neutron irradiation and the search for transuranics. Chadwick's discovery of the neutron (1932) opened a new experimental tool: neutrons have no charge, so they are not repelled by the nuclear Coulomb field and can enter heavy nuclei even at thermal energies. Fermi's group in Rome systematically bombarded elements with neutrons, hoping that n capture followed by β^- decay would produce elements beyond uranium (transuranics). Many new radioactivities appeared; their chemical identification was difficult [6, 2].

1938–1939: Hahn and Strassmann. Otto Hahn and Fritz Strassmann irradiated natural uranium with neutrons and performed meticulous radiochemistry. Among the products they found an element that behaved chemically like barium ($Z = 56$), not like a heavy transuranic homologue of radium. After repeated checks they concluded that barium itself was present: a medium-mass element had been produced from uranium ($Z = 92$) [6, 14].

1939: Meitner, Frisch, and Bohr–Wheeler. Lise Meitner and Otto Frisch interpreted the result as a binary split of the uranium nucleus, by analogy with a charged liquid drop that divides when deformed. They estimated the energy release from the mass defect and named the process *fission*. Niels Bohr and John Wheeler then developed the liquid-drop theory of the fission barrier (1939), which still organises the subject [6, 1, 14].

The chemical surprise (barium from uranium) forced a change of paradigm: neutron irradiation of heavy nuclei does not only build slightly heavier species; it can also *break* the nucleus into two pieces of intermediate mass.

23.3 Binding energy and the Q-value of fission

Write the binary fission of a nucleus ${}^{A}_{Z}X_N$ as

$${}^{A}_{Z}X_N \;\to\; {}^{A_1}_{Z_1}Y_{N_1} + {}^{A_2}_{Z_2}Z_{N_2}, \tag{23.3}$$

with

$$A = A_1 + A_2, \qquad Z = Z_1 + Z_2, \qquad N = N_1 + N_2. \tag{23.4}$$

(We omit free neutrons for the mass-energy bookkeeping; they can be restored later without changing the logic.) Nuclear masses in energy units are

$$Mc^2 = Z\,m_p c^2 + N\,m_n c^2 - E_B, \tag{23.5}$$

where E_B is the total binding energy. The reaction Q-value is the decrease in rest mass,

$$Q = (M_X - M_Y - M_Z)c^2. \tag{23.6}$$

Substitute (23.5) for each nucleus. The proton and neutron mass terms cancel by charge and nucleon conservation, leaving

$$Q = E_B(Y) + E_B(Z) - E_B(X). \tag{23.7}$$

Fission releases energy if and only if the sum of the fragment binding energies exceeds the parent binding energy: the fragments are more tightly bound per nucleon (and overall) than the parent.

Numerical estimate from B/A. Take $E_B(X) = (B/A)_X\, A$ with $(B/A)_X \approx 7.6\,\text{MeV}$ and $E_B(Y) + E_B(Z) \approx 8.5\,\text{MeV} \times (A_1 + A_2) = 8.5\,\text{MeV} \times A$. Then

$$Q \approx (8.5 - 7.6)\,\text{MeV} \times A = 0.9\,\text{MeV} \times A. \tag{23.8}$$

For $A \approx 240$,

$$Q \approx 0.9 \times 240 = 216\,\text{MeV}, \tag{23.9}$$

as anticipated in (23.2). Systems tend toward lower rest-mass energy; fission of heavy nuclei is therefore energetically favoured once the fragments sit on the high-B/A side of the binding curve [6, 1, 2].

Remark

Equation (23.7) is general: any rearrangement of nucleons that increases total binding releases energy. Fusion of light nuclei, treated in later lectures, uses the same identity with the opposite motion on the B/A curve.

23.4 Symmetric fission from the SEMF: full derivation

To estimate when symmetric fission is energetically possible without external energy, use one nuclear-mass SEMF convention throughout this chapter, neglecting pairing and replacing $Z(Z-1)$ by Z^2:

$$M(A,Z)c^2 = Zm_pc^2 + Nm_nc^2 - a_vA + a_sA^{2/3} + a_c\frac{Z^2}{A^{1/3}} + a_{\text{sym}}\frac{(A-2Z)^2}{A}. \tag{23.10}$$

All a_i are energies. We use $a_s = 17.8\,\text{MeV}$ and $a_c = 0.711\,\text{MeV}$ in every numerical illustration; modest coefficient changes shift the thresholds because this is a liquid-drop estimate, not evaluated barrier data [6, 1, 14]. Consider *symmetric* fission into two identical fragments,

$$A_1 = A_2 = \frac{A}{2}, \qquad Z_1 = Z_2 = \frac{Z}{2}. \tag{23.11}$$

The Q-value in energy units is

$$Q = M(A,Z)c^2 - 2\,M\left(\frac{A}{2}, \frac{Z}{2}\right)c^2. \tag{23.12}$$

23.4.1 Term-by-term cancellation

Volume term.

$$-a_v A - 2\left(-a_v \frac{A}{2}\right) = -a_v A + a_v A = 0.$$

Volume energy is extensive; it cancels for any binary split that conserves A.

Asymmetry term. For the parent,

$$a_{\text{sym}} \frac{(A-2Z)^2}{A}.$$

For one fragment,

$$a_{\text{sym}} \frac{\big(A/2 - 2(Z/2)\big)^2}{A/2} = a_{\text{sym}} \frac{\big((A-2Z)/2\big)^2}{A/2} = a_{\text{sym}} \frac{(A-2Z)^2}{4} \cdot \frac{2}{A} = a_{\text{sym}} \frac{(A-2Z)^2}{2A}.$$

Two fragments contribute

$$2 \times a_{\text{sym}} \frac{(A-2Z)^2}{2A} = a_{\text{sym}} \frac{(A-2Z)^2}{A},$$

which exactly cancels the parent asymmetry term. Thus the symmetry energy drops out of Q for *symmetric* charge and mass partition.

Surface term.

$$Q_s = a_s A^{2/3} - 2\, a_s \left(\frac{A}{2}\right)^{2/3} = a_s A^{2/3} \left[1 - 2 \cdot \left(2^{-1}\right)^{2/3}\right] = a_s A^{2/3} (1 - 2^{1/3}).$$

Since $2^{1/3} \approx 1.260$,

$$Q_s = a_s A^{2/3} (1 - 1.260) = -0.260\, a_s\, A^{2/3}. \tag{23.13}$$

The surface term is *negative* in Q: two spheres have more total surface area than one sphere of equal volume, so surface energy opposes fission.

Coulomb term.

$$Q_c = a_c \frac{Z^2}{A^{1/3}} - 2\, a_c \frac{(Z/2)^2}{(A/2)^{1/3}} = a_c \frac{Z^2}{A^{1/3}} - 2\, a_c \frac{Z^2/4}{A^{1/3}\, 2^{-1/3}} = a_c \frac{Z^2}{A^{1/3}} - a_c \frac{Z^2}{2A^{1/3}}\, 2^{1/3}.$$

Using $2^{1/3}/2 = 2^{-2/3} \approx 0.630$,

$$Q_c = a_c \frac{Z^2}{A^{1/3}} (1 - 2^{-2/3}) = a_c \frac{Z^2}{A^{1/3}} (1 - 0.630) = 0.370\, a_c\, \frac{Z^2}{A^{1/3}}. \tag{23.14}$$

The Coulomb term is *positive* in Q: separating the charge into two smaller nuclei lowers the electrostatic self-energy and favours fission.

23.4.2 Net Q and the condition $Q > 0$

Collecting surface and Coulomb contributions,

$$\boxed{Q = -0.26\, a_s\, A^{2/3} + 0.37\, a_c\, \frac{Z^2}{A^{1/3}}} \tag{23.15}$$

(pairing neglected; volume and symmetry cancelled). Fission is energetically allowed when $Q > 0$; whether it occurs spontaneously is a separate barrier-penetration question:

$$0.37\, a_c\, \frac{Z^2}{A^{1/3}} > 0.26\, a_s\, A^{2/3}, \tag{23.16}$$

or equivalently

$$\frac{Z^2}{A} > \frac{0.26\, a_s}{0.37\, a_c}. \tag{23.17}$$

With the stated coefficients,

$$\frac{0.26 \times 17.8}{0.37 \times 0.711} \approx \frac{4.63}{0.263} \approx 17.6. \tag{23.18}$$

Hence the classic energetic criterion

$$\boxed{\frac{Z^2}{A} > 17.6 \quad \text{(symmetric fission energetically possible).}} \tag{23.19}$$

For nuclei along the valley of stability, $Z/A \sim 0.4$–0.45 at large A, so $Z^2/A \sim 0.16\, A$ to $0.20\, A$. The inequality $Z^2/A > 17.6$ is then satisfied for

$$A \gtrsim 100 \tag{23.20}$$

(roughly). All nuclei with $A \gtrsim 100$ are, on pure rest-mass grounds, candidates for spontaneous fission into two medium fragments [6, 1, 14, 2].

Key idea

Symmetric fission Q from SEMF: $Q = -0.26\, a_s A^{2/3} + 0.37\, a_c Z^2/A^{1/3}$. $Q > 0$ when $Z^2/A > 17.6$. Energetically, $A \gtrsim 100$ can fission. Volume and symmetry terms cancel; surface opposes, Coulomb drives fission.

Caution

Energetically allowed does *not* mean observed. Uranium and thorium have $A > 100$ and $Q > 0$, yet they are long-lived. The reason is the fission *barrier*: intermediate deformed shapes cost energy even when the final fragments lie lower than the parent (Section 23.5).

23.5 Why no spontaneous fission? The fission barrier

If $Q > 0$, why do ^{235}U and ^{238}U exist? The nucleus cannot jump discontinuously from a single sphere to two well-separated fragments. It must pass through a continuous sequence of deformed shapes at essentially constant volume (nuclear matter is nearly incompressible). Along that path the total energy $E(r)$ of the liquid drop first *rises*, then falls toward the asymptotic Coulomb repulsion of two fragments. The maximum of $E(r)$ relative to the ground-state energy is the **activation energy** or **fission barrier height** E_f [6, 1, 14].

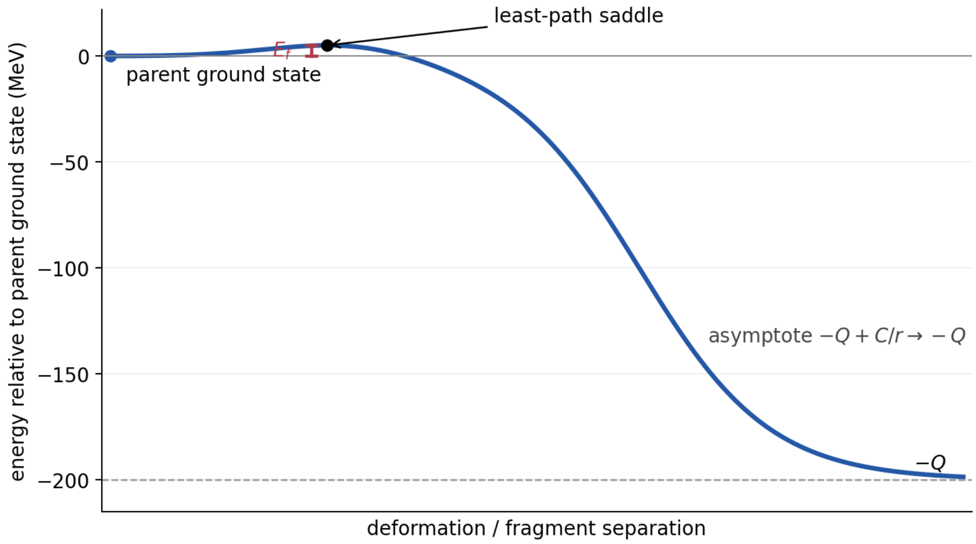


Figure 23.1: Schematic fission barrier: energy versus deformation (or fragment separation). The parent ground state sits in a well; a barrier of height E_f separates it from scission and the asymptote $-Q + C/r$. For $A \sim 240$, $E_f \sim 5.5$–$6.5\,\text{MeV}$. The curve is schematic rather than evaluated data.

Physical competition. As the drop elongates:

- **Surface energy increases** (larger surface area at fixed volume).
- **Coulomb energy decreases** (protons move farther apart on average).

Near sphericity the surface penalty dominates for moderate Z^2/A, so $E(r)$ rises. On the parent-ground-state energy scale used in the figure, the large-separation asymptote is

$$E(r) \longrightarrow -Q + \frac{Z_1 Z_2 e^2}{4\pi\varepsilon_0 r}, \qquad r \to \infty. \tag{23.21}$$

The curve approaches $-Q$, not zero; the positive $1/r$ term is the remaining fragment repulsion. Between these limits a barrier peak appears (Fig. 23.1).

Energy reference. If the zero is chosen as two fragments at rest at infinite separation, the intact parent lies $Q \sim 200\,\text{MeV}$ *above* that zero. In the more usual deformation-energy plot one instead sets the parent ground state to zero; the separated products then lie at $-Q$, and the saddle lies at $+E_f$. These are the same curve shifted by a constant. Confusing the two zeros makes the roughly 200 MeV energy release look like the roughly 6 MeV activation barrier.

Barrier height for heavy nuclei. Bohr and Wheeler estimated E_f from the liquid-drop energy as a function of deformation. For uranium-region nuclei one finds typically

$$E_f \sim 5.5\text{–}6.5\,\text{MeV}. \tag{23.22}$$

Spontaneous fission then requires either (i) thermal or quantum tunneling through a barrier of several MeV, which is extremely slow for $A \sim 240$, or (ii) supply of excitation energy comparable to E_f by a reaction (neutron capture, photon absorption, ...). That is why uranium is metastable on geological timescales despite $Q > 0$ [6, 1].

23.6 Deformation energy: ellipsoid calculation

A quantitative sketch of the barrier near sphericity follows Bohr and Wheeler. Start with a spherical nucleus of radius R and deform it into a prolate ellipsoid of semi-axes a (long) and $b = b$ (equal short axes), at *constant volume*:

$$\frac{4}{3}\pi R^3 = \frac{4}{3}\pi a b^2 \quad \Rightarrow \quad R^3 = ab^2. \tag{23.23}$$

Parametrise a small elongation by

$$a = R(1+\varepsilon), \qquad b = \frac{R}{\sqrt{1+\varepsilon}} \tag{23.24}$$

(so that $ab^2 = R^3$ exactly). For small ε,

$$b = R\left(1 - \frac{\varepsilon}{2} + \frac{3\varepsilon^2}{8} + \mathcal{O}(\varepsilon^3)\right).$$

The square root and the factor 2 in the resulting fissility criterion are both lost in the OCR transcription of the original source.

For a prolate ellipsoid define $e^2 = 1 - b^2/a^2$. Its exact area is

$$S = 2\pi b^2 \left(1 + \frac{a}{be}\sin^{-1} e\right).$$

Using $e^2 = 3\varepsilon - 6\varepsilon^2 + \mathcal{O}(\varepsilon^3)$, $\sin^{-1} e/e = 1 + e^2/6 + 3e^4/40 + \cdots$, and the volume constraint gives

$$S \approx S_0\left(1 + \frac{2}{5}\varepsilon^2\right), \tag{23.25}$$

where $S_0 = 4\pi R^2$ is the sphere area. The surface energy therefore becomes

$$E_{\text{surf}} = a_s A^{2/3}\left(1 + \frac{2}{5}\varepsilon^2\right). \tag{23.26}$$

For a uniformly charged ellipsoid the shape dependence can be written

$$\frac{E_{\text{Coul}}}{E^{(0)}_{\text{Coul}}} = \frac{R}{2}\int_0^\infty \frac{\mathrm{d}s}{(b^2+s)\sqrt{a^2+s}}.$$

Expanding the integrand with the same a, b and evaluating the elementary integrals gives

$$E_{\text{Coul}} = a_c \frac{Z^2}{A^{1/3}}\left(1 - \frac{1}{5}\varepsilon^2\right). \tag{23.27}$$

(Surface grows; Coulomb falls as charge spreads along the long axis.)

The change in energy relative to the sphere is

$$\Delta E(\varepsilon) = E(\varepsilon) - E(0) = a_s A^{2/3} \cdot \frac{2}{5}\varepsilon^2 - a_c \frac{Z^2}{A^{1/3}} \cdot \frac{1}{5}\varepsilon^2 = \frac{\varepsilon^2}{5}\left[2a_s A^{2/3} - a_c \frac{Z^2}{A^{1/3}}\right]. \qquad (23.28)$$

Here ΔE is the *increase in total liquid-drop energy* (equivalently the decrease in binding energy). This sign convention corrects a common OCR ambiguity in the source, where $B(\varepsilon) - B(0)$ and the deformation energy are interchanged. For the sphere to be *unstable* against infinitesimal deformation ($\Delta E < 0$ for any small $\varepsilon \neq 0$), one needs

$$2a_s A^{2/3} < a_c \frac{Z^2}{A^{1/3}}, \qquad (23.29)$$

or

$$\frac{Z^2}{A} > \frac{2a_s}{a_c}. \qquad (23.30)$$

With the chapter-wide coefficients,

$$\frac{2a_s}{a_c} \approx \frac{35.6}{0.711} \approx 50.1. \qquad (23.31)$$

Thus

$$\frac{Z^2}{A} \gtrsim 50 \qquad (23.32)$$

marks the loss of a barrier against small deformations: the nucleus is unstable at sphericity and fissions spontaneously with no activation energy. Taking $Z \approx 0.4A$ gives $Z^2/A \approx 0.16A$, so

$$A \gtrsim 300 \qquad (23.33)$$

for barrierless spontaneous fission. No such nuclei exist as long-lived species; they would fission on nuclear timescales. Between the energetic threshold $Z^2/A \sim 18$ ($A \sim 100$) and the absolute instability $Z^2/A \sim 50$ ($A \sim 300$), nuclei are metastable: $Q > 0$ but $E_f > 0$ [6, 1, 14].

Remark

The ratio $x = (Z^2/A)/(Z^2/A)_{\text{crit}}$ is the classical *fissility* parameter. For uranium, $x \sim 0.7$–0.8: a finite barrier remains, but it is only a few MeV high.

23.6.1 Fissility and a Bohr–Wheeler barrier estimate

For a spherical liquid drop define

$$E_s^{(0)} = a_s A^{2/3}, \qquad E_C^{(0)} = a_c \frac{Z^2}{A^{1/3}}, \qquad x \equiv \frac{E_C^{(0)}}{2E_s^{(0)}} = \frac{a_c}{2a_s}\frac{Z^2}{A}. \qquad (23.34)$$

Thus $x = 1$ is precisely the quadratic instability in Eq. (23.30). Keeping $Z(Z-1)$ rather than Z^2 gives the slightly more accurate classical expression but is immaterial for uranium. For ^{238}U, $Z^2/A = 35.56$; with $a_s = 17.8\,\text{MeV}$, $a_c = 0.711\,\text{MeV}$, one obtains $x \simeq 0.71$, so a finite barrier is expected.

The small-ε expansion proves local stability but cannot locate the saddle: a barrier height requires a family of shapes extending from a sphere to a necked drop. In the classical Bohr–Wheeler treatment one assumes

(i) an incompressible, uniformly charged drop with a sharp surface;

(ii) surface and direct Coulomb energies only (volume and symmetry terms are unchanged along a fixed-A, Z, constant-volume path);

(iii) axial, reflection-symmetric shapes and no shell, pairing, rotational, or dynamical corrections.

Writing the shape factors as

$$E_{\text{def}}(\alpha) = E_s^{(0)}[B_s(\alpha) - 1] + E_C^{(0)}[B_C(\alpha) - 1], \qquad B_s(0) = B_C(0) = 1, \tag{23.35}$$

the physical liquid-drop barrier is the lowest saddle separating the initial minimum from scission:

$$E_f^{(\text{LD})} = \min_{\mathcal{P}} \max_{\alpha \in \mathcal{P}} E_{\text{def}}(\alpha), \tag{23.36}$$

where $\mathcal{P}$ runs over continuous deformation paths from the ground state to scission. The inner maximum finds the saddle crossed by one path; the outer minimum selects the least-energy path. Numerical minimisation over shape coordinates is the barrier calculation. There is no universal elementary closed form because the result depends on the shape family. A frequently quoted historical interpolation in the actinide region is

$$\boxed{E_f^{(\text{LD})}(\text{MeV}) \simeq 19.0 - 0.36\,\frac{Z^2}{A}} \tag{23.37}$$

and gives $E_f \simeq 19.0 - 0.36(35.56) = 6.2\,\text{MeV}$ for ^{238}U. The constants 19.0 MeV and 0.36 MeV are fitted interpolation coefficients, not additional SEMF constants, so Eq. (23.37) must not be extrapolated to light nuclei or the superheavy region, nor presented as a universal Bohr–Wheeler derivation. Modern barriers are double-humped and include shell, pairing, equilibrium-deformation, and multidimensional-shape corrections. The robust Bohr–Wheeler result is the surface–Coulomb competition and $E_f \to 0$ as $x \to 1$, while the several-MeV value for a particular actinide is empirical/model-dependent [6, 14, 1].

23.7 Photofission and neutron-induced fission

Induced fission uses an external probe to supply the energy that spontaneous fission must borrow from the tunneling of a cold nucleus. Neutrons are especially effective in actinides because capture can form a compound nucleus already above the barrier, while photons couple primarily through the giant dipole resonance. Both channels are laboratory tools for mapping barrier heights and fission fragments [6, 1].

23.7.1 Photofission

A γ ray deposits energy without changing A or Z:

$$\gamma + {}^{A}X \;\to\; \text{fission products.} \tag{23.38}$$

Photon absorption in heavy nuclei is strongest in the giant dipole resonance (GDR), where the proton and neutron fluids oscillate against one another. Absorption forms an excited nucleus with the original A, Z, but fission is only one decay branch:

$$\sigma_{\gamma f}(E_\gamma) = \sigma_{\text{abs}}(E_\gamma)\frac{\Gamma_f}{\Gamma_f + \Gamma_n + \Gamma_\gamma + \cdots}.$$

Near threshold barrier penetration controls Γ_f; above particle threshold neutron emission can compete strongly, and the GDR line shape controls σ_{abs}. Consequently photofission need not rise monotonically or "saturate" above E_f. It probes both the barrier and channel competition [6, 1].

23.7.2 Neutron-induced fission and the compound nucleus

A neutron can induce fission by forming a compound nucleus that is already excited:

$$n + {}^{A}X \;\to\; {}^{A+1}X^* \;\to\; \text{fission products.} \tag{23.39}$$

The excitation energy of the compound system is essentially the capture Q-value plus the kinetic energy of the neutron (centre-of-mass corrections are small for heavy targets). If that excitation exceeds E_f, fission is likely; if not, the system must tunnel, and the fission cross section remains tiny. Competing channels include elastic scattering and radiative capture (n, γ).

Whether thermal neutrons suffice depends on the capture Q-value relative to the barrier of the compound nucleus. That comparison for ^{235}U versus ^{238}U is the central theme of the next chapter [6, 1, 2].

Takeaway

- Fission: heavy $X \to Y + Z$ with $A_Y \sim A_Z$; $Q \sim 200\,\text{MeV}$ for $A \sim 240$ from the rise in B/A.
- History: neutron discovery (1932); Hahn–Strassmann barium (1939); Meitner–Frisch interpretation; Bohr–Wheeler barrier theory.
- SEMF symmetric $Q = -0.26a_s A^{2/3} + 0.37a_c Z^2/A^{1/3}$; $Q > 0$ for $Z^2/A > 17.6$ ($A \gtrsim 100$).
- Barrier from surface vs Coulomb along the least-energy deformation path; $E_f \sim 6\,\text{MeV}$ for $A \sim 240$; no barrier for $Z^2/A \gtrsim 50$ in the stated SEMF convention.
- Induced fission: photofission reflects GDR absorption, barrier penetration, and decay competition; neutron capture forms B^* with excitation $\approx Q_{\text{capture}} + K_n$.

23.7.3 Multi-scale complexity of fission

Microscopic fission theory must cover deformation toward scission, fragment yields, prompt neutrons and photons, and the subsequent years of β feeding. Contemporary EDF and time-dependent approaches address that multi-scale problem; the liquid-drop barrier of this chapter remains the first quantitative filter before those tools are needed [56, 6, 14].

23.8 Deeper physics: fissility Z^2/A and the Bohr–Wheeler barrier

23.8.1 The fissility parameter

Liquid-drop fission balances Coulomb repulsion against surface tension. The dimensionless **fissility parameter**

$$\frac{Z^2}{A} \tag{23.40}$$

measures how strongly Coulomb destabilises a nucleus relative to its size [6, 14]. Empirically, spontaneous fission half-lives shorten rapidly once Z^2/A exceeds a critical value of order 45–47 [6, 1]. Light nuclei have small Z^2/A and are stable against fission; very heavy nuclei with large Z and moderate A sit in the fissile region where α decay, spontaneous fission, and cluster emission compete (Lectures 16–17).

Fissility of actinides and lead

For ^{235}U: $Z^2/A = 92^2/235 \approx 36.0$. For ^{238}U: $92^2/238 \approx 35.6$. For ^{208}Pb: $82^2/208 \approx 32.3$ [6]. Both uranium isotopes are fissionable with induced neutrons; ^{208}Pb lies well below the spontaneous-fission threshold. The small difference in Z^2/A between ^{235}U and ^{238}U combines with pairing effects to explain why thermal-neutron fission of ^{235}U dominates reactor physics (Lecture 24) [1, 35].

23.8.2 Bohr–Wheeler barrier estimate

In the Bohr–Wheeler picture, fission proceeds by deforming the nucleus from a sphere of radius $R = r_0 A^{1/3}$ into two touching spheres. The Coulomb energy rises and the surface energy falls; the barrier height is set by the competition. A schematic result is

$$B_f \approx a_S A^{2/3} \left(1 - \frac{Z^2/A}{a_S/a_C}\right)^2, \tag{23.41}$$

with surface and Coulomb coefficients a_S and a_C from the SEMF [6, 14]. When Z^2/A approaches the critical value, $B_f \to 0$ and fission becomes instantaneous on nuclear time scales. Shell corrections modify this picture: double-humped barriers and asymmetric yields require microscopic potential-energy surfaces (Lecture 24) [6, 56].

Key idea

Same tunneling language as α decay. Spontaneous fission can be viewed as tunneling through a multidimensional barrier, the heavy-nucleus analogue of the Gamow integral (Lecture 17). A 1 MeV uncertainty in barrier height can change the half-life by orders of magnitude, which is why modern fission theory emphasises collective inertia and dissipation alongside B_f [56, 6].

23.8.3 Induced fission and the compound nucleus

Neutron-induced fission proceeds through a compound nucleus $X + a \to C^* \to f_1 + f_2 + n + \cdots$ with $Q > 0$ for fissile targets [6, 1]. The saddle barrier B_f is an extra energy requirement beyond the Q-value: fast neutrons supply enough excitation to overcome B_f in ^{238}U, breeding ^{239}Pu after β^- decay (Lecture 24) [35]. The liquid-drop fissility Z^2/A explains why actinides fission while lead does not, but asymmetric yields require shell corrections beyond (23.41) [14, 56].

23.8.4 Double-humped barriers and shape isomers

For actinides the fission potential-energy surface often shows two barriers with a secondary minimum (shape isomer) at large deformation [6, 14]. Delayed fission and intermediate structure in fission cross sections are experimental signatures. The simple single-barrier sketch of the main lecture is the first approximation; microscopic calculations on multidimensional surfaces reproduce the double-humped structure and the asymmetric valleys that drive mass yields (Lecture 24) [56]. Spontaneous-fission half-life systematics correlate with Z^2/A but show large odd–even staggering from pairing [6, 20].

Shell-corrected barriers explain why some actinides are metastable against fission despite large Z^2/A. Odd nucleons often hinder spontaneous fission relative to even–even neighbours. Those systematics matter for SHE survival probabilities (Lecture 25) and for astrophysical fission cycling (Lecture 24).

23.9 Further physics from the literature

Asymmetric fission and shell corrections. The pure liquid drop predicts a preference for *symmetric* mass splits near the barrier, yet thermal-neutron fission of ^{235}U is strongly asymmetric (peaks near $A \sim 95$ and $A \sim 140$). Strutinsky shell corrections to the liquid-drop energy create secondary minima and drive the mass-asymmetric valleys observed experimentally. The following fission lecture returns to yield curves; the microscopic theory is developed in advanced texts [6, 14, 2].

Spontaneous fission half-lives. Even with $Q > 0$, spontaneous fission half-lives range from milliseconds (very heavy Z) to 10^{15} years and beyond. Systematics correlate roughly with Z^2/A: larger fissility, lower barrier, shorter half-life. Pairing and shell structure produce large odd–even staggering [6, 1, 20].

Double-humped barriers. For actinides the potential energy surface often shows two barriers with a secondary minimum (shape isomer). Delayed fission and intermediate structure in fission cross sections are experimental signatures. The simple single-barrier sketch of Fig. 23.1 is the first approximation [6, 14].

Historical context and data checks. Meitner and Frisch's 1939 note and Bohr and Wheeler's analysis remain the classical introductions. Nuclear masses and separation energies should be taken from NNDC QCalc (AME2020); fission yields and neutron multiplicities from IAEA/NDS or ENDF rather than inferred from the liquid-drop model alone [6, 1, 14, 27, 35, 33, 32].

23.9.1 Microscopic correction to the fissility estimate

The liquid-drop fissility parameter (23.40) captures the average trend of spontaneous-fission half-lives but misses shell structure. Strutinsky corrections add oscillatory terms to the potential-energy surface, creating double-humped barriers and asymmetric fission valleys [6, 56]. For ^{235}U, shell-stabilised fragments near $A \approx 95$ and 140 lower the asymmetric saddle relative to symmetric split, reversing the naive liquid-drop preference (Lecture 24) [14, 1].

WKB sensitivity of spontaneous-fission lifetime

Suppose a one-dimensional fission barrier has height $B_f = 6\,\text{MeV}$ and curvature radius $\hbar\omega \approx 1\,\text{MeV}$. A WKB estimate gives $\log_{10} t_{1/2} \propto B_f/\hbar\omega$. Raising B_f by 1 MeV can increase $t_{1/2}$ by ten to twenty orders of magnitude [6, 56]. This exponential sensitivity mirrors the Gamow factor in α decay (Lecture 17) and explains why microscopic barrier heights must be known to keV precision for quantitative half-life predictions.

23.9.2 From fission barriers to fusion and α competition

Very heavy nuclei sit in a competition zone: α decay (small Q_α lengthens lifetimes), spontaneous fission (large Z^2/A shortens them), and cluster emission (rare) [55, 6]. Superheavy synthesis chains terminate when both α and fission channels have short half-lives. Fusion-evaporation reactions (Lecture 25) populate the same Z, N region from the opposite direction: capture followed by neutron evaporation competes with quasifission [55, 54]. The Bohr–Wheeler picture is the macroscopic backbone; microscopic inertias set the dynamical timescale at scission.

Remark

Induced fission cross sections rise sharply once $E_{\text{CM}} > B_f$, analogous to sub-barrier fusion rising once tunneling through $V_{\max}$ becomes appreciable. Both are barrier-penetration phenomena, but fission involves a multidimensional saddle while fusion uses a one-dimensional entrance channel [56, 14].

23.9.3 Spontaneous fission versus α chains in superheavy nuclei

For $Z > 110$, spontaneous fission half-lives can fall below α-decay half-lives even when Q_α is favourable [55, 6]. SHE synthesis chains therefore terminate when neither channel allows further Z increase. The same Z^2/A diagnostic that classifies ^{235}U as fissionable predicts the onset of millisecond spontaneous fission in transfermium isotopes. Fusion evaporation (Lecture 25) and α emission (Lecture 16) are the two dominant de-excitation paths of the hot compound nucleus that sits above the fission saddle [55, 54].

23.9.4 Fragment kinetic energy and shell stabilisation

Total kinetic energy distributions in fission reflect Coulomb repulsion at scission and shell stabilisation of fragments [56, 6]. Models that agree on barrier height B_f can still disagree on kinetic energy and neutron multiplicity because scission dynamics and dissipation differ [56]. Asymmetric mass yields (Lecture 24) correlate with high kinetic energy in the heavy fragment peak near $A \approx 140$. The fissility parameter Z^2/A is therefore only the first diagnostic: fragment observables test dynamics beyond the static liquid-drop barrier [14, 1].

23.10 Deeper reading: Z^2/A, barriers, and microscopic dynamics

Bohr–Wheeler fissility compares Coulomb to surface energy. The dimensionless combination Z^2/A organises which nuclei have barriers against fission. Modern theory replaces a one-dimensional barrier with multidimensional potential-energy surfaces and collective inertias; half-lives remain exponentially sensitive to barrier height (Lecture 17).

Spontaneous fission of superheavy elements and fission recycling in the r-process both inherit this sensitivity. Evaluate Z^2/A for ^{235}U, ^{238}U, and ^{208}Pb as a standard exercise before reading yield distributions in Lecture 24.

Barrier systematics

Barrier heights decrease as Z^2/A increases, with shell corrections producing local islands of stability against fission. Odd-particle hindrance can add orders of magnitude to spontaneous-fission half-lives. Those systematics feed SHE survival estimates (Lecture 25).

Dissipation and scission

Post-barrier dynamics (friction, neck rupture) shape total kinetic energies and neutron multiplicities even when the static barrier is fixed. Models agreeing on B_f may still disagree on yields (Lecture 24). Ask which observables a microscopic fission paper was tuned to reproduce.

Fissility map

Compute Z^2/A for ^{232}Th, ^{235}U, ^{238}U, ^{246}Cm, and ^{208}Pb. Order them and discuss which have low fission barriers. Then sketch a one-dimensional barrier of height $B_f = 6\,\mathrm{MeV}$ versus $5\,\mathrm{MeV}$ and argue why thermal-neutron fission of ^{235}U is special once pairing and the barrier of the compound nucleus are considered (preview of Lecture 24).

Read a modern fission-theory abstract and list whether it constrains barriers, yields, kinetic energies, or half-lives. Different observables stress different parts of the multidimensional dynamics. Lecture 17's WKB lesson explains the half-life sensitivity; this chapter explains why the barrier itself depends on Z^2/A and shells.

Fissility organises barriers; microscopic dynamics organise yields and kinetic energies. Exponential lifetime sensitivity ties this chapter to Lecture 17.

In practice, rework one numerical estimate from the lecture with a different parameter set and compare the shift to the quoted experimental uncertainty. If the shift exceeds the uncertainty, the estimate is teaching a scale, not delivering a fit. That distinction keeps theoretical cartoons honest when they sit beside evaluated data tables and modern many-body calculations. Along the course arc, write one sentence linking this chapter backward and one forward: the discipline of those two sentences is as important as any single formula. Examination problems rarely ask for a literature review, but they do ask for the scale argument that literature reviews presuppose.

Synthesis problem

Write a half-page note that (i) restates the chapter's central scale or selection rule in one equation, (ii) evaluates it numerically for one concrete nucleus or energy, and (iii) names the neighbouring lecture that uses the result. Keep the note free of research-frontier speculation; the goal is fluency with the lecture physics itself. If an assumption fails (Born approximation, spherical mean field, allowed transition, one-dimensional barrier), state the failure mode in a final sentence. That habit converts passive reading into examination-ready understanding and makes the later research-frontier box read as an extension rather than a replacement.

Research frontier: microscopic fission dynamics and student projects

Bohr–Wheeler fissility (Lecture 23) explains the qualitative onset of fission through the competition of Coulomb and surface energies. Modern theory computes multidimensional potential-energy surfaces, collective inertias, and dissipative dynamics to predict barriers, half-lives, and fragment yields [56]. The same microscopic tools inform spontaneous fission of superheavy nuclei and fission recycling in the r-process [55, 54]. Remaining uncertainties in barrier heights of even 1 MeV can change half-lives by orders of magnitude; fragment-yield distributions for exotic actinides are still only partially constrained by experiment. Dissipation and scission dynamics remain active theory topics because different models can agree on barriers yet disagree on kinetic-energy and neutron-multiplicity observables. The lecture's fissility parameter Z^2/A remains the correct first diagnostic before those microscopic complications are introduced.

Yield and total kinetic energy distributions encode dynamics beyond the static barrier: shell-stabilised fragments, energy dissipation, and the timing of neck rupture. Microscopic models that agree on B_f can still disagree on those observables, which is why fission theory is not finished once a fissility formula is written down.

Models that match barriers but miss total kinetic energies are incomplete; dynamics beyond fissility remain part of the research frontier.

Student reviews should therefore ask which observables each microscopic model was tuned to reproduce.

Student directions. (1) Fissility exercise: evaluate Z^2/A for ^{235}U, ^{238}U, and ^{208}Pb and discuss the trend. (2) WKB insight: sketch a one-dimensional barrier and show why lifetimes are exponentially sensitive to barrier height. (3) Literature reading: from a fission-theory review, summarise two competing asymmetric yield modes [56].

23.11 Problems with solutions

Problem 23.1 — Q from B/A for $A = 238$. Estimate the fission Q-value for a nucleus with $A = 238$ if $(B/A)_{\text{parent}} = 7.6\,\text{MeV}$ and $(B/A)_{\text{fragments}} = 8.5\,\text{MeV}$. Express the result in MeV and comment on its magnitude relative to chemical combustion energies [6, 1].

Solution 23.1.

Step 1. From Eq. (23.7),

$$Q = E_B(Y) + E_B(Z) - E_B(X) \approx \left[(B/A)_{\text{frag}} - (B/A)_{\text{parent}}\right] A.$$

Step 2. Substitute the given values:

$$Q \approx (8.5 - 7.6)\,\text{MeV} \times 238 = 0.9 \times 238 = 214\,\text{MeV}.$$

Step 3. Chemical energies are a few eV per reaction. Fission releases $\sim 10^8$ times more energy per event. That is why a few kilograms of fissile material can power a reactor or a weapon [6].

Problem 23.2 — Symmetric SEMF Q algebra. Starting from $M(A, Z)$ in Eq. (23.10) (pairing omitted), derive $Q = -0.26\,a_s A^{2/3} + 0.37\,a_c Z^2/A^{1/3}$ for symmetric fission into two equal fragments. Show that volume and symmetry terms cancel [14, 1].

Solution 23.2.

Step 1. $Q = M(A, Z) - 2M(A/2, Z/2)$.

Step 2 (volume). $-a_v A - 2(-a_v A/2) = 0$.

Step 3 (symmetry). Parent: $a_{\text{sym}}(A - 2Z)^2/A$. One fragment: $a_{\text{sym}}(A/2 - Z)^2/(A/2) = a_{\text{sym}}(A - 2Z)^2/(2A)$. Two fragments: $a_{\text{sym}}(A - 2Z)^2/A$. Difference vanishes.

Step 4 (surface).

$$a_s A^{2/3} - 2a_s(A/2)^{2/3} = a_s A^{2/3}(1 - 2^{1/3}) = -0.260\,a_s A^{2/3}.$$

Step 5 (Coulomb).

$$a_c \frac{Z^2}{A^{1/3}} - 2a_c \frac{(Z/2)^2}{(A/2)^{1/3}} = a_c \frac{Z^2}{A^{1/3}}(1 - 2^{-2/3}) = 0.370\,a_c \frac{Z^2}{A^{1/3}}.$$

Step 6. Add surface and Coulomb contributions to obtain Eq. (23.15).

Problem 23.3 — Critical Z^2/A. Using $a_s = 17.8\,\text{MeV}$ and $a_c = 0.711\,\text{MeV}$, evaluate the threshold Z^2/A for $Q > 0$ in symmetric fission. For ^{238}U ($Z = 92$, $A = 238$), is $Q > 0$?

Solution 23.3.

Step 1. From Eq. (23.17),

$$\frac{Z^2}{A} > \frac{0.26\,a_s}{0.37\,a_c} = \frac{0.26 \times 17.8}{0.37 \times 0.711} = \frac{4.628}{0.2631} \approx 17.6.$$

Step 2. For ^{238}U,

$$\frac{Z^2}{A} = \frac{92^2}{238} = \frac{8464}{238} \approx 35.6 > 17.6.$$

Yes, $Q > 0$. Uranium is energetically free to fission but is held by the barrier E_f [6, 1].

Problem 23.4 — Barrierless fissility. From the small-deformation result $\Delta E \propto [2a_s A^{2/3} - a_c Z^2/A^{1/3}]$, show that the sphere is unstable when $Z^2/A > 2a_s/a_c$. With $a_s = 17.8\,\text{MeV}$ and $a_c = 0.711\,\text{MeV}$, estimate the critical A if $Z = 0.4A$.

Solution 23.4.

Step 1. Instability requires $\Delta E < 0$:

$$2a_s A^{2/3} < a_c Z^2/A^{1/3} \quad \Rightarrow \quad \frac{Z^2}{A} > \frac{2a_s}{a_c}.$$

Step 2.

$$\frac{2a_s}{a_c} = \frac{35.6}{0.711} \approx 50.1.$$

Step 3. With $Z = 0.4A$, $Z^2/A = 0.16A$. Set $0.16A = 50.1 \Rightarrow A \approx 313$. Order-of-magnitude result: nuclei with $A \gtrsim 300$ have no classical liquid-drop barrier in this convention and cannot exist as long-lived species [14, 6].

Problem 23.5 — Photofission competition. At one GDR energy a heavy nucleus has $\sigma_{\text{abs}} = 300\,\text{mb}$ and partial widths $\Gamma_f = 1.5\,\text{MeV}$, $\Gamma_n = 2.0\,\text{MeV}$, and $\Gamma_\gamma = 0.50\,\text{MeV}$. Estimate the photofission cross section. If Γ_n doubles at higher energy while the other quantities remain fixed, what happens? [6, 1]

Solution 23.5.

$$b_f = \frac{\Gamma_f}{\Gamma_f + \Gamma_n + \Gamma_\gamma} = \frac{1.5}{4.0} = 0.375, \qquad \sigma_{\gamma f} = \sigma_{\text{abs}} b_f = 112.5\,\text{mb}.$$

After Γ_n doubles,

$$b'_f = \frac{1.5}{1.5 + 4.0 + 0.5} = 0.250, \qquad \sigma'_{\gamma f} = 75.0\,\text{mb}.$$

Crossing the barrier does not force fission: neutron evaporation can reduce the fission branch even within a strong absorption region.

Problem 23.6 — Surface vs Coulomb for ε. Using Eqs. (23.26)–(23.28), compute ΔE (in MeV) for ^{238}U at $\varepsilon = 0.1$. Take $a_s = 17.8\,\text{MeV}$, $a_c = 0.711\,\text{MeV}$, $Z = 92$, $A = 238$. Is the sphere stable against this deformation?

Solution 23.6.

Step 1.

$$A^{2/3} = 238^{2/3} \approx 38.3, \qquad A^{1/3} \approx 6.20.$$

Step 2.

$$\begin{aligned}\Delta E &= \frac{(0.1)^2}{5}\left[2(17.8)(38.3) - 0.711\frac{8464}{6.20}\right] \\ &= 0.002(1363 - 971) = 0.002(392) \approx 0.78\,\text{MeV}.\end{aligned}$$

Step 3. $\Delta E > 0$: the sphere is stable against this small deformation; a finite barrier remains. (The full barrier height requires a global calculation beyond the quadratic expansion [6, 14].)

Problem 23.7 — Fissility and barrier scales. For ^{238}U, take $Z = 92$, $A = 238$, $a_s = 17.8\,\text{MeV}$, and $a_c = 0.711\,\text{MeV}$. (a) Compute the fissility $x = E_C^{(0)}/(2E_s^{(0)})$. (b) Use the actinide interpolation $E_f(\text{MeV}) \simeq 19.0 - 0.36Z^2/A$. (c) Explain why the positive symmetric-fission Q does not contradict the positive barrier.

Solution 23.7.

Step 1: charge-to-mass ratio.

$$\frac{Z^2}{A} = \frac{92^2}{238} = 35.563.$$

Step 2: dimensionless fissility.

$$x = \frac{a_c}{2a_s}\frac{Z^2}{A} = \frac{0.711}{2(17.8)}(35.563) = 0.710.$$

Because $x < 1$, the spherical drop has positive curvature against a small quadrupole deformation; it is not classically barrierless.

Step 3: barrier interpolation.

$$E_f \simeq 19.0 - 0.36(35.563) = 6.20\,\text{MeV}.$$

The numerical agreement with a uranium barrier should not be overinterpreted: the coefficients are an actinide fit and a real actinide has two saddles.

Step 4: energy landscape. The Q-value compares the initial ground state with the final separated fragments, whereas E_f compares the initial state with the highest intermediate deformation. Therefore $Q > 0$ and $E_f > 0$ can hold simultaneously: the final state is lower, but the continuous path first rises.

Problem 23.8 — Fissility parameter. Define $x = Z^2/A$ as a fissility measure. Compare x for ^{238}U and ^{56}Fe and interpret.

Solution 23.8.

$$x(^{238}\mathrm{U}) = \frac{92^2}{238} \approx 35.6, \qquad x(^{56}\mathrm{Fe}) = \frac{26^2}{56} \approx 12.1.$$

Coulomb-driven fissility is far higher for U; Fe sits near the B/A peak and is stable against spontaneous fission [6, 14].

Chapter 24

Nuclear Fission II

"Thermal neutrons fission ^{235}U *because the compound nucleus is born above the barrier;* ^{238}U *needs fast neutrons because it is born below."*

N. Bohr and J. A. Wheeler (1939)

The preceding lecture, *Nuclear Fission I*, established that heavy nuclei have $Q > 0$ for fission yet are protected by a barrier of several MeV. This chapter applies that picture to neutron-induced fission of uranium: why thermal neutrons fission ^{235}U but not ^{238}U, what the mass-yield curve looks like, how asymmetry reduces Q relative to the symmetric maximum, why prompt and delayed neutrons appear, how the fission energy is partitioned, and how a chain reaction can be sustained [6, 1, 14, 2].

24.1 Neutron capture on ^{235}U: fissile behaviour

Natural uranium is mostly ^{238}U ($\sim 99.3\%$) with ^{235}U ($\sim 0.7\%$). Thermal-neutron fission of the latter is the workhorse of many reactors. The capture reaction is

$$^{235}\mathrm{U} + n \;\rightarrow\; ^{236}\mathrm{U}^* \;\rightarrow\; \text{fission fragments} + \text{neutrons}. \tag{24.1}$$

The capture Q-value (excitation energy deposited in ^{236}U when the neutron has negligible kinetic energy) is

$$Q = [m_n + M(^{235}\mathrm{U}) - M(^{236}\mathrm{U})]c^2 \approx 6.545\,\mathrm{MeV}. \tag{24.2}$$

Thus thermal capture forms $^{236}\mathrm{U}^*$ at about 6.55 MeV above its ground state.

An illustrative one-barrier estimate for ^{236}U is

$$E_f(^{236}\mathrm{U}) \sim 6\,\mathrm{MeV}. \tag{24.3}$$

Using $E_f \simeq 6.2\,\mathrm{MeV}$ for orientation,

$$Q \approx 6.55\,\mathrm{MeV} \;>\; E_f \approx 6.2\,\mathrm{MeV}, \tag{24.4}$$

the compound nucleus is formed above the representative saddle scale. Real actinide barriers are double-humped and depend on spin, parity, and excitation; barrier transmission and compound-nuclear resonances therefore remain part of the measured cross section even when this simple energy comparison is favourable. The thermal-neutron fission cross section is nevertheless large (hundreds of barns). Materials that fission with thermal neutrons are called **fissile**; examples are ^{235}U, ^{239}Pu, and ^{233}U [6, 1, 2].

Key idea

Thermal $n + {}^{235}\mathrm{U} \to {}^{236}\mathrm{U}^*$: $S_n \sim 6.55\,\mathrm{MeV}$ exceeds a representative $E_f \sim 6.2\,\mathrm{MeV}$. This explains why ^{235}U is fissile, but a double-humped transmission calculation replaces the one-barrier picture in quantitative work.

24.1.1 Competing channels

Not every neutron that meets ^{235}U causes fission. Important competitors include:

(i) **Elastic scattering:** $n + {}^{235}\mathrm{U} \to n' + {}^{235}\mathrm{U}$.

(ii) **Radiative capture:** $n + {}^{235}\mathrm{U} \to {}^{236}\mathrm{U}^* \to {}^{236}\mathrm{U} + \gamma$. If enough γ energy is radiated before fission, the excitation drops below E_f and fission becomes improbable. Ground-state ^{236}U then lives for α decay with a multi-million-year half-life.

The branching between fission and (n, γ) is encoded in the fission cross section σ_f and the capture cross section σ_γ [6, 1].

24.2 Neutron capture on ^{238}U: fertile behaviour

For ^{238}U,

$$^{238}\mathrm{U} + n \;\to\; {}^{239}\mathrm{U}^*, \tag{24.5}$$

with capture Q-value

$$Q = [m_n + M(^{238}\mathrm{U}) - M(^{239}\mathrm{U})]c^2 \approx 4.806\,\mathrm{MeV}. \tag{24.6}$$

The effective fission barrier of ^{239}U is model-dependent and double-humped, but is of order 6 MeV. Using the 6.2 MeV one-barrier value adopted in the original lecture gives

$$Q \approx 4.806\,\mathrm{MeV} \;<\; E_f \approx 6.2\,\mathrm{MeV}: \tag{24.7}$$

thermal capture leaves the compound nucleus *below* the barrier. Fission requires tunneling through a residual barrier of order 1.4 MeV, so the thermal fission cross section is negligible.

If the neutron brings kinetic energy K_n, the excitation of $^{239}\mathrm{U}^*$ is approximately $Q + K_n$ (neglecting the small centre-of-mass correction). Fission becomes probable when

$$Q + K_n \gtrsim E_f \quad \Rightarrow \quad K_n \gtrsim E_f - Q \sim 1.4\,\mathrm{MeV}. \tag{24.8}$$

In practice a threshold near 1 MeV is observed: fast neutrons can fission ^{238}U, thermal neutrons cannot. ^{238}U is therefore called **fissionable** (by fast neutrons) but not fissile; it is also **fertile** because successive captures and β decays can breed ^{239}Pu [6, 1, 14].

Remark

The capture Q-values are neutron separation energies of the compound nuclei:

$$Q[n + {}^{235}\mathrm{U}] = S_n({}^{236}\mathrm{U}), \qquad Q[n + {}^{238}\mathrm{U}] = S_n({}^{239}\mathrm{U}).$$

The quoted 6.545 and 4.806 MeV values follow AME2020 masses (NNDC QCalc) [27, 21]. Barrier values are less sharply defined than masses because an actinide has inner and outer saddles; therefore $E_f - Q$ is an explanatory scale, not a precision prediction of the measured threshold.

Caution

The difference between ^{235}U and ^{238}U is not a gross difference in barrier height; both compound barriers are ~ 6 MeV. The decisive quantity is the neutron separation energy (capture Q) of the compound system: pairing makes even–even ^{236}U more tightly bound relative to ${}^{235}\mathrm{U} + n$ than odd ^{239}U is relative to ${}^{238}\mathrm{U} + n$.

Remark

Odd-A actinide targets (^{235}U, ^{239}Pu, ^{233}U) typically have larger capture Q values into even–even compound nuclei and are fissile. Even–even targets more often require fast neutrons [6, 1].

24.3 Fission fragment yields: asymmetric mass distribution

Once ${}^{236}\mathrm{U}^*$ is above the barrier, it does not split into a unique pair of fragments. Many partitions are open. Representative channels include

$$ {}^{236}\mathrm{U} \to {}^{140}\mathrm{Xe} + {}^{94}\mathrm{Sr} + 2n, \tag{24.9}$$
$$ {}^{236}\mathrm{U} \to {}^{139}\mathrm{Ba} + {}^{94}\mathrm{Kr} + 3n, \tag{24.10}$$
$$ {}^{236}\mathrm{U} \to {}^{141}\mathrm{Cs} + {}^{93}\mathrm{Rb} + 2n, \tag{24.11}$$
$$ {}^{236}\mathrm{U} \to {}^{137}\mathrm{I} + {}^{97}\mathrm{Y} + 2n. \tag{24.12}$$

Plotting the percentage yield against fragment mass number A produces the famous **double-humped mass-yield curve**: peaks near $A \sim 95$ and $A \sim 140$, with a deep valley at symmetric split $A \sim 118$. Symmetric fission is suppressed by roughly two orders of magnitude relative to the asymmetric peaks for thermal fission of ^{235}U, with the precise peak and valley values depending on whether independent, cumulative, or chain yields are plotted [6, 1, 2].

The pure liquid drop, with only surface and Coulomb energies, tends to favour symmetric scission near the barrier. The observed asymmetry is a **shell-structure** effect: fragments near closed shells (notably near $A \sim 132$, related to the doubly magic ^{132}Sn region) are preferentially formed. Liquid-drop theory remains excellent for the overall energetics and the existence of a barrier; it is incomplete for fine details of the yield curve [6, 14].

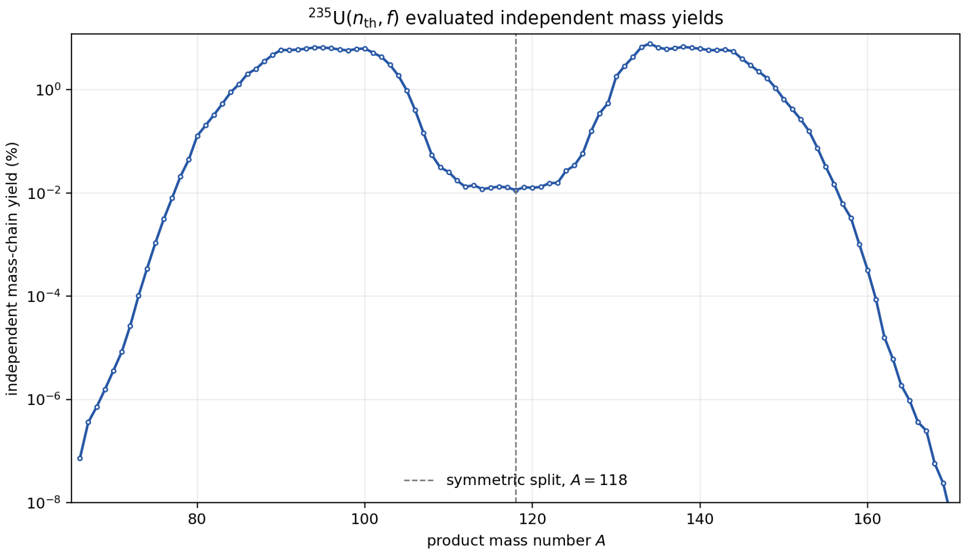


Figure 24.1: Independent mass-chain yields for thermal-neutron fission of ^{235}U, obtained by summing the ENDF/B-VIII.0 independent nuclide yields at fixed mass number A. The logarithmic ordinate faithfully displays both asymmetric peaks and the strongly suppressed symmetric region; the summed chain yield is two products per fission [35].

24.4 Energy release: B/A curve and asymmetry

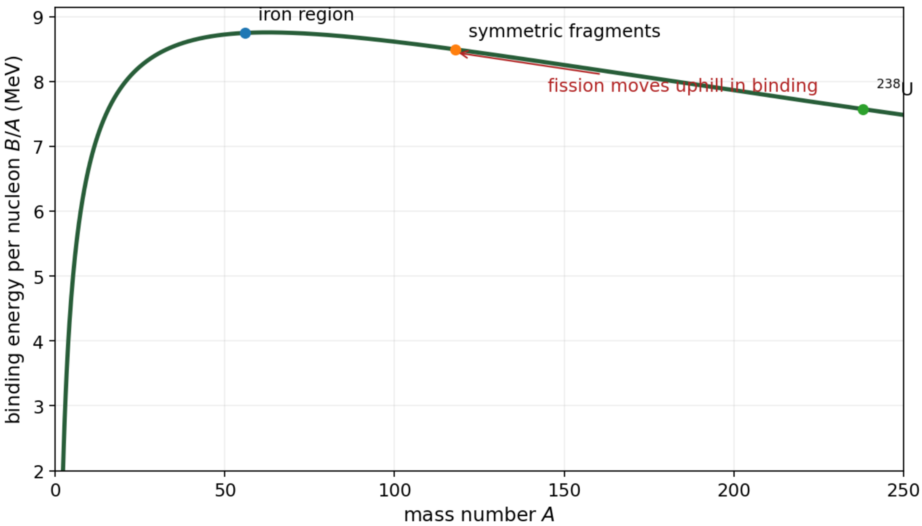


Figure 24.2: Binding energy per nucleon versus mass number. Heavy nuclei sit on the right-hand declining slope; fission products move toward the iron-peak region of higher B/A. The plotted curve is pedagogical rather than a mass evaluation.

Let the parent have mass number A and total binding B_{p}. Write the fragments as

$$A_1 = \frac{A}{2} - a, \qquad A_2 = \frac{A}{2} + a, \tag{24.13}$$

where a measures mass asymmetry and neutron emission is temporarily suppressed. Let

$F(x) = x\,b(x)$ be the total binding of a representative fragment of mass x, with $b = B/A$. Then

$$Q(a) = F(A/2 - a) + F(A/2 + a) - B_{\rm p}. \tag{24.14}$$

Interchanging the two fragments sends $a \to -a$ but cannot change the physical partition, so $Q(a) = Q(-a)$. Expanding both terms about $A/2$ makes the required cancellation explicit:

$$F(A/2 \pm a) = F(A/2) \pm F'(A/2)a + \frac{1}{2}F''(A/2)a^2 \pm \frac{1}{6}F'''(A/2)a^3 + \cdots, \tag{24.15}$$

$$Q(a) = Q(0) + F''(A/2)a^2 + \mathcal{O}(a^4). \tag{24.16}$$

There can be no fixed-δ term linear in a: such a term would assign different Q-values to the same unordered fragment pair merely because its labels were exchanged.

Symmetric estimate. Approximating the parent and symmetric fragments by $b_{\rm p} = E_0$ and $b(A/2) = E_1$ gives

$$Q(0) = (E_1 - E_0)A. \tag{24.17}$$

With $E_0 \approx 7.6\,\text{MeV}$, $E_1 \approx 8.5\,\text{MeV}$, and $A \approx 240$,

$$Q_{\rm sym} \approx 0.9 \times 240 = 216\,\text{MeV}. \tag{24.18}$$

Asymmetric case ($a \neq 0$). The sign of $F''(A/2)$, not an arbitrarily fixed slope offset, determines the local liquid-drop curvature. Moreover, pairing, prompt-neutron multiplicity, and fragment shell corrections all vary with partition, so the Taylor model does not predict the full measured $Q(A)$. Nature prefers asymmetric low-energy fission because shell-corrected potential-energy valleys and level densities favour those paths, while Q remains near 200 MeV [6, 1, 14].

Key idea

$Q(a) = Q(-a)$ because fragment labels are interchangeable. Hence the local expansion begins with a^2: $Q(a) = Q(0) + F''(A/2)a^2 + \mathcal{O}(a^4)$. Shell effects, rather than a symmetry-invalid linear term, drive the observed asymmetric yield.

24.5 Prompt neutrons, delayed neutrons, and β chains

Fission does not end at scission. The primary fragments are neutron-rich, hot, and still rearranging their nucleons. Prompt neutrons, β decay along mass chains, delayed neutrons, and γ cascades all follow. Reactor control hinges on the small delayed-neutron fraction: without it, a supercritical assembly would respond on the prompt-neutron lifetime alone and would be effectively uncontrollable [6, 1].

24.5.1 Why neutrons are emitted

Heavy nuclei are neutron-rich: for ^{236}U,

$$\frac{N}{Z} = \frac{144}{92} \approx 1.57. \tag{24.19}$$

Stable medium-mass nuclei have smaller N/Z (closer to 1.3–1.4). Immediately after scission the fragments are still neutron-rich and highly excited; they evaporate **prompt neutrons** on a timescale $\lesssim 10^{-14}$ s. For ^{235}U, the ENDF/B-VIII.0 evaluation at $E_n = 0.0253\,\text{eV}$ tabulates total, delayed, and prompt average multiplicities in MF=1, MT=452, 455, and 456, respectively. At thermal energy $\bar{\nu}_{\text{total}} \simeq 2.4355$; subtracting the evaluated delayed component gives approximately

$$\langle \nu_{\text{prompt}} \rangle \approx 2.42. \tag{24.20}$$

The quoted digits identify the evaluation and should not be read as a newly derived experimental uncertainty [35].

24.5.2 Example N/Z after a typical split

For $^{140}\text{Xe} + {}^{94}\text{Sr} + 2n$:

$$\left.\frac{N}{Z}\right|_{\text{Xe}} = \frac{86}{54} \approx 1.59, \qquad \left.\frac{N}{Z}\right|_{\text{Sr}} = \frac{56}{38} \approx 1.47. \tag{24.21}$$

Both fragments remain above the β-stability line after prompt neutron emission.

24.5.3 Beta-decay chains

Neutron-rich fragments decay by successive β^- transitions,

$$n \to p + e^- + \bar{\nu}_e, \tag{24.22}$$

until they reach a stable N/Z. Examples:

$$^{140}\text{Xe} \xrightarrow{\beta^-} {}^{140}\text{Cs} \xrightarrow{\beta^-} {}^{140}\text{Ba} \xrightarrow{\beta^-} {}^{140}\text{La} \xrightarrow{\beta^-} {}^{140}\text{Ce (stable)}, \tag{24.23}$$

$$^{94}\text{Sr} \xrightarrow{\beta^-} {}^{94}\text{Y} \xrightarrow{\beta^-} {}^{94}\text{Zr (stable)}. \tag{24.24}$$

24.5.4 Delayed neutrons

Some β daughters are still neutron-unstable and emit a neutron after a β delay of seconds to minutes. Classic example:

$$^{137}\text{I} \xrightarrow{\beta^-} {}^{137}\text{Xe}^* \to {}^{136}\text{Xe} + n \quad \text{(delayed neutron)}. \tag{24.25}$$

The average number of delayed neutrons per fission is only $\langle \nu_d \rangle \sim 0.016$, giving the *physical yield fraction* $\beta = \bar{\nu}_d/\bar{\nu}_{\text{total}} \simeq 0.0065$. Reactor kinetics instead uses the *effective delayed-neutron fraction* β_{eff}, which weights delayed and prompt neutrons by their different spectra, locations, and importance for sustaining the chain. Thus β_{eff} need not equal β. Delayed-neutron precursor half-lives introduce seconds-to-minutes timescales instead of prompt-generation times of order 10^{-4} s or less [6, 1, 2]. In reactor control this delay is essential: control-rod adjustment and feedback operate on human and mechanical timescales that would be useless if the neutron population could grow only on the prompt lifetime. Operating below prompt criticality ($\rho < \beta_{\text{eff}}$) keeps the delayed population in the kinetic equations and makes a power excursion slow enough to regulate [6, 1].

24.5.5 Prompt and delayed gamma rays

Fragments are born excited and emit **prompt γ rays** on the same fast timescale as prompt neutrons. Subsequent β daughters also radiate **delayed γ rays** as they de-excite.

24.6 Energy partition in fission

A typical total energy release of order 200 MeV per fission of ^{236}U is shared approximately as follows [6, 1]:

Channel	Energy (approx.)
Kinetic energy of fission fragments	167 MeV
Kinetic energy of prompt neutrons	5 MeV
Prompt γ rays	6 MeV
β particles from products	8 MeV
Delayed γ rays	7 MeV
Antineutrinos (β decays)	12 MeV

The bulk of the recoverable heat is the kinetic energy of the two fragments (Coulomb repulsion after scission), thermalised in the fuel. Antineutrinos escape and do not heat the reactor. The entries sum to about 205 MeV; delayed-neutron kinetic energy is negligible on this scale. Values are the rounded ^{236}U partition tabulated by evaluated-library totals depend on the precise definition (for example, whether later neutron-capture γ heating is included).

Remark

The fragment kinetic energies are nearly equal in the centre-of-mass frame for equal-mass splits; for asymmetric splits the light fragment carries more kinetic energy (same momentum, smaller mass).

24.7 Cross sections, barrier penetration, and chain reactions

24.7.1 Energy dependence of the fission cross section

Below the barrier, fission proceeds by tunneling; the cross section is exponentially small. As the compound excitation approaches and then exceeds E_f, the effective barrier thins and σ_f rises steeply, then saturates when the process is classically allowed. Resonances and intermediate structure appear when discrete compound states couple to the fission channels. For fissile nuclei,

σ_f is huge at thermal energies ($\propto 1/v$ capture into a dense compound spectrum already above E_f); for ^{238}U, σ_f is small until $K_n \sim 1\,\text{MeV}$ [6, 1].

24.7.2 Generations and multiplication

Each fission emits $\langle \nu \rangle \sim 2.4$ neutrons. If, on average, one or more of those neutrons induces another fission, a **chain reaction** results. A generation begins with the neutrons born from one set of fissions and ends when they are absorbed or leak. Define

$$k \equiv \frac{N_{g+1}}{N_g} = \bar{\nu}\, p_{\text{next}}, \qquad N_g = N_0 k^g, \tag{24.26}$$

where p_{next} is the net probability that a source neutron survives leakage and non-fission absorption and produces a next-generation fission:

- $k < 1$: subcritical (dies out);
- $k = 1$: critical (steady power);
- $k > 1$: supercritical (power rises).

In a thermal reactor, fast fission neutrons are slowed by a moderator (water, graphite, ...) to exploit the large thermal σ_f of ^{235}U; heat is removed continuously, engineered feedbacks and absorbers hold $k_{\text{eff}} \simeq 1$, and delayed neutrons permit regulation. Writing reactivity as $\rho = (k_{\text{eff}} - 1)/k_{\text{eff}}$, exact prompt criticality occurs at $\rho = \beta_{\text{eff}}$, equivalently

$$k_{\text{eff}} = \frac{1}{1 - \beta_{\text{eff}}}, \qquad k_{\text{prompt}} \equiv k_{\text{eff}}(1 - \beta_{\text{eff}}) = 1. \tag{24.27}$$

The often quoted $k_{\text{eff}} \simeq 1 + \beta_{\text{eff}}$ is only the first-order approximation for $\beta_{\text{eff}} \ll 1$. An explosive assembly instead undergoes a brief prompt-supercritical excursion with no steady heat-removal or control objective. This distinction is about kinetics and engineering purpose, not a different nuclear reaction; no design details are needed here. Detailed reactor physics (four-factor formula, moderation, poisoning) lies beyond this chapter; the nuclear ingredients are the large thermal σ_f of fissile isotopes, $\langle \nu \rangle > 1$, and the existence of delayed neutrons [6, 1, 14].

Takeaway

- $n + {}^{235}$U: $Q \sim 6.6\,\text{MeV} > E_f \sim 6.2\,\text{MeV}$ (fissile). $n + {}^{238}$U: $Q \sim 4.8\,\text{MeV} < E_f$ (needs $K_n \gtrsim 1\,\text{MeV}$).
- Mass yields: asymmetric peaks near $A \sim 95, 140$; liquid drop alone insufficient.
- Fragment exchange requires $Q(a) = Q(-a)$, so its local asymmetry expansion begins at a^2; the total release is $\sim 200\,\text{MeV}$, mostly fragment KE.
- $\langle \nu \rangle_{\text{total}} = 2.4355$ for thermal ^{235}U fission; delayed neutrons are only $\sim 0.65\%$; β is a yield fraction whereas β_{eff} is importance weighted.
- $N_g = N_0 k^g$: subcritical, critical, and supercritical correspond to $k < 1$, $k = 1$, and $k > 1$; delayed neutrons enable reactor control.

24.7.3 Prompt evaporation and delayed control

Primary fragments are born neutron-rich relative to stability and evaporate prompt neutrons within $\lesssim 10^{-14}$ s. A small delayed component follows β decay to neutron-unstable daughters and supplies the seconds-scale hold that makes reactor control possible—without it, kinetics would be governed only by the prompt lifetime [6, 1, 56].

24.8 Deeper physics: asymmetric fission yields and chain-reaction scale

24.8.1 Asymmetric mass yields

Thermal-neutron fission of ^{235}U produces two heavy fragments with a characteristic **asymmetric** mass distribution: yield peaks near $A \approx 95$ and $A \approx 140$, not at symmetric split $A \approx 117$ [6, 1]. The liquid drop favours symmetric fission near the saddle, but shell corrections stabilise fragments near closed proton and neutron shells ($Z \approx 38$, 50 and $N \approx 82$). The yield curve is therefore a probe of microscopic shell structure, not only of the average Q-value release [14, 56].

Energy release in $^{235}\mathrm{U}(n, f)$

Each fission releases about 200 MeV [1, 35]: roughly 170 MeV in fragment kinetic energy, ~ 30 MeV in prompt γ and neutron emission, with delayed energy from β decay of the fragments over minutes to years. The average prompt neutron multiplicity is $\bar{\nu} \approx 2.4$ for thermal neutrons [6, 35]. Those neutrons, moderated in a reactor, can induce further fissions in fissile material.

24.8.2 Chain reaction and k_{eff}

In a multiplying assembly, each fission generation produces on average k_{eff} fissions in the next generation. Steady operation requires $k_{\mathrm{eff}} \approx 1$. If each fission produces $\bar{\nu}$ neutrons and a fraction p survives to cause another fission,

$$k_{\mathrm{eff}} \approx \bar{\nu}\, p. \tag{24.28}$$

Delayed neutrons ($\sim 0.65\%$ of $\bar{\nu}$ for ^{235}U) arrive on second time scales and allow control by rods; prompt neutrons alone would make reactors uncontrollable [6, 1]. The scale separation between prompt ($\sim 10^{-14}$ s) and delayed (~ 1–100 s) neutron groups is what makes a thermal reactor an engineering possibility rather than an instantaneous energy burst [14].

Key idea

From fission yields to fusion fuel cycles. Fission releases ~ 200 MeV per heavy nucleus; fusion of light nuclei (Lectures 25–26) releases energy by moving up the same binding curve from the opposite direction. Both processes rely on the Q-value and barrier physics developed in the decay and reaction chapters [6, 2].

24.8.3 Pairing and the fissile isotope pattern

The large thermal-neutron fission cross section of ^{235}U and the small capture cross section leading to fission in ^{238}U reflect pairing as well as barrier height [6, 1]. Adding a neutron to odd-A ^{235}U forms even–even ^{236}U with extra pairing binding, increasing S_n and the capture Q. The same pattern makes ^{239}Pu and ^{233}U fissile. Asymmetric yields then reflect shell-stabilised fragments, not the average Q alone [35, 14]. Delayed neutrons from β-decaying fission products provide the control time scale that separates reactor operation from prompt criticality [6, 35].

24.8.4 Ternary fission and scission dynamics

Ternary fission emits a third light charged particle at scission at the 10^{-3} level and probes neck dynamics between the main fragments [6, 2]. The total kinetic energy partition and neutron multiplicity depend on how much energy is dissipated before scission, a topic of active microscopic theory [56]. Evaluated fission-product yields from ENDF/B libraries are used in reactor physics, safeguards, and r-process network calculations that terminate heavy-nucleus synthesis paths [35, 48].

Independent versus cumulative yields differ by β-decay feeding. Reactor antineutrino predictions need both fission yields and subsequent β branching ratios (Lectures 18–19). Safeguards applications similarly depend on consistent evaluated libraries, not only on a schematic asymmetric yield cartoon.

Library versus cartoon

After drawing the asymmetric yield cartoon, open an evaluated yield file and compare peak locations. Note how many isobars contribute under each peak once β feeding is included. The cartoon teaches the idea; the library teaches the data culture of reactors and safeguards.

24.9 Further physics from the literature

Pairing and the fissile pattern. The large capture Q of $n + {}^{235}$U relative to $n + {}^{238}$U is a pairing effect: adding a neutron to odd-A ^{235}U produces even–even ^{236}U with an extra pairing energy, increasing the neutron separation energy. The same pattern explains fissile ^{239}Pu and ^{233}U [6, 1, 14].

Ternary fission and light charged particles. Occasionally a third light charged particle (α, ^{3}H, ...) is emitted at scission (ternary fission), at the 10^{-3} level. It is a probe of the neck dynamics between the two main fragments [6, 2].

Reactor antineutrinos. Each fission leads to several β^- decays and therefore several $\bar{\nu}_e$. Reactor antineutrino spectra are used for oscillation experiments and for remote monitoring of reactor power and fuel evolution [14, 2].

Evaluated nuclear data. Fission yields, $\bar{\nu}(E)$, and cross sections $\sigma_f(E)$ are tabulated in ENDF/B, JEFF, and JENDL libraries and summarised in ENSDF/NuDat for decay properties of individual fragments [35, 20, 21, 33].

Provenance of the checked numbers. The separation energies $S_n(^{236}\mathrm{U}) = 6.545\,\mathrm{MeV}$ and $S_n(^{239}\mathrm{U}) = 4.806\,\mathrm{MeV}$ use AME2020 masses through NNDC QCalc [27, 21]. The IAEA Nuclear Data for Safeguards table gives $\bar{\nu}_{\mathrm{total}} = 2.4355 \pm 0.0023$ for thermal-neutron fission of ^{235}U; the commonly used delayed fraction is about 0.65%. The rounded 205 MeV partition in Section 24.6 follows the ^{236}U energy partition of [1]. These sources define quantities more carefully than the original OCR and explain small differences among values quoted in lecture notes [35, 33].

24.9.1 Quantitative asymmetric yield and Q release

The asymmetric fission yield $Y(A)$ integrates to two fragment peaks. The total energy release per event is approximately constant ($\sim 200\,\mathrm{MeV}$), but the partition between fragment kinetic energy, prompt neutrons, and γ rays depends on the mass split [1, 35]. Light fragments carry higher kinetic energy per nucleon because Coulomb repulsion accelerates them from a compact scission configuration. Delayed energy from fragment β decay adds $\sim 10\,\mathrm{MeV}$ per fission over minutes to years, relevant for reactor decay-heat calculations [6, 35].

Neutron multiplication scale in a thermal reactor

Thermal fission of ^{235}U emits $\bar{\nu} \approx 2.44$ prompt neutrons [35]. In an infinite medium with fission cross section $\sigma_f \approx 585\,\mathrm{b}$ and number density $n \approx 0.05\,\mathrm{fm}^{-3}$, the mean free path is $\lambda_f = 1/(n\sigma_f) \approx 34\,\mathrm{cm}$. If fraction $p \approx 0.4$ of neutrons cause another fission after moderation, $k_{\mathrm{eff}} \approx \bar{\nu}p \approx 1$ for critical operation (24.28) [6, 14]. Delayed neutrons stretch the generation time from nanoseconds to seconds, enabling control.

24.9.2 From fission chains to fusion fuel cycles

Fission releases $\sim 200\,\mathrm{MeV}$ per heavy nucleus; D–T fusion releases $\sim 17.6\,\mathrm{MeV}$ per event but with far higher specific energy per nucleon transferred to light products [1, 6]. Both processes move toward the iron peak on the binding-energy curve: fission from above, fusion from below (Lectures 25–26). Reactor antineutrinos from fragment β chains connect fission to the weak-interaction spectroscopy of Lectures 18–19 [14, 2]. r-process recycling through fission in neutron-star mergers uses the same yield libraries as reactor safeguards [48, 56].

Key idea

Arc through the course. Asymmetric yields (Lecture 24) reflect shell corrections to the fission saddle (Lecture 23). Chain-reaction scale sets reactor engineering; fusion reactivity $\langle \sigma v \rangle$ sets stellar lifetimes (Lecture 26). Both require barrier penetration developed in Lectures 16–17 and 25 [6, 36].

24.9.3 Decay heat and the β-γ cascade after fission

Prompt fission releases $\sim 200\,\mathrm{MeV}$ in fragments, neutrons, and γ rays; delayed energy from fragment β decay adds $\sim 7\,\mathrm{MeV}$ per fission on a $\sim 10\,\mathrm{min}$ scale and continues for years [1, 35]. Each β step is a continuous spectrum (Lecture 18) followed by γ cascades governed by multipole rules (Lecture 20) [6]. Reactor safety calculations require cumulative fission yields and decay-heat tables that are consistent with independent β branch data [35, 33]. The same fragment β chains produce reactor antineutrinos used in oscillation experiments [14, 2].

24.10 Deeper reading: yields, delayed neutrons, and r-process recycling

Asymmetric mass yields for thermal $^{235}\mathrm{U}(n, f)$ reflect shell stabilisation of fragments near $A \sim 95$ and $A \sim 140$. Symmetric yields appear at higher excitation energy. Delayed-neutron precursors control reactor timescales on the order of seconds; cumulative yields enter decay-heat and safeguards calculations.

In neutron-star merger ejecta, fission recycling terminates the r-process path beyond $A \sim 195$ and reshapes abundances. Microscopic yield models for exotic actinides remain uncertain compared with evaluated libraries for major actinides. The lecture chain-reaction discussion is the reactor face of the same nuclear data.

Energy dependence of yields

As excitation energy rises, shell-stabilised asymmetric valleys fill in and symmetric fission becomes more competitive. Multichance fission (neutron emission before fission) further complicates yield interpretation in hot heavy-ion reactions. Reactor applications mostly need thermal and fast-neutron libraries; astrophysics needs far-off-stability estimates.

Delayed neutrons and control

Groups of delayed-neutron precursors with different half-lives create the effective β_{eff} that makes reactors controllable. Connecting those groups to β-delayed neutron branching ratios from Lecture 18 closes the nuclear-data loop.

Yield and reactor drill

Using a published thermal yield table for $^{235}\mathrm{U}$, identify the light and heavy peak mass numbers and estimate the peak-to-valley ratio. Then list three delayed-neutron precursor isotopes and their half-life groups. Connect each precursor to a β-delayed neutron branch (Lecture 18).

Write a short paragraph on r-process fission recycling: where the path hits fission, fragments return to lower A and capture again, smoothing abundance patterns above $A \sim 195$. Emphasise that exotic actinide yields are much less constrained than reactor actinides, so astrophysical abundance predictions inherit nuclear-data uncertainties from this chapter's themes.

Yields connect nuclear structure to reactors and to r-process recycling. Evaluated libraries are the data backbone behind the asymmetric-yield cartoon.

In practice, rework one numerical estimate from the lecture with a different parameter set and compare the shift to the quoted experimental uncertainty. If the shift exceeds the uncertainty, the estimate is teaching a scale, not delivering a fit. That distinction keeps theoretical cartoons honest when they sit beside evaluated data tables and modern many-body calculations. Along the course arc, write one sentence linking this chapter backward and one forward: the discipline of those two sentences is as important as any single formula. Examination problems rarely ask for a literature review, but they do ask for the scale argument that literature reviews presuppose.

Synthesis problem

Write a half-page note that (i) restates the chapter's central scale or selection rule in one equation, (ii) evaluates it numerically for one concrete nucleus or energy, and (iii) names the neighbouring lecture that uses the result. Keep the note free of research-frontier speculation; the goal is fluency with the lecture physics itself. If an assumption fails (Born approximation, spherical mean field, allowed transition, one-dimensional barrier), state the failure mode in a final sentence. That habit converts passive reading into examination-ready understanding and makes the later research-frontier box read as an extension rather than a replacement.

Research frontier: fission yields, reactors, and r-process recycling

Mass-yield curves and chain-reaction ideas of Lecture 24 connect directly to evaluated fission-product libraries used in reactors and safeguards [35]. Astrophysically, fission recycling shapes the abundance pattern beyond $A \sim 195$ in neutron-star merger ejecta; yield models and β-delayed fission remain major uncertainties in r-process simulations [48, 53, 56]. Microscopic yield predictions are beginning to replace purely empirical systematics for exotic actinides, but they must still be validated against radiochemical and on-line mass-spectrometric data. Reactor applications add a second demand: delayed-neutron precursors and cumulative yields control reactivity and safeguards signatures. Open research questions include the energy dependence of yields for exotic fissioning systems and the coupling between fission and β-decay networks in merger outflows.

For safeguards and reactor modelling, cumulative and independent yields, decay heat, and delayed-neutron fractions must be consistent within evaluated libraries. Students who compare two library releases for the same fissioning system quickly see where nuclear data, not just reactor engineering, set the uncertainty budget.

Comparing two evaluated-library releases for the same fissioning system quickly reveals where nuclear data dominate the uncertainty budget.

That data-driven uncertainty view links the lecture yield curve to real reactor and astrophysics applications.

Student directions. (1) Data exercise: using an ENDF fission-yield table, identify the two asymmetric peaks for thermal $^{235}\mathrm{U}(n, f)$. (2) Reactor estimate: estimate the number of delayed-neutron precursors that matter for reactor control on the $\sim 10\,\mathrm{s}$ scale. (3) Short essay: explain how fission terminates the r-process path and which nuclear data are most urgently needed.

24.11 Problems with solutions

Problem 24.1 — Fissile vs fissionable. Given $Q(n+{}^{235}\mathrm{U}) = 6.55\,\mathrm{MeV}$, $E_f({}^{236}\mathrm{U}) = 6.2\,\mathrm{MeV}$, $Q(n + {}^{238}\mathrm{U}) = 4.80\,\mathrm{MeV}$, and $E_f({}^{239}\mathrm{U}) = 6.2\,\mathrm{MeV}$, determine for each case whether thermal neutrons can induce fission, and estimate the neutron kinetic energy threshold for ${}^{238}\mathrm{U}$ [6, 1].

Solution 24.1.

Step 1 (^{235}U). Excitation $E^* \approx 6.55\,\mathrm{MeV} > 6.2\,\mathrm{MeV}$. Thermal neutrons induce fission: fissile.

Step 2 (^{238}U). $E^* \approx 4.80\,\mathrm{MeV} < 6.2\,\mathrm{MeV}$. Thermal fission is blocked by a residual barrier $\approx 1.4\,\mathrm{MeV}$.

Step 3 (threshold). Need $Q + K_n \gtrsim E_f$, so

$$K_n \gtrsim 6.2 - 4.8 = 1.4\,\mathrm{MeV}.$$

(Laboratory thresholds are of this order; precise values include CM corrections and barrier tunneling tails.)

Problem 24.2 — Exchange-symmetric asymmetry expansion. Let $F(x)$ be the total binding of a fragment of mass x, so $Q(a) = F(120-a)+F(120+a)-B_{\mathrm{p}}$ for a parent with $A = 240$. Show that all odd powers of a cancel. If $Q(0) = 216\,\mathrm{MeV}$ and $F''(120) = -5.0 \times 10^{-3}\,\mathrm{MeV}$, estimate $Q(20)$ through quadratic order and verify that $a = 20$ and $a = -20$ give the same result.

Solution 24.2.

Step 1: expand both fragment bindings.

$$F(120 \pm a) = F(120) \pm F'(120)a + \frac{1}{2}F''(120)a^2 \pm \frac{1}{6}F'''(120)a^3 + \cdots .$$

Adding the two expansions cancels the linear, cubic, and all other odd terms:

$$Q(a) = Q(0) + F''(120)a^2 + \mathcal{O}(a^4) = Q(-a).$$

Step 2: evaluate the toy curvature.

$$Q(\pm 20) = 216 + (-5.0 \times 10^{-3})(20)^2 = 214\,\mathrm{MeV}.$$

The chosen negative curvature lowers the toy-model Q by $2\,\mathrm{MeV}$, but the sign and magnitude of the actual partition dependence require masses, prompt-neutron channels, and shell corrections [6, 14].

Problem 24.3 — Neutrons from N/Z. ^{236}U has $Z = 92$, $N = 144$. Suppose it splits into fragments with $Z_1 = 54$, $Z_2 = 38$ without emitting neutrons, sharing N in proportion to A. Argue why prompt neutron emission is expected, given that stable $A \sim 90$–140 nuclei have $N/Z \sim 1.3$–1.4.

Solution 24.3.

Step 1. Parent $N/Z = 144/92 \approx 1.57$.

Step 2. Conserving N/Z in each fragment leaves both with $N/Z \sim 1.57$, far above the stable valley at those Z.

Step 3. Excess neutrons are weakly bound in the hot fragments and evaporate as prompt neutrons until the neutron separation energy rises. Residual neutron richness is then removed slowly by β^- decay [1, 6].

Problem 24.4 — Energy partition. If 200 MeV is released per fission and 160 MeV appears as fragment kinetic energy, what fraction of the energy is carried by fragments? If a reactor fissions 10^{19} nuclei per second, estimate the fragment-heating power in megawatts.

Solution 24.4.

Step 1. Fraction: $160/200 = 0.80$ (80%).

Step 2. Energy per second from fragments:

$$P = 10^{19} \times 160\,\mathrm{MeV/s}.$$

Convert: $1\,\mathrm{MeV} = 1.602 \times 10^{-13}\,\mathrm{J}$, so

$$P = 10^{19} \times 160 \times 1.602 \times 10^{-13}\,\mathrm{W} \approx 2.56 \times 10^{8}\,\mathrm{W} \approx 256\,\mathrm{MW}.$$

(Total thermal power including other channels would be $\sim 320\,\mathrm{MW}$ before neutrino losses.)

Problem 24.5 — Multiplication factor and generations. In a certain assembly, each fission produces on average 2.4 neutrons, and the probability that a neutron induces another fission (rather than leaking or being captured non-fissilely) is $p = 0.40$. Find k, classify the assembly, and find N_{10}/N_0. Repeat the generation calculation for $p = 0.425$.

Solution 24.5.

$$k = \langle \nu \rangle\, p = 2.4 \times 0.40 = 0.96 < 1.$$

The assembly is subcritical: the chain dies out. Raising enrichment, reducing leakage, or improving moderation can increase p until $k = 1$ [6, 1].

Step 2.

$$\frac{N_{10}}{N_0} = k^{10} = 0.96^{10} = 0.665.$$

After ten generations about two-thirds of the original generation remains on average.

Step 3. If $p = 0.425$, then $k = 2.4(0.425) = 1.02$, and

$$\frac{N_{10}}{N_0} = 1.02^{10} = 1.219.$$

The same exponential law distinguishes a slowly decaying subcritical chain from a growing supercritical one; it does not by itself specify the elapsed time, which also requires the generation lifetime.

Problem 24.6 — Delayed fraction and prompt criticality. Suppose $\langle \nu_{\text{prompt}} \rangle = 2.43$ and $\langle \nu_{\text{delayed}} \rangle = 0.016$. What fraction β of neutrons are delayed? Explain why this yield fraction is not generally β_{eff}. If $\beta_{\text{eff}} = 0.0065$, find the exact prompt-critical k_{eff}.

Solution 24.6.

Step 1.

$$\beta = \frac{0.016}{2.43 + 0.016} \approx 0.0065 \quad (0.65\%).$$

Step 2. The physical β counts emitted neutrons. β_{eff} additionally weights their spectra and birth locations by their importance to the chain, so the two quantities need not be equal.

Step 3. Prompt criticality is $\rho = \beta_{\text{eff}}$, where $\rho = (k_{\text{eff}} - 1)/k_{\text{eff}}$. Hence

$$k_{\text{eff,prompt}} = \frac{1}{1 - \beta_{\text{eff}}} = \frac{1}{1 - 0.0065} = 1.00654.$$

The approximation $1 + \beta_{\text{eff}} = 1.0065$ is close but not exact. Below this prompt-critical threshold, delayed-neutron precursor timescales make mechanical control possible [6, 1].

Problem 24.7 — Separation energy and threshold. AME2020 gives $S_n(^{236}\text{U}) = 6.545\,\text{MeV}$ and $S_n(^{239}\text{U}) = 4.806\,\text{MeV}$. Using the original lecture's one-barrier value $E_f = 6.2\,\text{MeV}$, calculate the excess or deficit $E^* - E_f$ after capture of a negligible-energy neutron on ^{235}U and ^{238}U. Why should the second result not be quoted as a precision experimental threshold? [27, 21]

Solution 24.7.

Step 1: identify the capture energy. For a target ^{A}U,

$$E^* \simeq S_n(^{A+1}\text{U}) + E_{n,\text{cm}}.$$

For a thermal neutron $E_{n,\text{cm}}$ is negligible on the MeV scale.

Step 2: ^{235}U.

$$E^{*} - E_f = 6.545 - 6.2 = +0.345\,\mathrm{MeV}.$$

The compound $^{236}\mathrm{U}^{*}$ is formed above the illustrative barrier.

Step 3: ^{238}U.

$$E^{*} - E_f = 4.806 - 6.2 = -1.394\,\mathrm{MeV}.$$

Thus a fast-neutron energy of order 1 MeV is expected before fission becomes strong.

Step 4: limitation. The mass-derived separation energies are precise, but "the barrier" is a double-humped, spin-dependent transmission problem, and tunnelling gives a subthreshold tail. Centre-of-mass kinematics and competing channels also matter. Hence 1.394 MeV is a model estimate, not a sharp measured threshold.

Problem 24.8 — Delayed-neutron fraction. If $\bar{\nu}_{\mathrm{total}} = 2.44$ and $\bar{\nu}_d = 0.016$, compute the physical delayed fraction $\beta = \bar{\nu}_d/\bar{\nu}_{\mathrm{total}}$. Why is β_{eff} used in reactor kinetics instead?

Solution 24.8.

$$\beta = \frac{0.016}{2.44} \approx 0.0066 \sim 0.65\%.$$

β_{eff} weights precursors by worth (spectrum, location, importance) for the chain reaction, so it need not equal the raw yield fraction β [6, 1, 35].

Chapter 25

Nuclear Fusion I

"Fusion powers the stars: light nuclei climb the binding curve toward iron, releasing energy, if they can tunnel the Coulomb barrier."
H. A. Bethe (1939)

Fission exploits the decline of B/A for heavy nuclei. Fusion exploits the *rise* of B/A for light nuclei. Stars therefore burn from the hydrogen side of the iron peak; terrestrial fusion research aims at the same exothermic region under controlled plasma conditions, whereas fission reactors operate on the heavy-mass side. This chapter reads the binding-energy curve as a map of energy release, derives the fusion Q-value from binding energies with a neon–neon example, introduces the sharp-surface contact estimate $V_{\text{contact}} = Z_1 Z_2 e^2/[4\pi\varepsilon_0(r_1 + r_2)]$, evaluates its scale for Ne + Ne and $p + p$, and explains why stellar fusion proceeds by quantum tunneling at thermal energies far below the barrier [6, 1, 14, 2].

25.1 The binding-energy curve: fission right, fusion left

Figure 25.1 summarises nuclear energetics. The label "most stable" in the original slide should be read as "near the maximum of B/A": among measured nuclides ^{62}Ni has the largest binding energy per nucleon, whereas ^{56}Fe is especially abundant and is commonly used to mark the end of exothermic stellar burning.

- B/A rises steeply from the lightest nuclei toward $A \sim 56$–62 (the Fe–Ni region).
- After the peak, B/A decreases slowly through the heavy elements.

Any nuclear rearrangement that increases the total binding energy decreases the rest mass and releases energy $Q = (\Delta M)c^2$.

Fission (right of the peak). A heavy parent ($A \sim 240$, $B/A \sim 7.6\,\text{MeV}$) splits into medium-mass fragments nearer the peak ($B/A \sim 8.5\,\text{MeV}$). Energy is released; this was the subject of Lectures 23–24.

Fusion (left of the peak). Two light nuclei combine into a heavier product still left of or nearer the peak. The product has larger B/A than the reactants, so energy is released. The

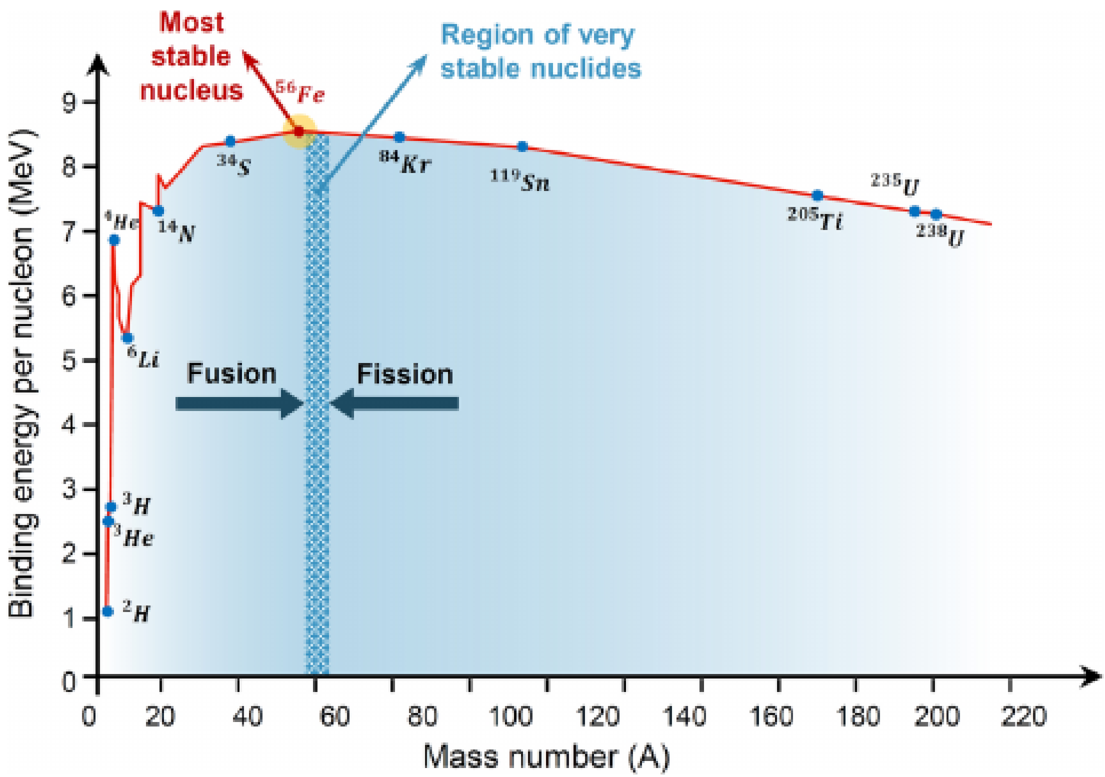


Figure 25.1: Binding energy per nucleon B/A versus mass number A. The peak near iron/nickel marks the most tightly bound nuclei. Fusion of light nuclei (left of the peak) and fission of heavy nuclei (right of the peak) both move systems toward higher B/A and release energy.

classic laboratory example is

$$\mathrm{D} + \mathrm{T} \;\to\; {}^{4}\mathrm{He} + n + 17.6\,\mathrm{MeV}, \tag{25.1}$$

and in stars the *pp* chain begins with $p + p \to \mathrm{d} + e^{+} + \nu_e$.

Key idea

Fusion: light nuclei join and move up the B/A curve toward iron. Fission: heavy nuclei split and move up the B/A curve from the other side. Both release energy; both are blocked by barriers (Coulomb for fusion, deformation for fission).

Beyond the Fe–Ni peak, charged-particle fusion generally ceases to be a steady energy source. Most nuclei heavier than iron are assembled instead by slow or rapid neutron capture followed by β^{-} decays; explosive environments can also drive selected endothermic charged-particle and photodisintegration sequences. Thus "fusion stops at iron" is a statement about stellar energy generation, not a claim that heavier nuclei cannot be synthesized [1, 2].

25.2 Fusion Q-value from binding energies

For a general fusion reaction

$$X + Y \to Z + \cdots, \tag{25.2}$$

the Q-value is

$$Q = [M_X + M_Y - M_Z - \cdots]c^2. \tag{25.3}$$

Writing each mass as $Mc^2 = \sum m_N c^2 - E_B$ and cancelling free nucleon masses (when nucleon number is conserved) yields the binding form

$$Q = E_B(\text{final}) - E_B(\text{initial}). \tag{25.4}$$

Fusion releases energy when the product is more tightly bound than the sum of the reactants: $E_B(\text{final}) > E_B(\text{initial})$.

25.2.1 Binding bookkeeping for the radiative neon channel

Consider

$$^{20}_{10}\text{Ne} + ^{20}_{10}\text{Ne} \;\rightarrow\; ^{40}_{20}\text{Ca} + \gamma. \tag{25.5}$$

(This is a valid radiative-capture channel for binding-energy bookkeeping, but it is not representative of the dominant decay of the highly excited $^{40}\text{Ca}^*$ compound system.)

Rest energy before fusion. One neon nucleus has rest energy

$$E(^{20}\text{Ne}) = (10m_p + 10m_n)c^2 - E_B(^{20}\text{Ne}). \tag{25.6}$$

Two neon nuclei contribute

$$E_{\text{before}} = 2(10m_p + 10m_n)c^2 - 2E_B(^{20}\text{Ne}). \tag{25.7}$$

Rest energy after fusion.

$$E(^{40}\text{Ca}) = (20m_p + 20m_n)c^2 - E_B(^{40}\text{Ca}). \tag{25.8}$$

***Q*-value.**

$$\begin{aligned} Q &= E_{\text{before}} - E_{\text{after}} \\ &= 2(10m_p + 10m_n)c^2 - 2E_B(\text{Ne}) - \left[(20m_p + 20m_n)c^2 - E_B(\text{Ca})\right] \\ &= E_B(^{40}\text{Ca}) - 2E_B(^{20}\text{Ne}). \end{aligned} \tag{25.9}$$

Equivalently, in terms of binding per nucleon,

$$Q = 40\left[\left(\frac{B}{A}\right)_{\text{Ca}} - \left(\frac{B}{A}\right)_{\text{Ne}}\right]. \tag{25.10}$$

Since $(B/A)_{\text{Ca}} > (B/A)_{\text{Ne}}$ on the rising part of the curve, $Q > 0$. The raw lecture estimated $Q \simeq 23\,\text{MeV}$ from a schematic B/A graph. Evaluated atomic masses give a more accurate result. Because both sides contain 20 electrons when neutral-atom masses are used, electron masses cancel:

$$\begin{aligned} Q &= \left[2M_{\mathrm{a}}(^{20}\mathrm{Ne}) - M_{\mathrm{a}}(^{40}\mathrm{Ca})\right] c^2 \\ &= [2(19.992\,440\,1762) - 39.962\,590\,863]\,\mathrm{u}\,c^2 \\ &= 0.022\,289\,4894\,\mathrm{u}\,c^2 = 20.76\,\mathrm{MeV}, \end{aligned} \tag{25.11}$$

where $1\,\mathrm{u}\,c^2 = 931.494\,\mathrm{MeV}$. Thus the original binding-energy logic is sound, but its graph-read value is about 2.2 MeV too large. For the idealized radiative-capture channel in Eq. (25.5), the energy ultimately appears in the photon and calcium recoil [27, 6].

At neon-burning energies the compound nucleus has open particle channels. Representative measured two-body exit channels include

$$^{20}\mathrm{Ne} + {}^{20}\mathrm{Ne} \to {}^{36}\mathrm{Ar} + \alpha, \tag{25.12}$$

$$^{20}\mathrm{Ne} + {}^{20}\mathrm{Ne} \to {}^{39}\mathrm{K} + p. \tag{25.13}$$

Additional multi-particle channels occur as energy increases. Their Q-values must be computed from the masses of the actual final products; the 20.76 MeV value above belongs only to the radiative $^{40}\mathrm{Ca} + \gamma$ channel [27, 1].

Remark

Equation (25.4) is the same identity used for fission ($Q = E_{B,\mathrm{frag}} - E_{B,\mathrm{parent}}$). Only the direction of motion on the B/A curve changes.

25.3 The Coulomb barrier

Although fusion of light nuclei is energetically favoured, the nuclei are positively charged. They repel until they approach within the range of the attractive nuclear force ($\sim$ a few fm). The electrostatic potential energy at centre-to-centre separation r is

$$V_{\mathrm{C}}(r) = \frac{Z_1 Z_2 e^2}{4\pi\varepsilon_0\, r}. \tag{25.14}$$

As r decreases, V_{C} rises. When the nuclear surfaces touch, the nuclear attraction rapidly dominates and the potential drops into a deep well. The value of V_{C} at contact provides a useful sharp-surface estimate:

$$\boxed{V_{\mathrm{contact}} = \frac{Z_1 Z_2 e^2}{4\pi\varepsilon_0\,(r_1 + r_2)}.} \tag{25.15}$$

The actual barrier V_B is the maximum of the combined Coulomb, nuclear, and centrifugal effective potential; it depends on diffuse nuclear density profiles, angular momentum, and the adopted ion–ion potential.

Nuclear radii follow the empirical law

$$r = r_0 A^{1/3}, \qquad r_0 \approx 1.2\,\mathrm{fm}. \tag{25.16}$$

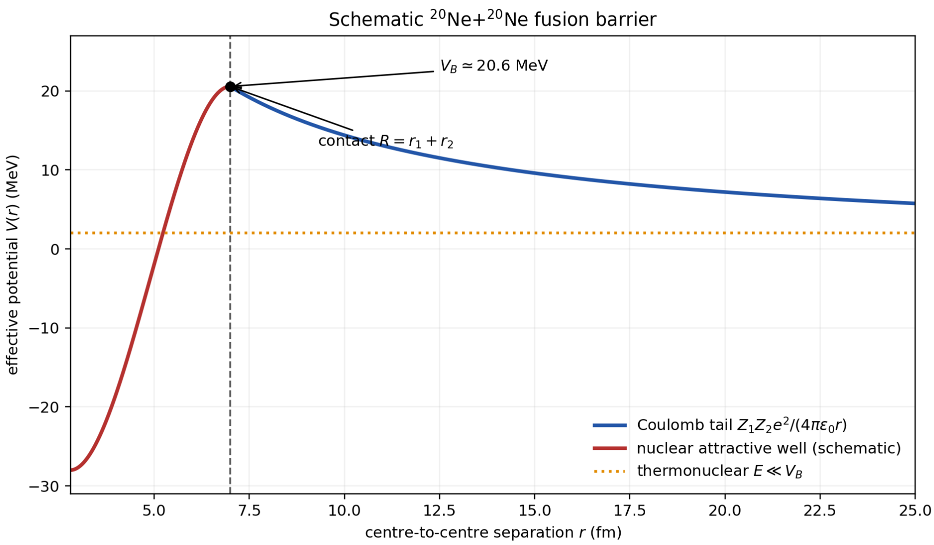


Figure 25.2: Schematic effective potential for $^{20}\mathrm{Ne} + {}^{20}\mathrm{Ne}$. The Coulomb tail rises toward the contact radius, while the short-range nuclear attraction produces an interior well. The contact estimate is $V_{\text{contact}} \simeq 20.6\,\mathrm{MeV}$; a quantitative barrier requires an ion–ion potential and partial-wave specification.

A convenient numerical constant is

$$\frac{e^2}{4\pi\varepsilon_0} \approx 1.44\,\mathrm{MeV \cdot fm}, \tag{25.17}$$

so

$$V_{\text{contact}} \approx \frac{Z_1 Z_2 \times 1.44\,\mathrm{MeV \cdot fm}}{r_1 + r_2}. \tag{25.18}$$

25.3.1 Example: neon–neon

For two ^{20}Ne nuclei, $Z_1 = Z_2 = 10,\ A = 20$:

$$r_1 = r_2 = 1.2 \times 20^{1/3}\,\mathrm{fm} \approx 1.2 \times 2.71\,\mathrm{fm} \approx 3.26\text{–}3.5\,\mathrm{fm} \tag{25.19}$$

(using the lecture's 3.5 fm estimate). Contact separation $r_1 + r_2 \approx 7.0\,\mathrm{fm}$, so

$$V_{\text{contact}} = \frac{10 \times 10 \times 1.44\,\mathrm{MeV \cdot fm}}{7.0\,\mathrm{fm}} \approx 20.6\,\mathrm{MeV} \sim 20\,\mathrm{MeV}. \tag{25.20}$$

Classically one needs $\sim 20\,\mathrm{MeV}$ of relative kinetic energy to surmount the barrier.

25.3.2 Example: proton–proton

For $p + p$, $Z_1 = Z_2 = 1$. If one adopts the conventional pedagogical contact scale $R = 2\,\mathrm{fm}$,

$$V_{\text{contact}} = \frac{1 \times 1 \times 1.44\,\mathrm{MeV \cdot fm}}{2\,\mathrm{fm}} = 0.72\,\mathrm{MeV}. \tag{25.21}$$

The proton has no sharp classical surface, so 0.72 MeV is a contact-scale estimate rather than a uniquely defined barrier height; changing the chosen strong-interaction radius changes this number. It nevertheless shows that the Coulomb scale is huge compared with thermal energies in the solar core (Section 25.4) [6, 1, 14, 2].

Key idea

$V_{\text{contact}} = Z_1 Z_2 e^2/[4\pi\varepsilon_0(r_1 + r_2)]$, $r = r_0 A^{1/3}$. Ne + Ne: $V_{\text{contact}} \sim 20\,\text{MeV}$. $p + p$: $V_{\text{contact}} \sim 0.72\,\text{MeV}$, subject to the chosen contact radius. The Coulomb scale grows rapidly with $Z_1 Z_2$.

25.4 Classical threshold vs quantum tunneling

Stars and terrestrial plasmas fuse far below the classical Coulomb barrier. The mean thermal energy in the solar core is only a few keV, while *pp* contact estimates sit near 0.7 MeV. Quantum tunneling through the barrier, averaged over the Maxwellian tail, makes fusion possible; the same Gamow physics that governs α decay (Lectures 16–17) runs here in reverse as capture [6, 14].

25.4.1 Classical requirement

Classically, fusion requires relative kinetic energy

$$E_k > V_B, \qquad V_B \sim V_{\text{contact}} = \frac{Z_1 Z_2 e^2}{4\pi\varepsilon_0(r_1 + r_2)} \tag{25.22}$$

in the sharp-surface estimate. Accelerators can supply tens of MeV and drive fusion reactions for nuclear physics studies, but the energy invested in the beams typically exceeds the energy recovered from Q: accelerators are not practical power plants.

25.4.2 Quantum tunneling and reduced mass

Separate the two-body motion into centre-of-mass and relative coordinates. The relative coordinate behaves as one particle of reduced mass

$$\mu = \frac{m_1 m_2}{m_1 + m_2}, \qquad E = \frac{1}{2}\mu v^2, \qquad v = |\boldsymbol{v}_1 - \boldsymbol{v}_2|. \tag{25.23}$$

For $E < V_{\max}$, WKB gives

$$P(E) \simeq \exp\left[-2\int_R^{r_c} \frac{\sqrt{2\mu[V_C(r) - E]}}{\hbar}\,\mathrm{d}r\right], \qquad r_c = \frac{K}{E}, \quad K = \frac{Z_1 Z_2 e^2}{4\pi\varepsilon_0}, \tag{25.24}$$

where $R \simeq r_1 + r_2$ and r_c are the inner and outer turning points. At thermonuclear energies $R/r_c \ll 1$. With $r = r_c \sin^2\theta$, the leading Coulomb integral is

$$\begin{aligned} 2\int_0^{r_c} \frac{\sqrt{2\mu(K/r - E)}}{\hbar}\,\mathrm{d}r &= \frac{4Er_c}{\hbar}\sqrt{\frac{2\mu}{E}}\int_0^{\pi/2}\cos^2\theta\,\mathrm{d}\theta \\ &= \frac{\pi K}{\hbar}\sqrt{\frac{2\mu}{E}} = \frac{2\pi K}{\hbar v} \equiv 2\pi\eta, \end{aligned} \tag{25.25}$$

because $\int_0^{\pi/2} \cos^2\theta \, d\theta = \pi/4$. The Sommerfeld parameter and Gamow energy are therefore

$$\eta = \frac{K}{\hbar v} = Z_1 Z_2 \alpha \frac{c}{v}, \qquad P(E) \propto e^{-2\pi\eta} = \exp\left[-\sqrt{\frac{E_{\mathrm{G}}}{E}}\right], \qquad E_{\mathrm{G}} = 2\mu c^2 (\pi\alpha Z_1 Z_2)^2. \quad (25.26)$$

Finite-R and nuclear-structure effects are later absorbed into the measured cross section and astrophysical S-factor. Since $2\pi\eta \propto 1/v \propto E^{-1/2}$, penetration rises exponentially with energy. This is the same physics as α decay (Lectures 16–17), run in reverse: instead of tunneling *out*, the nuclei tunnel *in* [6, 14].

25.4.3 Thermonuclear fusion

In a hot plasma the kinetic energy is thermal. The mean kinetic energy of a particle is

$$\langle E_k \rangle = \frac{3}{2} k_B T. \quad (25.27)$$

Fusion powered by the thermal motion of a plasma is called **thermonuclear fusion**. It is the energy source of main-sequence stars and the goal of magnetic and inertial confinement fusion devices.

Room temperature. At $T \approx 300\,\mathrm{K}$,

$$k_B T \approx 0.026\,\mathrm{eV}, \quad (25.28)$$

utterly negligible compared with even the *pp* barrier of 720 keV. The tunneling probability is zero for practical purposes.

Temperature for $E_k \sim 150\,\mathrm{keV}$. Set $\frac{3}{2} k_B T = 150\,\mathrm{keV}$, so $k_B T = 100\,\mathrm{keV}$:

$$T = \frac{100 \times 10^3\,\mathrm{eV}}{8.617 \times 10^{-5}\,\mathrm{eV/K}} \approx 1.16 \times 10^9\,\mathrm{K}. \quad (25.29)$$

This calculation only asks what temperature would make the *mean* single-particle energy 150 keV; it is not an ignition condition. Practical D–T plasmas use $k_B T$ of order 10 keV and still rely on tunneling.

Solar core. The Sun's core has $T \approx 1.5 \times 10^7\,\mathrm{K}$, so

$$k_B T \approx 1.3\,\mathrm{keV}, \qquad \langle E_k \rangle \sim 2\,\mathrm{keV} \quad (25.30)$$

(order 1 keV in the lecture's estimate). Thus $V_{\max}/k_B T \approx 720/1.3 \approx 550$. Fusion in the Sun is possible only because of tunneling, aided by two further facts:

(i) the Maxwell–Boltzmann distribution has a high-energy tail, so some protons have $E \gg \langle E_k \rangle$;

(ii) the solar core is extremely dense ($\rho \sim 150\,\mathrm{g\,cm^{-3}}$) and huge, so even a tiny rate per pair yields enormous luminosity.

Caution

Do not equate k_BT with the barrier height. Stars and fusion reactors operate with $k_BT \ll V_{\max}$; the Gamow tunneling factor and the Maxwell tail jointly set the rate (Lecture 26).

25.4.4 Salpeter plasma screening

The Coulomb field in a plasma is weakly screened by surrounding electrons and ions. In the Debye–Hückel limit the screening cloud lowers the free-energy cost of bringing charges Z_1e and Z_2e together. Salpeter's weak-screening result multiplies the unscreened thermonuclear rate by

$$f_{\mathrm{sc}} = \exp\left(\frac{Z_1Z_2e^2}{4\pi\varepsilon_0 R_D k_BT}\right) > 1, \qquad R_D^{-2} = \frac{e^2}{\varepsilon_0 k_BT}\sum_s n_s Z_s^2. \tag{25.31}$$

This is a modest rate enhancement under solar-core conditions, not removal of the Coulomb barrier. Its validity requires weak screening; that condition is stated explicitly in Lecture 26 [36].

25.5 Maxwell tail and barrier penetration

In a gas at temperature T the speed distribution is Maxwellian. A fraction of particles has kinetic energy many times k_BT:

$$E \sim 10\,k_BT,\ 20\,k_BT,\ \ldots \tag{25.32}$$

Those particles tunnel far more readily because $2\pi\eta \propto 1/v \propto 1/\sqrt{E}$. The fusion rate is therefore dominated by a compromise between the falling Maxwell factor e^{-E/k_BT} and the rising penetration factor $e^{-b/\sqrt{E}}$. The product peaks at the **Gamow peak**, derived properly in the next chapter.

The barrier penetration probability for energy E may be written as a WKB integral between classical turning points r_1 and r_2:

$$P \sim \exp\left(-2\int_{r_1}^{r_2}\sqrt{\frac{2\mu}{\hbar^2}\big(V(r)-E\big)}\,\mathrm{d}r\right), \tag{25.33}$$

with $V(r) = Z_1Z_2e^2/(4\pi\varepsilon_0 r)$ outside the nuclear well. Increasing E shrinks the integration range and the integrand, so P grows rapidly with energy [6, 14, 2].

25.6 Reaction rate preview

The number of fusion reactions per unit volume per unit time between species X and Y is

$$r_{XY} = \frac{n_Xn_Y}{1+\delta_{XY}}\,\langle\sigma v\rangle, \tag{25.34}$$

where n_X, n_Y are number densities, σ is the fusion cross section, v is the relative velocity, and $\langle \sigma v \rangle$ is the thermal average. The Kronecker delta supplies 1/2 for identical reactants, so each unordered pair is counted once. The energy generation rate is $r_{XY}Q$ per unit volume. Concentrations and temperature therefore control stellar luminosity and the performance of fusion devices. The full derivation of R from beam–target geometry, the $1/v^2$ low-energy cross section, and the S-factor form of the low-energy cross section, and the Gamow peak occupies Lecture 26.

Takeaway

- B/A peaks at Fe/Ni: fusion left of peak, fission right of peak.
- $Q = E_B(\text{final}) - E_B(\text{initial})$; for Ne + Ne → Ca, $Q = E_B(\text{Ca}) - 2E_B(\text{Ne}) = 20.76\,\text{MeV}$.
- The sharp-surface contact estimate is $V_{\text{contact}} = Z_1Z_2e^2/[4\pi\varepsilon_0(r_1 + r_2)]$; Ne + Ne $\sim$ 20 MeV, while the quoted $pp \sim 0.72\,\text{MeV}$ depends on the chosen contact radius.
- Thermonuclear fusion: $E \sim k_BT \ll V_{\max}$; tunneling $P \propto e^{-2\pi\eta}$; solar core $T \sim 1.5 \times 10^7\,\text{K}$, $E \sim 1\,\text{keV}$.
- Salpeter weak screening multiplies the unscreened rate by $f_{\text{sc}} > 1$ in a weakly coupled plasma.
- Rate $r_{XY} = n_Xn_Y\langle\sigma v\rangle/(1 + \delta_{XY})$ (derived next).

25.6.1 Coulomb barriers for fusion versus fission

Induced fission can exploit thermal neutrons because the compound system is already above the barrier. Fusion must climb a Coulomb barrier of several MeV even for hydrogen isotopes, so terrestrial and stellar burning both rely on tunneling; laboratory beam fusion is easy energetically per event but costly to sustain as a net power source [1, 6, 14].

25.7 Deeper physics: Coulomb barrier and sub-barrier tunneling

25.7.1 Contact Coulomb barrier

For two nuclei of charges Z_1e and Z_2e and radii $R_i = r_0A_i^{1/3}$, the Coulomb potential at contact is

$$V_C(R_1 + R_2) = \frac{Z_1Z_2e^2}{4\pi\varepsilon_0(R_1 + R_2)}. \tag{25.35}$$

This sets the scale of the fusion barrier height $V_{\max}$ in a one-dimensional model [6, 14]. Because $V_C \propto Z_1Z_2/A^{1/3}$, heavier-projectile combinations face higher barriers and require higher centre-of-mass energy for appreciable fusion [1, 55].

Barrier heights for light and heavy fusion

For $^{12}\text{C} +^{12}\text{C}$ with $R_1 + R_2 \approx 5.2\,\text{fm}$ and $Z_1Z_2 = 144$, $V_{\max} \approx 2.5\,\text{MeV}$ [6, 14]. For $^{48}\text{Ca} +^{248}\text{Cm}$ ($Z_1Z_2 = 4800$, $R_1 + R_2 \approx 12\,\text{fm}$), $V_{\max} \approx 30\,\text{MeV}$ [55]. Stellar burning never reaches these heavy-ion barriers in equilibrium; superheavy synthesis relies on sub-barrier tunneling and evaporation-residue survival.

25.7.2 Sub-barrier tunneling

Classically, fusion requires $E_{\mathrm{CM}} \geq V_{\max}$. Quantum mechanically, the colliding nuclei tunnel through the barrier. The Gamow penetration factor for fusion mirrors α decay (Lectures 16–17) with incoming rather than outgoing boundary conditions:

$$P \approx \exp\left[-\frac{2}{\hbar}\int_R^b \sqrt{2\mu(V(r)-E)}\,\mathrm{d}r\right]. \tag{25.36}$$

At energies well below $V_{\max}$, P rises steeply with E. Deformed targets and transfer channels can enhance sub-barrier fusion beyond the s-wave estimate, while quasifission removes flux before evaporation [55, 14]. Terrestrial D–T reactors operate at $T \sim 10^8$ K where the Maxwellian tail supplies enough $E > V_{\max}$ events, supplemented by tunneling (Lecture 26) [1, 2].

Key idea

Assumptions of the one-dimensional barrier model. The sketch neglects angular momentum (centrifugal barrier), nuclear absorption, and compound-nucleus lifetime. It remains the correct first estimate for why fusion rates are exponentially sensitive to $Z_1 Z_2$ and why the Sun burns hydrogen slowly enough to shine for gigayears [36, 6].

25.7.3 The S-factor connection to Lecture 26

Laboratory fusion data are reported as $S(E) = E\,\sigma(E)\exp(\sqrt{E_G/E})$, removing the steep Gamow energy dependence [14, 36]. The same WKB exponent that appears in α decay (Lectures 16–17) defines $E_G = 2\mu(\pi\alpha Z_1 Z_2 \hbar c)^2$. Sub-barrier heavy-ion fusion cross sections are therefore the incoming-channel counterpart of α-decay rates, with $S(E)$ playing the role of a nuclear structure factor analogous to the preformation factor S_α [6, 55]. Stellar rates integrate $S(E)$ over the Maxwellian Gamow peak (Lecture 26).

25.7.4 Lawson criterion and ignition temperature

Terrestrial fusion must exceed losses as well as penetrate the Coulomb barrier. A crude Lawson criterion requires the product of density n and energy confinement time τ_E to exceed a threshold that depends on temperature and fuel cycle [1, 6]. For D–T, $\langle\sigma v\rangle$ peaks near $T \sim 10^8$ K; for $p+p$ in the Sun, the peak is at much lower T but the weak *pp* step limits the overall burn rate (Lecture 26) [36, 14]. Magnetic and inertial confinement schemes both operate in the sub-barrier tunneling regime rather than at $E = V_{\max}$ [2, 1].

Coupled-channel barrier distributions replace a single barrier height with a weighted set of eigenbarriers. Measuring fusion excitation functions below the nominal barrier therefore constrains deformation and transfer couplings. For SHE, the evaporation-residue cross section folds fusion probability with survival against fission: both pieces are uncertain.

25.8 Further physics from the literature

Stellar versus terrestrial fusion. A star confines plasma gravitationally for millions to billions of years and can obtain useful luminosity from extraordinarily slow reactions. In particular, $p + p \to d + e^+ + \nu_e$ requires the weak interaction, which makes the Sun stable on gigayear scales. A terrestrial device has a much smaller fuel inventory and confinement time, so it chooses the much faster strong reaction D–T, heats the plasma externally, and must recover energy while replacing radiative, transport, and particle losses. "Lowest Coulomb barrier" alone therefore does not determine the fuel: nuclear matrix elements, resonances, density, and confinement matter as well.

Stellar burning stages. Hydrogen burning (*pp* chain and CNO cycle) dominates main-sequence stars. When core hydrogen is exhausted, contraction raises T until helium burning ($3\alpha \to {}^{12}$C) ignites, then carbon, neon, oxygen, and silicon burning in massive stars, ending at the iron peak where fusion no longer releases energy [14, 6, 2].

Screening and plasma effects. The derivation uses bare nuclei. In a plasma, correlations among electrons and ions lower the free-energy cost of bringing two reacting ions together. In weak screening the rate is multiplied by an enhancement $f_{\rm sc} > 1$; one should not describe this as simply replacing the Coulomb potential by a constant lower barrier over all radii. Laboratory targets also have bound electrons and require a distinct electron-screening correction before their data are extrapolated to a stellar plasma [14, 36].

Laboratory fusion approaches. Magnetic confinement (tokamaks, stellarators) and inertial confinement (laser or ion-beam driven pellets) both aim for thermonuclear D + T conditions. Neither is required to reach $E = V_{\max}$; both rely on tunneling at $T \sim 10^8$ K [1, 2].

Connection to α decay. The Gamow factor for fusion is the time-reversed cousin of the Gamow factor for α emission (Lectures 16–17). The same WKB integral appears with incoming versus outgoing boundary conditions [6, 14].

25.8.1 Sub-barrier fusion cross section scaling

Below the Coulomb barrier, the fusion cross section is dominated by tunneling. A useful parameterisation writes

$$\sigma(E) \approx \frac{S(E)}{E} \exp(-2\pi\eta(E)), \qquad \eta(E) = \frac{Z_1 Z_2 e^2}{4\pi\varepsilon_0 \hbar v}, \tag{25.37}$$

where η is the Sommerfeld parameter [14, 6]. As $E \to 0$, $\eta \to \infty$ and $\sigma \to 0$ exponentially. Coupled-channel calculations show that static deformation can enhance $\sigma(E)$ by an order of magnitude at fixed E because the effective barrier is lowered along the orientation that aligns the major axis [55].

Sub-barrier $^{12}C + ^{12}C$ fusion

At $E_{CM} = 2$ MeV, well below $V_{max} \approx 2.5$ MeV, measured fusion cross sections are $\sim 10^{-4}$ mb, many orders of magnitude below $\sigma_{geom} \approx 850$ mb [6, 14]. The same Gamow exponent that gives $t_{1/2} \sim 10^{-7}$ s for ^{212}Po α decay (Lecture 17) governs this incoming channel. Stellar ^{12}C burning occurs at temperatures where the Gamow peak sits near $E_0 \sim 1.5$ MeV (Lecture 26) [36].

25.8.2 From Coulomb barriers to stellar Gamow peak

Terrestrial magnetic confinement operates at $T \sim 10^8$ K where $k_BT \approx 10$ keV, comparable to D–T barrier penetration energies. Stellar cores at $T \sim 1.5 \times 10^7$ K have $k_BT \approx 1.3$ keV, far below the $p + p$ barrier, yet burn steadily because $\langle\sigma v\rangle$ integrates over a Maxwellian tail [36, 2]. Screening in the plasma enhances rates by $\exp(U_e/k_BT)$ with $U_e \sim 1$–2 keV at solar density [14, 36]. The lecture barrier estimate is therefore a zero-screening baseline.

Remark

Quasifission in heavy-ion fusion removes flux before compound-nucleus formation, analogous to hindrance factors suppressing α decay without changing the barrier integral. SHE synthesis cross sections are limited by both tunneling and survival against fission (Lecture 23) [55, 56].

25.8.3 Terrestrial D–T versus stellar *pp* burning

Terrestrial fusion targets D–T because $\langle\sigma v\rangle$ peaks near $T \sim 10^8$ K with a manageable confinement requirement [1, 6]. The solar *pp* chain relies on the same tunneling physics at $T \sim 1.5 \times 10^7$ K where the weak $p + p$ step sets the overall timescale (Lecture 26) [36, 14]. The Coulomb barrier for $p + p$ is lower than for D–T, but the weak interaction makes *pp* enormously slower than D–T at comparable temperature. This contrast explains why stars are stable for gigayears while fusion devices require external heating and choose the fastest exothermic strong-interaction channel available [2, 36].

25.8.4 Electron screening in laboratory versus stellar plasmas

Laboratory fusion experiments on solid targets require electron-screening corrections before extrapolating to bare-nucleus astrophysical rates [36, 14]. In stellar plasmas, weak Debye screening enhances rates by $f_{sc} \approx \exp(U_e/k_BT)$ without replacing the Coulomb barrier by a constant lower value at all radii [36]. The lecture contact-barrier estimate is the zero-screening baseline; Solar Fusion III curates recommended S-factors together with screening uncertainties for each light-ion channel [36, 6]. Sub-barrier heavy-ion fusion for SHE synthesis adds coupled-channel and quasifission complications beyond either screening correction [55, 56].

25.9 Deeper reading: Coulomb barriers, coupled channels, and SHE

The contact Coulomb barrier $Z_1 Z_2 e^2/(4\pi\varepsilon_0(R_1 + R_2))$ sets the scale for light-ion fusion and for heavy-ion synthesis. Sub-barrier fusion is enhanced or suppressed by deformation and transfer channels in coupled-channel models. Quasifission can remove flux before a compound nucleus forms, which is why SHE cross sections are not simple tunneling estimates.

Stellar fusion of light nuclei (Lecture 26) uses the same barrier physics at much lower energies, packaged as S-factors. Estimate barriers for $^{12}C + {}^{12}C$ and for a ^{48}Ca-induced SHE reaction to feel the gap between astrophysics and superheavy programmes.

Barrier distributions

Experimental fusion barrier distributions are extracted from derivatives of excitation functions. Peaks correspond to couplings to rotational or vibrational channels. Comparing a spherical projectile on a deformed target versus the reverse kinematics tests the coupling picture.

Survival probability

Forming a compound nucleus is not enough: it must survive fission to reach the ground state by neutron evaporation. Shell-stabilised residues have higher survival. That competition, not the Coulomb barrier alone, limits SHE synthesis rates.

Barrier comparison

Estimate contact barriers for pp, $^{12}C + {}^{12}C$, and $^{48}Ca + {}^{248}Cm$. The progression from kilo-electronvolt stellar scales (after tunneling) to hundreds of mega-electronvolts for SHE entrance channels shows why astrophysics and superheavy synthesis share formulas but not facilities.

Discuss one coupled-channel effect qualitatively: a deformed target presents a lower barrier at the tips, enhancing sub-barrier fusion. Then note quasifission as a competing dynamical process that can still suppress residue formation. Lecture 26 next packages the light-ion end of this physics into stellar rates; SHE research lives at the opposite extreme of the same tunneling idea.

Fusion tunneling unifies stellar burning and superheavy synthesis. Coupled channels and quasifission decide how far simple barrier estimates can be trusted.

In practice, rework one numerical estimate from the lecture with a different parameter set and compare the shift to the quoted experimental uncertainty. If the shift exceeds the uncertainty, the estimate is teaching a scale, not delivering a fit. That distinction keeps theoretical cartoons honest when they sit beside evaluated data tables and modern many-body calculations. Along the course arc, write one sentence linking this chapter backward and one forward: the discipline of those two sentences is as important as any single formula. Examination problems rarely ask for a literature review, but they do ask for the scale argument that literature reviews presuppose.

Synthesis problem

Write a half-page note that (i) restates the chapter's central scale or selection rule in one equation, (ii) evaluates it numerically for one concrete nucleus or energy, and (iii) names the neighbouring lecture that uses the result. Keep the note free of research-frontier speculation; the goal is fluency with the lecture physics itself. If an assumption fails (Born approximation, spherical mean field, allowed transition, one-dimensional barrier), state the failure mode in a final sentence. That habit converts passive reading into examination-ready understanding and makes the later research-frontier box read as an extension rather than a replacement.

Research frontier: sub-barrier fusion and superheavy synthesis

Coulomb-barrier tunneling (Lecture 25) governs both stellar burning and heavy-ion fusion used to synthesise superheavy elements. Coupled-channel models show that deformation and transfer channels can enhance or suppress sub-barrier fusion relative to a single-barrier estimate [55]. Synthesis of elements beyond oganesson requires new projectile–target combinations and higher beam intensities; theory and experiment remain tightly coupled [54]. For light systems, recommended astrophysical S-factors are curated in Solar Fusion III, providing a complementary low-energy application of the same tunneling physics [36]. Uncertainties that still limit SHE programmes include fusion-evaporation survival probabilities, the role of quasifission, and incomplete knowledge of optimal excitation energies. The lecture's barrier-penetration estimate is the right starting point before those coupled-channel and dynamical complications are added.

In SHE synthesis, the competition between fusion and quasifission means that a larger geometric overlap does not automatically yield more evaporation residues. Coupled-channel enhancement at sub-barrier energies can be offset by dynamical breakup of the composite system. That tension is the modern reading of the lecture tunneling estimate.

Quasifission can erase a coupled-channel fusion gain, which is why SHE cross sections remain hard to predict from barrier height alone.

Proposal writing for new SHE reactions therefore mixes tunneling estimates with dynamical transport simulations.

Student directions. (1) Barrier estimate: compute the contact Coulomb barrier for $^{12}\mathrm{C}+{}^{12}\mathrm{C}$ and for $^{48}\mathrm{Ca}+{}^{248}\mathrm{Cm}$. (2) Qualitative reasoning: explain why a deformed target can increase the sub-barrier fusion probability. (3) Literature status: summarise attempts to synthesise $Z = 119$ or 120 from a recent review [55].

25.10 Problems with solutions

Problem 25.1 — Q from B/A for D + T. The reaction $\mathrm{D}+\mathrm{T} \to {}^4\mathrm{He}+n$ has $Q = 17.6\,\mathrm{MeV}$. If the total binding energy of D + T is B_i and that of ^{4}He is $B_\alpha = 28.3\,\mathrm{MeV}$ (the neutron is unbound as a free particle, $B_n = 0$), find B_i from $Q = B_f - B_i$ and comment [6, 1].

Solution 25.1.

Step 1. Final binding: $B_f = B_\alpha + B_n = 28.3\,\mathrm{MeV}$.

Step 2.

$$Q = B_f - B_i \quad \Rightarrow \quad B_i = B_f - Q = 28.3 - 17.6 = 10.7\,\text{MeV}.$$

That matches $B(\text{d}) + B(\text{t}) \approx 2.2 + 8.5 = 10.7\,\text{MeV}$. Fusion increases total binding by 17.6 MeV.

Problem 25.2 — Neon fusion Q structure. Starting from rest energies expressed with binding energies, derive $Q = E_B(^{40}\text{Ca}) - 2E_B(^{20}\text{Ne})$ for $^{20}\text{Ne} + {}^{20}\text{Ne} \to {}^{40}\text{Ca} + \gamma$. Then use $M_a(^{20}\text{Ne}) = 19.992\,440\,1762\,\text{u}$, $M_a(^{40}\text{Ca}) = 39.962\,590\,863\,\text{u}$, and $1\,\text{u}c^2 = 931.494\,\text{MeV}$ to check the lecture's graph-read estimate.

Solution 25.2.

Step 1. Follow Eqs. (25.7)–(25.9): nucleon mass terms cancel, leaving $Q = E_B(\text{Ca}) - 2E_B(\text{Ne})$.

Step 2. Atomic masses may be used directly because 20 atomic electrons occur on each side:

$$\Delta M = 2(19.992\,440\,1762) - 39.962\,590\,863 = 0.022\,289\,4894\,\text{u}.$$

Step 3.

$$Q = \Delta M c^2 = (0.022\,289\,4894)(931.494) = 20.76\,\text{MeV}.$$

The corresponding exact difference in binding per nucleon is $Q/40 = 0.519\,\text{MeV}$, not the schematic 0.58 MeV. The original derivation is correct; only its low-resolution graph estimate needed correction [27].

Problem 25.3 — Coulomb barrier for $^{12}\text{C} + {}^{12}\text{C}$. Estimate $V_{\max}$ for two ^{12}C nuclei. Use $r = 1.2\,A^{1/3}\,\text{fm}$ and $e^2/(4\pi\varepsilon_0) = 1.44\,\text{MeV}\cdot\text{fm}$.

Solution 25.3.

Step 1.

$$r = 1.2 \times 12^{1/3} = 1.2 \times 2.29 \approx 2.75\,\text{fm}, \qquad r_1 + r_2 = 5.50\,\text{fm}.$$

Step 2.

$$V_{\max} = \frac{6 \times 6 \times 1.44}{5.50} = \frac{51.84}{5.50} \approx 9.4\,\text{MeV}.$$

Carbon burning in massive stars must tunnel a $\sim 10\,\text{MeV}$ barrier [6, 14].

Problem 25.4 — pp contact scale and solar k_BT. Show that $V_{\text{contact}}(pp) \approx 0.72\,\text{MeV}$ if the pedagogical contact radius is chosen as $r_1 + r_2 = 2\,\text{fm}$. At $T = 1.5 \times 10^7\,\text{K}$, compute k_BT in keV and the ratio V_{contact}/k_BT.

Solution 25.4.

Step 1.

$$V_{\text{contact}} = \frac{1.44\,\text{MeV}\cdot\text{fm}}{2\,\text{fm}} = 0.72\,\text{MeV}.$$

Step 2.

$$k_BT = (8.617\times10^{-5}\,\text{eV/K})\times1.5\times10^{7}\,\text{K} \approx 1.29\times10^{3}\,\text{eV} \approx 1.3\,\text{keV}.$$

Step 3.

$$\frac{V_{\text{contact}}}{k_BT} = \frac{720\,\text{keV}}{1.3\,\text{keV}} \approx 550.$$

The proton has no sharp classical surface, so this is a radius-dependent Coulomb scale, not a uniquely measured barrier. Its enormous ratio to k_BT still demonstrates why solar fusion requires tunneling [6, 2].

Problem 25.5 — Temperature for $150\,\text{keV}$. What temperature T gives $\frac{3}{2}k_BT = 150\,\text{keV}$? Compare with the solar core temperature.

Solution 25.5.

$$k_BT = 100\,\text{keV}, \qquad T = \frac{100\times10^{3}}{8.617\times10^{-5}} \approx 1.16\times10^{9}\,\text{K}.$$

This is about 80 times hotter than the solar core ($\sim 1.5\times10^{7}\,\text{K}$). Laboratory fusion concepts operate near 10^8 K with tunneling still essential [1, 14].

Problem 25.6 — Fusion vs fission energy per nucleon. Compare Q/A for (a) fission with $Q = 200\,\text{MeV}$, $A = 236$; (b) D + T fusion with $Q = 17.6\,\text{MeV}$ and 5 nucleons in the reactants. Which releases more energy per nucleon?

Solution 25.6.

(a)

$$\frac{Q}{A} = \frac{200}{236} \approx 0.85\,\text{MeV/nucleon}.$$

(b)

$$\frac{Q}{A} = \frac{17.6}{5} = 3.52\,\text{MeV/nucleon}.$$

Fusion of light nuclei releases several times more energy *per nucleon* than fission, which is why fusion fuels are attractive despite the barrier challenge [6, 1].

Problem 25.7 — Contact barrier for $p+p$. Estimate $V_{\text{contact}} = e^2/[4\pi\varepsilon_0(r_p+r_p)]$ with $r_p \approx 1\,\text{fm}$ and $e^2/(4\pi\varepsilon_0) = 1.44\,\text{MeV fm}$. Compare with k_BT in the solar core ($\sim 1\,\text{keV}$).

Solution 25.7.

$$V_{\text{contact}} \approx \frac{1.44}{2} \approx 0.72\,\text{MeV} \sim 700\,\text{keV},$$

hundreds of times $k_B T$. Fusion proceeds only by Gamow tunneling through the Coulomb barrier [6, 14].

Chapter 26

Nuclear Fusion II

"The fusion rate is $n_1 n_2 \langle \sigma v \rangle$: density squared times a thermal average that hides the Gamow peak."

G. Gamow; stellar thermonuclear rates

Lecture 25 introduced fusion energetics and the Coulomb barrier. This chapter derives the reaction rate from first principles (beam–target and plasma), derives the standard low-energy S-factor form, forms the thermal average, and identifies the Gamow peak that dominates thermonuclear burning. The discussion closes with the practical preference for D+T in laboratory fusion [6, 1, 14, 2].

26.1 Reaction rate from flux and cross section

Cross sections alone do not determine stellar energy generation or reactor yield. One must fold $\sigma(v)$ with the relative-velocity distribution of the fuel. In a beam–target experiment that distribution is nearly monoenergetic; in a plasma it is Maxwellian. Both cases reduce to a rate density of the form $n_1 n_2 \langle \sigma v \rangle$ once the average is defined [6, 14].

26.1.1 Beam on a fixed target

Consider a beam of nuclei Y with number density n_Y and speed v incident on a target containing nuclei X. The fusion (or reaction) cross section σ is an effective area: if a Y trajectory would pierce the area σ centred on an X, a reaction occurs with the probability built into the definition of σ.

In a time $\mathrm{d}t$, each X is exposed to all Y particles that lie in a cylinder of base area σ and length $v\,\mathrm{d}t$. The volume of that cylinder is $\sigma v\,\mathrm{d}t$, so the number of Y nuclei in it is

$$n_Y\,\sigma\,v\,\mathrm{d}t. \tag{26.1}$$

The reaction rate *per target nucleus* X is therefore

$$\lambda_X = n_Y\,\sigma\,v. \tag{26.2}$$

If the number density of X is n_X, the number of reactions per unit volume per unit time is

$$R = n_X\, n_Y\, \sigma\, v. \tag{26.3}$$

This is the fundamental connection between microscopic cross section and macroscopic rate.

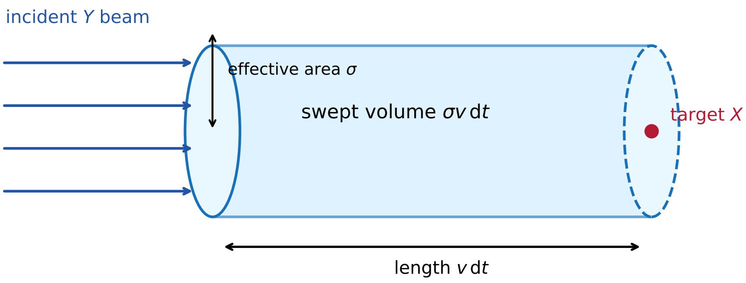


Figure 26.1: Vector redraw of the swept-volume construction. In time $\mathrm{d}t$, the effective reaction area σ sweeps the relative volume $\sigma v\,\mathrm{d}t$ past one target nucleus.

26.1.2 Thermonuclear plasma

In a fusion plasma both species move randomly; there is no laboratory "beam" and no fixed target. The same geometry applies in the *relative* frame: for any chosen X, construct the cylinder using the relative velocity v of approaching Y nuclei. Averaging over the thermal distribution of relative velocities replaces σv by its mean:

$$\boxed{r_{XY} = \frac{n_X n_Y}{1+\delta_{XY}}\,\langle\sigma v\rangle.} \tag{26.4}$$

For $X \neq Y$, there are $N_X N_Y$ unlike pairs in a volume. For $X = Y$, the number of unordered pairs is $N_X(N_X-1)/2 \simeq N_X^2/2$, not N_X^2; this proves the $1/(1+\delta_{XY})$ factor and identifies the double counting it removes. Once a fusion event occurs the participating nuclei are consumed; the instantaneous rate law is unchanged because the densities evolve in time.

The dimensions provide a useful check:

$$[n_X n_Y \langle\sigma v\rangle] = \mathrm{m}^{-6}\,(\mathrm{m}^2)(\mathrm{m\,s}^{-1}) = \mathrm{m}^{-3}\,\mathrm{s}^{-1}, \tag{26.5}$$

exactly reactions per unit volume per unit time. The reactivity $\langle\sigma v\rangle$ has SI units $\mathrm{m}^3\,\mathrm{s}^{-1}$ [6, 1, 14].

Key idea

Beam–target: $r = n_X n_Y \sigma v$ from the cylinder volume $\sigma v\,\mathrm{d}t$. Plasma: $r_{XY} = n_X n_Y\langle\sigma v\rangle/(1+\delta_{XY})$.

26.2 Thermal averaging of the fusion reactivity

For each Cartesian component, thermal velocities of the two species are independent Gaussians,

$$f_i(v_{i,x}) = \left(\frac{m_i}{2\pi k_B T}\right)^{1/2} \exp\left(-\frac{m_i v_{i,x}^2}{2k_B T}\right). \tag{26.6}$$

The relative component $v_x = v_{1,x} - v_{2,x}$ is therefore Gaussian with variance

$$\langle v_x^2 \rangle = k_B T \left(\frac{1}{m_1} + \frac{1}{m_2}\right) = \frac{k_B T}{\mu}, \qquad \frac{1}{\mu} = \frac{1}{m_1} + \frac{1}{m_2}. \tag{26.7}$$

Multiplying the three component distributions gives the isotropic relative-velocity density

$$F(\boldsymbol{v}) = \left(\frac{\mu}{2\pi k_B T}\right)^{3/2} \exp\left(-\frac{\mu v^2}{2k_B T}\right), \qquad \int F(\boldsymbol{v})\, \mathrm{d}^3 v = 1. \tag{26.8}$$

Integrating over solid angle, $\mathrm{d}^3 v = 4\pi v^2 \mathrm{d}v$, yields the Maxwell–Boltzmann *relative-speed* distribution

$$f(v)\, \mathrm{d}v = 4\pi v^2 \left(\frac{\mu}{2\pi k_B T}\right)^{3/2} \exp\left(-\frac{\mu v^2}{2k_B T}\right) \mathrm{d}v. \tag{26.9}$$

It is μ, not either individual mass, that must appear. Writing $f(v)\, \mathrm{d}v$ for the probability that the relative speed lies in $[v, v + \mathrm{d}v]$,

$$\langle \sigma v \rangle = \int_0^\infty \sigma(v)\, v\, f(v)\, \mathrm{d}v. \tag{26.10}$$

To change variables to relative kinetic energy,

$$E = \frac{1}{2}\mu v^2, \qquad v\, \mathrm{d}v = \frac{\mathrm{d}E}{\mu}, \qquad v^2 = \frac{2E}{\mu}. \tag{26.11}$$

The Maxwellian speed weight contains $v^2 \exp(-\mu v^2/2k_B T)\, \mathrm{d}v$. Rewrite

$$v^2\, \mathrm{d}v = v\,(v\, \mathrm{d}v) = \sqrt{\frac{2E}{\mu}}\, \frac{\mathrm{d}E}{\mu} = \frac{\sqrt{2}}{\mu^{3/2}}\, E^{1/2}\, \mathrm{d}E,$$

or equivalently, combining with the extra factor of v in $\sigma(v)\, v$,

$$v^3\, \mathrm{d}v = v^2\,(v\, \mathrm{d}v) = \frac{2E}{\mu} \cdot \frac{\mathrm{d}E}{\mu} = \frac{2E\, \mathrm{d}E}{\mu^2}.$$

Inserting the normalized relative Maxwellian into Eq. (26.10) and collecting constants yields

$$\langle \sigma v \rangle = \left(\frac{8}{\pi\mu}\right)^{1/2} \frac{1}{(k_B T)^{3/2}} \int_0^\infty \sigma(E)\, E\, e^{-E/k_B T}\, \mathrm{d}E. \tag{26.12}$$

The dimensions are $(\mathrm{kg}^{-1/2} \mathrm{J}^{-3/2})(\mathrm{m}^2\, \mathrm{J}^2) = \mathrm{m}^3\, \mathrm{s}^{-1}$, as required.

26.3 Low-energy fusion cross section

Two ingredients dominate $\sigma(v)$ at thermonuclear energies ($E \sim \text{keV} \ll V_{\max}$):

26.3.1 Barrier penetration

For two charges, define the Sommerfeld parameter

$$\eta(E) = \frac{Z_1 Z_2 e^2}{4\pi\varepsilon_0 \hbar v} = Z_1 Z_2 \alpha \sqrt{\frac{\mu c^2}{2E}}. \tag{26.13}$$

The Coulomb penetrability is

$$P_{\text{pen}} \propto e^{-2\pi\eta} = \exp\left[-\sqrt{\frac{E_G}{E}}\right], \qquad E_G = 2\mu c^2 (\pi\alpha Z_1 Z_2)^2. \tag{26.14}$$

There is no convention-dependent factor left: the full exponent is $2\pi\eta = \sqrt{E_G/E}$. In practical units,

$$E_G = 0.978\, Z_1^2 Z_2^2 \mu_{\text{u}}\,\text{MeV}, \qquad 2\pi\eta = 31.29\, Z_1 Z_2 \sqrt{\frac{\mu_{\text{u}}}{E_{\text{keV}}}}, \tag{26.15}$$

where μ_{u} is the reduced mass in atomic mass units and E_{keV} is the centre-of-mass energy in keV.

26.3.2 Geometrical and quantum low-energy factor

The original lecture motivates the entrance-channel factor through impact parameters. At low energy the reduced de Broglie wavelength $\bar{\lambda} = \hbar/(\mu v) = 1/k$ can be large compared with the nuclear radius R. Quantise the orbital angular momentum semiclassically, $L \simeq \mu v b = (\ell + \frac{1}{2})\hbar$. The exact partial-wave reaction cross section has the structure

$$\sigma_{\text{r}}(E) = \frac{\pi}{k^2} \sum_{\ell=0}^{\infty} (2\ell+1) T_\ell(E), \qquad k = \frac{\sqrt{2\mu E}}{\hbar}, \tag{26.16}$$

where $0 \le T_\ell \le 1$ is a transmission coefficient. Thus each partial-wave prefactor carries $\pi/k^2 \propto 1/E \propto 1/v^2$, but T_ℓ contains the dominant Coulomb penetrability and nuclear dynamics; the prefactor alone is not the fusion cross section. The slide's annular construction recovers the same scaling: Quantise the impact parameter by angular momentum $L = \ell\hbar = \mu v b_\ell$, so

$$b_\ell = \frac{\ell\hbar}{\mu v}. \tag{26.17}$$

The annular area between ℓ and $\ell + 1$ is

$$A_\ell = \pi b_{\ell+1}^2 - \pi b_\ell^2 = \frac{\pi\hbar^2}{\mu^2 v^2}(2\ell+1). \tag{26.18}$$

Summing sharp-cutoff annuli to $\ell_{\max} \approx \mu v R/\hbar$ gives the heuristic geometric estimate

$$\sigma_{\text{geom}} \sim \pi\left(R + \frac{\hbar}{\mu v}\right)^2. \tag{26.19}$$

At sufficiently low energy the s-wave often dominates. For a *nonresonant binary charged-particle reaction*, it is useful to factor the cross section into a kinematic $1/E$, Coulomb penetration, and residual nuclear factor. This does not imply that the observable cross section itself rises as $1/E$, because the exponential suppression overwhelms that prefactor as $E \to 0$.

26.3.3 Combined low-energy form

The standard nuclear-astrophysics parametrisation for such a reaction is

$$\boxed{\sigma(E) = \frac{S(E)}{E} \exp\left[-\sqrt{\frac{E_G}{E}}\right]}. \tag{26.20}$$

where $S(E)$ is the slowly varying **astrophysical S-factor** and $[S] = [\sigma E]$, commonly keV b. This form applies to nonresonant binary charged-particle reactions away from thresholds. A resonance can make $S(E)$ vary rapidly, and then a Breit–Wigner or multichannel treatment must replace the “slowly varying S” approximation [6, 14, 1].

Remark

The raw lecture’s statement that a large de Broglie wavelength by itself “increases” the fusion cross section is incomplete: the transmission coefficient contains the overwhelming factor $e^{-2\pi\eta}$. The standard $S(E)/E$ factorization keeps the kinematic $1/E$ and the Coulomb suppression separate.

26.4 The Gamow peak

Insert Eq. (26.20) into Eq. (26.12). The factor E in the Maxwellian integral cancels the $1/E$ in σ:

$$\langle \sigma v \rangle = \left(\frac{8}{\pi\mu}\right)^{1/2} \frac{1}{(k_B T)^{3/2}} \int_0^\infty S(E) e^{-\Phi(E)}\, \mathrm{d}E, \quad \Phi(E) = \frac{E}{k_B T} + \sqrt{\frac{E_G}{E}}. \tag{26.21}$$

The first term in Φ penalizes the rare high-energy Maxwell tail; the second penalizes low-energy Coulomb penetration. For a slowly varying $S(E)$, the maximum of the integrand is the minimum of Φ. Every calculus step is

- Differentiate:
$$\Phi'(E) = \frac{1}{k_B T} - \frac{1}{2}\sqrt{E_G}\, E^{-3/2}.$$
- Set $\Phi'(E_0) = 0$:
$$\frac{1}{k_B T} = \frac{\sqrt{E_G}}{2E_0^{3/2}} \implies E_0^{3/2} = \frac{\sqrt{E_G}\, k_B T}{2}.$$
- Raise both sides to 2/3:
$$\boxed{E_0 = \left[\frac{E_G (k_B T)^2}{4}\right]^{1/3}}. \tag{26.22}$$

At the stationary point, $\sqrt{E_G/E_0} = 2E_0/(k_BT)$, hence

$$\Phi(E_0) = \frac{3E_0}{k_BT}. \tag{26.23}$$

To obtain the width, differentiate again and use the stationary-point condition:

$$\Phi''(E) = \frac{3}{4}\sqrt{E_G}E^{-5/2}, \qquad \Phi''(E_0) = \frac{3}{2E_0k_BT}. \tag{26.24}$$

Taylor expansion through second order gives

$$e^{-\Phi(E)} \simeq e^{-3E_0/k_BT} \exp\left[-\frac{3(E-E_0)^2}{4E_0k_BT}\right]. \tag{26.25}$$

Define Δ as the full width between points where this Gaussian is $1/e$ of its maximum. Setting $E - E_0 = \pm\Delta/2$ gives

$$\frac{3(\Delta/2)^2}{4E_0k_BT} = 1 \quad \Longrightarrow \quad \boxed{\Delta = 4\sqrt{\frac{E_0k_BT}{3}}}. \tag{26.26}$$

Thus the effective nonresonant window is approximately $E_0 - \Delta/2 < E < E_0 + \Delta/2$. With $T_9 = T/(10^9\,\mathrm{K})$, $\mu_\mathrm{u} = \mu/\mathrm{u}$, and energies in MeV,

$$E_0 = 0.1220\left(Z_1^2Z_2^2\mu_\mathrm{u}T_9^2\right)^{1/3}\ \mathrm{MeV}, \tag{26.27}$$

$$\Delta = 0.2368\left(Z_1^2Z_2^2\mu_\mathrm{u}\right)^{1/6}T_9^{5/6}\,\mathrm{MeV}. \tag{26.28}$$

These constants use $\alpha = 1/137.036$, $1\,\mathrm{u}\,c^2 = 931.494\,\mathrm{MeV}$, and $k_B = 8.61733\times10^{-11}\,\mathrm{MeV\,K^{-1}}$.

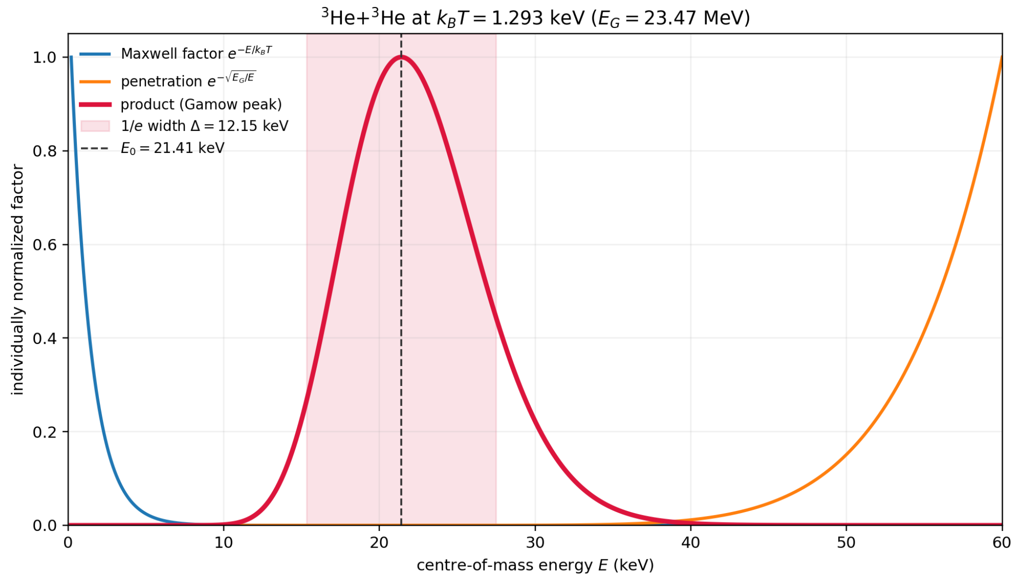


Figure 26.2: Maxwell factor, Coulomb penetration, and their product for $^3\mathrm{He}+{}^3\mathrm{He}$ at $k_BT = 1.293\,\mathrm{keV}$. With $Z_1Z_2 = 4$ and $\mu \simeq 1.50\,\mathrm{u}$, $E_G = 23.47\,\mathrm{MeV}$, $E_0 = 21.40\,\mathrm{keV}$, and the full $1/e$ width is $\Delta \simeq 12.15\,\mathrm{keV}$. Curves are individually normalized to show their competing energy dependence.

If $S(E) \simeq S(E_0)$, the Gaussian integral also yields the useful steepest-descent estimate

$$\langle \sigma v \rangle \simeq \left(\frac{32E_0}{3\mu}\right)^{1/2} \frac{S(E_0)}{k_B T} \exp\left(-\frac{3E_0}{k_B T}\right). \tag{26.29}$$

Key idea

Gamow peak: $E_0 = [E_G(k_BT)^2/4]^{1/3}$, $\Delta = 4\sqrt{E_0 k_B T/3}$, and $\Phi(E_0) = 3E_0/(k_BT)$. Fusion rates are exponentially temperature sensitive through this window.

The saddle-point formulas above assume a binary, nonresonant reaction with $S(E)$ nearly constant across the window. If a narrow resonance lies in or near that window, the rate is controlled by the resonance energy and strength rather than by $S(E_0)$. The D–T reaction is strongly enhanced by a broad compound-nuclear resonance, so fitted or evaluated reactivities are used for quantitative reactor calculations. The triple-alpha process is outside the binary formula altogether: it proceeds sequentially through the short-lived ^{8}Be system and the Hoyle-state resonance in ^{12}C [1, 14].

As temperature increases, E_0 shifts upward, the peak height grows rapidly, and $\langle \sigma v \rangle$ rises by many orders of magnitude. At extremely high T the simple low-energy form of σ eventually fails and other physics (resonances, different partial waves) enters; for solar and most laboratory fusion discussions the Gamow picture is the right first tool [14, 6, 2].

26.5 Putting it together: plasma reactivity

The volumetric reaction rate and locally deposited power density are

$$r_{XY} = \frac{n_X n_Y}{1+\delta_{XY}} \langle \sigma v \rangle (T), \tag{26.30}$$

$$p_{\text{dep}} = r_{XY}\, Q_{\text{dep}}. \tag{26.31}$$

Here $Q_{\text{dep}} = Q - \langle E_\nu \rangle - \cdots$ excludes energy carried away by weakly interacting particles or other escaping products; in a blanket, escaping fusion neutrons can instead be recovered as heat. The Salpeter factor introduced in Lecture 25 applies only in the weak-screening, Debye–Hückel limit

$$\epsilon_{\text{sc}} \equiv \frac{Z_1 Z_2 e^2}{4\pi\varepsilon_0 R_D k_B T} \ll 1, \qquad N_D \equiv \frac{4\pi}{3} R_D^3 \sum_s n_s \gg 1. \tag{26.32}$$

The first condition says that the screening free-energy correction is small compared with k_BT; the second ensures that a Debye sphere contains many particles. Strongly coupled or degenerate plasmas require a different screening treatment [36]. Design of a fusion reactor therefore requires:

(i) high density product $n_X n_Y$ (or n^2 for a single species);

(ii) high temperature so that $\langle \sigma v \rangle$ is appreciable;

(iii) sufficient confinement time for the plasma to react before cooling or disassembling (Lawson criterion: $n\tau_E$ above a threshold depending on T and the fuel cycle).

26.5.1 Why deuterium–tritium fuel?

Among light-ion reactions, deuterium–tritium fusion

$$^{2}\mathrm{H} + {}^{3}\mathrm{H} \;\to\; {}^{4}\mathrm{He} + n + 17.6\,\mathrm{MeV} \tag{26.33}$$

has the largest $\langle\sigma v\rangle$ at the lowest temperature of any practical fuel cycle: $Z_1 Z_2 = 1$ (same as *pp* or DD) but a more favourable nuclear matrix element and a broad ^{5}He compound-nuclear resonance. Consequently, the slowly varying-S Gamow approximation explains the Coulomb trend but is not a precision model of the D–T reactivity. Coulomb barriers for D + ^{3}He or pure proton fuels are higher or the weak interaction intervenes (*pp*), making them harder for terrestrial reactors. Stellar cores burn hydrogen via the slow *pp* chain precisely because weak interactions set the pace; stars have billions of years [6, 1, 14]. The rare solar hep branch,

$$^{3}\mathrm{He} + p \to {}^{4}\mathrm{He} + e^{+} + \nu_e, \tag{26.34}$$

produces the highest-energy standard solar neutrinos, with endpoint $E_\nu \simeq 18.77\,\mathrm{MeV}$, although its flux is tiny [36]. Two-body momentum conservation divides the 17.6 MeV into about 3.5 MeV for the α and 14.1 MeV for the neutron. Deuterium alone can burn through

$$\mathrm{D} + \mathrm{D} \to \mathrm{T} + p + 4.03\,\mathrm{MeV}, \tag{26.35}$$

$$\mathrm{D} + \mathrm{D} \to {}^{3}\mathrm{He} + n + 3.27\,\mathrm{MeV}, \tag{26.36}$$

with comparable low-energy branches, but substantially lower reactivity than D–T at reactor temperatures [1].

Caution

High T alone is not enough: without density and confinement, a hot plasma radiates and expands before fusing. The triple product $nT\tau_E$ is the usual figure of merit for ignition studies.

26.6 Summary of the fusion rate theory

Collecting the chain of reasoning:

(1) Geometry: each X sweeps a volume $\sigma v\,\mathrm{d}t$ relative to the Y gas; unordered-pair counting gives $r_{XY} = n_X n_Y \langle\sigma v\rangle/(1 + \delta_{XY})$.

(2) Thermal average: replace σv by $\langle\sigma v\rangle = \int \sigma(v) v f(v)\,\mathrm{d}v$.

(3) Nonresonant binary charged-particle cross section: $\sigma = S(E)E^{-1}e^{-\sqrt{E_G/E}}$; resonances require their own energy-dependent treatment.

(4) Maxwell × Gamow ⇒ Gamow peak at $E_0 \propto T^{2/3}$.

(5) Deposited power density $p_{\mathrm{dep}} = r_{XY} Q_{\mathrm{dep}}$; D + T maximises reactivity for laboratory T.

Takeaway

- $r_{XY} = n_X n_Y \langle \sigma v \rangle / (1 + \delta_{XY})$, with units $\mathrm{m^{-3}\,s^{-1}}$.
- $\langle \sigma v \rangle = \int \sigma(v) v f(v)\,\mathrm{d}v$ (Maxwellian).
- Barrier penetration $e^{-2\pi\eta} = e^{-\sqrt{E_G/E}}$.
- For nonresonant binary charged-particle reactions, $\sigma(E) = S(E) E^{-1} e^{-\sqrt{E_G/E}}$.
- Gamow peak: $E_0 = [E_G (k_B T)^2/4]^{1/3}$, $\Delta = 4\sqrt{E_0 k_B T/3}$.
- D + T is the most accessible laboratory fuel cycle, aided by broad resonant nuclear structure; triple-alpha is a sequential resonant process outside the binary formula.

26.6.1 Maxwellian rates and the astrophysical S-factor

Charged-particle fusion rates fold a Maxwellian tail with Gamow penetration. Factoring $\sigma(E) = S(E)E^{-1}e^{-2\pi\eta}$ defines a slowly varying astrophysical S-factor that can be extrapolated from laboratory energies to the stellar Gamow peak; modern evaluations (Solar Fusion) tabulate recommended $S(0)$ values and uncertainties for the solar *pp* chain and CNO cycles [36, 14, 6].

26.7 Deeper physics: Gamow peak, $\langle \sigma v \rangle$, and stellar rates

26.7.1 Thermal average and the Gamow peak

In a plasma at temperature T, the fusion rate per unit volume for species X and Y is $r = n_X n_Y \langle \sigma v \rangle$, where

$$\langle \sigma v \rangle = \int_0^\infty \sigma(v)\, v\, f(v)\,\mathrm{d}v \tag{26.37}$$

and $f(v)$ is the Maxwell–Boltzmann distribution of relative speeds at temperature T [14, 6]. At low energy, $\sigma(E) \propto E^{-1} e^{-b/\sqrt{E}}$ (Gamow factor), so the integrand $\sigma v f(v)$ peaks at a finite energy E_0 well below the Coulomb barrier. This **Gamow peak** is where most stellar fusion occurs: neither the mean thermal energy $k_B T$ nor the barrier height alone sets the rate [36, 2].

Gamow peak for $p + p$ at solar core temperature

At $T = 15 \times 10^6\,\mathrm{K}$, $k_B T \approx 1.3\,\mathrm{keV}$. For $p + p$ fusion, the Gamow peak sits near $E_0 \sim 6$–$7\,\mathrm{keV}$, roughly five times $k_B T$ [14, 36]. The cross section at E_0 is extraordinarily small ($\sim 10^{-47}\,\mathrm{cm^2}$), yet the solar core density ($\sim 150\,\mathrm{g\,cm^{-3}}$) and long confinement time yield a stable $4 \times 10^{26}\,\mathrm{W}$ luminosity. The weak $p + p$ step sets the overall timescale [6, 36].

26.7.2 The S-factor and practical rate formula

Laboratory data are quoted as the astrophysical S-factor,

$$\sigma(E) = \frac{S(E)}{E} \exp\left(-\sqrt{\frac{E_G}{E}}\right), \qquad E_G = 2\mu(\pi\alpha Z_1 Z_2 \hbar c)^2, \tag{26.38}$$

which removes the steep Gamow energy dependence [14, 36]. Near the Gamow peak, a useful approximation is

$$\langle\sigma v\rangle \approx \sqrt{\frac{8}{3\pi}}\,\frac{S(E_0)}{E_0^2}\,(k_BT)^{3/2}\,\exp(-3E_0/k_BT)\,, \tag{26.39}$$

showing the strong T dependence that makes CNO burning dominate in more massive stars [6, 1]. Recommended $S(E)$ values and uncertainties are maintained in Solar Fusion III [36].

Key idea

Closing the arc: decay, reactions, fission, fusion. The Gamow exponent first appeared in α decay (Lectures 16–17), reappears as fusion tunneling (Lecture 25), and sets stellar lifetimes through $\langle\sigma v\rangle$. Fission and fusion bracket the binding-energy curve: heavy nuclei split, light nuclei merge, both releasing Q when the products are more tightly bound [6, 14, 2].

26.7.3 Resonances and the Hoyle state

When a narrow resonance sits in the Gamow window, the thermal average is dominated by the Breit–Wigner contribution [14, 6]. The Hoyle state in ^{12}C at $E_r \approx 7.65$ MeV makes the 3α process possible in red giants: without it, carbon would not form in stars [2, 14]. The resonance contribution adds to, not replaces, the non-resonant $\langle\sigma v\rangle$ of (26.37). Screening in stellar plasmas enhances all channels by $f_{\mathrm{sc}} \approx \exp(U_e/k_BT)$ with $U_e \sim 1$–2 keV at solar core conditions [36, 14].

26.7.4 CNO cycle temperature sensitivity

The CNO-I loop regenerates ^{12}C and converts four protons to ^{4}He with the same net energy release as the *pp* chain [1, 36]. Because CNO reactions involve higher Z projectiles, their Coulomb barriers are higher and $\langle\sigma v\rangle$ is more temperature sensitive ($\sim T^{15}$ to T^{20} near main-sequence temperatures) than the *pp* chain ($\sim T^4$) [36, 6]. Massive stars with $T_{\mathrm{core}} \gtrsim 1.7\times10^7$ K therefore switch to CNO-dominated energy generation. Precision S-factors for $^{14}\mathrm{N}(p,\gamma)^{15}\mathrm{O}$ and related branches remain active research topics because solar and stellar models propagate their uncertainties into age and metallicity determinations [36, 53].

Screening corrections in laboratory targets enhance low-energy fusion relative to bare-nucleus S-factors; stellar plasma screening differs again. When comparing a measured low-energy cross section to Solar Fusion III recommendations, check whether screening was unfolded. Electron screening is a systematic, not a curiosity, at Gamow-window energies.

26.8 Further physics from the literature

Astrophysical S-factor. Writing $\sigma(E) = S(E)E^{-1}e^{-b/\sqrt{E}}$ removes the dominant Coulomb energy dependence so that $S(E)$ is nearly constant or slowly varying. Extrapolating $S(E)$ from laboratory energies (hundreds of keV) down to the Gamow window (tens of keV in the Sun) is a central task of nuclear astrophysics [14, 6, 36].

The solar *pp* chain. The initiating weak reaction and rapid deuterium capture are

$$p + p \to d + e^+ + \nu_e, \qquad Q_{\text{nuc}} = 0.420\,\text{MeV}, \tag{26.40}$$

$$d + p \to {}^3\text{He} + \gamma, \qquad Q = 5.493\,\text{MeV}. \tag{26.41}$$

The dominant *pp*-I termination is ${}^3\text{He} + {}^3\text{He} \to {}^4\text{He} + 2p$; the *pp*-II and *pp*-III branches proceed through ${}^7\text{Be}$, with ${}^8\text{B}$ beta decay producing the highest-energy solar neutrinos. All branches have the same net nuclear bookkeeping,

$$4p \to {}^4\text{He} + 2e^+ + 2\nu_e, \tag{26.42}$$

followed by positron annihilation in matter [6, 1, 36].

CNO catalysis. In the main CNO-I loop,

$${}^{12}\text{C}(p,\gamma){}^{13}\text{N}(\beta^+\nu_e){}^{13}\text{C}(p,\gamma){}^{14}\text{N}(p,\gamma){}^{15}\text{O}(\beta^+\nu_e){}^{15}\text{N}(p,\alpha){}^{12}\text{C}.$$

The ${}^{12}\text{C}$ catalyst is regenerated, so the net reaction is again four protons to one alpha plus two positrons and two neutrinos. Its stronger temperature sensitivity makes CNO burning increasingly important in hotter, more massive main-sequence stars; in the Sun it is subdominant [1, 36].

Neutrino energy accounting. Using neutral-atom masses, the complete conversion $4\,{}^1\text{H} \to {}^4\text{He}$ releases $Q = 26.73\,\text{MeV}$. This is the nuclear energy release, not all of the heat deposited in the star. Neutrinos escape with branch-dependent energies, so

$$Q_{\text{heat}} = 26.73\,\text{MeV} - \sum_\nu \langle E_\nu \rangle. \tag{26.43}$$

Positron annihilation energy is already accounted for consistently when atomic masses are used; adding it a second time would double count it [36].

Resonances. When a compound nuclear level sits in the Gamow window, $\sigma(E)$ can be resonantly enhanced (Breit–Wigner). The 3α process that forms carbon in red giants is a classic example (Hoyle state in ${}^{12}\text{C}$) [14, 2]. For an isolated narrow resonance at E_r, the Maxwellian average scales as

$$\langle \sigma v \rangle_r = \left(\frac{2\pi}{\mu k_B T} \right)^{3/2} \hbar^2 (\omega\gamma) e^{-E_r/k_B T}, \tag{26.44}$$

where $\omega\gamma$ is the resonance strength. This contribution must be added to, not hidden inside, the slowly varying nonresonant approximation.

Screening. For weak static screening with screening energy $U_e \ll E_0$, a useful local estimate is $f_{\text{sc}} \simeq \exp(U_e/k_B T)$, so that the bare-nucleus rate is multiplied by f_{sc}. Strongly coupled plasmas and laboratory electron-screening corrections require more careful treatments [14, 36].

Lawson criterion and ignition. A crude balance of fusion power against losses leads to a minimum $n\tau_E$ (or $nT\tau_E$) for net energy gain. Different fuel cycles and confinement schemes shift the required temperature and the peak of $\langle\sigma v\rangle$ [1, 6].

Neutron budgets in $\mathrm{D}+\mathrm{T}$. The 14 MeV neutron must be moderated and captured (often on lithium) to breed tritium if a closed fuel cycle is desired. Neutron damage to structural materials is a major engineering challenge [1, 2].

26.8.1 Locating the Gamow peak analytically

For $\sigma(E) \propto E^{-1}\exp(-\sqrt{E_G/E})$ and a Maxwellian weight $E^{1/2}\exp(-E/k_BT)$, the integrand in (26.37) peaks where

$$\frac{\mathrm{d}}{\mathrm{d}E}\left[E^{-1/2}\exp\left(-\sqrt{\frac{E_G}{E}}-\frac{E}{k_BT}\right)\right]=0. \tag{26.45}$$

The approximate solution is $E_0 \approx (E_G(k_BT)^2/4)^{1/3}$, the Gamow peak energy [14, 6]. The width of the peak is roughly $\Delta E \sim 4\sqrt{E_0k_BT/3}$. Reactions are sensitive to $S(E)$ evaluated at E_0, not at the mean thermal energy [36, 2].

Solar $p+p$ rate order of magnitude

With $S_{pp}(0) \approx 4\times10^{-25}\,\mathrm{MeV\,b}$ from Solar Fusion III, $E_G \approx 493\,\mathrm{keV}$ for $p+p$, and $T = 15\times10^6\,\mathrm{K}$, the practical formula (26.39) gives $\langle\sigma v\rangle_{pp} \sim 10^{-47}\,\mathrm{cm^3\,s^{-1}}$ [36, 14]. At $\rho \approx 150\,\mathrm{g\,cm^{-3}}$ and $X_p \approx 0.35$, the proton density is $n_p \sim 10^{26}\,\mathrm{cm^{-3}}$, so the local *pp* rate per unit volume is $\sim 10^{-21}\,\mathrm{cm^{-3}\,s^{-1}}$. Integrated over the solar core volume this sustains the observed luminosity [6, 36].

26.8.2 Closing the course arc: decay, reactions, fission, fusion

The Gamow exponent entered as α-decay tunneling (Lectures 16–17), reappeared in sub-barrier fusion (Lecture 25), and sets stellar lifetimes through $\langle\sigma v\rangle$. Weak β decay (Lectures 18–19) limits the *pp* chain rate. Multipole γ de-excitation (Lecture 20) follows reaction and decay cascades. Direct and compound reactions (Lecture 21) and two-body kinematics (Lecture 22) connect laboratory measurements to astrophysical networks. Fission (Lectures 23–24) terminates the heavy-nucleus stability region; fusion builds light nuclei up the binding curve. All share Q-value bookkeeping from the SEMF and mass tables [6, 27, 14].

Key idea

Temperature sensitivity. CNO burning scales roughly as $\langle\sigma v\rangle \propto T^{15}$ to T^{20} near main-sequence temperatures, far steeper than the *pp* chain $\propto T^4$ [36, 1]. This strong temperature dependence explains why massive stars switch to CNO-dominated cores and why terrestrial fusion devices must hold $T \sim 10^8\,\mathrm{K}$ for D–T ignition [6, 2].

26.8.3 Neutrino flux and the weak steps in the *pp* chain

The *pp* chain releases $Q = 26.73\,\mathrm{MeV}$ per $4\,{}^1\mathrm{H}$ to ${}^4\mathrm{He}$, but neutrinos carry 2–25% of that energy depending on the branch [36, 6]. The initiating $p + p \to d + e^+ + \nu_e$ step is weak (Lecture 18) and therefore slow; the subsequent strong captures proceed quickly once deuterium forms. Solar neutrino experiments test both the nuclear S-factors and the weak-interaction rate [36, 2]. Recommended rates and uncertainties in Solar Fusion III propagate directly into helioseismology and solar-age determinations [36, 14].

26.8.4 Lawson criterion and the D–T fuel choice

The Lawson product $n\tau_E$ required for net energy gain depends on $\langle \sigma v \rangle(T)$ and on loss mechanisms in the confinement scheme [1, 6]. D–T is chosen for terrestrial devices because $\langle \sigma v \rangle$ peaks at accessible temperatures near $10^8\,\mathrm{K}$, not because it has the lowest Coulomb barrier among all fuel cycles [2]. The *pp* chain in the Sun trades a lower Coulomb barrier for a weak-interaction bottleneck (Lecture 18), producing a stable gigayear burn rate [36, 14]. Reading Solar Fusion III tables with the Gamow-peak map from this lecture identifies which S-factors are experimentally anchored near stellar energies and which still rely on extrapolation [36, 54].

26.9 Deeper reading: stellar rates and the Gamow window

The reaction rate $\langle \sigma v \rangle$ weights the Maxwell–Boltzmann tail against tunneling through the Coulomb barrier. The Gamow peak energy is where that product maximises; most stellar burning occurs near that window, not at the Gamow energy of a single particle with mean thermal energy. Solar Fusion III tabulates recommended S-factors and uncertainties for *pp*-chain and CNO reactions.

A 10% change in a well-measured S-factor is not equivalent to a 10% change in an extrapolated one: neutrino-flux sensitivities make the distinction quantitative. Underground accelerators push laboratory energies toward the Gamow window. Link back to Lecture 25's barrier and to Lecture 6's stellar endpoints: nuclear rates are the microphysics inside the macroscopic stellar structure.

S-factor definition

Writing $\sigma(E) = S(E)E^{-1}e^{-2\pi\eta}$ removes most of the Coulomb exponential so that $S(E)$ varies slowly and can be extrapolated. Always state whether $S(0)$ or $S(E_0)$ at the Gamow peak is quoted. Polynomial or R-matrix extrapolations encode different nuclear-structure assumptions.

Neutrino diagnostics

Solar neutrino fluxes depend on core temperature and on specific nuclear rates. A change in ${}^7\mathrm{Be}(p,\gamma){}^8\mathrm{B}$ moves the high-energy neutrino flux almost linearly, whereas *pp* rate changes are partly compensated by stellar-structure feedback. Sensitivity studies in Solar Fusion III make that hierarchy explicit for students who only met $\langle \sigma v \rangle$ algebra in this chapter.

Gamow-peak numerics

For $p+p$ at $T = 15\times10^6$ K, estimate kT, the Gamow energy, and the width of the Gamow window using textbook formulas. Compare kT with the Gamow peak energy: they differ substantially, which is the whole point of the peak construction. Then open Solar Fusion III tables and record $S(0)$ and uncertainty for $^7\text{Be}(p,\gamma)^8\text{B}$ and for a CNO reaction.

Discuss how a 5% rate change propagates differently into *pp* and ^{8}B neutrino fluxes because of stellar-structure feedback. This sensitivity language turns the Lecture 26 algebra into the reason underground accelerators and solar-neutrino experiments are a single scientific programme.

The Gamow peak is why stars burn at energies far below the Coulomb barrier. Solar Fusion III turns that idea into recommended S-factors and neutrino-aware uncertainties.

In practice, rework one numerical estimate from the lecture with a different parameter set and compare the shift to the quoted experimental uncertainty. If the shift exceeds the uncertainty, the estimate is teaching a scale, not delivering a fit. That distinction keeps theoretical cartoons honest when they sit beside evaluated data tables and modern many-body calculations. Along the course arc, write one sentence linking this chapter backward and one forward: the discipline of those two sentences is as important as any single formula. Examination problems rarely ask for a literature review, but they do ask for the scale argument that literature reviews presuppose.

Synthesis problem

Write a half-page note that (i) restates the chapter's central scale or selection rule in one equation, (ii) evaluates it numerically for one concrete nucleus or energy, and (iii) names the neighbouring lecture that uses the result. Keep the note free of research-frontier speculation; the goal is fluency with the lecture physics itself. If an assumption fails (Born approximation, spherical mean field, allowed transition, one-dimensional barrier), state the failure mode in a final sentence. That habit converts passive reading into examination-ready understanding and makes the later research-frontier box read as an extension rather than a replacement.

Research frontier: stellar rates, neutrinos, and Solar Fusion III

The Gamow-peak rate $\langle\sigma v\rangle$ of Lecture 26 is the quantitative bridge from laboratory cross sections to stellar models. Solar Fusion III provides recommended S-factors and uncertainties for *pp*-chain and CNO reactions, driven by few-percent solar-neutrino measurements and by underground accelerator data that push closer to stellar energies [36]. The same nuclear inputs feed models of Population II stars, classical novae, and the early stages of massive-star evolution, while multimessenger astrophysics adds complementary constraints from compact-object mergers [48, 54]. Remaining uncertainties include several subdominant CNO and *pp*-chain S-factors, electron-screening corrections at stellar energies, and the propagation of nuclear-rate covariances into neutrino-flux predictions. Underground laboratories continue to reduce those errors by measuring closer to the Gamow window. The lecture's analytic Gamow peak therefore remains the conceptual map for reading Solar Fusion III tables.
When reading Solar Fusion III, students should note which S-factors are limited by experiment near the Gamow window and which still rely on extrapolation or theory. A 10%

change in a well-measured rate is not equivalent to a 10% change in a sparsely constrained one; neutrino-flux sensitivities make that distinction quantitative.
Not every 10% S-factor change is equal: neutrino-flux sensitivities weight well-measured and extrapolated rates differently.
Student directions. (1) Numerical exercise: locate the Gamow peak for $p{+}p$ at $T = 15{\times}10^6$ K. (2) Table lookup: using Solar Fusion III, quote the recommended $S(0)$ and uncertainty for $^7\mathrm{Be}(p,\gamma)^8\mathrm{B}$. (3) Sensitivity discussion: explain how a 10% change in one CNO S-factor propagates into the solar core temperature inferred from neutrinos.

26.10 Problems with solutions

Problem 26.1 — Derive $R = n_X n_Y \sigma v$. A beam of Y nuclei with density n_Y and speed v strikes a medium with target density n_X. Using the cylinder of area σ and length $v\,\mathrm{d}t$, derive the volumetric reaction rate R. Then state the thermal-plasma result for unlike species and for identical reactants [6, 14].

Solution 26.1.

Step 1. Volume swept relative to one X in time $\mathrm{d}t$: $\sigma v\,\mathrm{d}t$.

Step 2. Number of Y in that volume: $n_Y \sigma v\,\mathrm{d}t$. Rate per X: $\lambda = n_Y \sigma v$.

Step 3. Per unit volume there are n_X targets, so

$$R = n_X n_Y \sigma v.$$

In a plasma, average over relative velocities. For unlike species,

$$R_{XY} = n_X n_Y \langle \sigma v \rangle.$$

For identical reactants, unordered-pair counting supplies the required factor:

$$R_{XX} = \frac{1}{2} n_X^2 \langle \sigma v \rangle.$$

Problem 26.2 — Annular partial-wave area. Show that the area between impact parameters $b_\ell = \ell\hbar/(\mu v)$ and $b_{\ell+1}$ equals $\pi\hbar^2(2\ell+1)/(\mu^2 v^2)$. Explain why this $1/v^2$ partial-wave prefactor does *not* by itself imply that a charged-particle fusion cross section grows as $1/v^2$ at low energy.

Solution 26.2.

Step 1.

$$A_\ell = \pi\Big(\frac{(\ell+1)\hbar}{\mu v}\Big)^2 - \pi\Big(\frac{\ell\hbar}{\mu v}\Big)^2 = \frac{\pi\hbar^2}{\mu^2 v^2}\,[(\ell+1)^2 - \ell^2] = \frac{\pi\hbar^2}{\mu^2 v^2}(2\ell+1).$$

Step 2. Summing $\ell = 0 \ldots \ell_{\max}$ with $\ell_{\max} \sim \mu v R/\hbar$ gives $\sigma \sim \pi(R + \hbar/(\mu v))^2$. If $\hbar/(\mu v) \gg R$, $\pi\hbar^2/(\mu^2 v^2)$ is the entrance-channel area scale. The reaction probability also contains the transmission coefficient. For a nonresonant charged-particle reaction, $\sigma(E) = S(E)E^{-1}e^{-\sqrt{E_G/E}}$, so Coulomb suppression dominates as $E \to 0$ [14, 6].

Problem 26.3 — Gamow exponent with units. Starting from $\eta = Z_1 Z_2 \alpha c/v$ and $E = \frac{1}{2}\mu v^2$, derive $2\pi\eta = \sqrt{E_G/E}$ and $E_G = 2\mu c^2(\pi\alpha Z_1 Z_2)^2$. Check the dimensions.

Solution 26.3.

Step 1. $v = \sqrt{2E/\mu}$, so

$$2\pi\eta = 2\pi Z_1 Z_2 \alpha c \sqrt{\frac{\mu}{2E}} = \sqrt{\frac{2\mu c^2(\pi\alpha Z_1 Z_2)^2}{E}}.$$

Step 2. Identifying the numerator as E_G gives $2\pi\eta = \sqrt{E_G/E}$. Since μc^2 is an energy and π, α, Z_1, Z_2 are dimensionless, E_G has energy units.

Step 3. Larger $Z_1 Z_2$ raises $E_G \propto Z_1^2 Z_2^2$, making $P \propto e^{-\sqrt{E_G/E}}$ exponentially smaller. Heavy-ion fusion at thermonuclear energies is vastly harder than hydrogen fusion [6, 1].

Problem 26.4 — Gamow peak and width. Starting from $I(E) = \exp[-E/(k_B T) - \sqrt{E_G/E}]$, derive E_0 and the full $1/e$ width Δ. Evaluate both for $p + p$ at $T = 1.5 \times 10^7$ K, using $E_G = 489\,\mathrm{keV}$ and $k_B T = 1.293\,\mathrm{keV}$.

Solution 26.4.

Step 1. Minimise $\Phi(E) = E/(k_B T) + \sqrt{E_G}E^{-1/2}$:

$$\frac{\mathrm{d}\Phi}{\mathrm{d}E} = \frac{1}{k_B T} - \frac{\sqrt{E_G}}{2}E^{-3/2} = 0.$$

Step 2.

$$E_0 = \left[\frac{E_G (k_B T)^2}{4}\right]^{1/3} = \left[\frac{(489)(1.293)^2}{4}\right]^{1/3} = 5.89\,\mathrm{keV}.$$

Step 3. Since $\Phi''(E_0) = 3/(2E_0 k_B T)$,

$$\Phi(E) \simeq \Phi(E_0) + \frac{3(E - E_0)^2}{4E_0 k_B T}.$$

At the $1/e$ points $E - E_0 = \pm\Delta/2$, hence

$$\Delta = 4\sqrt{\frac{E_0 k_B T}{3}} = 4\sqrt{\frac{(5.89)(1.293)}{3}} = 6.38\,\mathrm{keV}.$$

The reacting window is well above $k_B T$ but far below the contact barrier [14, 36].

Problem 26.5 — Rate scaling with density. A plasma of identical fuel nuclei has density n and reactivity $\langle\sigma v\rangle$. Write R including the identical-particle factor $\frac{1}{2}$. If n is doubled at fixed T, by what factor does R change?

Solution 26.5.

$$R = \frac{1}{2} n^2 \langle\sigma v\rangle.$$

Doubling n multiplies R by 4. Fusion power density is quadratic in density at fixed temperature [6, 1].

Problem 26.6 — D + T energetics. In $\mathrm{D} + \mathrm{T} \to {}^4\mathrm{He} + n + 17.6\,\mathrm{MeV}$, the α and neutron share momentum. Show that the neutron carries about 14.1 MeV and the α about 3.5 MeV (nonrelativistic two-body kinematics in the CM frame).

Solution 26.6.

Step 1. In the CM frame, $p_\alpha = p_n = p$ and

$$\frac{p^2}{2m_\alpha} + \frac{p^2}{2m_n} = Q.$$

Step 2. With $m_\alpha \approx 4m_n$,

$$K_n = \frac{m_\alpha}{m_\alpha + m_n} Q = \frac{4}{5} Q = 0.8 \times 17.6 = 14.08\,\mathrm{MeV},$$

$$K_\alpha = \frac{m_n}{m_\alpha + m_n} Q = \frac{1}{5} Q = 3.52\,\mathrm{MeV}.$$

The neutron carries $\sim 80\%$ of Q; in magnetic fusion it escapes the confining field and heats a blanket, while the charged α can heat the plasma [6, 14].

Problem 26.7 — Astrophysical S-factor motivation. Why factor $\sigma(E) = S(E)\,E^{-1}\exp(-2\pi\eta)$ in charged-particle fusion rates? What part of the energy dependence is thereby removed for extrapolation?

Solution 26.7. The exponential Gamow penetrability and the $1/E$ from the de Broglie factor dominate $\sigma(E)$. Defining $S(E)$ removes those strongly energy-dependent pieces so that $S(E)$ varies slowly and can be extrapolated from laboratory energies to the stellar Gamow peak [6, 14, 36].

Chapter 27

Thermonuclear Reactors and Solar Neutrinos

"A plasma must be hot enough, dense enough, and confined long enough: that is the content of the Lawson criterion."

J. D. Lawson (1957)

Lectures 25–26 developed fusion energetics, Coulomb tunneling, the thermal rate $\langle \sigma v \rangle$, and the Gamow peak. Those tools explain *why* fusion releases energy and *how fast* it proceeds in a Maxwellian gas. This chapter turns to the two settings in which that rate must actually earn its keep: laboratory thermonuclear reactors, which must confine a multi-keV plasma without melting the vessel, and the solar core, whose neutrinos provide a clean, prompt test of the proton–proton chain [6, 1, 14, 36, 32].

27.1 The reactor problem

A fusion plasma stores thermal energy in ions and electrons, deposits fusion energy through charged reaction products, and simultaneously loses energy by radiation (chiefly electron–ion bremsstrahlung) and by particle and heat transport to the wall. Net thermonuclear gain therefore requires that the fusion energy deposited while the plasma remains hot exceed the energy that was invested to heat it, and, in a practical device, exceed continuing losses as well. That comparison is the content of Lawson's criterion [1, 6].

In symbols the fusion power density is

$$p_{\text{fus}} = \frac{n_X n_Y}{1 + \delta_{XY}} \langle \sigma v \rangle \, Q_{\text{dep}}, \tag{27.1}$$

with Q_{dep} the energy retained in charged products. Neutrons and neutrinos leave the plasma and do not heat it; in a $\mathrm{D} + \mathrm{T}$ machine the 14 MeV neutron must be caught in a blanket, while only the 3.5 MeV α heats the fuel on the spot [6, 1]. The engineering problem is to arrange density, temperature, and confinement time so that $\int p_{\text{fus}} \, \mathrm{d}t$ beats heating plus losses for a useful volume and duty cycle – either by magnetic bottles (tokamaks, stellarators) or by crushing a solid pellet so violently that inertia itself holds the fuel together for nanoseconds [6, 1].

27.2 Lawson criterion: every step

27.2.1 Setup (identical-species estimate)

Start from the identical-species D + D bookkeeping [1]. Let the ion density of each fuel species be $n/2$ in a plasma of total ion density n, with equal electron density $n_e = n$ for quasineutrality. The reaction rate density is

$$r = \left(\frac{n}{2}\right)\left(\frac{n}{2}\right)\langle\sigma v\rangle = \frac{n^2}{4}\langle\sigma v\rangle. \tag{27.2}$$

If the plasma is held at the required temperature and density for a time τ, the fusion energy deposited per unit volume is

$$E_f = r\,Q\,\tau = \frac{n^2}{4}\langle\sigma v\rangle\,Q\,\tau, \tag{27.3}$$

where Q is the energy that remains available to the plasma per fusion (the energy retained in the plasma per fusion). The same algebra with unequal densities n_1, n_2 replaces $n^2/4$ by $n_1 n_2$ (or $n_1 n_2/2$ for identical particles counted once), which is how the D + T case is written in the literature [1, 6].

27.2.2 Thermal energy to be recovered

Ions and electrons each contribute $\frac{3}{2}nk_BT$ per unit volume when $n_e = n$, so the thermal investment per unit volume is

$$E_{\text{th}} = 3\,n\,k_BT. \tag{27.4}$$

Above the bremsstrahlung "crossing" temperature one may first ignore radiation and ask only that fusion repay the heating bill; at lower T, or for fuels with smaller $\langle\sigma v\rangle$, radiation must be kept explicitly and tightens the required $n\tau$ [1, 6].

27.2.3 The inequality

Demand $E_f > E_{\text{th}}$:

$$\frac{n^2}{4}\langle\sigma v\rangle\,Q\,\tau > 3\,n\,k_BT. \tag{27.5}$$

Cancel one power of n (assuming $n > 0$) and rearrange:

$$\boxed{n\tau > \frac{12\,k_BT}{\langle\sigma v\rangle\,Q}}. \tag{27.6}$$

This is the **Lawson criterion**; $n\tau$ is the Lawson parameter [1, 6, 14]. The right-hand side depends on temperature through both the explicit T and the strongly temperature dependent reactivity $\langle\sigma v\rangle(T)$ from Lecture 26. Raising $\langle\sigma v\rangle$ at fixed T therefore lowers the required product: that is why D + T is the first fuel of every practical design discussion.

An order-of-magnitude estimate at $k_BT = 10\,\text{keV}$ for D + T gives $n\tau \gtrsim 10^{20}\,\text{s}\,\text{m}^{-3}$. For D + D the same temperature is already too cool because bremsstrahlung dominates; operating near $k_BT = 100\,\text{keV}$ pushes the Lawson product up by roughly two orders of magnitude [6]. Thus a D + D reactor is not merely "harder by Q": it demands a hotter plasma and a larger $n\tau$ before radiation allows net gain.

27.2.4 Triple product

Multiplying (27.6) by T defines the triple product $n\tau T$. Over the decade $T \sim 10$–$20\,\mathrm{keV}$ where $\mathrm{D}+\mathrm{T}$ is most reactive, $\langle\sigma v\rangle$ scales roughly as T^2, so the right-hand side becomes nearly constant. For $\mathrm{D}+\mathrm{T}$ the practical ignition ballpark is

$$n\,\tau\,T \gtrsim 3\times 10^{21}\,\mathrm{m}^{-3}\,\mathrm{keV\,s} \tag{27.7}$$

[1]. Modern campaigns therefore quote $n\tau T$ rather than $n\tau$ alone, because a device that is hotter but less well confined can still sit above the ridge if the product compensates.

27.2.5 Gain Q and ignition

Facility **gain** Q_{eng} is the ratio of fusion energy out to energy invested in the plasma (and, more broadly, in the plant). $Q_{\mathrm{eng}} \sim 1$ is scientific breakeven; a power plant needs much higher values once conversion efficiency and recirculating power are counted. **Ignition** means charged fusion products alone maintain the plasma temperature: external heating can be switched off and $Q_{\mathrm{eng}} \to \infty$ for the plasma itself [1, 6]. Because the $\mathrm{D}+\mathrm{T}$ neutron does not heat the fuel, lithium-bearing blankets both convert neutron energy to heat for electricity and breed tritium through $n + {}^6\mathrm{Li} \to \alpha + t$ (and related channels) [1, 2, 32].

Order-of-magnitude parameters for the three confinement philosophies appear in Table 27.1: magnetic bottles buy seconds at $n \sim 10^{20}\,\mathrm{m}^{-3}$; inertial capsules buy nanoseconds at $n \sim 10^{31}\,\mathrm{m}^{-3}$; the Sun buys geological times at solar-core density [14, 6].

Table 27.1: Order-of-magnitude confinement parameters [14].

Scheme	$n\,(\mathrm{m}^{-3})$	τ (s)	T (keV)
Tokamak (ITER-class)	10^{20}	~ 1	10–20
Inertial (laser capsule)	10^{31}	10^{-11}	~ 10
Solar core (gravity)	$\sim 10^{31}$	10^{17}	~ 1

27.3 Magnetic confinement

A multi-keV plasma melts any material wall and cools on contact. **Magnetic confinement** uses the Lorentz force $q\,\boldsymbol{v}\times\boldsymbol{B}$ so that charged particles spiral about field lines and, if the field geometry is clever enough, never hit the wall [6, 1, 2]. Figures 27.1 and 27.2 show a real magnetic-confinement machine – the Joint European Torus (JET) – as photograph and cutaway; Fig. 27.3 shows the much larger ITER tokamak under construction as an international burning-plasma experiment.

27.3.1 Motion in a magnetic field

Perpendicular to $\boldsymbol{B}$, a particle of mass m, charge q, and speed $v_\perp$ gyrates with cyclotron radius

$$\rho_c = \frac{mv_\perp}{|q|B}. \tag{27.8}$$

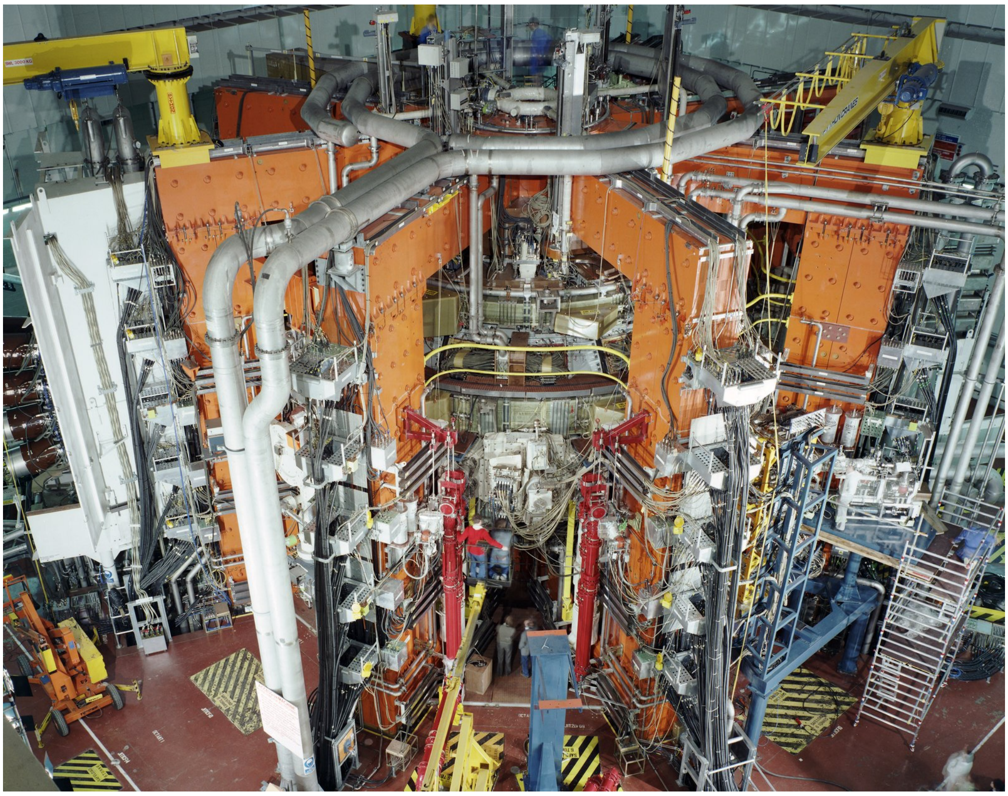


Figure 27.1: The Joint European Torus (JET) during magnetic fusion operation (1991). JET has been the leading European D–T capable tokamak and a principal pathfinder for ITER. Image: EUROfusion / EFDA (Wikimedia Commons), CC BY-SA 3.0.

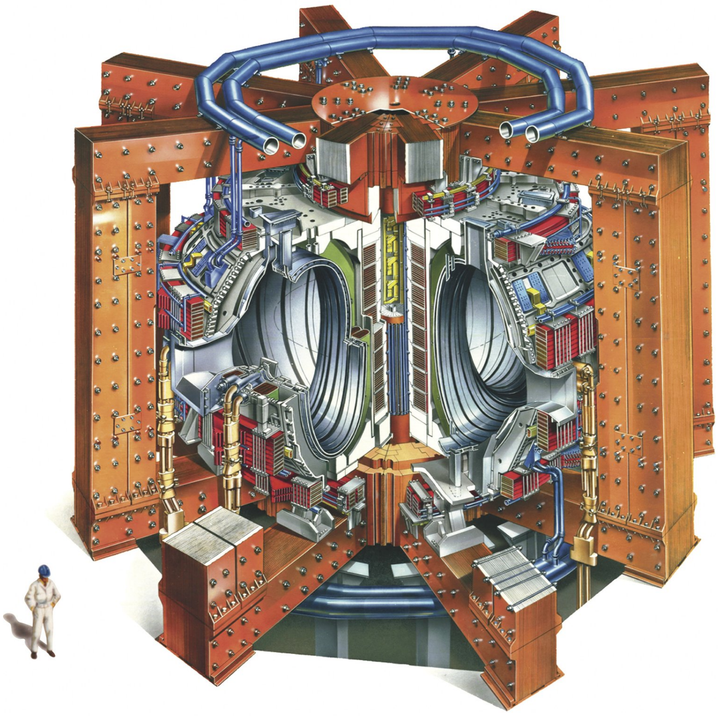


Figure 27.2: Cutaway drawing of JET (1980 design), showing the toroidal vacuum vessel, magnetic coils, and diagnostic ports around the plasma chamber. Image: EUROfusion (Wikimedia Commons), CC BY 4.0.

Parallel motion is free. A uniform field therefore confines only two spatial directions: particles stream out the ends of a finite solenoid. Textbooks give two remedies: bend the field into a **torus**, closing the field lines, or raise B at the ends of a linear bottle to form a **magnetic mirror** that reflects particles with sufficiently large magnetic moment [6, 1].

27.3.2 Why a pure torus still fails

Closing the field into a torus is not enough by itself. In a purely toroidal field, $|B|$ is stronger on the inside of the doughnut (smaller major radius R). Gradient and curvature drifts then send ions and electrons in opposite vertical directions. The resulting charge separation builds an electric field; the subsequent $\boldsymbol{E} \times \boldsymbol{B}$ drift drives the whole plasma into the outer wall [6, 14]. A particle spiralling in a real toroidal winding sees a weaker field at larger major radius, so the spiral opens toward the outer wall unless a poloidal field component is added [6].

27.3.3 Tokamak: toroidal plus poloidal fields

The tokamak cures the drift by twisting the field lines. External coils produce a strong **toroidal** field $\boldsymbol{B}_t$. A large toroidal **plasma current** (often mega-amperes) produces a **poloidal** field $\boldsymbol{B}_p$. The resultant $\boldsymbol{B} = \boldsymbol{B}_t + \boldsymbol{B}_p$ has helical field lines: particles streaming along $\boldsymbol{B}$ sample both the inner and outer major-radius regions and average away the vertical drift [6, 1, 2, 32]. Toroidal coils plus an axial plasma current produce a helical field with closed orbits [6].

In practice the plasma current is induced, transformer-style, by ohmic heating coils; that current also heats the plasma resistively until the conductivity rises and auxiliary heating (neutral beams, radio-frequency waves) takes over [6, 1]. Stellarators produce the rotational transform with shaped external coils and little net plasma current, trading coil complexity for reduced current-driven instability [1, 14]. ITER (Fig. 27.3) is designed as the step from present-day $Q_{\text{eng}} \sim$ few machines toward a sustained burning plasma with $Q_{\text{eng}} \gtrsim 10$, using superconducting magnets, remote-handling blankets, and tritium breeding technology that JET could only prototype.

27.4 Inertial confinement

Inertial confinement never builds a magnetic bottle. A millimetre-scale cryogenic D–T capsule is irradiated nearly isotropically by intense laser (or heavy-ion) beams. Ablation of the outer shell drives an implosion: the fuel is compressed and heated until a hotspot satisfies Lawson for a time set by hydrodynamic disassembly ($\tau \sim 10^{-11}$–10^{-9} s) [6, 1, 32]. Because τ is tiny, n must be enormous ($\sim 10^{31}\,\text{m}^{-3}$; see Table 27.1), so that the product $n\tau$ still clears the Lawson ridge of Table 27.1. Drive asymmetry and Rayleigh–Taylor growth of the shell are the classic failure modes: any seed perturbation that grows during acceleration can spoil the hotspot before burn begins [6, 1].

Figure 27.5 shows the NIF target chamber – a real inertial device whose scale is set by laser ports, debris shields, and diagnostics rather than by toroidal magnets. Confine by inertia, not by $\boldsymbol{B}$, and pay for the short τ with extreme density [6].

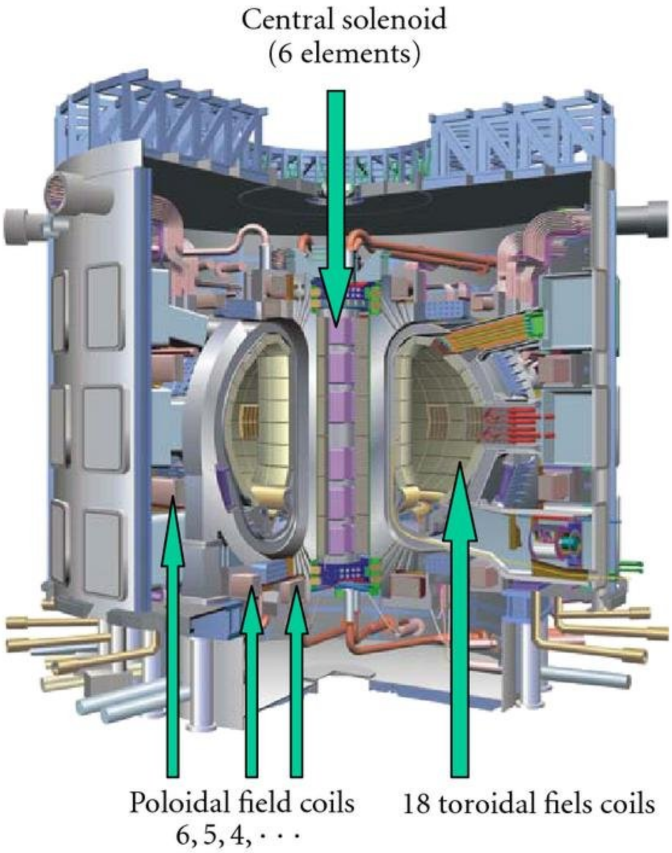


Figure 27.3: Sectional view of the ITER tokamak reactor, showing the vacuum vessel, blanket modules, and superconducting magnet systems that will confine a burning D–T plasma. Image: from Miri *et al.*, *Modelling and Simulation in Engineering* (2008), Wikimedia Commons, CC BY 4.0.

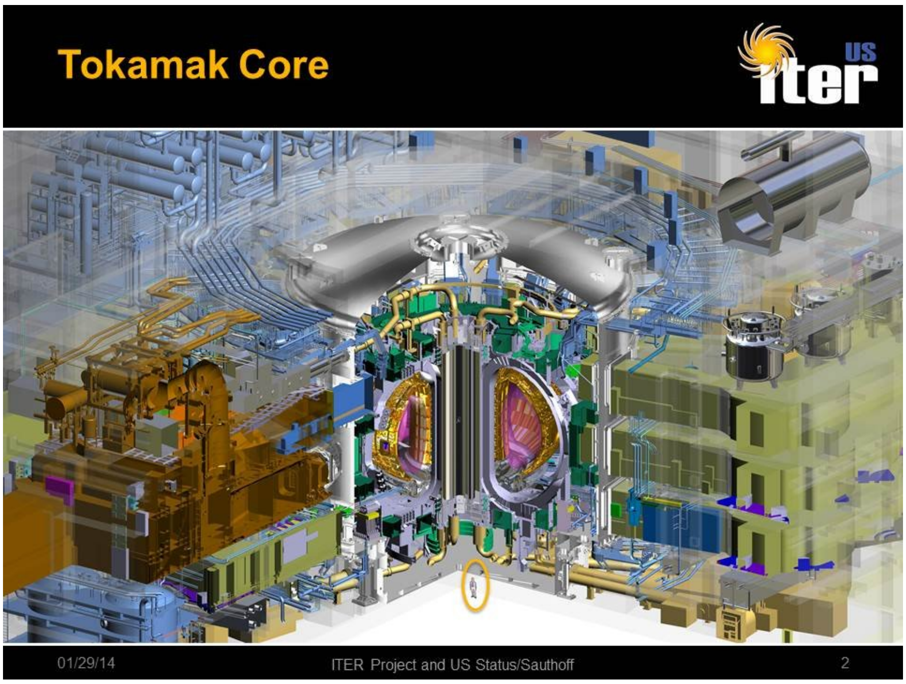


Figure 27.4: Interior of a tokamak-style fusion core mock-up / experiment (Oak Ridge National Laboratory photograph), illustrating the scale of the vacuum vessel relative to a person. Image: ORNL / Flickr via Wikimedia Commons, CC BY 2.0.

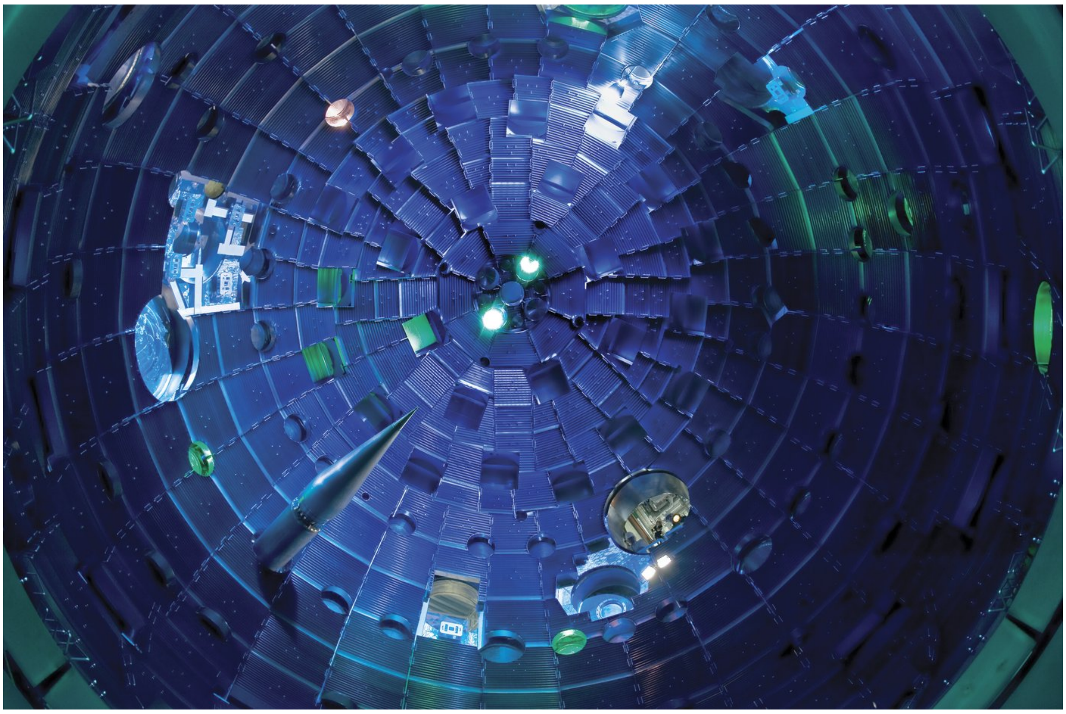


Figure 27.5: Target chamber of the National Ignition Facility (NIF) at Lawrence Livermore National Laboratory. Up to 192 ultraviolet laser beams enter the chamber to irradiate a cryogenic D–T capsule in an inertial-confinement geometry. Image: LLNL / L. Seaver (Wikimedia Commons), CC BY-SA 3.0.

27.5 Stellar hydrogen burning

Stars solve confinement with gravity. The solar core sits near $T \sim 1.3\,\mathrm{keV}$ ($\sim 1.5 \times 10^7\,\mathrm{K}$): cooler than a tokamak, but denser and longer-lived by many orders of magnitude, so the Gamow peak of the weak $p+p$ reaction still powers the Sun [14, 36, 6, 1].

27.5.1 Net stoichiometry

Every hydrogen-burning path obeys

$$4\,{}^1\mathrm{H} \;\to\; {}^4\mathrm{He} + 2e^+ + 2\nu_e, \tag{27.9}$$

with nuclear $Q \approx 26.7\,\mathrm{MeV}$ from atomic masses [1, 36]. Neutrinos escape, so the stellar heat deposit is $Q_{\mathrm{heat}} = Q - \sum\langle E_\nu\rangle$ [36, 1]. Photons random-walk for $\sim 10^4$–10^6 years to the photosphere; neutrinos free-stream from the core in minutes and therefore report on nuclear burning *now*. The light we see comes from the surface, while solar neutrinos come directly from the core reactions [6, 32].

27.5.2 The *pp* chain, branch by branch

Figure 27.6 shows the solar networks as drawn for open use. The dominant *pp* path proceeds as follows [1]:

(i) **Initiation (weak).** $p+p \to d+e^+ + \nu_e$. The cross section is of weak order and cannot be measured by colliding beams in the laboratory; the rate is inferred from theory calibrated to solar models and to the few available laboratory constraints on related reactions [14, 36].

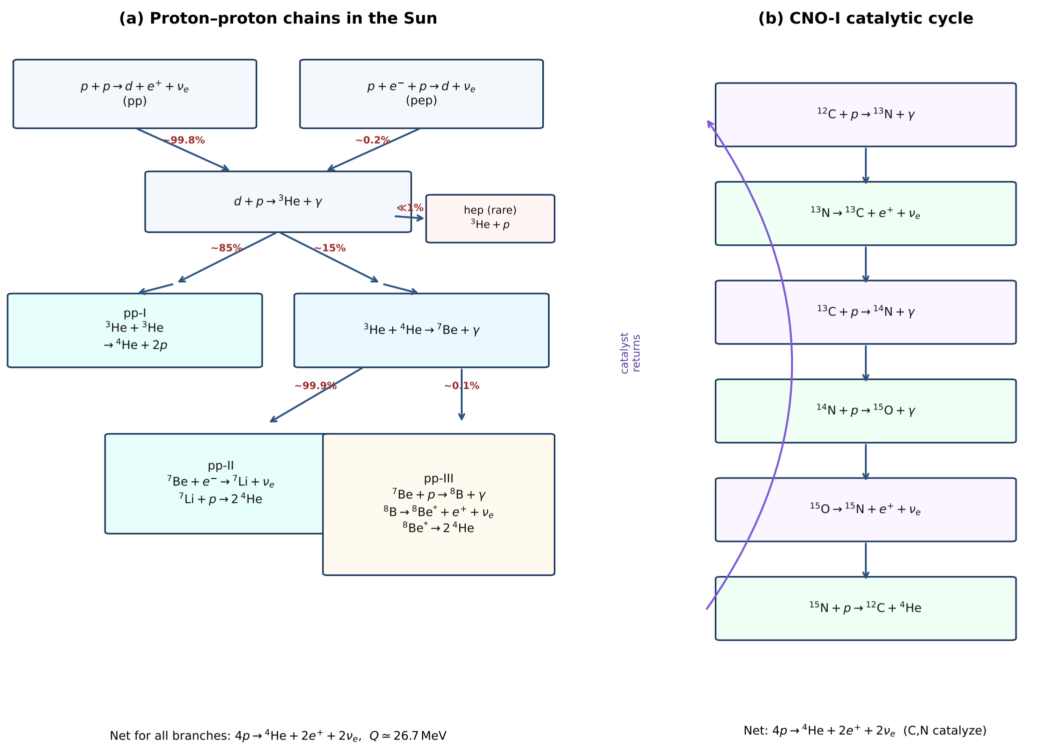


Figure 27.6: Solar hydrogen burning. (a) Proton–proton chains with approximate solar branching fractions: pp-I dominates; pp-II/III proceed through ^{7}Be; pep and hep are rare. (b) CNO-I catalytic cycle returning ^{12}C. Every branch has the same net stoichiometry $4p \to {}^4\mathrm{He} + 2e^+ + 2\nu_e$ with $Q \simeq 26.7\,\mathrm{MeV}$ [1, 36, 6].

(ii) **Deuterium burning.** $d + p \to {}^3\mathrm{He} + \gamma$ ($Q \approx 5.5\,\mathrm{MeV}$) is strong/electromagnetic and essentially instantaneous on stellar timescales.

(iii) **Termination.** *pp*-I: $^3\mathrm{He} + {}^3\mathrm{He} \to \alpha + 2p$ ($Q = 12.86\,\mathrm{MeV}$) [1]. *pp*-II/III: $^3\mathrm{He} + \alpha \to {}^7\mathrm{Be}$, then electron capture (*pp*-II) or p-capture to ^{8}B (*pp*-III). ^{8}B beta decay yields a hard neutrino continuum crucial for water Cherenkov detectors [6, 1, 36].

The net of four protons to one α, with two weak branches, releases $Q = 26.7\,\mathrm{MeV}$; only a small fraction leaves as neutrino kinetic energy [1].

27.5.3 CNO catalysis

When CNO nuclei are present, the CNO-I loop

$$
\begin{aligned}
&{}^{12}\mathrm{C}(p,\gamma)^{13}\mathrm{N}(\beta^+\nu_e)^{13}\mathrm{C}(p,\gamma)^{14}\mathrm{N} \\
&\qquad (p,\gamma)^{15}\mathrm{O}(\beta^+\nu_e)^{15}\mathrm{N}(p,\alpha)^{12}\mathrm{C}
\end{aligned}
$$

again converts four protons to one alpha. In the Sun it contributes only a few percent of the luminosity because of the larger Coulomb barriers; it dominates in hotter massive stars [1, 14, 36]. Solar neutrinos from CNO β^+ decays are therefore a small but increasingly measured correction on top of the *pp* flux hierarchy.

27.6 Solar neutrino experiments

27.6.1 Homestake chlorine detector

The most extensive early solar-neutrino programme was started in the late 1960s by Raymond Davis, Jr., in the Homestake Mine in South Dakota [1, 6]. A 380 m^3 tank of perchloroethylene C_2Cl_4 was placed 1478 m (4850 ft) underground to suppress cosmic-ray backgrounds (Figs. 27.7–27.8). Natural chlorine is about 24% ^{37}Cl. Electron neutrinos capture via

$$\nu_e + {}^{37}\mathrm{Cl} \to e^- + {}^{37}\mathrm{Ar}, \qquad E_{\mathrm{th}} = 0.814\,\mathrm{MeV}. \tag{27.10}$$

Inspection of the solar spectrum shows that Homestake is dominated by ^8B neutrinos, with smaller contributions from pep and ^7Be; the *pp* continuum lies almost entirely below threshold [1]. The daughter ^{37}Ar returns to ^{37}Cl by electron capture (half-life 35 d; mean life of order fifty days); argon is swept chemically from the tank on a one–two month cycle and counted via characteristic Auger electrons in a small proportional counter [1, 6].

Figure 27.7: Homestake Mine solar-neutrino detector: the large perchloroethylene tank of the Brookhaven / Davis chlorine experiment, photographed underground. Public-domain U.S. Department of Energy photograph (Wikimedia Commons, HD.6B.411).

The expectation is roughly one argon atom produced per day in Davis's tank, yet the observed rate was only about one-third of the Standard Solar Model prediction [6]. After approximately thirty years of running, the Homestake result is 2.55 ± 0.25 SNU against a predicted 8.0 ± 1.0 SNU (one SNU $= 10^{-36}$ captures per target nucleus per second), for a detector mass of 1.3×10^5 kg of ^{37}Cl nuclei [1]. That persistent deficit is the classic **solar neutrino problem**.

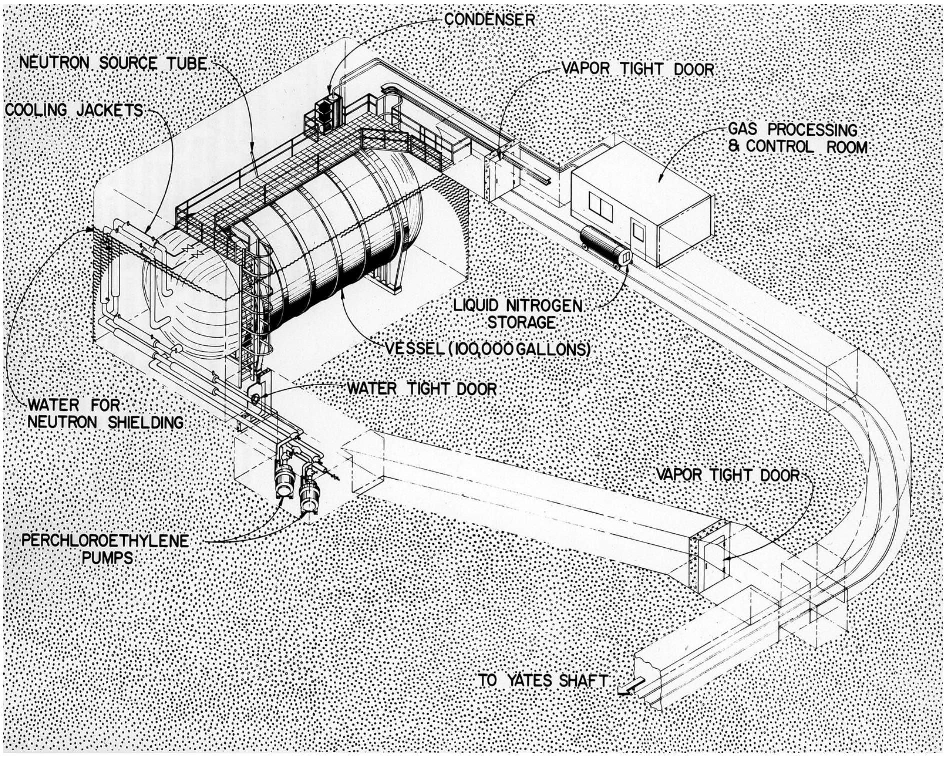


Figure 27.8: Contemporary diagram of the Brookhaven National Laboratory solar neutrino detector in the Homestake Mine (c. 1972), showing the tank, helium-sweep plumbing, and argon extraction system used to count ^{37}Ar. Public-domain U.S. Atomic Energy Commission / DOE diagram (Wikimedia Commons).

27.6.2 Gallium and water Cherenkov

Gallium experiments (SAGE in Russia; GALLEX and later GNO near Rome) use

$$\nu_e + {}^{71}\text{Ga} \to e^- + {}^{71}\text{Ge}, \tag{27.11}$$

with threshold $\approx 0.23\,\text{MeV}$, so *pp* neutrinos are admitted. Extracted ^{71}Ge decays back to ^{71}Ga with a lifetime of 16.5 d; periodic chemical extraction again provides a radiochemical rate. SAGE and GALLEX/GNO agree with each other but still fall below Standard Solar Model predictions, typically near half the expected rate [1].

Kamiokande and Super-Kamiokande detect νe elastic scattering by Cherenkov light in ultrapure water. The recoil electron points approximately back to the Sun, confirming the solar origin, but the channel is still dominated by high-energy ^{8}B neutrinos [1, 14, 2]. Figure 27.9 shows the interior of Super-Kamiokande: thousands of photomultiplier tubes viewing a vast water volume – the same detection principle used for solar νe scattering, atmospheric neutrinos, and supernova bursts.

27.6.3 Resolution: flavour oscillation and MSW

Radiochemical detectors count ν_e only. The Sudbury Neutrino Observatory (SNO) in Ontario used 1000 t of heavy water in a 12 m acrylic vessel, beginning physics data in 1999 [1]. On

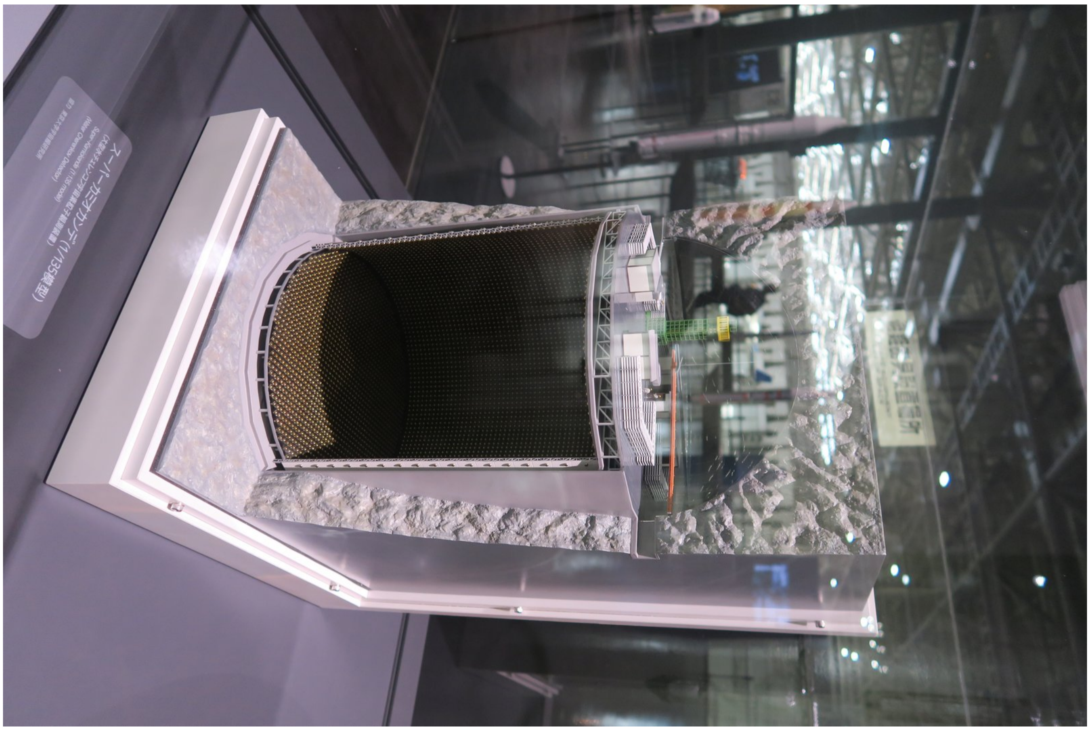


Figure 27.9: Interior of the Super-Kamiokande water Cherenkov detector (Kamioka, Japan), lined with photomultiplier tubes. Elastic scattering of solar neutrinos on electrons produces directional Cherenkov cones in such detectors. Image: Motohira (Wikimedia Commons), CC BY-SA 4.0.

deuterium it recorded three channels simultaneously:

$$\nu_e + d \to p + p + e^- \qquad \text{(charged current, } \nu_e \text{ only)}, \tag{27.12}$$

$$\nu_x + d \to p + n + \nu_x \qquad \text{(neutral current, all active flavours)}, \tag{27.13}$$

$$\nu_x + e^- \to \nu_x + e^- \qquad \text{(elastic scattering)}. \tag{27.14}$$

The neutral-current flux matched Standard Solar Model expectations while the charged-current ν_e flux remained suppressed: the total active neutrino flux from ^{8}B fusion is correct, and a large fraction of ν_e have oscillated into other flavours before detection [1, 14]. Neutrino flavour change (vacuum oscillations plus Mikheyev–Smirnov–Wolfenstein matter effects in the solar density profile) therefore resolves the deficit without overturning the nuclear S-factors. Figure 27.10 shows the energy-dependent electron- neutrino survival probability for solar neutrinos, with day/night curves and the pp, ^{7}Be, pep, and ^{8}B measurement windows marked.

Key idea

- Lawson: $n\tau > 12k_BT/(\langle\sigma v\rangle Q)$ (DD estimate); triple product $n\tau T \gtrsim 3 \times 10^{21}\,\mathrm{m^{-3}\,keV\,s}$ for DT ballpark [1].
- Tokamak (JET, ITER): $\boldsymbol{B}_t + \boldsymbol{B}_p$ helices kill toroidal drifts [6].
- Inertial (NIF): huge n, tiny τ [6].
- Sun: weak pp clock + Gamow tunneling; neutrinos probe the core [1, 6].

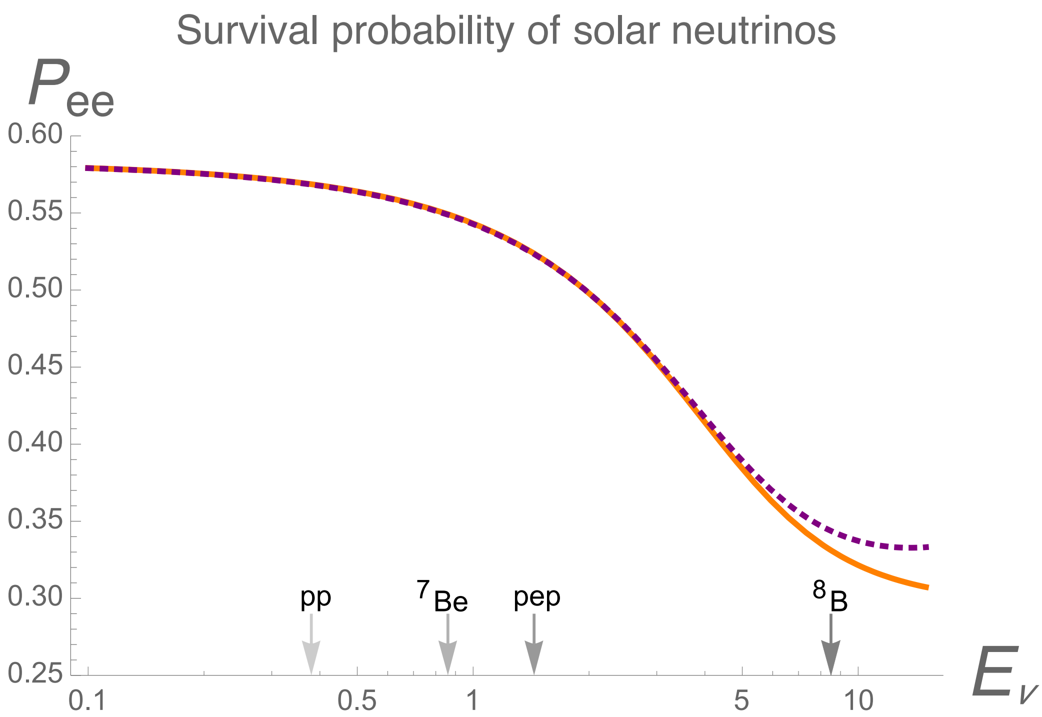


Figure 27.10: Electron-neutrino survival probability for solar neutrinos as a function of energy (day and night curves), with markers for *pp*, ^{7}Be, pep, and ^{8}B measurements. Adapted by F. Vissani from published oscillation analyses (Wikimedia Commons, CC BY-SA 4.0; after arXiv:1706.05435).

- Homestake 2.55 ± 0.25 SNU vs ~ 8 SNU predicted $\to$ flavour oscillation (SNO), not wrong S-factors [1].

27.7 Further physics from the literature

Bremsstrahlung floor. Electron–ion free–free radiation scales roughly as $n_e n_i Z_i^2 T^{1/2}$. Below a fuel-dependent temperature the radiation curve lies above the fusion power curve, so increasing density alone cannot ignite a pure plasma. High-Z wall impurities raise the floor sharply [6, 1].

Mirror and tandem-mirror concepts. Besides the tokamak torus, magnetic mirrors raise B at the ends of a linear device to reflect particles with large $v_\perp/v_\parallel$. Tandem-mirror and thermal-barrier variants are discussed in the standard surveys; they are less central to current reactor roadmaps than tokamaks and stellarators but illustrate the same cyclotron physics [6].

Tritium breeding. Neutron capture on lithium regenerates tritium,

$$^6\text{Li} + n \to {}^4\text{He} + {}^3\text{H} + 4.8\,\text{MeV}, \tag{27.15}$$

with an additional energetic channel on ^{7}Li [14, 1]. Without breeding, geological tritium cannot sustain a GW-scale D + T economy.

MSW effect. Electron density in the Sun modifies effective neutrino masses (MSW). Adiabatic level crossing converts a large fraction of core ν_e into other active flavours before they reach

Earth, explaining ν_e-only deficits once the total flux is known [1, 14].

27.8 Problems with solutions

Problem 27.1 — Derive the Lawson inequality. Starting from $r = (n/2)^2\langle\sigma v\rangle$, $E_f = rQ\tau$, and $E_{\text{th}} = 3nk_BT$, derive $n\tau > 12k_BT/(\langle\sigma v\rangle Q)$. State every cancellation.

Solution 27.1.

$$E_f = \frac{n^2}{4}\langle\sigma v\rangle Q\tau, \qquad E_{\text{th}} = 3nk_BT.$$

Require $E_f > E_{\text{th}}$, cancel $n > 0$, and rearrange to (27.6) [1].

Problem 27.2 — Lawson scaling with reactivity. Explain why raising $\langle\sigma v\rangle$ at fixed T lowers the required $n\tau_E$. For $\mathrm{D} + \mathrm{T}$ versus $\mathrm{D} + \mathrm{D}$ at $k_BT = 10\,\text{keV}$, which fuel needs the larger confinement product?

Solution 27.2. $\langle\sigma v\rangle$ sits in the denominator of (27.6). At 10 keV, $\langle\sigma v\rangle_{\text{DT}}$ greatly exceeds the $\mathrm{D} + \mathrm{D}$ value and radiation weighs more against $\mathrm{D} + \mathrm{D}$, so $\mathrm{D} + \mathrm{D}$ needs a much larger Lawson product [6, 1].

Problem 27.3 — Tokamak fields. Using the JET and ITER figures, explain why a purely toroidal magnetic field is insufficient and what role the plasma current plays in a tokamak.

Solution 27.3. Curvature/∇B drifts separate charges and drive $\boldsymbol{E} \times \boldsymbol{B}$ loss [6]. The plasma current supplies $\boldsymbol{B}_p$; together with $\boldsymbol{B}_t$ it twists field lines into closed helices so particles average over high- and low-B regions [6, 1].

Problem 27.4 — Inertial versus magnetic. Using Table 27.1 and the NIF chamber photograph, argue how inertial confinement can still satisfy Lawson despite $\tau \sim 10^{-11}\,\text{s}$.

Solution 27.4. Lawson constrains $n\tau$. With $n \sim 10^{31}\,\text{m}^{-3}$, even a $10^{-11}\,\text{s}$ disassembly time yields $n\tau \sim 10^{20}$ in SI units, comparable to a tokamak at $n \sim 10^{20}\,\text{m}^{-3}$ with $\tau \sim 1\,\text{s}$ [14, 6].

Problem 27.5 — Homestake threshold. Why did the Davis chlorine experiment miss most *pp* neutrinos? Which solar branches could it see? Quote the measured and predicted Homestake rates.

Solution 27.5. $E_{\text{th}} = 0.814\,\text{MeV}$ lies above the *pp* endpoint. Homestake sees mainly $^{8}\mathrm{B}$ (with pep and $^{7}\mathrm{Be}$ contributions). The values are $2.55 \pm 0.25\,\text{SNU}$ measured versus $8.0 \pm 1.0\,\text{SNU}$ predicted [1, 6].

Problem 27.6 — Neutrino versus photon information. Photons leave the photosphere; neutrinos leave the core. What does each messenger report?

Solution 27.6. Photons random-walk for 10^4–10^6 years and reflect transport-averaged surface conditions. Neutrinos free-stream from the core in minutes, so their spectrum probes nuclear burning rates today [6, 32].

Problem 27.7 — Why the deficit was not wrong S-factors. After SNO, how do we know the solar neutrino deficit was primarily a flavour effect? What does Fig. 27.10 add?

Solution 27.7. SNO's neutral-current channel counts all active flavours and matches Standard Solar Model fluxes, while ν_e-sensitive rates remain low: the total fusion rate is correct and ν_e have oscillated [1, 14]. The MSW survival curve shows that the suppression is energy dependent and consistent with matter-enhanced oscillation in the Sun.

Chapter 28

Nucleosynthesis of Elements in Stars

"The elements from carbon to nickel have been synthesized in stars."

F. Hoyle (1954)

Lecture 27 closed with hydrogen burning in the Sun and with solar neutrinos as core diagnostics. That process explains only the first step of elemental history: $4p \to {}^4\text{He}$. The Periodic Table contains roughly ninety stable elements and about three hundred stable nuclides. Their origins define **nucleosynthesis**: Big Bang production of the lightest species, followed by stellar fusion building nuclei up to the iron group, and neutron capture beyond the iron peak [6, 1, 14, 37, 32].

The logic of the chapter is causal rather than encyclopaedic. First one must understand why the early Universe stops at lithium; then why cosmic abundances show a gap at Li–Be–B and a peak at iron; then how helium burning bridges that gap via the Hoyle resonance; and finally how massive stars climb the binding-energy curve until fusion ceases and neutrons take over.

28.1 Two stages of nucleosynthesis

Hydrogen and helium still dominate the baryonic mass of the Universe, approximately 75% and 25% by mass, almost unchanged from the values set a few minutes after the Big Bang. Heavier species make up only about 1% of that budget, yet they include the carbon, oxygen, silicon and iron needed for planets and chemistry [14, 6]. Two physically distinct epochs produce them.

Big Bang nucleosynthesis (BBN) operates in an expanding, cooling radiation-dominated plasma. Temperatures fall from MeV to $\sim 0.1\,\text{MeV}$ in minutes. Reactions that require three-body collisions, long confinement, or slow penetration of a high Coulomb barrier simply do not have time. The stable products are almost only ${}^1\text{H}$ and ${}^4\text{He}$, with traces of d, ${}^3\text{He}$ and ${}^7\text{Li}$ ($A \leq 7$). Nothing beyond lithium is made in significant quantity [14, 6].

Stellar nucleosynthesis begins once gravity confines primordial gas in galaxies and stars. Hydrostatic and explosive burning then manufacture carbon through the iron group; neutron captures later reach the heavier species [37, 6]. Solar spectroscopy of dozens of photospheric elements samples galactic material already enriched by earlier stellar generations [14]. The Sun itself is a late generation star: its metals (everything heavier than helium, in astronomical usage) were forged elsewhere.

28.2 Cosmic abundances as a nuclear map

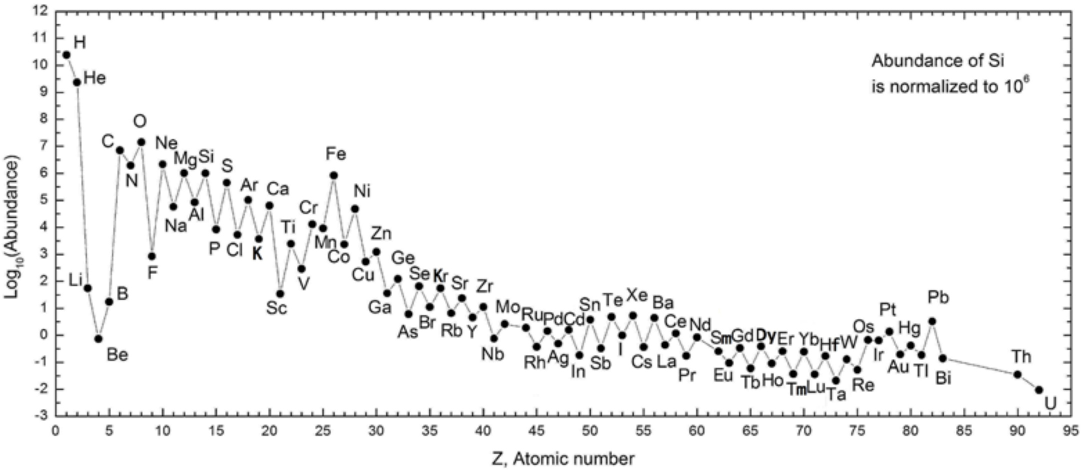


Figure 28.1: Solar-system elemental abundances (logarithmic scale). Hydrogen and helium dominate; Li–Be–B are strongly suppressed; even-Z peaks and an iron-group hump structure the rest of the chart. Image: Wikimedia Commons ("SolarSystemAbundances.png"), CC BY-SA 3.0 [14, 6].

Figure 28.1 is not merely a table of abundances; it is a nuclear fingerprint of the channels that operate in Nature [14, 6, 37].

Hydrogen and helium dominate. Even after $\sim 10^{10}$ years of stellar processing, baryonic matter remains mostly primordial. Stars convert H to He and He to heavier nuclei, but most of the cosmic baryon inventory never passes through a stellar core. That is why the BBN prediction $Y_{\mathrm{He}} \approx 0.25$ still matches the observed helium mass fraction to high accuracy.

Lithium, beryllium and boron are rare. Between helium ($A = 4$) and carbon ($A = 12$) there is no stable $A = 5$ or $A = 8$ nucleus that can serve as a steady stepping stone for charged-particle fusion. Cosmic LiBeB therefore cannot be primary ashes of hydrostatic helium burning; they arise mainly from cosmic-ray spallation of CNO nuclei and from limited Big Bang ^{7}Li. The deep trough at $Z = 3$–5 on Fig. 28.1 is the abundance signature of that mass gap.

Even–odd staggering. Even-Z and even-N nuclei are systematically more abundant than their odd neighbours. Pairing in the SEMF and the tendency of nucleons to cluster into α-like subunits favour even–even species; reaction networks that proceed by α capture inherit that preference.

Iron-group peak. The binding energy per nucleon B/A is maximal near $A \sim 56$–62. Charged-particle fusion therefore releases energy while climbing toward that peak and ceases to power a star beyond it (Sec. 28.7). The abundance hump at iron is the endpoint of exothermic fusion, visible already in the empirical B/A curve of Lecture 25.

Secondary abundance peaks near $A \sim 80$, 130 and 195 are not explained by charged-particle fusion; they are the fingerprints of neutron capture at closed neutron shells (previewed in Sec. 28.9).

28.3 Big Bang nucleosynthesis

28.3.1 Weak freeze-out and the n/p ratio

At temperatures $k_BT \gg 1\,\mathrm{MeV}$, neutrons and protons remain in weak equilibrium through

$$n + \nu_e \leftrightarrow p + e^-, \qquad n + e^+ \leftrightarrow p + \bar{\nu}_e. \tag{28.1}$$

Detailed balance then fixes

$$\frac{n_n}{n_p} = \exp\Big(-\frac{\Delta m\, c^2}{k_BT}\Big), \qquad \Delta m\, c^2 \approx 1.293\,\mathrm{MeV}. \tag{28.2}$$

As the Universe expands, the weak rates eventually fall below the Hubble rate (weak freeze-out, $k_BT \sim 0.8\,\mathrm{MeV}$). The ratio freezes near $n_n/n_p \sim 1/6$. Free-neutron β decay continues until nucleosynthesis begins, leaving a neutron mass fraction $X_n \equiv n_n/(n_n + n_p)$ of order 0.12–0.13.

28.3.2 Helium mass fraction

Almost every surviving neutron ends in ^{4}He. If two neutrons (with two protons) form one α particle, the helium mass is $2n_n$ while the total baryon number density is $n_n + n_p$, so

$$Y_{\mathrm{He}} \approx \frac{2X_n}{1 + X_n} \approx 0.25 \tag{28.3}$$

for $X_n \approx 1/7$–$1/8$ after the network is integrated [14, 6]. The complementary hydrogen fraction is $X_{\mathrm{H}} \approx 1 - Y_{\mathrm{He}} \approx 0.75$. Equation (28.3) is the cleanest example in the course of an observationally tested nuclear cosmology formula: the number of free neutrons at freeze-out sets the cosmic helium abundance.

28.3.3 Deuterium bottleneck and the $A \le 7$ freeze

The deuteron binding is only $B_d = 2.224\,\mathrm{MeV}$. While the radiation bath still contains a high-energy Maxwellian tail, photodisintegration $d + \gamma \to p + n$ destroys deuterium as fast as it forms (the **deuterium bottleneck**). Only when k_BT falls to $\sim 0.1\,\mathrm{MeV}$ can deuterium survive long enough for the strong network

$$d(p,\gamma)^3\mathrm{He}, \quad {}^3\mathrm{He}(d,p)^4\mathrm{He}, \quad {}^3\mathrm{He}({}^3\mathrm{He}, 2p)^4\mathrm{He}, \ldots \tag{28.4}$$

to lock remaining neutrons into ^{4}He, with tiny leftovers of d, ^{3}He and ^{7}Li [14, 6]. Expansion then freezes the network. Because no stable $A = 5, 8$ nucleus exists, BBN cannot climb to carbon: that step requires the sustained density of a stellar helium core.

28.4 From hydrogen burning to a helium core

Main-sequence stars convert hydrogen to helium by the *pp* chains and/or the CNO cycles of Lecture 27. In the Sun the *pp* path carries most of the luminosity; CNO contributes only a few percent at $T_c \sim 1.5 \times 10^7$ K. More massive (hotter) stars burn predominantly by CNO because the larger Coulomb barriers are offset by a higher Gamow peak [1, 36, 6].

As central hydrogen is exhausted, a helium ash core grows. Without nuclear energy generation that core contracts and heats under gravity (virial theorem), while the envelope may expand into a red-giant configuration [14]. Helium ignition occurs only if the core temperature and density reach the triple-α regime ($\sim 10^8$ K, $\rho \sim 10^5\,\mathrm{g\,cm^{-3}}$). A solar-mass star never proceeds through the full carbon–silicon sequence discussed below; those stages require more massive cores capable of successive gravitational heating after each fuel is exhausted [14, 6].

28.5 The triple-α process and helium burning

Figure 28.2 summarises the nuclear logic of helium burning: an equilibrium of unbound ^{8}Be, resonant capture into the Hoyle state of ^{12}C, and a rare radiative branch that locks carbon.

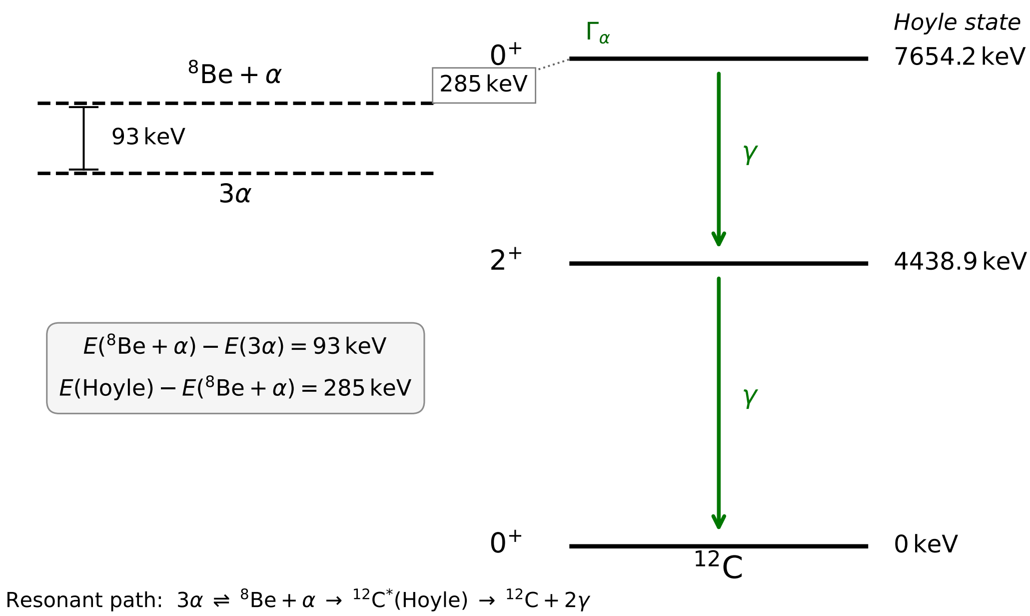


Figure 28.2: Level scheme for the triple-α process (redrawn; energies as in Wikimedia *Hoyle state and possible decay way*, CC BY-SA 4.0). Unbound ^{8}Be sits ≈ 92 keV above 2α; the Hoyle 0^+ of ^{12}C (7.654 MeV) lies ≈ 285 keV above the ^{8}Be + α threshold and radiates via 2^+ (4.439 MeV) to the ground state [38].

28.5.1 Why two-body α fusion fails

There is no exothermic two-body reaction among α particles alone. The mass of ^{8}Be exceeds twice the α mass by

$$|Q| = \left[m(^8\mathrm{Be}) - 2m(^4\mathrm{He})\right]c^2 \approx 92\,\mathrm{keV}, \tag{28.5}$$

so ^{8}Be decays back to 2α with $\tau \sim 10^{-16}$ s [14, 1]. Production is therefore reversible,

$$^4\mathrm{He} + {}^4\mathrm{He} \rightleftharpoons {}^8\mathrm{Be}. \tag{28.6}$$

Unlike the irreversible conversion of hydrogen to helium, helium burning must work with a tiny equilibrium population of an unbound intermediate.

28.5.2 Thermal equilibrium abundance of ^{8}Be

Applying chemical equilibrium (detailed balance) to (28.6) with reduced mass m of the α–α system yields the Saha-like ratio

$$\frac{n(^8\mathrm{Be})}{n(^4\mathrm{He})} = n(^4\mathrm{He}) \left(\frac{2\pi\hbar^2}{m\, k_B T}\right)^{3/2} \mathrm{e}^{-92\,\mathrm{keV}/k_B T} \tag{28.7}$$

(up to statistical-spin factors of order unity) [14]. At typical helium-core conditions $\rho \sim 10^5\,\mathrm{g\,cm^{-3}}$ and $k_B T \sim 15\,\mathrm{keV}$,

$$\frac{n(^8\mathrm{Be})}{n(^4\mathrm{He})} \sim 10^{-9}. \tag{28.8}$$

The exponential alone is $\mathrm{e}^{-92/15} \approx 2 \times 10^{-3}$; the remaining suppression is the density/thermal-wavelength prefactor. A non-resonant capture of a third α on that population would produce negligible cosmic carbon.

28.5.3 The Hoyle resonance and radiative branching

The capture

$$^8\mathrm{Be} + {}^4\mathrm{He} \to {}^{12}\mathrm{C}^* \to {}^{12}\mathrm{C} + \gamma \tag{28.9}$$

is rescued by the 0^+ **Hoyle state** of ^{12}C: 7654 keV above the ground state and only 287 keV above the $^8\mathrm{Be} + \alpha$ threshold. For $k_B T \sim 15\,\mathrm{keV}$ the Gamow energy is $E_\mathrm{G} \sim 200\,\mathrm{keV}$, so the resonance sits in the astrophysically sampled window [38, 14, 6]. Typical widths are $\Gamma_\alpha \approx 8.3\,\mathrm{eV}$ (return to $^8\mathrm{Be} + \alpha$) and $\Gamma_\gamma \sim 3 \times 10^{-3}\,\mathrm{eV}$ (electromagnetic decay toward bound ^{12}C), so only about one formation in 10^3–10^4 locks carbon permanently [14]. The net irreversible process is

$$3\,{}^4\mathrm{He} \to {}^{12}\mathrm{C}, \qquad Q = 7.27\,\mathrm{MeV}. \tag{28.10}$$

That resonance was predicted from the cosmic carbon abundance before it was measured in the laboratory [38, 6].

28.5.4 Helium-burning ashes: carbon and oxygen

Once ^{12}C is available, successive α captures compete with triple-α:

$$^{12}\mathrm{C} + {}^4\mathrm{He} \to {}^{16}\mathrm{O} + \gamma, \tag{28.11}$$

$$^{16}\mathrm{O} + {}^4\mathrm{He} \to {}^{20}\mathrm{Ne} + \gamma. \tag{28.12}$$

A helium-exhausted core is therefore typically a C/O mixture, with the C/O ratio sensitive to the still imperfectly known $^{12}\mathrm{C}(\alpha,\gamma)^{16}\mathrm{O}$ rate [14, 6, 1, 36]. That ratio later controls white-dwarf structure and Type Ia supernova yields.

28.6 Advanced burning in massive stars

When helium is exhausted, the C/O core contracts again. Only sufficiently massive stars reach carbon ignition and the subsequent stages (Fig. 28.3).

28.6.1 Carbon burning as a compound-nucleus branch

At $T \sim (5\text{–}9) \times 10^8\,\mathrm{K}$, two $^{12}\mathrm{C}$ nuclei form an excited $^{24}\mathrm{Mg}^*$ compound system that decays by γ, p or α emission [14]:

$$^{12}\mathrm{C} + {}^{12}\mathrm{C} \to {}^{20}\mathrm{Ne} + {}^{4}\mathrm{He} \quad (Q = 4.62\,\mathrm{MeV}), \tag{28.13}$$

$$\to {}^{23}\mathrm{Na} + p \quad (Q = 2.24\,\mathrm{MeV}), \tag{28.14}$$

$$\to {}^{24}\mathrm{Mg} + \gamma \quad (Q = 13.93\,\mathrm{MeV}). \tag{28.15}$$

The α and proton branches dominate over pure radiative capture. The liberated light particles are promptly absorbed by remaining carbon, feeding $^{13}\mathrm{N}$ and $^{16}\mathrm{O}$ and enriching the odd-A inventory.

28.6.2 Neon, oxygen and silicon burning

After carbon exhaustion the photon bath grows hard enough that photodisintegration begins to free α particles from neon and heavier ash. Neon burning is largely a rearrangement (γ, α) followed by α capture; oxygen burning again proceeds through compound channels analogous to carbon burning [14]. At still higher temperature ($T \sim 3 \times 10^9\,\mathrm{K}$) silicon burning becomes an α-particle quasi-equilibrium: photodisintegration maintains a population of α's that are captured in a ladder

$$^{28}\mathrm{Si} \xrightarrow{\alpha} {}^{32}\mathrm{S} \xrightarrow{\alpha} {}^{36}\mathrm{Ar} \cdots \xrightarrow{\alpha} {}^{56}\mathrm{Ni}, \tag{28.16}$$

after which $^{56}\mathrm{Ni}$ decays toward $^{56}\mathrm{Fe}$ [14, 37, 6]. Concurrent neutrino losses from pair and plasma processes shorten these advanced stages dramatically relative to hydrogen and helium burning [14].

Figure 28.3 is the geometric consequence: each exhausted fuel leaves an ash layer that ignites only after further gravitational heating, producing the classic onion of a pre-supernova massive star.

28.7 Why fusion stops at the iron peak

28.7.1 Binding-energy argument

The binding energy per nucleon B/A rises from light nuclei to a broad maximum near $A \sim 56$–62 and then falls slowly [6, 1]. For a fusion reaction $X + Y \to Z$,

$$Q = B(Z) - B(X) - B(Y) = A_Z\Big(\frac{B}{A}\Big)_Z - A_X\Big(\frac{B}{A}\Big)_X - A_Y\Big(\frac{B}{A}\Big)_Y. \tag{28.17}$$

If both reactants sit below the iron peak and the product remains near that peak, $Q > 0$. If both sit above it, $Q < 0$: hydrostatic fusion cannot power the star [6, 1, 14]. That is why Fig. 28.1

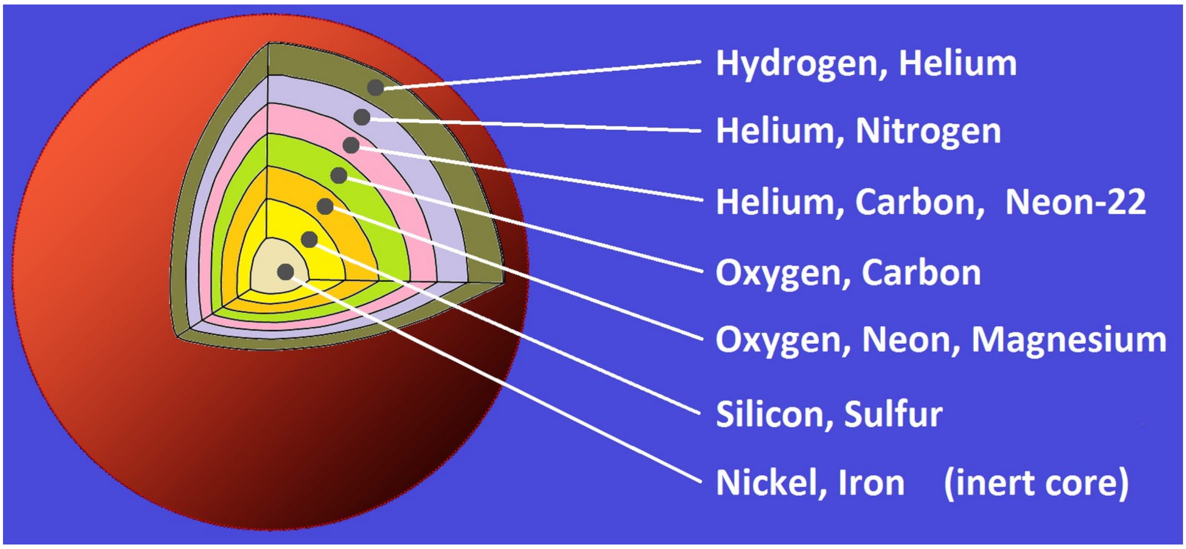


Figure 28.3: Original open-licence cutaway of a massive star near core collapse (onion-skin structure): nested burning shells of H/He, He/C, C/O, O/Ne/Mg, Si/S, surrounding an inert Ni/Fe core. Image: Wikimedia Commons ("Massive star cutaway pre-collapse"), CC0 [14, 6].

shows an iron-group abundance peak and why the onion of Fig. 28.3 ends in an inert Ni/Fe core.

28.7.2 Core collapse

Once silicon burning can no longer support the core against gravity, the iron core approaches the Chandrasekhar mass. Photodisintegration (γ, α and γ, n) and electron captures then cool and deleptonise the core, accelerating collapse. For a sufficiently massive progenitor the bounce and neutrino-driven explosion eject the outer onion and disseminate the synthesised nuclei into the interstellar medium, enriching the next generation of stars – including the Sun [14, 6, 1].

28.8 Side reactions: odd-A nuclei and free neutrons

α capture and even–even preference underproduce odd-A and odd-Z species unless side channels operate. Carbon-burning proton and α branches, followed by secondary captures, help populate those nuclides [14]. More important for the rest of the Periodic Table, some channels liberate free neutrons,

$$^{13}\mathrm{C} + {}^{4}\mathrm{He} \to {}^{16}\mathrm{O} + n, \qquad {}^{12}\mathrm{C} + {}^{12}\mathrm{C} \to {}^{23}\mathrm{Mg} + n \quad (Q = -2.6\,\mathrm{MeV}), \tag{28.18}$$

the second being endothermic but accessible at high temperature [14, 32]. Neutrons feel no Coulomb barrier: once a seed nucleus exists, successive (n, γ) reactions can increase A even when charged-particle fusion is forbidden by the iron peak.

28.9 Beyond iron: neutron capture (preview)

For $A \gtrsim 60$, charged-particle fusion is endothermic. Neutron capture followed by β^- decay changes both A and Z. Two limiting regimes are distinguished [37, 14, 6]:

- the **slow** (s) process, with capture timescales longer than β lifetimes, so the path hugs the valley of β stability;
- the **rapid** (r) process, with captures much faster than β decays, driving nuclei far to the neutron-rich side before they decay back after the neutron flux ends.

Abundance peaks near closed neutron shells ($N = 50, 82, 126$) are the signatures of both paths, shifted relative to each other because the r process β-decays into the valley from more neutron-rich progenitors. A minority of proton-rich rare isotopes require a separate p process (photodisintegration or neutrino processing of seed nuclei). The nuclear lesson of this chapter is that charged-particle stellar fusion builds the Periodic Table only through the iron group; neutrons carry nucleosynthesis the rest of the way [37, 6].

Key idea

- BBN: freeze-out $n/p \Rightarrow Y_{\mathrm{He}} \approx 2X_n/(1+X_n) \approx 0.25$; $A \leq 7$ only.
- Abundances map nuclear structure: LiBeB gap, even–odd staggering, iron peak, s/r shell peaks.
- Triple-α: ^{8}Be Saha equilibrium + Hoyle resonance (287 keV above threshold) $\Rightarrow$ carbon.
- Massive-star onion ends at inert Fe/Ni; Q from B/A explains why.
- Free neutrons enable s/r capture beyond iron.

28.10 Further physics

Detailed balance origin of (28.7). Equating the $\alpha\alpha$ formation rate through a continuum T-matrix with the inverse decay $^8\mathrm{Be} \to 2\alpha$ and integrating over thermal Maxwellians cancels the unknown matrix element and leaves a Saha-like ratio controlled only by $Q = -92\,\mathrm{keV}$, reduced mass, and density [14]. The same logic underlies Saha ionisation in stellar atmospheres.

Partial widths of the Hoyle state. Typical values are $\Gamma_\alpha \sim 8\,\mathrm{eV}$ into the $\alpha + {}^8\mathrm{Be}$ channel and $\Gamma_\gamma \sim 3 \times 10^{-3}\,\mathrm{eV}$ into electromagnetic decay toward bound ^{12}C. The radiative branching $\Gamma_\gamma/\Gamma \sim 4 \times 10^{-4}$ makes carbon production rare per formation of $^{12}\mathrm{C}^*$, but still cosmically decisive once integrated over helium-burning lifetimes [14, 38].

Primary versus secondary nuclei. Species produced from primordial H/He alone (e.g. C,O in massive stars) are primary; those requiring seed metals from earlier generations (many s-process nuclei, CNO catalysts in Population I stars) are secondary. Abundance trends with metallicity distinguish the two [37, 14].

Connection to solar neutrinos. Hydrogen-burning rates that set solar neutrino fluxes (Lecture 27) are the first link of the same chain that later builds CNO catalysts and seeds for capture nucleosynthesis [36, 1].

28.11 Problems with solutions

Problem 28.1 — Helium mass fraction from X_n. Derive $Y_{\mathrm{He}} \approx 2X_n/(1+X_n)$. Evaluate it for $X_n = 1/8$ and for $X_n = 0.12$.

Solution 28.1. If every two neutrons (with two protons) form one ^{4}He, the helium mass is $4n_n/2 = 2n_n$ while the total baryon mass is $n_n + n_p$. With $X_n = n_n/(n_n + n_p)$ one obtains (28.3). For $X_n = 1/8$, $Y = 2/9 \approx 0.222$; for $X_n = 0.12$, $Y \approx 0.214$ before small network corrections raise the standard BBN result near 0.25 [14].

Problem 28.2 — ^{8}Be equilibrium. Starting from detailed balance for $^4\mathrm{He} + {}^4\mathrm{He} \rightleftharpoons {}^8\mathrm{Be}$ with $Q = -92\,\mathrm{keV}$, explain the exponential factor in (28.7) and estimate why $n(^8\mathrm{Be})/n(^4\mathrm{He}) \sim 10^{-9}$ at $k_BT = 15\,\mathrm{keV}$.

Solution 28.2. The Boltzmann factor $\mathrm{e}^{-|Q|/k_BT}$ suppresses the unbound state; $\mathrm{e}^{-92/15} \approx \mathrm{e}^{-6.1} \approx 2 \times 10^{-3}$ already. The remaining $\sim 10^{-6}$ comes from the density and thermal wavelength prefactor in (28.7) at $\rho \sim 10^5\,\mathrm{g\,cm^{-3}}$ [14].

Problem 28.3 — Why the Hoyle state matters. The Hoyle 0^+ lies 283 keV above the $^8\mathrm{Be} + \alpha$ threshold. Why does that placement matter for helium burning at $k_BT \sim 15\,\mathrm{keV}$?

Solution 28.3. The Gamow peak for $\alpha + {}^8\mathrm{Be}$ sits near $E_{\mathrm{G}} \sim 200\,\mathrm{keV}$ at that temperature, so a resonance 283 keV above threshold lies in the astrophysically sampled window and enhances the rate by many orders of magnitude relative to a non-resonant extrapolation [38, 14, 6].

Problem 28.4 — Fusion Q from B/A. Using (28.17), argue schematically why $^{28}\mathrm{Si} + {}^{28}\mathrm{Si} \to {}^{56}\mathrm{Ni}$ can release energy while $^{56}\mathrm{Fe} + {}^{56}\mathrm{Fe} \to {}^{112}\mathrm{Cd}$ cannot power a star.

Solution 28.4. Near $A \sim 28$, B/A is still rising toward the iron peak, so the fused product has larger B/A and $Q > 0$. Near $A \sim 56$, B/A is maximal or already declining, so fusing two iron-group nuclei decreases the average binding and $Q < 0$ [6, 1].

Problem 28.5 — Reading the onion figure. Using Fig. 28.3, list the shells from outside in and state which stage of burning produced each ash layer.

Solution 28.5. Outer H/He (H burning ash and envelope); He/C/Ne (helium burning); C/O (carbon burning products); O/Ne/Mg (neon burning); Si/S (oxygen/silicon burning); central Ni/Fe (inert core). Each inner shell requires a higher ignition temperature than the one outside it [14].

Problem 28.6 — Why no carbon in BBN? Explain why Big Bang nucleosynthesis stops short of carbon even though helium is abundant.

Solution 28.6. No stable $A = 5, 8$ bridge exists; the ^{8}Be equilibrium needed for triple-α requires stellar densities sustained far longer than the minutes available in BBN [14, 6].

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ABOUT THIS BOOK

This book grew out of a nuclear and particle physics course. It expands each lecture into a self-contained chapter for continuous reading or use alongside formal study. The aim is to help students build a clear physical picture of the nucleus and of the processes that govern it, without losing the tools needed for examinations, research projects, and further study.

TOPICS

- Nuclear size, density, and binding
- Semi-empirical mass formula and mass parabolas
- Nuclear force, scattering, and meson exchange
- Shell model, radioactivity, and reactions
- Fission, fusion, and reaction rates

Bharat Kumar teaches nuclear and particle physics and works on the theory of dense nuclear matter and neutron stars. These notes reflect that classroom experience.

Nuclear and Particle Physics

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